The Institute of Chartered Accountants of Bangladesh (ICAB)

Corporate Reporting

Volume II



Workbook
For CA Advanced Level Exams

CA BANGLADESH

THE INSTITUTE OF CHARTERED ACCOUNTANTS OF BANGLADESH

www.icab.org.bd

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The Institute of Chartered Accountants of Bangladesh

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Questions within the Workbook should be treated as preparation questions, providing you with a firm foundation before you attempt the exam-standard questions.

**** Italic denotes changes***

Chapter 11

Earnings per share

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 EPS: overview of material covered in earlier studies

2 Basic EPS: weighted average number of shares

3 Basic EPS: profits attributable to ordinary equity holders

4 Diluted earnings per share

5 Diluted earnings per share: convertible instruments

6 Diluted EPS and options

7 Diluted EPS: contingently issuable shares

8 Retrospective adjustments, presentation and disclosure

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Explain and appraise accounting standards that relate to reporting performance: in respect of presentation of financial statements; revenue; operating segments continuing and discontinued operations; EPS; interim reporting
- Determine for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**EPS: overview of material covered in earlier studies** IAS 33, *Earnings per Share* seeks to provide guidance to ensure a consistent basis is followed so that a meaningful comparison can be made over time.	**Approach** This section will be revision from your earlier studies, so should not take long to work through. Do not overlook the brief subsection on auditing EPS. **Stop and think** Why is an adjustment needed for preference shares?	You may be asked for a simple EPS calculation, or more likely re-calculation once a series of adjustments have been made to the SPLOCI. Otherwise you are more likely to get complications that were not tested at Professional Level.	**IQ1: Adjustment for preference shares** This is a straightforward question, revising material from your earlier studies.
2	**Basic EPS: weighted average number of shares** This could be used in practice to compare companies for investment purposes.	**Approach** While you have done the weighted average calculations at Professional Level, you may not have come across all the factors that determine when shares are included in the weighted average.	These topics could come up as part of a financial analysis question.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		Stop and think What is the impact of bonus issues, rights issues and share consolidations on EPS?		
3	**Basic EPS: profits attributable to ordinary equity holders** This section focuses on the adjustments to earnings arising as a result of preference shares.	**Approach** Focus on the redeemable versus irredeemable distinction and examples rather than the increasing rate preference shares, which is less likely to come up. **Stop and think** Why does it matter whether the preference shares are redeemable or not?	This area has not yet been examined under the current syllabus.	**IQ2: Redeemable preference shares** This a short, straightforward question to reinforce the main point in this section.
4	**Diluted earnings per share** In addition to basic EPS, IAS 33 requires a diluted figure which takes into account the existence of convertible instruments already issued that could increase the number of shares.	**Approach** This section revises material from your earlier studies before moving on to more difficult topics. **Stop and think** What does 'antidilutive' mean?	Diluted earnings per share is often tested, together with basic EPS, in a question requiring adjustments to financial statements.	N/A
5	**Diluted EPS: convertible instruments** Convertible instruments are one of the most common reasons for dilution.	**Approach** Although this topic won't be entirely new, you may not have considered the indicators that a convertible instrument is antidilutive. **Stop and think** What is the impact of	This is a regular topic for examination and may arise in the context of financial instruments.	**IQ3: Test of dilution** This is a comprehensive question to enable a firm grasp of the dilutive/ antidilutive distinction,

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		convertibles on EPS?		
6	**Diluted EPS: options** Options are another reason for dilution. Options are dilutive when they are 'in the money'.	**Approach** The key method to take away is: Calculate the number of shares issued at market price. • Find the number of shares issued at nil consideration (the dilutive shares). • Add the dilutive shares to the number of shares under basic EPS. **Stop and think** What is the impact of options on EPS?	Next to convertibles, options are the most common way that the complications of IAS 33 will be tested.	**IQ4: Diluted earnings per share** This is a crucial question as options are new to Advanced Level.
7	**Diluted EPS: contingently issuable shares** These arise in a business combination, where consideration takes the form of shares which will only be issued if certain targets are met in the future. They are also issued to senior staff as a reward for meeting certain targets.	**Approach** Key points to remember: • Do not include the shares in basic EPS unless and until issued. • Include in DEPS only if the conditions have been met. • Include from the beginning of the period or the date of the agreement if later. **Stop and think** What is the impact of contingently issuable shares on EPS?	These are usually tested in the context of performance-based options.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
8	**Retrospective adjustments, presentation and disclosure** Additional EPS figures are often reported by entities based on what they consider to be sustainable earnings. These are intended to provide a more realistic measure of future performance.	**Approach** Note how IAS 33 interacts with IAS 8 in requiring adjustments adjusted for the effects of errors and those resulting from changes in accounting policies accounted for retrospectively. **Stop and think** Why are additional EPS measures often disclosed?	This topic is most likely to be tested by requiring DEPS for prior periods to be adjusted.	N/A

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 EPS: overview of material covered in earlier studies

Section overview

This section reviews the material on basic and diluted earnings per share covered at Professional Level.

1.1 Scope

IAS 33, *Earnings per Share* applies to entities whose ordinary shares are publicly traded or are in the process of being issued in public markets.

1.2 Basic earnings per share (EPS)

- Basic earnings per share (EPS) is calculated as:

  (Profit/(loss) attributable to ordinary equity holders of the parent) ÷ (Weighted average number of ordinary shares outstanding during the period)

- **Profit attributable to the ordinary equity holders** is based on profit after tax after the deduction of preference dividends and other financing costs in relation to preference shares classified as equity under IAS 32. Whether an adjustment is needed depends on the type of preference share:

Redeemable preference shares	No adjustment is required as these shares are classified as liabilities and the finance charge relating to them will already have been charged to profit or loss as part of finance charges.
Irredeemable preference shares	These shares are classified as equity and the dividend relating to them is disclosed in the statement of changes in equity. This dividend must be deducted from profit for the year to arrive at profit attributable to the ordinary shareholders.

- The **weighted average number of shares** should be adjusted for changes in the number of shares without a corresponding change in resources, for example a bonus issue, by assuming that the new number of shares had always been in issue.

  Shares should generally be included in the weighted average number of shares from the date the consideration for their issue is receivable.

- An entity is required to calculate and present a basic EPS amount based on the profit or loss for the period attributable to the ordinary equity holders of the parent entity. If results from 'continuing operations' and 'discontinued operations' are reported separately, EPS on these results should also be separately reported.

1.3 Diluted earnings per share

- A diluted EPS figure should also be reported by an entity. A dilution is a reduction in the EPS figure (or increase in a loss per share) that will result from the issue of more equity shares on the conversion of convertible instruments already issued.
- For the purpose of calculating diluted earnings per share, an entity shall adjust profit or loss attributable to ordinary equity holders of the parent entity, and the weighted average number of shares outstanding for the effects of all dilutive potential ordinary shares.

1.4 Presentation

- Basic and diluted EPS figures (and from continuing operations if reported separately) should be presented on the face of the statement of comprehensive income with equal prominence.
- Where changes in ordinary shares occur during the accounting period, an amendment is necessary to the number of shares used in the EPS calculations. In some situations, the EPS in prior periods will also have to be adjusted.
- Treasury shares are accounted for as a deduction from shareholders' funds. Since such shares are no longer available in the market, they are excluded from the weighted average number of ordinary shares for the purpose of calculating EPS.



Interactive question 1: Adjustment for preference shares

A company has issued £100,000 4% redeemable non-cumulative preference shares. Should the dividend be subtracted from the reported profit after tax figure for the calculation of EPS?

See **Answer** at the end of this chapter.

1.4.1 Auditing issues

Clearly, the main auditing issues with EPS can be summarised as follows:

- Ensuring that the stated EPS figures comply with IAS 33 (for example, obtaining a breakdown of the amounts and tracing each of them back to their source)
- Ensuring that both basic and diluted EPS figures are presented (so obtaining a copy of the draft financial statements to vouch this)
- Addressing any other disclosure issues as required (for example, treasury shares) by reference to the draft financial statements

2 Basic EPS: weighted average number of shares



Section overview

This section deals with certain adjustments to the number of shares used for the calculation of basic earnings per share.

IAS 33 requires that a time-weighted average number of shares should be used in the denominator of the earnings per share calculation. The basic idea of how to calculate such a weighted average has been covered at Professional Level. In this section we deal with issues relating to the treatment of share repurchases, partly paid shares, bonus and rights issues and the impact of consolidation.

2.1 Calculation of the weighted average number of shares

The use of the weighted average number of ordinary shares outstanding during the period reflects the possibility that the amount of shareholders' capital varied during the period.

The weighted average number of ordinary shares outstanding during the period is the number of ordinary shares outstanding at the beginning of the period, adjusted by the number of ordinary shares bought back or issued during the period multiplied by a time-weighting factor.

2.2 Time weights determination

Shares are usually included in the weighted average number of shares from the date consideration is receivable (which is generally the date of their issue), for example:

- Ordinary shares issued in exchange for cash are included when cash is receivable.
- Ordinary shares issued on the voluntary reinvestment of dividends on ordinary or preference shares are included when dividends are reinvested.
- Ordinary shares issued as a result of the conversion of a debt instrument to ordinary shares are included from the date that interest ceases to accrue.
- Ordinary shares issued in place of interest or principal on other financial instruments are included from the date that interest ceases to accrue.
- Ordinary shares issued in exchange for the settlement of a liability of the entity are included from the settlement date.
- Ordinary shares issued as consideration for the acquisition of an asset other than cash are included as of the date on which the acquisition is recognised.
- Ordinary shares issued for the rendering of services to the entity are included as the services are rendered.
- Ordinary shares issued as part of the cost of a business combination are included in the weighted average number of shares from the acquisition date. This is because the acquirer incorporates into its results the acquiree's profits and losses from that date.
- Ordinary shares that will be issued upon the conversion of a mandatorily convertible instrument are included in the calculation of basic earnings per share from the date the contract is entered into.
- Contingently issuable shares are included in the calculation of basic earnings per share only from the date when all necessary conditions for their issue are satisfied. Shares that are issuable solely after the passage of time are not contingently issuable shares, because the passage of time is a certainty.
- Outstanding ordinary shares that are contingently returnable (ie, subject to recall) are excluded from the calculation of basic earnings per share until the date the shares are no longer subject to recall.

Worked example: Weighted average number of ordinary shares

The following information is provided for an entity.

		Shares issued	Treasury shares	Shares outstanding
1 Jan 20X1	Balance at beginning of year	2,000	300	1,700
31 May 20X1	Issue of new shares for cash	800	–	2,500
1 Dec 20X1	Purchase of treasury shares for cash	–	250	2,250
31 Dec 20X1	Balance at year end	2,800	550	2,250

Requirement

Calculate the weighted average number of shares in issue during the year.

Solution

The calculation can be performed on a cumulative basis:

			Weighted average
1 Jan X1 – 30 May X1	1,700	× 5/12	708
Share issue	800		
31 May X1 – 30 Nov X1	2,500	× 6/12	1,250
Share purchase	(250)		
1 Dec X1 – 31 Dec X1	2,250	× 1/12	188
			2,146

or alternatively each issue or recall treated separately:

			Weighted average
1 Jan X1 – 31 Dec X1	1,700	12/12	1,700
31 May X1 – 31 Dec X1	800	7/12	467
1 Dec X1 – 31 Dec X1	(250)	1/12	(21)
			2,146

2.3 Partly paid shares

Where ordinary shares are issued but not fully paid, they are treated in the calculation of basic earnings per share as a fraction of an ordinary share to the extent that they were entitled to participate in dividends during the period relative to a fully paid ordinary share.

To the extent that partly paid shares are not entitled to participate in dividends during the period they are treated as the equivalent of warrants or options in the calculation of diluted earnings per share. The unpaid balance is assumed to represent proceeds used to purchase ordinary shares. The number of shares included in diluted earnings per share is the difference between the number of shares subscribed and the number of shares assumed to be purchased.

Worked example: Partly paid shares

At 1 January 20X5 an entity had 900 ordinary shares in issue. It issued 600 new shares at 1 September 20X5, at a subscription price of £4 per share. At the date of issue each shareholder paid
£2. The balance of £2 per share will be paid during 20X6. Each part-paid share will be entitled to dividends in proportion to the percentage of the issue price paid up on the share. The entity has a year end of 31 December.

Requirement

Calculate the weighted average number of shares for the year ended 31 December 20X5.

Solution

The new shares issued should be included in the calculation of the weighted average number of shares in proportion to the percentage of the issue price received from the shareholding during the period.

	Shares issued	Fraction of period	Weighted average shares
1 January 20X5 – 31 August 20X5	900	8/12	600
Issue of new shares for cash, part paid (2/4 × 600)	300		
1 September 20X5 – 31 December 20X5	1,200	4/12	400
Weighted average number of shares			1,000

2.4 The impact of bonus issues and share consolidations on the number of shares

The weighted average number of ordinary shares outstanding during the period must be adjusted for events that have changed the number of ordinary shares outstanding without a corresponding change in resources. These include the following:

- Bonus issues (capitalisation issues)
- Share consolidation

2.4.1 Bonus issue

In a bonus or capitalisation issue, or a share split, ordinary shares are issued to existing shareholders for no additional consideration. Therefore, the number of ordinary shares outstanding is increased without a corresponding increase in resources.

The number of ordinary shares outstanding **before** the event is adjusted for the proportionate change in the number of ordinary shares outstanding as if the event had occurred at the beginning of the earliest period presented. For example, on a 2 for 1 bonus issue, the number of ordinary shares outstanding before the issue is multiplied by three to obtain the new total number of ordinary shares, or by two to obtain the number of additional ordinary shares.

Bonus issue after the reporting date

Where a bonus issue takes place after the reporting date but before the financial statements are authorised for issue, the number of shares in the EPS calculation is adjusted for the current and prior periods as though the bonus issue took place during the current year.



Worked example: Bonus issue

The following information is given for an entity.

Profit attributable to ordinary equity holders for y/e 30 September 20X6	£300
Profit attributable to ordinary equity holders for y/e 30 September 20X7	£900
Ordinary shares outstanding until 30 September 20X7	200

Bonus issue 1 October 20X7 – two ordinary shares for each ordinary share outstanding at 30 September 20X7

Requirement

Calculate the basic earnings per share for 20X6 and 20X7.

Solution

The bonus issue arose in the period after the reporting date. It should be treated as if the bonus issue arose during 20X7, and EPS calculated accordingly:

Additional shares issued 200 × 2 = 400 Basic EPS 20X7

£900 ÷ (200 + 400) = £1.50

Basic EPS 20X6

£300 ÷ (200 + 400) = £0.50

2.4.2 Share consolidation

A consolidation of ordinary shares generally reduces the number of ordinary shares outstanding without a corresponding reduction in resources.

Sometimes, however, shares are repurchased at fair value, and in this instance, there is a corresponding reduction in resources. An example is a share consolidation combined with a special dividend. In this case, the weighted average number of ordinary shares outstanding for the period is adjusted for the reduction in the number of ordinary shares from the date the special dividend is recognised.



Worked example: Share consolidation

At the start of its financial year ended 31 December 20X5, an entity had 10 million ordinary shares in issue. On 30 April 20X5 it issued three million shares in consideration for the acquisition of a majority holding in another entity. On 31 August 20X5 it went through a share reconstruction by consolidating the shares in issue, on the basis of one new share for two old shares.

Requirement

Calculate the weighted number of shares in issue for the year to 31 December 20X5.

Solution

The three million new shares issued at the time of the acquisition should be weighted from the date of issue, but the consolidation should be related back to the start of the financial year (and to the start of any previous years presented as comparative figures).

The calculation of the weighted number of shares in issue is as follows:

Number		**Weighting**	
At 1 January 20X5			10,000,000
Effect of consolidation is to halve the number of shares (since one new share was issued for every two old			
Adjusted number			
shares held)	(5,000,000)		
	5,000,000	12/12	5,000,000

30 April 20X5 issue	3,000,000		
Effect of consolidation is to halve the number of shares	(1,500,000)		
	1,500,000	8/12	1,000,000
Weighted average shares in issue			6,000,000



Worked example: Special dividend and share consolidation

A company has issued 20,000 shares with a nominal value of 20p each. At the beginning of 20X7 it considers whether to launch a share repurchase of 2,000 shares at the current market price of £2 per share, or pay a special dividend of 20p per share to be followed by a share consolidation of 9 new shares for 10 old shares. The profit after tax for 20X6 and 20X7 is expected to be £4,000 for each year. Interest rates stand at 5% and the company tax rate is 20%.

Requirement

Calculate the basic EPS for the two alternatives.

Solution Basic EPS

(1) **Share repurchase at fair value**

	20X7	**20X6**
	£	**£**
Profit for the year	4,000	4,000
Loss of interest as cash paid out £4,000 × 0.05 × 0.80*	(160)	
Earnings	3,840	4,000
Number of shares outstanding	18,000	20,000
Earnings per share	21.33p	20.00p

(2) **Special dividend followed by share consolidation**

	20X7	**20X6**
	£	**£**
Profit for the year	4,000	4,000
Loss of interest on cash paid out as dividend £4,000 × 0.05 × 0.80*	(160)	
Earnings	3,840	4,000

The effect of share consolidation is to leave the total nominal value of outstanding shares the same, but to reduce the number of shares from 20,000 to 18,000, and raising the market price of a share from £2 to £2.22.

	20X7	**20X6**
Number of shares	18,000	20,000
Earnings per share	21.33p	20.00p

No adjustment to prior year's EPS is made for the share consolidation.

* 0.80 = (1 – tax rate)

2.5 Rights issue

A rights issue is an issue of shares for cash to the existing ordinary equity holders in proportion to their current shareholdings, at a discount to the current market price.

Because the issue price is below the market price, a rights issue is in effect a combination of **an issue at fair value and a bonus issue.**

In order to calculate the weighted average number of shares when there has been a rights issue, an adjustment factor is required:

Adjustment factor = Pre-rights issue price of shares ÷ Theoretical ex-rights price (TERP)

The TERP is the theoretical price at which the shares would trade after the rights issue and takes into account the diluting effect of the bonus element in the rights issue. It is calculated as:

TERP = (Total market value of original shares pre-rights issue + Proceeds of rights issue) ÷ Number of shares post rights issue

The adjustment factor is used to increase the number of shares in issue **before** the rights issue for the bonus element.

Where the rights are to be publicly traded separately from the shares before the exercise date, fair value for the purposes of this calculation is established at the close of the last day on which the shares are traded together with the rights.

Worked example: Rights issue

The following information is provided for an entity which is making a rights issue.

	20X4	20X5	20X6
Profit attributable to ordinary equity holders of the parent entity	£1,100	£1,500	£1,800

Shares outstanding before rights issue: 500 shares

Rights issue: One new share for each five outstanding shares (100 new shares total)

Exercise price: £5.00

Date of rights issue: 1 January 20X5

Last date to exercise rights: 1 March 20X5

Market price of one ordinary share immediately before exercise on 1 March 20X5: £11.00

Reporting date 31 December

Requirement

Calculate the theoretical ex-rights value per share and the basic EPS for each of the years 20X4, 20X5 and 20X6.

Solution

Calculation of theoretical ex-rights value per share

	No.	Price	Total
Pre-rights issue holding	5	£11	£55
Rights share	1	£5	£5
	6		£60

Therefore TERP = £60/6 = £10

(The TERP may also be calculated on the basis of all shares in issue ie, £6,000/600 shares.)

Calculation of adjustment factor

Adjustment factor = Fair value share before exercise of rights ÷ Theoretical ex-rights value per share

Calculation of basic earnings per share 20X4

20X4 basic EPS as originally reported: £1,100 ÷ 500 shares =	£2.20
20X4 basic EPS restated for rights issue in 20X5 accounts: £1,000 ÷ (500 shares) × (adjustment factor) =	£2.00

Alternatively the restated EPS may be calculated by applying the reciprocal of the adjustment factor to the basic EPS as originally reported:

£2.20 × 10/11 =	£2.00

20X5

Weighted average number of shares:

1 January – 28 February	500 × 11/10	× 2/12	92
Rights issue	100		
1 March – 31 December	600	× 10/12	500
			592

Basic EPS including effects of rights issue: £1,500 ÷ 592 shares = £2.53

20X6

Basic EPS: £1,800 ÷ 600 shares = £3.00

Worked example: Cash and rights issue

An entity had 14 million ordinary shares in issue on 1 January 20X4 and 20X5. In its financial year ended 31 December 20X5 it issued further shares, as follows:

- On 1 April 20X5, 4 million shares in consideration for the majority holding in another entity.
- On 1 July 20X5 a rights issue of 1 for 6 at £15 when the market price of the existing shares was £20. There were 18 million shares in issue at this date, another 3 million shares were therefore issued.

A profit of £17 million attributable to the ordinary equity holders was reported for 20X5 and £14 million for 20X4.

Requirement

Calculate the earnings per share for 20X5 and restate the comparative for 20X4.

Solution

As the shares issued on the acquisition were issued at full fair value, a time apportionment adjustment over the period they are in issue is required.

The rights issue shares require a time apportionment adjustment and an adjustment for the bonus element in the rights. The latter adjustment should be applied to the shares issued on 1 April as well as to those issued earlier.

To adjust for the bonus element the theoretical ex-rights fair value per share is required:

Computation of theoretical ex-rights price (TERP):

	No.	Price	Total
Pre-rights issue holding	6	£20	£120
Rights share	1	£15	£15
	7		£135

Therefore TERP = £135/7 = £19.29

The adjustment factor is therefore £20/£19.29

20X4 and earlier EPS figures would be adjusted by dividing the corresponding earnings figure by 1.037.

The weighted number of shares in issue in 20X5 is calculated as:

	Number	Weighting	Adjusted number
1 January to 31 March	14,000,000 × 20/19.29	3/12	3,628,823
Issue 1 April	4,000,000		
1 April to 30 June	18,000,000 × 20/19.29	3/12	4,665,630
Rights issue 1 July	3,000,000		
1 July to 31 December	21,000,000	6/12	10,500,000
Weighted average shares in issue			18,794,453

20X5 EPS:

£17m/18,794,453 shares = £0.90

20X4 restatement – original EPS: £14m ÷ 14m shares = £1.00

£1.00 × 19.29/20.00 = £0.96

3 Basic EPS: profits attributable to ordinary equity holders

Section overview

In this section we discuss the adjustments that are required to earnings as a result of preference shares, in order to calculate profits attributable to ordinary shareholders (equity holders).

As we have seen in earlier studies, for the purpose of calculating basic earnings per share, we must calculate the amounts attributable to ordinary equity holders of the parent entity in respect of profit or loss.

This is done in two steps:

- First the profit or loss which includes all items of income and expense that are recognised in a period, including tax expense, dividends on preference shares classified as liabilities or non-controlling interest is calculated according to IAS 1, *Presentation of Financial Statements*.
- In the second step the calculated profit or loss is adjusted for the after-tax amounts of preference dividends, differences arising on the settlement of preference shares, and other similar effects of preference shares classified as **equity** under IAS 32, *Financial Instruments: Presentation*.

3.1 Adjusting earnings for the impact of preference shares

Where an entity has preference shares in issue depending on their terms these will be classified under IAS 32 as either:

- **financial liabilities**; or
- **equity.**

3.1.1 Preference shares classified as equity

Any dividends and other appropriations (for example, amortised premium or discount) is debited directly to equity, in the statement of changes in equity. Therefore an adjustment is required to deduct these amounts from the profit for the period in order to derive the profit attributable to ordinary shareholders (equity holders).

3.1.2 Preference shares classified as liabilities

Any dividends or other appropriations are treated as finance costs in arriving at profit for the period and no adjustment is required.

In both the above cases the treatment is the same, as in both cases the amounts are deducted from profit attributable to ordinary shareholders. In the latter case the deduction will already have been made in arriving at reported profit and loss, however in the former case an adjustment to reported profits is required.

3.1.3 Cumulative preference shares

Where preference shares are **cumulative**, the dividends for the period need to be taken into account irrespective of whether these have been declared or not.

3.1.4 Non-cumulative preference shares

For non-cumulative preference dividends only the amount of dividend declared for the period should be deducted in arriving at profit or loss attributed to ordinary equity holders.



Interactive question 2: Redeemable preference shares

Turaco is a company listed on a recognised stock exchange. Given below is an extract from its statement of comprehensive income for the year ended 31 December 20X6.

	£
Profit before tax	500,000
Tax	150,000
Profit after tax	350,000

The company paid an ordinary dividend of £20,000 and a dividend on its redeemable preference shares of £70,000.

The company had £100,000 of £0.50 ordinary shares in issue throughout the year and authorised share capital of 1,000,000 ordinary shares.

Requirement

What is the basic earnings per share figure for the year according to IAS 33, *Earnings per Share*?

See **Answer** at the end of this chapter.

3.1.5 Other adjustments in respect of preference shares Increasing rate preference shares

These are preference shares that provide for a low initial dividend to compensate an entity for selling the preference shares at a discount, or an above-market dividend in later periods to compensate investors for purchasing preference shares at a premium.

Under IAS 32, *Financial Instruments: Presentation* and IFRS 9, *Financial Instruments* any original issue discount or premium on increasing rate preference shares is amortised using the effective interest method and treated as a preference dividend for the purposes of calculating earnings per share.

In addition, there may be other elements amortised such as transaction costs.

All these elements should be deducted in arriving at the earnings attributed to ordinary equity holders.



Worked example: Increasing rate preference shares

Dennison Co issued non-convertible, non-redeemable class A cumulative preference shares of £100 par value on 1 January 20X1. The class A preference shares are entitled to a cumulative annual dividend of £7 per share starting in 20X4.

At the time of issue, the market rate dividend yield on the class A preference shares was 7% a year. Thus, Dennison Co could have expected to receive proceeds of approximately £100 per class A preference share if the dividend rate of £7 per share had been in effect at the date of issue.

There was, however, to be no dividend paid for the first three years after issue. In consideration of these dividend payment terms, the class A preference shares were issued at £81.63 per share ie, at a discount of £18.37 per share. The issue price can be calculated by taking the present value of £100, discounted at 7% over a three-year period.

Requirement

Calculate the imputed dividends attributable to preference shares that need to be deducted from earnings to determine the profit or loss attributable to ordinary equity holders.

Solution

Because the shares are classified as equity, the original issue discount is amortised to retained earnings using the effective interest method and treated as a preference dividend for earnings per share purposes. To calculate basic earnings per share, the following imputed dividend per class A preference share is deducted to determine the profit or loss attributable to ordinary equity holders of the parent entity.

Year	Carrying amount of class A preference shares 1 Jan	Imputed dividend	Carrying amount of class A preference shares 31 Dec	Dividend paid
	£	£	£	£
20X1	81.63	5.71	87.34	–
20X2	87.34	6.12	93.46	–
20X3	93.46	6.54	100.00	–
Thereafter	100.00	7.00	107.00	(7.00)

Convertible preference shares

An entity may achieve early conversion of convertible preference shares by improving the original conversion terms or paying additional consideration.

- Where this is the case, then the excess amount transferred as a result of the improvement of conversion terms is treated as a return to the preference shareholders and so should be deducted in arriving at earnings attributable to ordinary equity holders.

Deduction	=	Fair value of ordinary shares issued/consideration paid	–	Fair value of ordinary shares issuable under original terms



Worked example: Cumulative convertible preference shares

An entity issued £100,000 2% cumulative convertible preference shares in 20X4 and the shares were due to be converted in the current year, 20X6.

The convertible shares were converted at the beginning of 20X6 and no dividend was accrued in respect of the year, although the previous year's dividend was paid immediately before conversion. The terms of conversion were also amended and the revised terms entitled the preference shareholders to a total additional 100 ordinary shares on conversion with a fair value of £300.

Requirement

If the profit attributable to ordinary equity holders for the year is £150,000 what adjustments need to be made for the purpose of calculating EPS in 20X6?

Solution

The excess of the fair value of additional ordinary shares issued on conversion of the convertible preference shares over fair values of the ordinary shares to which they would have been entitled under the original conversion terms is deducted from profit as it is an additional return to the convertible preference shareholders.

	£
Profits attributable to the ordinary equity holders	150,000
Fair value of additional ordinary shares issued on conversion of convertible preference shares	(300)
	149,700

There is no adjustment in respect of the preference shares as no dividend accrual was made in respect of the year. The payment of the previous year's cumulative dividend is ignored for EPS purposes as it will have been adjusted for in the prior year.

Repurchase of preference shares

- Where the fair value of consideration paid to preference shareholders exceeds the carrying value of the preference shares repurchased, the excess is a return to the preference shareholders and must be deducted in calculating profits attributable to ordinary equity holders.

- Where the carrying value of preference shares repurchased exceeds the fair value of consideration paid, the excess is added in calculating profit attributable to ordinary equity holders.

In respect of preference shares that are classified as liabilities, the above adjustments, where these are relevant, would have already been made in arriving at the profit or loss for the period.

Worked example: Repurchase of preference shares

An entity has issued £100,000 8% non-redeemable non-cumulative preference shares. Half way through the year, the entity repurchased half of the preference shares at a discount of £1,000. No dividends were paid on these shares in respect of the amounts repurchased or outstanding at the end of the year.

Requirement

If the profit attributable to ordinary equity holders for the year is £150,000, what adjustments should be made for the purpose of calculating EPS?

Solution

Adjustments	**£**
Profit for the year attributed to ordinary equity holders	150,000
Plus discount on repurchasing of preference shares	1,000
	151,000

The discount on repurchase of the preference shares has been credited to equity and it must therefore be adjusted against profit.

Had there been a premium payable on repurchase, the loss on repurchase would have been subtracted from profit.

No accrual for the dividend on the 8% preference shares is required as these are non-cumulative. Had a dividend been paid for the year it would have been deducted from profit for the purpose of calculating basic EPS as the shares are treated as equity and the dividend would have been charged to equity in the financial statements.

3.2 Participating securities and two-class ordinary shares

The equity of some entities includes:

- instruments that participate in dividends with ordinary shares according to a predetermined formula (for example, 2 for 1) with an upper limit on the extent of participation (for example, up to, but not beyond, a specified amount per share); or
- a class of ordinary shares with a different dividend rate from that of another class of ordinary shares.

Profit or loss for the period is allocated to the different classes of shares and participating equity instruments in accordance with their dividend rights or other rights to participate in undistributed earnings.

To calculate basic earnings per share:

- Profit or loss attributable to ordinary equity holders of the parent entity is adjusted as previously discussed.

- The remaining profit or loss is allocated to ordinary shares and participating equity instruments to the extent that each instrument shares in earnings as if all of the profit or loss for the period had been distributed. The total profit or loss allocated to each class of equity instrument is determined by adding together the amount allocated for dividends and the amount allocated for a participation feature.
- The total amount of profit or loss allocated to each class of equity instrument is divided by the number of outstanding instruments to which the earnings are allocated to determine the earnings per share for the instrument.



Worked example: Participating equity instruments

The following information is provided for an entity.

Profit attributable to equity holders of the parent entity	£100,000
Ordinary shares outstanding	10,000
Non-convertible preference shares	6,000
Non-cumulative annual dividend on preference shares (before any dividend is paid on ordinary shares)	£5.50 per share

After ordinary shares have been paid at a dividend of £2.10 per share, the preference shares participate in any additional dividends on a 20:80 ratio with ordinary shares.

Dividends on preference shares paid – £33,000 (£5.50 per share × 6,000 shares)

Dividends on ordinary shares paid – £21,000 (£2.10 per share × 10,000 shares)

Requirement

Calculate the earnings attributable to ordinary shares.

Solution

Basic earnings per share is calculated as follows

	£	£
Profit attributable to equity holders of the parent entity		100,000
Less dividends paid:		
Preference	33,000	
Ordinary	21,000	
		(54,000)
Undistributed earnings		46,000

Allocation of undistributed earnings

Let A be the allocation of undistributed earnings per ordinary share and B the allocation per preference share. That is:

(A × 10,000) + (B × 6,000) = £46,000

As B's entitlement is one quarter that of A's, we can eliminate B from the equation as follows:

(A × 10,000) + (1/4 × A × 6,000) = £46,000

10,000A + 1,500A = £46,000

11,500A = £46,000

A = £46,000/11,500

A = £4.00

Therefore B = £1.00

Basic per share amounts Basic per share amounts

	Per preference share	Per ordinary share
Distributed earnings	£5.50	£2.10
Undistributed earnings	£1.00	£4.00
Totals	£6.50	£6.10



Professional skills focus: Assimilating and using information

As we have seen, although the basic EPS calculation is straightforward, a great deal of information needs to be assimilated and understood, both regarding earnings (adjustments for preference shares) and weighted average number of shares (changes in share capital). The two interact.

4 Diluted earnings per share



Section overview

This section deals with the adjustments required to earnings in order to take into account the dilutive impact of potential ordinary shares.

The objective of diluted earnings per share is consistent with that of basic earnings per share; that is, to provide a measure of the interest of each ordinary share in the performance of an entity taking into account **dilutive** potential ordinary shares outstanding during the period.

4.1 Potential ordinary shares

Potential ordinary shares are financial instruments or other contracts that may entitle their holders to ordinary shares. Potential ordinary shares are as follows:

- Various financial liabilities or equity instruments, including preference shares that are convertible into ordinary shares
- Options
- Warrants
- Shares that would be issued on satisfaction of certain conditions that result from contractual arrangements, such as the purchase of a business or other assets

The conversion of potential ordinary shares will lead in the future to an increase in the weighted average number of ordinary shares outstanding by the weighted average number of additional ordinary shares that would have been outstanding assuming the conversion of all dilutive potential ordinary shares.

Conversion may also lead to consequential changes in income or expenses. For example, the reduction of interest expense related to convertible debt and the resulting increase in profit or reduction in loss may lead to an increase in the expense related to a non-discretionary employee profit-sharing plan.

For the purpose of calculating diluted earnings per share, profit or loss attributable to ordinary equity holders of the parent entity is adjusted for any such consequential changes in income or expense.

4.2 Dilutive and antidilutive potential ordinary shares

Potential ordinary shares are **dilutive** when their conversion to ordinary shares would decrease earnings per share or increase loss per share from continuing operations.

Potential ordinary shares are **antidilutive** when their conversion to ordinary shares would increase earnings per share or decrease loss per share from continuing operations. **Antidilution** is therefore the situation where the 'diluted' EPS is greater than the basic EPS (or where there is a lower loss per share). IAS 33 defines antidilution as follows.

Definition

Antidilution: An increase in earnings per share or a reduction in loss per share resulting from the assumption that convertible instruments are converted, that options or warrants are exercised, or that ordinary shares are issued upon the satisfaction of specified conditions.

In computing diluted EPS only potential ordinary shares that are dilutive are considered in the calculations. The calculation ignores the effects of potential ordinary shares that would have an antidilutive effect on earnings per share.

Determining whether potential ordinary shares are dilutive or antidilutive

In determining whether potential ordinary shares are dilutive or antidilutive, **each issue or series of potential ordinary shares is considered separately rather than in aggregate.**

A separate EPS calculation is performed for each potential share issue.

(a) Those individual EPS which exceed the entity's basic EPS are disregarded as they are antidilutive.

(b) Those individual EPS which are less than the entity's basic EPS are dilutive and are ranked from most to least dilutive. Options and warrants are generally included first because they do not affect the numerator of the calculation. These dilutive factors are added one by one into the DEPS calculation in order to identify the maximum dilution.

The calculation showing each issue or series of potential ordinary shares being considered separately is shown in the worked example Convertible loan stock 2.

4.3 Computation of diluted earnings

For the purpose of calculating diluted earnings per share, the profit or loss attributable to ordinary equity holders of the parent entity should be adjusted for the after-tax effect of:

- any dividends or other items related to dilutive potential ordinary shares deducted in arriving at profit or loss attributable to ordinary equity holders;
- any interest recognised in the period related to dilutive potential ordinary shares; and
- any other changes in income or expense that would result from the conversion of the dilutive potential ordinary shares.

After the potential ordinary shares are converted into ordinary shares, the dividends, interest and any other expenses associated with the potential ordinary shares will no longer arise. Instead, the new ordinary shares are entitled to participate in profit or loss attributable to ordinary equity holders of the parent entity. The expenses associated with potential ordinary shares include transaction costs and discounts accounted for in accordance with the effective interest method.

4.4 Calculation of the number of shares

In the calculation of diluted earnings per share, the number of ordinary shares in the denominator is the weighted average number of ordinary shares calculated for the basic earnings per share plus the weighted average number of ordinary shares that would be issued on the conversion of all the dilutive potential ordinary shares into ordinary shares. That is:

Number of shares in diluted earnings per share	=	Number of shares in basic earnings per share	+	Dilutive potential ordinary shares

- Dilutive potential ordinary shares shall be deemed to have been converted into ordinary shares at the **beginning of the period** or, if later, the **date of the issue** of the potential ordinary shares (ie, where the convertible instruments or options are issued during the current period).
- Potential ordinary shares are weighted for the period they are outstanding.
- Potential ordinary shares that are cancelled or allowed to lapse during the period are included in the calculation of diluted earnings per share only for the portion of the period during which they are outstanding.
- Potential ordinary shares that are converted into ordinary shares during the period are included in the calculation of diluted earnings per share from the beginning of the period to the date of conversion. From the date of conversion, the resulting ordinary shares are included in both basic and diluted earnings per share.
- The number of ordinary shares that would be issued on conversion of dilutive potential ordinary shares is determined from the terms of the potential ordinary shares. When more than one basis of conversion exists, the calculation assumes the **most advantageous** conversion rate or exercise price from the standpoint of the holder of the potential ordinary shares.

5 Diluted earnings per share: convertible instruments

Section overview

This section deals with the impact of convertible instruments on the diluted earnings per share.

5.1 Convertible instruments

Convertible instruments, such as convertible loan stock or convertible preference shares impact both the profit or loss attributed to ordinary equity holders, and the number of ordinary shares, on conversion.

Where this has a dilutive effect, the instrument should be taken into account when calculating diluted earnings per share (DEPS).

Indicators that convertible instruments are antidilutive

Convertible preference shares are antidilutive whenever the amount of the dividend on such shares declared in, or accumulated for, the current period per ordinary share obtainable on conversion exceeds basic earnings per share.

Similarly, convertible debt is antidilutive whenever its interest (net of tax and other changes in income or expense) per ordinary share obtainable on conversion exceeds basic earnings per share.



Worked example: Convertible loan stock 1

On 1 January 20X5 entity A had in issue:

- 24 million ordinary shares of £1 nominal value each; and
- £8 million of 8% convertible loan stock. These were issued on 1 January 20X5 and are convertible at any time from 1 January 20X8. The conversion terms are one ordinary share for each £2 nominal of loan stock.

The split accounting required for compound financial instruments per IAS 32 resulted in a liability element for the loan stock of £7 million and an effective interest rate of 10%.

After charging income tax at 20%, the entity reported profit attributable to the ordinary equity holders of £15 million for its year ended 31 December 20X5.

Requirement

Calculate the basic and diluted earnings per share for 20X5.

Solution

Basic EPS

£15m/24m shares = £0.63

Diluted EPS

To calculate the diluted earnings per share we need to consider the impact on both earnings and number of shares.

Impact on earnings:	On conversion, after tax earnings attributed to ordinary shareholders should be increased by the reduction in the interest charge payable to loan holders. Taking tax into account, the interest saved will be: £7m × 0.1 × (1 – 0.2) = £0.56m Therefore diluted earnings = £15m + £0.56m = £15.56m
Impact on number of shares:	On conversion the number of ordinary shares will increase by £8m/2 = 4 million shares, raising the number of ordinary shares after conversion to 28 million ordinary shares.

Therefore DEPS = £15.56m/28 million shares = £0.56



Professional skills focus: Structuring problems and solutions

The approach to calculating diluted EPS requires skills at structuring the problem. First basic EPS needs to be calculated. Then the profit or loss attributable to ordinary equity holders of the parent entity needs to be adjusted for dividends, interest or relating to the dilutive

potential of any other changes in income or expense that would result from the conversion of the dilutive potential ordinary shares. Then the basic EPS denominator needs to be adjusted for the weighted average number of ordinary shares that would be issued on the conversion of all the dilutive potential ordinary shares into ordinary shares.



Interactive question 3: Test of dilution

The issued share capital of Entity A at 31 December 20X5 was 2,000,000 ordinary shares of £1 each. On 1 January 20X6, Entity A issued £1,500,000 of 7% convertible loan stock for cash at par. (Ignore the requirement to split the value of a compound financial instrument.) Each £100 nominal of the loan stock may be converted into 140 ordinary shares at any time after 1 January 20X9.

The profit before interest and taxation for the year ended 31 December 20X6 amounted to £1,050,000 and arose exclusively from continuing operations. The rate of tax is 30%.

Requirement

Test whether the potential shares are dilutive.

See **Answer** at the end of this chapter.



Worked example: Convertible loan stock 2

On 1 January 20X5 Entity A had in issue:

(1) 20 million ordinary shares;

(2) £11 million of 6.5% convertible loan stock, convertible at any time from 1 January 20X7. The conversion terms are one ordinary share for each £2 nominal of loan stock, the 1 January carrying amount of the liability component is £10 million and the effective interest rate is 9%;

(3) £9 million of 6.75% convertible loan stock, convertible at any time from 1 January 20X8. The conversion terms are one ordinary share for each £2 nominal of loan stock, the 1 January carrying amount of the liability component is £8 million and the effective interest rate is 8%; and

(4) £12.6 million of 9% convertible loan stock, convertible at any time from 1 January 20X9. The conversion terms are one ordinary share for each £6 nominal of loan stock, the 1 January carrying amount of the liability component is £12 million and the effective interest rate is 12%.

The entity reported profit attributable to the ordinary equity holders of £4 million for its year ended 31 December 20X5.

Requirement

Ignoring taxes, calculate the diluted earnings per share.

Solution

The incremental earnings per share for each type of potential ordinary shares is shown below.

	Increase in earnings (interest saved) £	Increase in number of shares	Earnings per additional share £
£11 million of 6.5% convertible loan stock			
£10m × 9% convertible loan stock	900,000		
1 ordinary share for £2 nominal of loan stock		5,500,000	0.16
£9 million of 6.75% convertible loan stock			
£8m × 8% convertible loan stock	640,000		
1 ordinary share for £2 nominal of loan stock		4,500,000	0.14
£12.6 million of 9% convertible loan stock			
£12m × 12% convertible loan stock	1,440,000		
1 ordinary share for £6 nominal of loan stock		2,100,000	0.69

The earnings per share can be calculated adjusting both the earnings and the number of shares for each type of potential shares, and the results are shown below. Each issue of potential ordinary shares is added to the calculation at a time, taking the most dilutive factor first.

	Earnings	Number of shares	Earnings per share
	£		£
Shares already in issue	4,000,000	20,000,000	0.20
Including 6.75% convertible loan stock	4,640,000	24,500,000	0.189
Including 6.5% convertible loan stock	5,540,000	30,000,000	0.185
Including 9% convertible loan stock	6,980,000	32,100,000	0.217

The diluted earnings per share will be £0.185. The 9% convertible loan stock is antidilutive since it increases earnings per share, and it will not be taken into account in calculating diluted earnings per share.



Professional skills focus: Assimilating and using information

The above example is a good illustration of the skill of assimilating information (the position at the start, the effect on both earnings and number of shares of each type of potential shares), as well as the skill of using this information by presenting each effect in a logical layout in order to arrive at the resulting diluted earnings per share figure.

5.2 Testing for dilution

In some cases some convertible preference shares are redeemed or converted in a period, and others remain outstanding.

Where this occurs, any excess consideration paid on redemption or conversion is attributed to those shares which have been redeemed or converted.

Outstanding convertible preference shares are therefore tested for dilution as normal and without regard to this excess.

6 Diluted EPS and options

Section overview

This section deals with the impact of options on diluted earnings per share.

Definition

Options and warrants: Financial instruments that give the holder the right to purchase ordinary shares.

6.1 Options, warrants and their equivalents

Options and warrants are dilutive when they would result in the issue of ordinary shares for less than the average market price of ordinary shares during the period (ie, when they are 'in the money').

To calculate diluted EPS where there are options or warrants, the potential ordinary shares at less than average market price are treated as consisting of two elements:

- A contract to issue some shares at **average market price**. These shares are assumed to be fairly priced and to be neither dilutive nor antidilutive. They are ignored in the calculation of diluted EPS.
- A contract to issue the remaining ordinary shares for **no consideration.** These shares are dilutive and are added to the number of ordinary shares outstanding in the calculation of diluted EPS.

Average market price of ordinary shares

- Theoretically every market transaction for an entity's ordinary shares could be included in the determination of the average market price. It is however adequate to use a simple average of weekly or monthly prices.
- Generally, closing market prices are adequate for calculating the average market price. When prices fluctuate widely, however, an average of the high and low prices usually produces a more representative price.
- The method used to calculate the average market price must be used consistently unless it is no longer representative because of changed conditions. For example, an entity that uses closing market prices to calculate the average market price for several years of relatively stable prices might change to an average of high and low prices if prices start fluctuating greatly and the closing market prices no longer produce a representative average price.

Interactive question 4: Diluted earnings per share

At 31 December 20X6, the issued share capital of Entity A consisted of 3,000,000 ordinary shares of 20p each. Entity A has granted options that give holders the right to subscribe for ordinary shares between 20X8 and 20X9 at 50p each. Options outstanding at 31 December 20X7 were 600,000.

There were no grants, exercises or lapses of options during the year. The profit after tax attributable to ordinary equity holders for the year ended 31 December 20X7 amounted to £900,000 arising from continuing operations. The average market price of one ordinary share during year 20X7 was £1.50.

Requirement

Calculate the diluted earnings per share for 20X7.

See **Answer** at the end of this chapter.

6.2 Employee share options

The most common type of option that leads to earnings dilution is an employee share option granted by a company to its employees.

Employee share options give the right to the holder to acquire shares in the company at a price that is fixed when the options are issued.

Employee share options can normally be exercised after a certain time eg, once the employee has completed a period of service and within a certain period eg, over a period of five years after they become exercisable.

Employee share options that can be exercised are **vested** options, whereas options that cannot yet be exercised are **unvested** options.



Worked example: Vested options

The profit attributable to the ordinary equity holders of an entity for the year ended 31 December 20X5 was £30 million and the weighted average number of its ordinary shares in issue was 60 million. Its basic earnings per share was £0.50.

In addition, there was a weighted average of five million shares under options which had vested (ie, were able to be exercised). The exercise price for the options was £21 and the average market price per share over the year was £30.

Requirement

Calculate the diluted EPS.

Solution

The amount to be received on exercise is £21 × 5m = £105m

The number of shares issued at average market price is: £105m / £30 = 3.5m

The number of 'free' shares is: 5 million issued – 3.5 million issued at average market price = 1.5 million

Diluted earnings per share:

£30m/(60m + 1.5m) = £0.49

Where shares are **unvested**, the amount still to be recognised in profit or loss before the vesting date must be taken into account when calculating the number of 'free' shares.

Worked example: Unvested options

Assume the same information as in above example, except that:

- the options have not yet vested; and
- the amount to be recognised in relation to these options in the entity's profit or loss over future accounting periods up to date of vesting, as calculated according to IFRS 2, is £15 million.

Requirement

Calculate the diluted EPS.

Solution

The amount to be recognised in profit or loss is reduced to a per share amount: £15m/5m = £3 This is added to the exercise price: £21 + £3 = £24

The amount to be received on exercise: £24 × 5m = £120m

The number of shares issued at average market price: £120m/£30 = 4m

The number of 'free' shares: 5m – 4m = 1m

Diluted earnings per share: £30m/(60m + 1m) = £0.49

7 Diluted EPS: contingently issuable shares

Section overview

This section deals with the impact of contingently issuable shares on the number of ordinary shares used in the calculation of diluted earnings per share.

7.1 Contingently issuable shares

The consideration for acquisitions of other entities may partly be in the form of shares which will only be issued if certain targets are met in the future. The additional consideration is contingent consideration and the additional shares are described as contingently issuable.

Contingently issuable shares may also arise where senior staff members are issued shares as a performance reward.

- Until the shares are issued (if indeed they ever are), they should not be taken into account when calculating basic EPS.
- They should be taken into account when calculating diluted EPS **if and only if** the conditions leading to their issue have been satisfied. For these purposes the end of the accounting period is treated as the end of the contingency period.
- Contingently issuable shares are included from the beginning of the period (or from the date of the contingent share agreement, if later).

Worked example: Contingently issuable shares

The profit attributable to the ordinary equity holders of an entity for the year ended 31 December 20X5 was £20 million and the weighted average number of its ordinary shares in issue was 16 million. Basic earnings per share was therefore £1.25.

Under an agreement relating to a business combination, two million additional shares were to be issued if the share price on 30 June 20X6 was £8 or above. On 31 December 20X5 the

share price was £9. Assuming the end of the reporting period was the end of the contingency period, the condition would have been met.

Requirement

Determine the diluted EPS.

Solution

As the two million additional shares do not result in additional resources for the entity, they are brought into the diluted earnings per share calculation from the start of the 20X5 reporting period. The diluted earnings per share is therefore:

Diluted earnings per share = £20m/(16m + 2m) = £1.11

7.2 Conditions for issue

Contingently issuable shares are included within the calculation of diluted EPS where:

- the shares have not yet been issued
- the relevant performance criteria have been met

Future earnings

Achieving or maintaining a specified level of earnings for a particular period may be the condition for contingent issue.

In this case, if the effect is dilutive, the calculation of diluted EPS is based on the number of ordinary shares that would be issued if the amount of earnings at the end of the reporting period were the amount of earnings at the end of the contingency period.

Market price of shares

The number of ordinary shares contingently issuable may depend on the future market price of the ordinary shares.

In this case, if the effect is dilutive, the calculation of diluted EPS is based on the number of ordinary shares that would be issued if the market price at the end of the reporting period were the market price at the end of the contingency period.

If the condition is based on an average of market prices over a period of time that extends beyond the end of the reporting period, the average for the period of time that has lapsed is used.

Future earnings and market price of shares

The number of ordinary shares contingently issuable may depend on future earnings and future prices of the ordinary shares.

In such cases, the number of ordinary shares included in the diluted EPS calculation is based on both conditions (ie, earnings to date and the current market price at the end of the reporting period).

Contingently issuable ordinary shares are not included in the diluted earnings per share calculation unless both conditions are met

Other conditions

In other cases, the number of ordinary shares contingently issuable may depend on a condition other than earnings or market price (for example, the opening of a specific number of retail stores).

In this case, the contingently issuable ordinary shares are included in the calculation of diluted earnings per share according to the status at the end of the reporting period

Cumulative targets

Note that where performance criteria involve a cumulative target, no dilution is accounted for until the cumulative target has been met. For example, where the issue of shares is dependent upon average profits of £300,000 over four years, the cumulative target is £1,200,000. No dilution is accounted for until this cumulative target is met.



Worked example: Cumulative targets

A manufacturer has in issue 3,000,000 ordinary shares at 1 January 20X7. It agreed to issue 500,000 shares to its staff if factory output averages 100,000 units per annum over the period from 1 January 20X7 to 31 December 20X9. The shares are to be issued on 1 January 20Y0.

Results for the three periods are:

	Units produced	Profits
20X7	120,000	£780,000
20X8	99,000	£655,000
20X9	105,000	£745,000

Requirement

What are basic and diluted EPS in each of the years 20X7–20X9?

Solution

The cumulative target of 3 years × 100,000 units is not met in 20X7 and 20X8, therefore no dilution is accounted for.

In 20X9, the cumulative target is met as is the average target, therefore a diluted EPS is disclosed:

		Basic EPS		Diluted EPS
20X7	£780,000 ÷ 3,000,000 shares	£0.26		Not relevant
20X8	£655,000 ÷ 3,000,000 shares	£0.22		Not relevant
20X9	£745,000 ÷ 3,000,000 shares	£0.25	£745,000 ÷ 3,500,000 Shares	£0.21

7.3 Issue of contingently issuable shares

Where contingently issuable shares are issued at the end of a contingency period, they must be included within the calculation of basic EPS. Any outstanding contingently issuable shares are included in diluted EPS as discussed above.



Worked example: Contingently issuable shares

The profit attributable to the ordinary equity holders of an entity for the year ended 31 December 20X5 was £200 million and the number of its ordinary shares in issue at 1 January 20X5 was 80 million.

Under an agreement relating to a business combination, 12 million additional shares were to be issued each time the entity's products were ranked in the top three places in a consumer satisfaction survey conducted by a well-known magazine. A maximum of 36 million shares was

issuable under this agreement and the products appeared in the top three places in surveys dated 28 February and 30 September 20X5.

There were no other issues of ordinary shares.

Requirement

Determine the basic and diluted EPS.

Solution

Basic earnings per share

The 24 million additional shares are weighted by the period they have been in issue:

	Issued	Weighting	Adjusted number
1 January 20X5 – 28 February	80,000,000	2/12	13,333,333
Issued 28 February	12,000,000		
1 March – 30 Sept	92,000,000	7/12	53,666,667
Issued 30 September	12,000,000		
30 September – 31 December	104,000,000	3/12	26,000,000
Weighted average shares in issue			93,000,000

Basic earnings per share = £200m/93m = £2.15

Diluted earnings per share

As the 24 million additional shares do not result in any additional resources for the entity, the diluted calculation assumes all the new shares were issued at the start of the year (see section 2.4).

Diluted earnings per share = £200m/(80m + 12m + 12m) = £1.92

8 Retrospective adjustments, presentation and disclosure



Section overview

This section deals with retrospective adjustments to EPS and the provisions of IAS 33 concerning presentation and disclosure.

8.1 Retrospective adjustment

Bonus issues, share splits and share consolidations

If the number of ordinary or potential ordinary shares outstanding increases as a result of a capitalisation, bonus issue or share split, or decreases as a result of a share consolidation, the calculation of basic and diluted earnings per share for all periods presented must be adjusted retrospectively.

If these changes occur after the year end but before the financial statements are authorised for issue, EPS calculations for all periods presented must be based on the new number of shares.

The fact that EPS calculations reflect such changes must be disclosed.

Prior period adjustments and errors

Basic and diluted earnings per share of all periods presented must be adjusted for the effects of errors (IAS 8) and adjustments resulting from changes in accounting policies accounted for retrospectively.

DEPS

An entity does not restate diluted earnings per share of any prior period presented for changes in the assumptions used in earnings per share calculations or for the conversion of potential ordinary shares into ordinary shares.

8.2 Presentation

- Please note the following key points regarding presentation.
- Basic and diluted EPS for the year (and from continuing operations if reported separately) must be presented on the face of the statement of profit or loss and other comprehensive income with equal prominence for all periods presented.
- Where a separate statement of profit or loss is presented, basic and diluted EPS should be presented on the face of this statement.
- Earnings per share is presented for every period for which a statement of comprehensive income is presented.
- If diluted earnings per share is reported for at least one period, it shall be reported for all periods presented, even if it equals basic earnings per share.
- If basic and diluted earnings per share are equal, dual presentation can be accomplished in one line on the statement of comprehensive income.
- An entity that reports a discontinued operation shall disclose the basic and diluted amounts per share for the discontinued operation either on the face of the statement of comprehensive income or in the notes.
- An entity shall present basic and diluted earnings per share, even if the amounts are negative (ie, a loss per share).

8.3 Disclosure

An entity shall disclose the following:

- The amounts used as the numerators in calculating basic and diluted earnings per share, and a reconciliation of those amounts to profit or loss attributable to the parent entity for the period. The reconciliation shall include the individual effect of each class of instruments that affects earnings per share.
- The weighted average number of ordinary shares used as the denominator in calculating basic and diluted earnings per share, and a reconciliation of these denominators to each other. The reconciliation shall include the individual effect of each class of instruments that affects earnings per share.
- Instruments (including contingently issuable shares) that could potentially dilute basic earnings per share in the future, but were not included in the calculation of diluted earnings per share because they are antidilutive for the period(s) presented.
- A description of ordinary share transactions or potential ordinary share transactions, other than retrospective adjustments, that occur after the reporting date and that would have changed significantly the number of ordinary shares or potential ordinary shares outstanding at the end of the period if those transactions had occurred before the end of the reporting period.

Examples of transactions referred to in the paragraph above include the following:

- An issue of shares for cash
- An issue of shares when the proceeds are used to repay debt or preference shares outstanding at the reporting date
- The redemption of ordinary shares outstanding
- The conversion or exercise of potential ordinary shares outstanding at the reporting date into ordinary shares
- An issue of options, warrants, or convertible instruments
- The achievement of conditions that would result in the issue of contingently issuable shares

Earnings per share amounts are not adjusted for such transactions occurring after the reporting date because such transactions do not affect the amount of capital used to produce profit or loss for the period.

Financial instruments and other contracts generating potential ordinary shares may incorporate terms and conditions that affect the measurement of basic and diluted earnings per share. These terms and conditions may determine whether any potential ordinary shares are dilutive and, if so, the effect on the weighted average number of shares outstanding and any consequential adjustments to profit or loss attributable to ordinary equity holders. The disclosure of the terms and conditions of such financial instruments and other contracts is encouraged, if not otherwise required (refer also to IFRS 7, *Financial Instruments: Disclosures*).

8.4 Additional EPS

If an entity discloses, in addition to basic and diluted earnings per share, amounts per share using a reported component of profit other than one required by IAS 33, such amounts shall be calculated using the weighted average number of ordinary shares determined in accordance with this Standard.

Basic and diluted amounts per share relating to such a component shall be disclosed with equal prominence and presented in the notes.

An entity shall indicate the basis on which the numerator(s) is (are) determined, including whether amounts per share are before tax or after tax.

If a component of profit is used that is not reported as a line item in the statement of profit or loss and other comprehensive income, a reconciliation shall be provided between the component used and a line item that is reported in the statement of profit or loss and other comprehensive income.



Professional skills focus: Applying judgement

Judgement is needed as to whether the additional EPS potentially undermines the EPS as calculated in accordance with IAS 33. Although it is required by the standard to have equal prominence, the preparer of the financial statements needs to weigh up whether it is really necessary.



Context example: Allied Irish Bank plc

In addition to the required EPS under IAS 33, Allied Irish Bank plc reports an adjusted EPS measure excluding hedge volatility, profit on disposal of property and business and construction contract income. A comment to the disclosure states that the adjusted measure is presented to help better understand underlying business performance.

Allied Irish Bank plc 31 December 2006

Extract from Notes to the accounts:

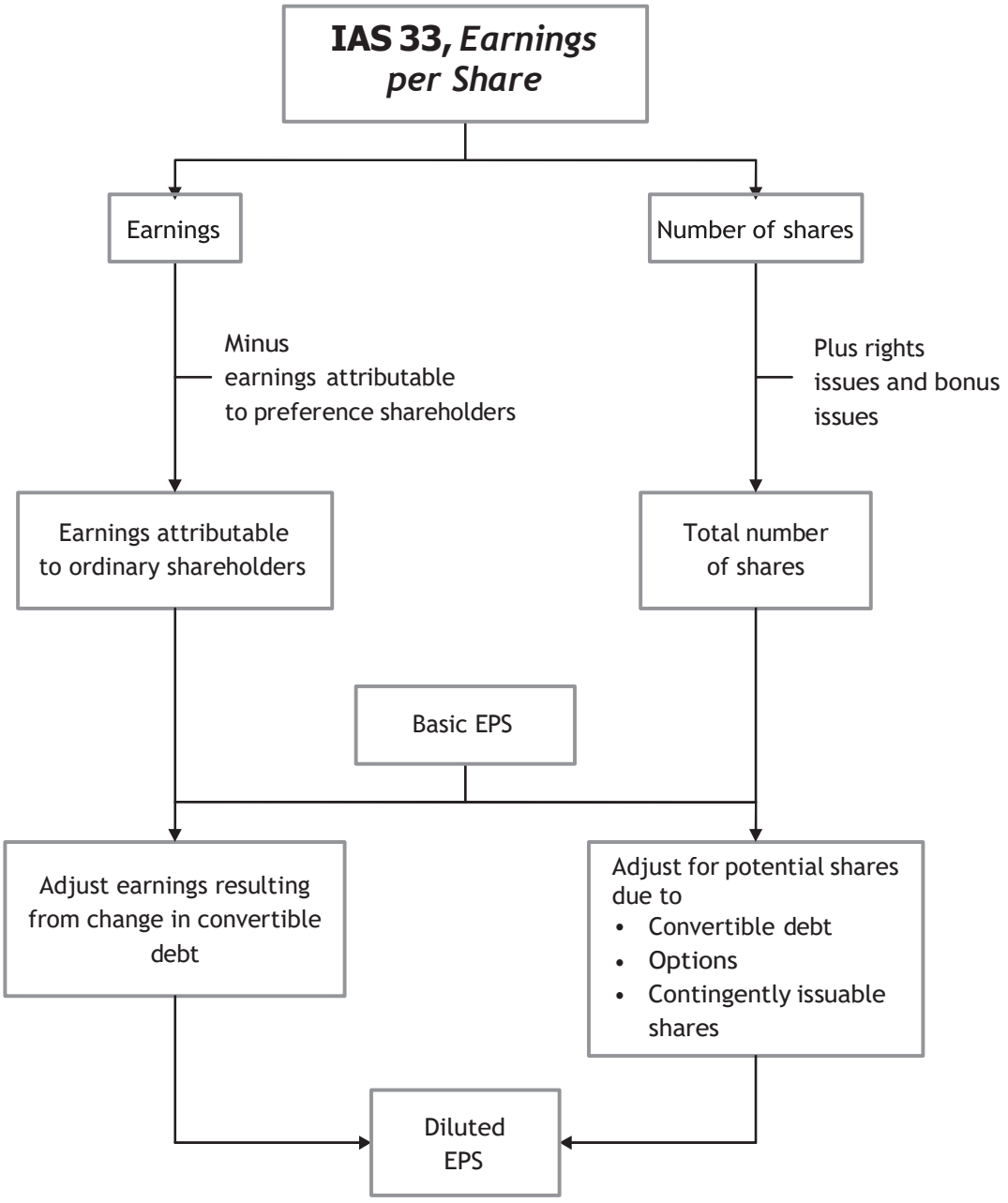
	Profit attributable		Earnings per share	
	2006	2005	2006	2005
19 Adjusted earnings per share	**€m**	**€m**	**€m**	**€m**
(a) Basic earnings per share				
As reported (Note 18(a))	2,147	1,305	246.8	151.0
Adjustments:				
- Construction contract income	(82)	(38)	(9.4)	(4.4)
- Hedge volatility	4	(6)	0.5	(0.7)
- Profit on disposal of property	(290)	–	(33.4)	–
- Profit on disposal of business	(189)		(21.7)	
	1,590	1,261	182.8	145.9

Although not required under IFRS® Standards, adjusted earnings per share is presented to help understand the underlying performance of the Group. The adjustments in 2006 and 2005 are items that management believe do not reflect the underlying business performance. The adjustment in respect of profit on sale of property relates only to the profit on sale of properties that are subject to sale and leaseback arrangements. The adjustments listed above are shown net of taxation.

Professional skills focus: Concluding, recommending and communicating

While earnings per share is a useful tool, users should be wary of relying solely on this figure. In particular, the context of the size of the entity, and the nature of its businesses should be taken into account.

Summary


IAS 33, Earnings per Share
Earnings
Number of shares
Minus earnings attributable to preference shareholders
Plus rights issues and bonus issues
Earnings attributable to ordinary shareholders
Total number of shares
Basic EPS
Adjust earnings resulting from change in convertible debt
Adjust for potential shares due to
• Convertible debt
• Options
• Contingently issuable shares
Diluted EPS

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	Can you calculate the weighted average number of shares for basic and diluted EPS? (Topic 2)
2.	Why is the distinction between redeemable and irredeemable preference shares significant? (Topic 3)
3.	What is meant by antidilutive? (Topic 4)
4.	Can you calculate EPS in the context of convertible shares and options? (Topic 7)
5.	Can you calculate EPS in the context of contingently issuable shares? (Topic 7)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Puffbird	A straightforward EPS question with a bonus issue and adjustment of the prior year figure, this is a good one to start on.
Whiting	An intermediate question, with a discontinued operation and convertible loan stock.
Citric	This is about as complicated as EPS gets, covering warrants and convertible bonds and test of dilution.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted the self-test questions, you can continue your studies by moving onto the next chapter.

In later chapters, we will recommend questions from the Question Bank for you to attempt. Earnings per share calculations often come at the end of scenario questions requiring adjustments to financial statements for financial reporting issues covered in later chapters.

Technical reference

1 Earnings

Amounts attributable to ordinary equity holders in respect of profit or loss for the period (and from continuing operations where reported separately) adjusted for the after tax amounts of preference dividends. – **IAS 33.12**

2 Shares

For the calculation of basic EPS the number of ordinary shares should be the weighted average number of shares outstanding during the period adjusted where appropriate for events, other than the conversion of shares, that have changed the number of ordinary shares outstanding without a corresponding change in resources. – **IAS 33.26**

3 Diluted earnings per share

For the purposes of calculating diluted earnings per share an entity shall adjust profit or loss attributable to ordinary equity holders and the weighted number of shares outstanding for the effects of dilutive potential ordinary shares. – **IAS 33.30, IAS 33.31**

4 Dilutive potential ordinary shares

Potential ordinary shares shall be treated as dilutive when, and only when, their conversion to ordinary shares could decrease earnings per share or increase loss per share from continuing operations. – **IAS 33.41**

5 Options, warrants and their equivalents – IAS 33.45

6 Convertible instruments – IAS 33.49

7 Contingently issuable shares – IAS 33.52

8 Retrospective adjustments

Basic and diluted EPS should be adjusted retrospectively for all capitalisations, bonus issues or share splits or reverse share splits that affect the number of shares in issue without affecting resources. – **IAS 33.64**

9 Presentation – IAS 33.66

10 Disclosure – IAS 33.70

Self-test questions

Answer the following questions.

1 Puffbird

Puffbird is a company listed on a recognised stock exchange. Its financial statements for the year ended 31 December 20X6 showed earnings per share of £0.95.

On 1 July 20X7 Puffbird made a 3 for 1 bonus issue.

Requirement

According to IAS 33, *Earnings per Share*, what figure for the 20X6 earnings per share will be shown as comparative information in the financial statements for the year ended 31 December 20X7?

2 Urtica

The Urtica Company is listed on a recognised stock exchange.

During the year ended 31 December 20X6, the company had 5 million ordinary shares of £1 and 500,000 6% irredeemable preference shares of £1 in issue.

Profit before tax for the year was £300,000 and the tax charge was £75,000

Requirement

According to IAS 33, *Earnings per Share*, what is Urtica's basic earnings per share for the year?

3 Issky

The following extracts relate to the Issky Company for the year ended 31 December 20X7.

Statement of comprehensive income

	£'000
Profit after tax	5,400

Statement of financial position

Ordinary shares of £1	8,400

In addition, the company had in issue throughout the year 1,800,000 share options granted to directors at an exercise price of £15. These were fully vested (ie, conditions required before these could be exercised were fulfilled and the options were exercisable) but had not yet been exercised. The market price for Issky's shares was £24 at 1 January 20X7, £30 at 31 December 20X7, and the average for 20X7 was £27.

Requirement

What is the diluted earnings per share for 20X7 according to IAS 33, *Earnings per Share*?

4 Whiting

The Whiting Company has the following financial statement extracts in the year ended 31 December 20X7.

Statement of comprehensive income

	£
Profit after tax	
Continuing operations	1,600,000
Discontinued operations	(400,000)
Total attributable to ordinary equity holders	1,200,000

Statement of financial position

Ordinary shares of £9,600,000 in £1 shares

On 1 January 20X7, Whiting issued £1.2 million of 7% redeemable convertible bonds, interest being payable annually in arrears on 31 December. The split accounting required of compound financial instruments resulted in the following classification.

	£
Equity component	100,000
Liability component	1,100,000
	1,200,000

The effective interest rate on the liability component is 10%. The bonds are convertible on specified dates in the future at the rate of one ordinary share for every £2 bond.

The tax regime under which Whiting operates gives relief for the whole of the charge based on the effective interest rate and applies a tax rate of 20%.

Requirement

Based upon the profit from continuing operations attributable to ordinary equity holders, what amount, if any, for diluted earnings per share should be presented by Whiting in its financial statements for the year ended 31 December 20X7 according to IAS 33, *Earnings per Share*?

5 Garfish

The Garfish Company had profits after tax of £3.0 million in the year ended 31 December 20X7.

On 1 January 20X7, Garfish had 2.4 million ordinary shares in issue. On 1 April 20X7 Garfish made a 1 for 2 rights issue at a price of £1.40 when the market price of Garfish's shares was £2.00.

Requirement

What is Garfish's basic earnings per share figure for the year ended 31 December 20X7, according to IAS 33, *Earnings per Share*?

6 Sakho

The Sakho Company has 850,000 ordinary shares in issue on 1 January 20X7 and had the following share transactions in the year ended 31 December 20X7.

(1) A 1 for 5 bonus issue on 1 May 20X7

(2) A 2 for 5 rights issue on 1 September 20X7 at £0.45 when the market price was £1.50

Requirement

Indicate whether the following statements are true or false according to IAS 33, *Earnings per Share*.

(1) The basic earnings per share for the year ended 31 December 20X6 has to be adjusted by a fraction of 5/6.

(2) For the calculation of 20X7 basic earnings per share, the number of shares in issue before the rights issue has to be adjusted by a rights fraction of 1.50/1.20.

7 Sardine

The Sardine Company operates in Moldania, a jurisdiction in which shares may be issued at a discount. It has profit after tax and before preference dividends of £200,000 in the year ended 31 December 20X7.

On 1 January 20X7 Sardine has in issue 500,000 ordinary shares, and on 1 January 20X7 issues £300,000 of £100 non-convertible, non-redeemable preference shares. Cash dividends of 8% per annum will only start to be paid on the preference shares from 1 January 20X9, so the shares are issued at a discount. The effective interest rate of the discount is 8%.

Requirement

According to IAS 33, *Earnings per Share*, what is the basic earnings per share for Sardine in the year ended 31 December 20X7?

8 Citric

The following information relates to The Citric Company for the year ended 31 December 20X7.

Statement of comprehensive income

Profit after tax – £100,000

Statement of financial position

Ordinary shares of £1 – 1,000,000

There are warrants outstanding in respect of 1.7 million new shares in Citric at a subscription price of £18.00. Citric's share price was £22.00 on 1 January 20X7, £24.00 on 30 June 20X7, £30.00 on 31 December 20X7 and averaged £25.00 over the year.

On 1 January 20X7 Citric issued £2 million of 6% redeemable convertible bonds, interest being payable annually in arrears on 31 December. The split accounting required of compound financial instruments resulted in a liability component of £1.75 million and effective interest rate of 7%. The bonds are convertible on specified dates many years into the future at the rate of two ordinary shares for every £5 bonds.

The tax regime under which Citric operates gives relief for the whole of the effective interest rate charge on the bonds and applies a tax rate of 25%.

Requirement

Determine the following amounts in respect of Citric's diluted earnings per share for the year ending 31 December 20X7 according to IAS 33, *Earnings Per Share*:

(1) The number of shares to be treated as issued for no consideration (ie, 'free' shares) on the subscription of the warrants

(2) The earnings per incremental share on conversion of the bonds, expressed in pence (to one decimal place)

(3) The diluted earnings per share, expressed in pence (to one decimal place)

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

No, the 4% preference shares are classed as liabilities under IAS 32. The dividend has been charged to profits as part of the finance cost and no adjustment is necessary.

Answer to Interactive question 2

£1.75

Being the total earnings £350,000 divided by the number of shares in issue (200,000).

The redeemable preference share dividend is included as a finance cost and deducted in arriving at profit before tax.

Answer to Interactive question 3

Basic EPS
Trading results

	20X6
	£
Profit before interest and tax	1,050,000
Interest on 7% convertible loan stock	(105,000)
Profit before tax	945,000
Taxation	(283,500)
Profit after tax	661,500
Number of shares outstanding	2,000,000
Basic EPS (£661,500/2,000,000 shares)	£0.33

Testing for dilutive impact

Increase in earnings = interest saved (£1,500,000 × 7% × (1 – 30%)) – £73,500

Increase in number of shares (£1,500,000/£100 × 140) – 2,100,000

EPS (£73,500/2,100,000) = 3.5p

This is less than basic EPS and therefore the convertible loan stock is dilutive.

Answer to Interactive question 4

Diluted EPS

Trading results

	20X7
	£
Profit after tax	900,000.00
Number of shares outstanding	3,000,000.00
Basic EPS	0.30
Number of shares under option	
Issued at full market price (600,000 × 50p)/£1.50	200,000.00

	20X7
	£
Issued at nil consideration 600,000 – 200,000	400,000.00
Total number of shares under option	600,000.00
Number of equity shares for basic EPS	3,000,000.00
Number of dilutive shares under option	400,000.00
Adjusted number of shares	3,400,000.00
Diluted EPS (£900,000/3,400,000)	0.26

Answers to Self-test questions

1 Puffbird

23.75 pence

Last year's EPS figure is adjusted by the reciprocal of the bonus fraction:

Bonus fraction = Number of shares post issue ÷ Number of shares pre issue = 4 ÷ 1 Therefore revised EPS = 95p × ¼ = 23.75p

2 Urtica

3.9 pence

IAS 33 12–13 define earnings for basic EPS as after tax and after dividends on irredeemable preference shares.

	£
Profit before tax	300,000
Tax	(75,000)
Profit after tax	225,000
Preference dividend (£500,000 × 6%)	(30,000)
Profit attributable to ordinary equity holders	195,000

Therefore BEPS = £195,000/5,000,000 shares = 3.9p

3 Issky

58.7 pence

Number of shares under option	
Issued at average market price (£15 × 1,800,000)/£27	1,000,000
Issued at nil consideration (1,800,000 – 1,000,000)	800,000
Number of equity shares for basic EPS	8,400,000
Number of dilutive shares under option	800,000
Adjusted number of shares	9,200,000
Diluted EPS (£5,400,000/9,200,000 shares) =	58.7p

According to IAS 33.46, the proceeds of the options should be calculated using the average market price during the year. The difference between the number of ordinary shares issued and the number that would have been issued at the average market price are the 'free' shares that create the dilutive effect.

4 Whiting

16.6 pence

Basic EPS	Based on continuing operations £1,600,000 ÷ 9,600,000	16.7 p
Incremental EPS	Increment to profits after conversion of bonds: £1,100,000 × 10% × (1 – 20%) = £88,000 Increase to number of shares: £1,200,000/£2 = 600,000 Therefore: £88,000 ÷ 600,000	 14.7 p
Diluted EPS	Based on continuing operations £(1,600,000 + 88,000) ÷ (9,600,000 + 600,000 shares)	 16.6 p

The shares issuable on conversion of the bonds are potentially dilutive, but IAS 33.41 only requires them to be taken into account if they dilute the basic EPS figure based on continuing operations.

5 Garfish

89.1 pence

Weighted average number of shares:

TERP	2	Shares @ £2.00 =	£4.00	
	1	Shares @ £1.40 =	£1.40	
	3		£5.40	Therefore £5.40/3 = £1.80

Adjustment factor: Market value of share ÷ TERP = £2.00 ÷ £1.80

1 Jan X7 – 31 March X7	2,400,000 × (2.00 ÷ 1.80) × 3/12	666,667
Rights issue	1,200,000	
1 April X7 – 31 Dec X7	3,600,000 × 9/12	2,700,000
		3,366,667

Basic EPS:

£3,000,000 ÷ 3,366,667 shares = 89.1p

6 Sakho

Statements

(1) False

(2) True

Bonus fraction = Number of shares post issue ÷ Number of shares pre issue = 6 ÷ 5

Rights adjustment factor = MV of share ÷ TERP = 1.50 ÷ 1.20

TERP	5	Shares @ £1.50 =	£7.50	
	2	Shares @ £0.45 =	£0.90	
	7		£8.40	Therefore £8.40/7 = £1.20

The basic EPS for the prior year is multiplied by the inverse of the rights factor and the bonus factor, so 1.20/1.50 × 5/6 = 2/3.

7 Sardine

35.9 pence

Issue price of preference shares = £300,000/1.08^2 = £257,202

Profit attributable to ordinary equity holders = £200,000 – (8% × £257,202) = £179,424

Basic EPS = £179,424 ÷ 500,000 shares = 35.9p

As no dividend is payable on the preference shares in 20X7, the discount on issue is amortised using the effective interest method and treated as preference dividend when calculating earnings for EPS purposes (IAS 33.15).

The £300,000 preference shares must be discounted at 8% for the two years between issue and the date when dividends commence. A dividend is then calculated at 8% per annum compound on that value.

8 Citric

Diluted EPS

(1) 476,000

(2) 11.5 pence

(3) 6.78 pence

Explanation

(1) Shares issued at average market price (1,700,000 × £18)/£25 = 1,224,000
Shares issued at nil consideration (1,700,000 – 1,224,000) = 476,000

(2) Incremental profits (£1,750,000 × 7% × (1 – 25%)) = £91,875
Increase in number of shares (£2,000,000/£5 × 2) = 800,000
Therefore incremental EPS (£91,875/800,000 shares) = 11.5p (3)

	Profits	Number of shares	EPS
Basic EPS	£100,000	1,000,000	10p
Add in options	£100,000	1,446,000	6.78p

The warrants (treated as issued for nil consideration) are more dilutive than the bonds, so are dealt with first under IAS 33.44. As the 11.5 pence earnings per incremental share on conversion of the bonds is antidilutive, under IAS 33.36 the conversion is left out of the calculation of diluted EPS.

Chapter 12

Reporting of assets

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Review of material from earlier studies

2 IAS 40, Investment Property

3 IAS 41, Agriculture

4 IFRS 6, Exploration for and Evaluation of Mineral Resources

5 IFRS 4, Insurance Contracts

6 Audit focus points

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Explain how different methods of recognising and measuring assets and liabilities can affect reported financial position and explain the role of data analytics in financial asset and liability valuation
- Explain and appraise accounting standards that relate to assets and non-financial liabilities for example: property, plant and equipment; intangible assets, held-for-sale assets; inventories; investment properties; provisions and contingencies
- Determine for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**Review of material from earlier studies** Section 1 gives a brief revision of standards covered in great detail in terms of content and application at Professional Level. They address the reporting of assets, impairment, liabilities, provisions and contingencies and events after the reporting period. These are likely to be examined in an integrated way with other areas at Advanced Level.	**Approach** Read the summary of knowledge brought forward and try the relevant questions. If you have any difficulty, go back to your earlier study material and revise it. Use the questions in the electronic Self-test bank for further revision. **Stop and think** What is the definition of an asset in the 2018 *Conceptual*	Just because these standards are tested at earlier levels does not mean that questions that include them will be easy. These are likely to be examined in an integrated way with other areas at Advanced Level. For example, a question on property may range from owner-occupied (IAS 16) to investment property (IAS 40) to leasehold property	**IQ2: Impairment loss** You should aim to do all the interactive questions because you will struggle to tackle a complex scenario question without a knowledge of the basics. This question is the one to focus on.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	Inventories can often be the most significant current asset in an entity's statement of financial position. In addition, whether an asset is capitalised or not, the value attributed to it and how it is depreciated or not has a significant impact on the financial statements.	*Framework*?	(IFRS 16, covered in Chapter 14).	
2	**IAS 40, *Investment Property*** Entities may invest in property (IAS 40) for the potentially attractive returns that are available from capital gains and rental income. Measuring such investment properties as a series of out-of-date historic costs is not an appropriate reflection of volatility caused by changes in property values.	**Approach** This is a new standard at Advanced Level, so work through carefully. Make sure you can distinguish investment property from owner-occupied property, that you know when to recognise investment property and how to measure it, and can deal with a change of use. **Stop and think** An investment property held under the fair value model is developed for future sale. How should it be accounted for?	Investment property comes up quite regularly as part of a longer scenario question and in the context of audit and fair value (IFRS 13). With IFRS 16, *Leases* being a topical standard, expect to see examined the interaction between this and IAS 40.	**IQ11: Replacement property** Although short, this is a useful question because it tests how IAS 40 interacts with other standards.
3	**IAS 41, *Agriculture*** Working in the context of the agricultural sector or of extractive industries you will need to make judgements on the	**Approach** This standard is set at level A, so this section should be studied carefully, even though the topic has not yet been examined. Do	This topic has yet to be examined.	**IQ13: Costs to sell** This is a quick, test-your-knowledge question.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	valuation of biological assets/the appropriate capitalisation of expenditure. Since the use of a historic cost model was not seen as wholly appropriate for accounting for agricultural activity IAS 41, *Agriculture* is based on a fair value model.	both interactive questions. **Stop and think** What are bearer biological assets?		
4	**IFRS 6, *Exploration for and Evaluation of Mineral Resources*** Entities such as those in the oil and gas industry follow a wide range of practices for the accounting of exploration and evaluation expenditure (IFRS 6).	**Approach** This is examinable at level D, so is not a key topic. Read through quickly. The key point is that the recognition criteria for impairment are different from IAS 36 but, once impairment is recognised, the measurement criteria are the same as for IAS 36. **Stop and think** How should these assets be measured on initial recognition?	This topic has yet to be examined.	N/A
5	**IFRS 4, *Insurance Contracts*** IFRS 4 provides **interim guidance** on accounting for insurance contracts. It will be replaced in 2023 by IFRS 17.	**Approach** This is examinable at level D, so should not be prioritised at the expense of key standards, such as IAS 40. **Stop and think** Focus more on other standards!	This topic has yet to be examined.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
6	**Audit focus points** This section focuses on the audit of investment properties, as the other key topics were covered at Professional Level.	**Approach** Read and learn the table in Section 6.2, which gives an approach to auditing classification, valuation and disclosure. **Stop and think** Why might investment property be more risky than owner-occupied property?	You are very likely to get investment property as part of an integrated financial reporting and auditing question.	**IQ14: Investment property and fair value** A typical scenario as part of Question 3, covering both audit issues and audit procedures in relation to IAS 40 and IFRS 13.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Review of material from earlier studies

Section overview

You should already be familiar with the standards relating to current and non-current assets from earlier studies. If not, go back to your earlier study material.

- IAS 2, *Inventories*
- IAS 16, *Property, Plant and Equipment*
- IAS 38, *Intangible Assets*
- IAS 36, *Impairment of Assets*

Read the summary of knowledge brought forward and try the relevant questions. If you have any difficulty, go back to your earlier study material and revise it.

Assets have been defined in many different ways and for many purposes. The definition of an asset is important because it directly affects the **treatment** of such items. A good definition will prevent abuse or error in the accounting treatment: otherwise some assets might be treated as expenses, and some expenses might be treated as assets.

In the current accounting climate, where complex transactions are carried out daily a definition that covers ownership and value is not sufficient, leaving key questions unanswered.

- What determines ownership?
- What determines value?

The definition of an asset in the IASB's *Conceptual Framework for Financial Reporting* from earlier studies is given below.

Definition

Asset: A present economic resource controlled by the entity as a result of past events. An economic resource is a right that has the potential to produce economic benefits. (*Conceptual Framework, para. 4.2*)

This definition ties in closely with the definitions produced by **other standard-setters**, particularly the FASB (USA) and the ASB (UK).

A general consensus seems to exist in the standard setting bodies as to the definition of an asset which encompasses **three important characteristics**.

- Future economic benefit
- Control
- The transaction to acquire control has already taken place

1.1 Property, plant and equipment

The following concepts will be familiar to you from your earlier studies.

- **Cost** is the amount of cash or cash equivalents paid or the fair value of the other consideration given to acquire an asset at the time of its acquisition or construction.
- **Residual value** is the estimated amount that an entity would currently obtain from disposal of the asset, after deducting the estimated costs of disposal, if the asset were already of the age and in the condition expected at the end of its useful life.

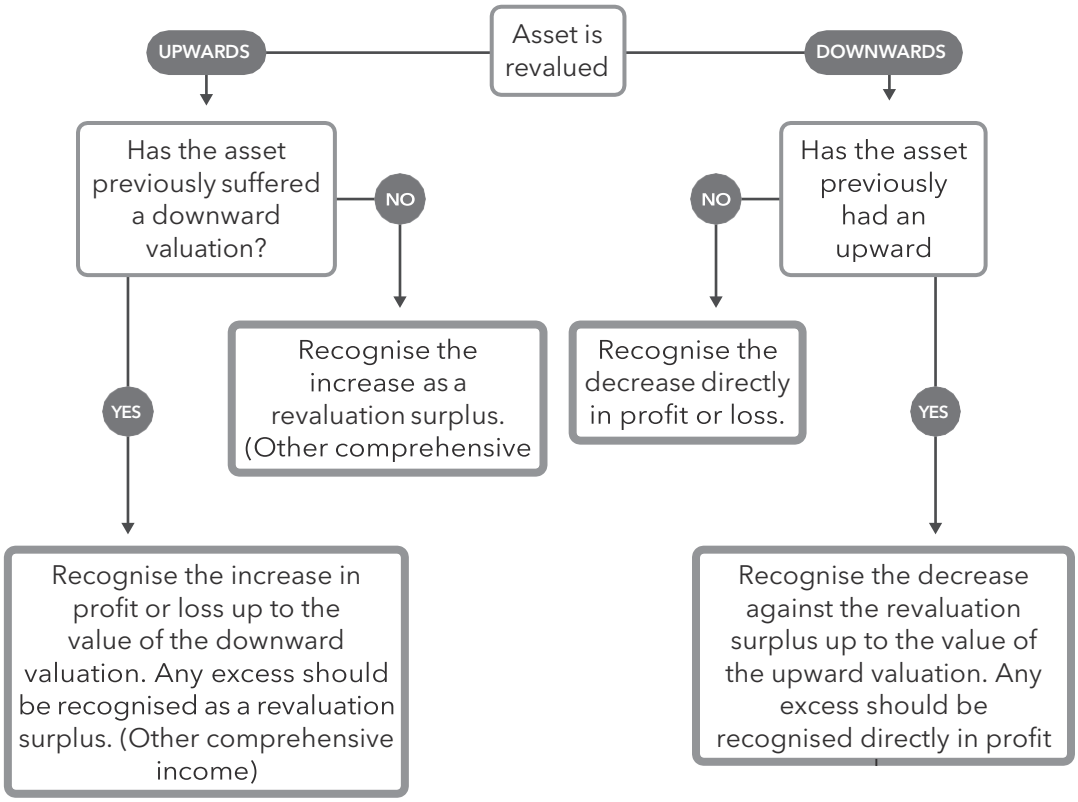
- **Fair value** is the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. (Note that this definition changed when IFRS 13, *Fair Value Measurement* came into force in January 2013.
- **Carrying amount** is the amount at which an asset is recognised after deducting any accumulated depreciation and accumulated impairment losses.

Accounting treatment

- As with all assets, **recognition** depends on two criteria:
  - It is probable that **future economic benefits** associated with the item will flow to the entity.
  - The cost of the item can be **measured reliably.**
- These recognition criteria apply to **subsequent expenditure** as well as costs incurred initially (ie, there are no longer separate criteria for recognising subsequent expenditure).
- Once recognised as an asset, items should **initially be measured at cost**.
- Cost is the **purchase price**, less trade discount/rebate plus:
  - **directly attributable costs** of bringing the asset to working condition for intended use; and
  - **initial estimate** of the **unavoidable cost of dismantling and removing the item** and **restoring the site** on which it is located.

  IAS 16, *Property, Plant and Equipment* also does the following:
- Provides additional guidance on directly attributable costs, including the cost of an item of property, plant and equipment
- States that income and related expenses of operations that are incidental to the construction or development of an item of property, plant and equipment should be recognised in profit or loss for the period
- Specifies that exchanges of items of property, plant and equipment, regardless of whether the assets are similar, are measured at **fair value**, unless the exchange transaction lacks commercial substance or the fair value of neither of the assets exchanged can be measured reliably. If the acquired item is not measured at fair value, its cost is measured at the carrying amount of the asset given up
- Permits a **choice of measurement models subsequent to initial recognition**
  - **Cost model**: carrying asset at cost less depreciation and any accumulated impairment losses
  - **Revaluation model**: carrying asset at revalued amount, ie, fair value less subsequent accumulated depreciation and any accumulated impairment losses. (IAS 16 makes clear that the revaluation model is available only if the fair value of the item can be measured reliably.) When using the revaluation model, a revaluation is required where the fair value of a revalued asset differs materially from its carrying amount. Rising inflation will result in a need for more frequent revaluations.

Figure 12.1: Revaluations



Professional skills focus: Structuring problems and solutions

The above flowchart is an example of how to structure a problem, in this case PPE revaluations. It should form part of your revision.

Interactive question 1: Revaluations

Binkie Co has an item of land carried in its books at £13,000. Two years ago a slump in land values led the company to reduce the carrying amount from £15,000. This was recorded as an expense. There has been a surge in land prices in the current year, however, and the land is now worth £20,000.

In the example given above assume that the original cost was £15,000, revalued upwards to £20,000 two years ago. The value has now fallen to £13,000.

Crinkle Co bought an asset for £10,000 at the beginning of 20X6. It had a useful life of five years. On 1 January 20X8 the asset was revalued to £12,000. The expected useful life has remained unchanged (ie, three years remain).

Requirements

1.1 Account for the revaluation in the current year

1.2 Account for the decrease in value

1.3 Account for the revaluation and state the treatment for depreciation from 20X8 onwards.

See **Answer** at the end of this chapter.

1.2 Inventories

Valuation Lower of: • Cost • Net realisable value Each item/group/category considered separately	**Allowable costs** Include: • Cost of purchase Exclude: • Cost of storage • Cost of selling
Determining cost • First in, first out (FIFO) • Weighted average cost	**Net realisable value** • Estimated cost of completion • Estimated costs necessary to make the sale (eg, marketing, selling and distribution)

1.3 IAS 38, *Intangible Assets*

- An **intangible asset** is an **identifiable non-monetary asset without physical substance**, such as a licence, patent or trademark.
- An intangible asset is **identifiable** if it is **separable** (ie, it can be sold, transferred, exchanged, licensed or rented to another party on its own rather than as part of a business) or it arises from **contractual or other legal rights**.
- An intangible asset should be recognised if it is probable that future economic benefits **attributable to the asset** will flow to the entity and the **cost** of the asset can be **measured reliably**.
- At recognition the intangible should be recognised at cost (purchase price plus directly attributable costs). After initial recognition an entity can **choose** between the **cost model** and the **revaluation model**. The revaluation model can only be adopted if an **active market** (as defined) exists for that type of asset.
- An intangible asset (other than goodwill recognised in the acquiree's financial statements) **acquired as part of a business combination** should initially be recognised **at fair value**.
- Internally generated goodwill should **not** be recognised.
- Expenditure incurred in the **research phase** of an internally generated intangible asset should be **expensed as incurred**.
- Expenditure incurred in the **development phase** of an internally generated intangible asset **mustbe capitalised** provided **certain tightly defined criteria are met**.
  - Expenditure incurred before the criteria being met may not be capitalised retrospectively.
- An intangible asset with a **finite useful life** should be **amortised** over its expected useful life, **commencing** when the asset is **available for use** in the manner intended by management.
- Residual values should be assumed to be nil, except in the rare circumstances when an active market exists or there is a commitment by a third party to purchase the asset at the end of its useful life.
- An intangible asset with an **indefinite life** should **not be amortised**, but should be reviewed for **impairment** on an **annual basis**.
  - There must also be an annual review of whether the indefinite life assessment is still appropriate.
- On disposal of an intangible asset the gain or loss is recognised in profit or loss.

1.4 IAS 36, *Impairment of Assets*

1.4.1 Key points

Recoverable amount	An asset's recoverable amount is the higher of **value in use** (net cash flows) and **fair value less costs of disposal**. Impairment losses occur where the **carrying amount** of an asset is **above its recoverable amount**.
Impairment indicators	An entity must do an impairment test when there are **impairment indicators**. These can be **internal** or **external**.
Cash-generating units	Where cash flows cannot be measured separately, the recoverable amount is calculated by reference to the **CGU**.
Recognition of impairment losses	Impairment losses are charged **first to other comprehensive income** (re any revaluation surplus relating to the asset) and then to profit or loss. In the case of a CGU, the credit is allocated **first against any goodwill** and then pro-rata over the other assets of the CGU.
After the impairment review	After the impairment review, depreciation/amortisation is allocated over the asset's **revised remaining useful life**.



Interactive question 2: Impairment loss

An entity has a single manufacturing plant which has a carrying value of £749,000. A new government elected in the country passes legislation significantly restricting exports of the product produced by the plant. As a result, and for the foreseeable future, the entity's production will be cut by 40%. Cash flow forecasts have been prepared derived from the most recent financial budgets/forecasts for the next five years approved by management (excluding the effects of general price inflation).

Year	1	2	3	4	5
	£'000	£'000	£'000	£'000	£'000
Future cash flows	230	211	157	104	233 (including disposal proceeds)

If the plant was sold now it would realise £550,000, net of selling costs.

The entity estimates the pre-tax discount rate specific to the plant to be 15%, after taking into account the effects of general price inflation.

Requirement

Calculate the recoverable amount of the plant and any impairment loss.

Note: PV factors at 15% are as follows.

Year	PV factor @15%
1	0.86957
2	0.75614
3	0.65752
4	0.57175
5	0.49718

See **Answer** at the end of this chapter.

Professional skills focus: Applying judgement

Judgement must be applied in considering whether an item is a cash-generating unit. Both in the exam, and in real life, it may not be obvious.

Interactive question 3: Cash-generating units

Discuss whether the following items would be cash-generating units in their own right, or part of a larger cash-generating unit.

(1) A pizza oven in a pizza restaurant

(2) A branch of a pizza restaurant in Warsaw

(3) A monorail that takes fee paying visitors to a theme park from its car park

(4) A monorail that transports fee paying commuters from a suburban part of town to the centre of town

(5) The internal large telephone network of a country's railway system, although its use is currently not permitted to anybody other than railway workers

See **Answer** at the end of this chapter.

1.4.2 Treatment of the non-controlling interest element of goodwill

The revision to IFRS 3 allows two methods of initially valuing the non-controlling (minority) interest in an entity:

- As a share of the net assets of the entity at the acquisition date; or
- at fair value.

The non-controlling interest is then taken into account in the goodwill calculation per the revised standard:

Purchase consideration	X
Non-controlling interest	X
	X
Total fair value of net assets of acquiree	(X)
Goodwill	X

This means that the resulting goodwill will represent:

- only the parent's share of total goodwill when valuing the non-controlling interest using the proportion of net assets method; and
- full goodwill (ie, the parent's share plus the non-controlling interest share) when using the fair value method.

Where the share of net assets method is used to value the non-controlling interest, the carrying amount of a CGU therefore comprises:

- the parent and non-controlling share of the identifiable net assets of the unit; and
- only the parent's share of the goodwill.

Part of the calculation of the recoverable amount of the CGU relates to the unrecognised share in the goodwill.

For the purpose of calculating the impairment loss, the carrying amount of the CGU is therefore notionally adjusted to include the non-controlling share in the goodwill by grossing it up.

The consequent impairment loss calculated is only recognised to the extent of the parent's share. Where the fair value method is used to value the non-controlling interest, no adjustment is required.



Interactive question 4: Allocation of impairment loss

Peter acquired 60% of Stewart on 1 January 20X1 for £450 million recognising net assets of £600 million, a non-controlling interest (valued as a proportion of total net assets) of £240 million and goodwill of £90 million. Stewart consists of a single cash-generating unit.

Due to adverse publicity, the recoverable amount of Stewart had fallen by 31 December 20X1. The depreciated value of the net assets at that date was £550 million (excluding goodwill). No impairment losses have yet been recognised relating to the goodwill.

Requirement

Show the allocation of the impairment losses:

(1) if the recoverable amount was £510 million at 31 December 20X1

(2) if the recoverable amount was £570 million at 31 December 20X1

See **Answer** at the end of this chapter.

1.4.3 Reversal of past impairments

A reversal for a CGU is allocated to the assets of the CGU, except for goodwill, pro rata with the carrying amounts of those assets.

However, the carrying amount of an asset is not increased above the lower of:

- its recoverable amount (if determinable); and
- its depreciated carrying amount had no impairment loss originally been recognised.

Any amounts left unallocated are allocated to the other assets (except goodwill) pro rata.

The reversal is recognised in profit or loss, except where reversing a loss recognised on assets carried at revalued amounts, which are treated in accordance with the applicable IFRS® Standard.

For example, an impairment loss reversal on property, plant and equipment first reverses the loss recorded in profit or loss and any remainder is credited to the revaluation surplus (IAS 16).

Goodwill

Once recognised, impairment losses on goodwill are **not reversed**.

1.4.4 Impairment and IFRS 5

IFRS 5 is covered in detail in Chapter 9. Regarding impairment, note the following:

- Immediately before initial classification as held for sale, the asset (or disposal group) is **measured in accordance with the applicable IFRS** (eg, property, plant and equipment held under the IAS 16 revaluation model is revalued).
- On classification of the non-current asset (or disposal group) as held for sale, it is **written down to fair value less costs to sell** (if less than carrying amount).

- Any impairment loss arising under IFRS 5 is **charged to profit or loss** (and the credit allocated to assets of a disposal group using the IAS 36 rules, ie, first to goodwill then to other assets pro rata based on carrying value).
- Non-current assets/disposal groups classified as held for sale are ***not* depreciated/amortised.**
- Any subsequent changes in fair value less costs to sell are **recognised as a further impairment** loss (or reversal of an impairment loss).
- However, gains recognised **cannot exceed cumulative impairment losses** to date (whether under IAS 36 or IFRS 5).

2 IAS 40, *Investment Property*



Section overview

- An investment property is land or buildings or both that is held by an entity to earn rentals and/or for its capital appreciation potential.
- Following initial measurement, investment property may be measured using the cost model or the fair value model.

One of the distinguishing characteristics of **investment property** is that it generates cash flows largely independent of the other assets held by an entity.

Owner-occupied property is not investment property and is accounted for under IAS 16, *Property, Plant and Equipment*.

2.1 Recognition

Investment property should be recognised as an asset when **two conditions** are met.

- It is **probable** that the **future economic benefits** that are associated with the investment property will **flow to the entity**.
- The **cost** of the investment property can be **measured reliably**.

2.2 Initial measurement and further aspects of recognition

An investment property should initially be measured at its cost, including transaction costs.

Further aspects of recognition	
Transaction costs	Expenses that are directly attributable to the investment property, for example professional fees and property transfer taxes. Cost does not include activities that, while related to the investment property, are not directly attributable to it. For example, start-up costs, abnormal amounts of wasted resources in constructing the property, relocation costs, losses incurred before full occupancy and the normal servicing of the property are not directly attributable.
Self-constructed investment properties	Property being self-constructed or under development for future use as an investment property qualifies itself as an investment property.

Leases	An investment property held by a lessee as a right-of-use asset must be recognised in accordance with IFRS 16, *Leases* (see Chapter 14).
Entity occupies part of property and leases out balance	If the two portions can be sold separately or leased separately, each is accounted for as appropriate. If not, entire property is an investment property only if **insignificant** portion is owner occupied.
Entity supplies services to the lessee of the property	An investment property only if the services are insignificant to the arrangement as a whole.
Property leased to and occupied by parent, subsidiary or other group company	An investment property in entity's own accounts but owner occupied from group perspective.



Interactive question 5: Identification of investment property

(1) An entity has a factory that has been shut down due to chemical contamination, worker unrest and strike. The entity plans to sell this factory.

(2) An entity has purchased a building that it intends to lease out under an operating lease.

(3) An entity has acquired a large-scale office building, with the intention of enjoying its capital appreciation. Rather than holding it empty, the entity has decided to try to recover its running costs by renting the space out for periods which run from one week to one year. To make the building attractive to potential customers, the entity has fitted the space out as small office units, complete with full-scale telecommunications facilities, and offers reception, cleaning, a loud speaker system and secretarial services. The expenditure incurred in fitting out the offices has been a substantial proportion of the value of the building.

(4) An entity acquired a site on 30 April 20X4 with the intention of building office blocks to let. After receiving planning permission, construction started on 1 September 20X4 and was completed at a cost of £10 million on 30 March 20X5 at which point the building was ready for occupation.

The building remained vacant for several months and the entity incurred significant operating losses during this period.

The first leases were signed in July 20X5 and the building was not fully let until 1 September 20X6.

Requirement

Do the buildings referred to in (1)–(4) above meet the definition of investment property?

See **Answer** at the end of this chapter.

2.3 Measurement subsequent to initial recognition

Following initial measurement, investment properties are held **either**:

- at cost less accumulated depreciation (the **cost model**); or
- measured at fair value (the **fair value model**).

2.3.1 Cost model

Where the cost model is adopted, the property should be accounted for in accordance with IAS 16, *Property, Plant and Equipment*.

2.3.2 Fair value model

- After initial recognition, an entity that chooses the **fair value model** should measure all of its investment property at fair value, except in the extremely rare cases where this cannot be measured reliably. In such cases it should apply the IAS 16 cost model.
- A gain or loss arising from a change in the fair value of an investment property should be recognised in net profit or loss for the period in which it arises.
- The fair value of investment property should reflect market conditions at the reporting date.
- A change in use of an investment property may lead to a change in classification.

Whatever policy the entity chooses should be applied to **all of its investment property**.



Interactive question 6: Estimating fair value

An entity with a 31 December year end owns the freehold of an office block standing on a city centre site on which there are four other similar buildings, none of which are owned by the entity. All the office buildings were constructed at the same time as the entity's building and the floors in all five buildings are let out on standard 25-year leases.

Requirements

Which of the following values could be used by the entity as a basis for estimating the fair value of its office building at 31 December 20X5, according to IAS 40?

6.1 The first of the other office buildings changed hands early in 20X5 for £5 million as a result of an auction which was widely publicised in the professional property press.

6.2 The second of the other office buildings changed hands late in 20X5 for £6 million as a result of a sale to an entity, 55% of whose shares were owned by the seller.

6.3 The third of the other office buildings changed hands late in 20X5 for £4.5 million as a result of sale to a financial institution to which the seller owed £3.5 million. It is understood that the seller had breached its banking covenants and had to raise cash by the end of 20X5.

6.4 The fourth of the other office buildings changed hands late in 20X5 for £5.5 million as a result of a sale to an overseas institution which was seeking to establish its first foothold in the country's property market. The offer of the office building was widely publicised in the professional property press although it is understood that local institutions were only prepared to offer in the region of £4.9 million.

See **Answer** at the end of this chapter.

2.4 Derecognition

When an investment property is derecognised, a gain or loss on disposal should be recognised in profit or loss. The gain or loss should normally be determined as the difference between the net disposal proceeds and the carrying amount of the asset.

Interactive question 7: Disposal of investment property

An entity purchased an investment property on 1 January 20X3, for a cost of £5.5 million. The property has a useful life of 50 years, with no residual value and at 31 December 20X5 had a fair value of £6.2 million. On 1 January 20X6 the property was sold for net proceeds of £6 million.

Requirement

Calculate the profit or loss on disposal under both the cost and fair value model.

See **Answer** at the end of this chapter.

2.5 Transfers following a change in use

A change in use may lead to recognition or de-recognition of an investment property. The following is a summary of such instances.

Evidence of change in use	Accounting treatment
Commencement of owner occupation	• Owner-occupied property recognised under IAS 16. • If fair value model was used, treat fair value as deemed cost.
Commencement of development with a view to sale	• Reclassify as inventory under IAS 2. • If fair value model was used, treat fair value as deemed cost.
Development with view to continue letting	• Continue to hold as an investment property.
End of owner occupation with view to let to third parties	• Transfer to investment properties under IAS 40. • If fair value model to be used, revalue at date of change and recognise difference as revaluation under IAS 16.
Property held as inventory now let to a third party	• Transfer to investment properties under IAS 40. • If fair value model to be used, revalue at date of change and recognise difference in profit or loss.
Commencement of lease to another party	• Transfer from property, plant and equipment to investment property under IAS 40.

Professional skills focus: Assimilating and using information

When considering a change of use, it is important to understand all the required evidence about its current use and its future use, including the model that is to be used. The current use affects the future accounting treatment.

Professional skills focus: Concluding, recommending and communicating

It is possible that you may, in an exam, come to the wrong conclusion about change of use, but you will gain credit for any valid arguments that led you to your conclusion, and which are clearly communicated.

Interactive question 8: Change of use

An entity with a 31 December year end purchased an office building, with a useful life of 50 years, for £5.5 million on 1 January 20X1. The amount attributable to the land was negligible. The entity used the building as its head office for five years until 31 December 20X5 when the entity moved its head office to larger premises. The building was reclassified as an investment property and leased out under a five-year lease.

Owing to a change in circumstances the entity took possession of the building five years later on 31 December 20Y0, to use it as its head office once more. At that date the remaining useful life of the building was confirmed as 40 years.

The fair value of the head office was as follows.

- At 31 December 20X5 £6 million
- At 31 December 20Y0 £7.5 million

Requirements

How should the changes of use be reflected in the financial statements on the assumption that:

(a) the entity uses the cost model for investment properties?

(b) the entity uses the fair value model for investment properties?

See **Answer** at the end of this chapter.

2.6 Summary of disclosure requirements

An entity shall disclose the following:

- Whether it has followed the fair value model or cost model
- Whether property interests held as leases are included in investment property
- Criteria for classification as investment property
- Assumptions in determining fair value
- Use of independent professional valuer (encouraged but not required)
- Rental income and expenses
- Any restrictions or obligations

2.6.1 Fair value model – additional disclosures

An entity that adopts this must also disclose a **reconciliation** of the carrying amount of the investment property at the beginning and end of the period.

2.6.2 Cost model – additional disclosures

These relate mainly to the depreciation method, rates and useful lives used as well as a reconciliation of the carrying amount at the beginning and end of the period. In addition, an entity which adopts the cost model **must disclose the fair value** of the investment property.

Interactive question 9: Installation of new equipment 1

An entity owns the freehold of an office building which was acquired on 31 December 20X0 for £17 million, £2 million of which was attributable to the land. The freehold is an investment property measured under the cost model with the building's useful life estimated at 30 years. The building was fully equipped with an air-conditioning system. No separate value was placed on the air conditioning unit as this was not something that was required by accounting standards at the time of acquisition.

On 31 December 20X5 the entity replaced the air-conditioning system for £1.2 million, which has an estimated useful life of 10 years. As no more reliable information was available, it used this cost as an indication of the cost of the old system.

Requirement

How should the replacement of the air-conditioning be accounted for?

See **Answer** at the end of this chapter.



Interactive question 10: Installation of new equipment 2

An entity with a 31 December year end owns an office building which is recognised as an investment property. The lift system is an integral part of the office building. The entity uses the fair value model for measurement of investment properties.

The lift system was purchased on 1 January 20X0 for £400,000 and is being depreciated at 12.5% per annum on cost. Its carrying amount has been accepted as a reasonable value at which to include it within the fair value of the office building as a whole.

Early in December 20X5 a professional valuer determined the fair value of the office building, including the lift system, to be £3 million. The lift system failed on 28 December 20X5 and was immediately replaced on 31 December 20X5 with a new system costing £600,000.

Requirement

How should the lift system be recognised?

See **Answer** at the end of this chapter.



Interactive question 11: Replacement property

An entity with a 31 December year end owns an investment property which it measures using the fair value model. At 31 December 20X4, the property's carrying amount is £4 million. On 30 June 20X5, an explosion close to the property causes major damage to the property. In July 20X5, the entity makes a number of insurance claims as a result, one of which is for the rebuilding cost, estimated at £3.7 million.

Although the property is repairable, the entity decides to sell it in its present state and buy a replacement property. This decision is made on 30 September 20X5, on which date the damaged property meets the criteria for classification as held for sale. Its fair value on that date is £350,000 and the costs to sell are £35,000. The fair value does not change between 30 September 20X5 and 31 December 20X5. The sale is completed in the middle of 20X6 for £375,000, with selling costs of £40,000.

On 1 March 20X6, the entity acquires a replacement property for £3.8 million.

The entity's insurers contest the claim relating to the building on the basis of an exclusion clause. The entity disagrees with the insurers' interpretation and in February 20X6 initiates legal proceedings. Negotiations are protracted and it is not until the end of 20X7 that the insurers agree to settle for £3.9 million.

Requirement

How should the entity recognise these transactions?

See **Answer** at the end of this chapter.

3 IAS 41, *Agriculture*

Section overview

IAS 41 sets out the accounting treatment, including presentation and disclosure requirements, for agricultural activity.

3.1 Definitions

Definitions

Agricultural activity: Defined as the management of the biological transformation of biological assets for sale, into agricultural produce, or into additional biological assets.

Agricultural activities include, for example, raising livestock, forestry and cultivating orchards and plantations.

In its simplest form a biological transformation is the process of growing something such as a crop, although it also incorporates the production of agricultural produce such as wool and milk.

Biological transformation: Comprises the processes of growth, degeneration, production and procreation that cause qualitative or quantitative changes in a biological asset.

Biological asset: A living plant or animal.

Agricultural produce: The harvested produce of an entity's biological assets.

IAS 41 considers the classification of biological assets and how their characteristics, and hence value, change over time. The standard applies to agricultural produce up to the point of harvest, after which IAS 2, *Inventories* is applicable. A distinction is made between the two because IAS 41 applies to biological assets throughout their lives but to agricultural produce only at the point of harvest.

IAS 41 includes a table of examples which clearly sets out three distinct stages involved in the production of biological assets. For example, we can identify dairy cattle as the biological asset, milk as the agricultural produce and cheese as the product that is processed after the point of harvest.

Calves and cows are biological assets as they are living animals whereas beef and milk are agricultural produce.

Agricultural activities may be quite diverse, but all such activities have similar characteristics as described below.

Common characteristics of agricultural activities

- **Capacity to change** – living animals and plants are capable of changing. For example, a sapling grows into a fruit tree which will bear fruit and a sheep can give birth to a lamb;
- **Management of change** – the biological transformation relies on some form of management input, ensuring, for example, the right nutrient levels for plants, providing the right amount of light or assisting fertilisation; and
- **Measurement of change** – the changes as a result of the biological transformation are measured and monitored. Measurement is in relation to both quality and quantity.

Context example: Common characteristics of agricultural activities

An entity is involved in the production and sale of raw materials for food products, in the sale of fish reared at its own 'fish farms' and in the sale of fish caught in the Northern seas by ocean-going trawlers. An analysis of the processes involved is as follows.

Food products raw materials

The entity owns farmland on which it grows annual crops of corn for sale to food manufacturers. The growing process is aided by the careful application of a range of nutrients, while additives are administered to the underlying land immediately after harvesting has ended. This is an agricultural activity; the corn growing each year is the biological transformation; growth is encouraged by management's activities and the change is monitored, to identify the time at which harvest should commence.

Fish farming

The entity leases a number of privately-owned lakes into which it puts underwater tanks for the rearing of fish. This is an agricultural activity, because the fish are grown to the size suitable for sale and their feed must be provided for them since the amount available naturally in the tanks will be insufficient.

Ocean fishing

The entity owns a number of trawlers. The trawlers go to sea in search of suitable fish. This is not an agricultural activity. Although there is biological transformation on the part of the fish, there is no management intervention in the process; neither is there any routine measurement of the amount of any change which has taken place. The trawlers harvest what the seas yield naturally.

3.1.1 Bearer biological assets

An amendment was issued to IAS 41 in 2014 regarding plant-based bearer biological assets, which would include trees grown in plantations, such as grape vines, rubber trees and oil palms.

These plants are used solely to grow crops over several periods and are not in themselves consumed. When no longer productive they are usually scrapped.

It was decided that fair value was not an appropriate measurement for these assets as, once they reach maturity, the only economic benefit they produce comes from the agricultural produce they create. In this respect, they are similar to assets in a manufacturing activity.

Consequently, these assets have been removed from the scope of IAS 41 and should be accounted for under IAS 16, *Property, Plant and Equipment*. They are measured at accumulated cost until maturity and are then subject to depreciation and impairment charges. The IAS 16 revaluation model could also be applied. Agricultural produce from these plants continues to be recognised under IAS 41.

Interactive question 12: Classification

Into which category would the following items be classified according to IAS 41, *Agriculture*?

12.1 Wool

12.2 Sugar

See **Answer** at the end of this chapter.

3.2 Recognition and measurement

3.2.1 Recognition criteria

A biological asset or agricultural produce should only be recognised when:

- the entity controls the asset as a result of past events, for example the acquisition of dairy cattle. The past event is the purchase, and control is obtained as the entity is now the legal owner;
- it is probable that future economic benefits will flow to the entity, for example because the dairy cattle will produce milk which can be sold or processed into cheese and sold; and
- fair value, or cost, of the asset can be measured reliably.

3.2.2 Measurement of biological assets

A biological asset should initially be measured at its **fair value less estimated costs to sell**, such as duty and commission to brokers or dealers.

Costs to sell

Costs to sell do not include any costs that are necessary to get the asset to a market, for example transport. These should, however, be deducted in determining fair value.

Fair value

Fair value is 'the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date'.

Where an active market exists for a biological asset the quoted price in the market is the appropriate fair value.

Where an active market does not exist, then fair value may be derived by using:

- the most recent transaction in the market, assuming that similar economic conditions exist at the time of the transaction and at the reporting date;
- market prices for similar assets with appropriate adjustments to reflect differences; and
- sector-based benchmarks, for example the value of meat per kilogram.

IAS 41 includes the presumption that it will be possible to fair value a biological asset. But if fair value cannot be measured reliably at the time of initial recognition then the biological asset should be recognised at cost less accumulated depreciation and impairment cost (i.e., the decrease in the recoverable amount of an asset). Fair value should then be used as soon as a reliable measurement can be made.

At subsequent reporting dates a biological asset should continue to be measured at its fair value. Once a biological asset has been measured at fair value it is not possible to revert to cost.

3.2.3 Measurement of agricultural produce

Agricultural produce should be measured at its fair value, less estimated costs to sell at the point of harvest. Subsequent measurement is by reference to IAS 2. It will always be possible to fair value the agricultural produce since, by its very nature, there must be a market for it.



Interactive question 13: Costs to sell

Which of the following expenses would be classified as costs to sell when valuing biological assets and agricultural produce?

A Commission to brokers

B Transfer taxes and duties

C Transport costs

D Advertising costs

See **Answer** at the end of this chapter.



Worked example: Fair value

An entity rears animals to be sold in a local market. The market is 50 km away, and transport to market costs £1 per animal. At the measurement date the open market value is £60. The auctioneers charge a sales commission of 2% of market value and there is a government levy, based on market value, of 1% on purchases and 3% on sales.

Requirement

How should the fair value less cost to sell be calculated?

Solution

The fair value less costs to sell is calculated as:

	£
Market value	60.00
Transport to market costs	(1.00)
Fair value	59.00
Costs to sell	
Auctioneers' commission – 2% of £60	(1.20)
	£
Government levy – 3% of £60	(1.80)
Fair value less costs to sell	56.00

3.2.4 Grouping assets

Grouping biological assets, or agricultural produce, according to significant attributes, such as age or quality, may help to establish fair value. Such groupings should be consistent with attribute groupings that are used in the market as a basis for pricing, for example, by wine vintage.

Some biological assets are physically attached to land, for example tree plantations, and it is necessary to value the land and biological assets together as one asset, even though agricultural land is not within the scope of IAS 41. To obtain the fair value of the biological assets, the fair value of the land element should be deducted from the combined fair value.

Land is dealt with under IAS 16 or IAS 40.

3.3 Gains and losses

Gains or losses arising on the initial recognition at fair value of a biological asset and agricultural produce should be reported directly in profit or loss for the period to which they relate, for example a gain may arise on the birth of a calf. Subsequent changes in the fair value will also be reported directly in profit or loss.

Gains or losses on the initial recognition of agricultural produce should also be included in profit or loss in the period in which they arise. Such gains or losses may arise as a result of

harvesting, because the harvested crop may be worth more than the unharvested crop. In this case a gain would arise.

Worked example: Changes in fair value

A herd of five four-year old animals was held on 1 January 20X3. On 1 July 20X3, a 4.5-year old animal was purchased. The fair values less estimated costs to sell were as follows.

4. year-old animal at 1 January 20X3	£200
4.5-year-old animal at 1 July 20X3	£212
5. year-old animal at 31 December 20X3	£230

Requirement

Show the reconciliation of the changes in fair value.

Solution

The movement in the fair value less estimated costs to sell of the herd can be reconciled as follows.

	£
At 1 January 20X3 (5 × £200)	1,000
Purchased	212
Change in fair value (the balancing figure)	168
At 31 December 20X3 (6 × £230)	1,380

The entity is encouraged to disclose separately the amount of the change in fair value less estimated costs to sell arising from physical changes and price changes.

If it is not possible to measure biological assets reliably and they are instead recognised at their cost less depreciation and impairment an explanation should be provided of why it was not possible to establish fair value. A full reconciliation of movements in the net cost should be presented with an explanation of the depreciation rate and method used.

4 IFRS 6, *Exploration for and Evaluation of Mineral Resources*

Section overview

IFRS 6, *Exploration for and Evaluation of Mineral Resources* has been effective since 2006 and essentially deals with two matters.

- Allows entities to use existing accounting policies for exploration and evaluation assets.
- Requires entities to assess exploration and evaluation assets for impairment. The recognition criteria for impairment are different from IAS 36 but, once impairment is recognised, the measurement criteria are the same as for IAS 36.

4.1 Scope

The standard deals with the accounting of expenditures on the exploration for and evaluation of mineral resources (that is, minerals such as gold, copper, etc, oil, natural gas and similar resources), except:

- expenditures incurred before the acquisition of legal rights to explore; and
- expenditures incurred following the assessment of technical and commercial feasibility.

IAS 8, *Accounting Policies, Changes in Accounting Estimates and Errors* still applies to such industries in helping them determine appropriate accounting policies. Thus, accounting policies must present information that is relevant to the economic decision needs of users. Entities may change their policies under IFRS 6 as long as the new information comes closer to meeting the IAS 8 criterion.

4.2 Measurement at recognition

At recognition, exploration and evaluation assets must be measured at cost.

Entities must determine which expenditures to recognise and **apply their policy consistently**. Such expenditure may include acquisition of rights to explore, exploratory drilling, sampling, studies and activities relating to commercial evaluation.

Expenditure related to the **development of mineral resources** is outside the scope of IFRS 6. This comes under IAS 38.

4.3 Measurement after recognition

Entities must apply either the **cost model** or the **revaluation model** in IAS 16.

4.4 Changes in accounting policies

These may be made if the change makes the financial statements more relevant to users. IAS 8 criteria need to be applied.

4.5 Classification and reclassification

Exploration and evaluation assets are classified as **tangible or intangible according to the nature** of the assets acquired. The classification must be applied consistently. They should no longer be classified as such when the technical feasibility and commercial viability of extracting a mineral resource are demonstrable.

4.6 Impairment

The difficulty with respect to exploratory activities is that future economic benefits are generally very uncertain and hence forecasting future cash flows, for example, is difficult. IFRS 6 modifies IAS 36 to state that impairment tests are required:

- When the technical and commercial viability of extraction is demonstrable, at which point IFRS 6 is no longer relevant to the asset.
- When other facts indicate that the carrying amount exceeds recoverable amounts, such as:
  - exploration rights have expired;
  - substantive expenditure on further exploration for and evaluation of mineral resources in the specific area is neither budgeted nor planned;
  - there has been no success in finding commercially viable mineral resources and the entity has decided to discontinue exploratory activities within a specific area; and
  - estimates suggest that the carrying amounts of assets are unlikely to be recovered in full following successful development of the mineral resource.

In such circumstances, impairment is undertaken in accordance with IAS 36.

4.7 Presentation and disclosure

Exploration and evaluation assets are recorded as tangible or intangible assets, as appropriate. Once benefits are demonstrable, assets dealt with under IFRS 6 are superseded by other appropriate standards and are reclassified accordingly.

Disclosure relates to the following:

- A description of the accounting policies applied
- The amounts relating to assets, liabilities, income and expense, and operating and investing cash flows arising from exploration for and evaluation of mineral resources

5 IFRS 17, *Insurance Contracts*

Section overview

IFRS 17 became effective in 2023, replacing the previous Standard, IFRS 4, *Insurance Contracts*. It introduces a comprehensive financial reporting framework for insurance contracts which aims to achieve greater comparability and consistency in financial reporting by insurers.

5.1 What is an insurance contract?

Definition

Insurance contract: A contract between two parties, where one party, the insurer, agrees to compensate the other party, the policyholder, if it is adversely affected by an uncertain future event.

An uncertain future event exists where at least one of the following is uncertain at the inception of an insurance contract:

- the occurrence of an insured event;
- the timing of the event; or
- the level of compensation that will be paid by the insurer if the event occurs (IFRS 17 Appendix B).

Some insurance contracts may offer **payments-in-kind** rather than **compensation** payable to the policyholder directly. For example, an insurance repair contract may pay for a washing machine to be repaired if it breaks down; the contract will not necessarily pay monetary compensation.

In identifying an insurance contract it is important to make the distinction between **financial risk** and **insurance risk**. A contract that exposes the issuer to financial risk without significant insurance risk does not meet the definition of an insurance contract.

Financial risk is where there is a possible change in a financial or non-financial variable, for example a specified interest rate, commodity prices, an entity's credit rating or foreign exchange rates.

Definition

Insurance risk: A risk that is not a financial risk, transferred from the holder of a contract to the issuer. The risk in an insurance contract is whether an event will occur (rather than arising from

a change in something), for example a theft, damage against property, or product or professional liability.

Examples of insurance contracts

Appendix B to IFRS 17, which forms an integral part of the standard, includes an extensive list of examples of insurance contracts including:

- life insurance and prepaid funeral plans. It is the timing of the event that is uncertain here, for example certain life cover plans only pay out if death occurs within a specified period of time;
- disability and medical cover;
- credit insurance, covering the policyholder for non-recoverable receivables; and
- travel cover to provide against any loss suffered while travelling.

Examples of an insurer taking on insurance risk are as follows:

- An insurance contract issued to a policyholder against the escalation of claims from faulty motorcycles. The fault was discovered a year ago and the extent of total claims is yet to be established. This is an insurance contract since the insured event is the discovery of the ultimate cost of the claims.
- A gas boiler repair service available from a supplier who, for the payment of a fixed fee, will fix the malfunctioning boiler. This is an insurance contract as it is a payment-in-kind contract, with the uncertain event being whether the boiler will break down and the policyholder will be adversely affected.

Examples which are not insurance contracts

It is important to distinguish between **insurance contracts** and **other contracts** that are not covered by IFRS 17 but which might look like insurance contracts. To provide clarification IFRS 17 specifically identifies a number of areas where its provisions do not apply, for example:

- the provision of product warranties given directly by the manufacturer, dealer or retailer;
- employers' assets and liabilities in relation to employee benefit plans and obligations under a defined benefit plan;
- a contractual right, or obligation, that is contingent on the right to use a non-financial item, for example some licences;
- a lease that contains a residual value guaranteed by the lessee, ie, a specified value for the asset at the end of the lease is guaranteed by the lessee;
- financial guarantees within the scope of IFRS 9, *Financial Instruments;*
- contingent consideration that has arisen as a result of a business combination; and
- insurance contracts that the entity holds as policyholder.
- *credit card contracts that meet the definition of an insurance contract if, and only if, the entity does not reflect an assessment of the insurance risk associated with a particular customer when setting the price of the contract for that customer.*

5.2 *Recognition and measurement*

IFRS 17 requires that its accounting provisions are applied to portfolios of insurance contracts, being groups of insurance contracts that are exposed to similar risks and are managed together.

The underlying issue that the Standard addresses is when an insurer should recognise a liability for insurance contracts and how that liability should be measured.

A group of insurance contracts that have been issued by an entity should be recognised from the earliest of:

- *The beginning of the coverage period of the group of contracts; and*
- *The date when the first payment from a policyholder in the group becomes due.*

There are two measurement models within IFRS 17, the general measurement model and a simplified model, known as the premium allocation approach (PAA). An awareness of the general measurement model is sufficient in Corporate Reporting.

5.3 *IFRS 17*

5.3.1 *General measurement model*

The general measurement model uses a building block approach to measure insurance contracts. This general method initially measures a group of insurance contracts as the sum of:

- *Fulfilment cash flows, which are risk-adjusted and probability weighted estimates of the present value of expected cash inflows and outflows that will arise as the insurer fulfils the contracts; and*
- *The contractual service margin (CSM), which is the amount that represents the unearned profit that the entity will recognise in profit or loss as services are provided.*

For a group of profitable insurance contracts, the CSM is an equal and opposite amount to fulfilment cash flows (a net cash inflow) so ensuring that no asset or liability and no income or expense arises at initial recognition.

For a group of loss making (onerous) contracts, the fulfilment cash flows are a net cash outflow and this loss is recognised in profit or loss immediately and the CSM is zero.

Subsequent to initial recognition, the insurance contract liability comprises:

- *A liability for remaining coverage, being the fulfilment cash flows that relate to future coverage to be provided and the remaining CSM, plus*
- *A liability for incurred claims, being the fulfilment cash flows for claims and expenses already incurred but not yet paid.*

Fulfilment cash flows are remeasured at each reporting date and changes in estimates are recognised in profit or loss or other comprehensive income depending on their nature. These are presented as insurance revenue, insurance service expenses and insurance finance income or expenses.

5.3.2 *Reinsurance contracts*

The measurement of reinsurance contracts follows the same measurement principles and mirrors that of the primary insurance contract(s) to which it is related.

5.3.3 *Implementation challenges*

The implementation of IFRS 17 is likely to represent a significant systems challenge to insurers, and auditors will need to perform additional work in relation to these systems and the related internal controls. Implementation is likely to be a complex, lengthy process, with the following implications:

(a) *Changes to profit recognition patterns*

(b) *Increased volatility of profit and equity*

(c) *Increased options and requirement for judgement*

Set against this, it can be argued that the changes will bring greater transparency through disclosure.

6 Audit focus points

Section overview

The audit of investment properties should focus on:

- whether the property has been correctly classified;
- whether the valuation is materially correct (either under the cost model or the FV model); and
- whether the disclosure complies with IAS 40/IFRS 13.

Other areas will have been tested at the Professional Level in Audit and Assurance but may still require some key questions to be answered by auditors in the Corporate Reporting exam.

6.1 General approach to auditing assets

You should be familiar with the auditing of tangible and intangible assets from the Certificate Level Assurance exam. If you do not feel confident in this area, we would recommend that you revise it.

Your earlier studies should give you a good understanding of the audit techniques, and the sources of audit evidence, that an auditor can use in a range of different scenarios. Your knowledge of the financial reporting standards will inform your specific approach in the exam. For example:

- testing for tangible non-current assets should consider both over- and under-statement, as well as valuation and impairment
- intangible assets can vary significantly, but in most cases, the auditor is on the lookout for creative accounting and instances where expenditure might not be recognised when it should be, leading to overstated assets and understated expenses
- inventory is usually going to be a material balance, so auditors must concentrate on the two key issues of quantity (using the inventory count) and valuation (lower of cost and net realisable value)

6.2 Auditing investment properties

Definition

Investment property: Property (land or a building – or part of a building – or both) held (by the owner or by the entity as a right-of-use asset) to earn rentals or for capital appreciation or both, rather than for:

- use in the production or supply of goods or services or for administrative purposes; or
- sale in the ordinary course of business.

The following would be non-investment properties:

- Property held for sale in the ordinary course of business
- Property being constructed or developed on behalf of third parties
- Owner-occupied property

Investment property is initially measured at its **cost** (including transaction costs and directly attributable expenditure). After recognition it is measured at **either depreciated cost or fair value**.

Audit evidence

Issue	Evidence
Classification as an investment property	Confirm that all investment properties are classified in accordance with the IAS 40 definition. This will include: • a building owned by the entity and leased out under one or more operating leases; and • a building that is vacant but is held to be leased under one or more operating leases.
	Verify rental agreements, ensuring that the occupier is not a connected company and that the rent has been negotiated at arm's length If the building has recently been built, check the architect's certificates to ensure that cost/fair value is reasonable
Valuation	If cost model adopted, check compliance with IAS 16. If fair value model adopted: • Check that fair value has been measured in accordance with IFRS 13 • Where current prices in an active market are not available confirm that alternative valuation basis is reasonable and in accordance with IFRS 13, and document the relevant audit evidence • Agree valuation to valuer's certificate • Recalculate gain or loss on change in fair value and agree to amount in statement of profit or loss and other comprehensive income • If fair value cannot be measured reliably confirm use of cost model • Consider the use of an auditor's expert to review the appropriateness of the underlying market assumptions and valuation methodology used
Disclosure	Confirm compliance with IAS 40/IFRS 13, for example: • Disclosure of policy adopted • If fair value model adopted disclosure of a reconciliation of carrying amounts of investment property at the beginning and end of the period • Disclosure of the inputs used to measure fair value based on the fair value hierarchy (Level 1, Level 2 or Level 3) • Where Level 2 or Level 3 inputs are used, a description of the valuation techniques and inputs used (interest rates, net reversionary yield, stabilised net rental value, costs to complete and developer's profit) • For fair value measurements using significant unobservable inputs (Level 3), the effect of the measurements on profit or loss and other comprehensive income for the period

Note: IAS 40 states that fair value should be measured in accordance with IFRS 13.

Interactive question 14: Investment property and fair value

Propertyco, an investment property company, has a portfolio of properties, including the following:

Property A	This is used as the company head office.
Property B	This is held as a right-of-use asset and is currently rented out to a non-group company under an operating lease.
Property C	This was acquired in the year at a cost of £3 million including legal fees. It is currently vacant but a tenant is being actively sought.
Property D	This has been owned by Propertyco for a number of years and is currently rented out to a non-group company.

Propertyco uses the fair value model in accordance with IAS 40. Currently all the above properties are recorded in the financial statements at fair value.

Requirements

Based on the information above:

(a) Identify the audit issues which the auditor would need to consider

(b) List the audit procedures you would perform regarding fair values

See **Answer** at the end of this chapter.

Context example: Big Yellow Group plc

Deloitte, the statutory auditor of the Big Yellow Group plc – the self-storage solutions business listed on the FTSE 250 – noted in the auditor's report in the group's financial statements for the year ended 31 March 2015 that investment properties constituted an area of particular audit risk. The nature of the risk and the auditor's responses to the risk were described in the following except from the auditor's report. The full annual report can be found at the following URL: http://html.investis.com/B/Big-yellow-plc/reports/ar2015/pdfs/Big_Yellow_AR2015.pdf

Risk

At 31 March 2015, the Group held wholly owned investment properties and investment properties under construction valued at £1,022.8 million.

Investment properties are held at fair value on the balance sheet. The net valuation gain, relating to Group held properties was £64.5 million, which was recognised through the Consolidated Income Statement during the year. The fair values at 31 March 2015 are calculated using actual and forecast inputs, such as occupancy, capitalisation rates, an assessment of cost to complete for investment properties under construction and net rent per square foot by property. In addition, the valuers apply professional judgement concerning market conditions and factors impacting individual properties.

The valuation process is inherently judgemental, which is why we consider this to be a risk of material misstatement. In particular, changes in assumptions such as the capitalisation rates, forecast rent per square foot, forecast occupancy levels and, in the case of investment property under construction, cost to complete can lead to significant movements in the value of the property, as can changes in the underlying market conditions.

How the scope of our audit responded to the risk

(a) We assessed the design and implementation of controls around the property valuations by considering the level of management oversight and review of the valuations prepared by the external valuation specialists engaged by management, who have been named in note 14;

(b) We tested the integrity of the information provided by management to the valuers by agreeing key inputs such as actual occupancy and net rent per square foot to underlying records and source evidence;

(c) We modelled eight years of valuations and key valuation inputs to the investment property portfolio, to understand the historical trends of key inputs and compared these against the key forecast assumptions included in the property valuation;

(d) We met with the valuers. We assessed their independence, the scope of the work they were requested to perform by management, and the valuation methodology applied. For each property we identified as having significant or unusual valuation movements (compared to market data or previous periods), we challenged the valuers on the key assumptions applied. Our challenge was informed by input from our internal valuation specialists, utilising their knowledge and expertise in the market at a macro level and the relevant geographies to challenge the key judgemental inputs noted adjacent. We also researched comparable transactions and understood trends in analogous industries. We understood the rationale for outlying valuations or movements and obtained corroborative evidence. We also assessed the valuations for a sample of other properties; and

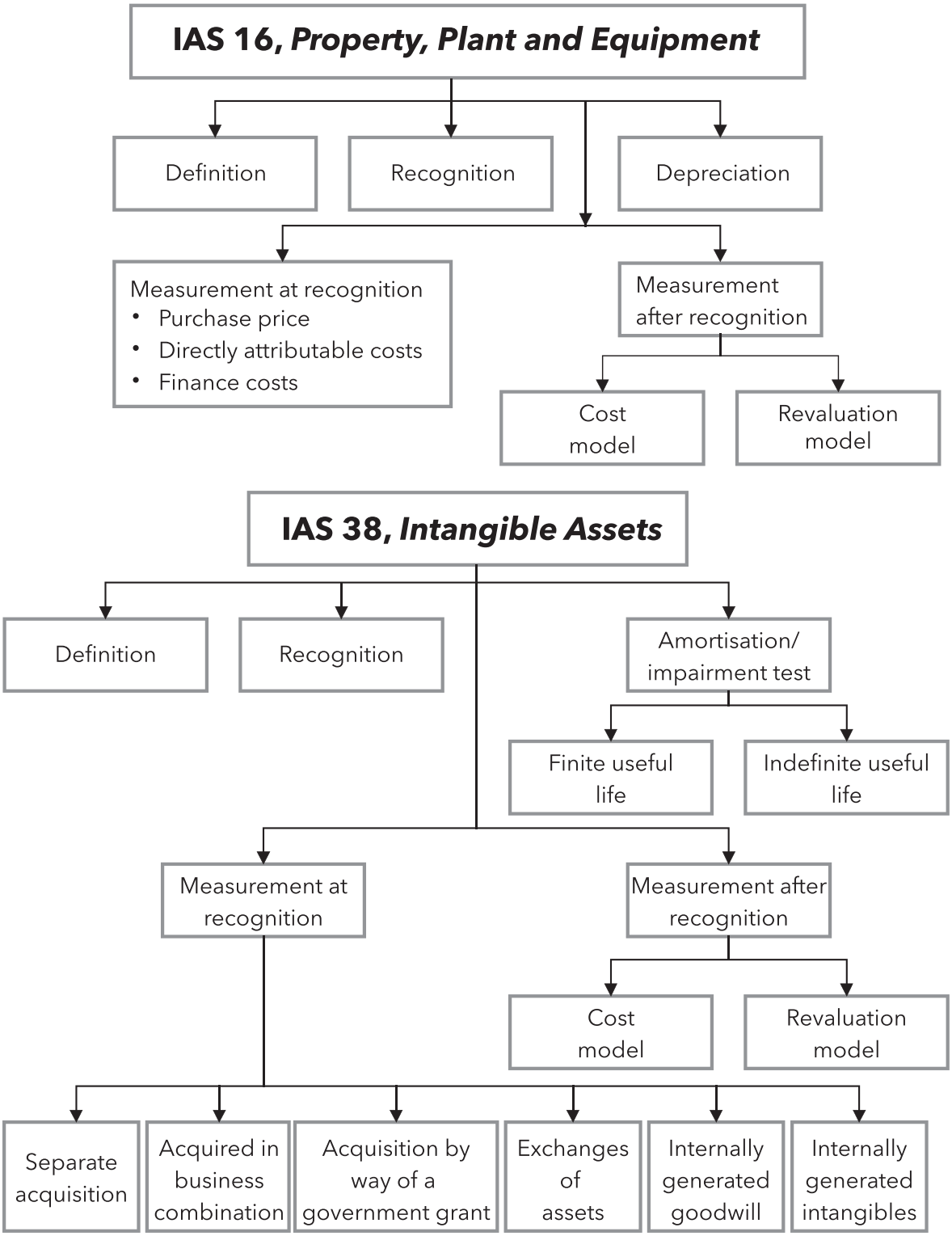
(e) We visited a sample of properties to assess the condition of the buildings.

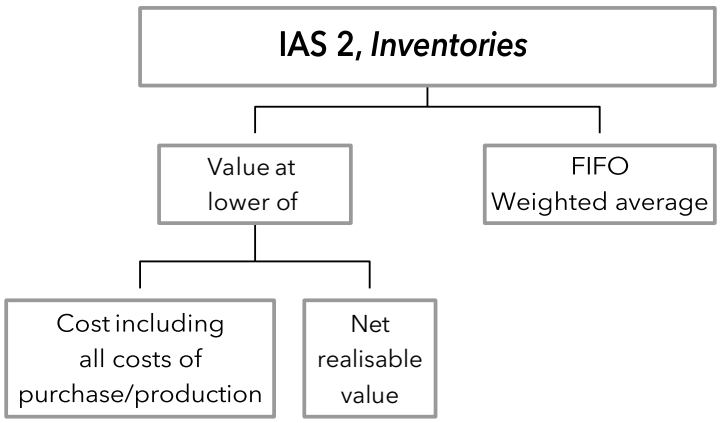


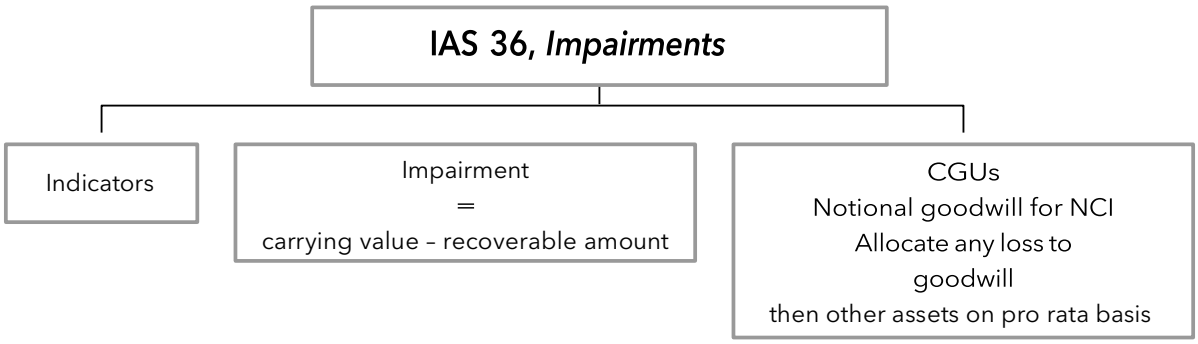
Professional skills focus: Concluding, recommending and communicating

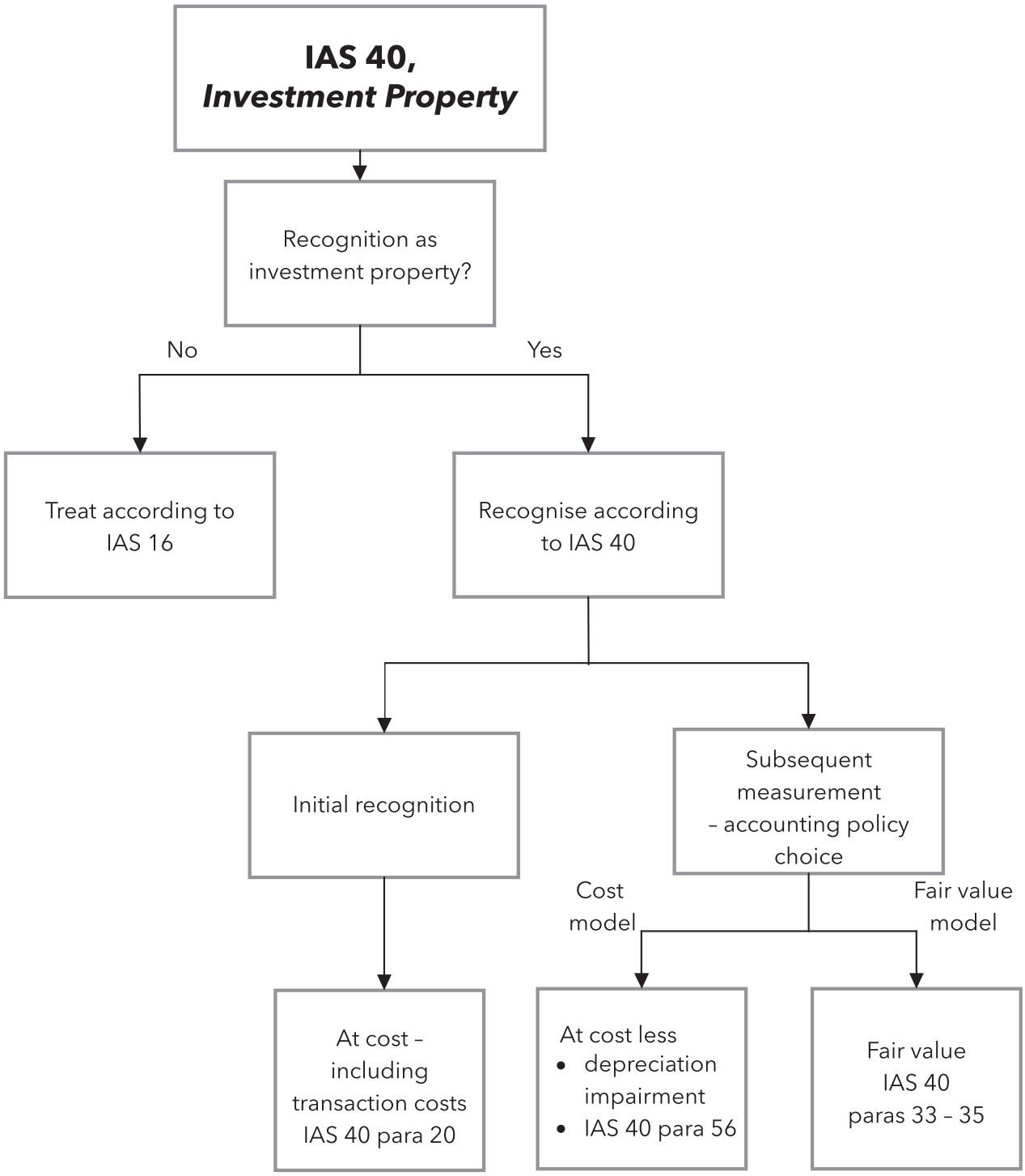
Investment property can be risky and subject to judgement on the part of valuers. Effective communication on the part of the auditors is therefore particularly important. The above case study shows the auditors acting with professional scepticism while maintaining sufficient trust to elicit the necessary information.

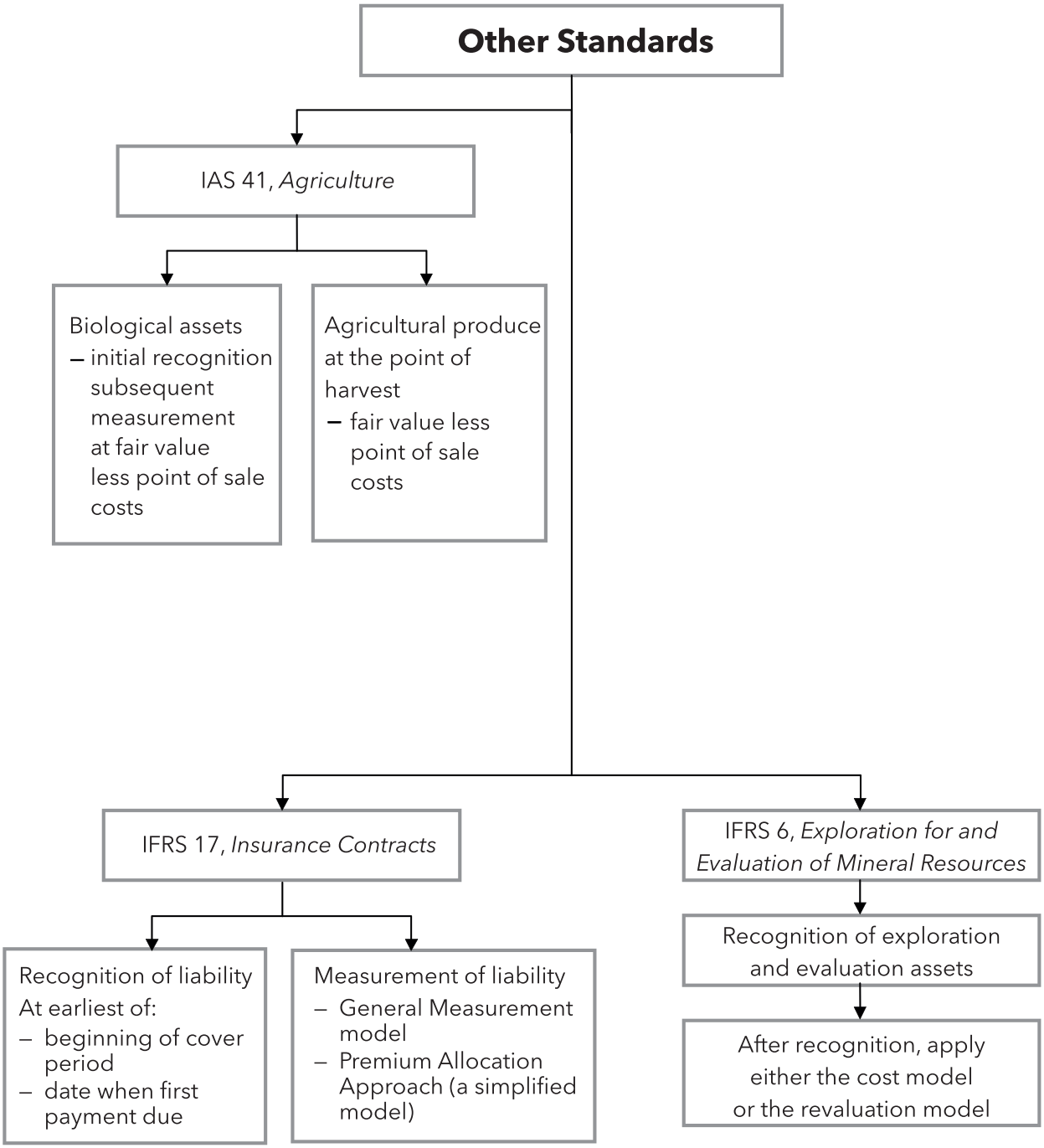
Summary


IAS 16, Property, Plant and Equipment
Definition
Recognition
Depreciation
Measurement at recognition
• Purchase price
• Directly attributable costs
• Finance costs
Measurement after recognition
Cost model
Revaluation model
IAS 38, Intangible Assets
Definition
Recognition
Amortisation/ impairment test
Finite useful life
Indefinite useful life
Measurement at recognition
Measurement after recognition
Cost model
Revaluation model
Separate acquisition
Acquired in business combination
Acquisition by way of a government grant
Exchanges of assets
Internally generated goodwill
Internally generated intangibles


IAS 2, Inventories
Value at lower of
FIFO Weighted average
Cost including all costs of purchase/production
Net realisable value


IAS 36, Impairments
Indicators
Impairment = carrying value – recoverable amount
CGUs
Notional goodwill for NCI
Allocate any loss to goodwill
then other assets on pro rata basis


IAS 40,
Investment Property
Recognition as investment property?
No
Yes
Treat according to IAS 16
Recognise according to IAS 40
Initial recognition
Subsequent measurement – accounting policy choice
Cost model
Fair value model
At cost – including transaction costs IAS 40 para 20
At cost less
depreciation
impairment
IAS 40 para 56
Fair value IAS 40 paras 33 – 35


Other Standards
IAS 41, Agriculture
Biological assets
– initial recognition
subsequent measurement at fair value less point of sale costs
Agricultural produce at the point of harvest
– fair value less point of sale costs
IFRS 17, Insurance Contracts
Recognition of liability
At earliest of:
– beginning of cover period
– date when first payment due
Measurement of liability
– General Measurement model
– Premium Allocation Approach (a simplified model)
IFRS 6, Exploration for and Evaluation of Mineral Resources
Recognition of exploration and evaluation assets
After recognition, apply either the cost model or the revaluation model

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	Can you list the key points in IAS 36, *Impairment of Assets*? (Topic 1)
2.	How should change in use be dealt with under IAS 40? (Topic 2)
3.	How should biological assets be measured under IAS 41? (Topic 3)
4.	How should exploration and evaluation assets be measured under IFRS 5? (Topic 4)
5.	What should be the auditor's main focus when auditing investment properties? (Topic 6)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Ruapehu	You should have no trouble with this quick revision of inventory from your earlier studies.
Oruatua	This is on PPE, and is rather more complex than you will have met in your earlier studies.
Acetone	This tests impairment of goodwill, including 'notional' goodwill, which you may have found tricky in your Professional Level studies.
Ramshead	Investment property is new at Advanced Level, so you need a substantial question to get you started. This one tests the interaction of IAS 40 with IFRS 5, which you covered in an earlier chapter.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted the self-test questions, you can continue your studies by moving onto the next chapter. In later chapters, we will recommend questions from the Question Bank for you to attempt.

Technical reference

1 IAS 16, *Property, Plant and Equipment*

- Recognition – **IAS 16.7**
- Initial costs – **IAS 16.11**
- Subsequent costs – **IAS 16.12**
- Measurement at recognition – **IAS 16.15**
- Measurement after recognition – **IAS 16.29**
- Derecognition – **IAS 16.67-72**
- Disclosure – **IAS 16.73-79**

2 IAS 38, *Intangible Assets*

- Scope – **IAS 38.2**
- Definition – **IAS 38.8**
- Intangible assets – **IAS 38.9-10**
- Identifiability – **IAS 38.11-12**
- Control – **IAS 38.13**
- Future economic benefits – **IAS 38.17**
- Recognition and measurement – **IAS 38.18-67**
- Recognition of an expense – **IAS 38.68**
- Measurement after recognition – **IAS 38.72**
- Cost model – **IAS 38.74**
- Revaluation model – **IAS 38.75-87**
- Useful life – **IAS 38.88-96**
- Intangible assets with finite useful lives – **IAS 38.97-106**
- Intangible assets with indefinite useful lives – **IAS 38.107-110**
- Recoverability of the carrying amount - impairment losses – **IAS 38.111**
- Retirements and disposals – **IAS 38.112**
- Disclosure – **IAS 38.118**

3 IAS 2, *Inventories*

- Measurement and disclosure but not recognition – **IAS 2.1**
- Measured at lower of cost and net realisable value – **IAS 2.9**
- Cost = expenditure incurred, in bringing the items to their present location and condition, so the cost of purchase and the cost of conversion – **IAS 2.10**
  - Fixed costs included by reference to normal levels of activity – **IAS 2.13**
- Cost formula: FIFO or Weighted average – **IAS 2.25**
- NRV includes costs to complete and selling costs – **IAS 2.6**
- Disclosures include accounting policies, carrying amounts and amounts recognised as an expense – **IAS 2.36-38**

4 IAS 36, *Impairment of Assets*

(1) **Indications**

- At each reporting date assess whether indication of impairment: – **IAS 36.9**
  - If so, estimate recoverable amount (RA)
  - RA is higher of fair value less costs to sell and value in use (present value of future cash flows in use and on disposal) – **IAS 36.6**
  - Review both external and internal information for evidence of impairment – **IAS 36.12**
- Impairment loss where carrying amount exceeds RA – **IAS 36.59**

(2) **Fair value less costs to sell**

- The way in which fair value is determined depends on whether there is a binding sale agreement and/or an active market – **IAS 36.25-27**
- Examples of disposal costs – **IAS 36.28**

(3) **Value in use**

- Calculation involves the estimation of future cash flows as follows: – **IAS 36.39**
  - Cash flows from continuing use
  - Cash flows necessarily incurred to generate cash inflows from continuing use
  - Net cash flows receivable/payable on disposal
- These should reflect the current condition of the asset – **IAS 36.44**
- The discount rate should reflect:
  - The time value of money – **IAS 36.55**
  - Risks specific to the asset for which the future cash flow estimates have not been adjusted

(4) **Cash-generating units**

- Estimate recoverable amount of CGU if not possible to assess for an individual asset – **IAS 36.66**
- Identification of an asset's CGU involves judgement – **IAS 36.68**
- Goodwill should be allocated to each of the acquirer's CGUs that are expected to benefit – **IAS 36.80**
- Goodwill that cannot be allocated to a CGU on a non-arbitrary basis is allocated to the group of CGUs to which it relates – **IAS 36.81**
- Annual impairment review required for any CGU which includes goodwill – **IAS 36.90**
- Corporate assets should be allocated on a reasonable and consistent basis – **IAS 36.102**

(5) **Impairment losses**

- If the asset is held under the cost model the impairment should be recognised in profit or loss – **IAS 36.60**
- If the asset has been revalued the impairment loss is treated as a revaluation decrease – **IAS 36.60**
- An impairment loss for a CGU should be allocated: – **IAS 36.104**
  - To goodwill then
  - To all other assets on a pro rata basis

- When a CGU is a non-wholly owned subsidiary and non-controlling interest is measured at acquisition date at share of net assets, notionally gross up goodwill for that part attributable to the non-controlling interest – **IAS 36 (Appendix C)**

(6) **Reversals**

- An impairment loss recognised for goodwill should not be reversed – **IAS 36.124**

(7) **Disclosures**

- All impairments – **IAS 36.126**
- For a material impairment on an individual asset – **IAS 36.130**
- For a material impairment on a CGU – **IAS 36.130**

5 IAS 40, *Investment Property*

- Definition of investment property – **IAS 40.5**
- Definition of fair value – **IFRS 13.9**
- Property held as a right-of-use asset may be investment property – **IAS 40.6**
- Fair value model – **IAS 40.33-35, IAS 40.38**
- Cost model – **IAS 40.56**

6 IAS 41, *Agriculture*

- Scope – **IAS 41.1**
- Agricultural activities – **IAS 41.5, 8**
  - Biological assets
  - Agricultural produce at the point of harvest
  - Government grants
- Recognition and measurement – **IAS 41.10, 12-13**
- Gains and losses – **IAS 41.26, 28**
- Government grants – **IAS 41.34-35**
- Disclosure – **IAS 41.40, 41, 46-50, 54-57**

7 IFRS 6, *Exploration for and Evaluation of Mineral Resources*

- Scope – **IFRS 6.3-5**
- Measurement at recognition – **IFRS 6.8-11**
- Measurement after recognition – **IFRS 6.12**
- Changes in accounting policies – **IFRS 6.13**
- Impairment – **IFRS 6.18**

8 IFRS 17, *Insurance Contracts*

- Objective – **IFRS 17.1**
- Scope – **IFRS 17.3-7**
- Recognition– **IFRS 17.5**
- Measurement at initial recognition – **IFRS 17.32**
- General measurement approach - **IFRS 17.40-42**
- Premium allocation approach - **IFRS 17.53-55**

Self-test questions

Answer the following questions

1 IAS 40

Which of the following properties fall under the definition of investment property and therefore within the scope of IAS 40, *Investment Property*?

A Property occupied by an employee paying market rent

B A building owned by an entity and leased out under an operating lease

C Property being constructed on behalf of third parties

D Land held for long-term capital appreciation

2 Ruapehu

The Ruapehu Company manufactures a single type of concrete mixing machine, which it sells to building companies. Ruapehu is currently considering the value of its inventories at 31 December 20X7. The following data are relevant at this date:

	Cost per item £
Variable production costs	200,000
Fixed production costs	40,000
	240,000

There are 85 mixing machines held in inventory.

The company has a contract to sell 15 concrete mixing machines at £225,000 each to a major local building company in January 20X8. The normal selling price is £260,000 per machine. Selling costs are minimal.

Requirement

What is the value of Ruapehu's inventory at 31 December 20X7, according to IAS 2, *Inventories*?

3 Utah

The Utah Company manufactures motors for domestic refrigerators. A major customer is The Bushbaby Company, which is a major international electrical company making refrigerators as one of its products.

Utah is currently preparing its financial statements for the year to 31 December 20X7 and it expects to authorise them for issue on 3 March 20X8.

Utah holds significant inventories of motors (which are unique to the Bushbaby contract) as Bushbaby requires them to be supplied on a just-in-time basis and has variable production schedules.

On 3 January 20X8, Bushbaby announced that it was fundamentally changing the design of its refrigerators and that, while this had been planned for some time, it had not been possible to

warn Utah for reasons of commercial confidentiality. As a consequence, it would cease to use Utah's motors from 30 April 20X8 and would reduce production before that date. Details for Utah are as follows:

Number of motors held in inventory at 31 December	4,000 motors
Expected sales in the four months to 30 April 20X8	1,600 motors
Net selling price per motor sold to Bushbaby	£50
Net selling price per motor unsold at 30 April 20X8	£10
Cost per motor	£25

Requirement

At what value should the inventories of motors be stated by Utah in its statement of financial position at 31 December 20X7 according to IAS 2, *Inventories*, and IAS 10, *Events after the Reporting Period*?

4 Niobium

The Niobium Company operates in the petrol refining industry. A fire at a competitor using similar plant has revealed a safety problem and the Government has introduced new regulations requiring the installation of new safety equipment in the industry. The refinery had a carrying amount of £30 million before the installation of the safety equipment. The new safety equipment cost £5 million and was fully operational at 31 December 20X7, but it does not generate any future economic benefits. The refinery would, however, be closed down without such equipment being installed.

At 31 December 20X7 the net selling price of the refinery was estimated at £33 million. In determining its value in use, the directors have determined that the refinery would generate annual cash flows of £3.2 million from next year in perpetuity, to be discounted at 10% per annum.

Requirement

According to IAS 16, *Property, Plant and Equipment*, what is the carrying amount of the refinery in Niobium's statement of financial position at 31 December 20X7?

5 Oruatua

The Oruatua Company acquired a piece of machinery for £800,000 on 1 January 20X6. It identified that the asset had three major components as follows:

Component	Useful life	Cost (£'000)
Pump	5 years	110
Filter	4 years	240
Engines	15 years	450

Under the terms of the 15-year licence agreement for the use of the machinery, the engines (but not the other components) were to be dismantled at the end of the licence period. The machinery contained three engines, and dismantling costs for all three engines were initially estimated at a total cost of £480,000 (ie, £160,000 per engine) payable in 15 years' time. Oruatua's discount rate appropriate to the risk specific to this liability is 7% per annum.

One of the three engines developed a fault on 1 January 20X7 and had to be sold for scrap for £40,000. A replacement engine was purchased at a cost of £168,000 on 1 January 20X7, for use until the end of the licence period, when dismantling costs on this engine estimated at £150,000 would be payable.

At a rate of 7% per annum the present value of £1 payable in 15 years' time is 0.3624 and of £1 payable in 14 years' time is 0.3878.

Requirements

Calculate the following figures for inclusion in Oruatua's financial statements for the year ended 31 December 20X7 according to IAS 16, *Property, Plant and Equipment*, and IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*.

(a) The carrying amount of the machinery at 31 December 20X6;

(b) The profit/loss on the disposal of the faulty engine;

(c) The carrying amount of the machinery at 31 December 20X7.

6 Antimony

The Antimony Company acquired its head office on 1 January 20W8 at a cost of £5.0 million (excluding land). Antimony's policy is to depreciate property on a straight-line basis over 50 years with a zero residual value.

On 31 December 20X2 (after five years of ownership) Antimony revalued the non-land element of its head office to £8.0 million. Antimony does not transfer annual amounts out of revaluation reserves as assets are used: this is in accordance with the permitted treatment in IAS 16, *Property, Plant and Equipment*.

In January 20X8, localised flooding occurred and the recoverable amount of the non-land element of the head office property fell to £2.9 million.

Requirement

What impairment charge should be recognised in the profit or loss of Antimony arising from the impairment review in January 20X8 according to IAS 36, *Impairment of Assets*?

7 Sundew

The Sundew Company is a vertically integrated manufacturer of chainsaws. It has two divisions. Division X manufactures engines, all of which are identical. Division Y assembles complete chainsaws and sells them to third party dealers.

Division X, a cash-generating unit, sells to Division Y at cost price but sells to other chainsaw manufacturers at cost plus 50%. Details of Division X's budgeted revenues for the year ending 31 December 20X7 are as follows:

	Engines	Price per engine
Sales to Division Y	2,500	£1,000
Third party sales	1,500	£1,500

Requirement

What are the 20X7 cash inflows which should be used in determining the value in use of Division X according to IAS 36, *Impairment of Assets*?

8 Cowbird

The Cowbird Company operates in the television industry. It acquired a licence to operate in a particular region for 20 years at a cost of £10 million on 31 December 20X3. Cowbird's policy was to amortise the fee paid for the licence on a straight-line basis.

By 31 December 20X5 it had become apparent that Cowbird had overpaid for the licence and, measuring recoverable amount by reference to value in use, it recognised an impairment charge of £4.05 million, leaving a carrying amount of £4.95 million.

At 31 December 20X7 the market place had improved, such that the conditions giving rise to the original impairment no longer existed. The recoverable amount of the licence by reference to value in use was now £11 million.

Requirement

What should be the carrying amount of the licence in the statement of financial position of Cowbird at 31 December 20X7, according to IAS 36, *Impairment of Assets*?

9 Acetone

The Acetone Company is testing for impairment two subsidiaries which have been identified as separate cash-generating units.

Some years ago, Acetone acquired 80% of The Dushanbe Company for £600,000 when the fair value of Dushanbe's identifiable assets was £400,000. As Dushanbe's policy is to distribute all profits by way of dividend, the fair value of its identifiable net assets remained at £400,000 on 31 December 20X7. The impairment review indicated Dushanbe's recoverable amount at 31 December 20X7 to be £520,000.

Some years ago Acetone acquired 85% of The Maclulich Company for £800,000 when the fair value of Maclulich's identifiable net assets was £700,000. Goodwill of £205,000 (£800,000 – (£700,000 × 85%)) was recognised. As Maclulich's policy is to distribute all profits by way of dividend, the fair value of its identifiable net assets remained at £700,000 on 31 December 20X7. The impairment review indicated Maclulich's recoverable amount at 31 December 20X7 to be £660,000.

It is Acetone group policy to value the non-controlling interest using the proportion of net assets method.

Requirements

Determine the following amounts in respect of Acetone's consolidated financial statements at 31 December 20X7 according to IAS 36, *Impairment of Assets*.

(a) The carrying amount of Dushanbe's assets to be compared with its recoverable amount for impairment testing purposes.

(b) The carrying amount of goodwill in respect of Dushanbe after the recognition of any impairment loss.

(c) The carrying amount of the non-controlling interest in Maclulich after recognition of any impairment loss.

10 Titanium

On 1 January 20X7 The Titanium Company acquired the copyright to four similar magazines, each with a remaining legal copyright period for 10 years. At the end of the legal copyright period, other publishing companies will be allowed to tender for the copyright renewal rights.

At 31 December 20X7 the following information was available in respect of the assets:

Publication name	Copyright cost at 1 January 20X7	Remaining period over which publication is expected to generate cash flows at 1 January 20X7	Value in active market at 31 December 20X7
Dominoes	£900,000	6 years	£700,000
Billiards	£1,200,000	16 years	£1,150,000
Skittles	£1,700,000	8 years	Unknown
Darts	£1,400,000	Indefinite	£2,100,000

Titanium uses the revaluation model as its accounting policy in relation to intangible assets.

Requirement

What is the total charge to profit or loss for the year ended 31 December 20X7 in respect of these intangible assets per IAS 38, *Intangible Assets*?

11 Lewis

The following issues have arisen in relation to business combinations undertaken by the Lewis Company.

(1) Lewis acquired the trademark of a type of wine when it acquired 80% of the ordinary share capital of The Calcium Company on 1 April 20X7. This wine is produced from a vineyard that is exclusively used by Calcium.

(2) When Lewis bought a football club on 1 May 20X7, it acquired the registrations of a group of football players.

(3) Lewis acquired a 75% share in the Stilt Company during 20X7. At the acquisition date Stilt was researching a new pharmaceutical product which is expected to produce future economic benefits.

The cost of these assets can be measured reliably.

Requirement

Indicate which of the above items should or should not be recognised as assets separable from goodwill in Lewis's statement of financial position at 31 December 20X7, according to IAS 38, *Intangible Assets*.

12 Diversified group

The following issues have arisen within a diversified group of businesses.

(1) The Thrasher Company has signed a three-year contract with a team of experts to write questions for a computer based examination on International Financial Reporting Standards. The contract states that the experts cannot work on similar projects for rival entities. Thrasher incurred costs of £5,000 in training the experts to use the software, and believes that the product developed by the team will be a market leader.

(2) The Curium Company has a loyalty card scheme for customers. Every customer purchase is recorded in such a way that Curium is able to create a profile of spending amounts and habits of customers, and uses this to target them with special offers and discounts to encourage repeat business. The database has cost £60,000 to create and Curium has been approached by another company wishing to buy the contents of the database.

Requirement

Which of the above items should be classified as intangible assets per IAS 38, *Intangible Assets*?

13 Cadmium

The Cadmium Company produces a globally recognised dog food that is a market leader. The trademark was established over 50 years ago and is renewable every eight years. The last renewal was effective from 1 January 20X2 and cost £65,000. Cadmium intends to continue to renew the trademark in future years.

Cadmium uses the revaluation model where allowed for measuring intangible assets, in accordance with IAS 38, *Intangible Assets*. A valuation of £50 million was made by an independent valuation expert on 31 December 20X7, who charged £650,000 for the valuation report.

Requirement

What is the carrying amount of the trademark in Cadmium's statement of financial position at 31 December 20X7 per IAS 38, *Intangible Assets*?

14 Piperazine

The Piperazine Company's financial reporting year ends on 31 December. It has adopted the revaluation model for intangible assets and revalues them on a regular three-year cycle. For intangibles with a finite life Piperazine transfers the relevant amount from revaluation reserve to retained earnings each year.

During 20X4 Piperazine incurred £70,000 on the process of preparing an application for licences for 15 taxis to operate in a holiday resort where, in order to prevent excessive traffic pollution, the licensing authority only allowed a small number of taxis to operate. The outcome of its application was uncertain up to 30 November 20X4 when the local authority accepted its application. In December 20X4 Piperazine incurred a total cost of £9,000 in registering its licences. The licences were for a period of nine years from 1 January 20X5. The licences are freely transferable and an active market in them exists. The fair value of the licences at 31 December 20X4 was £9,450 per taxi and Piperazine carried them at fair value in its statement of financial position at 31 December 20X4.

At 31 December 20X7, Piperazine undertook its regular revaluation. On that date the licensing authority announced that it would triple the number of licences offered to taxi operators and there were transactions in the active market for licences with six years to run at £4,500.

Requirements

Determine the following amounts in respect of the revaluation reserve in respect of these taxi licences in Piperazine's financial statements according to IAS 38, *Intangible Assets*.

14.1 The balance at 31 December 20X4
14.2 The balance at 31 December 20X7 before the regular revaluation

14.3 The balance at 31 December 20X7 after the regular revaluation

15 Boron

The Boron Company is an investment property company. On 31 December 20X6, it purchased a retirement home as an investment at a cost of £600,000. Legal costs associated with the acquisition of this property were a further £50,000.

At 31 December 20X7 Boron adopted the fair value model. The fair value of the retirement home at this date was £700,000 and costs to sell were estimated at £40,000.

Requirement

What amount should appear in the statement of profit or loss and other comprehensive income of Boron in the year ending 31 December 20X7 in respect of the retirement home under IAS 40, *Investment Property*?

16 Laburnum

The Laburnum Company is an investment property company. One of its properties is a warehouse which has the specialist use of storing tropical plants at high temperatures. As a result, the central heating system is an important and integral part of the warehouse building. Laburnum uses the fair value model for investment properties.

The central heating system was purchased on 1 January 20X2 for £80,000. It is being depreciated at 10% per annum on cost and it has been agreed by the valuer that the carrying amount of the central heating system is a reasonable value at which to include it in the fair value of the entire warehouse.

In December 20X7, the valuer initially determined the fair value of the warehouse, including the central heating system, to be £1,250,000. Unfortunately, the central heating system completely failed on 25 December 20X7 and was immediately scrapped and replaced with a new heating system costing £140,000 on 31 December 20X7.

Requirement

According to IAS 40, *Investment Property*, at what value should the warehouse, including the heating system, be recognised in the financial statements of Laburnum in the year ending 31 December 20X7?

17 Ramshead

On 1 January 20X6, The Ramshead Company acquired an investment property for which it paid £3.1 million and incurred £100,000 agency and legal costs. The property's useful life was estimated at 20 years, with no residual value; its fair value at 31 December 20X6 was estimated at £3.45 million and agency and legal costs to dispose of the property at that date were estimated at £167,500.

On 1 July 20X7, Ramshead decided to dispose of the property. The criteria for being classified as held for sale were met on that date, when the property's fair value was £3.5 million. Agency and legal costs to dispose of the property were estimated at £160,000.

On 1 October 20X7, the property was sold for a gross price of £3.7 million, with agency and legal costs of £165,000 being incurred.

Requirements

Calculate the following amounts in respect of Ramshead's financial statements for the year ended 31 December 20X7 in accordance with IAS 40, *Investment Property* and IFRS 5, *Non-current Assets Held for Sale and Discontinued Operations*.

17.1 The gain or loss arising in 20X6 from the change in carrying amount if the fair value model is used to account for the property

17.2 The gain or loss on disposal arising in 20X7 if the cost model is used

17.3 The increase or decrease, compared with the cost model, in the gain or loss on disposal arising in 20X7 if the fair value model is used

18 Arapawanui

The Arapawanui Company keeps a flock of sheep on its land, selling the milk outputs to make cheese. The day after its production, the milk is collected on behalf of the purchasers and revenue from its sale is recognised.

On 30 June 20X7 300 animals were born, all of which survived and were still owned by Arapawanui at 31 December 20X7. 10,000 litres of milk were produced in the year to 31 December 20X7.

The following market data is available in respect of the sheep.

Type of animal	At 30 June 20X7 Fair value per animal	At 31 December 20X7 Fair value per animal
	£	£
Newborn	22	23
6 months old	25	26

The animal fair values are based on transactions prices in the local markets. Auctioneers' commission is 1.5% of the transaction price and the government sales levy is 0.5% of that price.

The production cost, including overheads, of the milk was £0.08 per litre and the fair values were £0.13 per litre throughout 20X7 and £0.14 per litre throughout 20X8. Costs to sell were estimated at 4%.

Requirement

What gain should be recognised in respect of the newborn sheep and the milk in Arapawanui's financial statements for the year to 31 December 20X7, according to IAS 41, *Agriculture*?

19 Tepev

The Tepev Company bought a flock of 400 sheep on 1 December 20X7. The cost of each sheep was £80, which represented fair value at that date. Auctioneers' fees on sale are 5% of fair value, and the cost of transporting each sheep to market is £4.00. An agricultural levy of £2.00 is payable on each sheep sold.

At 31 December 20X7 all of the sheep are still held and fair value has increased to £90 per sheep. No other costs have changed. Tepev has a contract to sell the sheep on 31 March 20X8 for £100 each.

Requirement

What is the carrying amount of the flock in the statement of financial position at 31 December 20X7, according to IAS 41, *Agriculture*?

20 Saving

The Saving Company bought a flock of 500 sheep on 1 December 20X7. The cost of each sheep was £95, which represented fair value at that date. Auctioneers' fees on sale are 5% of fair value, and the cost of transporting each sheep to market is £3.00. An agricultural levy of £2.00 is payable on each sheep sold.

At 31 December 20X7 all of the sheep are still held and fair value has increased to £107 per sheep. No other costs have changed. Saving has a contract to sell the sheep on 31 March 20X8 for £119 each.

Requirement

What is the gain arising in relation to the flock between the date of initial recognition as an asset and 31 December 20X7, according to IAS 41, *Agriculture*?

21 Monkey

The Monkey Company has the following information in relation to a cattle herd in the year ended 31 December 20X7.

	£'000
Cost of herd acquired on 1 January 20X7 (which equates to fair value)	1,800
Auctioneers' sales fees	2% of sale price
Loan obtained at 8% to finance acquisition of herd	1,500
Fair value of herd at 31 December 20X7	2,500
Transport cost to market	35
Government transfer fee on sales – no fee on purchases	50

Requirement

What is the loss arising on initial recognition of the herd as biological assets and the gain arising on its subsequent remeasurement under IAS 41, *Agriculture*, in the year ended 31 December 20X7?

22 Blackbuck

The Blackbuck Company has in issue unit-linked contracts which pay benefits measured by reference to the fair value of the pool of investments supporting the contracts. The terms of the contracts include the following.

(1) On surrender by the holder or on maturity, the benefits shall be the full fair value of the relevant proportion of the investment.

(2) In the event of the holder's death before surrender or maturity, the benefits shall be 120% of the full value of the relevant proportion of the investments.

Blackbuck's accounting policies do not otherwise require it to recognise all the obligations under any deposit component within these contracts.

Blackbuck's financial controller is unclear whether these contracts should be accounted for under IFRS 4, *Insurance Contracts*, or under IFRS 9, *Financial Instruments*.

Requirement

Explain how these contracts should be accounted for.

23 *Traore*

The Traore Company (TTC) issued 1,000 insurance contracts each with a coverage period of three years, starting when the insurance contracts are issued. TTC expects to receive premiums totaling £12,000 immediately after initial recognition and estimates that the annual cash

outflows at the end of each year will be £4,000. Using a discount rate of 5%, determined in line with the provisions of IFRS 17, the present value of total future cash outflows is £14,800. The present value of net cash outflows is subject to an upwards adjustment of £1,500 to reflect non-financial risk.

Requirement

How is the group of contracts initially measured and recognised in accordance with IFRS 17, Insurance Contracts?

24 Evaluation and exploration

Give examples of circumstances that would trigger a need to test an evaluation and exploration asset for impairment.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 The double entry is:

DEBIT	Asset value (statement of financial position)	£7,000
CREDIT	Profit or loss	£2,000
CREDIT	Revaluation surplus (other comprehensive income)	£5,000

The case is similar for a **decrease in value** on revaluation. Any decrease should be recognised as an expense, except where it offsets a previous increase taken as a revaluation surplus in other comprehensive income. Any decrease greater than the previous upwards increase in value must be recorded as an expense in profit or loss.

1.2 The double entry is:

DEBIT	Revaluation surplus (other comprehensive income)	£5,000	
DEBIT	Profit or loss	£2,000	
CREDIT	Asset value (statement of financial position)		£7,000

There is a further complication when a **revalued asset is being depreciated**. An upward revaluation means that the depreciation charge will increase. Normally, a revaluation surplus is only realised when the asset is sold, but when it is being depreciated, part of that surplus is being realised as the asset is used. The amount of the surplus realised is the difference between depreciation charged on the revalued amount and the (lower) depreciation which would have been charged on the asset's original cost. **This amount can be transferred to retained (ie, realised) earnings but not through profit or loss**.

1.3 On 1 January 20X8 the carrying value of the asset is £10,000 – (2 × £10,000 ÷ 5) = £6,000. For the revaluation:

DEBIT	Asset value (statement of financial position)	£6,000	
CREDIT	Revaluation surplus (other comprehensive income)		£6,000

The depreciation for the next three years will be £12,000 ÷ 3 = £4,000 compared to depreciation on cost of £10,000 ÷ 5 = £2,000. Each year the extra £2,000 is treated as realised and transferred to retained earnings:

DEBIT	Revaluation surplus	£2,000	
CREDIT	Retained earnings		£2,000

This is a movement within reserves, not an item in profit or loss.

Answer to Interactive question 2

The fair value less costs to sell of the plant is below its carrying value so it may be impaired. It is now necessary to find the value in use in order to determine whether an impairment has occurred and to quantify any impairment loss.

Year	Future cash flows	PV factor at 15%	Discounted future cash flows
	£'000		£'000
1	230	0.86957	200
2	211	0.75614	160

Year	Future cash flows	PV factor at 15%	Discounted future cash flows
	£'000		£'000
3	157	0.65752	103
4	104	0.57175	59
5	233	0.49718	116
			638

To calculate the impairment loss, compare the carrying value of £749,000 with the higher of value in use (£638,000) and fair value less costs to sell (£550,000). The impairment loss is therefore £749,000 – £638,000 = £111,000.

Answer to Interactive question 3

The key issue is whether the cash-generating unit produces cash flows which are independent of other assets or not.

The CGUs which appear to have cash flows independent of the other assets (and can therefore be subject to reliable assessment of their recoverable value) are:

(2) a branch of a pizza restaurant in Warsaw; and

(4) a commuter monorail.

Options (1) and (3) are not generators of independent cash flows and are therefore too small to be CGUs in their own right. In the case of (3) the CGU is the theme park as one entity.

Additionally (5) is a CGU in its own right as there is an external active market for its services, even though these are not openly available (IAS 36.71).

Answer to Interactive question 4

In both scenarios:

	(1)	(2)
	£m	£m
Recognised goodwill	90	90
Notional goodwill (£90m × 40/60)	60	60
Carrying amount of net assets	550	550
	700	700
Recoverable amount	510	570
Impairment loss	190	130
Allocation of impairment loss:		
	£m	£m
Recognised goodwill	90	90
Notional goodwill	60	40
Other assets pro rata	40	
	190	130

Carrying value after impairment:

	£m	£m
Goodwill (90 – (150 × 60%))/(90 – (130 × 60%))	–	12
Other net assets (550 – 40)	510	550
	510	562

Answer to Interactive question 5

For each scenario:

(1) The factory is not an investment property. It should be classified as property held for sale and accounted for under IFRS 5.

(2) The building would qualify as an investment property under IAS 40, as the entity intends to earn rentals from it under an operating lease.

(3) The provisions offered over and above the office space itself fall within what IAS 40 describes as 'ancillary services'. Considering the nature and extent of these services, it would be unlikely that they could be described as 'insignificant' in relation to the arrangements as a whole. The building is, in essence, being used for the provision of serviced offices and therefore does not meet the definition of an investment property.

Although the entity's main objective in acquiring the building is its potential capital appreciation, the building should be recognised and measured in accordance with IAS 16 rather than IAS 40.

(4) The property should be recognised as an investment property on 30 March 20X5 when the offices were ready to be occupied. Costs incurred, and consequently operating losses, after this date should be expensed even though the entity did not start to receive rentals until later in 20X5. Losses incurred during this 'empty' period are part of the entity's normal business operations and do not form part of the cost of the investment property.

Answer to Interactive question 6

6.1 The £5 million value could be used as a basis of fair value, because the price was agreed between market participants.

6.2 The £6 million value could not be used as a basis of fair value, because the sale transaction cannot be presumed to be between market participants in an orderly transaction.

6.3 The £4.5 million value could not be used as a basis of fair value, because the sale transaction would appear to have been made by a forced, not willing, seller, and therefore not an orderly transaction.

6.4 The £5.5 million value could not be used as a basis of fair value, because the sale transaction would appear to have been made to a buyer who was not knowledgeable of local market conditions, and therefore not a market participant in an orderly transaction.

Answer to Interactive question 7

Profit or loss on disposal

The cost model	**£m**
Net proceeds	6.00
Carrying amount £5,500,000 × 47/50	(5.17)
Profit on sale	0.83
The fair value model	**£m**
Net proceeds	6.0
Fair value	(6.2)
Loss on sale	(0.2)

Answer to Interactive question 8

(a) The changes of use will be reflected in the financial statements based on whether the entity uses the cost model or the fair value model for investment properties as follows.

The cost model for investment properties

At 31 December 20X5, the building has a carrying amount of:

£5.5m × 45/50 years = £4.95 million in accordance with IAS 16.

On 1 January 20X6 the property will be recognised as an investment property at its IAS 16 carrying amount of £4.95 million and will continue to be depreciated over its remaining 45-year life.

At 31 December 20Y0, the building has a carrying amount of:

£4.95m × 40/45 years = £4.4 million in accordance with IAS 40.

On 1 January 20Y1 the property will be recognised as property, plant and equipment at its IAS 40 carrying amount of £4.4 million and will continue to be depreciated over its remaining 40- year life.

(b) **The fair value model for investment properties**

At 31 December 20X5, the building has a carrying amount of £4.95 million in accordance with IAS 16 (as set out above).

On 1 January 20X6, the property will be recognised as an investment property. However, the property should be revalued to fair value at 31 December 20X5, and any change in value should be recognised in accordance with IAS 16.

The property will therefore be recognised at a carrying amount of £6 million and the difference of £1.05 million should be recognised as a revaluation surplus (other comprehensive income).

During the period between 1 January 20X6 and 31 December 20Y0 the building is measured at fair value with any gain or loss recognised directly in profit or loss. At the end of 20Y0 the cumulative gain is £1.5 million.

At 31 December 20Y0, the building has a carrying amount of £7.5 million being its fair value and this is the amount that should be recognised as its carrying amount under IAS 16. The carrying amount will be depreciated over the building's remaining 40-year useful life.

Answer to Interactive question 9

£1 million is derecognised being the depreciated cost of the replaced system: £1.2 million × (25/30 years)

£1.2 million is capitalised as the cost of the new system and will be depreciated over its estimated useful life of 10 years.

Answer to Interactive question 10

The carrying amount of the failed system should be derecognised:

Carrying amount is £100,000 (£400,000 less six years' depreciation at 12.5%) The replacement system should be recognised:

Total carrying amount of the office building is £3,500,000

(£3m – £100,000 + £600,000)

Answer to Interactive question 11

The entity recognises these transactions and events as follows.

20X5

The property continues to be measured under the fair value model on classification as held for sale on 30 September. An impairment of £3.65 million is recognised (£4 million less £350,000).

At 31 December the property is presented as held for sale within current assets at £350,000.

20X6

The replacement property is recognised at a cost of £3.8 million and a loss on disposal is recognised of £15,000 being (proceeds of £375,000 less selling costs of £40,000 less carrying amount of property of £350,000).

20X7

The insurance proceeds of £3.9 million are recognised in profit or loss.

Note: The requirement to measure an asset 'held for sale' at the lower of carrying amount and fair value less costs to sell does not apply to investment properties measured at fair value (IFRS 5.5). IAS 40.37 states that costs to sell should not be deducted from fair value.

Answer to Interactive question 12

12.1 Wool: Agricultural produce

12.2 Sugar: Products that are the result of processing after harvest

Answer to Interactive question 13

The correct answers are:

A Commission to brokers

B Transfer taxes and duties

Commissions to brokers and transfer taxes and duties are recognised costs to sell in the standard.

Answer to Interactive question 14

(a) **Audit issues**

(1) **Classification as an investment property**

Property A: As this property is owner occupied it does not fall within the definition of an investment property in accordance with IAS 40 (IAS 40(9)).

Property B: While this property is not legally owned it is held as a right-of-use asset under IFRS 16 and therefore can be treated as an investment property (IAS 40(5)).

Property C: Although this property is currently vacant, on the basis that it is being held for investment purposes it can be classified as an investment property (IAS 40(8d)).

Property D: This is an investment property as it is legally owned by Propertyco and is let out to a non-group company (IAS 40(5)).

(2) **Valuation**

Property A: Should be valued in accordance with IAS 16 ie, cost less accumulated depreciation unless the revaluation model is to be adopted.

Property B: Would have been recognised in accordance with IFRS 16 at the inception

of the lease at the present value of the future lease payments, plus payments made at or before commencement less lease incentives. After initial recognition it would be valued at fair value in accordance with company policy in respect of investment properties.

Property C: Should initially be recognised at cost including transaction costs. In this case the asset should initially be recognised at £3 million. As the fair value model is adopted by Propertyco the value will then be revised to fair value.

Property D: Should be recognised at fair value in accordance with IAS 40 and the accounting policy adopted by Propertyco. Changes in fair value should be recognised in profit or loss for the period.

(b) **Audit procedures**

- Evaluate the control environment and the process by which Propertyco establishes fair values.
- Determine the basis on which fair values have been calculated. (In accordance with IAS 40/IFRS 13 this should be the price that would be received to sell an asset in an orderly transaction between market participants at the measurement date.) Current prices per square metre in an active market for similar property in the same location and condition are likely to provide the best evidence or observable market rents. (IAS 40.40 states that the fair value must reflect rental income from current leases and other assumptions that participants would use when pricing investment property under current market conditions).
- Where external valuers have been used assess the extent to which they can be relied on in accordance with the principles of using the work of a management's expert in ISA 500, *Audit Evidence*.
- If fair values have been based on discounted cash flows ie, discounted future rental incomes compare predicted cash flows to current rental agreements and assess whether this is the most appropriate basis for estimating fair value in accordance with IFRS 13. Review the basis on which the interest rate applied has been selected and any other assumptions built into this calculation eg, consider management's history of carrying out its intentions.
- Review any documentation to support assumptions.
- Agree level of disclosure is in accordance with IAS 40/IFRS 13.

Answers to Self-test questions

1 IAS 40

The correct answers are:

A building owned by an entity and leased out under an operating lease D

Land held for long-term capital appreciation

IAS 40.8 and 9 give examples of types of investment property.

2 Ruapehu

£20,175,000

IAS 2.31 requires that NRV should take into account the purpose for which inventory is held. The NRV for the contract is therefore determined separately from the general sales, thus:

	£
Contract: NRV is lower than cost thus use NRV, so (£225,000 × 15) =	3,375,000
General: Use cost as this is less than NRV, so (£240,000 × (85 – 15)) =	16,800,000
	20,175,000

3 Utah

£64,000

IAS 2.30 requires the NRV of inventories to be calculated on the basis of all relevant information, including events after the reporting period. This is supported by the example in IAS 10.9(b).

Thus:

	£
£25 × 1,600 expected to be sold to Bushbaby:	40,000
£10 × 2,400 remainder	24,000
Total	64,000

4 Niobium

£33 million

IAS 16.11 requires the capitalisation of essential safety equipment even if there are no future economic benefits flowing directly from its operation.

It does however subject the total value of all the related assets to an impairment test. In this case the recoverable amount is £33 million, as the net selling price £33 million is greater than the value in use £32 million (ie, £3.2m/0.1). As this is less than the total carrying amount of £35 million (£30m + 5m), the assets are written down to £33 million.

5 Oruatua

(a) £850,355

The initial cost of the asset must include the dismantling cost at its present value, where the time value of the money is material (IAS 16.16(c) and IAS 37.45). The present value of these costs at 1 January 20X6 is £173,952 (£480,000 × 0.3624), making the total cost of the engines £623,952. Each part of the asset that has a cost which is significant in relation to the total asset cost should be depreciated separately (IAS 16.43). Therefore, at the end of 20X6 the carrying amount of the asset is £850,355 (£110,000 × 4/5) + (£240,000 × 3/4) + (£623,952 × 14/15).

(b) £(154,118)

The loss on disposal is (per IAS 16.71) the difference between the carrying amount of an individual engine at 1 January 20X7 of £194,118 (£623,952/3 × 14/15)) and the scrap sale proceeds of £40,000, to give a loss of £154,118.

(c) £756,521

The replacement engine is capitalised at cost of £226,170 (£168,000 + £150,000 × 0.3878), and then depreciated over the remaining length of the licence of 14 years. The carrying amount of the asset at 31 December 20X7 is therefore £756,521 (£110,000 × 3/5) + (£240,000 × 2/4) + (£623,952 × 2/3 × 13/15) + (£226,170 × 13/14).

6 Antimony

£0.7 million

IAS 36.60 and 61 (also IAS 16.40) require that an impairment that reverses a previous revaluation should be recognised through the revaluation reserve to the extent of that reserve. Any remaining amount is recognised through profit or loss. Thus:

- The carrying amount at 31 December 20X2 is 45/50 × £5.0m = £4.5 million.
- The revaluation reserve created is £3.5 million (ie, £8.0m – £4.5m).
- The carrying amount at 31 December 20X7 is 40/45 × £8.0m = £7.1 million.
- The recoverable amount at 31 December 20X7 is £2.9 million.
- The total impairment charge is £4.2 million (ie, £7.1m – £2.9m).
- Of this, £3.5 million is a reversal of the revaluation reserve, so only £0.7 million is recognised through profit or loss.

7 Sundew

£6,000,000

IAS 36.70 requires that in determining value in use where internal transfers are made, then a best estimate should be made of prices that would be paid in an orderly transaction between market participants at the measurement date.

Thus revenues are 4,000 × £1,500 = £6,000,000

8 Cowbird

£8.0 million

IAS 36.110 requires consideration of whether an impairment loss recognised in previous years has reversed or decreased.

IAS 36.117 and 118 restrict the recognition of any such reversal to the value of the carrying amount at the current reporting date had the original impairment not taken place. Thus:

Carrying amount under original conditions = £10m × 16/20 years = £8.0m.

9 Acetone

(a) £750,000

Book value of Dushanbe's net assets	400,000
Goodwill recognised on acquisition £600,000 – (80% × £400,000)	280,000
Notional goodwill (£280,000 × 20/80)	70,000
	750,000

(b) £96,000

The impairment loss is the total £750,000 less the recoverable amount of £520,000 = £230,000. Under IAS 36.104 this is firstly allocated against the £350,000 goodwill. (As the impairment loss is less than the goodwill, none is allocated against identifiable net assets.)

As only the goodwill relating to Acetone is recognised, only its 80% share of the impairment loss is recognised:

	£
Carrying value of goodwill	280,000
Impairment (80% × 230,000)	(184,000)
Revised carrying amount of goodwill	96,000

(c) £99,000

	£
Carrying amount of Maclulich's net assets	700,000
Recognised goodwill	205,000
Notional goodwill (15/85 × £205,000)	36,176
	941,176
Recoverable amount	(660,000)
Impairment loss	281,176
Allocated to:	
Recognised and notional goodwill	241,176
Other net assets	40,000

Therefore the non-controlling interest is (£700,000 – £40,000) × 15% = £99,000.

10 Titanium

£672,500

IAS 38.94 deals with the identification of the useful life of an intangible asset arising from legal rights.

The **Dominoes** publication has a useful life of six years, and so should be amortised over this period. At the year end the carrying amount of £750,000, (900,000 × 5/6), exceeds the active market value, so an impairment of £50,000 is required. This gives a total charge of £200,000 (£150,000 amortisation plus £50,000 impairment charge)

The **Billiards** publication is initially amortised over the period of 10 years to the end of the copyright arrangement, as there is no certainty that the company can publish the magazine after this date. This gives a charge of £120,000.

The **Skittles** publication is amortised over the period it is expected to generate cash flows of eight years, giving a charge of £212,500.

The **Darts** publication has an indefinite period over which it is expected to generate cash flows. Under normal circumstances it would be automatically subject to an annual impairment review. However, because the copyright arrangement does have a finite period, amortisation should take place over 10 years, and so a charge of £140,000 is required.

The total charge is £672,500.

11 Lewis

For each statement:

(1) Recognised

The vineyard trademark is not separable because it could only be sold with the vineyard itself. But under IAS 38.36, the combination of the vineyard and the trademark should be recognised.

(2) Recognised

The footballers' registrations represent a legal right which meets the identifiability criterion in IAS 38.12.

(3) Recognised

The research project should be treated as a separate asset, as on a business combination it meets the definition of an asset and is identifiable (IAS 38.34).

12 Diversified group

For each item:

(1) Not an intangible

The training costs would not satisfy the definition of an intangible asset. This is because Thrasher has insufficient control over the expected future benefits of the team of experts (IAS 38.15).

(2) An intangible

The database would be classified as an intangible asset because the willingness of another party to buy the contents provides evidence of a potential exchange transaction for the relationship with customers and that the asset is separable (IAS 38.16).

13 Cadmium

£16,250

The revaluation model cannot be used for this trademark, because for a unique item there cannot be the active market required by IAS 38.75. (A professional valuation does not rank as a value by reference to an active market.) IAS 38.81 requires the cost model to be applied to such an item, even if it is in a class for which the revaluation model is used.

The cost of renewal should be treated as part of the cost of an intangible, under IAS 38.28(b), but the valuation expenses should be charged directly to profit or loss, as administration overheads (IAS 38.29(c)).

The trademark is therefore carried at the cost of renewal, depreciated for six of the eight years' life since last renewal, so £65,000 × 2/8 = £16,250.

14 Piperazine

14.1 £132,750

Under IAS 38.21 the £70,000 spent in 20X4 in applying for the licences must be recognised in profit or loss, because the generation of future economic benefits is not yet probable. The £9,000 incurred in December 20X4 in registering the licences is treated as the cost of the licences because the economic benefits are then probable. The carrying amount of the licences under the revaluation model at 31 December 20X4 is £141,750 (£9,450 × 15), so the balance on the revaluation reserve is the £132,750 uplift (IAS 38.75 & 85).

14.2 £88,500

After three years the accumulated amortisation based on the revalued amount is £47,250 (£141,750 × 3/9), whereas the accumulated amortisation based on the cost would have been £3,000 (£9,000 × 3/9). So £44,250 will have been transferred from the revaluation reserve to retained earnings (IAS 38.87). The remaining balance before the regular revaluation is £88,500 (£132,750 – £44,250).

14.3 £61,500

The carrying amount of the licences immediately before the revaluation is £94,500 (£141,750 – £47,250). The revalued carrying amount is £67,500 (£4,500 × 15). The deficit of £27,000 is recognised in the revaluation reserve, reducing the balance to £61,500 (IAS 38.86).

15 Boron

£50,000

Under the fair value model IAS 40.33 requires investment properties to be measured at fair value, while IAS 40.37 requires fair value to be determined excluding transaction costs that may be incurred on sale or other disposal. IAS 40.35 requires changes in fair value to be recognised in profit or loss.

IAS 40.20 requires transaction costs, such as legal costs, to be included in the initial measurement. So the change in fair value is £700,000 – (£600,000 + £50,000) = £50,000.

16 Laburnum

£1,358,000

IAS 40.19 and 68 require derecognition of the carrying amount of the failed system & inclusion of the replacement.

Thus £1,250,000 – (£80,000 × 4/10) + £140,000 = £1,358,000

17 Ramshead

17.1 £250,000 gain

Transaction costs should be included in the initial measurement of investment properties (IAS 40.20). Under the fair value model an investment property is subsequently carried at fair value without any deduction for costs to sell (IAS 40.33 & 5). The gain recognised in profit or loss is £250,000 (£3.45m – (£3.1m + £0.1m)).

17.2 £575,000 gain

Any asset classified as held for sale is measured in accordance with applicable IFRS immediately before classification. So if the cost model is used, the carrying amount before initial classification is cost less depreciation to the date of classification, so £2.96 million (£3.2m less 18 months' depreciation at 5% per annum). On initial classification, the property is

measured at the lower of this carrying amount and the £3.34 million (£3.5m – £160,000) fair value less costs to sell (IFRS 5.15) so £2.96 million. There is no subsequent depreciation (IFRS 5.25), so the carrying amount will be the same at the date of disposal. The profit on disposal is net disposal proceeds less the carrying amount (IAS 40.69), so net sales proceeds of £3.535 million (£3.7m – £165,000) less £2.960 million gives a profit on disposal of £575,000.

17.3 £540,000 decrease

If the fair value model is used, then the carrying amount immediately before initial classification will be the £3.5 million fair value. The requirement to measure an asset 'held for sale' at the lower of carrying amount at fair value less costs to sell does not apply to investment properties measured at fair value (IFRS 5.5) and so the property continues to be measured at fair value. IAS 40.37 states that costs to sell should not be deducted from fair value, so the property continues to be measured at £3.5 million. Profit on disposal will be net sales proceeds of £3.535m less £3.5m = £35,000. This is a reduction of £540,000 on the cost model gain. If the fair value model is used, then the carrying amount immediately before initial classification will be the £3.5 million fair value. The requirement to measure an asset 'held for sale' at the lower of carrying amount at fair value less costs to sell does not apply to investment properties measured at fair value (IFRS 5.5) and so the property continues to be measured at fair value.

IAS 40.37 states that costs to sell should not be deducted from fair value, so the property continues to be measured at £3.5 million. Profit on disposal will be net sales proceeds of £3.535m less £3.5m = £35,000. This is a reduction of £540,000 on the cost model gain.

18 Arapawanui

The newborn sheep are biological assets and should be measured at fair value less costs to sell, both on initial recognition and at each reporting date (IAS 41.12). The gains on initial recognition and from a change in this value should be recognised in profit or loss (IAS 41.26). As the animals are six months old at the year end, the total gain in the year (being the initial gain based on a newborn fair value of £22 plus the year-end change in value by £4 to £26) is £7,644 (300 × £26 × (100% – 1.5% – 0.5%)).

The milk is agricultural produce and should be recognised initially under IAS 41 at fair value less costs to sell (IAS 41.13). (At this point it is taken into inventories and dealt with under IAS 2.) The gain on initial recognition should be recognised in profit or loss (IAS 41.28). The gain is £1,248 (10,000 litres × £0.13 × (100% – 4)).

Total gain is £8,892.

19 Tepev

£33,400

Biological assets should be measured at fair value less costs to sell (IAS 41.12). Costs to sell include sales commission and regulatory levies but exclude transport to market (IAS 41.14). Transport costs are in fact deducted from market value in order to reach fair value. In this question fair value of £90 is provided; it is assumed that this is calculated as a market value of £94 less the quoted transport costs of £4. Contracts to sell agricultural assets at a future date should be ignored (IAS 41.16).

The statement of financial position carrying amount per sheep is:

	£
Fair value	90.00
Costs to sell (£90 × 5%) + £2.00	(6.50)
Value per sheep	83.50

For the flock of 400 sheep, the amount is £33,400.

20 Saving

£5,700

Biological assets should be measured at fair value less costs to sell, both on initial recognition and at each reporting date (IAS 41.12). Costs to sell include sale commission and regulatory levies but exclude transport to market (IAS 41.14). Transport costs are in fact deducted from market value in order to reach fair value. Contracts to sell agricultural assets at a future date should be ignored (IAS 41.16).

	£
FV at reporting date (£107 – commission (£107 × 5%) – levy £2.00)	99.65
Initial FV per sheep (£95 – commission (£95 × 5%) – levy £2.00)	(88.25)
Gain per sheep	11.40

There is, therefore, a gain on the flock of 500 sheep of £5,700.

21 Monkey

£86,000 loss on initial recognition

£686,000 gain on subsequent measurement

	£'000
Cost of herd	1,800
Recognised at FV – costs to sell (£1.8m – fees (£1.8m × 2%) – gvnmt fee £50,000)	(1,714)
Initial loss on recognition	86

On acquisition of the herd, the cattle are initially recognised as biological assets at fair value less costs to sell (IAS 41.27), which in this case is less than cost by the costs to sell which are immediately deducted (IAS 41.27). Acceptable costs to sell include auctioneers' fees and government transfer fees (IAS 41.14) but exclude transport to market costs (IAS 41.14). The interest on the loan taken out to finance the acquisition is not a cost to sell (IAS 41.22).

The value is then restated to fair value less costs to sell at each reporting date (IAS 41.12)

	£'000
Fair value at 31 December 20X7	2,500
Costs to sell: auctioneers fees (£2.5m × 2%)	(50)
Government fees	(50)
Carrying value	2,400
Less initial recognition value	(1,714)
Gain	686

22 Blackbuck

The extra payable on death before surrender/maturity should be accounted for under IFRS 4 and the remainder under IAS 39.

Given the entity's accounting policies in relation to the recognition of obligations under the deposit components, IFRS 4.10 requires the insurance component and the deposit component to be unbundled; IFRS 4.12 requires the insurance component to be accounted for under IFRS 4 and the deposit component under IAS 39.

23 *Traore*

The general measurement model of IFRS 17 must be applied since the contracts' term is more than 12 months.

The present value of net future cash outflows is £2,800 (£14,800 expected cash outflow less £12,000 cash inflow). This is adjusted upwards by £1,500 to reflect non-financial risk, giving fulfilment cash outflows of £4,300. The contract is expected to be loss making and therefore no contractual service margin is recognised. Instead the £4,300 is recognised as a loss in profit or loss immediately.

24 Evaluation and exploration

Circumstances that would trigger a need to test an evaluation and exploration asset for impairment:

- The expiration or anticipated expiration in the near future of the period for which the entity has the right to explore the relevant area, unless the right is expected to be renewed.
- The lack of available planned or budgeted expenditure for further exploration and evaluation of the specific area.
- A decision to discontinue evaluation activities in the exploration and specific area when commercially viable resources have not been identified.

Chapter 13

Reporting of non-financial liabilities

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 IAS 10, Events After the Reporting Period

2 IAS 37, Provisions, Contingent Liabilities and Contingent Assets

3 Audit focus

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Identify and explain current and emerging issues in corporate reporting
- Explain how different methods of recognising and measuring assets and liabilities can affect reported financial position, and explain the role of data analytics in financial asset and liability valuation
- Explain and appraise accounting standards that relate to assets and non-financial liabilities for example: property, plant and equipment; intangible assets, held-for-sale assets; inventories; investment properties; provisions and contingencies
- Justify and conclude for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**IAS 10, *Events After the Reporting Period*** Regardless of how quickly the financial statements are published after the end of the year, subsequent events and transactions can be so significant that they need to be considered for either disclosure or adjustment or (in extreme situations) for their effect on the underlying principle of going concern. IAS 10, *Events After the*	**Approach** This section should be mainly revision from your earlier studies. Make sure you do all the questions and look back to your earlier studies if you are still unsure. **Stop and think** What type of events could be classified as adjusting or non-adjusting respectively? What type of events could affect the underlying principle of going concern?	Although this standard was tested at Level A at Professional Level, questions at Advanced Level will be more complex and there will be more interaction between standards. Ethical issues may arise, for example if a sale of an asset is treated incorrectly as an adjusting event, rather than non-adjusting, causing revenue and profit to be recognised too early.	**IQ2: Significant events** This short, straightforward question will test whether you have remembered this topic from your earlier studies.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	Reporting Period deals with the treatment of these issues. You will need to identify the appropriate cut-off date which will depend on the process of authorisation and the reporting entity's jurisdiction.			
2	**IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*** You will be required to identify circumstances in which provisions will be required and when contingent liabilities or contingent assets should be disclosed.	**Approach** The key point is that provisions need to be recognised on the basis of obligations rather than management intentions. **Stop and think** A provision may be required in relation to cyber security breaches.	IAS 37 appears regularly in exam questions within a scenario that tests other standards, for example IAS 19 and IFRS 15. Possible ethical issues may arise, such as the directors wishing to make a lower provision than is required.	**IQ6: Restructuring** This tests IAS 37 in conjunction with IAS 36. **IQ7: Obligation to dismantle** Tests onerous contracts and discounting.
3	**Audit focus** This section focuses on provisions and contingencies, and also covers ISA 501, *Audit Evidence – Specific Considerations for Selected Items* which provides guidance on procedures regarding litigation and claims.	**Approach** The audit focus section will help you with the integrated questions on this area. **Stop and think** Why would the auditors need to communicate with the company's lawyers?	Integrated questions have generally focused on contraventions of IAS 10 and what the auditors should do in such cases.	**IQ8: Contingencies** This is an integrated question with an FR part on disclosures and an audit part on procedures to be carried out to determine whether a company will have to pay damages and if so, how much.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 IAS 10, *Events After the Reporting Period*

Section overview

- Events after the reporting period are split into adjusting and non-adjusting events. Those events that may affect the going concern assumption underlying the preparation of the financial statements must be considered further.
- The following is a summary of the material covered in earlier studies.

1.1 Overview of earlier studies

Events after the reporting period are split into **adjusting** and **non-adjusting events**.

1.1.1 Adjusting events after the reporting period

Adjusting events are events that provide evidence of conditions that existed at the reporting date, and the financial statements should be adjusted to reflect them. Examples include the following:

- Settlement of a court case that confirms that the entity had an obligation at the reporting date
- Evidence that an asset was impaired at the reporting date eg:
  - Bankruptcy of a customer
  - Selling prices achieved for inventory
- Determination of profit-sharing or bonus payments relating to the year
- Finalisation of prices for assets sold or purchased before year end
- The discovery of fraud or errors (where material) that show that the financial statements are misstated
- An adjustment to the disclosed earnings per share (EPS) for transactions such as bonus issues, share splits or share consolidations where the number of shares altered without an increase in resources. The additional shares are thus treated as having been in issue for the whole period

1.1.2 Non-adjusting events after the reporting period

Non-adjusting events are events that are indicative of conditions that arose after the reporting date. Disclosure should be made in the financial statements where the outcome of a non-adjusting event would influence the economic decisions made by users of the financial statements. Examples are as follows:

- A major business combination after the reporting date (IFRS 3 or the disposing of a major subsidiary)
- Announcement of plan to discontinue an operation
- Major purchases and disposals of assets
- Classification of assets as held for sale
- Expropriation of assets by government
- Destruction of assets, for example by fire or flood
- Announcing or commencing the implementation of a major restructuring
- Major ordinary share transactions (unless these involve transactions such as capitalisation and bonus issues where there is a change in the number of shares without an inflow or

outflow of resources, see adjusting events above. Such transactions require EPS to be restated as if the new number of shares was in issue for the whole year)

- Decline in the market value of investments including investment properties after the reporting date. These should reflect the fair value at the reporting date and should not be affected by hindsight

1.1.3 Going concern basis

Financial statements are prepared on the '**going concern**' basis. Where an entity goes into liquidation after the reporting date, it is no longer considered to be a going concern and the financial statements should not be prepared on this basis.
Where the going concern basis is clearly not appropriate, a basis other than the going concern basis should be adopted, for example the 'break-up basis'. The break-up basis measures the assets at their recoverable amount in a non-trading environment, and a provision is recognised for future costs that will be incurred to 'break-up' the business.

Where the financial statements are not prepared on a going concern basis, this should be fully disclosed, along with the actual basis of preparation used.

Management is required to make an explicit assessment of the entity's ability to continue as a going concern by considering a number of financial, operating and other indicators. Indicative of inability to continue as a going concern would be major restructuring of debt, adverse key financial ratios, substantial sale of non-current assets not intended to be replaced, loss of key staff or major markets.

1.1.4 The period of review

The cut-off date for the consideration of events after the reporting period is the date on which the financial statements are authorised for issue. Events that occur after the reporting date but before the financial statements are authorised for issue need to be considered, regardless of what financial information has been made publicly available during this period.

Normally the financial statements are authorised by the directors before being issued to the shareholders for approval; the authorisation date is the date these are authorised for issue to the shareholders, and not the date they are approved by the shareholders.

Where a supervisory board is made up wholly of non-executive directors, the financial statements will first be authorised by the executive directors for issue to that supervisory board for its approval. The relevant cut-off date for the review of events that have occurred after the reporting date is the date on which the financial statements are authorised for issue to the supervisory board.

The date on which the financial statements were authorised for issue should be disclosed, since events occurring after that date will not be reflected in the financial statements.

1.1.5 Treatment of errors

Errors identified before the authorisation date will be adjusted in the current financial statements. Those identified after the financial statements have been published should be dealt with in a subsequent period under IAS 8, *Accounting Policies, Changes in Accounting Estimates and Errors*. IAS 8 requires an error relating to a prior period to be treated as an adjustment to the comparative information presented in the subsequent financial statements.

If a significant event occurs after the authorisation of the financial statements but before the annual report is published, the entity is not required to apply the requirements of IAS 10. However, if the event was so material that it affects the entity's business and operations in the future, the entity may wish to discuss the event in the narrative section at the front of the Annual Review but outside the financial statements themselves.

1.1.6 Equity dividends

These should only be recognised as a liability where they have been declared before the reporting date, as this is the date on which the entity has an obligation. Where equity dividends are declared after the reporting date, this fact should be disclosed but no liability recognised at the reporting date.

1.1.7 Further points to note

Information on customers and suppliers

- Information after the reporting date on either a customer or supplier may not only affect amounts that have been recorded in the financial statements but also impact on the future trading of the entity.
- This should be assessed and disclosure made if the liquidation of a major customer or supplier is likely to influence the economic decisions of users of the financial statements.
- Significant customer and supplier relationships are fundamental where an entity relies on one major supplier. An example of a significant supplier/customer relationship is Intel Corporation and Dell Inc, where until recently Dell computers have used only Intel microprocessors. Dell relies almost totally on the ongoing supplier/customer relationship with Intel, and the success of Intel is vitally important to the future trade of Dell itself.

1.1.8 Contingent liabilities

Evidence may come to light regarding a contingent liability or provision that an entity was unaware of at the reporting date. The distinction between a contingent liability and a provision is discussed in IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*. An example of an unknown provision is where, because of a major fault with goods which were purchased before the reporting date, an electrical retail chain has had the goods returned after the reporting date. The fault may raise safety issues and the retailer may have to recall all such items sold within a period of time in order to repair the fault. In such circumstances, a provision should be recognised for the repair of all items that have been sold before the reporting date. The entity may not have been aware of the problem at the reporting date but, as it existed at that date, a provision should be recognised in light of the new information.



Interactive question 1: Various events

The Roach Company is completing the preparation of its draft financial statements for the year ended 31 May 20X6.

On 24 July 20X6, an equity dividend of £200,000 was declared and a contractual profit share payment of £35,000 was made, both based on the profits for the year to 31 May 20X6.

On 20 June 20X6, a customer went into liquidation having owed the company £31,000 for the past six months. No provision had been made against this debt.

On 17 July 20X6, a manufacturing plant was destroyed by fire resulting in a financial loss of £200,000.

Requirement

According to IAS 10, *Events After the Reporting Period*, which amounts should be recognised in Roach's financial statements for the year to 31 May 20X6 to reflect adjusting events after the reporting period?

See **Answer** at the end of this chapter.

Interactive question 2: Significant events

An entity's draft financial statements for the year ended 31 December 20X3 were completed on 30 May 20X4, approved by the finance director on 7 June 20X4, authorised for issue on 20 June 20X4 and approved by the shareholders on 5 July 20X4.

The following events occurred after the reporting date (assume all amounts are significant to the entity):

Requirements

How should the entity treat these events in its financial statements?

(a) Notification on 18 February 20X4 that a customer owing £100,000 as at 31 December 20X3 has gone into liquidation. The financial statements already include a specific provision of £20,000 for this customer and the entity does not make general provisions.

(b) A rights issue on 6 April 20X4 to raise £1,500,000 for an acquisition.

(c) Confirmation on 28 May 20X4 from the entity's insurer that they will pay £500,000 for inventories that were destroyed in a fire on 24 December 20X3. The entity had claimed £650,000 and included this as a receivable in the financial statements.

See **Answer** at the end of this chapter.

Interactive question 3: Dividends proposed and declared

The recent financial calendar of an entity with a 31 December year end has included the following:

	Authorised by directors for issue	Approved in annual general meeting
Financial statements for 20X2	28 February 20X3	3 May 20X3
Financial statements for 20X3	28 February 20X4	4 May 20X4

Dividends on ordinary shares	Proposed by directors	Declared by directors	Approved in annual general meeting
20X2 final	28 Feb 20X3	no	yes
20X3 interim	31 Aug 20X3	yes	no
20X3 final	28 Feb 20X4	no	yes

Requirement

How will the dividends be dealt with in the entity's financial statements?

See **Answer** at the end of this chapter.

Professional skills focus: Concluding, recommending and communicating

While IAS 10 was covered at earlier levels, the focus at Advanced Level will be on more complex scenarios, where conclusions will need to be drawn.

2 IAS 37, Provisions, Contingent Liabilities and Contingent Assets

Section overview

The following is a summary of the material covered in earlier studies.

2.1 Overview of earlier studies

Definition

Provision: A liability where there is uncertainty over its timing or the amount at which it will be settled.

2.1.1 Recognition

- A provision should be recognised when:
  - an entity has a present obligation (legal or constructive) as a result of a past event;
  - it is probable that there will be an outflow of resources in the form of cash or other assets; and
  - a reliable estimate can be made of the amount.
- A provision should not be recognised in respect of future operating losses since there is no present obligation arising from a past event.

2.1.2 Onerous contracts

- If future benefits under a contract are expected to be less than the unavoidable costs under it, the contract is described as onerous. The excess unavoidable costs should be provided for at the time a contract becomes onerous.
- A lease agreement that becomes onerous is only within the scope of IAS 37, and therefore results in the creation of a provision, if simplified accounting is applied, so that no lease liability has been recognised. This is only the case where a lease is short-term or for an asset with a low value.

Definition

Unavoidable costs: Unavoidable costs of meeting an obligation are the lower of:

- the cost of fulfilling the contract; and
- any penalties from failure to fulfil the contract

2.1.3 *Climate change provisions*

As companies take action to address climate change, new provisions may be recognised and contingent liabilities disclosed. Where legislation is introduced to combat climate change, an obligation usually only arises when the legislation is enacted. However, the past event that triggers recognition of a provision only occurs when a company takes action to comply with the legislation eg, by fitting particular systems and so being obliged to pay for them. The fact that there is a legal requirement to incur expenditure does not itself result in recognition of a provision as a company may choose not to comply. If non-compliance results in fines or penalties, these are provided for.

A company may make a public commitment to behave in a certain way e.g, by announcing its transition to net zero plans. In this case careful assessment is required to determine whether a constructive obligation does exist.

2.1.4 Restructuring costs

- A constructive obligation, requiring a provision, only arises in respect of restructuring costs where the following criteria are met:
  - A detailed formal plan has been made, identifying the areas of the business and number of employees affected with an estimate of likely costs and timescales
  - An announcement has been made to those who will be affected by the restructuring

2.1.5 Contingent liability

- A contingent liability arises where a past event may lead to an entity having a liability in the future but the financial impact of the event will only be confirmed by the outcome of some future event not wholly within the entity's control.
- A contingent liability should be disclosed in the financial statements unless the possible outflow of resources is thought to be remote.

2.1.6 Contingent asset

- A contingent asset is a potential asset that arises from past events but whose existence can only be confirmed by the outcome of future events not wholly within an entity's control.
- A contingent asset should be disclosed in the financial statements only when the expected inflow of economic benefits is probable.

2.1.7 Reimbursement

- An entity may be entitled to reimbursement from a third party for all or part of the expenditure required to settle a provision. In these circumstances, an entity generally retains the contractual obligation to settle the expenditure. A provision and reimbursement are therefore recognised separately in the statement of financial position. A reimbursement should be recognised only when it is virtually certain that an amount will be received.

2.1.8 Recognition and disclosure

- A full reconciliation of movements in provisions should be presented in the financial statements. Detailed narrative explanations should also be provided in relation to provisions, contingent liabilities and contingent assets. The narrative should include an estimate of the financial amount in relation to contingent liabilities and assets as well as indications of uncertainties.
- The required disclosures have already been covered at Professional Level. In particular, in relation to discounting, any increase in the value of the discounted amount arising from the passage of time or the effect of any change in the discount rate need to be disclosed.

2.1.9 Disclosure let out

- IAS 37 permits reporting entities to avoid disclosure requirements relating to provisions, contingent liabilities and contingent assets if they would be expected to be **seriously prejudicial** to the position of the entity in dispute with other parties. However, this should only be employed in **extremely rare** cases. Details of the general nature of the provision/contingency must still be provided, together with an explanation of why it has not been disclosed.

2.1.10 Discounting to present value

- Where the time value of money is material, the amount of provision should be the present value of the expenditures required to settle the obligation. The main types of provision where the impact of discounting may be significant are those relating to decommissioning and other environmental restoration liabilities. For most other provisions, no discounting will be required, as the cash flows are not sufficiently far into the future.
- The discount rate to be used should reflect current market assessments of the time value of money and the risks specific to the liability (ie, it would be a risk-adjusted rate). In practice it may be more appropriate to use a risk-free rate and adjust the cash flows for risk. For further guidance on the risk-free rate you may refer to your Strategic Business Management Workbook. Whichever method is adopted, it is important not to double count risk.

2.1.11 Unwinding the discount

- Where discounting is used, the carrying amount of the provision increases each period to reflect the passage of time and this is recognised as a finance cost in profit or loss.



Professional skills focus: Assimilating and using information

As the above overview from earlier studies shows, deciding whether a provision should be made and if so, how much, may require complex information to be assimilated. In particular, when considering the discount rate, the risks specific to the liability must be taken into account.



Interactive question 4: Constructive obligation

On 25 September 20X7, further to a decision made earlier in the year by the board of directors, Industrial plc publicly announced a decision to reduce the level of harmful emissions from its manufacturing plants.

The directors had reached their decision to proceed with the project after appraising the investment using discounted cash flow techniques and an annual discount rate of 8%.

The directors estimated that the future cash payments required to meet their stated objective would be as follows:

- £20 million on 30 September 20X8
- £25 million on 30 September 20X9
- £30 million on 30 September 20Y0

No contracts were entered into until after the start of the new accounting year on 1 October 20X7; however, the entity has a reputation of fulfilling its financial commitments after it has publicly announced them. Industrial included a provision for the expected costs of its proposal in its financial statements for the year ended 30 September 20X7. The actual expenditure in September 20X8 was £20 million as expected.

The average remaining useful lives of the factories on 30 September 20X7 (the reporting date) was 30 years and depreciation is computed on a straight-line basis and charged to cost of sales.

Requirements

4.1 Compute the appropriate provision in the statements of financial position in respect of the proposed expenditure at 30 September 20X7 **and** 30 September 20X8, and explain why the directors decided to recognise the provision.

4.2 Compute the two components of the charge to profit in respect of the proposal for the year ended 30 September 20X8. You should explain how each component arises and identify where in the statement of profit or loss and other comprehensive income each component is reported.

See **Answer** at the end of this chapter.

Professional skills focus: Structuring problems and solutions

The interactive question you have just done is an example of structuring problems, both on your part and the part of the directors in the scenario. First, an investment appraisal is carried out by the directors, then the provision is announced, then calculated by you.

Interactive question 5: Unwinding the discount

A company has a present obligation at 31 December 20X0, which it expects to settle in four years' time for £200,000. It calculates that the present value of the obligation is £136,603, discounted at 10%.

The unwinding of the discount in 20X1, 20X2, 20X3 and 20X4 is shown in the table below.

Provision

	Cr balance b/f 1 Jan	Cr unwinding discount @ 10% 31 Dec	Cr balance c/f 31 Dec
	£	£	£
20X1	136,603	13,660	150,263
20X2	150,263	15,026	165,289
20X3	165,289	16,529	181,818
20X4	181,818	18,182	200,000

Requirement

What are the accounting entries for the above for 20X1?

See **Answer** at the end of this chapter.

Professional skills focus: Applying judgement

In the case of provisions, it is likely that the discount rate will be given to you. However, in other cases and in practice, choosing an appropriate discount rate is an area where judgement may need to be exercised. Selecting an appropriate discount rate is important as different discount rates can have a significant effect on the calculation.

Interactive question 6: Restructuring

How should these matters be recognised in the statement of profit or loss and other comprehensive income?

Requirements

6.1 An entity has a 31 December year end. The directors approved a major restructuring programme on 1 December 20X5 and announced the details on the entity's intranet and to the media on 2 December 20X5. The programme involves two stages.

Stage 1

Closure of three production lines during 20X6, the redundancy of 3,000 employees on 31 March 20X6, and the transfer during 20X6 of 500 employees to continuing parts of the business. All associated costs would be settled during 20X6.

Stage 2

Probable closure of four more production lines during 20X7, with probable redundancies of 3,500 employees during 20X7. Other staff will be transferred to continuing businesses. All associated costs would be settled during 20X7. Assume that the details of Stage 2 were formally confirmed on 1 November 20X6.

6.2 The entity had some years ago signed a 'take or pay' contract with a supplier, in order to ensure the reliable supply each year of 100,000 tonnes of critical raw materials to each of the seven production lines affected by the restructuring programme. Under the contract, the entity must pay for the 700,000 tonnes each year, even if it decides not to take delivery. This contract falls due for renewal on 1 January 20X8.

See **Answer** at the end of this chapter.



Interactive question 7: Obligation to dismantle

A company is awarded a contract to build and operate a nuclear power station on 1 January 20X1. The power station comes into operation on 31 December 20X3 and the operating licence is for 30 years from that date.

The construction cost of the power station was £450 million. Part of the agreement for the contract was that, in addition to building and operating the power station, the company is obliged to dismantle it at the end of its 30-year life and make the site safe for alternative use. At 31 December 20X3, the estimated cost of the obligation was £50 million.

An appropriate discount rate reflecting market assessments of the time value of money and risks specific to the power station is 8%.

Requirement

Explain the treatment of the cost of the power station and obligation to dismantle it as at 31 December 20X3 and for the year ended 31 December 20X4.

Work to the nearest £0.1 million.

See **Answer** at the end of this chapter.

3 Audit focus



Section overview

Auditors will carry out specific procedures on provisions and contingencies.

3.1 Auditing provisions and contingencies

Much of the audit work here is focused on ensuring that the recognition and treatment of these items is in accordance with IAS 37, which we looked at in section 2 of this chapter.

The audit procedures that should be carried out on provisions and contingent assets and liabilities are as follows.

- **Obtain details** of all **provisions** which have been included in the **accounts** and all **contingencies** that have been disclosed.
- **Obtain** a **detailed analysis** of all **provisions** showing opening balances, movements and closing balances.
- **Determine** for each material provision **whether** the **company** has a **present obligation** as a result of past events by:
  - **reviewing** of **correspondence** relating to the item; and
  - **discussing** with the **directors**. Have they created a valid expectation in other parties that they will discharge the obligation?
- **Determine** for each material provision **whether** it is **probable** that a **transfer of economic benefits** will be required to settle the obligation by:
  - **confirming** whether any **payments** have been **made** after the end of the reporting period in respect of the item;
  - **reviewing correspondences** with solicitors, banks, customers, insurance company and suppliers both pre and post year end for evidence of the existence of any liabilities;
  - **sending** a **letter** to the **solicitor** to obtain their views (where relevant);
  - **discussing** the **position** of similar **past provisions** with the directors. Were these provisions eventually settled?; and
  - **considering** the **likelihood** of **reimbursement**.
- **Recalculate** all **provisions** made.
- **Compare** the **amount provided** with any post year end payments and with any amount paid in the past for similar items.
- In the event that it is not possible to estimate the amount of the **provision**, confirm that this **contingent liability** is **disclosed** in the accounts.
- **Consider** the **nature** of the **client's business**. Would you expect to see any other provisions, for example warranties?
- **Consider** whether disclosures of **provisions, contingent liabilities and contingent assets** are correct and sufficient.

3.2 Procedures regarding litigation and claims

3.2.1 Introduction

- ISA 501, *Audit Evidence – Specific Considerations for Selected Items* provides guidance on procedures regarding litigation and claims.

A summary of the procedures regarding litigation and claims is provided below.

3.2.2 Litigation and claims

Litigation and claims involving the entity may have a material effect on the financial statements, and so will require adjustment to or disclosure in those financial statements.

The auditor shall **design and perform procedures** in order to identify any litigation and claims involving the entity which may give rise to a risk of material misstatement. (ISA 501.9)

Such procedures would include the following:

- Make appropriate **inquiries of management and those charged with governance** including obtaining **representations**.
- **Review board minutes** and **correspondence** with the entity's lawyers.
- Examine **legal expense account**.
- Use **any information** obtained regarding the entity's business including information obtained from discussions with any in-house legal department.

When litigation or claims have been identified or when the auditor believes they may exist, the auditor must seek **direct communication with the entity's lawyers**. (ISA 501.10)

This will help to obtain **sufficient, appropriate audit evidence** as to whether potential material litigation and claims are known and management's estimates of the financial implications, including costs, are reliable.

Form of the letter of inquiry

The letter, which should be prepared by management and sent by the auditor, should request the lawyer to **communicate directly** with the auditor.

If it is thought unlikely that the lawyer will respond to a general inquiry, the letter should specify the following.

- A list of litigation and claims
- Management's assessment of the outcome of the litigation or claim and its estimate of the financial implications, including costs involved
- A request that the lawyer confirm the reasonableness of management's assessments and provide the auditor with further information if the list is considered by the lawyer to be incomplete or incorrect

The auditors must consider these matters **up to the date of their report** and so a further, updating letter may be necessary.

A meeting between the auditors and the lawyer may be required, for example where a complex matter arises, or where there is a disagreement between management and the lawyer. Such meetings should take place only with the permission of management, and preferably with a management representative present.

If management refuses to give the auditor permission to communicate with the entity's lawyers or if the lawyer refuses to respond as required and the auditor can find no alternative sufficient evidence, this would mean that the auditor is unable to obtain sufficient, appropriate evidence and should ordinarily lead to a qualified opinion or a disclaimer of opinion. (ISA 501.11)



Interactive question 8: Contingencies

In February 20X7 the directors of Newthorpe Engineering suspended the managing director. At a disciplinary hearing held by the company on 17 March 20X7 the managing director was dismissed for gross misconduct, and it was decided the managing director's salary should stop from that date and no redundancy or compensation payments should be made.

The managing director has claimed unfair dismissal and is taking legal action against the company to obtain compensation for loss of his employment. The managing director says he has a service contract with the company which would entitle him to two years' salary at the date of dismissal.

The financial statements for the year ended 30 April 20X7 record the resignation of the director. However, they do not mention his dismissal and no provision for any damages has been included in the financial statements.

Requirements

8.1 State how contingent liabilities should be disclosed in financial statements according to IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*.

8.2 Describe the audit procedures you will carry out to determine whether the company will have to pay damages to the director for unfair dismissal, and the amount of damages and costs which should be included in the financial statements.

Note: Assume the amounts you are auditing are material.

See **Answer** at the end of this chapter.



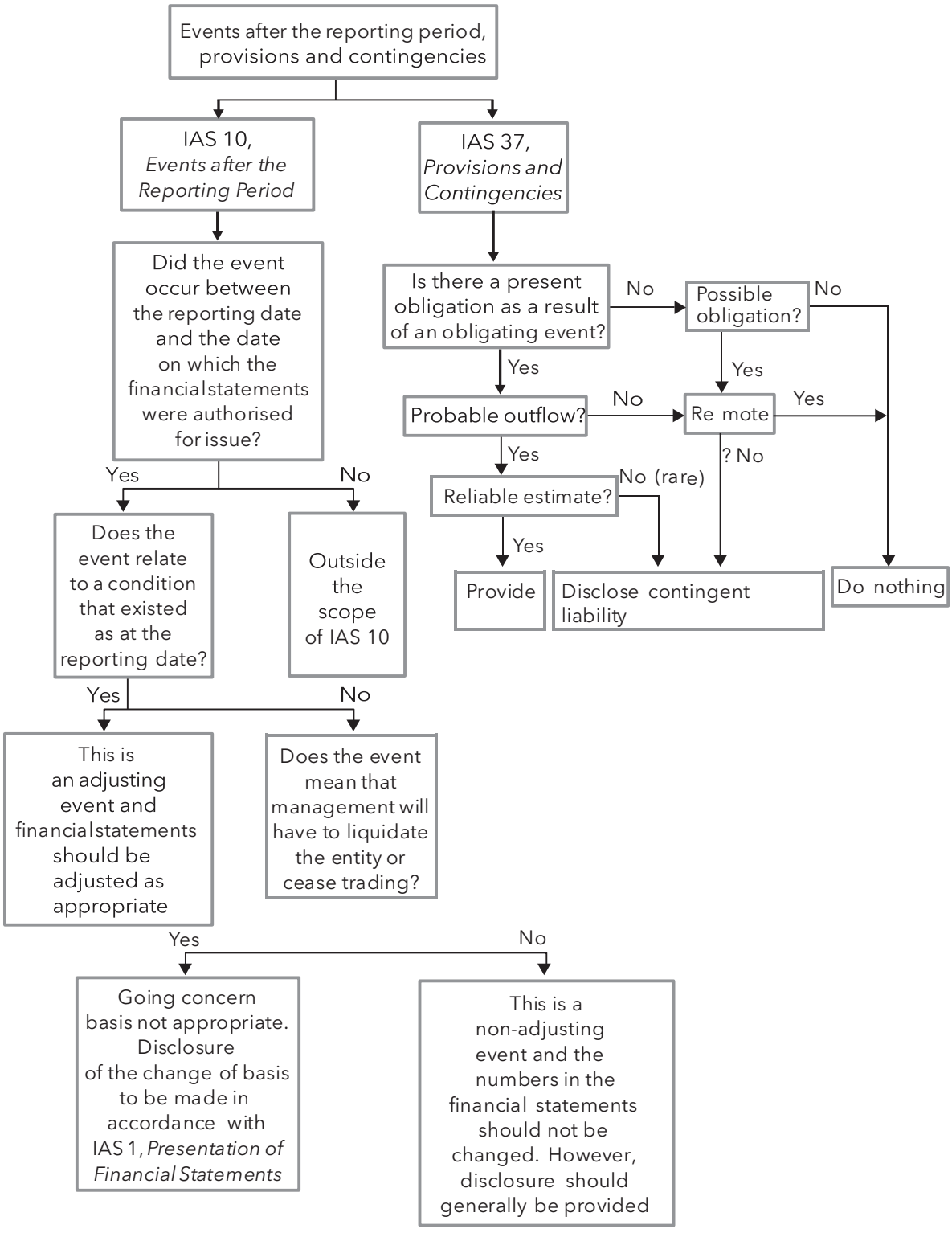
Professional skills focus: Applying judgement

Provisions are an area where judgement is required, for example in assessing the likelihood of success of a legal claim. In an exam, you are generally told whether a claim is likely to be successful, but in practice it could be difficult to determine.

3.3 Procedures regarding events after the reporting period

ISA 560, *Subsequent Events* sets out the audit requirements in relation to events occurring after the reporting period. Please refer to earlier chapters for a more detailed discussion.

Summary


Events after the reporting period, provisions and contingencies
IAS 10, Events after the Reporting Period
IAS 37, Provisions and Contingencies
Did the event occur between the reporting date and the date on which the financial statements were authorised for issue?
Yes
No
Does the event relate to a condition that existed as at the reporting date?
Outside the scope of IAS 10
Yes
No
This is an adjusting event and financial statements should be adjusted as appropriate
Does the event mean that management will have to liquidate the entity or cease trading?
Yes
No
Going concern basis not appropriate. Disclosure of the change of basis to be made in accordance with IAS 1, Presentation of Financial Statements
This is a non-adjusting event and the numbers in the financial statements should not be changed. However, disclosure should generally be provided
Is there a present obligation as a result of an obligating event?
No
Possible obligation?
No
Yes
Yes
Probable outflow?
No
Re mote
Yes
Yes
? No
Reliable estimate?
No (rare)
Yes
Provide
Disclose contingent liability
Do nothing

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	What is an adjusting event (IAS 10)? Give examples (Topic 1)
2.	What are IAS 10's requirements in relation to dividends? (Topic 1)
3.	When should a provision be recognised? (Topic 2)
4.	When, if at all, should a contingent liability be recognised? (Topic 2)
5.	When, if at all, should a contingent asset be recognised? (Topic 2)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Saimaa	This question revises knowledge of IAS 10, but requires application to an outcome that is uncertain.
Quokka	This question is good practice because it tests the interaction of two standards: IAS 10 and IAS 2.
Wilcox	In this question your brought forward knowledge of IAS 37, including discounting, is tested.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Upstart Records (Exhibit 3)	This tests IAS 37 in the context of a restructuring plan in considerably more depth and complexity than at earlier levels.
Newpenny (Exhibit 1 (1))	You will need to determine whether a provision is required for additional payment if a set volume of purchases is not met.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving onto the next chapter.

Technical reference

1 IAS 10, *Events After the Reporting Period*

- Authorisation
  - Process of authorisation of financial statements – **IAS 10.4**
  - Authorisation date is the date on which financial statements are authorised for issue to shareholders – **IAS 10.5, 10.6**
  - The relevant cut-off date for consideration of events after the reporting period is the authorisation date – **IAS 10.7**
- Adjusting events
  - Amounts recognised in financial statements should be adjusted to reflect adjusting events after the reporting date – **IAS 10.8, 10.9**
  - Examples of adjusting events include
    - Outcome of court case that confirms obligation at reporting date
    - Receipt of information on recoverability or value of assets
    - Finalisation of profit sharing or bonus payments
    - Discovery of fraud or errors
- Non-adjusting events
  - An entity should not adjust amounts recognised in financial statements for non-adjusting events after the reporting period (an example is the subsequent decline of market value of investments) – **IAS 10.10**
  - Non-adjusting events may need to be disclosed – **IAS 10.10**
  - Dividends proposed or declared on equity instruments after the reporting date cannot be recognised as a liability at the reporting date – **IAS 10.12**
  - Dividends proposed or declared after the reporting date should be disclosed – **IAS 10.13**
- Going concern basis
  - Financial statements are not to be prepared on going concern basis if management intends to liquidate entity or cease trading – **IAS 10.14**
  - If going concern assumption no longer appropriate, disclosures required in accordance with IAS 1 – **IAS 10.16**
- Disclosure
  - Date of authorisation to be disclosed – **IAS 10.17**
  - Disclosures relating to information after the reporting date to be updated in the light of new information – **IAS 10.19**
  - For material non-adjusting events after the reporting period an entity shall disclose: – **IAS 10.21**
    - Nature of event
    - Estimate of financial effect

2 IAS 37, Provisions, Contingent Liabilities and Contingent Assets

- Scope – **IAS 37.1**
- Definitions – **IAS 37.10**
- Recognition of provisions – **IAS 37.14–15**
- Contingent liabilities – **IAS 37.27**
- Contingent assets – **IAS 37.31**
- Measurement best estimate – **IAS 37.36**
- Risks and uncertainties – **IAS 37.42**
- Present value – **IAS 37.45–47**
- Future events – **IAS 37.48**
- Onerous contracts – **IAS 37.66**
- Restructuring – **IAS 37.70–72, IAS 37.78–80**

3 ISA 501

- Audit procedures in respect of litigation and claims – **ISA 501.9–12**

Self-test questions

Answer the following questions.

1 ABC International

ABC International Inc is a company that deals extensively with overseas entities and its financial statements include a substantial number of foreign currency transactions. The entity also holds a portfolio of investment properties.

Requirements

Discuss the treatment of the following events after the reporting date.

(a) Between the reporting date of 31 December 20X6 and the authorisation date of 20 March 20X7, there were significant fluctuations in foreign exchange rates that were outside those normally expected.

(b) The entity obtained independent valuations of its investment properties at the reporting date based on current prices for similar properties. On 15 March 20X7, market conditions which included an unexpected rise in interest rates and the expectation of further rises resulted in a fall in the market value of the investment properties.

(c) A competitor introduced an improved product on 1 February 20X7 that caused a significant price reduction in the entity's own products.

2 Saimaa

The Saimaa Company operates in the banking industry. It is attempting to sell one of its major administrative office buildings and relocate its employees.

Saimaa has found a potential buyer, The Nipigon Company, which operates a chain of retail stores. Nipigon would like to convert the building into a new retail store but would require planning permission for this change of use. Nipigon may, however, still consider purchasing the building and using it for its own administrative offices if planning permission is declined. It is estimated that there is approximately a 50% probability of planning permission being granted.

A contract for sale of the building is to be drawn up in November 20X7 and two alternatives are available:

Contract 1: This sale contract would be made conditional on planning permission being granted. Thus the contract would be void if planning permission is not granted but it would otherwise be binding.

Contract 2: This contract would be unconditional and binding, except that the price would vary according to whether or not planning permission is granted.

The financial statements of Saimaa for the year to 31 December 20X7 are authorised for issue on 28 March 20X8. A decision on planning permission will be made in February 20X8.

Requirement

With respect to the financial statements of Saimaa for the year to 31 December 20X7, and according to IAS 10, *Events After the Reporting Period*, indicate whether the granting of planning permission on each of the contracts is an adjusting event.

3 Quokka

The Quokka Company manufactures balers for agricultural use. The selling price per baler, net of selling expenses, at 31 December 20X7 is £38,000. Due to increasing competition, however, Quokka decides to reduce the selling price by £5,000 on 3 January 20X8.

On 4 January 20X8 a health and safety report was delivered to Quokka by the government, showing that some of its balers were toppling over on moderate gradients. £9,000 per baler would need to be incurred by Quokka to correct the fault. No further sales could be made without the correction.

Quokka had been unaware of any problem or health and safety investigation until the report was delivered.

The financial statements of Quokka for the year to 31 December 20X7 are to be authorised for issue on 23 March 20X8.

The cost of manufacture for each baler was £36,000 and there were 80 balers in inventory at 31 December 20X7.

Requirement

After adjustment (if any) for the above events, what should be the carrying amount of the inventory in the financial statements of Quokka at 31 December 20X7, in accordance with IAS 10, *Events After the Reporting Period* and IAS 2, *Inventories*?

4 Labeatis

The financial statements of the Labeatis Company for the year to 31 December 20X7 were approved and issued with the authority of the board of directors on 6 March 20X8. However, the financial statements were not presented to the shareholders' meeting until 27 March 20X8.

The following events took place:

Event 1: On 18 February 20X8 the Government announced a retrospective increase in the tax rate applicable to Labeatis's year ending 31 December 20X7.

Event 2: On 19 March 20X8 a fraud was discovered which had had a material effect on the financial statements of Labeatis for the year ending 31 December 20X7.

Requirement

State which event (if any) is an adjusting event according to IAS 10, *Events After the Reporting Period*.

5 Scioto

The Scioto Company's financial statements for the year ended 30 April 20X7 were approved by its finance director on 7 July 20X7 and a public announcement of its profits for the year was made on 10 July 20X7.

The board of directors authorised the financial statements for issue on 15 July 20X7 and they were approved by the shareholders on 20 July 20X7.

Requirement

Under IAS 10, *Events After the Reporting Period*, after which date should consideration no longer be given as to whether the financial statements to 30 April 20X7 need to reflect adjusting and non- adjusting events?

6 Fushia

The Fushia Company sells electrical goods covered by a one-year warranty for any defects.

Of sales of £60 million for the year, the company estimates that 3% will have major defects, 6% will have minor defects and 91% will have no defects.

The cost of repairs would be £5 million if all the products sold had major defects and £3 million if all had minor defects.

Requirement

What amount should Fushia provide as a warranty provision?

7 Wilcox

The Wilcox Company has been lead mining in Valovia for many years. To clean the lead, Wilcox uses toxic chemicals which are then deposited back into the mines. Historically there has not been any legislation requiring environmental damage to be cleaned up. The company has a policy of only observing its environmental responsibilities when legally obliged.

In December 20X7, the Government of Valovia introduced legislation on a retrospective basis, forcing mining companies to rectify environmental damage that they have caused.

Wilcox estimates that the damage already caused will cost £27 million to rectify, but the work would not be paid for until December 20X9. It also estimates that damage caused by its operations each year for the remaining four years of the mines' lifespan will be £3 million, payable at the end of the relevant year.

17% is the pre-tax rate that reflects the time value of money and the risk specific to these liabilities.

Requirement

To the nearest £1 million, what provision should be shown in the statement of financial position of Wilcox at 31 December 20X7 under IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*?

8 Noble

The Noble Company operates a fleet of commercial aircraft. On 1 April 20X7, a new law was introduced requiring all operators to use aircraft fitted with fuel-efficient engines only.

At 31 December 20X7 Noble had not fitted any fuel-efficient engines and the total cost of fitting them throughout the fleet was estimated at £4.2 million.

Under the terms of the legislation, the company is liable for a fine of £1 million for non-compliance with legislation for any calendar year, or part of a year, in which the law has been broken. The Government rigorously prosecutes all violations of the new law.

The effect of the time value of money is immaterial.

Requirement

State the provision required in Noble's financial statements for the year ended 31 December 20X7 under IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

The £35,000 profit share payment and the £31,000 bad debt expense are adjusting events and should be recognised in the financial statements.

See IAS 10.9, 10.12 and 10.22.

Answer to Interactive question 2

(a) This is an adjusting event, as it provides more up to date information about a provision that was recognised at the reporting date. The £100,000 receivable should be written off.

(b) This is a disclosable non-adjusting event. The rights issue occurred after the reporting date, but is considered to be of significant importance and should be disclosed in the financial statements.

(c) This is an adjusting event since it is in relation to an asset that was recognised at the reporting date. The receivable should be reduced to £500,000.

Answer to Interactive question 3

These dividends will be dealt with in the entity's financial statements for 20X2, 20X3 and 20X4 as follows:

Financial statements for:	20X2	20X3	20X4
20X2 final dividend	Disclosed in the notes	Charged to statement of changes in equity	–
20X3 interim dividend	–	Charged to statement of changes in equity	–
20X3 final dividend	–	Disclosed in the notes	Charged to statement of changes in equity

Answer to Interactive question 4

4.1 **Provision at 30 September 20X7**

Expenditure on:		**£'000**
30 September 20X8	20,000 × 0.926	18,520
30 September 20X9	25,000 × 0.857	21,425
30 September 20Y0	30,000 × 0.794	23,820
		63,765

Provision at 30 September 20X8

Expenditure on:		**£'000**
30 September 20X9	25,000 × 0.926	23,150
30 September 20Y0	30,000 ×0.857	25,710
		48,860

A provision should be recognised where:

(1) there is a present obligation as a result of a past event;

(2) there is a probable outflow of economic benefits; and

(3) the amount can be measured reliably.

(2) and (3) are clearly met, as Industrial will incur expenditure and the detailed estimates of the amounts have been prepared.

By announcing the plan to reduce emissions publicly, Industrial has created a constructive obligation to carry out the project. Therefore, although there is no legal obligation, Industrial should record a provision for the estimated (and discounted) costs of the project.

4.2 The charge to profit or loss for the year ended 30 September 20X8 consists of:

(1) Depreciation (£63,765,000 ÷ 30) = £2,125,500 This is reported in cost of sales.

The provision of £63,765,000 also represents an **asset**, as it gives rise to future economic benefits (it enhances the performance of the factories). This is **capitalised and depreciated over 30 years** (the average useful life of the factories).

(2) Unwinding of the discount (see working) = £5,095,000 This is reported as a **finance cost**.

WORKING

	£'000
Provision at 1 October 20X7	63,765
Expenditure on 30 September 20X8	(20,000)
Unwinding of discount (balancing figure)	5,095
Provision at 30 September 20X8	48,860

Answer to Interactive question 5

The accounting entry to record the unwinding of the discount in 20X1 will be:

DEBIT	Finance costs	£13,660	
CREDIT	Provisions		£13,660

Answer to Interactive question 6

6.1 Detailed information was made available about who would be affected by Stage 1 and when the various steps in the first stage of the closure programme would take place.

The announcement about Stage 2 was more of an overview. It was not until 1 November 20X6 that information was announced in respect of Stage 2 in as much detail as that provided in December 20X5 about Stage 1. There is a constructive obligation in respect of Stage 1 on 2 December 20X5; no such obligation in respect of Stage 2 is made until 1 November 20X6. Although the announcement is made as a single restructuring programme, there will be two entirely separate restructuring provisions.

The costs of this major programme will be recognised in profit or loss for the years ending 31 December 20X5–20X7 as follows:

	Reason	Stage 1	Stage 2

Termination payments to those taking voluntary redundancy	Restructuring provision	20X5	20X6
Termination payments to those being made compulsorily redundant	Restructuring provision	20X5	20X6
Employment costs during closing down activities and selling off inventory	Restructuring provision	20X5	20X6
One-off payments to employees agreeing to move to continuing parts of the business	Continuing activities	20X6	20X7
Cost of moving plant and equipment to continuing parts of the business	Continuing activities	20X6	20X7
Cost of moving saleable inventory to continuing parts of the business	Continuing activities	20X6	20X7
Impairment losses on non-current assets	See Note	20X5 & 20X6	20X5 to 20X7
Losses on disposal of non-current assets	Year when loss on disposal incurred	20X6	20X7
Revenue less expenses up to date of closure, other than itemised expenses	Year when operating losses incurred	20X6	20X7

Note: The announcement of a restructuring programme is an indicator of impairment under IAS 36, so an impairment test should be carried out at the time of the first announcement for all relevant non-current assets. Despite Stage 2 only being 'probable', its assets should still be tested for impairment in 20X5.

6.2 A provision should be recognised in respect of all contracts when they become onerous, regardless of whether this is associated with a restructuring programme. If the entity's production lines were loss-making before the restructuring announcement, then this contract may already have been classified as onerous.

If it had not already been identified as being onerous then, at a minimum, a provision should be made for the 300,000 tonnes per annum for the three Stage 1 production lines, for the period from when they cannot take any further supplies through to the end of the contract on 31 December 20X7.

A provision should be made for the remaining 400,000 tonnes per annum for the four Stage 2 production lines for the period from when they cannot take any further supplies through to the end of the contract on 31 December 20X7.

Note: This answer includes a comprehensive list of issues to be considered under restructuring programmes, included for learning purposes.

Answer to Interactive question 7

At 31 December 20X3

The discounted amount of the provision would be included in the initial measurement of the cost of the power station as at 31 December 20X3:

£m

Cost	450.0
Provision (£50m × $1/1.08^{30}$)	5.0
	455.0

Year ended 31 December 20X4

The power plant would be depreciated over its 30-year life resulting in a charge of £455.0m/30 = £15.2m to profit or loss and a carrying amount of £455m – £15.2m = £439.8m.

The provision would begin to be compounded resulting in an interest charge of £5.0 × 8% = £0.4m and an outstanding provision of £5.0 + £0.4 = £5.4m in the statement of financial position.

Any change in the expected present value of the provision would be made as an adjustment to the provision and to the asset value (affecting future depreciation charges).

Answer to Interactive question 8

8.1 IAS 37 states that a provision should be recognised in the accounts if:

- an entity has a **present obligation** (legal or constructive) as a result of a past event;
- a **transfer** of **economic benefits** will **probably** be **required** to settle the obligation; and
- a **reliable estimate** can be **made** of the amount of the obligation.

Under IAS 37 contingent liabilities should not be recognised. However, they should be disclosed unless the prospect of settlement is remote. The entity should disclose:

- the **nature** of the liability;
- an estimate of its **financial effect**;
- the **uncertainties** relating to any possible payments; and
- the likelihood of any **reimbursement**.

8.2 The following procedures should be carried out to determine whether the company will have to pay damages and the amount to be included in the financial statements.

- **Review** the director's **service contract** and **ascertain** the **maximum amount** to which he would be entitled and the **provisions** in the service contract that would **prevent** him making a **claim**, in particular those relating to grounds for justifiable dismissal.
- **Review** the results of the **disciplinary hearing. Consider** whether the company has acted in accordance with **employment legislation** and its **internal rules**, the **evidence** presented by the **company** and the defence made by the **director**.
- **Review correspondence** relating to the case and **determine** whether the **company** has **acknowledged** any **liability** to the director that would mean that an amount for compensation should be accrued in accordance with IAS 37.
- **Review correspondence** with the company's **solicitors** and **obtain legal advice**, either from the company's solicitors or another firm, about the likelihood of the claim succeeding.
- **Review** correspondence and contact the company's solicitors about the likely **costs** of the case.
- **Consider** the **likelihood** of **costs** and **compensation** being **reimbursed** by **reviewing** the company's **insurance arrangements** and contacting the insurance company.
- **Consider** the **amounts** that should be **accrued** and the **disclosures** that should be made in the accounts. Legal costs should be accrued, but compensation payments should only be accrued if the company has admitted liability or legal advice indicates that the company's chances of success are very poor. However, the claim should be disclosed unless legal advice indicates that the director's chance of success appears to be remote.

Answers to Self-test questions

1 ABC International

(a) Details of the abnormal fluctuations in exchange rates should be disclosed as a non-adjusting event after the reporting period.

(b) The decline in the value of investment properties is a non-adjusting event, as it does not reflect the state of the market at the reporting date. The valuation should reflect the state of the market at the reporting date and should not be affected by hindsight or events at a later date.

(c) The improved product issued by the competitor is likely to have been developed over a period of time. The value of inventories should be reviewed and adjusted to their net realisable value where appropriate. Non-current assets may need to be reviewed for possible impairment. This is an adjusting event, as it reflects increased competitive conditions which existed at the reporting date even if the entity was not fully aware of them.

2 Saimaa

Contract 1 is a non-adjusting event. Contract 2 is an adjusting event.

Under IAS 10.3, events after the reporting period are those which occur after the reporting period but before the financial statements are authorised for issue. The planning permission decision is such an event, because it is to be made before the financial statements are to be authorised for issue on 28 March 20X8. Adjusting events are those providing evidence of conditions that existed at the reporting date and non-adjusting events are those indicative of conditions that arose after that date.

Under Contract 1, the uncertainty surrounding the contract at the reporting date would be such that no sale could be recognised as at that date. So there is no transaction for which the planning permission decision could provide evidence and there would not be an adjusting event.

Under Contract 2, there would be an unconditional sale recognised at the reporting date, with only the consideration needing to be confirmed after the reporting date. According to IAS 10.9(c) the planning permission decision would provide additional evidence of the proceeds and would be an adjusting event.

3 Quokka

£1,920,000

IAS 2.9 states that inventories should be stated at the lower of cost and NRV, and under IAS 10.9(b)(ii) the sale of inventories after the reporting date may give evidence of NRV at the reporting date.

The 3 January price reduction is a response to competitive conditions which would have existed at the reporting date. So even though it comes after the year end, it is an adjusting event. Similarly, the safety report received after the year end relates to conditions at the year end (as the balers in inventories were defective at this date) and is an adjusting event.

The carrying amount is 80 balers at the lower of cost (£36,000) and NRV £(38,000 – 5,000 – 9,000). So 80 × £24,000 = £1,920,000

4 Labeatis

Event 1 is a non-adjusting event. Event 2 happened after the statements were authorised, so does not constitute an event after the reporting period.

Applying IAS 10.3, events after the reporting period are those which occur after the 31 December 20X7 reporting date but before the financial statements are authorised for issue on 6 March 20X8.

Event 1 occurs before 6 March 20X8 but is a non-adjusting event, because in accordance with IAS 10.22(h) this change was not enacted before the reporting date. This is the case even though the announcement has a retrospective effect on the financial statements still being prepared.

Event 2, the discovery of fraud, would have been an adjusting event per IAS 10.9(e), had it occurred before the date of authorisation for issue. As it was not discovered until 19 March 20X8, it is not even an event after the reporting period, let alone an adjusting event.

5 Scioto

15 July is the correct answer.

IAS 10.7 states that the authorisation date is the date on which the financial statements are authorised for issue, even if this is after a public announcement of profit. IAS 10.5 confirms that it is not the date on which the shareholders approve the financial statements.

6 Fushia

£330,000

Provision must be made for estimated future claims by customers for goods already sold. The expected value (£5m × 3%) + (£3m × 6%) is the best estimate of this amount (IAS 37.39).

7 Wilcox

£20 million

At the year end a legal obligation exists – through the retrospective legislation – as a result of a past event (the environmental damage caused in the past) (IAS 37.14). The company should therefore create a provision for the damage that has already been caused.

It should not now set up a provision for the future damage, because that will be caused by a future event (the company could close down the mines and therefore not cause further damage to the environment).

Because the effect of discounting at 17% over two years is material, the cost should be discounted to present value (IAS 37.45).

So, the provision is £27m/1.17^2 = £20 million (to the nearest £m)

8 Noble

No provision is required for the fitting of the engines. This is because the present obligation as a result of the past event required by IAS 37.14 does not exist. The company can choose not to fit the engines and then not to operate the aircraft.

However, a provision of £1.0 million is required in relation to the fines, because at the reporting date there is a present obligation in respect of a past event (the non-compliance with legislation).

Chapter 14

Leases, government grants and borrowing costs

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Identifying a lease

2 Lessee accounting

3 Sale and leaseback transactions

4 Lessor accounting

5 IAS 20, Accounting for Government Grants and Disclosure of Government Assistance

6 IAS 23, Borrowing Costs

7 Statements of cash flows

8 Audit focus

Summary

Further question practice

Technical references

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Determine and calculate how different bases for recognising, measuring and classifying financial assets and financial liabilities can impact upon reported performance and position
- Appraise and evaluate cash flow measures and disclosures in single entities and groups
- Explain and appraise accounting standards that relate to an entity's financing activities which include: financial instruments; leasing; cash flows; borrowing costs; and government grants
- Justify and conclude for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**Identifying a lease** Lease finance is a common route to acquiring assets. It is also topical, being the subject of a recent standard, IFRS 16, *Leases*. In a working context you may have to decide whether an arrangement constitutes a lease.	**Approach** Make sure you know the detailed definition of a lease by heart and are able to apply it to any scenario. **Stop and think** Do leases always need to be recognised in the statement of financial position?	'Is it a lease?' is a question that could well be asked in an exam, perhaps in the context of ethics, where a director wants to reduce gearing by claiming it isn't. Right of use is a key test in this respect.	**IQ2: Is it a lease?** This is an essential interactive question, covering a wide range of transactions that may or may not be leases.
2	**Lessee accounting** For lessees, all leases other than short leases and leases of low value assets must be	**Approach** As this is a recent, topical standard and ripe for examination, it is necessary to go	Once it is established that it is a lease, any aspect of IFRS 16 could be tested. A regular way of	**IQ3: Payments in arrears** This is a relatively straightforward question and should be done in

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	recognised in the statement of financial position. 'Creative' accountants may try to argue that an arrangement is not a lease in order to avoid showing it in the SOFP and thereby reducing gearing.	through all worked examples and interactive questions very carefully. **Stop and think** How is the right-of-use asset made up?	testing this is a scenario in which the lease has been incorrectly treated, for example by just showing the lease payments as an expense in the SPL, rather than an asset in the SOFP. You would need to explain why this is wrong, calculate the correct figures and often adjust the financial statements for this and other errors.	conjunction with the rather trickier next question. **IQ4: Payments in advance** Doing these questions together will show clearly how the treatment differs.
3	**Sale and leaseback transactions** Sale and leaseback arrangements can provide entities with capital funding by releasing capital caught up in the business for investment in other core opportunities. You will later encounter sale and leaseback transactions again in practical real life situations in later Chapters on financial analysis.	**Approach** Ensure you understand the different scenarios relating to sale and leaseback. And the formula to calculate the right-of-use asset and the gain on the sale. **Stop and think** Why is it important to consider whether an arrangement constitutes a lease?	Sale and leaseback is a regular topic for examination. IFRS 16 interacts with IFRS 15 as the sale has to meet the IFRS 15 criteria for a sale in order to be treated as a genuine sale and leaseback.	**IQ6: Sale and leaseback where the transfer is a sale** This is a comprehensive question on this topic and will prepare you for most exam scenarios.
4	**Lessor accounting** It is common for a company to rent out a building, as you saw earlier in your studies of IAS 40, *Investment Property*.	**Approach** The key point to note is that for lessors, a distinction is made between finance leases, which transfer the risks and rewards of ownership, and operating leases, which do not. Finance leases are	Lessor accounting is less commonly examined than lessee accounting, but it does come up occasionally. It is very important in an exam that you consider carefully whether the company is a lessee or a lessor, as the	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		recognised in the SOFP; operating lease receipts are credited to P/L **Stop and think** Why is the unguaranteed residual value of a leased asset important?	financial reporting treatment is very different in each case.	
5	**IAS 20, *Accounting for Government Grants and Disclosure of Government Assistance*** This is revision from your earlier studies.	**Approach** Revise IAS 20 through interactive questions and earlier material if necessary. **Stop and think** It is important to distinguish grants related to assets from grants related to income.	IAS 20 has not been tested under the current syllabus	**IQ8: Government grants** This is a quick revision question asking for an explanation of the two methods of recognising government grants.
6	**IAS 23, *Borrowing Costs*** This is revision from your earlier studies.	**Approach** Revise IAS 23 through interactive questions and earlier material if necessary. **Stop and think** Borrowing costs must be capitalised as part of the cost of the asset if they are directly attributable to acquisition/ construction/ production. Other borrowing costs must be expensed.	Borrowing costs are likely to be tested as part of a scenario also testing other PPE standards.	**IQ10: Borrowing costs 2** A detailed question covering any complications you are likely to encounter.
7	**Statements of cash flows** This section is for revision of single company statements of cash flows. You will cover	**Approach** Start off with the interactive question, referring to the summary only if you get stuck.	The statement of cash flows in an exam could be a consolidated one. The most likely context is a financial analysis question or	**IQ11: Single company statement of cash flows** You should be able to get this question almost completely right. If not, have a

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	consolidated statements of cash flows in later Chapters.	**Stop and think** What are the benefits of a statement of cash flows?	analytical procedures question.	go at Self-test question 9 and practise questions from your earlier studies.
8	**Audit focus** This section is concerned with the audit of leases, following the structure of the financial reporting sections.	**Approach** Read and learn the table in Section 8.1, which takes you through the audit of lessees.	Exam questions could focus on the financial reporting of leases. Alternatively there could be an audit issue relating to leases which gives rise to a financial	N/A
		Stop and think With lessors, the main concern for the auditor will be establishing whether the arrangement is an operating lease or a finance lease.	reporting issue. In addition, an ethical issue may arise whereby a company claims that a transaction is not a lease, or, where the company is a lessor, that the lease is an operating lease rather than a finance lease.	

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Identifying a lease

Section overview

- A contract to obtain the use of an asset is within the scope of IFRS 16 if it meets the definition of a lease contract.
- IFRS 16 provides optional exemptions from the application of the standard lessee accounting model for short-term leases and leases of low-value assets.
- A contract may contain both a lease component and a non-lease component. IFRS 16 requires entities to account for the lease component of the contract separately from the non-lease component.

Tutorial note

You have studied leasing at Professional Level. If you feel confident in your knowledge, you may wish to skim-read the explanations and go straight to the examples and questions, coming back to the detail if you get stuck.

1.1 Leasing agreements and IFRS 16

As you learned in your Professional level studies, one way by which businesses can obtain the use of an asset is by a leasing agreement. A contract has to meet the definition of a lease contract to be in the scope of IFRS 16, *Leases*.

IFRS 16 was published in 2016. IFRS 16 takes the approach of a lessee measuring the 'right of use' of an asset, and recognising it on the statement of financial position.

Definitions

Lease: A contract, or part of a contract, that conveys the right to use an asset (the underlying asset) for a period of time, in exchange for consideration.

Underlying asset: An asset that is the subject of a lease, for which the right to use that asset has been provided by a lessor to a lessee.

In a leasing transaction, there is a contract between the lessor and the lessee for the hire of an asset.

- The **lessor** is owner and supplier of the underlying asset.
- The **lessee** is the entity who has the right to use the underlying asset.

The lessor retains legal ownership but transfers to the lessee the right to use the asset for an agreed period of time in return for specified payments.

Definition

Right-of-use asset: An asset that represents a lessee's right to use an underlying asset for the lease term.

Professional skills focus: Structuring problems and solutions

The below guidance provides a structure for identifying whether an arrangement is a lease. You should expect that at Advanced Level the issue may not be clear cut, so the guidance will provide a useful tool in making the decision.

1.2 Identifying a lease

Under IFRS 16, a contract is deemed to contain a lease if it conveys the right to control the use of an **identified asset** for a **period of time**, in return for consideration (usually cash, although this purposely doesn't restrict the definition).

The right to control the use of the underlying asset depends on the lessee having both:

- the right to obtain substantially all of the **economic benefits** from using the asset; and
- the **right to direct the use** of the identified asset.

Note: If the supplier can substitute the asset for an equivalent during the lease, the lessee is not deemed to have the right to use an identified asset.

Identified asset

- The asset must be specified in the lease.
- May be part of an asset (such as part of an office facility).
- If the supplier has the practical ability to substitute the asset for an equivalent during the lease, and would benefit economically from doing so (ie, they have a substantive substitution right), there is no identified asset and the contract is not a lease.

Period of time is either:

- a period of time, such as years; or
- the amount of use, based upon the production of the asset, such as number of units produced.

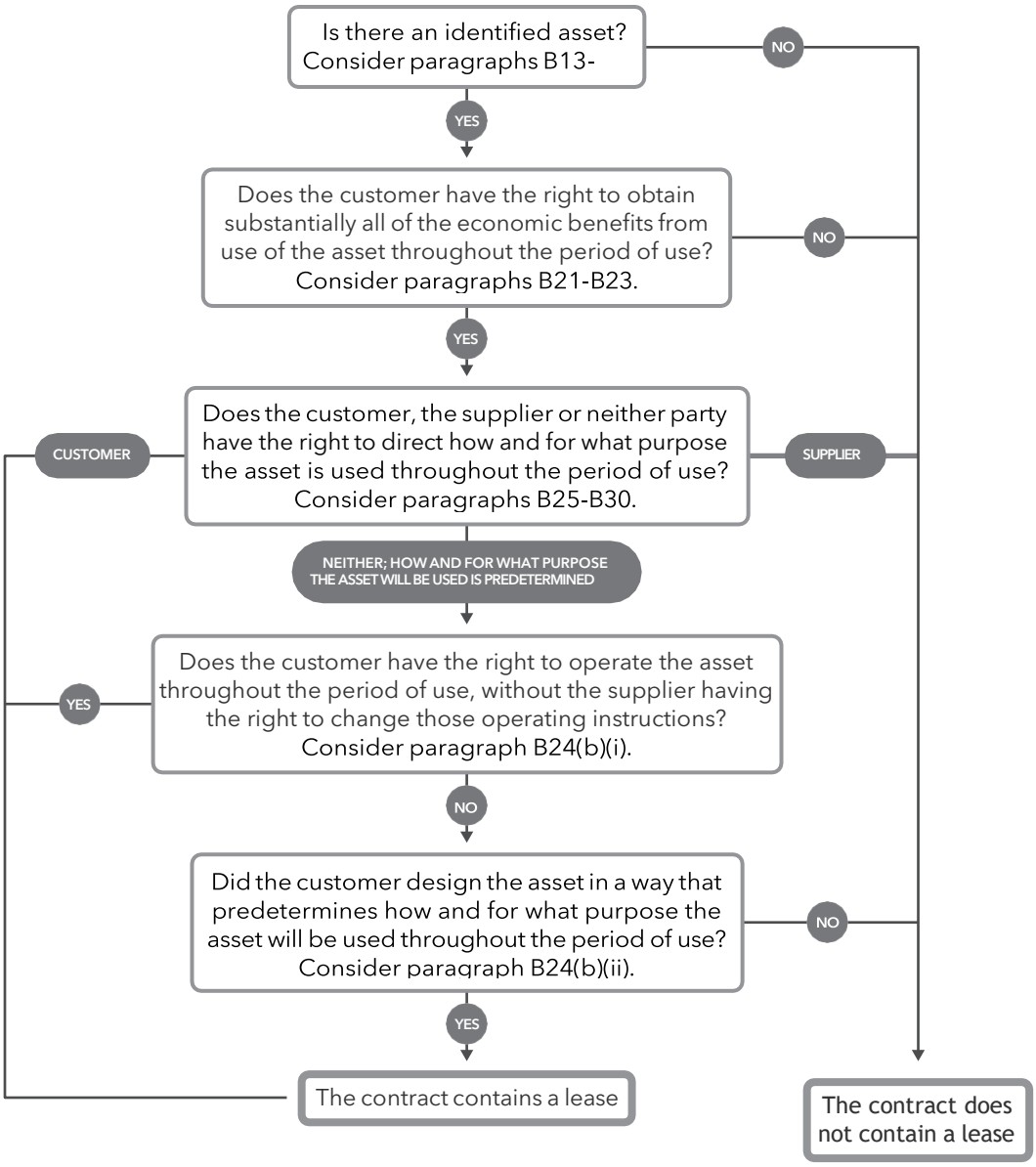
Right to obtain substantially all of the economic benefits

- Can be achieved through holding, using or sub-leasing the asset.
- Assessed only within the parameters of the contract eg, a lease contract for a car may restrict its use to within one country, in which case only economic benefits within that country are considered.

Right to direct the use of the asset (control)

- Lessee can direct how and for what purpose the asset is used throughout the period of use.
- Assessment is unaffected by protective rights within the terms of the contract that are designed to protect the asset eg, prohibiting a truck from being used to carry explosives.

The following flowchart, taken from IFRS 16, Appendix B, paragraph B31, may assist you in determining whether a lease should be identified in the examples that follow:



Worked example: Is it a lease 1?

Big Farm Ltd has three Fassey Mergason tractors under contract from Agrirental Ltd. The tractors are leased for five years, during which time Big Farm Ltd uses and keeps them on the farm, and they cannot be retrieved by Agrirental Ltd (except in the case of default of payment). Agrirental Ltd will supply a replacement tractor if any of the three requires servicing or repair. The tractors are returned to Agrirental Ltd after five years.

Requirement

Is this a lease?

Solution

This is a lease. There are **identifiable assets** (the three tractors), and Big Farm Ltd has the **right to direct their use** for the period of the contract in order to obtain **substantially all economic benefits** from them. Agrirental Ltd cannot substitute or swap one of the tractors, unless it

requires a repair or servicing; this is not a substantive substitution right. Therefore, the contract is classified as a lease.



Worked example: Is it a lease 2?

Big Farm Ltd also has the use of two combine harvesters under contract from Agrirental Ltd. The contract agrees that Big Farm Ltd will have access to combine harvesters which have VARIO cutters, however, Agrirental Ltd can decide on the brand and specific machine supplied to Big Farm Ltd, albeit with VARIO cutters. As harvesting is not a year round activity, Agrirental Ltd retains the combine harvesters at its depot, but will supply them as requested within 24 hours.

Requirement

Is this a lease?

Solution

This is not a lease. There are **no identifiable assets** (the combine harvesters), and although Big Farm Ltd has the **right to use** the machines for the period of the contract, Agrirental Ltd can decide upon the actual combines (provided they have the VARIO cutters) provided to Big Farm Ltd.

Therefore the contract is not classified as a lease, and the rental payments should be expensed to profit or loss.



Professional skills focus: Assimilating and using information

In the above examples, the background is the same and some of the details are similar. Your task is to take account of all the information and focus on the differences.



Worked example: Is it a lease 3?

Outandabout Co provides tours round places of interest in the tourist city of Sightsee. While these tours are mainly within the city, it does the occasional day trip to visit tourist sites further away. Outandabout Co has entered into a three-year contract with Fastcoach Co for the use of one of its coaches for this purpose. The coach must seat 50 people, but Fastcoach Co can use any of its 50- seater coaches when required.

Requirement

Is this arrangement a lease within the scope of IFRS 16, *Leases*?

Solution

This is not a lease. There is no identifiable asset. Fastcoach can substitute one coach for another, and would derive economic benefits from doing so in terms of convenience. Therefore Outandabout should account for the rental payments as an expense in profit or loss.

Worked example: Is it a lease 4?

Coketown Council has entered into a five-year contract with Carefleet Co, under which Carefleet Co supplies the council with 10 vehicles for the purposes of community transport. Carefleet Co owns the relevant vehicles, all 10 of which are specified in the contract. Coketown Council determines the routes taken for community transport and the charges and eligibility for discounts. The council can choose to use the vehicles for purposes other than community transport. When the vehicles are not being used, they are kept at the council's offices and cannot be retrieved by Carefleet unless Coketown Council defaults on payment. If a vehicle needs to be serviced or repaired, Carefleet is obliged to provide a temporary replacement vehicle of the same type.

Requirement

Is this a lease?

Solution

This is a lease. There is an identifiable asset, the 10 vehicles specified in the contract. The council has a right to use the vehicles for the period of the contract. Carefleet Co does not have the right to substitute any of the vehicles unless they are being serviced or repaired. Therefore Coketown Council would need to recognise an asset and liability in its statement of financial position.

Worked example: Is it a lease 5?

This example is taken from IFRS 16 illustrative example 3.

Kabal enters into a 10-year contract with a utilities company (Telenew) for the right to use three specified, physically distinct dark fibres within a larger cable connecting North Town to South Town. Kabal makes the decisions about the use of the fibres by connecting each end of the fibres to its electronic equipment (ie, Kabal 'lights' the fibres and decides what data, and how much data, those fibres will transport). If the fibres are damaged, Telenew is responsible for the repairs and maintenance. Telenew owns extra fibres, but can substitute those for Kabal's fibres only for reasons of repairs, maintenance or malfunction (and is obliged to substitute the fibres in these cases).

Requirement

Is this a lease?

Solution

This is a lease. The contract contains a lease of dark fibres. Kabal has the right to use the three dark fibres for 10 years.

There are three identified fibres. The fibres are explicitly specified in the contract and are physically distinct from other fibres within the cable. Telenew cannot substitute the fibres other than for reasons of repairs, maintenance or malfunction (IFRS 16: Para B18).

Kabal has the right to control the use of the fibres throughout the 10-year period of use because:

(1) Kabal has the right to obtain substantially all of the economic benefits from use of the fibres over the 10-year period of use and Kabal has exclusive use of the fibres throughout the period of use.

(2) Kabal has the right to direct the use of the fibres because IFRS 16: Para B24 applies:

  (a) The customer has the right to direct how and for what purpose the asset is used during the whole of its period of use, or

(b) The relevant decisions about use are pre-determined and the customer can operate the asset without the supplier having the right to change those operating instructions.

Kabal makes the relevant decisions about how and for what purpose the fibres are used by deciding (a) when and whether to light the fibres and (b) when and how much output the fibres will produce (ie, what data, and how much data, those fibres will transport). Kabal has the right to change these decisions during the 10-year period of use.

Although Telenew's decisions about repairing and maintaining the fibres are essential to their efficient use, those decisions do not give Telenew the right to direct how and for what purpose the fibres are used. Consequently, Telenew does not control the use of the fibres during the period of use.



Professional skills focus: Concluding, recommending and communicating

The above example, taken from IFRS 16, analyses a complex situation and comes to a conclusion. In an exam, if you missed one of the details and came to a different conclusion, you would still gain credit for parts of your analysis that were correct.

1.3 Your exam

IFRS 16 was introduced because lease liabilities had been kept off the statement of financial position. There is still an incentive to do this, and preparers of financial statements who wish to do this may argue that a transaction is not a lease when in fact it is, and therefore should appear on the SOFP. An exam question could present an ethical dilemma, where the auditor or junior accounting staff member is under pressure to accept this incorrect financial reporting treatment.

1.4 Recognition exemptions

IFRS 16 provides optional exemptions from the application of the standard lessee accounting model. Instead of applying the recognition requirements of IFRS 16 described below, a lessee may **elect to** account for lease payments as an expense on a straight-line basis over the lease term or another systematic basis for the following two types of leases (IFRS 16: paras. 5, 6 and 8). These are available for:

- **Short term leases,** which are those for less than 12 months that do not contain a purchase option. The election to use the exemption must be made by asset class.
- **Leases for low value assets,** being assets with a low value when new (such as laptop computers, mobile phones or small items of office furniture). An asset can only be of low value if:
  - The lessee can benefit from using the underlying asset on its own or with other available resources; and
  - The underlying asset is not highly dependent on or interrelated with other assets, for example, a circuit board costing £200 but which is required to operate an item of plant worth £40,000.

This election may be made on a lease-by-lease basis.

Leases of low-value assets are leases of assets with a value, when new, of $5,000 or less (IFRS 16, Basis for Conclusions, para. 100).

Interactive question 1: Short lease

Oscar Co is preparing its financial statements for the year ended 30 June 20X6. On 1 May 20X6, Oscar made a payment of £32,000 for an eight-month lease of a milling machine.

Requirement

How should the transaction be recognised in the financial statements? See **Answer** at the end of this chapter.

Context example: Leases of low-value assets and portfolio application

IFRS 16 Illustrative Example 11 is of a lessee in the pharmaceutical manufacturing and distribution industry with leases including the following:

(a) Leases of IT equipment for use by individual employees (such as laptop computers, desktop computers, hand held computer devices, desktop printers and mobile phones)

(b) Leases of servers, including many individual modules that increase the storage capacity of those servers. The modules have been added to the mainframe servers over time as the lessee has needed to increase the storage capacity of the servers

(c) Leases of office equipment:

  (1) Office furniture such as desks, chairs and partitions

  (2) Water dispensers

The company determines that the leases of IT equipment to individual employees and the office furniture and water dispensers qualify as leases of low-value assets on the basis that the underlying assets, when new, are individually of low value.

However, although each module within the servers, if considered individually, might be an asset of low value, the leases of modules within the servers do not qualify as leases of low-value assets. This is because each module is highly interrelated with other parts of the servers. The lessee would not lease the modules without also leasing the servers. Accordingly, the company would apply the recognition and measurement requirements of IFRS 16 to the servers.

Interactive question 2: Is it a lease?

Consider whether the following meet the criteria to be classified as leases.

Question	Fill in your answer
A company leases a laptop computer for their financial controller over a three-year period.	
A company leases a photocopier. The copier will remain on the client site, and will only be returned to the lessor in the event of a major repair.	
A company leases cars for their team of sales representatives for a three-year period. The cars are required to be estate models, but from time to time, the car may be changed by the lessor.	
A company acquires three architect's drawing desks which are paid for over 10 months. The company already has four of these desks, and no elections have been made.	

See **Answer** at the end of this chapter.

1.5 Separating components of a contract

A contract may contain both a lease component and a non-lease component. In other words, it may include an amount payable by the lessee for activities and costs that do not transfer goods or services to the lessee (IFRS 16: Para B33). These activities and costs might, for example, include maintenance, repairs or cleaning.

IFRS 16 requires entities to **account for the lease component of the contract separately from the non-lease component.** The entity must split the rental or lease payment and:

- account for the lease component under IFRS 16; and
- account for the service element separately, generally as an expense in profit or loss.

The consideration in the contract is **allocated on the basis of the stand-alone prices** of the lease component(s) and the non-lease component(s).



Context example: Separating components of a contract

Livery Co leases a delivery van from Bettalease Co for three years at £12,000 per year. This payment includes servicing costs.

Livery could lease the same make and model of van for £11,000 per year and would need to pay £2,000 a year for servicing.

Livery Co would allocate £10,154 (£12,000 × £11,000 ÷ £(11,000 + 2,000)) to the lease component and account for that as a lease under IFRS 16.

Livery Co would allocate £1,846 (£12,000 × £2,000 ÷ £(11,000 + 2,000)) to the servicing component and recognise it in profit or loss as an expense.

CHAPTER 14

2 Lessee accounting



Section overview

- IFRS 16 recognises an asset and corresponding liability for all leases (unless the lease is for less than 12 months, or the underlying asset is of low value).
- A lease liability is initially recognised at the present value of future lease payments; interest accrues subsequently.
- A right-of-use asset is initially recognised at the amount of the lease liability, subject to some adjustments; it is subsequently depreciated.
- In the statement of financial position, a right-of-use asset and lease liability are recognised in respect of leased assets.
- Lease payments reduce the lease liability and cover any accrued interest.
- Interest accrues on the outstanding liability in each period of the lease.
- The finance charge is recognised in profit or loss.
- Where a contract is a lease, a right-of-use asset and a lease liability are recognised at the commencement date.

2.1 Initial measurement of the right-of-use asset

The right-of-use asset is measured at cost, which includes:

- the amount of the initial measurement of the lease liability;
- any lease payments made at or before the commencement date, less any lease incentives received;
- any initial direct costs incurred by the lessee related to the asset; and
- an estimation of any costs or penalties to be incurred by the lessee at the end of the lease term, such as dismantling costs or removal costs.



Worked example: Initial measurement of the right-of-use asset

Beta Ltd commenced a five-year lease to acquire plant on 1 January 20X8. Beta Ltd has paid an initial deposit of £800, and received lease incentives of £250. At the commencement date, the present value of the future lease payments was £12,500. Beta Ltd has incurred legal costs of acquiring the lease of £1,200. The plant will produce a new product, the XR3, and marketing costs for the new product have been incurred of £300.

At the end of the lease, Beta Ltd will have to pay £750 (at present value) to remove the plant.

Requirement

What is the initial measurement of the right-of-use asset?

Solution

Right-of-use asset

	£
PVFLP	12,500
Initial deposit	800
Direct costs	1,200
End of term costs to remove the plant	750
Less: lease incentives	(250)
Right-of-use asset	15,000

Note: The marketing costs for the new product to be made by the plant are not included as these costs are not directly attributable to the lease. This is consistent with IAS 16 (s.19(b)), where marketing costs cannot be capitalised as part of the non-current asset.

2.2 Subsequent measurement of the right-of-use asset

Subsequently, the right-of-use asset will be measured at cost less accumulated depreciation and impairment losses in line with IAS 16, *Property, Plant and Equipment*.

The right-of-use asset will be depreciated from the commencement date to the **earlier** of the end of its useful life or the end of the lease term **unless** the asset is expected to be transferred to the lessee at the end of the lease term. In that case, the asset will be depreciated over the useful life of the asset.

Worked example: Subsequent measurement of the right-of-use asset

Taking the information from the previous example, Beta has estimated that the useful life of the plant will be six years. There is no option to purchase at the end of the lease term.

Requirement

What is the carrying amount of the right-of-use asset at 31 December 20X8?

Solution

Right-of-use asset is initially measured at £15,000.

The depreciation period is the shorter of the useful life (six years) or the lease term (five years).

£15,000/5 years = £3,000 depreciation charge.

Carrying amount of the right-of-use asset at 31 December 20X8 is £12,000 (£15,000 – £3,000).

2.3 Depreciating the asset

DEBIT	Depreciation expense	The asset should be depreciated over the shorter of the lease term or the asset's useful life
CREDIT	Accumulated depreciation	

The asset should be depreciated over the shorter of the lease term or the asset's useful life.

Points to note

- Depreciation policies adopted should be consistent with other non-current assets of the same class.
- If any impairment reviews are required, these must be conducted in accordance with IAS 36, *Impairment of Assets*.
- If there is reasonable certainty that the lessee will eventually own the asset, then it should be depreciated over its estimated useful life.
- The lease term comprises the period for which the lessee has contracted to lease the asset and any further terms for which there is reasonable certainty at the inception of the lease that the lessee will exercise the option.

2.4 Initial measurement of the lease liability

At the commencement of the lease, the lease liability is measured at the **present value** of **future lease payments** including any payments expected at the end of the lease.

Lease liability will include:

- fixed payments less any lease incentives receivable;
- variable lease payments that depend on an index (eg, the consumer price index) or rate (eg, market rental rates);
- amounts expected to be payable by the lessee under residual value guarantees;
- purchase options (if reasonably certain to be exercised); and
- penalties for early termination.

The discount rate used is the interest rate implicit in the lease, or if this is not available, the lessee's incremental borrowing rate.

Definition

Lease payments: Payments made by a lessee to a lessor in order to use an underlying asset during the lease term, less any lease incentives.

IFRS 16 requires lease payments to also include the exercise price of a purchase option if the lessee is reasonably certain to exercise that option, and any penalty payments for terminating the lease.

Context example: Present value of future lease payments

Alpha Ltd has commenced a four-year lease on 1 January 20X8. Payments are £10,000 p.a. in arrears and the interest rate implicit in the lease is 10%. The present value of the future lease payments can be calculated using the PV function in excel. The PV excel function is used to calculate the present value (PV) of a series of equal cash flows (annuities). Type **=PV(** to begin the function entry then insert the rate of return required (this is the interest rate implicit in the lease), the number of periods and the payment made in each period, finally insert a closed bracket).

B4 =PV(B1,B2,B3)

	A	B
1	Rate	0.10
2	Number	4
3	payment	10,000
4	Present value of future lease payments	31,698

Therefore, at 1 January 20X8, the lease liability is measured at £31,698.

Professional skills focus: Structuring problems and solutions

Once an excel file like the one in the above example has been set up, it can be used repeatedly with different rates, lease periods and payment amounts. The structure is there and just needs to be populated with details.

2.5 Interest rate implicit in the lease

You can also use an excel file to calculate the interest rate implicit in the lease from a given payments schedule and the present value of the future lease payments.

2.6 Subsequent measurement of the lease liability

In future periods, the lease liability is amortised.

- Interest will accrue on the outstanding lease, at the rate stated in the lease contract.
- Payments made in respect of the lease by the lessee will reduce the outstanding liability.

Context example: Context example: Year 2 lease liability

Taking the example of Alpha Ltd in the previous example.

In the previous example, Alpha Ltd had calculated the initial lease liability to be £31,698 on 1 January 20X8.

Year	£
1.1.X8: Lease liability (PVFLP)	31,698
31.12.X8: Interest 10%	3,170
31.12.X8: Instalment in arrears	(10,000)
Liability at 31.12.X8	24,868

Therefore, at 1 January 20X9, the lease liability will be £24,868.

2.7 Setting up accounts in the statement of financial position

IFRS 16 requires that, when an asset is leased, the accounting treatment should reflect the **substance of the transaction**. In the lessee's books therefore:

DEBIT	Right-of-use asset account	The amount to be recorded in this way is the present value of the future lease payments
CREDIT	Payables: Lease liabilities	

The **initial** deposit, if any, counts as one of the lease payments and hence is included in the cost of the asset.

Points to note

- The entries are made at the **commencement** of the lease term, with the **values** determined at the start of the lease (see sections above).
- The present value of the future lease payments is derived by discounting them at the **interest rate implicit in the lease**. If it is not practicable to determine the interest rate implied in the lease, then the lessee's **incremental borrowing rate** can be used.
- Initial direct costs can be treated as part of the cost of the asset – provided they are directly attributable to activities performed by the lessee to obtain the lease.
- Although interest is payable under the lease, this is accrued over time. The justification is that the capital could, in theory, be paid off at any time, with cancellation charges. These charges could be avoided, so they are not a 'true' long term liability. **Interest is therefore recognised as it accrues**.

2.8 Making the Payment

Every period, the payments are accounted for as follows.

DEBIT	Payables: Lease liabilities	Each lease payment is comprised partly of a repayment of capital and partly of an interest charge for the period.
CREDIT	Cash	

2.9 Finance charge

The finance charge is dealt with as follows.

DEBIT	Profit or loss: Finance cost	With the amount of the interest accrued over the period
CREDIT	Payables: Lease liabilities	

2.10 Allocating the interest charge to accounting periods

IFRS 16 requires the total finance charge to be allocated to each period during the lease term so as to produce a **constant periodic rate of interest** on the outstanding lease obligation. As the lessee pays off the capital sum, the total capital owed falls from period to period. You would therefore also expect a reduction in the total interest payable too, on the outstanding balance.

For example, if you owe £10,000 and pay 15%, the interest will be £1,500. After you have paid off, say, £8,000 of the capital, interest would be £300 (on £2,000). The monthly payments remain the same, but the mix of interest and capital changes over the life of the loan.

Therefore, in the earlier years, the finance charge (interest) is a higher proportion of the annual lease payments. Towards the end of the lease, the finance charge will be smaller as the outstanding lease liability is smaller.

2.11 Instalments in arrears



Worked example: Lease payments in arrears

A Ltd has a year-end of 31 December.

A lease commences on 1 January 20X1. Lease payments comprise three payments of £10,000 annually, commencing on 31 December 20X1.

There is no transfer of asset at the end of the lease and no purchase option. The implicit interest rate is 10%.

Requirement

You are required to calculate the interest charge and the year-end liability for each year of the lease. Round your discounted amounts to the nearest £10.

Solution

The lease liability is measured at the present value of the three payments:

B4 =PV(B1,B2,B3)

	A	B
1	Rate	0.10
2	Number	3
3	Payment	10,000
4	Present value of future lease payments	24,869

	£
20X1 balance b/f	24,869
Interest 10%	2,487
Payment 31/12/X1	(10,000)
20X2 balance b/d	17,356
Interest 10%	1,736
Payment 31/12/X2	(10,000)
20X3 balance b/d	9,092
Interest 10%	909
Payment 31/12/X3	(10,000)

Note: The balance doesn't come down to exactly the outstanding amount due to rounding.

2.12 Instalments in advance

As we have seen in the examples above, interest accrues over time and is included in the payment at the end of each period of borrowing. However, where instalments are **paid in advance**:

- The first instalment **repays capital only** as no time has yet elapsed for interest to accrue.
- At the end of each accounting period the year-end liability will include **capital and interest** that has accrued to date but which has not been paid.
- The initial payment in advance (and/or any deposit) is **not included in the calculation of the lease liability. It is added to the right-of-use asset.**

Worked example: Lease payments in advance

[This is based on IFRS 16 Illustrative example 13.]

A lessee enters into a five-year lease of a building which has a remaining useful life of 10 years. Lease payments are £50,000 per annum, payable at the beginning of each year.

The lessee incurs initial direct costs of £20,000 and receives lease incentives of £5,000. There is no transfer of the asset at the end of the lease and no purchase option.

The interest rate implicit in the lease is not immediately determinable but the lessee's incremental borrowing rate is 5%.

Requirement

Calculate the initial lease liability and calculate the value of the right-of-use asset.

Solution

At the commencement date the lessee pays the initial £50,000, incurs the direct costs and receives the lease incentives. The lease liability is measured at the present value of the **remaining four payments**, ie, the **future lease payments**:

B4 =PV(B1,B2,B3)

	A	B
1	Rate	0.05
2	Number	4
3	Payment	50,000
4	Present value of future lease payments	177,297

Assets and liabilities will initially be recognised as follows:

		Debit	Credit
		£	£
Right-of-use asset:			
Initial payment	50,000		
Discounted liability	177,297		
Initial direct costs	20,000		
Incentives received	(5,000)		
		242,297	

Lease liability			177,297
Cash	(50,000 + 20,000 – 5,000)		65,000
		242,297	242,297

2.13 Disclosure: lease liability

IFRS 16 requires that there is presentation in the statement of financial position for the following items:

- the **right-of-use asset**
- the **lease liability**.

Right-of-use assets may be presented as a separate line item in the statement of financial position under the heading non-current assets. Alternatively, they may be included in the total of the relevant class of assets, and in this case the carrying amount of right-of-use assets is disclosed in the notes to the financial statements.

IFRS 16 makes no specific requirement to split lease liabilities between non-current and current liabilities. However, in accordance with the *Conceptual Framework's* fundamental principle of faithful representation, they should be presented in this way as best practice, so that the users of the financial statements can understand the impact of the lease liability in future years.

Interactive question 3: Payments in arrears

Using the facts from the Worked Example: Lease payments in arrears, show the presentation of the lease liability at the end of 20X1.

The lease liability working from the Worked example was as follows:

	£
20X1 balance b/f	24,869
Interest 10%	2,487
Payment 31/12/X1	(10,000)
20X2 balance b/d (current period)	17,356
Interest 10%	1,736
Payment 31/12/X2	(10,000)
20X3 balance b/d (future periods)	9,092
Interest 10%	909
Payment 31/12/X3	(10,000)

Total lease liability at 31 December 20X1 = £ ____

Capital > 1 year = £ ____

< 1 year (β) = £ ____

3.2 Prepare journals to show accounting entries in respect of the lease during 20X1.

See **Answer** at the end of this chapter.

Interactive question 4: Payments in advance

Using the facts from the Worked Example: Lease payments in advance, show the split of the lease liability at the end of year 1.

See **Answer** at the end of this chapter.

2.14 Other disclosures

Note: The leased assets and lease liabilities should not be netted off against each other.

Non-current assets

- Disclosure must be made of the carrying amount of right-of-use assets, by class of underlying asset, either on the face of the statement of financial position, or in a note.
- Depreciation charge for right-of-use assets
- Additions to right-of-use assets
- Carrying amount of right-of-use assets at the end of the reporting period by class of underlying asset
- Gains/losses resulting from any sale and leaseback transactions (see section 3)

Liabilities

- Interest expense on lease liabilities

Other disclosures

- In the case of most entities, the IAS 1, *Presentation of Financial Statements* requirement to disclose significant accounting policies would result in the disclosure of the policy in respect of leases.
- Expenses relating to short-term and low-value leases
- Total cash outflow for leases

Interactive question 5: Summary

Summer Ltd leases an asset on 1 January 20X1. The terms of the lease are to pay a non-refundable deposit of £575 followed by seven annual instalments of £2,000 payable in arrears. The asset is expected to have a useful life of seven years.

The present value of the future lease payments (excluding the deposit) is £11,164. There is no option to purchase the asset at the end of the lease term.

The interest rate implicit in the lease is 6%.

Requirement

Calculate the value of the lease liability and the right-of-use asset at 31 December 20X1 and 31 December 20X2.

Note: (Round your answers to the nearest pound)

See **Answer** at the end of this chapter.

Professional skills focus: Applying judgement

IFRS 16 has tightened the rules on leasing to the extent that there is limited scope for judgement. The standard was brought in to remedy abuses and loopholes. However, there remains some scope, for example in electing to show potentially exempt leases in the statement of financial position, or in the estimation of the useful economic life of the underlying asset.

3 Sale and leaseback transactions

Section overview

- A sale and leaseback transaction involves the sale of an asset and the subsequent leasing back of the same asset.
- The treatment of a sale and leaseback transaction will depend on whether the transfer is a sale as defined by IFRS 15, *Revenue from Contracts with Customers*.

3.1 Introduction

Companies can raise finance in a number of different ways. These include short-term measures such as a bank overdraft, medium-term measures such as loans or leases and longer-term measures including secured loans.

Another option is a **sale and leaseback transaction**. This is a common feature of certain industries, including retailing and hotels. It involves **the original owner of the asset selling it**, typically to a finance house or bank, **and immediately leasing it back**, thereby raising cash and retaining the use of the asset. Such arrangements provide entities with the opportunity to release capital caught up in the business for investment in other opportunities or to return it to shareholders. In essence, an entity acquires cash in exchange for a commitment to make regular lease payments without losing use of the asset.

Under IFRS 16, the key initial assessment is whether or not the transfer of the asset constitutes a genuine sale. For this to be the case, the requirements of IFRS 15, *Revenue from Contracts with Customers* for determining when a performance obligation is satisfied must be met.

3.2 IFRS 15 criteria for a sale

The transfer of an asset qualifies as a sale if the customer obtains **control** of the asset. In determining this, the following should be considered (IFRS 15, para. 38):

(a) Whether the selling entity has a present right to payment for the asset

(b) Whether the selling entity has transferred legal title to the asset

(c) Whether the selling entity has transferred physical possession of the asset

(d) Whether the selling entity has transferred the risks and rewards of ownership of the asset to the customer

(e) Whether the purchaser has accepted the asset

Criterion 4, relating to the risks and rewards of ownership, is likely to be the most important of the above criteria in considering whether control has passed to the customer and therefore whether the IFRS 15 criteria for a sale have been satisfied.

Context example: Transfer is a sale (1)

[This is based on IFRS 16 Illustrative example 13.]

A lessee enters into a five-year lease of a building which has a remaining useful life of 10 years. Lease payments are £50,000 per annum, payable at the beginning of each year.

If the transfer of the asset is a sale:

Stage 1

The seller/lessee measures the **right-of-use asset arising from the leaseback** as a proportion of the previous carrying amount of the asset that relates to the right of use retained by the seller/lessee:

Carrying amount × (PV of future lease payments at transfer date ÷ Fair value of asset at transfer date)

Stage 2

The seller/lessee only recognises the **amount of any gain or loss** on the sale that relates to the **rights transferred to the buyer**.

Calculating the gain or loss on sale:

WORKINGS

Calculate the total gain on the sale:

	£
Fair value	X
Less: carrying amount	(X)
Total gain/(loss) on the sale	X/(X)

Then **determine the gain** that relates to the rights retained

Gain × (PV of future lease payments ÷ Fair value of asset at transfer date) = Gain related to rights retained

Stage 3

The gain relating to the rights transferred is the balancing figure:

Gain on rights transferred = total gain (W1) – gain on rights retained (W2)

The right-of-use asset continues to be depreciated as normal, although there may be a revision of the useful life required (see 2.3).

Worked example: Transfer is a sale (2)

Hill plc owned a machine with a carrying amount of £5,040,000 on 1 July 20X5. On this date Hill plc sold the property to Dale plc for £7,920,000, which was the fair value of the machine, and then undertook to lease it back under a 20-year lease. Annual lease payments were £240,000 in arrears and the present value of future lease payments at the 6% interest rate implicit in the lease was £2,752,800 at 1 July 20X5.

The machine was transferred to Dale plc's premises, but Hill plc has the right to use it. Hill plc has no option to repurchase the machine at the end of the lease period.

Requirement

How should the transaction be accounted for by Hill plc in the year ended 30 June 20X6?

Solution

The fact that the machine has been transferred to the new lessor's premises, and the fact that Hill plc has no right to repurchase the machine both indicate that control has transferred to Dale plc, and that this qualifies as a sale under IFRS 15.

Part of the carrying amount of the asset is allocated to be a right-of-use asset retained. This is calculated based on the right-of-use asset (lease liability) as a proportion of fair value:

Right of use asset = £5,040,000 × (2,752,800 ÷ 7,920,000) = £1,751782

The remaining carrying amount of £3,288,218 (5,040,000 – 1,751,782) represents the transferred asset.

The overall gain on disposal is £2,880,000 (£7,920,000 – 5,040,000); only that part of the gain relating to the transferred asset is recognised:

Gain relating to retained rights = £2,880,000 × (2,752,800 ÷ 7,920,000) = £1,001,018

Therefore the recognised gain relating to the transferred rights is £1,878,982 (2,880,000 – 1,001,018).

At 1 July 20X5, the following entries are required:

DEBIT	Right-of-use asset	£1,751,782	
DEBIT	Bank	£7,920,000	
CREDIT	PPE		£5,040,000
CREDIT	Gain on disposal		£1,878,982
CREDIT	Lease liability		£2,752,800

The right-of-use asset is subsequently depreciated over the lease term of 20 years; therefore in the year ended 30 June 20X6:

DEBIT	Depreciation expense (1,751,782/20)	£87,589	
CREDIT	Right-of-use asset		£87,589

The lease liability is amortised:

	£
1 July 20X5	2,752,800
Interest at 6%	165,168
Lease payment	(240,000)
28 June 20X6	2,677,968

Amortisation for the year ended 30 June 20X6 is recognised by: £87,589

DEBIT	Finance charge	£165,168	
DEBIT	Lease liability	£74,832	
CREDIT	Bank		£240,000



Interactive question 6: Sale and leaseback where the transfer is a sale

Frayn plc sold an asset (with a carrying amount at 31 December 20X0 of £70,000) and entered into a sale and leaseback arrangement on 1 January 20X1, when:

- the sale proceeds were at fair value of £120,000
- the remaining useful life of the asset was four years

The lease provided for five annual rentals of £25,000 payable in arrears on 31 December of each year. The interest rate implicit in the lease was 5% and the present value of the future lease payments was £108,225.

Requirement

Set out the journal entries at the date of disposal and calculate the amounts to be recognised in profit or loss in the year to 31 December 20X1 and in the statement of financial position at that date.

See **Answer** at the end of this chapter.



Professional skills focus: Structuring problems and solutions

Sale and leaseback is a clear illustration of the importance of structuring your approach. Use the three-stage process above to focus your answer.

3.3 Transfer is not a sale

If the transfer is not a sale under IFRS 15, *Revenue from Contracts with Customers*, ie, the performance obligation is not satisfied, then the seller/lessee will continue to recognise the transferred asset and also recognises a financial liability equal to the transfer proceeds, accounting for it using IFRS 9, *Financial Instruments*.

The treatment is reversed in the books of the buyer/lessor:

Seller/lessee	Buyer/lessor
Continue to recognise the transferred asset	No physical asset recognised
Recognise a financial liability measured at transfer proceeds	Recognise a financial asset measured at transfer proceeds



Worked example: Transfer is a not a sale

Green plc has a year end of 31 December. The company owned a machine with a carrying amount of £720,000 on 1 January 20X3. On this date Green plc sold the machine to Blue Bank plc for £480,000, and then undertook to lease it back under a five-year lease. The asset is retained at Green plc's premises and the company continues to use it. Green plc is also responsible for insuring the machine.

The annual rental is £120,000 payable in arrears and the interest rate implicit in the lease is 8%.

Requirements

1 Show how this transaction should be accounted for by Green plc.

2 Show the journal entries in relation to this transaction for the year ended 31 December 20X3.

Solution

1 Green plc retains the machine at its premises and continues to use it. Green plc also insures the machine, which means that the risks and rewards of ownership have not been transferred to the customer. Both these matters indicate that the buyer has not obtained

control of the machine and therefore that the transfer is not a sale in accordance with IFRS 15.

Therefore on 1 January 20X3 the sale proceeds are recognised as a financial liability as follows:

At the date of sale/inception of the lease

DEBIT	Cash	£480,000	
CREDIT	Financial liability		£480,000

The liability is accounted for in line with IFRS 9 at amortised cost (because it is not held for trading):

Year ended	B/f	Interest at 8%	Payment	C/f
	£	£	£	£
31 December 20X3	480,000	38,400	(120,000)	398,400
31 December 20X4	398,400	31,872	(120,000)	310,272
31 December 20X5	310,272	24,822	(120,000)	215,222
31 December 20X6	215,222	17,218	(120,000)	112,439
31 December 20X7	112,439	8,995*	(120,000)	–

* rounding difference £1,434

2 In the year ended 31 December 20X3, the amortisation of the liability is recognised as follows:

At the date of sale/inception of the lease

DEBIT	Financial liability	£81,600	
DEBIT	Finance charge	£38,400	
CREDIT	Bank		£120,000

3.4 Transfer is a sale not on market terms

If the fair value of the consideration for the sale does not equal the fair value of the asset, or if the lease payments are not at market rates, adjustments should be made to the accounting entries as this is deemed to be a method of additional financing:

Below market terms	Above market terms
Treat as a prepayment of lease payments	Treat as additional financing by the lessor/buyer
	Adjust the PVFLP by the excess of consideration over the fair value of the asset

In practice the consideration for the sale is much more likely to be above market terms, and this is the scenario that will be tested in the exam. IFRS 16 gives detailed guidance on this situation. A sale below market price is unlikely to happen, and may even be an indicator of fraud.

Worked example: Sale and leaseback not at market rates

Sydney plc has a year end of 31 December.

The company sold a building on 1 January 20X5 for £1.2 million. The fair value of the building was £1 million.

The carrying amount of the building at 31 December 20X4 was £900,000.

Sydney plc arranges to leaseback the building for a period of six years, and the remaining useful life of the building is 20 years.

The lease payments will be made in arrears starting 31 December 20X5 of £180,000 p.a. The implicit interest rate of the lease is 5%.

Requirement

Produce the journal entries for the financial statements as at 31 December 20X5 reflecting the above transactions.

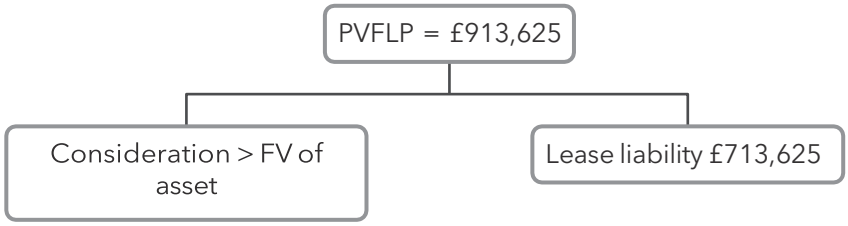
Solution

Step 1

Calculate the value of the lease liability

The present value of future lease payments is £913,625 (W1).

However, the consideration (£1,200,000) was in excess of the fair value (£1,000,000). This excess of £200,000 is considered as additional financing, therefore:



Step 2

Calculate the right-of-use asset

Carrying amount × (PVFLP ÷ FV) (adjusted for excess consideration)

£900,000 × (£713,625 ÷ £1,000,000) = £642,262

Initial measurement of the right-of-use asset is £642,262.

Depreciation will be charged over six years as this is the shorter of the useful life (20 years) and the lease term (six years).

Therefore, charge for year ended 31 December 20X5 is £642,262/6 = £107,044.

Step 3

Calculate the gain on sale and leaseback

The gain is calculated based on the fair value of the transferred asset rather than full proceeds. Sydney plc can only recognise the amount of gain relating to the rights transferred.

- Stage 1 (calculate the gain): £1,000,000 – £900,000 = £100,000
- Stage 2 (gain relating to rights retained): £100,000 × £713,625/£1,000,000 = £71,363
- Stage 3 (gain on the rights transferred): £100,000 – £71,363 = £28,637

Journal entry

		£	£
DEBIT	Cash	1,200,000	
DEBIT	Depreciation expense	107,044	
DEBIT	Right-of-use asset	642,262	
CREDIT	Building (carrying amount)		900,000
CREDIT	Accumulated depreciation		107,044
CREDIT	Lease liability		913,625
CREDIT	Gain on rights transferred		28,637
		1,949,306	1,949,306

WORKING

Lease payment	Discount	£
180,000	1.05^1	171,429
180,000	1.05^2	163,265
180,000	1.05^3	155,491
180,000	1.05^4	148,086
180,000	1.05^5	141,035
180,000	1.05^6	134,319
Total PVFLP		913,625



Interactive question 7: Sale and leaseback

On 1 April 20X2, Wigton Co bought an injection moulding machine for £600,000. The carrying amount of the machine as at 31 March 20X3 was £500,000. On 1 April 20X3, Wigton Co sold it to Whitehaven Co for £740,000, its fair value. Wigton Co immediately leased the machine back for five years, the remainder of its useful life, at £160,000 per annum payable in arrears. The present value of the annual lease payments is £700,000 and the transaction satisfies the IFRS 15 criteria to be recognised as a sale.

Requirement

What gain or loss should Wigton Co recognise for the year ended 31 March 20X4 as a result of the sale and leaseback?

See **Answer** at the end of this chapter.

3.5 *Variable lease payments in a sale and leaseback transaction that meets the IFRS 15 criteria to be a sale*

IFRS 16, Leases was amended in 2023 to address the issue of variable lease payments in a sale and leaseback transaction that meets the IFRS 15 criteria to be a sale. This amendment is effective from 1 January 2024. In some sale and leaseback transactions, all future lease payments are variable payments that do not depend on a rate or index. Since IFRS 16 does not include these payments in the measurement of a lease liability, the initial lease liability would be zero in this situation. It follows that when applying the sale and leaseback accounting described above, the underlying asset would be derecognised in its entirety and the whole gain or loss

on disposal would be recognised. This does not, however, provide useful information. The amendment therefore requires that future lease payments are determined (and so the lease liability measured) so that the seller (lessee) does not recognise a gain or loss relating to their retained right to use the underlying asset. This effectively means that variable payments that do not depend on a rate or index are included in the initial measurement of a lease liability in a sale and leaseback transaction. The recognised and unrecognised part of the gain or loss on disposal is then calculated in the normal way.

4 Lessor accounting



Section overview

- For **lessor accounting**, IFRS 16 **makes a distinction between finance leases and operating leases**.
- **Finance leases:** record the amount due to the lessor in the statement of financial position at the net investment in the lease, recognise finance income to give a constant periodic rate of return.
- **Operating leases:** record the asset as a non-current asset and depreciate over useful life, record income on a straight-line basis over the lease term.

Tutorial Note

You have not studied lessor accounting at Professional Level, so all the material in this section must be studied carefully.

4.1 Introduction

Several definitions are relevant to lessor accounting in particular, all from IFRS 16 Appendix A.



Definitions

Finance lease: A lease that transfers substantially all the risks and rewards incidental to ownership of an underlying asset.

Operating lease: A lease that does not transfer substantially all the risks and rewards incidental to ownership of an underlying asset.

Unguaranteed residual value: That portion of the residual value of the underlying asset, the realisation of which by the lessor is not assured or is guaranteed solely by a party related to the lessor.

Gross investment in the lease: The sum of:

(a) the lease payments receivable by the lessor under a finance lease; and

(b) any unguaranteed residual value accruing to the lessor.

Net investment in the lease: The gross investment in the lease discounted at the interest rate implicit in the lease.

Unearned finance income: The difference between:

(a) the gross investment in the lease; and

(b) the net investment in the lease.

4.2 Risks and rewards

The classification of leases is based on the extent to which risks and rewards incidental to ownership of a leased asset lie with the lessee or the lessor. When we talk of **risks** here, we specifically mean the risks of ownership, not other types of risk. Risks of **ownership** include the possibility of losses from idle capacity or technological obsolescence, or variations in return due to changing economic conditions. The **rewards** are represented by the expectation of profitable operation over the asset's economic life, and also any gain from appreciation in value or realisation of a residual value.

There are some specific examples that suggest a finance lease such as the lessee receives ownership of the asset at the end of the lease term, the lessee is expected to exercise the option to purchase, the lease term forms the majority of the asset's economic life, the asset is substantially loaned whereby the present value of lease payments is almost all of the fair value of the asset and/or the asset is specialised for use by the lessee or entities authorised by the lessee.

When a lease includes both **land and building** elements, the classification of each element as an operating or finance lease should be made separately. In determining whether the land element is an operating or finance lease, an important consideration is that land normally has an indefinite economic life.

4.3 Operating leases

A lessor continues to recognise an asset that is leased out by way of an operating lease, and charges depreciation over the useful life of the asset. Lease income from the operating lease is recognised on a straight-line basis over the lease term, unless another systematic and rational basis is more representative of the time pattern in which the benefit from the leased asset is receivable.

The cost of incentives provided to the lessee is recognised as a reduction of the rental income over the lease term, generally on a straight-line basis.

If a lessor incurs costs to negotiate and arrange an operating lease, these costs are capitalised within the carrying amount of the underlying asset that is the subject of the lease. They are amortised over the lease term on the same basis as lease income.



Worked example: Operating lease

At 1 April 20X3, SupaTruck plc owns a tow truck costing £75,000 with an economic life of 10 years. The company entered into a two years lease agreement to lease the truck to Dimble Recovery plc for £10,000 per year. SupaTruck continues to retain ownership of the tow truck and charges depreciation on a straight line basis, with no residual value.

Requirement

What journal entries should SupaTruck record in the year ended 31 March 20X4?

Solution

SupaTruck recognises a non-current asset at historical cost less depreciation at year-end. The depreciation charge is £75,000/10 years = £7,500 and rental income of £10,000 is also recognised with appropriate disclosures.

The required journal entries in the year ended 31 March 20X4 are (£):

		£	£
DEBIT	Depreciation expense	7,500	
CREDIT	PPE (accumulated depreciation)		7,500
and			
DEBIT	Cash/receivable	10,000	
CREDIT	Rental income		10,000

4.4 Finance leases

Although a lessor is the legal owner of an asset leased to another party under a finance lease, it has passed the risks and rewards of ownership to the lessee in return for a cash flow stream. As a result, the lessor derecognises the asset and instead recognises a receivable equal to the net investment in the lease. This is the present value of the aggregate of:

(a) the lease payments receivable by the lessor; and

(b) any unguaranteed residual amount accruing to the lessor.

An unguaranteed residual amount is that part of the residual value of a leased asset that is not assured or is guaranteed only by a party related to the lessor.

Initial direct costs incurred in negotiating and arranging the lease are added to the receivable.

Upon receipt of cash payments from the lessee, the amount received is allocated between the repayment of principal and finance income. The finance income should be recognised so as to generate a constant periodic rate of return on the outstanding net investment.



Worked example: Finance lease

Leighton Machinery plc leased a machine to a customer with effect from 1 January 20X4. The fair value of the machine at this date (equal to the net investment in the lease) was £14,875 and arrangement fees amounted to £125. The lease term was eight years and rentals of £2,750 are payable annually in advance. The interest rate implicit in the lease is 12.8%.

Requirement

What amounts are recognised in Leighton Machinery's financial statements in the year ended 31 December 20X4 in respect of the lease?

Solution

The receivable is initially recognised at £14,875 + £125 = £15,000.

Interest accrues to the receivable and payments reduce it as follows over the first two years of the term:

y/e	Balance b/f	Repayment	Balance c/f	Finance income	Balance c/f
	£	£	£	£	£
31.12.X4	15,000	(2,750)	12,250	1,568	13,818
31.12.X5	13,818	(2,750)	11,068	1,417	12,485

Therefore in the statement of financial position of Leighton Machinery, a current and non-current asset are recognised.

Non-current asset: Net investment in finance lease: £11,068 Current asset: Net investment in finance lease: £2,750

Interest receivable of £1,568 is recognised in the statement of profit or loss and other comprehensive income.

4.4.1 Summary

For lessor accounting, IFRS 16 makes a distinction between finance leases and operating leases.

5 IAS 20, *Accounting for Government Grants and Disclosure of Government Assistance*



Section overview

This section gives a very brief overview of the material covered in earlier studies.



Definitions

Government assistance: Action by government designed to provide an economic benefit specific to an entity or range of entities qualifying under certain criteria.

Government grants: Assistance by government in the form of transfers of resources to an entity in return for past or future compliance with certain conditions relating to the operating activities of the entity. They exclude those forms of government assistance which cannot reasonably have a value placed upon them and transactions with government which cannot be distinguished from the normal trading transactions of the entity.

Grants related to assets: Government grants whose primary condition is that an entity qualifying for them should purchase, construct or otherwise acquire long-term assets. Subsidiary conditions may also be attached restricting the type or location of the assets or the periods during which they are to be acquired or held.

Grants related to income: Government grants other than those related to assets.

Forgivable loans: Loans which the lender undertakes to waive repayment of under certain prescribed conditions.

Accounting treatment

- **Recognise government grants and forgivable loans** once conditions complied with and receipt/waiver is assured.
- Grants are recognised under the **income approach**: recognise grants as income to match them with related costs that they have been received to compensate.
- Use a **systematic basis** of matching over the relevant periods.
- Grants for **depreciable assets** should be recognised as income on the same basis as the asset is depreciated.
- Grants for **non-depreciable assets** should be recognised as income over the periods in which the cost of meeting the obligation is incurred.

- A grant may be **split into parts** and allocated on different bases where there are a series of conditions attached.
- Where **related costs have already been incurred**, the grant may be recognised as income in full immediately.
- A grant in the form of a **non-monetary asset** may be valued at fair value or a nominal value.
- **Grants related to assets** may be presented in the statement of financial position **either** as **deferred incomeor** deducted in arriving at the carrying value of the asset.
- **Grants related to income** may be presented in the statement of profit or loss and other comprehensive income (in profit or loss) **either** as a **separate creditordeducted** from the related expense.
- Repayment of government grants should be accounted for as a **revision of an accounting estimate**.

Disclosure

- **Accounting policy** note
- **Nature and extent** of government grants and other forms of assistance received
- **Unfulfilled conditions** and other contingencies attached to recognised government assistance

Interactive question 8: Government grants

IAS 20 suggests that there are two approaches to recognising government grants: a capital approach (credit directly to shareholders' interests) and an income approach. IAS 20 requires the use of the income approach.

Requirement

What are the arguments in support of each method? See **Answer** at the end of this chapter.

6 IAS 23, *Borrowing Costs*

Section overview

- This section gives a very brief overview of the material covered in earlier studies.
- IAS 23 deals with the treatment of borrowing costs, often associated with the construction of **self- constructed assets**, but which can also be applied to an asset purchased that takes time to get ready for use/sale.

Definitions

Borrowing costs: Interest and other costs incurred by an entity in connection with the borrowing of funds.

Qualifying asset: An asset that necessarily takes a substantial period of time to get ready for its intended use or sale.

6.1 Accounting treatment

Note the following key points, which should be familiar from your earlier studies.

- Borrowing costs must be **capitalised** as part of the cost of the asset if they are directly attributable to acquisition/construction/production. Other borrowing costs must be expensed.
- **Borrowing costs eligible for capitalisation** are those that would have been avoided otherwise. Use judgement where a range of debt instruments is held for general finance.
- **Amount of borrowing costs available for capitalisation** is actual borrowing costs incurred less any investment income from temporary investment of those borrowings.
- For borrowings obtained generally, apply the **capitalisation rate** to the expenditure on the asset (weighted average borrowing cost). It must not exceed actual borrowing costs.
- **Capitalisation is suspended** if active development is interrupted for extended periods. (Temporary delays or technical/administrative work will not cause suspension.)
- **Capitalisation ceases** (normally) when physical construction of the asset is completed. When an asset is comprised of separate stages, capitalisation should cease when each stage or part is completed.
- Where the recoverable amount of the asset falls below carrying amount, it must be **written down/off**.

6.2 Disclosure

- Amount of **borrowing costs capitalised** during the period
- **Capitalisation rate** used to determine borrowing costs eligible for capitalisation

Interactive question 9: Borrowing costs 1

On 1 January 20X8, Rechno Co borrowed £15 million to finance the production of two assets, both of which were expected to take a year to build. Production started during 20X8. The loan facility was drawn down on 1 January 20X8 and was used as follows, with the remaining funds invested temporarily.

	Asset X	Asset Y
	£m	£m
1 January 20X8	2.5	5.0
1 July 20X8	2.5	5.0

The loan rate was 10% and Rechno Co can invest surplus funds at 8%.

Requirement

Ignoring compound interest, calculate the borrowing costs which must be capitalised for each of the assets and consequently the cost of each asset as at 31 December 20X8.

See **Answer** at the end of this chapter.

Interactive question 10: Borrowing costs 2

Zenzi Co had the following loans in place at the beginning and end of 20X8.

	1 January 20X8	31 December 20X8
	£m	£m
10.0% bank loan repayable 20Y3	120	120
9.5% bank loan repayable 20Y1	80	80
8.9% debenture repayable 20Y8	–	150

The 8.9% debenture was issued to fund the construction of a qualifying asset (a piece of mining equipment), construction of which began on 1 July 20X8.

On 1 January 20X8, Zenzi Co began construction of a qualifying asset, a piece of machinery for a hydroelectric plant, using existing borrowings. Expenditure drawn down for the construction was: £30 million on 1 January 20X8, £20 million on 1 October 20X8.

Requirement

Calculate the borrowing costs to be capitalised for the hydroelectric plant machine. See

Answer at the end of this chapter.

7 Statements of cash flows



Section overview

This section briefly revises single company statements of cash flows, which was covered at Professional Level. Try the interactive questions here to make sure you are comfortable with this topic before revising consolidated statements of cash flows in later Chapters.

7.1 Basic statements of cash flows – revision

(a) A statement of cash flows prepared in accordance with IAS 7, *Statement of Cash Flows* provides information about the historical changes in an entity's cash and cash equivalents. This information is presented in a statement that classifies cash flows between operating activities, investing activities and financing activities.

(b) Cash, as defined in IAS 7, includes not only cash itself but also any instrument that can be converted into cash so quickly that it is in effect equivalent to cash.

(c) 'Operating activities' are the principal revenue-generating activities of an entity, together with any other activities which are not identified as being investing or financing in nature.

(d) The cash flows from an entity's operating activities can be presented using two methods:

(1) The **direct method**, which discloses the major classes of gross cash receipts and payments; or

(2) The **indirect method**, where the entity starts with the net profit or loss for the period and adjusts it for non-cash transactions, deferrals or accruals of income and expenditure and items that will form part of the entity's investing and financing activities.

(e) 'Investing activities' are acquisitions and disposals of long-term assets and investments, other than cash and cash equivalents. Examples include: cash paid or received to acquire or sell an item of property, plant or equipment, a receipt of cash from the sale of a business, and cash advanced as a loan to another entity.

(f) 'Financing activities' are activities that change the amount and composition of an entity's equity capital and borrowings. Examples include: cash proceeds from issuing shares, cash paid to repay debt instruments, and the capital element in a finance lease payment.

(g) Investing and financing activities that do not impact on cash, for example the conversion of debt to equity, should not be included in the statement of cash flows.

Interactive question 11: Single company statement of cash flows

Elida is a publicly listed company. The following financial statements of Elida are available:

Statement of profit or loss and other comprehensive income for year ended 31 March 20X8

	£'000
Revenue	5,740
Cost of sales	(4,840)
Gross profit	900
Income from and gains on investment property	60
Distribution costs	(120)
Administrative expenses (Note (ii))	(350)
Finance costs	(50)
Profit before tax	440
Income tax expense	(160)
Profit for the year	280
Other comprehensive income	
Gains on property revaluation	100
Total comprehensive income	380

Statements of financial position as at:

	31 March 20X8		31 March 20X7	
	£'000	£'000	£'000	£'000
ASSETS				
Non-current assets (Note (1))				
Property, plant and equipment		2,880		1,860
Investment property		420		400
		3,300		2,260
Current assets				
Inventory	1,210		810	
Trade receivables	480		540	
Income tax asset	nil		50	
Bank	10	1,700	nil	1,400
Total assets		5,000		3,660
EQUITY AND LIABILITIES				
Equity				
Equity shares of 20p each (Note (3))		1,000		600
Share premium	600		nil	
Revaluation reserve	150		50	

Retained earnings	1,440	2,190	1,310	1,360
		3,190		1,960
Non-current liabilities				
6% loan notes (Note (2))	nil		400	
Deferred tax	50	50	30	430
Current liabilities				
Trade payables	1,410		1,050	
Bank overdraft	nil		120	
Warranty provision (Note (4))	200		100	
Current tax payable	150	1,760	nil	1,270
Total equity and liabilities		5,000		3,660

The following supporting information is available:

(1) An item of plant with a carrying amount of £240,000 was sold at a loss of £90,000 during the year. Depreciation of £280,000 was charged (to cost of sales) for property, plant and equipment in the year ended 31 March 20X8.

Elida uses the fair value model in IAS 40, *Investment Property*. There were no purchases or sales of investment property during the year.

(2) The 6% loan notes were redeemed early incurring a penalty payment of £20,000 which has been charged as an administrative expense in the statement of profit or loss.

(3) There was an issue of shares for cash on 1 October 20X7. There were no bonus issues of shares during the year.

(4) Elida gives a 12-month warranty on some of the products it sells. The amounts shown in current liabilities as warranty provision are an accurate assessment, based on past experience, of the amount of claims likely to be made in respect of warranties outstanding at each year end. Warranty costs are included in cost of sales.

(5) A dividend of 3p per share was paid on 1 January 20X8.

Requirement

Prepare a statement of cash flows for Elida for the year to 31 March 20X8 in accordance with IAS 7, *Statement of Cash Flows*.

See **Answer** at the end of this chapter.

8 Audit focus



Section overview

Matters that the auditor should consider when auditing leases for lessees include:

- correctly identifying whether a lease exists;
- confirming the amounts to be included in the financial statements; and
- whether the disclosure is in line with applicable accounting standards.

Auditing issues for lessors require the distinction between operating and finance leases.

8.1 Auditing leases for lessees

In auditing leases recognised at fair value (eg, investment properties), the auditor must evaluate whether the fair value is appropriate. We will cover the auditing of fair value in further detail in the Audit Focus Sections in later Chapters.

The table below summarises the areas of audit focus when auditing leases in accordance with IFRS 16, and provides some examples of audit evidence required.

Issue	Evidence
Identifying right-of-use assets	As the key test here is correctly identifying an asset as a right-of-use asset, the auditor should consider the following: • Obtain contract for relevant assets and determine whether lessee has control (via evidence of receiving economic benefits and the ability to direct the use of the asset concerned) • Review assets for any below the threshold (eg, $5,000 or 12 months in duration) to determine any exemptions • Review contracts for any separate components (such as maintenance contracts) and confirm each component is separately identified and the correct costs are used
Confirming the **correct accounting treatment** of right-of-use assets	Obtain details of the costs of any leased right-of-use assets and confirm they include attributable costs (such as incentives or advance payments) by reference to terms of contract and any supporting transactions Confirm correct treatment of recognition throughout the asset's life by challenging management on the depreciation policy used (for example, is the asset life appropriate?)
Lease **liabilities**	Amounts recognised as liabilities should be confirmed by reference to the contract and challenged for attributable amounts (such as amounts paid in advance or arrears)
Sale and leaseback arrangements	As these are likely to be complex, auditors should obtain an understanding of the sale and leaseback arrangements via the contract and the client to ensure they reflect the reality of the transaction (for example, does it satisfy the IFRS 15 conditions required to be recognised as a sale?)
Disclosure in the financial statements	Review the IFRS 16 disclosures in the financial statements to determine whether they are consistent and complete

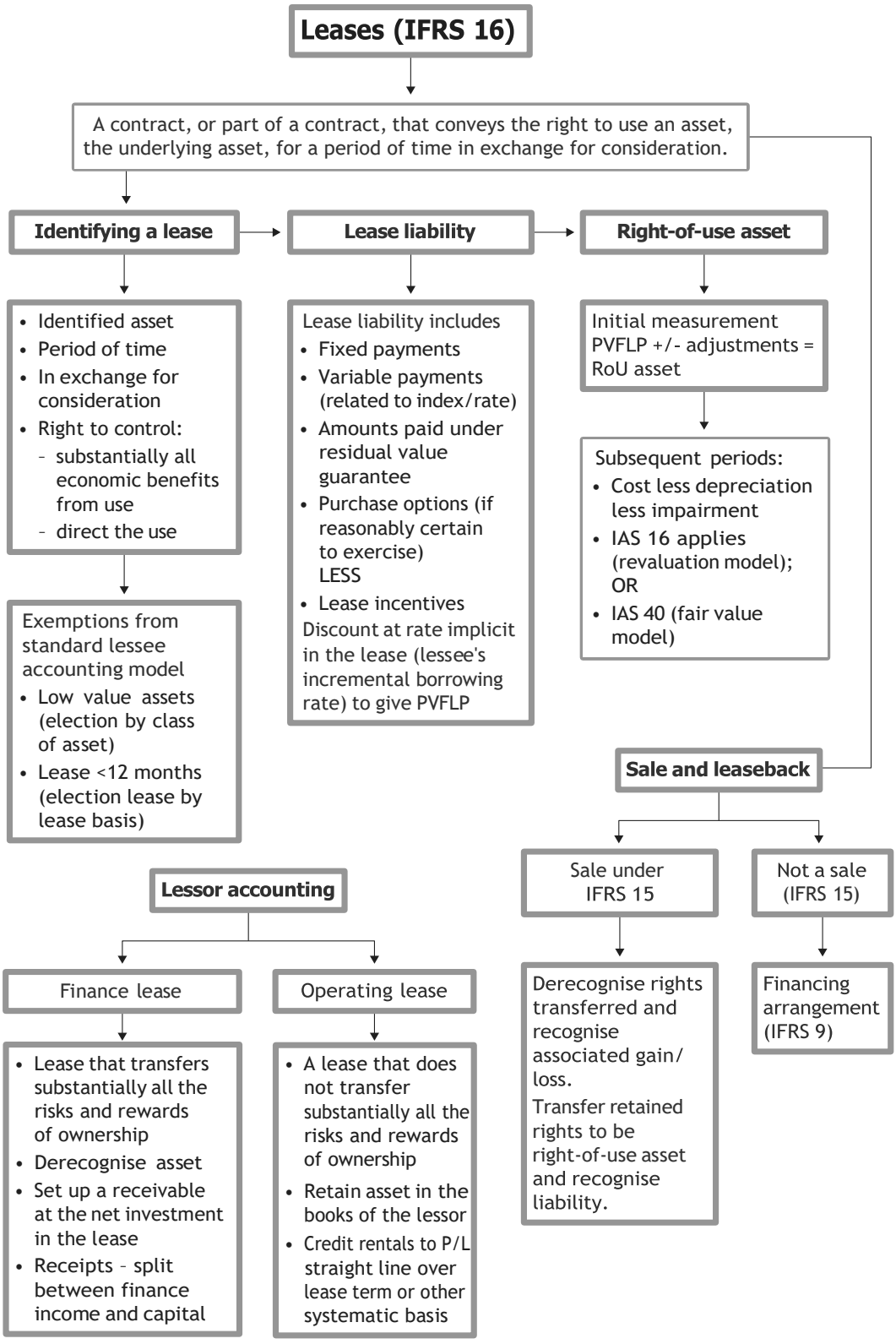
8.2 Auditing leases for lessors

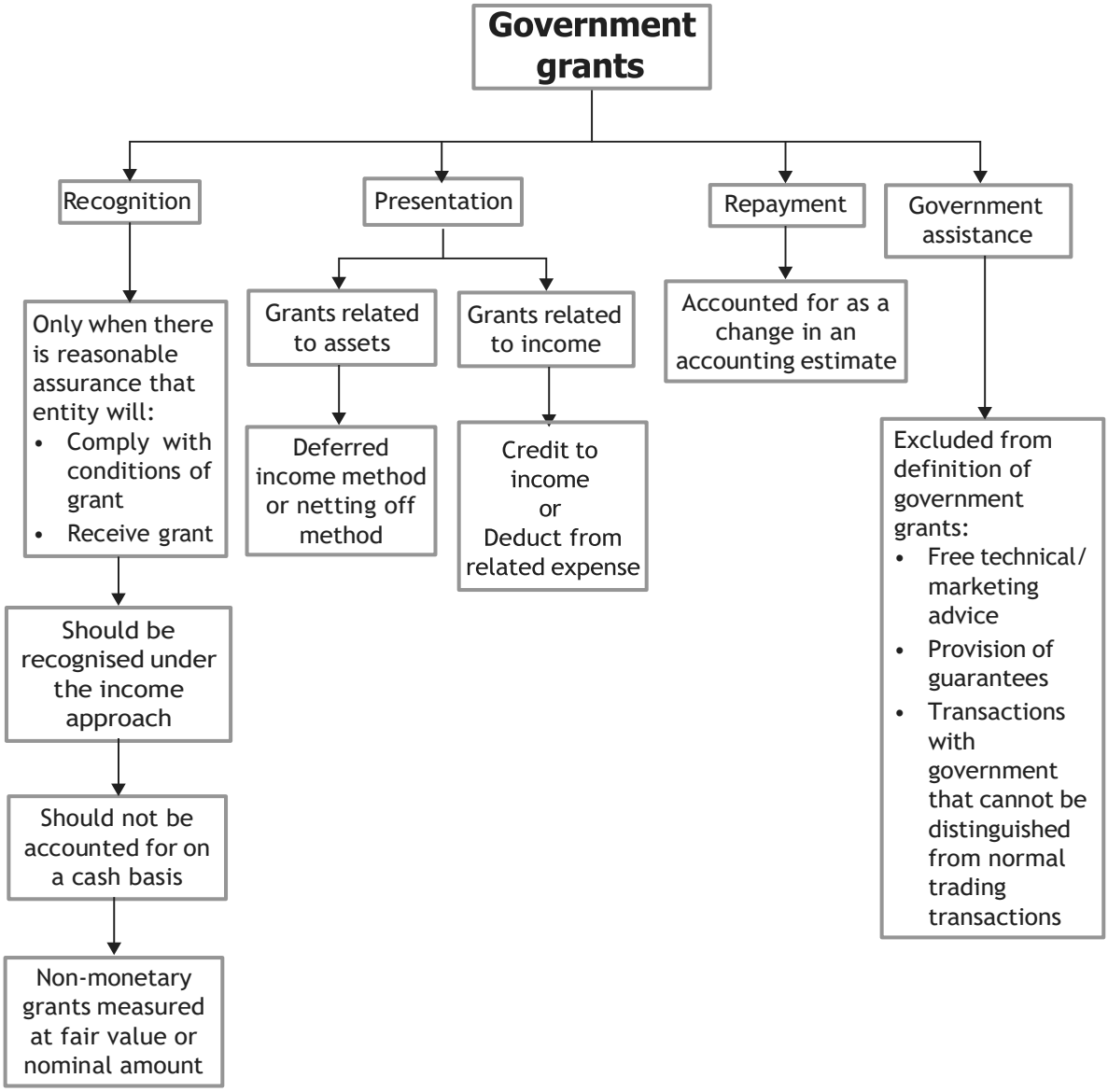
Central to the work of the auditor for a lessor is being able to confirm whether any leases are finance or operating leases. The key procedure here is to obtain a copy of the contract in place for any such lease and to confirm who bears the ultimate risk and reward (the contract will not specify this, so the auditor will need to consider this by analysing the terms stated).

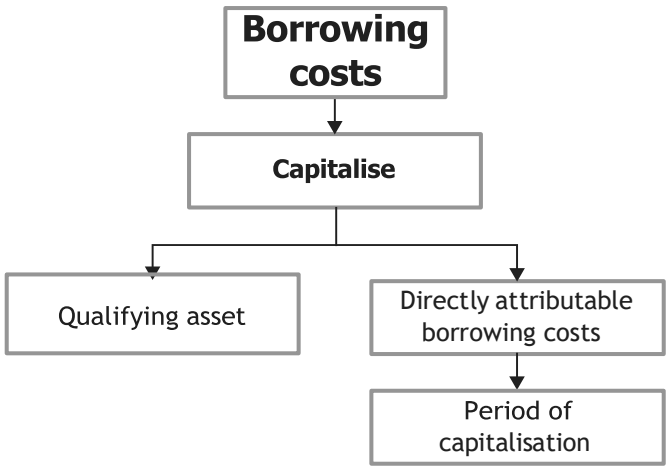
8.3 Further auditing issues

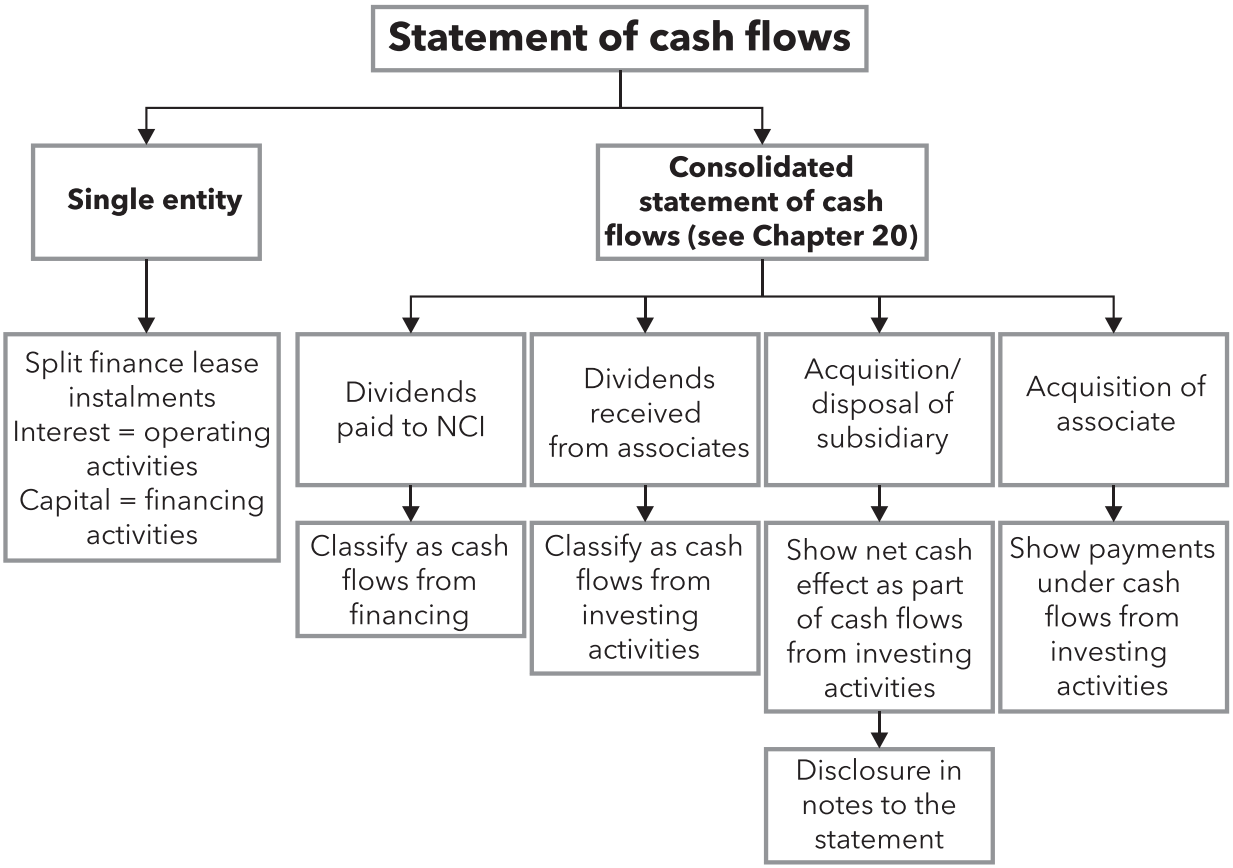
- Financial statements may be deliberately misstated to present short term or low value assets inappropriately to avoid the need to show lease liabilities.
- Sale and leaseback arrangements require an accounting treatment that supports IFRS 15 and auditors will need to ensure their clients have followed this treatment in such cases.

Summary


Leases (IFRS 16)
A contract, or part of a contract, that conveys the right to use an asset, the underlying asset, for a period of time in exchange for consideration.
Identifying a lease
Lease liability
Right-of-use asset
• Identified asset
• Period of time
• In exchange for consideration
• Right to control:
- substantially all economic benefits from use
- direct the use
Exemptions from standard lessee accounting model
• Low value assets (election by class of asset)
• Lease <12 months (election lease by lease basis)
Lease liability includes
• Fixed payments
• Variable payments (related to index/rate)
• Amounts paid under residual value guarantee
• Purchase options (if reasonably certain to exercise)
LESS
• Lease incentives
Discount at rate implicit in the lease (lessee's incremental borrowing rate) to give PVFLP
Initial measurement
PVFLP +/- adjustments = RoU asset
Subsequent periods:
• Cost less depreciation less impairment
• IAS 16 applies (revaluation model); OR
• IAS 40 (fair value model)
Sale and leaseback
Sale under IFRS 15
Not a sale (IFRS 15)
Derecognise rights transferred and recognise associated gain/loss.
Transfer retained rights to be right-of-use asset and recognise liability.
Financing arrangement (IFRS 9)
Lessor accounting
Finance lease
Operating lease
• Lease that transfers substantially all the risks and rewards of ownership
• Derecognise asset
• Set up a receivable at the net investment in the lease
• Receipts - split between finance income and capital
• A lease that does not transfer substantially all the risks and rewards of ownership
• Retain asset in the books of the lessor
• Credit rentals to P/L straight line over lease term or other systematic basis


Government grants
Recognition
Presentation
Repayment
Government assistance
Only when there is reasonable assurance that entity will:
• Comply with conditions of grant
• Receive grant
Grants related to assets
Grants related to income
Accounted for as a change in an accounting estimate
Deferred income method or netting off method
Credit to income or Deduct from related expense
Excluded from definition of government grants:
• Free technical/ marketing advice
• Provision of guarantees
• Transactions with government that cannot be distinguished from normal trading transactions
Should be recognised under the income approach
Should not be accounted for on a cash basis
Non-monetary grants measured at fair value or nominal amount


Borrowing costs
Capitalise
Qualifying asset
Directly attributable borrowing costs
Period of capitalisation


Statement of cash flows
Single entity
Consolidated statement of cash flows (see Chapter 20)
Split finance lease instalments
Interest = operating activities
Capital = financing activities
Dividends paid to NCI
Dividends received from associates
Acquisition/ disposal of subsidiary
Acquisition of associate
Classify as cash flows from financing
Classify as cash flows from investing activities
Show net cash effect as part of cash flows from investing activities
Show payments under cash flows from investing activities
Disclosure in notes to the statement

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	What is meant by 'control' in the context of IFRS 16, *Leases?* (Topic 1)
2.	Can you identify a lease as defined in IFRS 16? (Topic 1)
3.	How should the right-of-use asset be measured on initial recognition and subsequently? (Topic 2)
4.	How should the lease liability be measured on initial recognition and subsequently? (Topic 2)
5.	Can you account for sale and leaseback transactions? (Topic 3)
6.	How does lessor accounting differ from lessee accounting? (Topic 4)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Isaac	This is a useful leasing question, because you are asked for the total charge to profit or loss without being told how this is made up.
Snow	Here you are tested on two aspects of leasing that regularly come up: the carrying amount of the right-of-use asset and sale and leaseback.
Noname	This recaps provisions and emphasises that the treatment of onerous leases falls under IFRS 16, rather than IAS 37.
Tonto	The single company statement of cash flows is revised from your earlier studies, in preparation for consolidated statements of cash flows introduced in a later chapter.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Telo (note 3 issue only)	This deals with lessor accounting, which is less commonly tested than lessee accounting.
Solvit (sale and leaseback only)	You have answered other parts of this question in earlier chapters. The issue of sale and leaseback gets tested regularly and the calculations will get easy with practice.
UHN (issue 1 only)	This tests a more complicated sale and leaseback. Note that you are later required to show the impact of this and other adjustments on the gearing and interest cover ratio, but you can't do this just from one adjustment.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving onto the next chapter.

Technical references

1 IFRS 16, *Leases*

(1) **Identification of a lease**

- An assessment of whether the contract contains a lease by considering the following elements: – **IFRS 16 (9)**
  - A right to control an asset – **IFRS 16 (B24)**
  - Use of an identified asset – **IFRS 16 (B13)**
  - Use for a period of time in exchange for consideration – **IFRS 16 (10)**
  - At the commencement date, the lessee shall recognise a 'right-of-use' asset – **IFRS 16 (23)**

(2) **Measurement**

- Initial measurement of the right-of-use asset includes: – **IFRS 16 (24), IAS 16 (18–19)**
  - Initial measurement of the lease liability
  - Lease payments made before the commencement date
  - Less any lease incentives received
  - Any initial direct costs incurred by the lessee
  - Estimate of costs required to dismantle and remove the asset at the end of the lease term to be incurred by the lessee
  - Depreciate asset over its useful life, or the lease term if shorter and no reasonable certainty that lessee will obtain ownership at end of lease – **IFRS 16 (30)**
  - Consider whether IAS 36 impairment procedures are needed – **IFRS 16 (33)**
  - Disclosures:
  - Disclose information about the leases either in a single note or in a separate section of the financial statements – **IFRS 16 (52)**
  - Disclosure is required of the following: – **IFRS 16 (53)**
    - Depreciation charge for the right-of-use assets – **IFRS 16 (53)(a)**
    - Interest expense on lease liabilities – **IFRS 16 (53)(b)**
    - Additions to right-of-use assets – **IFRS 16 (53)(h)**
    - Gains or losses arising from sale and leaseback transactions – **IFRS 16 (53)(i)**
    - Carrying amount of right-of-use assets at the end of the reporting period by class of underlying asset – **IFRS 16 (53)(j)**
    - Details of low value and short-term transactions – **IFRS 16 (53)(c)(d)**

(3) **Sale and leaseback**

- If the fair value of the consideration for the sale of an asset does not equal the fair value, below market terms are accounted for as a prepayment of lease payments and any above market terms are accounted for as additional financing provided by the buyer – **IFRS 16 (101)**
- Any potential adjustment is calculated on the difference between the fair value of the consideration and the fair value of the asset – **IFRS 16 (102)**

(4) **Lessor accounting**

Finance lease:

- A lease that transfers substantially all the risks and rewards incidental to ownership of an underlying asset – **IFRS 16 AppA**
- Recognise a receivable measured at an amount equal to the net investment in the lease – **IFRS 16.67**
- Net investment in the lease is the present value of the payments receivable by the lessor plus the value of any unguaranteed residual value accruing to the lessor – **IFRS 16 AppA**
- Include initial direct costs incurred – **IFRS 16.69**
- Recognition of finance income should be based on a pattern reflecting a constant periodic rate of return on the lessor's net investment – **IFRS 16.75**
- Finance income should be allocated on a systematic and rational basis – **IFRS 16.76**

Operating lease:

- A lease that does not transfer substantially all the risks and rewards incidental to ownership of an underlying asset – **IFRS 16 AppA**
- The asset should be recorded in the statement of financial position according to its nature – **IFRS 16.71**
- Operating lease income should be recognised on a straight-line basis over the lease term, unless another basis is more appropriate – **IFRS 16.81**
- Asset should be depreciated as per other similar assets – **IFRS 16.84**
- IAS 36 should be applied to determine whether the asset is impaired – **IFRS 16.85**
- Disclosures – **IFRS 16.89**

2 IAS 20, *Accounting for Government Grants and Disclosure of Government Assistance*

(1) **Treatment**

- Should only be recognised if reasonable assurance that: – **IAS 20.7**
  - Entity will comply with conditions
  - Grant will be received
- Manner in which received does not affect accounting method adopted – **IAS 20.9**
- Should be recognised as income over periods necessary to match with related costs – **IAS 20.12**
- Income approach, where grant is taken to income over one or more periods should be adopted – **IAS 20.13**
- Grants should not be accounted for on a cash basis – **IAS 20.16**
- Grants in recognition of specific expenses are recognised as income in same period as expense – **IAS 20.17**
- Grants related to depreciable assets usually recognised in proportion to depreciation – **IAS 20.17**
- Grants related to non-depreciable assets requiring fulfilment of certain obligations should be recognised as income over periods which bear the cost of meeting obligations – **IAS 20.18**
- Grant received as compensation for expenses already incurred recognised in period in which receivable – **IAS 20.20**
- Non-monetary grants should be measured at fair value or a nominal amount – **IAS 20.23**

(2) **Presentation of grants related to assets**

- Can be presented in the statement of financial position by: – **IAS 20.24**
  - Setting up the grant as deferred income or
  - Netting it off against the carrying amount of the asset

(3) **Presentation of grants related to income – IAS 20.29**

- Either:
  - Recognised in profit or loss as income separately or under a general heading or
  - Deducted in arriving at the amount of the related expense recognised in profit or loss

(4) **Repayment of government grants**

- Accounted for as a revision to an accounting estimate – **IAS 20.32**

(5) **Government assistance**

- The following forms of government assistance are excluded from the definition of government grants: – **IAS 20.34–35**
  - Assistance which cannot reasonably have a value placed on it
  - Transactions with government which cannot be distinguished from the normal trading transactions of the entity

(6) **Disclosures**

- Required disclosures – **IAS 20.39**

3 IAS 23, *Borrowing Costs*

- Core principle – **IAS 23.1 and 8**
- Qualifying asset – **IAS 23.5 and 7**
- Directly attributable borrowing costs – **IAS 23.10–11**
- Eligible borrowing costs – **IAS 23.12–15**
- Excess of carrying amount over recoverable amount of asset – **IAS 23.16**
- Commencement of capitalisation – **IAS 23.17–19**
- Suspension of capitalisation – **IAS 23.20–21**
- Cessation of capitalisation – **IAS 23.22–25**
- Disclosure – **IAS 23.26**

4 IAS 7, *Statement of Cash Flows*

- Objective of the statement of cash flows
  - The statement of cash flows should show the historical changes in cash and cash equivalents
  - Cash comprises cash on hand and demand deposits – **IAS 7.6**
  - Cash equivalents are short-term, highly liquid investments that are readily convertible to known amounts of cash and which are subject to an insignificant risk of changes in value – **IAS 7.6**
- Presentation of a statement of cash flows – **Appendix A**
  - Cash flows should be classified by operating, investing and financing activities – **IAS 7.10**
  - Cash flows from operating activities are primarily derived from the principal revenue-producing activities of the entity – **IAS 7.13–14**

- Cash flows from investing activities are those related to the acquisition or disposal of any non- current assets, or trade investments together with returns received in cash from investments (ie, dividends and interest) – **IAS 7.16**

- Financing activities include: – **IAS 7.17**
  - Cash proceeds from issuing shares
  - Cash proceeds from issuing debentures, loans, notes, bonds, mortgages and other short- or long-term borrowings
  - Cash repayments of amounts borrowed
  - Dividends paid to shareholders / the non-controlling interest
  - Principal receipts of amounts advanced under leases
  - Cash flows from operating activities
  - There are two methods of presentation allowed:
    - Direct method – **IAS 7.19**
    - Indirect method – **IAS 7.20**

Self-test questions

Answer the following questions

1 Henry

Henry acquired an asset on a lease. The details were as follows.

Date of acquisition	1 January 20X1
Present value of future lease payments	£7,210
Annual lease payments in arrears	£2,000

The interest rate implicit in the lease is 12% p.a. The payments are made on the last day of each year. There is no option to purchase the asset at the end of the lease.

Requirement

In accordance with IFRS 16, *Leases* what is the lease liability at 31 December 20X2? Round your answer to the nearest pound.

2 Sam plc

Sam plc acquired a machine on a lease. The details were as follows.

Date of commencement	1 July 20X6
Present value of future lease payments	£24,300
Deposit (including the first payment)	£8,000
Remaining annual lease payments (in advance)	4 @£8,000
Interest rate implicit in the lease	12%

The useful life of the machine is eight years. There is no option to purchase the machine at the end of the lease.

Requirement

What is the carrying amount of the right-of-use asset as at 30 June 20X7 in accordance with IFRS 16,
Leases?

3 Isaac Co

Isaac Co acquired an item of plant under a lease on 1 January 20X7. The present value of the future lease payments at the commencement date was £7,731,000 and three lease payments of £3 million p.a. are due to be paid in arrears on 31 December each year.

The useful life of the plant is deemed to be six years. There is no option to buy the asset at the end of the lease term.

The interest rate implicit in the lease is 8% p.a.

Requirement

What is the total charge to the statement of profit or loss in respect of this lease for the year ended 31 December 20X7?

4 Alpha plc

Alpha plc enters into a lease with Omega Ltd for a plastics moulding machine.

The terms of the 10 year lease require Alpha plc to make 10 annual lease payments of £36,000 in arrears. The discounted value of the lease payments at the implicit rate within the lease of 6% is £264,960.

There is an option to buy the machine on the final date of the lease for £25,000 (discounted value of £13,950), which Alpha plc is expected to exercise.

The useful life of the machine is 15 years.

Requirements

Calculate the following values:

(a) The initial measurement of the right-of-use asset at the commencement of the lease on 1 January 20X1.

(b) The carrying amount of the machine at 31 December 20X1.

(c) The lease liability brought forward at 1 January 20X2.

5 Snow plc

On 1 January 20X1 Snow plc entered into the following lease agreements.

(1) **Snow machine**

To lease a snow machine for five years from Slush plc. The snow machine is estimated to have a useful life of eight years.

Snow plc has agreed to make five annual payments of £35,000, payable in arrears, commencing on 31 December 20X1. The present value of future discounted lease payments is £151,515. There is no option to purchase the machine at the end of the lease. Lease incentives of £3,000 were received by Snow plc in respect of this asset.

The interest rate implicit in the lease is 5%.

(2) **Head office and warehouse**

Snow plc has agreed to sell its Head Office and warehouse to Slush plc on 1 January 20X1 for £6 million. Slush plc is to lease the building back to Snow plc over a period of 10 years.

The carrying amount of the building in Snow plc's books on the date of the sale was £5.2 million, and its fair value was £6 million.

The present value of the lease payments was calculated to be £4.2 million, with the remaining useful life of the building being 15 years.

The transaction constitutes a sale in accordance with IFRS 15.

Requirements

5.1 Calculate the carrying amount of the right-of-use asset at 31 December 20X1 in respect of the snow machine.

5.2 What is the initial measurement of the right-of-use asset in respect of the leased building?

5.3 What is the gain on the sale that should be recognised on 1 January 20X1 in the financial statements of Snow plc?

6 Sidcup plc

On 1 January 20X6, Sidcup plc sold its head office building to Eltham Co for £3 million and immediately leased it back on a 10-year lease. On that date, the carrying value of the building was £2.6 million and its fair value was £3 million. The present value of the lease payments was calculated as £2.1 million. The remaining useful life of the building at 1 January 20X6 was 15 years. The transaction constituted a sale in accordance with IFRS 15.

Requirements

6.1 A right-of-use asset must be recognised in respect of the leased building. At what value should this right-of-use asset be recognised on 1 January 20X6 in the financial statements of Sidcup plc?

6.2 What is the gain on the sale that may be recognised on 1 January 20X6 in the financial statements of Sidcup plc?

7 Noname

The Noname Company decided to carry out a fundamental restructuring of its papermaking division which operates in Hyberia. The effect was that most activities carried out in this location would cease with a number of employees being made redundant, whereas other activities and employees would relocate to Sidonia where there was unused capacity. Negotiations with landlords and employee representatives were concluded on 30 December 20X7 and a formal announcement was made to all employees on 31 December 20X7.

The restructuring budget approved by the board of directors in November 20X7 included the following amounts:

	£
Payments to employees:	
Termination payments to those taking voluntary redundancy	90,000
Termination payments to those being made compulsorily redundant	180,000
One-off payments to employees agreeing to move to Sidonia	37,000
Employment cost for closing down activities in Hyberia in preparation for the move to Sidonia	50,000
Lease costs:	
5 years remaining of a lease which can immediately be sublet for £70,000 per annum	45,000 per annum
7 years remaining of a lease which can immediately be sublet for £35,000 per annum	90,000 per annum
Cost of moving plant and equipment from Hyberia to Sidonia	26,000
Impairment losses on non-current assets under IAS 36, *Impairment of Assets*	110,000
Trading transactions in Hyberia up to date of closure, other than those itemised above:	
Revenue	850,000
Expenses	1,150,000

None of these amounts has yet been recognised in Noname's financial statements. The effect of the time value of money is immaterial.

Requirement

Determine the amounts to be included in the financial statements for the Noname Company for the year ending 31 December 20X7 according to IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*.

8 Tonto

The following information relates to the draft financial statements of Tonto.

Summarised statements of financial position as at:

	31 March 20X1		31 March 20X0	
	£'000	£'000	£'000	£'000
ASSETS				
Non-current assets				
Property, plant and equipment (Note (1))		19,000		25,500
Current assets				
Inventory		12,500		4,600
Trade receivables		4,500		2,000
Tax refund due		500		nil
Bank		nil		1,500
Total assets		36,500		33,600
EQUITY AND LIABILITIES				
Equity				
Equity shares of £1 each (Note (2))		10,000		8,000
Share premium (Note (2))	3,200		4,000	
Retained earnings	4,500	7,700	6,300	10,300
		17,700		18,300
Non-current liabilities				
10% loan note (Note (3))	nil		5,000	
Lease obligations	4,800		2,000	
Deferred tax	1,200	6,000	800	7,800
Current liabilities				
10% loan note (Note (3))	5,000		nil	
Tax	nil		2,500	
Bank overdraft	1,400		nil	
Lease obligations	1,700		800	
Trade payables	4,700	12,800	4,200	7,500
Total equity and liabilities		36,500		33,600

Summarised statements of profit or loss for the years ended:

	31 March 20X1	31 March 20X0
	£'000	£'000
Revenue	55,000	40,000
Cost of sales	(43,800)	(25,000)
Gross profit	11,200	15,000
Operating expenses	(12,000)	(6,000)
Finance costs (Note (4))	(1,000)	(600)
Profit (loss) before tax	(1,800)	8,400
Income tax relief (expense)	700	(2,800)
Profit (loss) for the year	(1,100)	5,600

The following additional information is available:

(1) Property, plant and equipment is made up of:

As at:	31 March 20X1	31 March 20X0
	£'000	£'000
Leasehold property	nil	8,800
Owned plant	12,500	14,200
Leased plant	6,500	
	19,000	25,500

During the year Tonto sold its leasehold property for £8.5 million and entered into an arrangement to rent it back from the purchaser. There were no additions to or disposals of owned plant during the year. The depreciation charges (to cost of sales) for the year ended 31 March 20X1 were:

	£'000
Leasehold property	200
Owned plant	1,700
Leased plant	1,800
	3,700

(2) On 1 July 20X0 there was a bonus issue of shares from share premium of one new share for every 10 held. On 1 October 20X0 there was a fully subscribed cash issue of shares at par.

(3) The 10% loan note is due for repayment on 30 June 20X1. Tonto is in negotiations with the loan provider to refinance the same amount for another five years.

(4) The finance costs are made up of:

For year ended:	31 March 20X1	31 March 20X0
	£'000	£'000
Lease charges	300	100
Overdraft interest	200	nil
Loan note interest	500	500
	1,000	600

Requirement

Prepare a statement of cash flows for Tonto for the year ended 31 March 20X1 in accordance with IAS 7, *Statement of Cash Flows*, using the indirect method.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

The lease is for eight months, which counts as a short-term lease, and so it does not need to be recognised in the statement of financial position. If the short-term lease exemption is adopted, then the amount charged to profit or loss for the year ended 30 June 20X6 is £32,000 × 2/8 = £8,000.

If the short-term lease exemption is not adopted, then the standard IFRS 16 model must be applied; an asset and liability will be recognised for the leased asset at 30 June 20X6.

Answer to Interactive question 2

Question	Fill in your answer
A company leases a laptop computer for their financial controller over a three-year period.	This can be classified as a lease, or the company may make an election for this lease to be classified as a low value item, and therefore exempt from the IFRS 16 standard model.
A company leases a photocopier. The copier will remain on the client site, and will only be returned to the lessor in the event of a major repair.	Lease: The arrangement involves an identified asset which the customer has the right to control. Substitution rights are not substantive.
A company leases cars for their team of sales representatives for a three-year period. The cars are required to be estate models, but from time to time, the car may be changed by the lessor.	Not a lease: The lease is not for identified assets, as the lessor can change the exact model of car provided during the lease.
A company acquires three architect's drawing desks which are paid for over 10 months. The company already has four of these desks, and no elections have been made.	These would qualify as a short-term lease, however, IFRS 16 requires the election to be made for the whole class of assets, and this has not been done, so the full standard should be applied.

Answer to Interactive question 3

3.1 Total lease liability at 31 December 20X1 = £ **17,356**

Capital > 1 year = £ **9,092**

< 1 year (β) = £ 8,264

3.2 **Journals**

		£	£
DEBIT	Right-of-use asset	24,869	
CREDIT	Lease liability		24,869
	Being recognition of right-of-use asset and lease liability at commencement of lease.		
DEBIT	Finance charge (profit or loss)	2,487	
CREDIT	Lease liability		2,487
	Being interest on lease		
DEBIT	Lease liability	10,000	
CREDIT	Cash		10,000
	Being payment in respect of lease		

Answer to Interactive question 4

At the end of year 1 the liability will be measured as:

	£
Opening balance	177,297
Interest 5%	8,865
	186,162
Current liability	50,000
Non-current liability (bal. fig).	136,162
	186,162

Answer to Interactive question 5

Calculations

(1) **Calculation of the right-of-use asset**

	£
PVFLP	11,164
Deposit	575
	11,739

Depreciate the asset over seven years

11,739/7 = £1,677 per annum

Carrying amount at 31 December 20X1: £11,739 – £1,677 = £10,062

Carrying amount at 31 December 20X2: £10,062 – £1,677 = £8,385

(2) **Calculation of the lease liability**

Balance of the lease liability at 31 December 20X1: £9,834

Balance of the lease liability at 31 December 20X2: £8,424

	Cr	Cr	Dr	Cr
	Bal b/f 1 Jan	Interest accrued at 6%	Payment 31 Dec	Bal c/f 31 Dec
	£	£	£	£
20X1	11,164	670	(2,000)	9,834
20X2	9,834	590	(2,000)	8,424

Note: You can do these calculations in vertical format (as in the chapter body) or horizontal, whichever is easiest. The principles are the same.

Answer to Interactive question 6

Journal entries at date of disposal

		£	£
DEBIT	Cash	120,000	
CREDIT	Non-current asset (carrying amount)		70,000
CREDIT	Gain on rights transferred (W2)		4,906
DEBIT	Right-of-use asset (W1)	63,131	
CREDIT	Lease liability (PVFLP)		108,225

WORKINGS

(1) **Calculate the right-of-use asset**

Carrying amount × (PVFLP ÷ FV) = £70,000 × (£108,225 ÷ £120,000) = £63,131

(2) **Calculate the gain on rights transferred**

As follows:

a. Calculate the gain on the sale:

Fair value – carrying amount = £120,000–£70,000 = £50,000

b. Calculate the gain relating to the rights retained by the seller/lessee:

Gain × PVFLP/fair value = £50,000 × £108,225/£120,000 = £45,094

c. Calculate the gain relating to the rights transferred:

Total gain (1) – Gain relating to the rights retained (2) = £50,000 – £45,094 = £4,906

(3) **Lease liability and interest accrued**

		£
1.1.X1	Lease liability (PVFLP)	108,225
31.12.X1	Interest accrued 108,225 × 5%	5,411
31.12.X1	Instalment paid	(25,000)
Lease liability carried down at 31.12.X1		88,636
31.12.X2	Interest accrued 88,636 × 5%	4,432
31.12.X2	Instalment paid	(25,000)
Lease liability carried down at 31.12.X2		68,068

Therefore, current liabilities (<12 months) at 31.12.X1 would be £88,636 – £68,068 = £20,568, of which £4,432 is the finance charge (interest payable).

Alternative presentation:

	Balance	Finance cost at 5%	Lease payment	Balance
	£	£	£	£
20X1	108,225	5,411	(25,000)	88,636
20X2	88,636	4,432	(25,000)	68,068

Statement of profit or loss

	£
Depreciation (63,131/4)	(15,783)
Interest (108,225 × 5%) (W3)	(5,411)

Statement of financial position

	£
Non-current assets	
Carrying amount at 1 January 20X1 (W1)	63,131
Depreciation (W3)	(15,783)
Carrying amount at 31 December 20X1	47,348
Non-current liabilities	
Obligations under leases (W3)	68,068
Current liabilities	
Obligations under leases (88,636 – 68,068)	20,568

Answer to Interactive question 7

The answer is £12,973

Stage 1: Gain on sale: £740,000 – £500,000 = £240,000

Stage 2: Gain relating to rights retained = £(240,000 × 700,000/740,000) = £227,027

Stage 3: Gain relating to rights transferred = £240,000 – £227,027 = £12,973

Answer to Interactive question 8

IAS 20 gives the following arguments in support of each method.

Capital approach

(1) The grants are a **financing device**, so should go through the statement of financial position. In the statement of profit or loss and other comprehensive income they would simply offset the expenses which they are financing. No repayment is expected by the Government, so the grants should be credited directly to shareholders' interests.

(2) Grants are **not earned**, they are incentives without related costs, so it would be wrong to record them in profit or loss.

Income approach

(1) The grants are **not received from shareholders** so should not be credited directly to shareholders' interests.

(2) Grants are **not given or received for nothing**. They are earned by compliance with conditions and by meeting obligations. There are therefore associated costs with which the grant can be matched in the statement of profit or loss and other comprehensive income, as these costs are being compensated by the grant.

(3) Grants are an extension of **fiscal policies** and so, as income and other taxes are charged against income, grants should be credited to income.

Answer to Interactive question 9

Cost of each asset as at 31 December 20X8

	Asset X	Asset Y
Borrowing costs	**£'000**	**£'000**
£5.0m/£10m × 10%	500	1,000
Less investment income		
To 30 June 20X8: £2.5m/£5.0m × 8% × 6/12	(100)	(200)
	400	800
Cost of assets		
Expenditure incurred	5,000	10,000
Borrowing costs	400	800
	5,400	10,800

Answer to Interactive question 10

Capitalisation rate = weighted average rate = (10% × (120 ÷ (120 + 80))) + (9.5% × (80 ÷ (120 +80))) = 9.8%

Borrowing costs = (£30m × 9.8%) + (£20m × 9.8% × 3/12) = £3.43m

Answer to Interactive question 11

Elida – statement of cash flows for the year to 31 March 20X8

	£'000	£'000
Cash flows from operating activities		
Profit before tax		440
Loss on sale of plant		90
Depreciation		280
Early redemption penalty		20
Finance costs		50
Investment income		(60)
Increase in warranty provision (200 – 100)		100
		920
Increase in inventory (1,210 – 810)		(400)
Decrease in receivables (480 – 540)		60
Increase in trade payables (1,410 – 1,050)		360
Cash generated from operations		940
Interest paid		(50)

	£'000	£'000
Tax refund received (W1)		60
Net cash from operating activities Cash flows from investing activities		950
Proceeds of sale of plant (240 – 90)	150	
Purchase of plant (W2)	(1,440)	
Income from investment property (60 – 20)	40	
Net cash used in investing activities Cash flows from financing activities		(1,250)
Share issue ((1,000 – 600) + 600)	1,000	
Loan notes repaid	(400)	
Early redemption penalty	(20)	
Dividend paid (1,000 × 5 × 0.03)	(150)	
Net cash from financing activities		430
Net increase in cash and cash equivalents		130
Cash and cash equivalents at beginning of period		(120)
Cash and cash equivalents at end of period		10

WORKINGS

(1) **Income tax payable**

	£'000		£'000
Bal b/d (current tax)	50	Bal b/d (deferred tax)	30
Bal c/d (current tax)	150	Profit or loss charge	160
Bal c/d (current tax)	50	Cash received (balancing figure)	60
	250		250

(2) **Property, plant and equipment**

	£'000		£'000
Bal b/d	1,860	Disposal	240
Revaluation (150 – 50)	100	Depreciation	280
Additions (balancing figure)	1,440	Bal c/d	2,880
	3,400		3,400

Answers to Self-test questions

1 Henry

Liability at 31.12.X2 is £4,804

Date	B/f	Interest 12%	Payment	C/f
	£	£	£	£
31.12.X1	7,210	865	(2,000)	6,075
31.12.X2	6,075	729	(2,000)	4,804

2 Sam plc

The carrying amount of the machine is 32,300 – 6,460 = £25,840

	£
Lease liability (PVFLP)	24,300
Deposit payments at the start of the lease (deposit)	8,000
Right-of-use asset	32,300

Depreciate the machine over the shorter of the lease term (5 years) and the useful life (8 years)

Depreciation charge for the year £32,300/5 = £6,460

3 Isaac Co

Total lease liability

	£	Finance charge
Initial liability (PV of future lease payments)	7,731,000	
Interest 8% (£7,731,000 × 8%)	618,480	618,480
Payment	(3,000,000)	
Total lease liability at 31.12.X8	5,349,480	

Depreciation is charged based on the shorter of the lease term (three years) and the useful life (six years) as there is no option to purchase the asset the end of the lease period.

		Depreciation charge
Right-of-use asset	7,731,000	
Depreciation charge 7,731,000/3	(2,577,000)	2,577,000
Carrying amount	5,154,000	

Charge to the profit and loss is £618,480 + £2,577,000 = £3,195,480

4 Alpha plc

(a) **Right-of-use asset**

	£
Discounted future lease payments	264,960
Option to purchase (discounted)	13,950
Lease liability and right-of-use asset	278,910

(b) **Carrying amount at 31 December 20X1**

	£
Right-of-use asset	278,910
Less Depreciation (W1)	(18,594)
Carrying amount	260,316

WORKING

278,910/15 years = £18,594

The asset is depreciated over 15 years as there is an option to purchase the asset which Alpha plc is expected to exercise.

(c) **Lease liability brought forward at 1 January 20X2**

	£
Lease liability	278,910
Interest at 6%	16,735
Less payment	(36,000)
Liability	259,645

5 Snow plc

5.1

Right-of-use asset at 31 December 20X1

Lease liability

	B/f	Interest @ 5%	Payment	C/f
	£	£	£	£
31 December 20X1	151,515	7,576	(35,000)	124,091

	£
Right-of-use asset	
Present value of future lease payments	151,515
Less lease incentive	(3,000)
Right-of-use asset	148,515
Less depreciation 148,515/5 years	(29,703)
Carrying amount 31 December	118,812

5.2 Calculation of the right-of-use asset

Building

Calculation of the right-of-use asset:

Carrying amount × (present value of lease payments ÷ fair value of asset)

£5.2m × (£4.2m ÷ £6m) = £3.64m

5.3 **Stage 1:** Gain on sale: £6m – £5.2m = £0.8m

Stage 2: Gain that relates to rights retained = £0.8m × (£4.2m ÷ £6m) = £0.56m

Stage 3: Gain relating to rights transferred = £0.8m – £0.56m = £0.24m

6 Sidcup plc

6.1 £1,820,000

IFRS 16 requires that, at the start of the lease, Sidcup should measure the right-of-use asset arising from the leaseback of the building at the proportion of the previous carrying amount of the building that relates to the right of use retained. This is calculated as carrying amount × discounted lease payments/fair value. The discounted lease payments were given in the question as £2.1 million.

The right-of-use asset is therefore: £2.6m × £2.1m/£3m = £1,820,000.

6.2 £120,000

Sidcup only recognises the amount of gain that relates to the rights transferred.

Stage 1: Gain is £3,000,000 – £2,600,000 = £400,000

Stage 2: Gain relating to rights retained £(400,000 × 2,100,000/3,000,000) = £280,000

Stage 3: Gain relating to rights transferred £(400,000 – 280,000) = £120,000

7 Noname

The total amount recognised in profit or loss under IAS 37 is the £270,000 restructuring provision + the £110,000 impairment losses = £380,000.

The five-year lease is not onerous because the premises can be sublet at profit. The seven-year lease is onerous and under IFRS 16 will lead to an impairment review of the right-of-use asset for the premises; however following IFRS 16 there is no longer a provision to be recognised under IAS 37.

Under IAS 37.80 all the payments to employees should be included in the restructuring provision, with the exception of the employment costs of £50,000 in preparation for the move to Sidonia and £37,000 payable to those moving to Sidonia – this relates to the ongoing activities of the business, so is disallowed by IAS 37.80(b). For the same reason the costs of moving plant and equipment is disallowed. Impairment losses reduce the carrying amount of the relevant assets rather than increasing the restructuring provision and revenue less expenses are trading losses which are disallowed by IAS 37.63. So provision = £270,000.

8 Tonto

Tonto statement of cash flows for the year ended 31.3.20X1

	£'000	£'000
Cash flows from operating activities		
Loss before tax	(1,800)	
Depreciation (W1)	3,700	
Interest expense	1,000	
Loss on disposal of leasehold property (8,500 – (8,800 – 200))	100	
Increase in inventories (12,500 – 4,600)	(7,900)	
Increase in receivables (4,500 – 2,000)	(2,500)	

	£'000	£'000
Increase in payables (4,700 – 4,200)	500	
Cash used in operations	(6,900)	
Interest paid	(1,000)	
Tax paid (W2)	(1,900)	
Net cash used in operating activities		(9,800)
Cash flows from investing activities		
Cash from sale of leasehold property	8,500	
Net cash from investing activities Cash flows from financing activities		8,500
Dividends paid (6,300 – 4,500 – 1,100)	(700)	
Payments made under leases (W3)	(2,100)	
Share issue (10,000 + 3,200) – (8,000 + 4,000)	1,200	
Net cash used in financing activities		(1,600)
Net decrease in cash and cash equivalents		(2,900)
Cash and cash equivalents at 31.3.20X0		1,500
Cash and cash equivalents at 31.3.20X1		(1,400)

WORKINGS

(1) **PPE – CARRYING AMOUNT**

	£'000		£'000
B/d	25,500	Disposal (8,800 – 200)	8,600
Lease plant additions		Depreciation (β)	3,700
(6,500 – 2,500 + 1,800)	5,800	C/d	19,000
	31,300		31,300

(2) **INCOME TAX**

	£'000		£'000
Profit or loss	700	31.3.X0 – Current tax	2,500
31.3.X1 – Deferred tax	1,200	31.3.X0 – Deferred tax	800
Tax paid β	1,900	Refund due	500
	3,800		3,800

(3) **LEASE LIABILITY**

	£'000		£'000
Payments made β	2,100	Balance b/f (2,000 + 800)	2,800
		Additions (W1)	5,800
Balance c/f (4,800 + 1,700)	6,500		
	8,600		8,600

Chapter 15

Financial instruments: presentation and disclosure

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Overview of material from earlier studies

2 IFRS 7: Objective and scope

3 Disclosures in financial statements

4 Other disclosures

5 Financial instruments risk disclosure

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Identify and explain current and emerging issues in corporate reporting
- Determine and calculate how different bases for recognising, measuring and classifying financial assets and financial liabilities can impact upon reported performance and position
- Explain and appraise accounting standards that relate to an entity's financing activities which include: financial instruments; leasing; cash flows; borrowing costs; and government grants

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**Overview of material from earlier studies** The use of financial instruments by business for funding, investment and risk management purposes is an essential part of operations. You will need to prepare detailed disclosures for financial instruments and establish the correct presentation of preference shares and convertible instruments between debt and equity.	**Approach** Although you will have a detailed knowledge of IAS 32 from earlier studies, the importance of this topic cannot be overestimated. Make sure you understand the issues relating to the accounting of compound financial instruments. **Stop and think** When might a contract require delivery of the entity's own equity instruments but be classified as a financial liability?	At Advanced Level, you will be concerned with more complex issues than at Professional Level, for example the sometimes difficult distinction between debt and equity. An ethical dilemma may arise whereby a company wishes to reduce gearing by claiming that an instrument is equity rather than a liability.	**IQ7: Purchase of own equity instruments** This question has the finance director of a company claiming that an instrument is equity rather than a liability, and you have to discuss whether they are correct.
2	**Objective and scope** This is a short section on the objective and scope	**Approach** The key point to remember from this short section is that the objective of IFRS	An exam question would not ask for a list of instruments that IFRS 7 does not apply to. However,	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	of the standard that will take up the rest of the chapter: IFRS 7, *Financial Instruments: Disclosures*	7 is for the user to evaluate: • the **significance** of financial instruments for the entity's **financial position and performance**; and • the **nature and extent of risks** arising from financial instruments. **Stop and think** IFRS 7 applies to recognised and unrecognised financial instruments.	you would be expected to know that disclosures relating to certain instruments are covered by other standards, for example IFRS 2, *Share-based payment*.	
3	**Disclosures in financial statements** The detailed qualitative and quantitative disclosures required by IFRS 7 are intended to provide to the users of financial statements an understanding of the significance of financial instruments for the entity's position and performance, and an analysis of the risks to which the entity is exposed and how it manages them.	**Approach** There is no short cut to reading through and understanding these disclosures. Rather than learn them by heart, it is best to know where to find them in the open books. **Stop and think** The individual classes of financial instruments will mean more to you once you have studied recognition and measurement in Chapter 16.	As part of an audit question you may need to mention procedures to check whether financial instruments have been disclosed in the financial statements in accordance with IFRS 7.	N/A
4	**Other disclosures** This section contains summaries of and references to quantitative and qualitative	**Approach** Skim through **Stop and think** IFRS 13, *Fair Value Measurement* affects	This section will not be examined specifically; it is for awareness only to link to other material.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	disclosures in the financial statements which are covered elsewhere.	a large number of areas, particularly financial instruments.		
5	**Financial instruments risk disclosure** IFRS 7 also provides an analysis of the risks to which the entity is exposed and how it manages them.	**Approach** Study the table in Section 5.1. You should focus on credit risk, liquidity risk and market risk. **Stop and think** What qualitative disclosures are required by IFRS 7 in relation to risk?	Risk disclosures are likely to be examined in the context of audit risks and procedures.	N/A However, you can try Self-test questions 2 and 3, which deal with risk and sensitivity analysis.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Overview of material from earlier studies

Section overview

- IAS 32 applies to all entities and all types of financial instruments except where explicitly covered by another standard, such as IFRS 10 for subsidiaries and IAS 28 for associates and joint ventures.
- A financial instrument should be classified as either equity, financial liability or financial asset. This classification is made at the time the financial instrument is issued and not changed subsequently.
- Compound financial instruments, ie, instruments that contain both a liability and an equity component, should be split into their component parts at the date they are issued.
- Interest, dividends, losses and gains arising from financial instruments classified as financial liabilities are recognised in the profit or loss for the year. Dividends paid to holders of a financial instrument classified as equity are charged directly against equity.
- Financial assets and financial liabilities are presented as separate items in the statement of financial position with offset being allowed only in limited cases.
- Treasury shares are deducted from equity with the consideration paid or received recognised directly in equity. Gains or losses on the purchase, sale, issue or cancellation of equity instruments are not recognised in profit or loss.

1.1 Refresh your memory

The definition of a financial instrument is consistent throughout the accounting standards covering financial instruments. It is introduced in IAS 32.

Definition

Financial instrument: Any contract that gives rise to a financial asset of one entity and a financial liability or equity instrument of another entity.

Note that a financial instrument has **two parties**. It should be recognised as an **asset** by one party and either a **liability or equity** by the other. The classification of a financial instrument as a financial liability or equity is particularly important as it will have an effect on gearing.

1.2 Test your memory

Try these questions to see how well you remember the material covered in your earlier studies.

Interactive question 1: Financial instruments

Why do you think that physical assets and prepaid expenses do not qualify as financial instruments?

See **Answer** at the end of this chapter.

Interactive question 2: Liability or equity?

During the financial year ended 31 December 20X5, Kim issued the financial instrument described below. Identify whether it should be classified as liability or equity, giving reasons for your choice.

Redeemable preference shares with a coupon rate 5%. The shares are redeemable on 31 December 20X9 at premium of 20%.

See **Answer** at the end of this chapter.



Interactive question 3: Convertible bond 1

An entity issues a convertible bond for £1,000. The bond is convertible into equity shares of the issuer at the discretion of the holder at any time in the next 10 years. The bond converts into a variable number of shares equal to the value of the liability.

The entity also issues £7,000 of 8% convertible redeemable preference shares. In five years' time the preference shares will either be redeemed or converted into 5,000 equity shares of the issuer, at the option of either the holder or issuer.

Requirement

How should the entity account for the instruments according to IAS 32?

See **Answer** at the end of this chapter.



Interactive question 4: Convertible bond 2

An entity issued 5,000 8% convertible bonds at par value of £10 on 1 January 20X5. Each bond is convertible into three ordinary shares on 31 December 20X6. Interest is payable annually in arrears. The entity incurred transaction costs of £1,000. On the date of issue the market interest rate for similar debt without the conversion option was 10%.

Requirement

Calculate the liability and equity component of the convertible bond on issue.

See **Answer** at the end of this chapter.



Interactive question 5: Options contract

An entity enters into an options contract to acquire 100 ounces of platinum in 90 days' time. The entity will settle the contract by delivering as many of its own shares as are equal to the cash value of
£1,000 on the purchase date.

Requirement

Explain whether the options contract should be classified as a financial asset, financial liability or equity.

See **Answer** at the end of this chapter.



Interactive question 6: Offsetting

An entity issues debt with a variable rate of interest linked to SONIA. SONIA (Sterling Overnight Index Average) is a benchmark rate that replaced LIBOR in 2021. It enters into a corresponding, receive floating, pay fixed interest rate swap for the period of the debt. The effect of the two instruments is to synthesise a fixed-rate long-term loan.

Requirement

Explain whether the two instruments should be presented separately or whether they should be offset.

See **Answer** at the end of this chapter.



Professional skills focus: Assimilating and using information

This section was for revision from earlier studies. At Advanced Level, the scenarios will be much more complex and you will need to sift through detail and pick out the important principles that you need to apply. So if you're at all shaky on the above examples, it is a good idea to practise questions on this area from your earlier studies.

1.3 More complex issues

1.3.1 Debt/equity distinction

The separation of debt and equity components of a financial instrument should be familiar to you from your Professional Level studies, and you should not have had any serious difficulty with the convertible bond question above. However, at Advanced Level you will need to consider more complex, 'real world' issues. In practice, the debt/equity distinction may not be clear cut. Classification of financial instruments as debt or equity can have a significant effect on the financial statements. Guidance is provided in IAS 32, *Financial Instruments: Presentation* (and other standards for items outside IAS 32's scope), but there are sometimes areas where it is difficult to determine whether a transaction is debt or equity.

A contract is not an equity instrument solely because it may result in the receipt or delivery of the entity's own equity instruments. It is necessary to consider whether the settlement results in receipt or delivery of variable or fixed number of the entity's own equity instruments.

If the contract results in delivery of variable number of equity instruments, the fair value of the equity instruments equals to the amount of the fixed contractual right or obligation. Such a contract does not evidence a residual interest in the entity's assets after deducting all of its liabilities.

The following contracts require delivery of the entity's own equity instruments but are **classified as financial liability**:

- A contract to deliver as many of the entity's own equity instruments as are equal in value to £6,000
- A contract to deliver as many of the entity's own equity instruments as are equal in value to 200 ounces of gold

Contingent settlement provisions

A financial instrument may require the entity to deliver cash or another financial asset in the event of **occurrence or non-occurrence of uncertain future events** that are beyond the control of both the issuer and holder of the instrument. These are known **as contingent settlement provisions** and could relate to changes in stock market index, consumer price index, interest rate, issuer's future profits or revenues. Such instruments are classified as financial liability. This is because the issuer **does not have an unconditional right to avoid** delivering cash or another financial asset.

The financial instrument would be an equity instrument if it has equity-like features, for example, obligation arises only in the event of liquidation of the issuer.

Context example: Redemption of preference shares

High Growth Bank issues 7% fixed preference shares. The dividends on the preference shares are cumulative. The terms and conditions of the issue of preference shares indicate that they will be redeemed if the net profit of the bank increases by more than 50% in the next three years.

In this case, the contingent event is outside the control of both the bank and the holder. If the net profit of the bank increases by more than 50% in the next three years, it does not have the unconditional right to avoid delivering cash. The preference shares also pay a fixed cumulative dividend and hence are classified as financial liability.

Context example: Contingent convertible bonds (CoCos)

Contingent convertible bonds issued by Stable Bank give it the right to convert the debt to equity in the event of its capital ratio falling below a pre-set level. To offset the risk the holder is undertaking, the yields that Stable Bank offers on its CoCos are 6.6% compared to an average yield to maturity on other medium-term bonds of 3.5%.

Stable Bank must classify the bonds as a hybrid instrument with both debt and equity features.

1.3.2 Practical implications

The classification of financial instruments as debt versus equity is particularly important with items that are financial liabilities or equity as the presentation of the two items and associated financial effects are very different.

Consider the following examples.

Worked example: Debt or equity?

1 Acquittie issued 40 million non-redeemable £1 preference shares at par value. Under the terms attaching to the preference shares, a dividend is payable on the preference shares only if Acquittie also pays a dividend on its ordinary shares relating to the same period.

2 Acquittie entered into a contract with a supplier to buy a significant item of equipment. Under the terms of the agreement the supplier will receive ordinary shares with an equivalent value of £5 million one year after the equipment is delivered.

3 The directors of Acquittie, on becoming directors, are required to invest a fixed agreed sum of money in a special class of £1 ordinary shares that only directors hold. Dividend payments on the shares are discretionary and are ratified at the annual general meeting (AGM) of the company. When a director's service contract expires, Acquittie is required to repurchase the shares at their nominal value.

Solution

1 IAS 32 requires a financial instrument to be classified as a liability if there is **a contractual obligation** to deliver cash or another financial asset to another entity.

In the case of the preference shares, as they are **non-redeemable**, there is **no obligation to repay the principal**.

In the case of the **dividends**, because of the condition that preference dividends will only be paid if ordinary dividends are paid in relation to the same period, the **preference shareholder has no contractual right** to a dividend. Instead, the distributions to holders of the preference shares are at the discretion of the issuer as **Acquittie can choose** whether or not to pay an ordinary dividend and therefore a preference dividend. Therefore, there is **no contractual obligation** in relation to the dividend.

As there is no contractual obligation in relation to either the dividends or principal, the **definition of a financial liability has not been met** and the preference shares should be treated as **equity** and initially recorded at **fair value** ie, their par value of £40 million.

The treatment of **dividends** should be consistent with the classification of the shares and should therefore be charged directly to **retained earnings**.

2 The price of the equipment is fixed at £5 million one year after delivery. In terms of recognition and measurement of the equipment, the £5 million price would be discounted back one year to its present value.

The company is paying for the equipment by issuing shares. However, this is **outside the scope of IFRS 2**, *Share-based Payment* because the payment is not dependent on the value of the shares, it is fixed at £5 million.

This is an example of a contract that "will or may be settled in an entity's own equity instruments and is a non-derivative for which the entity is or may be obliged to deliver a **variable** number of the entity's own equity instruments" (IAS 32.11) ie, a financial liability.

It is the number of shares rather than the amount paid that will vary, depending on share price. Therefore it should be classed as a **financial liability** and initially measured at the **present value** of the £5 million.

Subsequently, as it is not measured at fair value through profit or loss (as it is not held for short- term profit-taking or a derivative), it should be measured at **amortised cost**.

As a result, **interest** will be applied to the discounted amount over the period until payment and recognised in **profit or loss** with a corresponding increase in the financial liability.

3 Most ordinary shares are treated as equity as they do not contain a contractual obligation to deliver cash.

However, in the case of the directors' shares, a **contractual obligation to deliver cash exists** on a specific date as the shares are **redeemable** at the end of the service contract.

The redemption is **not discretionary**, and Acquittie has no right to avoid it. The mandatory nature of the repayment makes this capital a financial liability. The financial liability will initially be recognised at its fair value ie, the present value of the payment at the end of the service contract. It will be subsequently measured at amortised cost and effective interest will be applied over the period of the service contract.

Dividend payments on the shares are **discretionary** as they must be ratified at the AGM. Therefore, no liability should be recognised for any dividend until it is ratified. When recognised, the classification of the **dividend** should be consistent with the classification of the shares and therefore any dividends are classified as a **finance cost** rather than as a deduction from retained earnings.

Professional skills focus: Applying judgement

The classification of a financial instrument as debt or equity is an area where judgement must be applied. This area is still under consideration by the IASB in the form of a Discussion Paper.

Professional skills focus: Structuring problems and solutions

The debt/equity distinction also exercises the student/practitioner's ability to structure problems and approach them logically. For example, just because it may result in the receipt

or delivery of the entity's own equity instruments does not necessarily follow it is an equity instrument.

Interactive question 7: Purchase of own equity instruments

Emporium is a listed retail group, and has a year end of 31 October. On 21 October 20X8, Emporium carried out a bonus issue where the shareholders of Emporium received certain rights. The shareholders are able to choose between the following:

- receiving newly issued shares of Emporium, which could be traded on 30 November 20X8; or
- transferring their rights back to Emporium by 10 November 20X8 for a fixed cash price which would be paid on 20 November 20X8.

While preparing the financial statements at 31 October 20X8, the finance director of Emporium argued that the criteria for the recognition of a financial liability as regards the second option were not met at 31 October 20X8 because it was impossible to reliably determine the full amount to be paid until 10 November 20X8.

Requirement

Discuss whether the finance director is correct regarding the recognition of a financial liability.

See **Answer** at the end of this chapter.

2 IFRS 7: Objective and scope

Section overview

This section discusses the objectives and sets out the scope of IFRS 7, *Financial Instruments: Disclosures.*

2.1 Objective

The principles of IFRS 7 complement the principles for recognising, measuring and presenting financial assets and financial liabilities in IAS 32, *Financial Instruments: Presentation* and IFRS 9, *Financial Instruments.*

IFRS 7 requires entities to provide **disclosures** in their financial statements that enable users to **evaluate**:

- the **significance** of financial instruments for the entity's **financial position and performance**; and
- the **nature and extent of risks** arising from financial instruments to which the entity is exposed during the period and at the reporting date, and how the entity manages those risks.

The main presentation and disclosure requirements as detailed in IFRS 7 and IAS 32 together with certain aspects of recognition and measurement of IFRS 9 have already been covered at Professional Level. This chapter extends the coverage of the disclosure requirements of IFRS 7 and the presentation requirements.

2.2 Scope

IFRS 7 applies to all entities and to **all types** of financial instruments**, except instruments that are specifically covered by other standards**. Examples of financial instruments not covered by IFRS 7 include the following:

- Interests in subsidiaries, associates and joint ventures that are accounted for in accordance with IFRS 10, *Consolidated Financial Statements* or IAS 28, *Investments in Associates and Joint Ventures*
- Employers' rights and obligations arising from employee benefit plans, to which IAS 19, *Employee Benefits* applies
- Insurance contracts as defined in IFRS 4, *Insurance Contracts*
- Financial instruments, contracts and obligations under share-based payment transactions to which IFRS 2, *Share-based Payment* applies

IFRS 7 applies to recognised and unrecognised financial instruments. Recognised financial instruments include financial assets and financial liabilities that are within the scope of IFRS 9. Unrecognised financial instruments include some financial instruments that, although outside the scope of IFRS 9, are within the scope of IFRS 7 (such as some loan commitments).

IFRS 7 also applies to contracts to buy or sell a non-financial item that are within the scope of IFRS 9 because they can be settled net and there is not the expectation of delivery, receipt or use in the ordinary course of business.

2.3 General considerations

2.3.1 Classes of financial instruments and level of disclosures

IFRS 7 requires that certain disclosures should be given by class of financial instruments. The classes of financial instruments that will be disclosed should be appropriate to the nature of the information disclosed and should take into account the characteristics of those financial instruments. An entity should provide sufficient information to permit reconciliation to the line items presented in the statement of financial position.

In deciding how to disclose the classes of financial instruments, an entity should not necessarily adopt the classification of IFRS 9. The classes should be determined by the entity, but at the minimum it should:

- provide distinctive classes for financial instruments at amortised cost and financial instruments at fair value; and
- provide a separate class or classes for financial instruments outside the scope of IFRS 7.

Note: A 'class' is not the same as a classification under IFRS 9 (investment in equity instruments, financial assets held at amortised cost, financial assets at fair value through profit or loss and financial assets at fair value through other comprehensive income).

2.3.2 Significance of financial instruments for financial position and performance

An entity must disclose information that enables users of its financial statements to evaluate the significance of financial instruments for its financial position and performance.

2.3.3 Risks from financial instruments

IFRS 7 requires qualitative and quantitative disclosure about the following risks associated with financial instruments:

- **Market risk**. This is the risk of changes in the market value of a financial instrument. When changes in the market value can be attributed to changes in interest rates then the market risk is normally called interest rate risk, and when it can be attributed to changes in exchange rates, market risk is called currency risk.
- **Credit risk**. This is the risk that one party to a financial instrument will fail to fulfil the obligations that arise for the financial instrument causing loss to the other party.
- **Liquidity risk**. This is the risk that an entity will encounter difficulty in meeting obligations associated with financial liabilities.

3 Disclosures in financial statements



Section overview

This section discusses the basic disclosures required by IFRS 7 in the financial statements.

3.1 Statement of financial position

3.1.1 Categories of financial assets and financial liabilities

The carrying amounts of each of the following categories, as defined in IFRS 9, must be disclosed either in the statement of financial position or in the notes:

- Financial assets at fair value through profit or loss, showing separately:
  - those designated as such on initial recognition; and
  - those classified as held for trading in accordance with IFRS 9.
- Financial assets at fair value through other comprehensive income
- Financial assets held at amortised cost
- Investments in equity instruments
- Financial liabilities at fair value through profit or loss, showing separately:
  - those designated as such on initial recognition; and
  - those classified as held for trading in accordance with IFRS 9.
- Financial liabilities measured at amortised cost

3.1.2 Loans and receivables at fair value through profit or loss

(a) **Disclosures**

If the entity has designated a loan or receivable (or group of loans or receivables) as at fair value through profit or loss, it should disclose the following:

(1) The maximum exposure to credit risk of the loan or receivable (or group of loans or receivables) at the reporting date

(2) The amount by which any related credit derivatives or similar instruments mitigate that maximum exposure to credit risk

(3) The amount of change, during the period and cumulatively, in the fair value of the loan or receivable (or group of loans or receivables) that is attributable to changes in the credit risk of the financial asset

(4) The amount of the change in the fair value of any related credit derivatives or similar instruments that has occurred during the period and cumulatively since the loan or receivable was designated

(b) **Calculation**

The amount of any change in the fair value attributable to credit risk can be calculated as the amount of the change not attributed to market risk. That is:

[Amount of change in fair value attributed to credit risk] = [Total amount of change in fair value] – [Amount of change in fair value attributed to market risk]

(c) **Alternative method**

The standard allows the employment of an alternative method to calculate the amount of change in the fair value attributed to credit risk if the entity believes that such a method more faithfully represents the amount of change in its fair value that is attributable to changes in the credit risk of the asset.

(d) **Market risk**

Changes in market conditions that give rise to market risk include changes in an observed (benchmark) interest rate, commodity price, foreign exchange rate or index of prices or rates.

3.1.3 Financial liabilities at fair value through profit or loss

(a) **Disclosures**

If the entity has designated a financial liability as at fair value through profit or loss it should disclose the following:

(1) The amount of change, during the period and cumulatively, in the fair value of the financial liability that is attributable to changes in the credit risk of that liability

(2) The difference between the financial liability's carrying amount and the amount the entity would be contractually required to pay at maturity to the holder of the obligation

(b) **Calculation**

The amount of any change in the fair value attributable to credit risk can be calculated as the amount of the change not attributed to market risk. That is:

[Amount of change in fair value attributed to credit risk] = [Total amount of change in fair value] – [Amount of change in fair value attributed to market risk]

(c) **Alternative method**

The standard allows the employment of an alternative method to calculate the amount of change in the fair value attributed to credit risk if the entity believes that such a method is more accurate.

(d) **Market risk**

Changes in market conditions that give rise to market risk include changes in a benchmark interest rate, the price of another entity's financial instrument, a commodity price, a foreign exchange rate or an index of prices or rates. For contracts that include a unit-linking feature, changes in market conditions include changes in the performance of the related internal or external investment fund.

The example below illustrates how an entity can arrive at the change in fair value attributable to credit risk by estimating the amount of change in fair value attributable to risks other than credit risk.



Context example: Credit risk and change in value

On 1 January 20X6, an entity issues a five-year bond with a par value of £200,000, and an annual fixed coupon rate of 7%. The coupon rate reflects the market SONIA rate and the credit spread associated with the bond at the time of the issue. At the time of the issue SONIA was 5%, implying a credit spread of 2%.

The price of the bond will subsequently change either due to change in SONIA (market risk) or due to a change in the credit spread (credit risk).

Thus a change in the fair value of the bond attributed to credit risk can be calculated by subtracting from the total change in the fair value the changes due to market risk (ie, due to changes in SONIA).

Suppose that on 31 December 20X6, the value of the bond has decreased to £196,651, as the SONIA has increased to 5.25%. The yield to maturity for the bond has now risen to 7.50%. The credit spread has now increased to 2.25% implying deterioration in the credit quality of the bond.

In order to calculate the change in the value of the bond due to changes in SONIA alone, we shall calculate the fair value of the bond, at the new SONIA of 5.25% assuming that the credit spread has remained at 2%. This means that we need to discount the remaining four payments using a discount rate of 7.25%.

Using this discount factor produces:

$(14,000/1.0725) + (14,000/1.0725^2) + (14,000/1.0725^3) + (14,000 + 200,000/1.0725^4) =$ £198,316

Total change in market value (£200,000 – £196,651)	£3,349
Change in market value due to market risk (£200,000 – £198,316)	£1,684
Difference in value due to credit risk (£198,316 – £196,651)	£1,665

3.1.4 Reclassification

If an entity has reclassified a financial asset, previously measured at fair value as measured at cost or amortised cost or vice versa, it should **disclose the amount reclassified** into and out of each category and the **reason for that reclassification**.

3.1.5 Derecognition

An entity may have transferred financial assets in such a way that **part or all of the financial assets do not qualify for** derecognition. In such a case, the entity should **disclose** the following for each class of financial assets:

- The **nature** of the assets
- The nature of the **risksand rewards of ownership** to which the entity remains exposed
- When the entity continues to recognise all of the assets, the **carrying amounts** of the assets and of the associated liabilities
- When the entity continues to recognise the assets to the extent of its continuing involvement, the **total carrying amount of the original assets**, the **amount** of the assets that the entity **continues** to recognise, and the carrying amount of the **associated liabilities**

3.1.6 Collateral

An entity should disclose the following:

- The **carrying amount** of financial assets it has pledged as collateral for liabilities or contingent liabilities, including amounts that have been reclassified
- The **terms and conditions** relating to its pledge

When an entity holds collateral (of financial or non-financial assets) and is permitted to sell or re-pledge the collateral in the absence of default by the owner of the collateral, it shall disclose the following:

- The fair value of the collateral **held**
- The fair value of any such collateral **sold or re-pledged**, and whether the entity has an obligation to return it
- The **terms and conditions** associated with its use of the collateral

3.2 Profit or loss for the year

Items of income, expense, gains or losses

An entity should disclose the following items of income, expense, gains or losses either in the statement of profit or loss and other comprehensive income or in the notes:

(a) **Net gains or net losses** on:

(1) Financial assets or financial liabilities at fair value through profit or loss, showing separately those on financial assets or financial liabilities designated as such on initial recognition, and those on financial assets or financial liabilities that are classified as held for trading in accordance with IFRS 9

(2) Investments in equity instruments where gains are recognised in profit or loss for the period

(3) Held-to-maturity investments

(4) Loans and receivables

(5) Financial liabilities measured at amortised cost

(b) **Total interest income and total interest expense** (calculated using the effective interest method) for financial assets or financial liabilities that are not at fair value through profit or loss

(c) **Fee income and expense** (other than amounts included in determining the effective interest rate) arising from:

- financial assets or financial liabilities that are not at fair value through profit or loss; and
- trust and other fiduciary activities that result in the holding or investing of assets on behalf of individuals, trusts, retirement benefit plans and other institutions.

(d) **Interest income on impaired financial assets**

(e) **The amount of any impairment** loss for each class of financial asset

4 Other disclosures

Section overview

This section discusses additional quantitative and qualitative disclosures in the financial statements.

4.1 Accounting policies

An entity should disclose, in the summary of significant accounting policies, the measurement basis (or bases) used in preparing the financial statements and the other accounting policies used that are relevant to an understanding of the financial statements.

4.2 Hedge accounting

Hedge accounting is covered in later Chapters. The disclosures are dealt with in later Chapters.

4.3 Fair value

IFRS 7 retains the following general requirements in relation to the disclosure of fair value for those financial instruments **measured at amortised cost**:

- For each class of financial assets and financial liabilities an entity should disclose the **fair value of that class of assets and liabilities** in a way that permits it to be compared with its carrying amount.
- In disclosing fair values, an entity should group financial assets and financial liabilities into classes, but should **offset them only to the extent that their carrying amounts are offset in the statement of financial position**.

It also states that disclosure of fair value is not required where carrying amount is a reasonable approximation of fair value.

IFRS 13 provides disclosure requirements in respect of the fair value of financial instruments measured at fair values. It requires that information is disclosed to help users assess:

- For assets and liabilities measured at **fair value after initial recognition**, **the valuation techniques and inputs used to develop those measurements**
- For **recurring fair value measurements** (ie, those measured at each period end) using significant unobservable (Level 3) inputs, the **effect of the measurements on profit or loss** or other comprehensive income for the period

In order to achieve this, the following should be **disclosed as a minimum** for each class of financial assets and liabilities measured at fair value:

- The fair value measurement at the end of the period
- The level of the fair value hierarchy within which the fair value measurements are categorised in their entirety
- For assets and liabilities measured at fair value at each reporting date (recurring fair value measurements), the amounts of any transfers between Level 1 and Level 2 of the fair value hierarchy and reasons for the transfers
- For fair value measurements categorised within Levels 2 and 3 of the hierarchy, a description of the valuation techniques and inputs used in the fair value measurement, plus details of any changes in valuation techniques
- For recurring fair value measurements categorised within Level 3 of the fair value hierarchy:
  - A reconciliation from the opening to closing balances
  - The amount of unrealised gains or losses recognised in profit or loss in the period and the line item in which they are recognised
  - A narrative description of the sensitivity of the fair value measurement to changes in unobservable inputs
- For recurring and non-recurring fair value measurements categorised within Level 3 of the fair value hierarchy, a description of the valuation processes used by the entity

An entity should also disclose its policy for determining when transfers between levels of the fair value hierarchy are deemed to have occurred.



Context example: Fair value disclosures

For financial assets and liabilities measured at fair value at the end of the reporting period, IFRS 13 requires quantitative disclosures about the fair value measurements for each class of financial assets and liabilities.

An entity might disclose the following for financial assets and financial liabilities (excludes comparatives and further quantitative disclosures in relation to fair value hierarchy):

Description	31.12.X5	Fair value measurements at the end of the reporting period using		
		Level 1 inputs	Level 2 inputs	Level 3 inputs
	£'000	£'000	£'000	£'000
Trading portfolio assets	95	95		
Non-trading equity securities	73			73
Corporate bonds	175	25	150	
Designated at fair value	33		31	2
Investments in equity instruments	41	11	25	5
Derivatives – interest rate swaps	55		55	
Financial assets at fair value	472	131	261	80
Trading portfolio liabilities	60	40	20	
Designated at fair value	12		9	3
Derivatives – interest rate swaps	55		55	
Financial liabilities at fair value	127	40	84	3

5 Financial instruments risk disclosure



Section overview

An entity should disclose information that enables users of its financial statements to evaluate the nature and extent of risks arising from financial instruments to which the entity is exposed at the reporting date. These risks include, but are not limited to, credit risk, liquidity risk and market risk.

5.1 Types of risk

In undertaking transactions in financial instruments, an entity may assume or transfer to another party one or more different types of financial risk as defined below. The disclosures required by the standard show the extent to which an entity is exposed to these different types of risk, relating to both recognised and unrecognised financial instruments.

Credit risk	The risk that one party to a financial instrument will cause a financial loss for the other party by failing to discharge an obligation.
Currency risk	The risk that the fair value or future cash flows of a financial instrument will fluctuate because of changes in foreign exchange rates.
Interest rate risk	The risk that the fair value or future cash flows of a financial instrument will fluctuate because of changes in market interest rates.

Liquidity risk	The risk that an entity will encounter difficulty in meeting obligations associated with financial liabilities.
Loans payable	Loans payable are financial liabilities, other than short-term trade payables on normal credit terms.
Market risk	The risk that the fair value or future cash flows of a financial instrument will fluctuate because of changes in market prices. Market risk comprises three types of risk: currency risk, interest rate risk and other price risk.
Other price risk	The risk that the fair value or future cash flows of a financial instrument will fluctuate because of changes in market prices (other than those arising from interest rate risk or currency risk), whether those changes are caused by factors specific to the individual financial instrument or its issuer, or factors affecting all similar financial instruments traded in the market.
Past due	A financial asset is past due when a counterparty has failed to make a payment when contractually due.

5.2 Qualitative disclosures

For each type of risk arising from financial instruments, an entity must disclose the following:

(a) The **exposures** to risk and how they arise

(b) Its **objectives**, policies and processes for managing the risk and the methods used to measure the risk

(c) **Any changes** in (a) or (b) from the previous period

5.3 Quantitative disclosures

For each financial instrument risk, summary quantitative data about risk exposure must be disclosed. This should be based on the information provided internally to key management personnel. More information should be provided if this is unrepresentative.

Information about credit risk must be disclosed by class of financial instrument:

(a) **Maximum exposure** at the year end

(b) Any **collateral** pledged as security

(c) In respect of the amount disclosed in (b), a **description** of collateral held as security and other credit enhancements

(d) Information about the **credit quality** of financial assets that are neither past due nor impaired

(e) Financial assets that are **past due or impaired**, giving an age analysis and a description of collateral held by the entity as security

(f) **Collateral and other credit enhancements** obtained, including the nature and carrying amount of the assets and policy for disposing of assets not readily convertible into cash

For **liquidity risk**, entities must disclose the following:

- A **maturity analysis** of financial liabilities
- A **description** of the way risk is managed

Disclosures required in connection with **market risk**:

- **Sensitivity analysis**, showing the effects on profit or loss of changes in each market risk
- If the sensitivity analysis reflects interdependencies between risk variables, such as interest rates and exchange rates, the method, **assumptions and limitations** must be disclosed

5.3.1 Credit risk disclosures in more detail

Information about credit risk must be disclosed by class of financial instrument:

(a) **Maximum exposure** at the year end, which for a financial asset is the gross carrying amount, net of:

- any amounts offset in accordance with IAS 32; and
- any impairment losses recognised in accordance with IFRS 9.

The maximum exposure to credit risk at the reporting date should be without taking account of any collateral held or other credit enhancements (eg, netting agreements that do not qualify for offset in accordance with IAS 32).

(b) In respect of the amount disclosed in (a), a **description** of collateral held as security and other credit enhancements.

(c) Information about the **credit quality** of financial assets that are neither past due nor impaired.

(d) Financial assets that are **past due or impaired**, giving an age analysis and a description of collateral held by the entity as security.

(e) **Collateral and other credit enhancements** obtained, including the nature and carrying amount of the assets and policy for disposing of assets not readily convertible into cash.

(f) **Analysis of financial assets** that are **individually determined to be impaired** as at the reporting date, including the factors the entity considered in determining that they are impaired.

The **activities that give rise to credit risk** and the associated maximum exposure to credit risk include, but are not limited to the following:

- **Grant of loans** to customers and placing deposits with other entities: Maximum exposure to credit risk is the carrying amount of related financial assets.
- **Derivative contracts**: When the resulting financial asset is measured at fair value, the maximum exposure to credit risk at the end of the reporting period will equal the carrying amount.
- **Grant of financial guarantees**: Maximum exposure to credit risk is the maximum amount the entity could have to pay if the guarantee is called on, which may be significantly greater than the amount recognised as a liability.
- **Making a loan commitment** that is irrevocable over the life of the facility or is revocable only in response to a material adverse change: If the issuer cannot settle the loan commitment net in cash or another financial instrument, the maximum credit exposure is the full amount of the commitment.

Additional disclosures

These include disclosures such as the following:

- Reconciliation of loss allowances
- Explanation of how significant changes in the gross carrying amount of financial instruments during the period contributed to changes in loss allowance
- Inputs, assumptions and techniques in estimating 12-month and lifetime expected credit losses including how forward looking information has been incorporated into the determination of expected credit losses
- Separate disaggregation by credit risk rating grades of the gross carrying amount
- Information about collateral, modified financial assets and write offs still subject to enforcement activity

5.4 Sensitivity analysis: more detail

The standard requires that an entity should disclose a sensitivity analysis for each type of market risk to which the entity is exposed.

The standard gives **two options** regarding disclosure of the sensitivity analysis.

Option 1

An entity should disclose the following:

(a) Sensitivity analysis for each type of market risk to which the entity is exposed at the reporting date, showing how profit or loss and equity would have been affected by changes in the relevant risk variable that were reasonably possible at that date. Risk variables that are relevant to disclosing market risk include, but are not limited to:

- the yield curve of market interest rates, for example the SONIA rate. It may be necessary to consider both parallel and non-parallel shifts in the yield curve;
- foreign exchange rates;
- prices of equity instruments; and
- market prices of commodities.

(b) The methods and assumptions used in preparing the sensitivity analysis

(c) Changes from the previous period in the methods and assumptions used, and the reasons for such changes

The effect on profit or loss and equity of reasonably possible changes in the relevant risk variables may include changes in the prevailing interest rates for interest-sensitive instruments, and/or changes in currency rates for foreign currency financial instruments.

For interest rate risk, the sensitivity analysis might show separately the effect of a change in market interest rates on interest income and expenditure, on other items of profits and on equity.

Option 2

Alternatively, if an entity prepares a sensitivity analysis, such as value at risk, that reflects interdependencies between risk variables (eg, interest rates and exchange rates) **and uses it to manage financial risks**, it may use that sensitivity analysis in place of the analysis specified in the previous paragraph. In this case the entity should also disclose:

- an explanation of the method used in preparing such a sensitivity analysis, and of the main parameters and assumptions underlying the data provided; and
- an explanation of the objective of the method used and of limitations that may result in the information not fully reflecting the fair value of the assets and liabilities involved.

It is important for auditors to test this sensitivity analysis and document this on their files. For example, they should document how they gained comfort over the underlying assumptions and sensitivity analysis methodology. Auditing financial instruments is covered in later Chapters.

5.5 Other market risk disclosures

Other market risk is **any market risk which is not currency or interest rate risk**. When the disclosure of sensitivity analyses discussed above are unrepresentative of a risk inherent in a financial instrument (for example because the year-end exposure does not reflect the exposure during the year), the entity shall disclose that fact and the reason it believes the sensitivity analyses are unrepresentative.

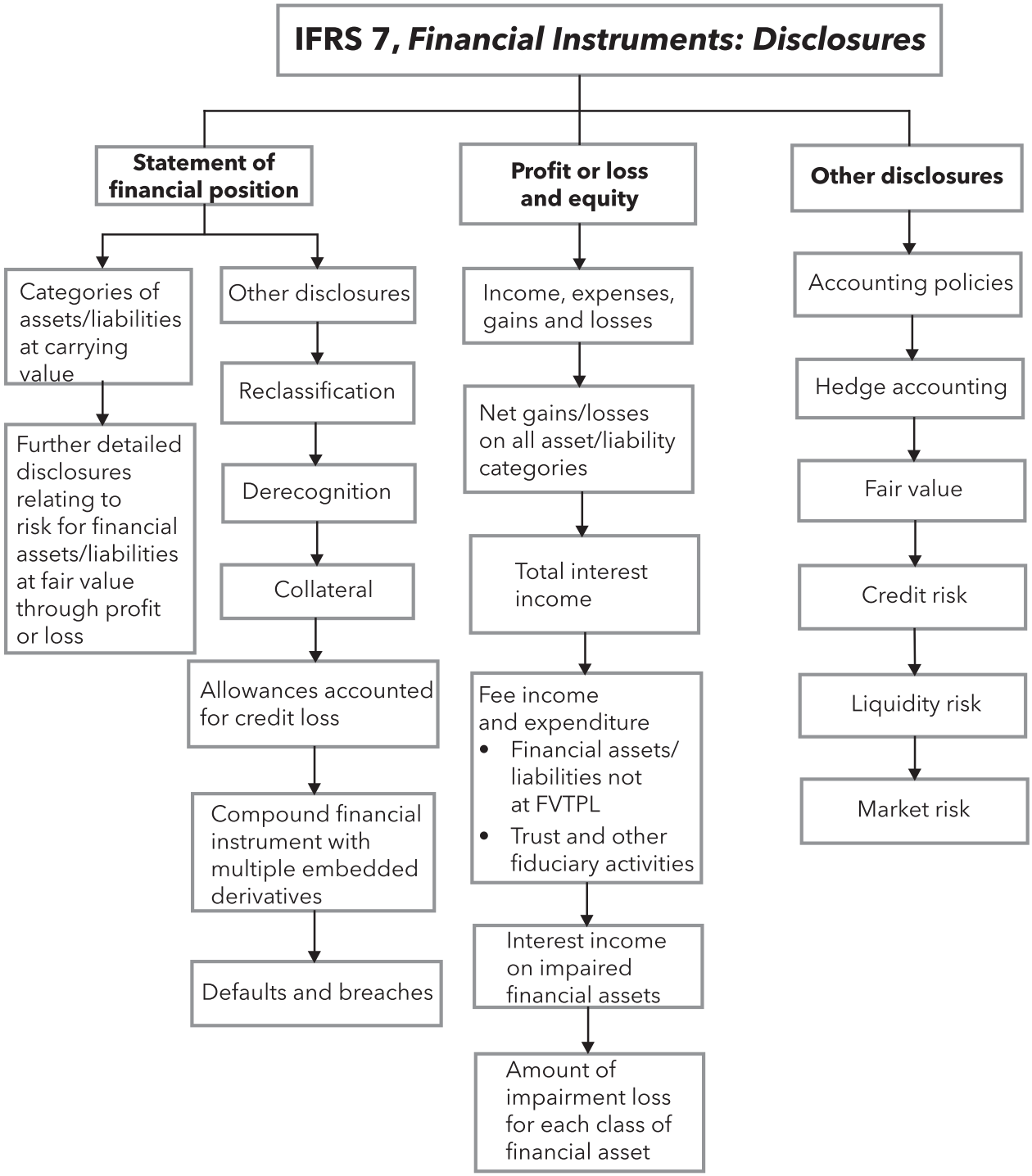
Professional skills focus: Assimilating and using information

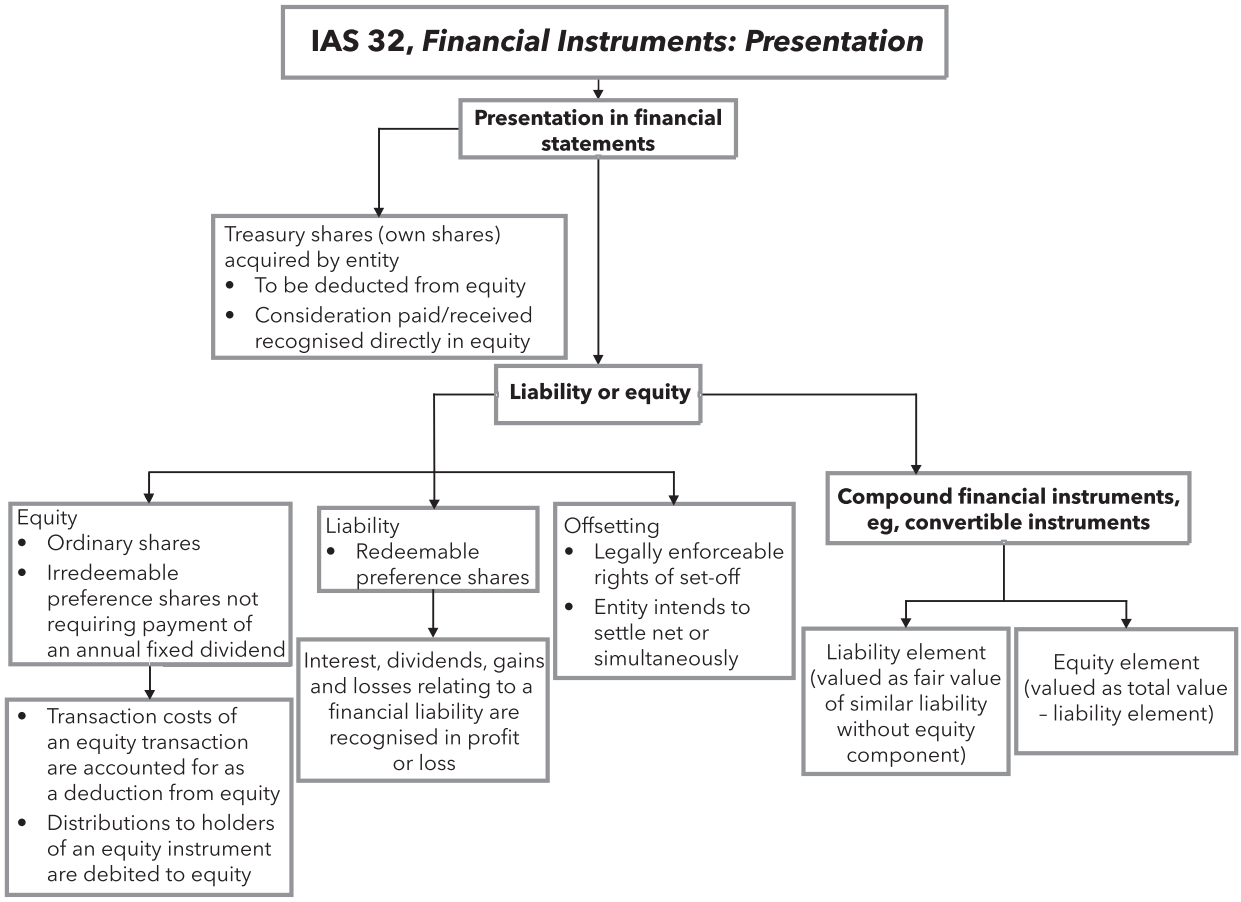
It can be seen that a great deal of information needs to be assimilated and applied in order to carry out a sensitivity analysis.

Professional skills focus: Concluding, recommending and communicating

Many financial instruments, particularly those held at fair value, are risky. Risk is not necessarily a negative, provided it is communicated to users and disclosures are transparent.

Summary


IFRS 7, Financial Instruments: Disclosures
Statement of financial position
Profit or loss and equity
Other disclosures
Categories of assets/liabilities at carrying value
Further detailed disclosures relating to risk for financial assets/liabilities at fair value through profit or loss
Other disclosures
Reclassification
Derecognition
Collateral
Allowances accounted for credit loss
Compound financial instrument with multiple embedded derivatives
Defaults and breaches
Income, expenses, gains and losses
Net gains/losses on all asset/liability categories
Total interest income
Fee income and expenditure
• Financial assets/ liabilities not at FVTPL
• Trust and other fiduciary activities
Interest income on impaired financial assets
Amount of impairment loss for each class of financial asset
Accounting policies
Hedge accounting
Fair value
Credit risk
Liquidity risk
Market risk


IAS 32, Financial Instruments: Presentation
Presentation in financial statements
Treasury shares (own shares) acquired by entity
• To be deducted from equity
• Consideration paid/received recognised directly in equity
Liability or equity
Equity
• Ordinary shares
• Irredeemable preference shares not requiring payment of an annual fixed dividend
• Transaction costs of an equity transaction are accounted for as a deduction from equity
• Distributions to holders of an equity instrument are debited to equity
Liability
• Redeemable preference shares
Interest, dividends, gains and losses relating to a financial liability are recognised in profit or loss
Offsetting
• Legally enforceable rights of set-off
• Entity intends to settle net or simultaneously
Compound financial instruments, eg, convertible instruments
Liability element (valued as fair value of similar liability without equity component)
Equity element (valued as total value – liability element)

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	What issues arise when distinguishing debt and equity? (Topic 1)
2.	IFRS 7 requires certain disclosures relating to risk. What are the three main types of risk identified by the standard? (Topic 2)
3.	What disclosures does IFRS 7 require in relation to loans and receivables at fair value through profit or loss? (Topic 3)
4.	What are IFRS 7's minimum disclosure requirements in respect of the fair value of financial instruments measured at fair value, on recognition and subsequently? (Topic 4)
5.	In relation to risk, can you distinguish qualitative disclosures and quantitative disclosures? (Topic 5)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Short-answer question	This is a quick test of knowledge of IFRS 7 disclosures.
Warburton	This is revision of the IAS 32 requirement regarding convertible bonds.
Erubus	This is a further exercise in separating the liability and equity components, but slightly more complex.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted the self-test questions, you can continue your studies by moving onto the next chapter. In later chapters, we will recommend questions from the Question Bank for you to attempt.

Technical reference

1 IAS 32, *Financial Instruments: Presentation*

Presentation of equity and liabilities

- Classification as financial asset, financial liability or equity instrument – **IAS 32.15–16**
- Definitions – **IAS 32.11**
- Contractual obligation and substance of instrument – **IAS 32.17–18**
- Settlement options – **IAS 32.26–27**
- Treasury shares – **IAS 32.33–34**
- Interest, dividends, losses and gains – **IAS 32.35–36**
- Offsetting – **IAS 32.42**

Compound instruments

- Recognising liability and equity elements – **IAS 32.28**
- Example of convertible bonds – **IAS 32.29–30**
- Calculation of liability and equity elements – **IAS 32.31–32**

2 IFRS 7, *Financial Instruments: Disclosures*

- Statement of financial position disclosures – **IFRS 7.8–19**
- Statement of profit or loss and other comprehensive income and statement of changes in equity disclosures – **IFRS 7.20**
- Nature and extent of risks arising from financial instruments
  - Purpose of disclosures – **IFRS 7.31–32**
  - Qualitative disclosures – **IFRS 7.33**

Self-test questions

Answer the following questions.

1 Disclosure

What is the main objective of the disclosure requirements of IFRS 7?

2 IFRS 7

How does IFRS 7 define the following?

2.1 Liquidity risk

2.2 Market risk

3 Sensitivity analysis

Why does IFRS 7 require entities to disclose sensitivity analysis to market risk?

4 Warburton

The Warburton Company issued £10 million of convertible bonds at par on 31 December 20X7. Interest is payable annually in arrears at a rate of 7%. The bonds are redeemable on 31 December 20X9. The bonds can be converted at any time up to maturity into 12.5 million ordinary shares.

At the time of issue, the market interest rate on debt with a similar credit status and the same cash flows, but without conversion rights, was 10% per annum.

Requirement

What carrying amount should be recognised for the liability in the statement of financial position of Warburton at 31 December 20X7 in respect of the convertible bond, in accordance with IAS 32, *Financial Instruments: Presentation*?

5 Erubus

The Erubus Company issued £15 million of 6% convertible bonds at par on 31 December 20X7. The bonds are redeemable at 31 December 20Y1. The bonds can be converted by their holders any time up to maturity into ordinary shares of Erubus.

At 31 December 20X7 the present value of the future capital and interest payments discounted at the prevailing market interest rate for similar bonds without the conversion rights is £13 million.

The transaction costs directly attributable to the issue of the convertible bonds were £400,000. These costs are deductible against Erubus's taxable profits. Erubus's tax rate is 25%.

Requirement

What increase in equity should be recognised in the statement of financial position of Erubus at 31 December 20X7 as a result of the issue of the convertible bonds, in accordance with IAS 32, *Financial Instruments: Presentation*?

Note: Some of the requirements of IAS 32 relate to derivatives and embedded derivatives. These will be tested after you have covered those topics in Chapter 16.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Refer to the definitions of financial assets and liabilities.

- **Physical assets**: Control of these creates an opportunity to generate an inflow of cash or other assets, but it does not give rise to a present right to receive cash or other financial assets.
- **Prepaid expenses, etc**: The future economic benefit is the receipt of goods/services rather than the right to receive cash or other financial assets.

Answer to Interactive question 2

Liability. The preference shares require regular distributions to the holders but more importantly have the debt characteristic of being redeemable. Therefore, according to **IAS 32**, *Financial Instruments: Presentation* they must be classified as liability.

Answer to Interactive question 3

The convertible bond is not a compound financial instrument, as it is not settled in a fixed amount of shares. It should instead be wholly classified as a liability.

The convertible redeemable preference shares are compound instruments. They have a financial liability component, as there is an obligation to deliver cash through dividends and on redemption in five years' time to deliver either cash or equity instruments (through the holder's right to convert into equity).

Answer to Interactive question 4

The liability component is computed as the present value of the maximum potential cash flows discounted at 10%.

Time	Cash flow	Discount factor @ 10%	PV
	£		£
1	4,000	0.909	3,636
2	54,000	0.826	44,604
			48,240

The equity component of the gross proceeds is therefore (£50,000 – £48,240) £1,760.

The issue costs of £1,000 are split in the ratio 48,240:1,760 ie, £965 is netted against the liability and £35 is netted against the equity.

The entries are therefore:

		£	£
DEBIT	Cash	50,000	
CREDIT	Liability		48,240
CREDIT	Equity		1,760

and

CREDIT	Cash		1,000
DEBIT	Liability	965	
DEBIT	Equity	35	

The net liability initially recognised is £47,275. This is then amortised to £50,000 over the next 2 years at an effective interest rate of 11.19% (the IRR of the cash flows of £47,275, –£4,000 and –£54,000) as follows:

Year	B/fwd	Interest expense at 11.19%	Cash flow	C/fwd
	£	£	£	£
1	47,275	5,290	(4,000)	48,565
2	48,565	5,435	(4,000)	50,000

Answer to Interactive question 5

The contract is a financial asset or financial liability, even though the entity must settle it by issuing its own equity. It is not an equity instrument, as it is settled using a variable number of the entity's own equity instruments.

Answer to Interactive question 6

Offsetting is not appropriate.

Each of the financial instruments has its own terms and conditions and may be transferred or settled separately. The risks of the financial instruments are different.

They should not be offset unless they meet the criteria in IAS 32 (legal right of set-off and intention to settle net), which is unlikely. However, disclosure of the relationship between the two financial instruments would provide useful information to users of the financial statements.

Answer to Interactive question 7

Emporium's finance director is incorrect. A financial liability for the present value of the maximum amount payable to shareholders should be recognised in the financial statements as of 31 October 20X8. At 31 October 20X8, the rights are equivalent to a written put option because they represent for Emporium a purchase obligation which gives shareholders the right to sell the entity's own equity instruments for a fixed price. The fundamental principle of IAS 32 *Financial Instruments: Presentation* is that a financial instrument should be classified as either a financial liability or an equity instrument according to the substance of the contract, not its legal form, and the definitions of financial liability and equity instrument. IAS 32 states that a contract which contains an entity's obligation to purchase its own equity instruments gives rise to a financial liability, which should be recognised at the present value of its redemption amount. IAS 32 also states that a contractual obligation for an entity to purchase its own equity instruments gives rise to a financial liability for the present value of the redemption amount even if the obligation is conditional on the counterparty exercising a right to redeem, as is the case with the bonus issue of Emporium.

Answers to Self-test questions

1 Disclosure

The main objective of the disclosure requirements of IFRS 7 is to show the significance of financial instruments for an entity's financial position and financial performance and qualitative and quantitative information about exposure to risks arising from financial instruments.

2 IFRS 7

2.1 **Liquidity risk** is defined as the risk that an entity will encounter difficulty in meeting the obligations associated with its financial liabilities.

2.2 **Market risk** is the risk that the fair value or future cash flows of a financial instrument will fluctuate because of changes in market prices.

3 Sensitivity analysis

Sensitivity analysis helps users of financial statements to evaluate the effect of possible changes in the entity's financial position and financial performance due to changes in market risk factors.

4 Warburton

£9,479,339

IAS 32.28 requires the separation of the compound instrument into its liability and equity elements. IAS 32.31 and .32 explain how this separation should be made. IAS 32.AG30 – AG35 explain the application of this principle.

Thus the liability is (£0.7m/1.10) + (£10.7m/1.10^2) = £9,479,339.

5 Erubus

£1,960,000

IAS 32.28 requires the separation of a compound instrument into liability and equity elements where this is appropriate.

IAS 32.35 and 32.37 require that transaction costs of an equity transaction shall be deducted from equity net of tax.

IAS 32.38 requires that transaction costs directly attributable to a compound financial instrument should be allocated to the liability and equity components in proportion to the allocation of the proceeds.

Liability component of gross proceeds: £13m

Equity component of gross proceeds: £2m

Issue costs are £400,000, allocated:

Liability component (13/15) £346,667

Equity component (2/15) £53,333

The initial liability recognised is therefore (£13m – £346,667) £12,653,333.

The equity component is (£2m – (£53,333 – 25% tax relief on £53,333) £1,960,000

Note that tax relief on the issue costs allocated to the liability component will be given, as they are amortised against profit or loss. Initially they attract no relief and so there is no tax adjustment for them in the original allocation of the issue costs.

Requires the separation of a compound instrument into liability and equity elements where this is appropriate.

IAS 32.35 and 32.37 require that transaction costs of an equity transaction shall be deducted from equity net of tax.

IAS 32.38 requires that transaction costs directly attributable to a compound financial instrument should be allocated to the liability and equity components in proportion to the allocation of the proceeds.

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Equity component of gross proceeds: £2m

Issue costs are £400,000, allocated:

Liability component (13/15) £346,667

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The initial liability recognised is therefore (£13m – £346,667) £12,653,333.

The equity component is (£2m – (£53,333 – 25% tax relief on £53,333) £1,960,000

Note: Tax relief on the issue costs allocated to the liability component will be given, as they are amortised against profit or loss. Initially they attract no relief and so there is no tax adjustment for them in the original allocation of the issue costs.

Chapter 16

Financial instruments: recognition and measurement

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Introduction and overview of earlier studies

2 Recognition, classification and derecognition

3 Measurement

4 Credit losses (impairment)

5 Application of IFRS 13 to financial instruments

6 Derivatives and embedded derivatives

7 Current developments

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Identify and explain current and emerging issues in corporate reporting
- Determine and calculate how different bases for recognising, measuring and classifying financial assets and financial liabilities can impact upon reported performance and position
- Evaluate the impact of accounting policies and choice in respect of financing decisions for example hedge accounting and fair values
- Explain and appraise accounting standards that relate to an entity's financing activities which include: financial instruments; leasing; cash flows; borrowing costs; and government grants

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**Introduction and overview of earlier studies** Financial instruments, especially derivatives, can significantly change the risk profile of organisations. The IASB has therefore paid a great deal of attention to developing IFRS 9, *Financial Instruments*	**Approach** Although you have studied IFRS 9 at Professional Level, the subject is treated in much more depth and complexity. It is essential, therefore that you have a clear grasp of the basics, so do not skip or skim read this section. **Stop and think** Section 1.2 is a 'stop and think' section, stepping back before looking at the detail.	All the topics covered in this section are assumed knowledge and can be tested in the exam.	**IQ3: Loan** This is a very straightforward question. If you struggle with it, you need to go back to your material from earlier studies and revise.
2	**Recognition, classification and derecognition** This section builds on your previous	**Approach** Note that IFRS 9 requires that financial assets are classified as	This is a key area and frequently tested. You could get a financial reporting	**IQ4: Contractual cash flows and selling financial assets** While this question

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	studies, which only dealt with financial assets and liabilities at amortised cost.	measured at either: • amortised cost; or • fair value through other comprehensive income; or • fair value through profit or loss. Your earlier studies only dealt with amortised cost. **Stop and think** What is the basis for classification?	question requiring adjustments to the financial statements with a substantial number of adjustments based around financial instruments. You must know the business model test in detail, as it could easily come up.	is only short it is essential you do it as it does test whether you have understood the issue.
3	**Measurement** This section deals with initial measurement (cost = fair value), the treatment of transaction costs and subsequent measurement. Subsequent measurement depends on how the instrument has been classified on initial recognition (amortised cost, FVTPL or FVTOCI).	**Approach** Work through this section very carefully, paying particular attention to the worked examples, especially the debt instrument and FVTOCI. With liabilities make sure you fully understand section 3.9.2 on credit risk, which may seem counterintuitive. **Stop and think** How are transaction costs recognised?	Any of the topics in the worked examples could come up in the exam and will be of the same level of difficulty.	**IQ8: Investment in listed shares** Use the summary table in paragraph 3.6 to help you with this and other questions. This table will be useful in revising for your exam.
4	**Credit losses (impairment)** This deals with the complex area of credit losses (impairment). IFRS 9 uses an expected loss model, which is a change from the earlier incurred loss model.	**Approach** Make sure you understand what is meant by: • credit loss • expected credit loss • lifetime expected credit loss • 12 month expected credit loss	This area could be examined within the context of a scenario question requiring adjustments to the financial statements.	**IQ12: Modification** This deals with a situation where the terms of the loan are modified and credit risk increases.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		Stop and think What is meant by Stages 1, 2 and 3?		
5	**Application of IFRS 13 to financial instruments** IFRS 13 works its way into many standards, particularly IFRS 9, because so many financial instruments are measured at fair value.	**Approach** Skim quickly through para. 5.1, which should be familiar to you. Focus more on fair value of financial assets (5.2 and financial liabilities). **Stop and think** The approach to fair valuing financial liabilities and own equity shares may differ from financial assets.	The interaction of IFRS 9 and IFRS 13 has been tested in depth, for example, the valuation of unquoted shares, and also in the context of quoted shares, for example in July 2020.	**IQ13: Fair value of liability** This question requires fair valuing a liability in the books of two companies with different credit ratings.
6	**Derivatives and embedded derivatives** Derivatives are financial instruments whose value changes in response to a change in the value of an underlying security, commodity, currency, index or other financial instrument(s). They normally require a zero, or small, initial net investment and are settled at a future date.	**Approach** Work carefully through section 6 on derivatives including all the worked examples, interactive questions and illustrations. Note in particular the need to separate the host instrument from the embedded derivative, unless the host contract is a financial asset within the scope of IFRS 9, in which case the whole contract is measured at fair value through profit or loss. **Stop and think** What instruments qualify as	Derivatives commonly examined include swaps and options.	**IQ15: Embedded derivatives** This question has two embedded derivatives, one which needs to be separated from the host contract and one that doesn't. There is a useful summary diagram in paragraph 6.6 that will guide you with this.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		derivatives for accounting purposes? What is the accounting treatment of derivatives?		
7	**Current developments** Now that IFRS 9 is fully examinable, there is less going on in the area of current developments, the main one being financial instruments with the characteristics of equity.	**Approach** Skim read this for background only. **Stop and think** Why is the debt/equity distinction important?	This will not be examined in detail as it is only at the Discussion Paper stage.	N/A

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Introduction and overview of earlier studies

Section overview

This section gives a chapter overview and summarises the material covered at Professional Level in order to consolidate student knowledge before more advanced issues are covered.

- **IFRS 9, *Financial Instruments***
  - Financial assets and financial liabilities are classified on initial recognition. This classification drives subsequent measurement of the instruments.
  - Financial assets are classified as either measured at amortised cost, fair value through other comprehensive income or fair value through profit or loss.
  - Reclassifications are permitted only if there is a change in the entity's business model for holding the financial asset.
  - The financial statements should reflect the general pattern of deterioration or improvement in the credit quality of financial instruments within the scope of IFRS 9. The impairment model in IFRS 9 is based on the premise of providing for expected losses.
- **IAS 39, *Financial Instruments: Recognition and Measurement***
  - The IAS 39 rules on hedging may still be applied, and are therefore covered in next Chapter.

1.1 Introduction

The purpose of this chapter is to provide thorough coverage of the accounting treatment of financial instruments. The main presentation and disclosure requirements as detailed in IAS 32, *Financial Instruments: Presentation* and IFRS 7, *Financial Instruments: Disclosures* together with certain aspects of recognition and measurement were covered at Professional Level and revisited in earlier chapter. This chapter extends the coverage of recognition and derecognition of financial assets and liabilities, and their initial and subsequent measurement and impairment, and finally discusses particular issues relating to the definition of derivatives and the accounting treatment of derivatives and embedded derivatives.

1.2 What are financial instruments?

Before looking at formal definitions and technical details, it is worth stepping back and thinking clearly about what financial instruments are and what they are for. We can start by demystifying some of the definitions:

Financial asset: an asset can be defined in simple terms as something you own or a favour that you can call in; similarly a financial asset can be defined as money that you own, or are owed. You can also own shares in a company. You have paid for the shares, and the company you have invested in owes you something of monetary value in return.

Financial liability: the party who owes you the money has a liability equal to your financial asset.

Equity investment: if the company is solvent, this is your share of the amount left over after deducting the liabilities from the assets.

It follows from the above that a **financial asset of one entity** is **either** a **financial liability or equity of another entity.** This is reflected in the formal definition of a financial instrument: "**A financial instrument** is any **contract** that gives rise to a **financial asset of one entity and a financial liability or equity instrument of another entity**" (IAS 32: para. 11). The new word in

the formal definition is **contract:** this is what distinguishes financial instruments from other assets and liabilities.

Types of financial instruments: financial instruments can be divided into three basic types according to the rights and obligations they carry:

- Debt instruments (including receivables)
- Equities (for example shares)
- **Derivatives:** these are contracts with negligible or zero initial net value and subsequent fair value changes depending on the value of the underlying assets. Derivatives can be **either assets or liabilities** depending on whether there is a gain or loss on the contract.

Two further terms are worth introducing at this stage, as they are not particularly intuitive.

Bond holder = Lender (Imagine, if it helps, holding in your hand the contract in which the debtor agrees to pay you back.)

Bond issuer = Borrower (As with shares, a bond is issued to raise cash. You owe this to the lender, who is holding your contract.)

Finally, a term that comes up in many definitions concerning financial instruments is **accounting mismatch.** This is a measurement or recognition inconsistency that would otherwise arise from measuring assets or liabilities or recognising gains or losses on them on different bases. IFRS® Standards occasionally have rules which enable an accounting mismatch to be eliminated, for example hedging, or designating a financial asset at fair value through profit or loss if this would eliminate the mismatch.

1.2.1 Summary: features of financial instruments

Category	Rights or obligations	Asset or (liability) for:	Example
Debt	Defined amount and timing	Lender/holder (Borrower/payer)	Bond Receivable
Equity	Residual rights (in assets less liabilities), obligation only while solvent	Investor/shareholder (No liability within scope of IFRS 9)	Share
Derivative	Fair value of underlying	Gain (Loss)	Options, swaps, futures

1.2.2 Summary: recognition of financial assets and liabilities

The category of financial instrument and the rights and/or obligations associated with it, determine how it is measured on recognition and subsequently. We will look at recognition and measurement in detail below. For now, here are summaries to relate to the above table. As we have seen, debt instruments and derivatives can be financial assets or financial liabilities, but an equity instrument can only be a financial asset:

Financial assets

		Initial measurement (IFRS 9: para. 5.1.1)	Subsequent measurement (IFRS 9: paras. 4.1.2–4.1.5, 5.7.5)
1	**Investments in debt instruments** Business model approach: (a) Held to collect contractual cash flows; and cash flows are solely principal and interest	Fair value + transaction costs	Amortised cost
	(b) Held to collect contractual cash flows and to sell; and cash flows are solely principal and interest	Fair value + transaction costs	Fair value through other comprehensive income (with reclassification to profit or loss (P/L) on derecognition) NB: interest revenue calculated on amortised cost basis recognised in P/L
2	**Investments in equity instruments not 'held for trading'** (optional irrevocable election on initial recognition)	Fair value + transaction costs	Fair value through other comprehensive income (no reclassification to P/L on derecognition) NB: dividend income recognised in P/L
3	**All other financial assets** (and any financial asset if this would eliminate or significantly reduce an ' accounting mismatch')	Fair value (transaction costs expensed in P/L)	Fair value through profit or loss

Financial liabilities

		Initial measurement (IFRS 9: para. 5.1.1)	Subsequent measurement (IFRS 9: para. 4.2.1)
1	**Most financial liabilities** (eg, trade payables, loans, preference shares classified as a liability)	Fair value less transaction costs	Amortised cost
2	**Financial liabilities at fair value through profit or loss** • 'Held for trading' (short-term profit making) • Derivatives that are liabilities • Designated on initial recognition at 'fair value through profit or loss' to eliminate/significantly reduce an 'accounting mismatch'	Fair value (transaction costs expensed in P/L)	Fair value through profit or loss

	• A group of financial liabilities (or financial assets and financial liabilities) managed and performance evaluated on a fair value basis in accordance with a documented risk management or investment strategy		

Professional skills focus: Assimilating and using information

In exam questions on financial instruments at Advanced Level, you will often be presented with a great deal of seemingly complex information, which you must analyse. It helps to have the principles in the above summary in mind or to hand in order to avoid getting lost or overwhelmed.

Professional skills focus: Structuring problems and solutions

The above summary may also play a useful role in structuring problems. What is the nature and purpose of the financial instrument? How is it valued?

1.2.3 Further points forward from Professional Level

(a) A **compound financial instrument** (that is, one that has features of both debt and equity) should be **split into its component parts** according to their substance at the date that it is issued.

(b) **Interest, dividends, losses or gains** relating to a financial instrument (or a component) that is a financial liability should be **recognised** as income or expense **in profit or loss**.

(c) **Dividend distributions** paid to holders of an equity instrument should be debited **directly to equity, net of any related income tax benefit**. These should be presented in the statement of changes in equity.

(d) **Financial assets and financial liabilities** should generally be **presented as separate items** in the statement of financial position. No offsetting is allowed except where it is required because an entity has a legally enforceable right to set off recognised amounts and the entity intends to settle on a net basis, or to realise the asset and settle the liability simultaneously.

1.3 Practical implications for key ratios

The type of financial instrument, how it is measured and where changes in value are measured have important effects on key ratios. The classification of financial instruments as debt versus equity is particularly important with items that are financial liabilities or equity as the presentation of the two items and associated financial effects are very different.

If a financial instrument is classified as a financial liability (debt) it will be reported within current or non-current liabilities. Non-current liabilities are relevant in determining an entity's gearing ie, the proportion of debt finance versus equity finance of a business, and therefore risk to ordinary equity holders.

Distributions relating to instruments classified as financial liabilities are categorised as finance cost, having an impact on reported profitability.

If a financial instrument is classified as equity, reported gearing will be lower than if it were classified as a financial liability. However, classification as equity is sometimes viewed negatively as it can be seen as a dilution of existing equity interests.

Distributions on instruments classified as equity are charged to equity and therefore do not affect reported profit.

If a company elects for changes in value on an equity instrument to be recorded in other comprehensive income, rather than profit or loss, this will affect key profitability ratios. The requirement for such an election to be irrevocable prevents manipulation by booking gains to profit or loss one year and losses to other comprehensive income in the following year.

Classification of instruments can also have financial implications for businesses. For example, debt covenants on loans from financial institutions often contain clauses that reported gearing cannot exceed a stated figure, with penalties or call-in clauses if it does. Companies with high gearing may also find it harder to get financing or financing may be at a higher interest rate.

High gearing is particularly unpopular in the current economic climate and there have been high profile cases of companies that have been pressed to sell off parts of their business to reduce their 'debt mountains' eg, Telefónica SA, the Spanish telecoms provider, selling O_2 Ireland to Hutchison Whampoa (the owner of the '3' telephone network).



Interactive question 1: Classification

For each of the below instruments, identify whether it should be classified as a financial liability or as part of equity, explaining the reason for your choice.

1.1 Redeemable preference shares with a coupon rate of 8%. The shares are redeemable on 28 February 20X9 at a premium of 10%.

1.2 A grant of share options to senior executives. The options may be exercised from 28 February 20X8.

See **Answer** at the end of this chapter.



Interactive question 2: Transactions covered by IFRS 9

Should the following be recognised under IFRS 9?

2.1 A guarantee to replace or repair goods sold by a business in the normal course of business

2.2 A firm commitment (order) to purchase a specific quantity of cocoa beans for use in manufacturing

2.3 A forward contract to purchase cocoa beans at a specified price and quantity on a specified date

See **Answer** at the end of this chapter.

1.4 Amortised cost: revision

The **amortised cost** category applies to an asset held within a business model whose objective is to hold financial assets in order to collect contractual cash flows and where the contractual terms of the asset give rise to cash receipts of interest and capital on specific dates. This applies to most debt instruments. Such assets are measured at **amortised cost** using the **effective interest method**.

Amortised cost is:

- the initial amount recognised for the financial asset
- less any repayments of the principal sum
- plus any amortisation

The amount of amortisation should be calculated by applying the effective interest method to spread the financing cost (that is the difference between the initial amount recognised for the financial asset and the amount receivable at maturity) over the period to maturity. The amount amortised in respect of a financial asset should be recognised as income in profit or loss.

Definition

Effective interest rate: The rate that exactly discounts estimated future cash payments or receipts through the expected life of the instrument or, when appropriate, a shorter period to the net carrying amount of the financial asset or financial liability.

You may be required to calculate the effective interest rate in the exam, so it is important that you study the worked example below carefully.

Worked example: Effective interest rate

An entity acquires a 6% coupon bond paying interest on an annual basis; it is redeemable in four years' time at its par value of £100. The current market price of this bond is £105 per £100 nominal value.

Requirement

Calculate the effective interest rate of this bond.

Solution

The RATE excel function is used to calculate the effective interest rate of a bond. To calculate the RATE, the following variables need to be input to the RATE function:

Nper = the number of periods

Pmt = the amount (of interest) paid in any single period

Pval = the present value of the asset (its market price), **inserted as a negative number**

Fval = the future value (the amount paid at maturity). Type and guess **can be left blank.**

This will give the effective interest rate (known as the yield to maturity) over a given period.

Alternatively, type **=RATE(** to begin the function entry then insert the number of periods, the payment made in each period, the present value (as a negative value) and the future value, and end by inserting a closed bracket).

=RATE(B1,B2,B3,B4)

	A	B
1	Nper = the number of periods	4
2	Pmt = the amount (of interest) paid in any single period	6
3	Pval = the present value of the asset (its market price)	–105

4	Fval = the future value (the amount at maturity)	100
5	Effective interest rate	0.046

In the Corporate Reporting exam, the amount amortised in respect of a financial asset should be recognised as income in profit or loss using the effective rate of interest (here 4.6%) as this represents the true return on the bond to maturity. The 6% coupon rate in this example is not the true return on the bond as it does not take account of the difference between the initial amount recognised for the financial asset (£105) and the amount receivable at maturity (£100).



Worked example: Amortised cost

An entity acquires a zero coupon bond with a nominal value of £20,000 on 1 January 20X6 for £18,900. The bond is quoted in an active market and broker's fees of £500 were incurred in relation to the purchase. The bond is redeemable on 31 December 20X7 at a premium of 10%. The effective interest rate on the bond is 6.49%.

Requirement

Set out the journals to show the accounting entries for the bond until redemption if it is classified as being held at amortised cost. The entity has a 31 December year end.

Solution

On 1 January 20X6

		£	£
DEBIT	Financial asset (£18,900 plus £500 broker fees)	19,400	
CREDIT	Cash		19,400

On 31 December 20X6

		£	£
DEBIT	Financial asset (£19,400 × 6.49%)	1,259	
CREDIT	Interest income		1,259

On 31 December 20X7

		£	£
DEBIT	Financial asset ((£19,400 + £1,259) × 6.49%)	1,341	
CREDIT	Interest income		1,341
DEBIT	Cash	22,000	
CREDIT	Financial asset		22,000

1.4.1 Loan

A financial asset classified as a loan should also be measured at amortised cost using the effective interest method.

Amortisation should be recognised as income in profit or loss.

Most financial assets that meet this classification are simple receivables and loan transactions.

Interactive question 3: Loan

Hallowes plc has agreed to lend a customer £9,500 on 1 January 20X2 subject to the following terms:

- The loan is repaid on 31 December 20X4 in full.
- Three interest payments of £1,000 are paid on 31 December each year.

Hallowes plc incurred £250 of legal fees in agreeing the loan documentation with the customer. The effective rate of interest on the loan is 9.48%.

Requirement

Demonstrate by journal entries how the loan should be recorded in the financial statements of Hallowes plc for the year ended 31 December 20X2 and subsequent years.

See **Answer** at the end of this chapter.

2 Recognition, classification and derecognition

Section overview

- This section deals with recognition, classification and derecognition of financial assets and financial liabilities.
- IFRS 9 requires an entity to recognise financial assets and financial liabilities when it **becomes a party** to the contractual provisions of the instrument rather than when the contract is settled.
- IFRS 9 requires that financial assets are **classified as measured** at either:
  - **amortised cost**;
  - **fair value through other comprehensive income**; or
  - **fair value through profit or loss**.

Subsequent measurement depends on the category into which financial assets and financial liabilities are classified on origination.

- There is an **option to designate** a financial asset **at fair value through profit or loss** to **reduce or eliminate an 'accounting mismatch'** (measurement or recognition inconsistency).
- Financial assets are measured at **amortised cost** if: the asset is held within a business model whose objective is to collect contractual cash flows; and the cash flows are solely payments of principal and interest on the principal amount outstanding.
- Holdings of debt instruments are measured at **fair value through other comprehensive income** if: the asset is held within a business model whose objective is achieved by both collecting contractual cash flows and selling financial assets; and cash flows are solely payments of principal and interest on the principal amount outstanding.
- Financial **liabilities** are classified as being measured at **fair value through profit or loss**, or **amortised cost**.
- Reclassification of financial assets **is permitted only if the business model within which they are held changes**.
- Financial assets are derecognised when the **contractual rights** to the cash flows **expire** or the entity **passes** substantially all the **risks and rewards** of ownership to another party.

Rights and obligations are recognised to reflect continuing involvement with the financial asset that has been transferred. Financial liabilities are derecognised when they are **extinguished** ie, when they are discharged, expired or cancelled.

- Regular way purchase and sale transactions are recognised using either **trade date** accounting or **settlement date** accounting for financial assets.

2.1 Introduction

When an entity first recognises a financial asset or financial liability, it must classify it into an appropriate category. This classification determines how the financial instrument will be subsequently measured.

A financial asset or financial liability should be **derecognised**, that is removed, from an entity's statement of financial position, **when the entity ceases to be a party to the financial instrument's contractual provisions**.

2.2 IFRS 9 definitions

Before we look in more detail at recognition, classification and derecognition, here are the IFRS 9 definitions that are applicable.

Definitions

Financial asset: Any asset that is:

- cash;
- an equity instrument of another entity;
- a contractual right:
  - to receive cash or another financial asset from another entity; or
  - to exchange financial assets or financial liabilities with another entity under conditions that are potentially favourable to the entity; or
- a contract that will or may be settled in the entity's own equity instruments and which is:
  - a non-derivative for which the entity is or may be obliged to receive a variable number of the entity's own equity instruments; or
  - a derivative that will or may be settled other than by exchange of a fixed amount of cash or another financial asset for a fixed number of the entity's own equity instruments. For this purpose the entity's own equity instruments do not include instruments that are themselves contracts for the future receipt or delivery of the entity's own equity instruments.

Financial liability: Any liability that is:

- a contractual obligation:
  - to deliver cash or another financial asset to another entity; or
  - to exchange financial assets or financial liabilities with another entity under conditions that are potentially unfavourable to the entity; or
- a contract that will or may be settled in the entity's own equity instruments and which is:
  - a non-derivative for which the entity is or may be obliged to deliver a variable number of the entity's own equity instruments; or
  - a derivative that will or may be settled other than by exchange of a fixed amount of cash or another financial asset for a fixed number of the entity's own equity instruments. For this purpose the entity's own equity instruments do not include instruments that are themselves contracts for the future receipt or delivery of the entity's own equity instruments.

Equity instrument: Any contract that evidences a residual interest in the assets of an entity after deducting all of its liabilities.

2.3 Initial recognition and measurement

An entity will recognise a financial asset or financial liability on its statement of financial position **when it becomes a party to the contractual provisions of the instrument** rather than when the contract is settled. An important consequence of this is that all derivatives should be recognised in the statement of financial position. This is because even if there is no initial investment or the initial investment is relatively low, derivative contracts expose the entity to risks and rewards due to changes in value of the underlying. In most cases, the date on which an entity becomes a party to a financial instrument's contractual obligations is fairly obvious. For example, option contracts are recognised as assets or liabilities when the holder or writer becomes a party to the contract.

2.4 Classification of financial assets

2.4.1 Introduction

On recognition, IFRS 9 requires that financial assets are classified as measured at either:

- **amortised cost**;
- **fair value through other comprehensive income**; or
- **fair value through profit or loss**.

2.4.2 Basis of classification

The IFRS 9 classification is made on the basis of both:

(a) the **entity's business model** for managing the financial assets; and

(b) the **contractual cash flow** characteristics of the financial asset.

Amortised cost

A financial asset is classified as measured at **amortised cost** where:

(a) the objective of the business model within which the asset is held is to hold assets in order to collect contractual cash flows; and

(b) the contractual terms of the financial asset give rise on specified dates to cash flows that are solely payments of principal and interest on the principal outstanding.

Fair value through other comprehensive income

A debt instrument **must** be classified and measured **at fair value through other comprehensive income** (unless the asset is designated at fair value through profit or loss under the fair value option) if it meets both the following criteria:

(a) The financial asset is held within a business model whose objective is achieved by both collecting contractual cash flows and selling financial assets.

(b) The contractual terms of the financial asset give rise on specified dates to cash flows that are solely payments of principal and interest on the principal amount outstanding.

Fair value through profit or loss

All other debt instruments must be measured at **fair value through profit or loss.**

Fair value through profit or loss option to avoid an 'accounting mismatch'

Even if an instrument meets the above criteria for measurement at amortised cost or fair value through other comprehensive income, IFRS 9 allows such financial assets to be **designated, at initial recognition, as being measured at fair value through profit or loss. This concession is limited to cases where a recognition or measurement inconsistency** (an 'accounting mismatch') would **otherwise arise** from measuring assets or liabilities or recognising the gains and losses on them on different bases.

Equity instruments

Equity instruments may not be classified as measured at amortised cost and must be measured at fair value. This is because contractual cash flows on specified dates are not a characteristic of equity instruments. However, if an equity instrument is **not held for trading**, an entity can make an **irrevocable election** at initial recognition to measure it at **fair value through other comprehensive income** with only dividend income recognised in profit or loss.

This is **different from the treatment of debt instruments**, where the **fair value through other comprehensive income classification is mandatory** for assets meeting the criteria, unless the fair value through profit or loss option is chosen.

(IFRS 9.4.1.1–.4.1.2)



Professional skills focus: Applying judgement

IFRS 9 is very specific about how instruments may be classified, which means there is limited scope for judgement. However, entities have a choice about the irrevocable FVTOCI election for equity instruments not held for trading.

2.4.3 Business model test in more detail

IFRS 9 contains a business model test that requires an entity to assess whether its **business objective for a debt instrument is to collect the contractual cash flows of the instrument as opposed to realising its fair value change from sale prior to its contractual maturity**. Note the following key points:

(a) The assessment of a 'business model' is not made at an individual financial instrument level.

(b) The assessment is based on how key management personnel actually manage the business, rather than management's intentions for specific financial assets.

(c) An entity may have more than one business model for managing its financial assets and the classification need not be determined at the reporting entity level. For example, it may have one portfolio of investments that it manages with the objective of collecting contractual cash flows and another portfolio of investments held with the objective of trading to realise changes in fair value. It would be appropriate for entities like these to carry out the assessment for classification purposes at portfolio level, rather than at entity level.

(d) Although the objective of an entity's business model may be to hold financial assets in order to collect contractual cash flows, the entity need not hold all of those assets until maturity. Thus an entity's business model can be to hold financial assets to collect contractual cash flows even when sales of financial assets occur.

The following examples, from the Application Guidance to IFRS 9 (IFRS 9: AG, B4.1.1–B4.1.26), are of situations where the **objective of an entity's business model may be to hold financial assets to collect the contractual cash flows.**

Context example: Collecting contractual cash flows 1

A Co holds investments to collect their contractual cash flows but would sell an investment in particular circumstances, perhaps to fund capital expenditure, or because the credit rating of the instrument falls below that required by A Co's investment policy.

Although A Co may consider, among other information, the financial assets' fair values from a liquidity perspective (ie, the cash amount that would be realised if A Co needs to sell assets), A Co's objective is to hold the financial assets and collect the contractual cash flows. Some sales would not contradict that objective. If sales became frequent, A Co might be required to reconsider whether the sales were consistent with an objective of collecting contractual cash flows.

Context example: Collecting contractual cash flows 2

B Co has a business model with the objective of originating loans to customers and subsequently to sell those loans to a securitisation vehicle. The securitisation vehicle issues instruments to investors.

B Co, the originating entity, controls the securitisation vehicle and thus consolidates it. The securitisation vehicle collects the contractual cash flows from the loans and passes them on to its investors in the vehicle.

It is assumed for the purposes of this example that the loans continue to be recognised in the consolidated statement of financial position because they are not derecognised by the securitisation vehicle.

The consolidated group originated the loans with the objective of holding them to collect the contractual cash flows.

However, B Co has an objective of realising cash flows on the loan portfolio by selling the loans to the securitisation vehicle, so for the purposes of its separate financial statements it would not be considered to be managing this portfolio in order to collect the contractual cash flows.

Context example: Collecting contractual cash flows 3

C Co's business model is to purchase portfolios of financial assets, such as loans. Those portfolios may or may not include financial assets that are credit impaired. If payment on the loans is not made on a timely basis, C Co attempts to extract the contractual cash flows through various means – for example, by contacting the debtor through mail, telephone, and so on.

In some cases, C Co enters into interest rate swaps to change the interest rate on particular financial assets in a portfolio from a floating interest rate to a fixed interest rate.

The objective of C Co's business model is to hold the financial assets and collect the contractual cash flows. The entity does not purchase the portfolio to make a profit by selling them.

The same analysis would apply even if C Co does not expect to receive all of the contractual cash flows (eg, some of the financial assets are credit impaired at initial recognition).

Moreover, the fact that C Co has entered into derivatives to modify the cash flows of the portfolio does not in itself change C Co's business model.

2.4.4 Contractual cash flow test in more detail

The requirement in IFRS 9 to assess the contractual cash flow characteristics of a financial asset is based on the concept that **only instruments with contractual cash flows of principal and interest on principal may qualify for amortised cost measurement**. By interest, IFRS 9 means consideration for the time value of money and the credit risk associated with the principal outstanding during a particular period of time.

Measurement at amortised cost is permitted when the cash flows on a loan are entirely fixed (eg, a fixed interest rate loan or zero coupon bond), or where interest is floating (eg, a GBP loan where interest is contractually linked to GBP SONIA), or combination of fixed and floating (eg, where interest is SONIA plus a fixed spread). Note that SONIA, Sterling Overnight Index Average is a benchmark rate that replaced LIBOR in 2021.

2.4.5 Examples of instruments that pass the contractual cash flows test

The following instruments satisfy the IFRS 9 criteria.

(a) A variable rate instrument with a stated maturity date that permits the borrower to choose to pay three-month SONIA for a three-month term or one-month SONIA for a one-month term

(b) A fixed term variable market interest rate bond where the variable interest rate is capped

(c) A fixed term bond where the payments of principal and interest are linked to an unleveraged inflation index of the currency in which the instrument is issued

2.4.6 Examples of instruments that do not pass the contractual cash flows test

The following instruments do not satisfy the IFRS 9 criteria:

- A bond that is convertible into equity instruments of the issuer
- A loan that pays an inverse floating interest rate (eg, 8% minus SONIA)

2.4.7 Business model of both collecting contractual cash flows and selling financial assets

The following examples, from the Application Guidance to IFRS 9 (IFRS 9: AG, B4.1.1 – B4.1.26), are of situations where the **objective of an entity's business model is achieved by both collecting contractual cash flows and selling financial assets**.



Context example: Both collecting contractual cash flows and selling financial assets 1

D Co expects to incur capital expenditure in a few years' time. D Co invests its excess cash in short and long-term financial assets so that it can fund the expenditure when the need arises. Many of the financial assets have contractual lives that exceed D Co's anticipated investment period.

D Co will hold financial assets to collect the contractual cash flows and, when an opportunity arises, it will sell financial assets to re-invest the cash in financial assets with a higher return.

The remuneration of the managers responsible for the portfolio is based on the overall return generated by the portfolio.

The objective of the business model is achieved by **both collecting contractual cash flows and selling financial assets**. D Co decides on an ongoing basis whether collecting contractual cash flows or selling financial assets will maximise the return on the portfolio until the need arises for the invested cash.

Context example: Both collecting contractual cash flows and selling financial assets 2

Logan plc holds financial assets to meet its everyday liquidity needs. The company actively manages the return on the portfolio in order to minimise the costs of managing those liquidity needs. That return consists of collecting contractual payments, as well as gains and losses from the sale of financial assets.

To this end, Logan plc holds financial assets to collect contractual cash flows, and sells financial assets to reinvest in higher yielding financial assets or to better match the duration of its liabilities. In the past, this strategy has resulted in frequent sales activity and such sales have been significant in value. This activity is expected to continue in the future.

The objective of the business model is to maximise the return on the portfolio to meet everyday liquidity needs and Logan plc achieves that objective by **both collecting contractual cash flows and selling financial assets.** In other words, **both collecting contractual cash flows and selling financial assets are integral** to achieving the business model's objective.

Interactive question 4: Contractual cash flows and selling financial assets

E Co expects to pay a cash outflow in 10 years to fund capital expenditure and invests excess cash in short-term financial assets. When the investments mature, E Co reinvests the cash in new short-term financial assets. E Co maintains this strategy until the funds are needed, at which time E Co uses the proceeds from the maturing financial assets to fund the capital expenditure. Only sales that are insignificant in value occur before maturity (unless there is an increase in credit risk).

Requirement

How is the business model of E Co classified under IFRS 9?

See **Answer** at the end of this chapter.

2.5 Classification of financial liabilities

On **recognition**, IFRS 9 requires that financial liabilities are **classified as measured** either:

(a) at **fair value through profit or loss**; or

(b) at **amortised cost**.

A financial liability is classified at fair value through profit or loss if:

(a) it is **held for trading**; or

(b) upon initial recognition it is **designated at fair value through profit or loss**. This is permitted when it results in more relevant information because:

(1) it eliminates or significantly reduces a measurement or recognition inconsistency ('accounting mismatch'); or

(2) it is a group of financial liabilities or financial assets and liabilities and its performance is evaluated on a fair value basis, in accordance with a documented risk management or investment strategy.

Derivatives are always measured at fair value through profit or loss.

2.6 Reclassification of financial assets

IFRS 9 requires that when an entity changes its business model for managing financial assets, it should reclassify all affected financial assets. This is expected to be a rare event.

A change in an entity's business model will occur only when an entity begins or ceases to perform an activity that is significant to its operations – for example, when the entity has acquired, disposed of or terminated a business line.

The examples of **change in a business model** include the following:

- An entity has a portfolio of commercial loans that it holds to sell in the short term. The bank acquires an entity that manages commercial loans and has a business model that holds the loans in order to collect the contractual cash flows. The portfolio of commercial loans is no longer for sale, and the portfolio is now managed together with the acquired commercial loans and all are held to collect contractual cash flows.
- An entity decides to shut down its retail mortgage business. That business no longer accepts new business and the bank is actively marketing its mortgage loan portfolio for sale.

If an entity reclassifies financial assets, the **reclassification is applied prospectively** from the first day of the first reporting period following the change in business model (**the reclassification date**).

The impact of reclassification of financial assets is as follows:

Asset category	Reclassified to:	Impact of reclassification
Amortised cost	Fair value through profit or loss	• Fair value is measured at the date of reclassification • Difference between amortised cost and fair value is recognised in profit or loss
Amortised cost	Fair value through other comprehensive income	• Fair value is measured at the date of reclassification • Difference between amortised cost and fair value is recognised in other comprehensive income
Fair value through other comprehensive income	Fair value through profit or loss	• Continue to measure at fair value • Cumulative gain or loss in other comprehensive income is reclassified to profit or loss at the reclassification date
Fair value through other comprehensive income	Amortised cost	• Fair value at the reclassification date is the new gross carrying amount • Cumulative gain or loss is removed from other comprehensive income (removed from equity and adjusted against fair value at the reclassification date but is not a reclassification adjustment hence does not affect profit or loss)
Fair value through profit or loss	Fair value through other comprehensive income	• Continue to measure at fair value • Subsequent gains and losses are recognised in OCI
Fair value through profit or loss	Amortised cost	• Fair value at the reclassification date is the new gross carrying amount • Effective interest rate is determined on the basis of the fair value at the reclassification date

Reclassification is not permitted for derivatives, financial liabilities and equity investments that are designated as at fair value through other comprehensive income on initial recognition.

When financial instruments are reclassified, disclosures are required under IFRS 7.

2.7 Derecognition of financial assets

2.7.1 Criteria for derecognition

Derecognition is the removal of a previously recognised financial instrument from an entity's statement of financial position.

An entity should derecognise a **financial asset** when:

(a) the **contractual rights** to the cash flows from the financial asset **expire;** or

(b) the entity **transfers substantially all the risks and rewards of ownership** of the financial asset to another party.

Derecognition of a financial asset is often straightforward, as the above criteria can be implemented easily. For example, a bond should be derecognised when an entity has collected all the payments of coupon and principal amount – ie, the bond has matured. The collection of payment signifies the end of any exposure to risks or any continuing involvement.

There may, however, be more complex transactions which involve the transfer of legal title to another entity but only a partial transfer of risks and rewards. In such instances, further analysis is required to evaluate whether the asset should be fully derecognised, and whether a separate asset or liability recognised, to reflect the remaining rights or obligations.

2.7.2 Accounting treatment

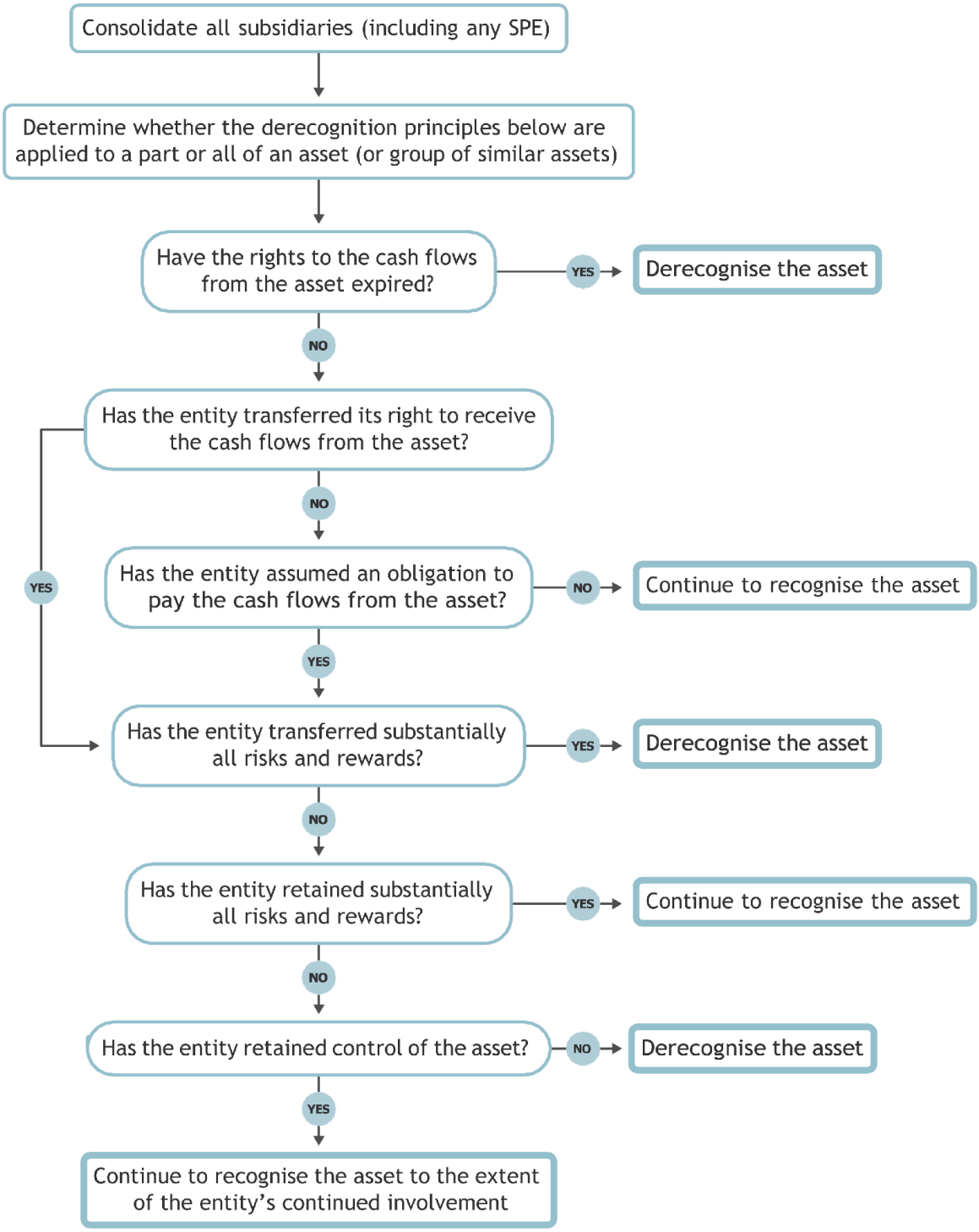
On derecognition of a financial asset the difference between the **carrying amount and any consideration received should be recognised in profit or loss**. For investments in debt instruments that are measured at fair value through other comprehensive income, any accumulated gains or losses that have been recognised in other comprehensive income should be reclassified to profit or loss on derecognition of the asset.

2.7.3 The IFRS 9 derecognition steps

The complexity of financial transactions and the difficulty of establishing whether the transfer of legal title leaves residual risk and reward exposures as well as control and involvement, has prompted the IASB to produce a fairly prescriptive set of rules to aid companies in the derecognition of financial assets.

The following flowchart, included in the application guidance which accompanies IFRS 9 (IFRS 9: Application Guidance, Appendix B, 3.2.1), summarises the evaluation of whether, and to what extent, a financial asset should be derecognised.

Figure 16.1: Derecognition of financial assets (IFRS 9.B3.2.1) The following points relate to this flowchart:



- If the **contractual rights** to receive the cash flows from the asset have **expired** or have been wholly transferred, the whole of the asset should be derecognised. This is also the case if the contractual rights have been retained by the entity but it has assumed a contractual obligation to pay the cash flows to one or more recipients. Such an obligation is only assumed if:

- there is no obligation to pay unless amounts are actually collected;
- the entity is forbidden to sell or pledge the original asset other than to the recipient of the cash flows; and
- the entity must remit the cash flows collected, including any interest earned during temporary investment between the date of collection to date of remittance, without material delay.

- If an entity has sold just a portion of the cash flows arising from an asset, only part of the asset should be derecognised.
- If substantially **all the risks and rewards** of ownership have been **transferred**, the financial asset should be derecognised and separate assets or liabilities should be recognised for any rights or obligations created in the transfer.
- If the entity has neither retained nor transferred all the risks and rewards of ownership, it should determine whether it has **retained control** of the financial asset. If it has, it continues to recognise the asset to the extent of its continuing involvement.

Repurchase agreements may be employed in order to try to remove assets from the statement of financial position.

Remember always to apply the principle of substance over legal form.

2.7.4 Repurchase agreements

In a repurchase agreement, a financial asset such as a bond is sold with a simultaneous agreement to buy it back at some future date at a specified price, which may be the future market price.



Worked example: Repurchase agreements

1 Green Co sells a bond carried on its statement of financial position to Red Co for £1,000. Green Co commits to buy back the bond in three months for £1,025.

Has Green Co transferred substantially all the risks and rewards of ownership?

2 Would your answer be different if Green Co commits to buyback in three months at market price?

Solution

1 IFRS 9 includes examples with regard to these situations:

This is a sale and repurchase transaction where the repurchase price is a fixed price or at the sale price plus a lender's return. Green Co has not transferred substantially all the risks and rewards of ownership and hence the bond is not derecognised. Green Co will recognise a loan liability of £1,000 and interest expense of £25 to reflect the collateralised borrowing.

2 This is a sale of a financial asset together with commitment to repurchase the financial asset at its fair value at the time of repurchase. Because any repurchase is at the then fair value, all risks and rewards of ownership are with the buying party and hence Green Co will derecognise the bond.

2.8 Derecognition of financial liabilities

An entity should derecognise a **financial liability** when it is **extinguished** ie, when the obligation specified in the contract is discharged or cancelled or expires.

- An entity **discharges** its obligation by paying in cash, other financial assets or by delivering other goods or services to the counterparty.

- An obligation may **expire** due to passage of time as, for example, an unexercised written option.
- An obligation is **cancelled** when through the process of law, or via negotiation with a creditor, an entity is released from its primary obligation to pay the creditor.

When a liability is extinguished, the difference between its carrying amount and the consideration paid including any non-cash assets transferred and any new liabilities assumed is recognised in profit or loss.



Context example: Extinguishing a financial liability

Entity A has borrowed £10 million from a bank to invest in commercial development. However, due to a recession, the shops built did not yield the expected rental income and Entity A cannot service the debt. It negotiates with the bank to transfer the ownership of the development to the bank in settlement of the outstanding debt. The market value of the development is £7 million. The development's carrying amount was £7 million as it was measured at fair value.

As a result of the transfer, Entity A should extinguish the liability but it should also recognise a gain of £3 million in profit or loss, arising from the difference between the carrying amount of the liability (£10 million) and the value of the development (£7 million) that was transferred to the bank.

2.9 Partial derecognition of financial assets and financial liabilities

It is possible for only **part** of a financial asset or liability to be derecognised. This is the case if the part comprises:

(a) only specifically identified cash flows; or

(b) only a fully proportionate (pro rata) share of the total cash flows.

For example, if an entity holds a bond, it has the right to two separate sets of cash inflows: those relating to the principal and those relating to the interest. It could sell the right to receive the interest to another party while retaining the right to receive the principal.

On **derecognition**, the amount to be **included in profit or loss** for the year is calculated as follows:

	£
Carrying amount (measured at the date of derecognition) allocated to the part derecognised	X
Less consideration received for the part derecognised (including any new asset obtained less any new liability assumed)	(X)
Difference to profit or loss	X

Where only part of a financial asset is derecognised, the carrying amount of the asset should be allocated between the part retained and the part transferred based on their relative fair values on the date of transfer. A gain or loss should be recognised based on the proceeds for the portion transferred.



Worked example: Partial derecognition

Westerly plc purchased £100,000 of bonds which were classified as designated at fair value through profit or loss. One year later, 25% of the bonds were sold for £37,500. Total cumulative gains previously recognised in profit or loss in respect of the asset were £6,250.

Requirement

In accordance with IFRS 9, *Financial Instruments*, what is the amount of the gain on the disposal to be recognised in profit or loss?

Solution

When a part of a financial asset is derecognised, the amount recognised in profit or loss should be the difference between the carrying amount allocated to the part derecognised and the sum of:

(1) the consideration received for the part derecognised; and an

(2) any cumulative gain or loss allocated to it that had been recognised in other comprehensive income.

The previous gains had been recognised in profit or loss and so are not included in the calculation.

	£
Carrying amount of the assets sold/derecognised (25% of £106,250)	26,562.50
Proceeds from sale of 25% of bonds	37,500.00
Gain recognised in the period of sale	10,937.50



Interactive question 5: Sale of cash flows from debt instrument

During the year ended 31 December 20X0, Jones sold to a third party the right to receive the interest cash flows on a fixed maturity debt instrument it holds and will continue to legally own up to the date of maturity. The debt instrument is quoted in an active stock market. The entity has no obligation to compensate the third party for any cash flows not received.

Requirement

Discuss whether the debt instrument should be derecognised.

See **Answer** at the end of this chapter.



Interactive question 6: Derecognition of financial assets and liabilities

Discuss whether the following financial instruments should be derecognised.

6.1 ABC sells an investment in shares, but retains a call option to repurchase those shares at any time at a price equal to their current market value at the date of repurchase.

6.2 DEF enters into a stock lending agreement where an investment is lent to a third party for a fixed period of time for a fee.

6.3 XYZ sells title to some of its receivables to a debt factor for an immediate cash payment of 90% of their value. The terms of the agreement are that XYZ has to compensate the factor for any amounts not recovered by the factor after six months.

See **Answer** at the end of this chapter.

2.10 Exchange or modification of debt

If a new loan is agreed between a borrower and a lender, or the two parties agree revised terms for an existing loan, the accounting depends on whether the original liability should be derecognised and a new liability recognised, or whether the original liability should be treated as modified.

A new liability should be recognised if the new terms are substantially different from the old terms. The terms are substantially different if the present value of the cash flows under the new terms, including any fees payable/receivable, discounted at the original effective interest rate, is 10% or more different from the present value of the remaining cash flows under the original terms. There is said to be an 'extinguishment' of the old liability. In these circumstances:

- the difference between the carrying amount extinguished and the consideration paid should be recognised in profit or loss; and
- the fees payable/receivable should be recognised as part of that gain or loss.

If the difference between the two present values is below this cut-off point, there is said to be a modification of the terms. In these circumstances:

- the existing liability is **not** derecognised; and
- its carrying amount is adjusted by the fees payable/receivable and amortised over the remaining term of the modified liability.



Worked example: Modification of debt

On 1 May 20X4 Delta plc issues £10 million 10% loan stock redeemable at par after 10 years. Interest is paid annually in arrears. On 1 May 20X9, Delta plc and the loan stockholders agree to a modification with the following terms:

- No further interest payments are made
- The loan stock is redeemed on the original date for £17 million

Legal and other fees relating to the modification amount to £500,000. Assume the borrowing cost of Delta has not changed over the years.

Requirement

How is this modification accounted for by Delta plc?

Solution

The present value of remaining cash flows is compared with the present value of cash flows under the new terms.

The present value of the cash flows arising on the old terms is £10 million:

	£
Interest 30.4.Y0 (£1m/1.1)	909,091
Interest 30.4.Y1 (£1m/1.1_2)	826,446
Interest 30.4.Y2 (£1m/1.1_3)	751,315
Interest 30.4.Y3 (£1m/1.1_4)	683,013
Interest and principal 30.4.Y4 (£11m/1.1_5)	6,830,135
	10,000,000

The present value of cash flows arising after the modification is £11,055,662:

	£
Principal (£17m/1.1$_5$)	10,555,662
Fees incurred	500,000
	11,055,662

The increase in cash flows represents more than a 10% change from £10 million. Therefore the original financial liability is derecognised and a new financial liability on the new terms is recognised. The new financial liability is initially measured at fair value in accordance with IFRS 9 and any difference between this amount and the derecognised financial liability is recognised in profit or loss.

2.11 Regular way transactions

Note: As regards purchases, this issue relates to recognition rather than derecognition; however, it was necessary to understand the classification aspects in detail before studying this section.

Most financial markets set out regulations for 'regular way' transactions whereby purchases and sales are contractually deliverable (and therefore settled) on a specified date. Settlement date is later than the contractual date of the transaction (the 'trade date'). A regular way purchase or sale is the acquisition or disposal of a financial asset under a contract requiring delivery within a **specified time frame**. The time frame may be established through regulation or convention in the market.

Regular way purchases and sales of financial instruments should be recognised using either the **trade date** or the **settlement date**. The method chosen should be applied consistently for all financial assets. A derivative contract is not a regular way contract, since it can be settled on a net basis.

For purchases, **trade date accounting** requires the recognition of an asset and the liability to pay for it at the trade date. After initial recognition, the financial asset is subsequently measured either at amortised cost or at fair value, depending on its initial classification. For sales of financial assets, the asset is derecognised and the receivable from the buyer, together with any gain or loss on disposal, are recognised on the trade date.

With **settlement date accounting**, an asset purchased is not recognised until the date on which it is received. Movements in fair value of the contract between the trade date and settlement date are recognised in the same way as the acquired asset. For assets classified as assets at fair value through profit or loss, the change is recognised in profit or loss. For assets classified at fair value through other comprehensive income, the change is recognised in other comprehensive income.

For sales of financial instruments, the asset is derecognised and the receivable from the buyer, together with any gain or loss on disposal, are recognised on the day that it is delivered by the entity. Any change in the fair value of the asset between the trade date and settlement date is not recognised, as the sale price is agreed at the trade date, making subsequent changes in fair value irrelevant from the seller's perspective.

Worked example: Regular way purchase of a financial asset

IDB Bank entered into a contractual commitment on 27 December 20X8 to purchase a financial asset for £2,500. On 31 December 20X8, the bank's reporting date, the fair value was

£2,513. The transaction was settled on 5 January 20X9 when the fair value was £2,519. The bank has classified the asset at fair value through profit or loss.

Requirement

How should the transactions be accounted for under trade date accounting and settlement date accounting?

Solution

Trade date accounting

- On 27 December 20X8, the bank should recognise the financial asset and the liability to the counterparty at £2,500.
- At 31 December 20X8, the financial asset should be re-measured to £2,513 and a gain of £13 recognised in profit or loss.
- On 5 January 20X9, the liability to the counterparty of £2,500 will be paid in cash. The fair value of the financial asset should be re-measured to £2,519 and a further gain of £6 recognised in profit or loss.

Settlement date accounting

- No transaction should be recognised on 27 December 20X8.
- On 31 December 20X8, a receivable of £13 should be recognised (equal to the fair value movement since the trade date) and the gain recognised in profit or loss.
- On 5 January 20X9, the financial asset should be recognised at its fair value of £2,519. The receivable should be derecognised, the payment of cash to the counterparty recognised and the further gain of £6 recognised in profit or loss.

Interactive question 7: Regular way sale of a financial asset

High Growth Bank acquired a financial asset on 1 January 20X8 for £3,000. On 27 December 20X8, it entered into a contract to sell the asset for £3,250. On 31 December 20X8, the bank's reporting date, the fair value of the asset was £3,293. The transaction was settled on 5 January 20X9. The bank classified the asset as at fair value through other comprehensive income.

Requirement

How should the transactions be accounted for under trade date accounting and settlement date accounting?

See **Answer** at the end of this chapter.

3 Measurement

Section overview

- **Financial assets** should initially be measured at **cost** = **fair value**.
- **Transaction costs increase this amount for financial assets** classified as measured at amortised cost, or where an irrevocable election has been made to take all gains and losses through other comprehensive income and **decrease this amount for financial liabilities** classified as measured at amortised cost.

- **Subsequent measurement** of both financial assets and financial liabilities depends on how the instrument is classified: at amortised cost or fair value.

3.1 Initial measurement: financial assets

Financial instruments are initially measured at the transaction price, that is the **fair value** of the consideration given (IFRS 9.5.1.1).

An **exception** is where part of the consideration given is for something other than the financial asset. In this case, the financial asset is initially measured at fair value evidenced by a quoted price in an active market for an identical asset (ie, an IFRS 13 level 1 input) or based on a valuation technique that uses only data from observable markets. The difference between the fair value at initial recognition and the transaction price is recognised as a gain or loss.

In the case of financial assets classified as measured at **amortised cost or at fair value through other comprehensive income**, **transaction costs** directly attributable to the acquisition of the financial asset are **added** to this amount.

3.2 Initial measurement: financial liabilities

IFRS 9 requires that financial liabilities are initially measured at transaction price, i.e., the fair value of consideration received except where part of the consideration received is for something other than the financial liability. In this case the financial liability is initially measured at fair value measured as for financial assets (see above). Transaction costs are deducted from this amount for financial liabilities classified as measured at amortised cost (IFRS 9:Para. 5.1.1).

3.3 Subsequent measurement: financial assets

Under IFRS 9, financial assets are measured subsequent to recognition either:

- at **amortised cost**, using the **effective interest method**, or
- at **fair value through other comprehensive income**, or
- at **fair value through profit or loss**.

(IFRS 9.5.2.1)

3.4 Financial assets measured at amortised cost

Definitions

Amortised cost: The amount at which the financial asset or liability is measured at initial recognition minus principal repayments, plus or minus the cumulative amortisation using the effective interest method of any difference between that initial amount and the maturity amount and, for financial assets, adjusted for any loss allowance.

Effective interest method: A method of calculating the amortised cost of a financial instrument and of allocating the interest income or interest expense over the relevant period.

Worked example: Financial asset at amortised cost

On 1 January 20X4, Beta plc purchases a debt instrument for its fair value of £1,000. The debt instrument is due to mature on 31 December 20X8. The instrument has a principal amount of £1,250 and carries fixed interest at 4.72% that is paid annually. (The effective interest rate is 10%.)

Requirement

How should Beta plc account for the debt instrument over its five-year term?

Solution

Beta plc will receive interest of £59 (1,250 × 4.72%) each year and £1,250 when the instrument matures.

Beta must allocate the discount of £250 and the interest receivable over the five-year term at a constant rate on the carrying amount of the debt. To do this, it must apply the effective interest rate of 10%.

The following table shows the allocation over the years:

Year	Amortised cost at beginning of year	Profit or loss: Interest income for year (@10%)	Interest received during year (cash inflow)	Amortised cost at end of year
	£	£	£	£
20X4	1,000	100	(59)	1,041
20X5	1,041	104	(59)	1,086
20X6	1,086	109	(59)	1,136
20X7	1,136	113	(59)	1,190
20X8	1,190	119	(1,250 + 59)	–

Each year the carrying amount of the financial asset is increased by the interest income for the year and reduced by the interest actually received during the year.

3.5 Financial assets measured at fair value

Where a financial asset is classified as measured at fair value, fair value is established at each period end in accordance with IFRS 13, *Fair Value Measurement*.

Fair value is the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date.

IFRS 13 provides extensive guidance on how the fair value of assets and liabilities should be established.

This standard requires that the following are considered in determining fair value:

(a) The asset or liability being measured

(b) The principal market (ie, that where the most activity takes place) or where there is no principal market, the most advantageous market (ie, that in which the best price could be achieved) in which an orderly transaction would take place for the asset or liability

(c) The highest and best use of the asset or liability, and whether it is used on a standalone basis or in conjunction with other assets or liabilities

(d) Assumptions that market participants would use when pricing the asset or liability

Having considered these factors, IFRS 13 provides a hierarchy of inputs for arriving at fair value. It requires that Level 1 inputs are used where possible:

Level 1: Quoted prices in active markets for identical assets that the entity can access at the measurement date

Level 2: Inputs other than quoted prices that are directly or indirectly observable for the asset

Level 3: Unobservable inputs for the asset

If an entity has investments in **equity instruments that do not have a quoted price** in an active market, they should also be measured at fair value. IFRS 13 should be applied, for example the

market approach, using prices and other relevant information that have been generated by market transactions that involve identical or comparable assets (IFRS 13 para B5).

The fair value on initial recognition is normally the transaction price. However, if part of the consideration is given for something other than the financial instrument, then the fair value should be estimated using a valuation technique.

Any changes in fair value are normally recognised in profit or loss. There are three **exceptions** to this rule:

(a) The asset is **part of a hedging relationship** (see next Chapter).

(b) The financial asset is an investment in an **equity instrument not held for trading**. In this case, the entity can make an **irrevocable election** to recognise changes in the fair value in **other comprehensive income**.

(c) It is a **financial asset measured at fair value through other comprehensive income** because it meets the criteria above, that is the financial asset is held within a business model whose objective is achieved by both collecting contractual cash flows and selling financial assets.

Note that direct costs of acquisition are capitalised only in the case of a financial asset or financial liability **not** held at fair value through profit or loss. If the asset or liability is held at fair value through profit or loss, the costs of acquisition are expensed. This means that in the case of **financial assets held at amortised cost, costs of acquisition are capitalised.** They would be added to the asset and deducted from the liability amount. Similarly, if an **irrevocable election** has been made to take **gains and losses** on the financial asset **to other comprehensive income**, costs of acquisition should be **added to the purchase cost.**

Investments whose **fair value cannot be reliably measured** should be measured at cost.



Worked example: Asset measurement

On 8 February 20X8, Orange Co acquires a quoted investment in the shares of Lemon Co with the intention of holding it in the long term. The investment cost £850,000. At Orange Co's year end of 31 March 20X8, the market price of an identical investment is £900,000.

Requirement

How is the asset initially and subsequently measured?

Orange Co has elected to recognise changes in the fair value of the equity investment in other comprehensive income.

Solution

Measurement

- The asset is initially recognised at the fair value of the consideration, being £850,000.
- At the period end it is re-measured to £900,000.
- This results in the recognition of £50,000 in other comprehensive income.



Interactive question 8: Investment in listed shares

In January 20X6 Wolf purchased 10 million £1 listed equity shares in Hall at a price of £5 per share. Transaction costs were £3 million. Wolf's year end is 30 November.

At 30 November 20X6, the shares in Hall were trading at £6.50. On 31 October 20X6 Wolf received a dividend from Hall of 20p per share.

Requirements

Show the financial statement extracts of Wolf at 30 November 20X6 relating to the investment in Hall on the basis that:

(a) the shares were bought for trading;

(b) the shares were bought as a source of dividend income and were the subject of an irrevocable election at initial recognition to recognise them at fair value through other comprehensive income; and

(c) the shares were bought as a source of dividend income and were the subject of an irrevocable election at initial recognition to recognise them at fair value through other comprehensive income. However, on 1 June 20X6, Wolf sold half of its shareholding in Hall when the shares were trading at £6.

See **Answer** at the end of this chapter.

3.6 Summary of measurement rules

The following table summarises the IFRS 9 measurement rules. It is a useful reference point for your exam.

	Amortised cost	Fair value through other comprehensive income: equity instrument	Fair value through other comprehensive income: debt instrument	Fair value through profit or loss
Interest/ dividend income	Profit or loss	Profit or loss	Profit or loss	Profit or loss
Expected credit losses (see section 4)	Profit or loss	Profit or loss	Profit or loss	Profit or loss
Foreign exchange gains/losses	Profit or loss	Profit or loss	Profit or loss	Profit or loss
Other gains/losses on remeasurement	–	Other comprehensive income	Other comprehensive income	Profit or loss
Gain/loss on derecognition	Profit or loss	Profit or loss but OCI is not reclassified*	Profit or loss Amounts previously recognised in OCI are reclassified to profit or loss	Profit or loss

Note: *While the gain/loss on derecognition of an equity instrument at FVTOCI is recognised in profit or loss, in practice this should normally be a nil amount, assuming the disposal is at fair value. The asset's carrying amount is remeasured to fair value at the date of derecognition (IFRS 9: para. 3.2.12) immediately prior to the disposal. Any change resulting from such a remeasurement is recognised in OCI.

Worked example: Debt instrument at fair value through other comprehensive income

On 1 January 20X1 Gemma plc purchases a quoted debt instrument for its fair value of £1,000. The debt instrument is due to mature on 31 December 20X5 and the entity has the intention to hold the debt instrument until that date in order to collect contractual cash flows. The instrument has a principal amount of £1,250 and carries fixed interest at 4.72% that is paid annually. (The effective interest rate is 10%.)

Gemma plc holds the debt instrument within a business model with the intention of collecting contractual cash flows and selling assets. The fair value of the debt instrument is £1,210 at 31 December 20X1, £1,340 at 31 December 20X2, £1,400 at 31 December 20X3 and £1,445 at 31 December 20X4.

Requirements

1 How should Gemma plc account for the debt instrument over its five-year term?

2 How should Gemma plc account for the debt instrument on derecognition?

Solution

1 The financial asset is classified as measured at fair value through other comprehensive income since it is held within a business model to collect contractual cash flows and sell assets. It gives rise to contractual cash flows on specified dates that are solely payments of principal and interest on the principal outstanding.

The following table shows the measurement over the term.

Year	Carrying amount b/f	Profit or loss: Interest income for year (@ 10%) £	Interest and/or principal received during year (cash inflow) £	Gain or loss on remeasurement to fair value/ derecognition £	Fair value £
20X1	1,000	100	(59)	169	1,210
20X2	1,210	104	(59)	85	1,340
20X3	1,340	109	(59)	10	1,400
20X4	1,400	113	(59)	(9)	1,445
20X5	1,445	119	(1,309)	(255)	0

Note that interest income is calculated based on amortised cost and interest received is calculated based on the principal amount.

The required journal entries in 20X1 are:

		£
DEBIT	Financial asset through other comprehensive income	1,000
CREDIT	Cash/bank	1,000

To recognise the financial asset on 1 January 20X1

		£
DEBIT	Financial asset through other comprehensive income	210
DEBIT	Cash/bank	59
CREDIT	Profit or loss (interest income)	100
CREDIT	Other comprehensive income	169

To recognise interest income at the effective rate and the change in fair value.

2 On derecognition of a debt instrument measured at fair value through other comprehensive income, any amounts previously recognised as other comprehensive income are reclassified to profit or loss. Note that this is different from equity instruments held at FVTOCI, where gains and losses previously recognised through OCI are not reclassified.

Therefore in the example above, a loss of £255 is recognised in profit or loss on derecognition:

DEBIT	Profit or loss	£255
CREDIT	Financial asset	£255

And a cumulative net gain of £255 (169 + 85 + 10 – 9) is reclassified by:

DEBIT	Other comprehensive income	£255
CREDIT	Profit or loss	£255

3.7 Subsequent measurement of financial liabilities

After initial recognition, all financial liabilities should be measured at **amortised cost**, with the exception of financial liabilities at fair value through profit or loss (including most derivatives). These should be measured at **fair value**, but where the fair value **is not capable of reliable measurement**, they should be measured at **cost** (IFRS 9.5.3.1).

3.8 Financial liabilities measured at amortised cost

The definitions of amortised cost, effective interest method and effective interest rate that are used for measurement of financial assets are also used for financial liabilities.

Worked example: Liability measurement

Galaxy Co issues a bond for £503,778 on 1 January 20X2. No interest is payable on the bond, but it will be redeemed on 31 December 20X4 for £600,000. The effective interest rate of the bond is 6%.

Requirement

Calculate the charge to profit or loss of Galaxy Co for the year ended 31 December 20X2 and the balance outstanding at 31 December 20X2.

Solution

The bond is a 'deep discount' bond and is a financial liability of Galaxy Co. It is measured at amortised cost. Although there is no interest as such, the difference between the initial cost of the bond and the price at which it will be redeemed is a finance cost. This must be allocated over the term of the bond at a constant rate on the carrying amount.

The effective interest rate is 6%.

The charge to profit or loss for the year is £30,227 (503,778 × 6%).

The balance outstanding at 31 December 20X2 is £534,004 (503,778 + 30,226).

Interactive question 9: Measurement of liability

On 1 January 20X3 Deferred issued £600,000 loan notes. Issue costs were £200. The loan notes do not carry interest, but are redeemable at a premium of £152,389 on 31 December 20X4. The effective finance cost of the loan notes is 12%.

Requirement

What is the finance cost in respect of the loan notes for the year ended 31 December 20X4?

A £72,000
B £76,194
C £80,613
D £80,640

See **Answer** at the end of this chapter.

3.9 Financial liabilities at fair value through profit or loss

Financial liabilities which are **held for trading** are re-measured to fair value each year in accordance with IFRS 13, *Fair Value Measurement* with any gain or loss recognised in profit or loss (IFRS 9.5.7.1).

3.9.1 Exceptions

The exceptions to the above treatment of financial liabilities are as follows:

(a) It is part of a hedging arrangement.

(b) It is a financial liability designated as at fair value through profit or loss and the entity is required to present the effects of changes in the liability's **credit risk** in other comprehensive income (see 3.9.2 below).

3.9.2 Credit risk

IFRS 9 requires that financial liabilities which are **designated as measured at fair value through profit or loss are treated differently**. In this case the gain or loss in a period must be classified into:

- gain or loss **resulting from credit risk**; and
- **other** gain or loss.

This provision of IFRS 9 was in response to an anomaly regarding changes in the credit risk of a financial liability.

Changes in a financial liability's credit risk affect the fair value of that financial liability. This means that when an entity's creditworthiness deteriorates, the fair value of its issued debt will decrease (and *vice versa*). For financial liabilities measured using the fair value option, this causes **a gain (or loss) to be recognised in profit or loss for the year**. For example:

Statement of profit or loss and other comprehensive income (extract) – Profit or loss for the year

	£'000
Liabilities at fair value (except derivatives and liabilities held for trading)	
Change in fair value	100
Profit (loss) for the year	100

Many users of financial statements found this result to be **counter-intuitive** and confusing.

Accordingly, IFRS 9 requires the gain or loss as a result of credit risk to be recognised in other comprehensive income, unless it creates or enlarges an **accounting mismatch** (see 3.9.3), in which case it is recognised in profit or loss. The other gain or loss (not the result of credit risk) is recognised in profit or loss.

On derecognition any gains or losses recognised in other comprehensive income are **not** transferred to **profit or loss**, although the cumulative gain or loss may be transferred within equity.



Context example: IFRS 9 presentation

Statement of profit or loss and other comprehensive income (extract) Profit or loss for the year
Liabilities at fair value (except derivatives and liabilities held for trading)

£'000	
Change in fair value not attributable to credit risk	90
Profit (loss) for the year	90
Other comprehensive income (not reclassified to profit or loss)	
	£'000
Fair value loss on financial liability attributable to change in credit risk	10
Total comprehensive income	100

3.9.3 Accounting mismatch

IFRS 9 allows the recognition of the full amount of change in the fair value in the profit or loss only if the recognition of changes in the liability's **credit risk** in other comprehensive income would **create** or **enlarge** an **accounting mismatch** in profit or loss. That determination is made at initial recognition and is not reassessed (IFRS 9.4.1.5).

An accounting mismatch is a measurement or recognition inconsistency arising from measuring assets or liabilities or recognising the gains or losses on them on different bases.

4 Credit losses (impairment)



Section overview

- This section covers the key points in the IFRS 9 expected credit loss impairment model.
- The previous standard, IAS 39, required an impairment loss to be recognised if and only if there was objective evidence of impairment. This approach was criticised after the Global Financial Crisis of 2007/08 as recognising impairments 'too little, too late'. IFRS 9 therefore requires the expected credit loss model to ensure timely recognition of credit losses.
- On initial recognition of the financial asset, an impairment allowance is recognised based on 12- month expected credit losses.
- If there is a significant increase in credit risk after initial recognition, impairment allowances are recognised based on lifetime expected credit losses (LEL).

- The general or three-stage approach requires the financial asset to be classified in Stage 1 on initial recognition. If there is a significant increase in credit risk, the asset is moved to Stage 2. On default the financial asset transfers to Stage 3. Interest revenue is calculated on the net carrying amount of the asset (after the deduction of the impairment allowance) in Stage 3.
- Assessment of a significant increase in credit risk may be based on quantitative indicators, qualitative indicators or a mixture of both. It requires the use of reasonable and supportable information available without undue cost or effort, including use of forward looking macro- economic data.
- If a financial asset is purchased or originated credit-impaired, expected credit losses shall be discounted using the credit-adjusted effective interest rate determined at initial recognition.
- Simplified approaches and practical expedients may be used by non-financial entities for operational simplification in applying the impairment model.

4.1 Indications of impairment

The following are indications that a financial asset or group of assets may be impaired (IFRS 9: Appendix A).

(a) Significant financial difficulty of the issuer

(b) A breach of contract, such as a default in interest or principal payments

(c) The lender granting a concession to the borrower that the lender would not otherwise consider, for reasons relating to the borrower's financial difficulty

(d) It becomes probable that the borrower will enter bankruptcy

(e) The disappearance of an active market for that financial asset because of financial difficulties

(f) The purchase or origination of a financial asset at a deep discount that reflects the incurred credit losses

It is not always possible to single out one particular event; rather, several events may combine to cause an asset to become credit-impaired.

4.2 Background and scope of IFRS 9 impairment model

IFRS 9, *Financial Instruments* was published in response to an urgent need to improve the accounting for financial instruments. IFRS 9 requires an **expected credit loss** model to ensure earlier and **timely recognition of credit losses**. This has a significant impact on banks as both incurred and future expected credit losses are considered in the measurement of impairment. A single set of impairment requirements applies to all instruments within the scope of the new standard. Financial liabilities are not subject to impairment.

The following instruments are in the **scope of IFRS 9 impairment** requirements:

- Financial assets that are debt instruments measured at amortised cost or fair value through other comprehensive income
- Loan commitments and financial guarantee contracts not accounted for at fair value through profit or loss under IFRS 9
- Contract assets under IFRS 15, *Revenue from Contracts with Customers*
- Lease receivables under IFRS 16, *Leases*

All instruments measured at fair value through profit or loss are not required to be assessed for impairment because any fair value movements are automatically reflected in profit or loss.

4.3 IFRS 9 impairment model

4.3.1 Initial recognition of financial asset

An entity is required to create a **credit loss allowance/impairment allowance on initial recognition** of the financial asset. This is calculated by multiplying the probability of a default occurring in the next **12 months** by the total lifetime **expected credit losses** that would result from that default. Note that this is not the same as the expected cash shortfalls over the next 12 months.

4.3.2 Subsequent treatment of financial asset

An entity may continue to provide for 12-month expected credit losses if there is not a significant change in credit risk. However, the probability of default occurring in the next 12 months may have changed and the credit loss allowance would have to be adjusted to reflect this.

If the **credit risk increases significantly since initial recognition** the 12-month expected credit impairment allowance is replaced by **lifetime expected credit losses**. If the credit quality subsequently improves and the lifetime expected credit losses criterion is no longer met, the 12- month expected credit loss basis is reinstated which means that the credit impairment allowance will reduce.



Definitions

Credit loss: The difference between all contractual cash flows that are due to an entity in accordance with the contract and all the cash flows that the entity expects to receive discounted at the original effective interest rate.

Expected credit losses (ECL): The weighted average of credit losses with the respective risks of the default occurring as the weights.

Lifetime expected credit losses: Those that result from all possible default events over the expected life of a financial instrument.

12-month expected credit losses: The portion of lifetime expected credit losses which represent the expected credit losses that result from default events on a financial instrument that are possible within the 12 months after the reporting date.

4.3.3 General or 3-stage approach

The amount of the impairment to be recognised on financial assets **depends on whether or not they have significantly deteriorated** since their initial recognition. For this purpose, financial instruments are classified into three stages as follows:

Step 1 Financial assets on initial recognition and financial assets where credit quality has not significantly deteriorated since initial recognition. Stage 1 contains loans from all risk classes except 'credit-impaired' loans (section 4.4).

Step 2 Financial assets whose credit quality has significantly deteriorated since their initial recognition.

Step 3 Financial assets for which there is objective evidence of impairment at the reporting date.

For Stage 1 financial instruments, the impairment represents the present value of expected credit losses that will result if a default occurs in the 12 months after the reporting date **(12 months' expected credit losses)**.

For financial instruments classified as Stage 2 or 3, an impairment is recognised at the present

value of expected credit shortfalls over their remaining life **(lifetime expected credit loss)**. Entities are required to reduce the gross carrying amount of a financial asset in the period in which they no longer have a reasonable expectation of recovery.

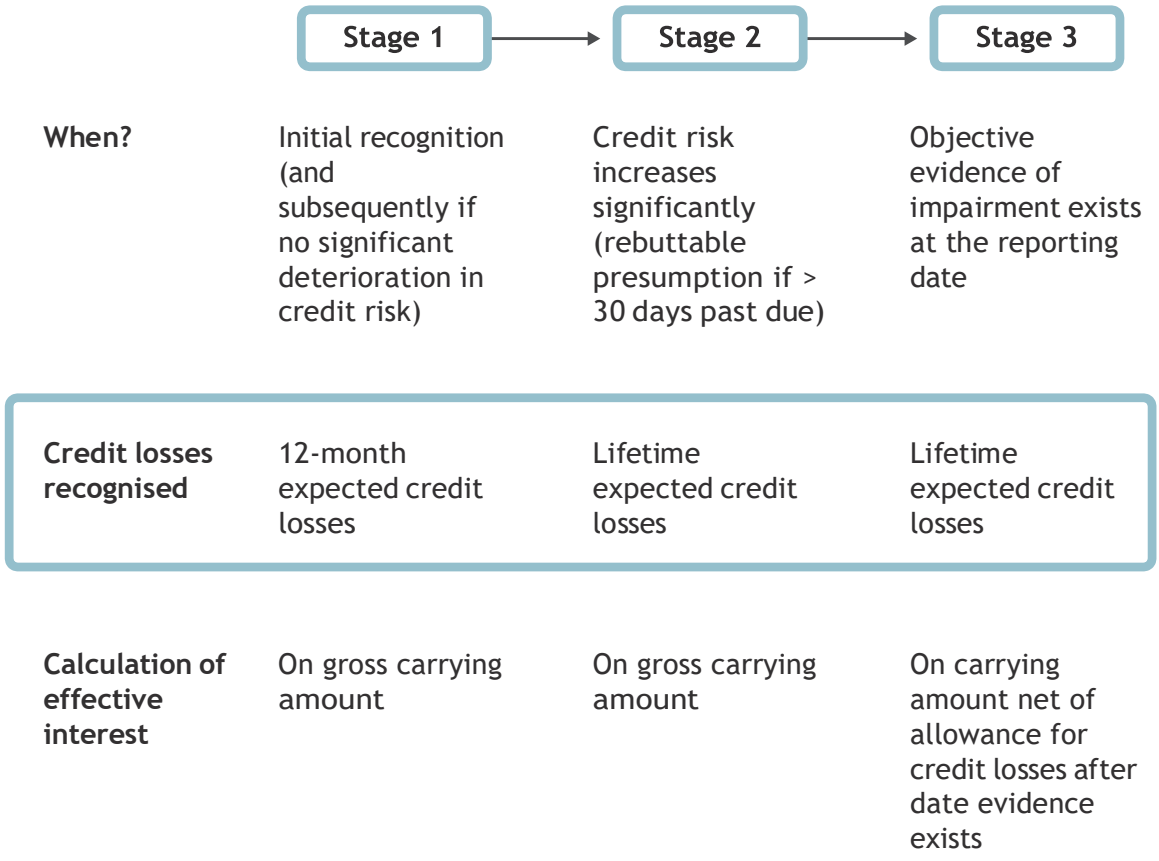
4.3.4 Interest

For Stage 1 and 2 instruments interest revenue will be calculated on their gross carrying amounts, whereas interest revenue for Stage 3 financial instruments would be recognised on a net basis (ie, after deducting expected credit losses from their carrying amount).

4.3.5 Summary

The following figure gives a useful summary of the process.

Figure 16.2: The three-stage approach



Worked example: Impairment loss

On 1 January 20X2 Dexter Lee plc originated a loan of £500,000. It classified the financial asset as measured at amortised cost. The loan is fully repayable at the end of Year 5. The effective interest rate is 4% per annum (payable at the end of each year).

On initial recognition, the loan has a low credit risk and the probability of default in the next 12 months is 1% with lifetime credit losses estimated at £125,000.

At 31 December 20X2, there has been no significant deterioration in credit quality and the loan is considered to be low credit risk. The probability of default increases to 1.5% due to marginal increase in credit risk of the borrower. The lifetime credit losses are still estimated at £125,000.

At 31 December 20X3, there has been a significant deterioration of credit quality but there is no objective evidence of impairment loss. The expected credit losses over the remaining life of the loan are estimated at £50,000.

At 31 December 20X4, there is evidence of loss event and the loan defaults. The actual impairment loss is estimated at £125,000.

Requirements

Explain how the loan should be accounted for under IFRS 9 including the following:

(a) Impairment loss allowance to be recognised on origination of the loan and at the end of each of the year ended 31 December 20X2, 20X3 and 20X4 with relevant journal entries.

(b) Amount of interest income to be recognised in each of the years ended 31 December 20X2 to 20X5.

Solution

(a) Impairment loss recognised on origination and adjusted subsequently for changes in credit risk.

Initial recognition

		£
DEBIT	Loan receivable	500,000
CREDIT	Cash	500,000

		£
DEBIT	Impairment loss (P/L)	1,250
CREDIT	Loan asset/loss allowance	1,250

(Based on 12-month expected credit losses = 1% of £125,000)

Loan is classified in Stage 1

Year ended 31 December 20X2

		£
DEBIT	Impairment loss (P/L)	625
CREDIT	Loan asset/loss allowance	625

(Based on 12-month expected credit losses = 1.5% of £125,000 less £1,250 previously recognised)

Loan remains in Stage 1 as there is no significant increase in credit risk

Interest amount = 4% of £500,000 = £20,000.

Year ended 31 December 20X3

		£
DEBIT	Impairment loss (P/L)	48,125
CREDIT	Loan asset/loss allowance	48,125

(Based on lifetime expected credit losses = £50,000 less £1,875 previously recognised)

Loan is moved to Stage 2 as there is a significant increase in credit risk

Interest amount = 4% of £500,000 = £20,000.

Year ended 31 December 20X4

		£
DEBIT	Impairment loss (P/L)	75,000
CREDIT	Loan asset/loss allowance	75,000

(Based on lifetime expected credit losses = £125,000 less £50,000 previously recognised)

Loan is moved to Stage 3 as it is credit-impaired

Interest amount = 4% of £500,000 = £20,000.

(b) **Year ended 31 December 20X5**

Interest amount in the year ended 31 December 20X5 = 4% of £375,000 = £15,000

The loan has moved to Stage 3 at the end the year ended 31 December 20X4. The interest in the year ended 31 December 20X5 is calculated on the net carrying amount of the loan.

4.3.6 Simplified approach for trade and lease receivables

For trade receivables that do not have an IFRS 15 financing element, the loss allowance is measured at the lifetime expected credit losses, from initial recognition.

For other trade receivables and for lease receivables, the entity can choose (as a separate accounting policy for trade receivables and for lease receivables) to apply the three-stage approach or to recognise an allowance for lifetime expected credit losses from initial recognition.



Worked example: Trade receivable provision matrix

On 1 June 20X4, Kredco sold goods on credit to Detco for £200,000. Detco has a credit period with Kredco of 60 days. Kredco applies IFRS 9, and uses a pre-determined matrix for the calculation of allowances for receivables as follows.

Days overdue	Expected loss provision
Nil	1%
1 to 30	5%
31 to 60	15%
61 to 90	20%
90 +	25%

Detco had not paid by 31 July 20X4, and so failed to comply with its credit term, and Kredco learned that Detco was having serious cash flow difficulties due to a loss of a key customer. The finance controller of Detco has informed Kredco that they will receive payment.

Ignore sales tax.

Requirement

Show the accounting entries on 1 June 20X4 and 31 July 20X4 to record the above, in accordance with the expected credit loss model in IFRS 9.

Solution

On 1 June 20X4

The entries in the books of Kredco will be:

		£
DEBIT	Trade receivables	200,000
CREDIT	Revenue	200,000

Being initial recognition of sales

An expected credit loss allowance, based on the matrix above, would be calculated as follows:

		£
DEBIT	Expected credit losses	2,000
CREDIT	Allowance for receivables	2,000

Being expected credit loss: £200,000 × 1%

On 31 July 20X4

Applying Kredco's matrix, Detco has moved into the 5% bracket, because it has exhausted its 60-day credit period (note that this does not equate to being 60 days overdue!). Despite assurances that Kredco will receive payment, the company should still increase its credit loss allowance to reflect the increased credit risk.

Kredco will therefore record the following entries on 31 July 20X4:

		£
DEBIT	Expected credit losses	8,000
CREDIT	Allowance for receivables	8,000

Being expected credit loss: £200,000 × 5% – £2,000



Worked example: Impairment of trade receivable

Included in Samipa's trade receivables at 31 October 20X8 is an amount due from its customer Dasgupta of £51,542,000. This relates to a sale which took place on 31 October 20X6, payable in three annual instalments of £20,000,000 commencing 31 October 20X7 discounted at a market rate of interest adjusted to reflect the risks of Dasgupta of 8%. Based on previous sales where consideration has been received in annual instalments, the directors of Samipa estimate a lifetime expected credit loss in relation to this receivable of £14.4 million. The probability of default over the next 12 months is estimated at 25%. For trade receivables containing a significant financing component, Samipa chooses to follow the three-stage approach for impairments (rather than always measuring the loss allowance at an amount equal to lifetime credit losses). No loss allowance has yet been recognised in relation to this receivable.

Requirement

How should the receivable be treated in the financial statements?

Solution

A loss allowance for the trade receivable should be recognised at an amount equal to 12-month expected credit losses. Although IFRS 9 offers an option for the loss allowance for trade receivables with a financing component to always be measured at the lifetime expected losses, Samipa has chosen instead to follow the three-stage approach of IFRS 9.

The 12-month expected credit losses are calculated by multiplying the probability of default in the next 12 months by the **lifetime** expected credit losses that would result from the default. Here this amounts to £3.6 million (£14.4m × 25%).

Adjustment:

		£m
DEBIT	Expected credit loss	3.6
CREDIT	Allowance for receivables (this is offset against trade receivables)	3.6



Worked example: Impairment review

Blacksmith holds a financial asset: an investment in debt instruments, measured at amortised cost, which has a carrying amount of £430,000 (before adjustments for impairment and/or fair value changes) at 31 December 20X3. Impairment indicators require an impairment test to be conducted on each asset at 31 December 20X3. The investment in debt instruments has an impairment allowance brought forward at 1 January 20X3 of £6,000 as a Stage 1 12-month expected credit loss based on a probability of default of 2%.

Revised estimated future cash flows, measured as at 31 December 20X3 are:

	20X4	20X5	20X6	20X7	20X8
	£	£	£	£	£
Cash inflows	80,000	80,000	80,000	100,000	100,000

Cash flows are assumed to occur on 31 December each year. The original contractual cash flows for the investment in debt instruments were £100,000 on each of the above dates. The probability of default has increased to 2.5% because of the reduced cash flows.

The investment in debt instruments is held at amortised cost, using an effective interest rate of 5.25%. Its fair value at 31 December 20X3 (due to a rise in market interest rates) is £360,000. It would cost £1,000 in transaction fees to sell the instruments.

Requirement

Discuss how the asset should be accounted for in the financial statements of Blacksmith for the year ended 31 December 20X3.

Solution

An impairment test on financial assets is only required for investments in debt instruments measured at amortised cost or at fair value through other comprehensive income. An impairment allowance is recognised on initial recognition of the debt instrument based on 12-month expected credit losses (ie, lifetime expected credit losses multiplied by the probability of a default arising in the next 12 months). Debt instruments at amortised cost hold an impairment allowance on the statement of financial position (with movements taken to profit or loss). Debt instruments at fair value through other comprehensive income recognise impairment allowances in other comprehensive income (with the movement taken to profit or loss).

An impairment test is performed annually to assess any changes in credit risk. If there is no change in credit risk, the 12-month expected credit losses are recalculated using latest estimates and the asset remains in Stage 1. If there has been a significant increase in credit risk, the asset moves to Stage 2 and lifetime expected credit losses must be calculated and recognised. A significant increase in credit risk is presumed for debts more than 30 days past due. If there is 'objective evidence' of impairment (such as a default in interest or capital payments) the asset moves to Stage 3, lifetime expected credit losses continue to be

recognised and interest is calculated on the asset net of the impairment allowance. Default is presumed for debts more than 90 days past due.

Worked example: Portfolio of mortgages and personal loans

Credito Bank operates in South Zone, a region in which clothing manufacture is a significant industry. The bank provides personal loans and mortgages in the region. The average loan to value ratio for all its mortgage loans is 75%.

All loan applicants are required to provide information regarding the industry in which they are employed. If the application is for a mortgage, the customer must provide the postcode of the property which is to serve as collateral for the mortgage loan.

Credito Bank applies the expected credit loss impairment model in IFRS 9, *Financial Instruments*. The bank tracks the probability of customer default by reference to overdue status records. In addition, it is required to consider forward-looking information as far as that information is available.

Credito Bank has become aware that a number of clothing manufacturers are losing revenue and profits as a result of competition from abroad, and that several are expected to close.

Requirement

How should Credito Bank apply IFRS 9 to its portfolio of mortgages in light of the changing situation in the clothing industry?

Solution

Credito Bank should segment the mortgage portfolio to identify borrowers who are employed by clothing manufacturers and suppliers and service providers to the clothing manufacturers. This segment of the portfolio may be regarded as being 'in Stage 2', that is having a significant increase in credit risk. Lifetime credit losses must be recognised.

In estimating lifetime credit losses for the mortgage loans portfolio, Credito Bank will take into account amounts that will be recovered from the sale of the property used as collateral. This may mean that the lifetime credit losses on the mortgages are very small even though the loans are in Stage 2.

Interactive question 10: Particular defaults identified

Later in the year, more information emerged, and Credito Bank was able to identify the particular loans that defaulted or were about to default.

How should Credito Bank treat these loans?

See **Answer** at the end of this chapter.

Interactive question 11: Mortgage loans

A bank makes mortgage loans to clients. Interest charged to these clients is SONIA (Sterling Overnight Index Average) + 1%, reset monthly.

The bank recognises that in its portfolio of clients there will be some clients who will experience financial difficulties in the future and will not be able to keep up mortgage payments.

Under the mortgage agreement, the bank takes first legal charge over the mortgaged property and, in the event of a default where payments cannot be rescheduled, the property would be sold to cover unpaid debts.

Requirement

Discuss how revenue relating to the above would be accounted for under IFRS 9.

See **Answer** at the end of this chapter.

4.4 Purchased credit-impaired approach

A financial asset is described as 'credit-impaired' when one or more events that have a negative effect on future expected cash flows have already occurred.

Where financial assets are already credit-impaired when they are purchased/originated, the following approach is taken:

(a) Purchased credit-impaired financial assets are initially measured at the transaction price without an allowance for expected contractual cash shortfalls that are implicit in the purchase price.

(b) Lifetime credit losses are included in the estimated cash flows for the purposes of calculating the effective interest rate.

(c) Interest revenue is calculated on the net carrying amount at the credit-adjusted effective interest rate.

(d) Expected credit losses are discounted using the credit-adjusted effective interest rate determined at initial recognition.

(e) Subsequent changes from the initial expected credit losses are recognised immediately in profit or loss.

4.5 Recognition of credit losses

Impairment losses are recognised in profit or loss with a corresponding credit entry as follows:

Financial asset	Credit entry
Financial assets at amortised cost	Credit an allowance account. This is offset against the carrying amount of the financial asset so that a net position is presented in the statement of financial position.
Financial assets at fair value through other comprehensive income	Credit an 'accumulated impairment amount' in other comprehensive income. The carrying amount remains at fair value in the statement of financial position.
Loan commitments and financial guarantee contracts	Credit a provision account, which is presented as a separate liability.

4.6 Modifications to existing financial assets

Loans and advances may be **subject to modification**, for example, if forbearance terms are offered such as a payment holiday or a reduction in the rate of interest charged.

The modification of cash flows **may lead to derecognition of the existing financial asset** if there is a **substantial change in terms**. Judgement will need to be applied regarding what constitutes a substantial change.

When the modification results in the derecognition of the existing financial asset, the modified financial asset is considered to be a new financial asset. The new financial asset is therefore treated as any other financial asset on initial recognition and a loss allowance equal to 12-month expected credit losses is created.

In rare circumstances, the new asset may be credit-impaired at initial recognition and would be treated according to section 4.4. This could occur if a distressed asset was subject to a substantial modification.

When the modification does not result in the derecognition of the existing financial asset, the entity must assess whether there has been a significant increase in credit risk since initial recognition using the principles outlined in section 4.3.2.

Evidence that the criteria for the recognition of lifetime expected credit losses are no longer met, and that the asset may return from Stage 2 to Stage 1, may include a history of up-to-date and timely payments against modified contractual terms. A customer would need to demonstrate consistently good payment behaviour over a period of time before the credit risk is considered to have decreased. For example, a history of missed payments would not be erased simply by making one payment on time following a modification of the contractual terms.



Worked example: Modification

Melrose Bank originated a loan of £2.3 million to Cosima Ltd on 1 January 20X8. The terms of the loan were that interest at SONIA plus 2.5% would be paid annually in arrears and a bullet capital repayment was due in three years' time. The 12-month expected credit losses on initial recognition were £200,000.

Cosima Ltd suffered cash flow problems during 20X8 and informed Melrose Bank on 31 December 20X8 that it could not make its first interest payment on 31 December 20X8. Melrose Bank assessed Cosima Ltd's cash flow and profit forecasts and offered it forbearance on the loan. Forbearance terms are such that a fixed rate of interest will be charged to enable Cosima to manage its cashflows and the loan term will extend to five years term with a premium on redemption.

Requirement

How should Melrose Bank recognise impairment on the loan to Cosima Ltd in its financial statements for the year ended 31 December 20X8 based on the three-stage general model for expected credit losses?

Solution

The loan to Cosima Ltd should initially be in Stage 1 of the IFRS 9 three-stage general model, which requires 12-month expected credit losses of £200,000 to be recognised.

When forbearance is offered to Cosima Ltd, Melrose Bank must assess whether there is a substantial change in the terms of the loan.

The changes are from variable to fixed interest rate and to the term of the loan and there will be a premium on redemption. If the changes to the terms are considered substantial, the loan should be derecognised on 31 December 20X8 and a new loan recognised on which 12-month expected credit losses are provided. The assessment of subsequent increases in credit risk is based on the date the new loan is recognised. If the changes to the terms are not considered substantial, the original loan continues to be recognised.

Interactive question 12: Modification

Southwold Co originated a 4% fixed rate loan of $3.5 million to Framlingham Inc on 1 May 20X8. The term of the loan was three years, at which point the loan would be repaid in full. Interest is paid annually in arrears.

Framlingham Inc approached Southwold Co on 1 June 20X9 to ask for an extension to the loan of a further two years, making the total loan term five years. Interest continues to accrue at the fixed rate of 4% for the additional two years and is payable at the end of each year.

Requirement

Explain how the modification of the loan to Framlingham Inc should be accounted for in Southwold Co's financial statements for the year ending 31 December 20X9.

Would the accounting treatment change if interest no longer accrued in the additional two years of the loan term?

See **Answer** at the end of this chapter.

5 Application of IFRS 13 to financial instruments

Section overview

The use of fair value accounting is permitted, or required in some instances, by IFRS 9. Additional guidance is provided in IFRS 13 on how the standard is applied to financial assets and liabilities, and own equity instruments.

5.1 Introduction

IFRS 13, *Fair Value Measurement* gives extensive guidance on how the fair value of assets and liabilities should be established. It sets out to:

- define fair value
- set out in a single IFRS a framework for measuring fair value
- require disclosures about fair value measurements

IFRS 13 was covered in Chapter 2 and referred to in section 3 of this chapter. This section gives more detail of its application to financial instruments. Below is a reminder of the definition of fair value and the three-level valuation hierarchy.

IFRS 13 states that valuation techniques must be those which are appropriate and for which sufficient data are available. Entities should maximise the use of relevant **observable inputs** and minimise the use of **unobservable inputs**.

The standard establishes a three-level hierarchy for the inputs that valuation techniques use to measure fair value:

Level 1: Quoted prices (unadjusted) in active markets for identical assets or liabilities that the reporting entity can access at the measurement date.

Level 2: Inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly or indirectly eg, quoted prices for similar assets in active markets or for identical or similar assets in non-active markets or use of quoted interest rates for valuation purposes.

Level 3: Unobservable inputs for the asset or liability ie, using the entity's own assumptions about market exit value.

5.2 Financial assets

The following should be considered where a financial asset is being fair valued:

(a) If a quoted item has a bid price (the price that buyers are willing to pay) and an ask price (the price that sellers are willing to achieve), the price within the bid-ask spread that is most representative of fair value is used to measure fair value. The use of bid prices for financial assets and the use of ask prices for financial liabilities is permitted but not required. IFRS 13 does not preclude the use of mid-market pricing.

(b) In the case of equity shares, a control premium is considered when measuring the fair value of a controlling interest. Similarly, any non-controlling interest discount is considered when measuring a non-controlling interest.

(c) If an entity holds a position in a single asset or liability (including a position comprising a large number of identical assets or liabilities, such as a holding of financial instruments) and the asset or liability is traded in an active market, the fair value of the asset or liability shall be measured within Level 1 as the product of the quoted price for the individual asset or liability and the quantity held by the entity. That is the case even if a market's normal daily trading volume is not sufficient to absorb the quantity held, and placing orders to sell the position in a single transaction might affect the quoted price.

(d) A subsidiary, associate or joint venture investment in quoted equity shares is always measured as a multiple of the share price, with no adjustment for a premium associated with influence or control.

(e) The valuation of unlisted equity investments involves significant judgment and different valuation techniques are likely to result in different fair values, however this does not mean that any of the techniques are incorrect. Certain techniques are better suited to particular types of business, for example an asset-based approach is relevant to property companies whilst an income-based approach is more relevant to service businesses. It is likely that valuation will be based on some unobservable inputs and as a result the overall fair value will be classified as a Level 3 measurement.

5.3 Liabilities and own equity instruments

Liabilities and own equity instruments must be measured on the assumption that the liability or equity is transferred to a market participant at the measurement date and therefore:

- a liability would remain outstanding and the market participant would be required to fulfil the obligation; and
- an entity's own equity instrument would remain outstanding and the market participant would take on the rights and responsibilities associated with the instrument.

This differs (sometimes significantly so) from a measurement that is based on the assumed settlement of a liability or cancellation of an entity's own equity instrument.

IFRS 13 further requires that the fair value of a liability must factor in non-performance risk. Anything that could influence the likelihood of an obligation being fulfilled is considered a non-performance risk, including an entity's own credit risk.



Worked example: Fair value of liabilities

Crossley Co has a bank loan with a nominal value of £1 million that attracts a market rate of interest. Due to market concern regarding non-performance risk of Crossley Co, the market value of the loan to the bank is just £800,000. The bank will not agree to discount the amount paid by Crossley Co to extinguish the loan.

Miller Co is seeking similar financing to the bank loan and has a similar credit profile to Crossley Co. Miller Co is indifferent to obtaining a new bank loan or assuming Crossley Co's bank loan.

Requirement

What is the fair value (transfer value) of Crossley Co's bank loan?

Solution

As Miller Co has the same credit profile as Crossley Co, if it were to take out a bank loan, the bank would lend only £800,000 (the market value of Crossley Co's loan) in return for the same cash flow as are outstanding in respect of Crossley Co's loan. This is because the bank would require a higher rate of interest to compensate for the increased credit risk.

Therefore the transfer value (fair value) of Crossley Co's loan is £800,000.



Interactive question 13: Fair value of liability

Morden Co and Merton Co individually enter into legal obligations to each pay £200,000 to Wallington Co in seven years' time in exchange for some goods.

Morden Co has a very good credit rating and can borrow at 4%. Merton Co's credit rating is lower and it can borrow at 8%.

Requirement

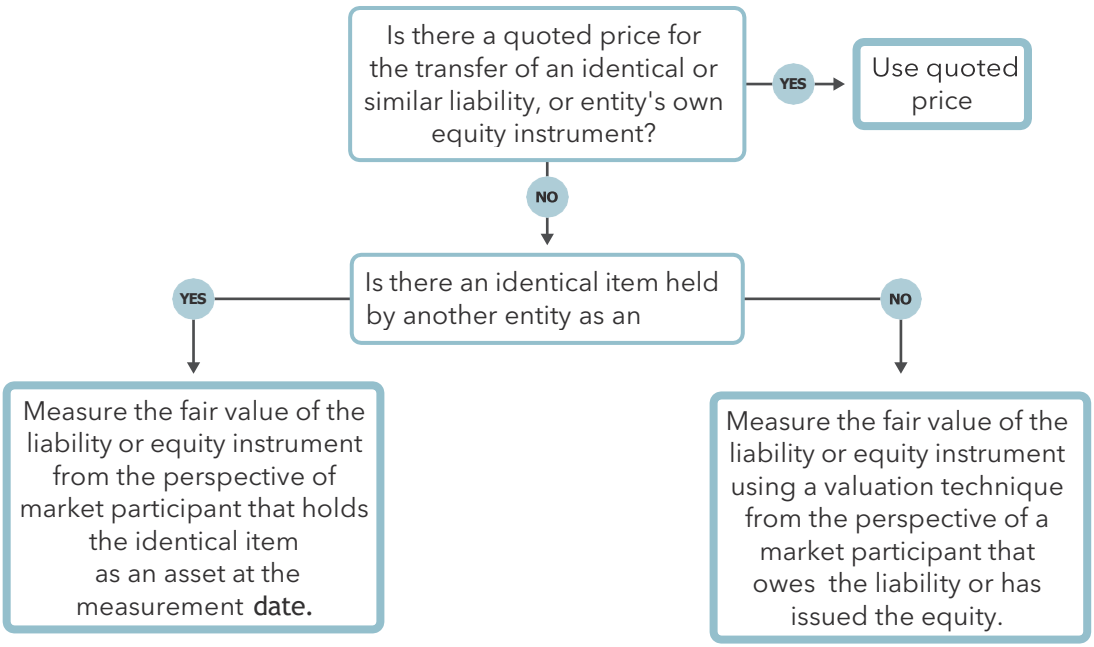
What is the fair value of the legal obligation that Morden Co and that Merton Co must record in their financial statements?

See **Answer** at the end of this chapter.

5.4 Liabilities and own equity instruments

The specific approach to fair value liabilities and an entity's own equity instruments sometimes differs from the concepts to fair value an asset and is summarised in the following flowchart:

Figure 16.3: Fair value of liabilities



6 Derivatives and embedded derivatives

Section overview

- Derivatives are financial instruments whose value changes in response to a change in the value of an underlying security, commodity, currency, index or other financial instrument(s). They normally require a zero, or small, initial net investment and are settled at a future date.
- IFRS 9 requires derivatives to be recognised when the entity becomes a party to the contractual provisions of the contract, rather than when the contract is settled.
- Derivatives are measured at fair value through profit or loss (except for derivatives used as hedging instruments in certain types of hedges).
- An embedded derivative is a component of a hybrid instrument that also includes a non-derivative host contract, and which causes some of the cash flows of the combined instrument to vary in a way similar to a standalone derivative.
- Where the host contract is a financial asset within the scope of IFRS 9, the whole contract is measured at fair value through profit or loss.
- If the host contract is not an asset within the scope of IFRS 9, the embedded derivative should be separated from the host contract and recognised separately as a derivative if:
  - economic characteristics and risks are not closely related to the host contract;
  - a separate instrument with the same terms as the embedded derivative would meet the definition of a derivative; and
  - the hybrid instrument is not measured at fair value through profit or loss

6.1 Definition of a derivative – further examples

The definition of derivative was set out earlier in the summary of the material covered at Professional Level. A quick reminder – it is a financial instrument:

- whose **value changes in response to the change in price of an underlying security, commodity, currency, index or other financial instrument(s)**;
- where the **initial net investment** is zero or is small in relation to the value of the underlying security or index; and
- that is **settled at a future date.**

A derivative normally has a notional amount, such as a number of shares or other quantity specified in the contract. For example, a forward currency contract has a quoted amount of currency even though neither the holder nor, writer is required to invest or receive this amount at the inception of the contract. However, this is not a requirement and a derivative could require a fixed payment or a variable payment based on the outcome of some future event that is unrelated to the notional amount. For example, a contract that requires the fixed payment of £1,000 if a commodity price increases by 5% or more is a derivative.

Common types of derivatives include the following:

- Swaps
- Purchased or written options (call and put)
- Futures
- Forwards

Underlying variables

Examples of underlying variables attaching to the derivatives include the following:

- Interest rates
- Currency rates
- Commodity prices
- Equity prices
- Credit-related variables

In determining whether a derivative exists, the substance of the transaction should be considered. Non-derivative transactions should be aggregated and treated as derivatives when the transactions result, in substance, in derivatives. For example, if Entity A grants a fixed rate loan to Entity B and in return Entity B grants a variable rate loan of the same amount and maturity, both entities should treat the arrangement as an interest rate swap and account for it as a derivative.

a. A defining characteristic of a derivative is that it **requires either no initial investment or an initial net investment smaller than would be required for other types of contracts** that would be expected to have a similar response to changes in market factors. A purchased call option contract, for example, meets this definition because the premium is less than the investment that would be required to obtain the underlying financial instrument to which the option is linked.

b. An interest rate swap in which the parties settle on a net basis qualifies as a derivative instrument. This is because the definition of a derivative makes reference to future settlement, but not to the method of settlement.

c. If a party prepays its obligation under a **pay-fixed, receive-variable interest rate swap** at inception, the swap should be classified as a **derivative**.

d. However, if the fixed rate payment obligation is **prepaid subsequent to initial recognition** this would be regarded as a **termination of the old swap** and an origination of a new instrument.

e. A **prepaid pay-variable**, **receive-fixed interest rate swap is not a derivative if it is prepaid at inception**, and it is no longer a derivative if it is prepaid after inception because it provides a return on the prepaid (invested) amount comparable to the return on a debt instrument with fixed cash flows. The prepaid amount fails the 'no initial net investment or an initial net investment that is smaller than would be required for other types of contracts that would be expected to have a similar response to changes in market factors' criterion of a derivative.

f. An **option** which is **expected not to be exercised**, for example because it is 'out of the money', **still qualifies as a derivative**.

   This is because an option is settled upon exercise or at its maturity. Expiry at maturity is a form of settlement even though there is no additional exchange of consideration.

g. Many derivative instruments, such as futures contracts and exchange-traded written options, require margin accounts.

   The **margin account** is **not part of the initial net investment in a derivative** instrument. Margin accounts are a form of collateral for the counterparty or clearing house and may take the form of cash, securities or other specified assets, typically liquid assets. Margin accounts are separate assets that are accounted for separately.

h. A derivative **may have more than one underlying variable**.

i. A **contract to buy or sell a non-financial asset** is a derivative if:

   (1) it can be settled net in cash or by exchanging another financial instrument; and

   (2) the contract was **not** entered for the purpose of receipt or delivery of the non-financial item to meet the entity's expected purchase, sale or usage requirements.

   An example would be a gas supply contract in the UK (where there is an active market) where the supplier or purchaser has the right to refuse delivery or receipt of the gas for financial reasons, for example because they can get a better price in the market. However, if the right to refuse delivery on the part of the seller can only be invoked for operational reasons (ie, they do not have the gas available to supply), this does not on its own make the contract a derivative.

6.1.1 Currency swaps

A currency swap (or cross-currency swap) is an interest rate swap with cash flows in different currencies. It is an agreement to make a loan in one currency and to receive a loan in another currency. With the currency swap there are three sets of cash flows. Initially, the underlying principals are exchanged when the swap starts; interest payments are then made over the life of the swap; and finally the underlying principal amounts are re-exchanged.

A currency swap could also be interpreted as issuing a bond in one currency (and paying interest on this bond) while investing in a bond in another currency (and receiving interest on this bond).



Worked example: Currency swap

Two entities enter into a 10-year fixed currency swap of euros (€) and pounds (£). Current interest rates relating to € and £ are 4% and 6% respectively. At inception of the contract the current rate of exchange is €2 per £1. The contract requires the initial exchange of €2,000 and £1,000.

Annual interest payments are made between the parties without netting of €80 (4% × €2,000) and £60 (6% × £1,000). After 10 years, the swap terminates and the original principal amounts are returned.

Requirement

Does the instrument fit the definition of a derivative?

Solution

The currency swap meets the definition of a derivative, as the exchange of the initial fair values means there is zero initial investment, its value changes in response to a specified exchange rate and it is settled at a future date.



Interactive question 14: Loan agreement as derivatives

Two entities make loans to each other for the same amount and on the same terms except that one is based on a fixed rate of interest and the other on a variable rate of interest. There are no transfers of principal at inception of the transaction since the two entities have a netting agreement.

Requirement

Does the transaction fit the definition of a derivative?

See **Answer** at the end of this chapter.



Worked example: Forward contract to buy commodity

SML, a tools manufacturer, entered into a contract to buy 50 tonnes of steel in 12 months' time, in accordance with its expected use requirements. The contract permits SML to take physical delivery of the steel or to pay or receive a net settlement in cash based on the change in the market price of steel.

Requirement

Is the contract a derivative?

Solution

The contract meets some of the criteria of a derivative; that is, there is no initial investment and it is to be settled at a future date. However, because the underlying is a non-financial asset, classification as a derivative will depend on whether the contract was entered into in order to benefit from short-term price fluctuations by selling it. If SML intends to take delivery of the steel and use it as an input in its production process, then the contract is not a derivative.

6.2 Accounting for derivatives

As noted previously, derivatives are classified as held for trading (unless they are hedging instruments – see the next chapter), so they should be **measured at fair value and changes in fair value should be recognised in profit or loss**. The following example highlights the accounting treatment of derivatives.

Context example: Accounting treatment of purchased option

On 31 December 20X0, Theta purchases put options over 100,000 shares in Omega which expire on 31 December 20X2. The exercise price of the option is £2, the market price on 31 December 20X0 and the premium paid is £11,100.

The intrinsic value of the option (i.e., the exercise price less the price per share, times the number of shares specified in the option contract) is zero at acquisition. The cost of £11,100 reflects the time value of the option which depends on the time to expiration, the price of the stock and its volatility.

If the stock price falls below £2 the put becomes in the money by the amount below the £2 strike price times the number of option shares. For instance, if the price of Omega stock fell to £1.90, the intrinsic value gain on the put option is £0.10 per share. If the stock price rises and stays above £2 for the term of the contract, the put option expires worthless to the buyer because it is out of the money. The purchaser of the put option loses the premium which is kept by the seller (writer).

Economic assumptions

The value of the shares in Omega and the put options are shown in the table below. The value of the put option increases as the stock price decreases.

	31.12.20X0	30.6.20X1	31.12.20X1
Omega shares	**£**	**£**	**£**
Price per share	2.00	1.90	1.85
Value of put option	11,100	13,500	15,000

On 31 December 20X1, Theta sells the option.

Accounting entries under IFRS 9:

	Debit	Credit
31 December 20X0	**£**	**£**
Financial asset – put option	11,100	
Cash		11,100
(To record the purchase of the put option)		
30 June 20X1		
Financial asset – put option (13,500 – 11,100)	2,400	
Profit or loss – gain on put option		2,400
(To record the increase in the fair value of the put option)		
31 December 20X1		
Financial asset – put option (15,000 – 13,500)	1,500	
Profit or loss – gain on put option		1,500
(To record the increase in the fair value of the put option)		
Cash	15,000	
Financial asset – put option		15,000
(To record the sale of the put option on 31.12.20X1)		

6.3 Embedded derivatives

Certain contracts that are not themselves derivatives (and may not be financial instruments) include derivative contracts that are 'embedded' within them.

Definition

Embedded derivative: A component of **a hybrid (combined) instrument** that also includes a **non- derivative host contract** – with the effect that some of the cash flows of the combined instrument vary in a way similar to a stand-alone derivative.

6.4 Embedded derivatives: the basics

Below we look at some basic examples and their treatment. Then we will look in detail at more complex issues.

Examples of host contracts

Possible examples include the following:

(a) An investment in a debt instrument that is convertible into ordinary shares of the issuer. The debt instrument is the host contract and the conversion option is the embedded derivative.

(b) A debt instrument with an extension option with the interest rate in the extension period reset to 1.2 times the market rate. The debt instrument is the host contract with embedded extension option.

(c) A lease contract has a rent adjustment clause based on changes in the local inflation rate. The lease is the host contract with inflation-related rentals being the embedded derivative.

Examples of embedded derivatives

Possible examples include:

(a) A construction contract priced in a foreign currency. The construction contract is a non-derivative contract, but the changes in foreign exchange rate is the embedded derivative.

(b) A bond which is redeemable in five years' time, with part of the redemption price being based on the increase in the FTSE 100 Index:

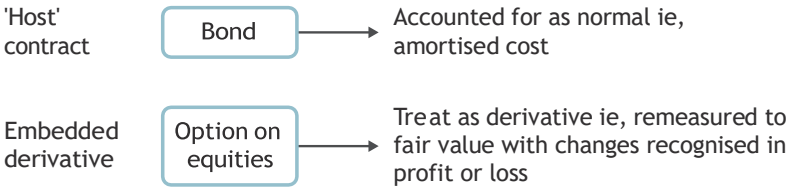


Figure 16.4: Embedded derivatives

Accounting treatment of embedded derivatives

The basic rule for accounting for an embedded derivative is that, if the host contract is a financial asset within the scope of IFRS 9, the whole contract is measured as at fair value through profit or loss.

Context example: Embedded derivative – host is a financial asset

Myrtle plc has made an investment in a bond which has interest payments linked to the price of platinum.

The bond investment is a financial asset within the scope of IFRS 9, therefore the entire contract is measured at fair value through profit or loss.

If the host contract is not a financial asset within the scope of IFRS 9, the embedded derivative should be separated from its host contract and accounted for separately as a derivative. The purpose is to ensure that the embedded derivative is measured at fair value and any changes in its fair value are recognised in profit or loss. But this separation should only be made when the following conditions are met:

(a) The **economic characteristics** and risks of the embedded derivative are **not closely related** to the economic characteristics and risks of the **host contract**.

(b) A **separate instrument** with the same terms as the embedded derivative **would meet the definition** of a **derivative**.

(c) The hybrid (combined) instrument is **not measured at fair value with changes in fair value recognised inprofit or loss** (if changes in the fair value of the total hybrid instrument are recognised in profit or loss, then the embedded derivative is already accounted for on this basis, so there is no benefit in separating it out).

The meanings of **'closely related'** and **'not closely related'** are dealt with in more detail below.

Note that an entity may, subject to conditions, **designate a hybrid contract as at fair value through profit or loss**, thereby avoiding the need to measure the fair value of the embedded derivative separately from that of the host contract.

The conditions for classifying the entire hybrid contract as at fair value through profit or loss are as follows:

(a) If the host contract is not an asset within the scope of IFRS 9, an entity may designate the whole contract as at fair value through profit or loss unless:
   (1) the embedded derivative does not significantly modify the host contract's cash flows; or
   (2) it is clear with little or no analysis that separation of the embedded derivative is prohibited, such as a prepayment option embedded in a loan that permits the holder to prepay the loan for approximately its amortised cost.

(b) If the entity is required to separate the embedded derivative but it cannot be measured separately, the entity may designate the entire contract as at fair value through profit or loss.

If the fair value of the embedded derivative cannot be determined due to the complexity of its terms and conditions, but the value of the hybrid and the host can be determined, then the **value of the embedded derivative** should be determined as the **difference between the value of the hybrid** and the **value of the host contract**.



Context example: Embedded derivative – host is not a financial asset (1)

Zainab Co leases machinery to Kool Ltd to produce a new product. The payments under the lease are determined by reference to the sales Kool Ltd makes of the new product.

The variable lease payments are created by an embedded derivative within the lease host contract. The embedded derivative is based on the underlying sales of the new product.

The host contract (the lease) is not a financial asset under IFRS 9. Zainab Co may not separate the embedded derivative from the host contract because it is closely related to the host. The whole hybrid instrument may be designated at fair value through profit or loss.



Context example: Embedded derivative – host is not a financial asset (2)

Ayia Co has a lease contract with a rent adjustment clause based on changes in the inflation rate. The lease contract (outside the scope of IFRS 9) contains an embedded derivative based on inflation.

The embedded derivative is not separated from the lease contract because it is considered to be closely related to the host lease contract.

If the lease contract has a rent adjustment clause based on leveraged inflation (say twice the change in the inflation rate), the embedded derivative must be separated because it is not closely related to the host contract.

6.5 Key characteristics of embedded derivatives

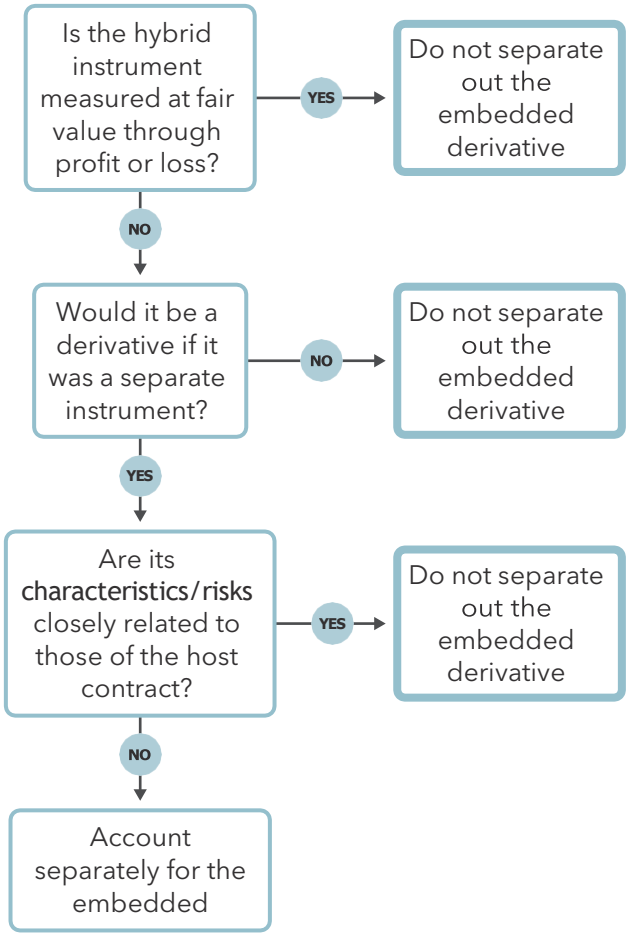
An embedded derivative **causes some or all of the cash flows of the host contract to be modified**, based on a specified interest rate, financial instrument price, commodity price, foreign exchange rate, index of price or rates, credit rating or credit index or other variables. As a result, the financial payoffs of the hybrid instrument will resemble those of a standalone derivative.

A **key characteristic** of embedded derivatives is that the **embedded derivative cannot be transferred** to a third party independently of the instrument. For example, the embedded equity option in a convertible bond cannot be exercised with the bond being retained. If the conversion option is exercised, the bond will have to be derecognised. This is different from a bond with a detachable warrant, which gives the right to the owner to exercise the warrant and buy shares while retaining the bond. This is not a hybrid or combined instrument. The warrant is a separate financial instrument, not an embedded derivative.

6.6 Separation of host and embedded derivatives

If the host contract within a hybrid instrument is not a financial asset under IFRS 9, the embedded derivative may need to be separated out. Whether there is a requirement to separate the host contract from its embedded derivative can require complex analysis. The three conditions above must be met. This diagram shows how these three conditions should be tested for.

Figure 16.5: Establishing the treatment of embedded derivatives



6.7 Identification of embedded derivatives

Embedded derivatives can be structured deliberately, as has been already mentioned, or they **may arise inadvertently**. An example of this would be a **currency that is different from the functional currencies of both the buyer and seller of goods**. Multiple embedded derivatives are treated as if they were a single embedded derivative, unless they relate to different risk exposures which can be identified and separated, in which case they can be accounted for separately.

Although, theoretically, the host of an embedded derivative could be any type of contract that is not recorded at fair value, in practice there is a small number of contracts that have derivatives embedded in them, the most common of which are:

- debt instruments
- equity instruments
- leases
- insurance contracts
- executory contracts such as purchase and sale contracts

The identification of embedded derivatives requires the entity to consider all executory contracts such as purchases and sales contracts, and commitments.

For example, an embedded derivative may be identified if contracts contain:

- rights or obligations to **exchange** at some time in the future;
- rights or obligations to **buy or sell**;
- provisions for **adjusting the cash flows** according to some interest rate, price index or specific time period;
- options which permit either party to **do something not closely related** to the contract; and/or
- unusual **pricing terms** (e.g., a lease which charges interest at rates linked to the FTSE 100 (Financial Times Stock Exchange) yield contains an embedded swap).

Finally, a comparison of the terms of a contract (such as maturity, cancellation or payment provisions) with the terms of another similar, non-complex contract may indicate the existence, or not, of an embedded derivative.

6.8 Examples of embedded derivatives not closely related

Examples where the embedded derivative is **not** closely related to the **host instrument** include the following:

- The option to **extend the term** on a fixed rate debt instrument without resetting the interest rate to market rates
- **Credit derivatives** in a debt instrument
- A **put option** embedded in an instrument that enables the holder to require the issuer to reacquire the instrument for an amount of cash or other assets, which varies on the basis of the change in an equity or commodity price or index
- **Equity-indexed interest or principal payments** embedded in a host debt instrument or insurance contract
- **Commodity-indexed interest or principal payments** embedded in a host debt instrument or insurance contract
- A **call, put, or prepayment option** embedded in a host debt contract or host insurance contract
- Credit derivatives that are embedded in a host debt instrument allowing the transfer of credit risk from one party to another.

In all of the above circumstances, the embedded derivative is separated out from the host contract for accounting purposes.

6.9 Examples of closely related embedded derivatives

No definition of 'closely related' is included in IFRS 9. In general, an embedded derivative is considered to be **closely related if it modifies** the **inherent risk of the combined contract but leaves the instrument substantially unaltered**. Some common examples of closely related embedded derivatives are given below.

- An embedded derivative based on an interest rate (or interest rate index) that can **change the amount of interest otherwise paid or received** on an interest-bearing debt or insurance host contract.
- An **embedded floor or cap** on the interest rate on a debt contract or insurance contract, provided the floor is at or below, and the cap is at or above, the market rate of interest when the contract is issued, and the cap or floor is not leveraged in relation to the host contract.
- An **embedded foreign currency derivative** that provides a stream of principal or interest payments that are denominated in a foreign currency and is embedded in a host **debt instrument** (e.g., a dual currency bond). Such a derivative is not separated from the host instrument because IAS 21 requires foreign currency gains and losses on monetary items to be recognised in profit or loss.

- An **embedded foreign currency derivative** in a host contract that is an **insurance contract** or not a financial instrument (such as a contract to purchase a non-financial item where the price is denominated in a foreign currency), provided the payment is to be in the functional currency of one of the substantial parties to the contract, or in the currency in which the contracted good or service is routinely denominated (such as the US dollar for crude oil transactions).
- An **embedded prepayment option** in an interest-only or principal-only strip is closely related to the host contract provided the host contract initially resulted from separating the right to receive contractual cash flows of a financial instrument that, in and of itself, did not contain an embedded derivative, and does not contain any terms not present in the original host debt contract.
- An embedded derivative in a host lease contract is closely related to the host contract if the embedded derivative is an **inflation-related index** such as an index, of lease payments to a consumer price index (provided that the lease is not leveraged and the index relates to inflation in the entity's own economic environment), contingent rentals based on related sales or contingent rentals based on variable interest rates.
- A **unit-linking feature** embedded in a host financial instrument or host insurance contract is closely related to the host instrument or host contract if the unit-denominated payments are measured at current unit values that reflect the fair values of the assets of the fund. A unit-linking feature is a contractual term that requires payments denominated in units of an internal or external investment fund.
- A derivative **embedded in an insurance contract** is closely related to the host insurance contract if the embedded derivative and host insurance contract are so interdependent that an entity cannot measure the embedded derivative separately (i.e., without considering the host contract).

In all of the above circumstances, the hybrid contract is accounted for as a whole – the embedded derivative is not separated out.

6.10 Accounting for embedded derivatives

A common transaction involving an embedded derivative is the purchase or sale of goods at a price denominated in a foreign currency – for example, the purchase by an entity of 100 items for $10 each with settlement in 90 days, where the functional currency of the purchasing entity is the pound (£).

Under the guidance set out above, the embedded derivative should be separated out unless:

- the vendor's functional currency is the dollar ($); or
- the items in the transaction are routinely denominated in dollars in international commerce.

The terms of non-option-based derivatives, such as forwards and swaps, should be determined such that the derivative has a fair value of nil at inception. Option-based derivatives, such as puts, swaptions (an option on a swap), and caps, should be separated based on the terms in the contract. Multiple embedded derivatives are required to be separated as a single compound embedded derivative.



Interactive question 15: Embedded derivatives

Moorgate plc is based in the UK and has sterling (£) as its functional currency. In the ordinary course of business it entered into the following contracts:

Contract (1)

Supply of services to Blue Co which operates in Andlay, whose functional currency is the CU. Under the contract Moorgate plc will supply services for a fixed price of INR12 million and is due to receive INR3 million in each of the next four years. The INR is infrequently used as the measure of contract prices in UK or Andlay.

Contract (2)

Issue of debt in £ with interest payable at SONIA plus credit spread of 100 basis points. SONIA at issuance is 1.5% so the rate at inception of the debt is 2.5%. The debt contract provides that if SONIA were to rise, the rate payable on the debt would not rise above 4%.

Neither contract is measured at fair value through profit or loss.

Requirement

Explain in respect of each contract whether there is an embedded derivative and, if so, whether it should be recognised separately in the financial statements of Moorgate plc.

See **Answer** at the end of this chapter.

6.11 Reassessment of embedded derivatives

IFRS 9 addresses the question of whether it is necessary to reassess the treatment of an embedded derivative throughout the life of a contract if certain events occur after an entity first becomes a party to the contract. **It concludes that reassessment is not permitted unless there is a significant change to the terms of the contract.**

6.12 Impact on financial statements

Derivatives and embedded derivatives are measured at fair value with changes in fair value recognised in profit or loss. Derivative contracts are entered into for both trading and risk management purposes. Derivatives enable entities to hedge against fair value or cash flow exposures. Banks should be clear about the objectives of originating such exposures that are likely to have a significant impact on the financial statements. It is important to regularly review the exposures and accounting treatment of derivatives as their values are subject to changes in value of an underlying, which is generally beyond the control of the entity. The fair value movements can have a significant impact on the financial performance and financial position of the entity.

The auditors have to verify that the principles in IFRS 9 are appropriately applied to identify and account for embedded derivatives. An assessment of management judgement (on whether an embedded derivative is closely related to the host contract) is required since this is a factor in determining whether the embedded derivative should be separated or not. The auditors may need to review in detail the terms and conditions of the hybrid contract.



Professional skills focus: Concluding, recommending and communicating

This complex area involving management judgement requires considerable communication skills. Professional scepticism is important, and if the auditors disagree with management's assessment, they will need to support their reasons for doing so.

7 Current developments



Section overview

This section deals with a new discussion paper on financial instruments with the characteristics of equity.

7.1 Financial instruments with the characteristics of equity

The issue of distinguishing, in some cases, between equity and debt has been an IASB project since 2010. The significance of the distinction was discussed in Chapter 15 in connection with IAS 32, *Financial Instruments: Presentation*. In practice, the debt/equity distinction may not be clear cut. Classification of financial instruments as debt or equity can have a significant effect on the financial statements.

The project is linked to the *Conceptual Framework* as the complexity of questions around distinguishing between liabilities and equity meant that these were excluded from the project to revise the *Conceptual Framework*. The revised framework therefore includes a revised definition of a liability and new supporting guidance, but the definition of equity remained unchanged.

7.1.1 2018 Discussion Paper

In 2018, the IASB issued a Discussion Paper, *Financial Instruments with Characteristics of Equity*. The main proposals are as follows:

(a) At least initially, the main principle of IAS 32 is to remain unchanged. In other words, equity is a **residual** that remains if the characteristics of a financial liability are not fulfilled.

(b) Accordingly, a financial instrument must be classified as a financial liability if its contractual terms contain an unavoidable obligation:

  (1) to transfer cash or another financial asset at a specified time other than at liquidation (**timing feature**); and/or

  (2) for an amount independent of the entity's available economic resources (**amount feature**).

The IASB met in June 2022 to continue its discussions on the reclassification of financial instruments issued by an entity as financial liabilities or equity instruments when the substance of the contractual terms changes without a modification to the contract. Its decisions so far are tentative.

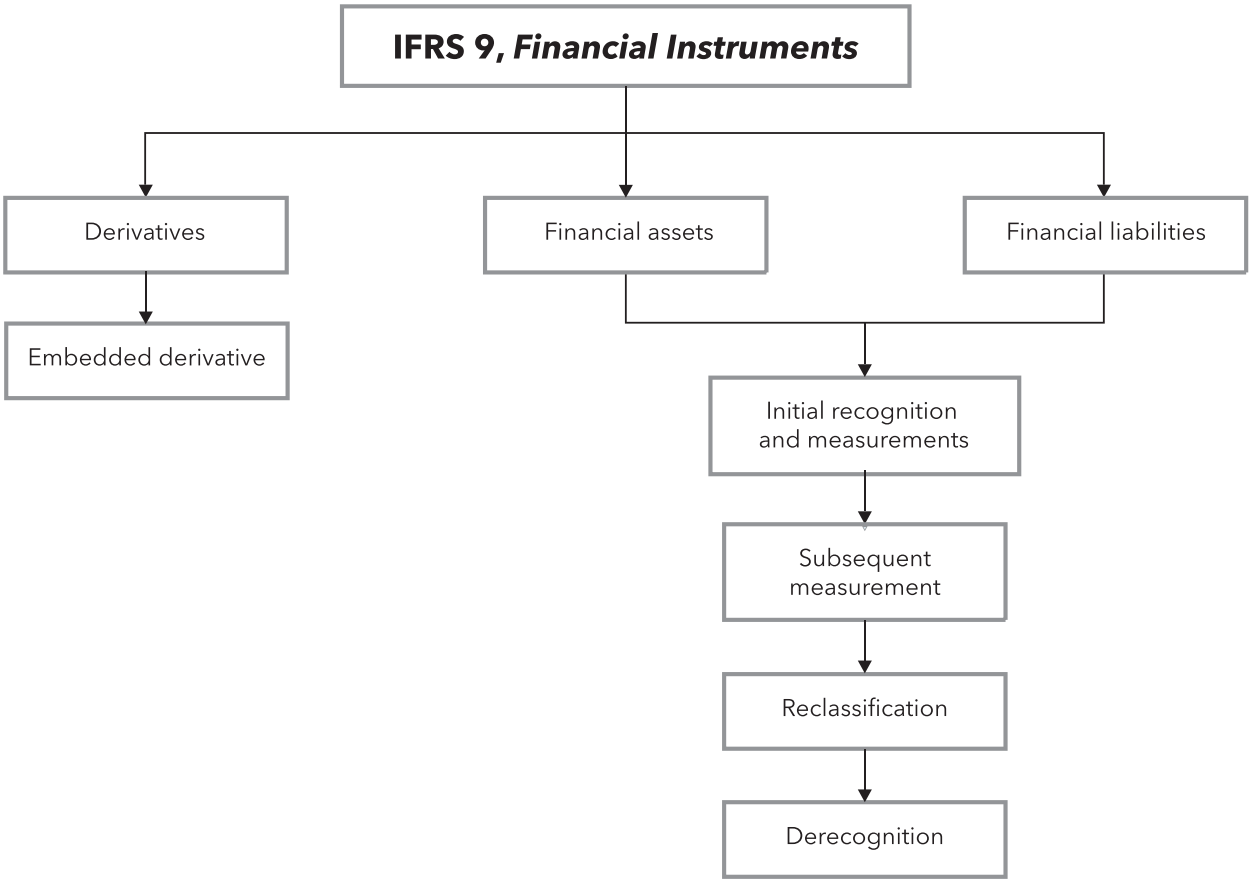
7.2 *Classification of financial instruments*

A Post-Implementation Review of the classification and measurement requirements of IFRS 9 was completed in 2022 and identified that, whilst the requirements of IFRS 9 generally work well, additional guidance in certain areas would be useful. One of these areas was the classification of financial assets with Environmental, Social and Governance (ESG) linked features eg, loans whose interest rate is reduced if the borrower meets pre-determined ESG targets such as water usage or CO2 emissions. In particular the issue of whether the contractual cash flows test is met by this type of financial asset is unclear. The contractual cash flows test must be met in order for an asset to be classified as measured at amortised cost or FVTOCI.

The IASB has proposed to amend IFRS 9 to clarify that where a change to contractual cash flows is contingent upon an ESG target, the contractual cash flows test is met if the contingent event is specific to the borrower. Therefore:

- *In the case of a loan whose interest rate is periodically adjusted if the borrower achieves a contractually specified reduction in greenhouse gases, the contractual cash flows test is met, however*
- *In the case of a loan whose interest rate is periodically adjusted if a market-determined carbon price index reaches a contractually defined threshold, the contractual cash flows test is not met.*

Summary


IFRS 9, *Financial Instruments*
Derivatives
Financial assets
Financial liabilities
Embedded derivative
Initial recognition and measurements
Subsequent measurement
Reclassification
Derecognition

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	Can you distinguish between debt instruments at amortised cost and debt instruments at FVTOCI? (Topic 1)
2.	Which financial liabilities are valued at FVTOCI? (Topic 1)
3.	Can you apply the business model test? (Topic 2)
4.	Can you apply IFRS 13, *Fair Value Measurement* to a financial instrument? (Topic 4)
5.	Can you apply the correct financial reporting treatment for embedded derivatives? (Topic 6)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Trowbridge	This is an amortised cost calculation requiring journal entries and knowledge of the IFRS 9, *Financial Instruments* business model. It builds on simpler calculations at Professional Level.
Longridge	One of the trickier areas of IFRS 9 is impairment and the expected credit loss model. This question should be a good stepping stone to this topic should it come up in the exam.
Gaia	This provides further practice at expected credit losses with the emphasis on explaining rather than calculating.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Robicorp	This question tests derecognition, fair value of an equity instrument and fair value of an interest-free loan to employees.
Expando	You are required to show the correct treatment – the company has treated it incorrectly – and to consider auditing issues.

Verloc	This requires a detailed discussion on derecognition of a financial liability.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving onto the next chapter.

Technical reference

1 IFRS 9, *Financial Instruments*

Recognition and measurement

- Initial recognition – **IFRS 9.3.1.1**
- Regular way purchase or sale – **IFRS 9.3.1.2, B3.1.3–B3.1.6**
- Derecognition of financial assets – **IFRS 9.3.2**
- Initial measurement – **IFRS 9.5.1**
- Subsequent measurement of financial assets – **IFRS 9.5.2**
- Subsequent measurement of financial liabilities – **IFRS 9.5.3**
- Amortised cost measurement – **IFRS 9.5.4**
- Embedded derivatives – **IFRS 9.4.3**

Impairment

- IFRS 9 impairment of financial assets
  - General approach – **IFRS 9.5.5.1–5.5.8**
  - Significant increase in credit risk – **IFRS 9.5.5.9–11, B5.5.15–24**
  - Collective and individual assessment – **IFRS 9.B5.5.1–6**
  - Purchased or originated credit-impaired assets – **IFRS 9.5.5.13**
  - Simplified approach – **IFRS 9.5.5.15–16**
  - Measurement of expected credit losses – **IFRS 9.5.5.17–20, B5.5.28–35**
  - Reasonable and supportable information – **IFRS 9.B5.5.49–5**

Self-test questions

Answer the following questions.

1 Stripe Co

On 6 November 20X3 Stripe Co acquires an equity investment with the intention of holding it in the long term. The investment cost £500,000. At Stripe Co's year end of 31 December 20X3, the market price of the investment is £520,000.

Stripe Co has elected to present the equity investment at FVTOCI.

Requirement

How is the asset initially and subsequently measured?

2 Trowbridge Co

On 1 January 20X8, Trowbridge Co purchased 1,000 bonds issued by Spectra Tech Limited for £97,327. The remaining period to maturity on these bonds is three years and the bonds will be held until maturity when they will be redeemed at par. The par value of each bond is £100.

The annual coupon on these bonds, receivable in arrears, is 5% and the effective interest rate is 6%.

Requirement

Present journal entries to show how the bond asset and related income are recognised over the three years ending 31 December 20X8, 20X9 and 20Y0 if the bonds are in the business model of being held to collect contractual cash flows.

3 Purple Company

During 20X1, The Purple Company invested in 80,000 shares in a stock market quoted company. The shares were purchased at £4.54 per share. The broker collected a commission of 1% on the transaction.

Purple elected to measure these shares at fair value through other comprehensive income.

The quoted share price at 31 December 20X1 was £4.22–£4.26.

Purple decided to 'bed and breakfast' the shares to realise a tax loss, and therefore sold the shares at market price on 31 December 20X1 and bought the same quantity back the following day. The market price did not change on 1 January 20X2. The broker collected a 1% commission on both the transactions.

Requirement

Explain the IFRS 9 accounting treatment of the above shares in the financial statements of Purple for the year ended 31 December 20X1 including relevant calculations.

4 Marland

The Marland Co is preparing its financial statements for the year ended 30 April 20X5. Included in Marland's trade receivables is an amount due from its customer, Metcalfe, of £128.85 million. This relates to a sale which took place on 1 May 20X4, payable in three annual

instalments of £50 million commencing 30 April 20X5 discounted at a market rate of interest adjusted to reflect the risks of Metcalfe of 8%. Based on previous sales where consideration has been received in annual instalments, the directors of Marland estimate a lifetime expected credit loss of £75.288 million in relation to this receivable balance. The probability of default over the next 12 months is estimated at 25%. For trade receivables containing a significant financing component, Marland chooses to follow the three-stage approach for impairments (rather than always measuring the loss allowance at an amount equal to lifetime credit losses). No loss allowance has yet been recognised in relation to this receivable.

Requirement

Explain, with supporting calculations, how this receivable should be accounted for in the financial statements of Marland for the year ended 30 April 20X5.

5 Longridge Co

Longridge Co purchased a bond on 1 January 20X8 for its par value of £100,000 and measures it at fair value through other comprehensive income. The instrument has a contractual term of five years and an interest rate of 5%, payable annually in arrears on 31 December. The 12-month expected credit losses on origination are £1,000.

On 31 December 20X8, the fair value of the debt instrument has decreased to £96,000 as a result of changes in market interest rates. There has been no significant increase in credit risk since initial recognition, and expected credit losses are measured at an amount equal to 12-month expected credit losses, amounting to £1,500.

On 1 January 20X9, the company sells the bond for its fair value of £96,000.

Requirement

Explain the impact of the above transactions in 20X8 and 20X9 on profit or loss, other comprehensive income and the statement of financial position under IFRS 9.

6 Gaia Bank

Gaia Bank holds a portfolio of credit card loans held at £460 million on its statement of financial position at 31 December 20X8. Gaia Bank intends to collect contractual cash flows of interest and principal repayments from its customers. The average balance on each credit card is approximately £1,200. On average, the probability of default on initial recognition is 18% and the loss given default is 90%. Impairment allowances are assessed collectively for unsecured lending on credit cards.

The probability of default remains the same over the product life. Gaia Bank does not classify credit card loans and receivables as Stage 3 until required payments are outstanding for more than 120 days.

Requirement

Explain the correct financial reporting treatment of the credit card loans and advances under IFRS 9.

7 Pike

The Pike Company issued £18 million of convertible bonds at par on 31 December 20X7. Interest is payable annually in arrears at a rate of 11%. The bondholders can convert into 8 million ordinary shares after 31 December 20Y1.

The bond has no fixed maturity and contains a call option whereby Pike can redeem the bond at any time at par value.

At 31 December 20X7, a bond with a similar credit status and the same cash flows as the one issued by Pike, but without conversion rights or a call option, is valued in the market at £11 million.

Using an option pricing model, it is estimated that the value of the call option on a similar bond without conversion rights would be £3 million.

Requirement

What carrying amount should be recognised for the liability in respect of the convertible bond in the statement of financial position of Pike at 31 December 20X7, in accordance with IAS 32, *Financial Instruments*: *Presentation*?

8 Mullet

The Mullet Company issued £55 million of convertible bonds at par on 31 December 20X7. Interest is payable annually in arrears at a rate of 8%. The bondholders can convert into 20 million ordinary shares after 31 December 20Y1.

The bond has no fixed maturity and contains a call option whereby Mullet can redeem the bond at any time at par value.

At 31 December 20X7, a bond with a similar credit status and the same cash flows as the one issued by Mullet but without conversion rights or a call option is valued in the market at £52 million.

Using an option pricing model it is estimated that the value of the call option on a similar bond without conversion rights would be £1 million.

Requirement

What carrying amount should be recognised for the equity element of the convertible bond in the statement of financial position of Mullet at 31 December 20X7, in accordance with IAS 32, *Financial Instruments*: *Presentation*?

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 The redeemable preference shares require regular distributions to the holders, but more importantly have the debt characteristic of being redeemable. Therefore, according to IAS 32 they should be classified as debt (a financial liability).

1.2 According to IFRS 2, *Share-based Payment* the grant of share options must be recognised in equity. Share options are an alternative to cash as remuneration, so an expense should be measured in profit or loss with a credit to equity.

Answer to Interactive question 2

2.1 A guarantee to replace or repair goods sold by a business in the normal course of business does not fall within the definition of a financial liability, so it should be dealt with under IAS 37 *Provisions, Contingent Liabilities and Contingent Assets*.

2.2 A firm commitment (order) to purchase a specific quantity of cocoa beans for use in manufacturing is not a financial liability. This is a normal operating purchase which is not recognised until delivery when there is a contractual obligation on the part of the purchaser to pay for the cocoa beans.

2.3 A forward contract such as this falls within the definition of a derivative, so in principle it does fall within IFRS 9. The only exception would be if the contract is for the entity's expected usage of cocoa beans in its business (outside the scope of IFRS 9). This would be accounted for as a normal purchase on delivery, as in requirement 2.

Answer to Interactive question 3

1 January 20X2

		£	£
DEBIT	Loan (£9,500 + £250 legal fees)	9,750	
CREDIT	Cash		9,750

31 December 20X2

		£	£
DEBIT	Cash	1,000	
CREDIT	Interest income (£9,750 × 9.48%)		924
CREDIT	Loan (bal fig)		76

Note: Loan balance is now £9,674 (£9,750 – £76).

31 December 20X3

		£	£
DEBIT	Cash	1,000	
CREDIT	Interest income (£9,674 × 9.48%)		917
CREDIT	Loan (bal fig)		83

Note: Loan balance is now £9,591 (£9,674 – £83).

31 December 20X4

		£	£
DEBIT	Cash (£9,500 + £1,000)	10,500	
CREDIT	Interest income (£9,591 × 9.48%)		909
CREDIT	Loan (bal fig)		9,591

Answer to Interactive question 4

The objective of E Co's business model is to hold financial assets to collect contractual cash flows. Selling financial assets is only incidental to E Co's business model.

Answer to Interactive question 5

This is a derecognition issue. Legally the debt instrument remains the property of the entity. However, in order to determine whether the investment in the debt instrument should be derecognised, the entity needs to establish if substantially all the risks and rewards of ownership have been transferred.

In this case, the risks and rewards relating to the interest cash flows generated by the asset have been transferred because the entity has no obligation to compensate the third party for any cash flows not received ie, the third party suffers the risk. This is not the case for the ultimate maturity cash flow (the principal).

Under IFRS 9, where an entity transfers substantially all the risks and rewards of part of a financial asset, that part is derecognised providing that part comprises only specifically identified cash flows. An interest rate strip is given as an example by the standard.

Any difference between the proceeds received and the carrying amount (measured at the date of derecognition) of the interest cash flows derecognised is recognised in profit or loss. The amount derecognised is calculated by multiplying the carrying amount of the debt instrument by the proportionate fair value of the interest flows versus the whole fair value of the debt instrument, both at the date of the transfer. This leaves a 'servicing asset' (the principal element) which continues to be recognised.

Answer to Interactive question 6

6.1 ABC should derecognise the asset as its option to repurchase is at the prevailing market value.

6.2 DEF should not derecognise the asset as it has retained substantially all the risks and rewards of ownership. The stock should be retained in its books even though legal title is temporarily transferred.

6.3 XYZ has received 90% of its transferred receivables in cash, but whether it can retain this amount permanently is dependent on the performance of the factor in recovering all of the receivables. XYZ may have to repay some of it and therefore retains the risks and rewards of 100% of the receivables amount. The receivables should not be derecognised. The cash received should be treated as a loan.

The 10% of the receivables that XYZ will never receive in cash should be treated as interest over the six-month period; it should be recognised as an expense in profit or loss and increase the carrying amount of the loan.

At the end of the six months, the receivables should be derecognised by netting them against

the amount of the loan that does not need to be repaid to the factor. The amount remaining is bad debts which should be recognised as an expense in profit or loss.

Answer to Interactive question 7

Trade date accounting

The financial asset should be derecognised on 27 December 20X8 and a receivable of £3,250 recognised. At the same date, a gain of £250 should be recognised in profit or loss, which includes any previous gains recognised in other comprehensive income which are now reclassified to profit or loss.

On 5 January 20X9, the counterparty pays the £3,250 to clear the receivable.

Settlement date accounting

- The financial asset should be re-measured at the fair value of £3,250 on 27 December 20X8. The cumulative gain of £250 should be recognised in other comprehensive income.
- No further entries are made on 31 December 20X8, as the entity has no right to further fair value movements.
- On 5 January 20X9 the financial asset should be derecognised and the gain of £250 reclassified to profit or loss.

Answer to Interactive question 8

(a) **Statement of profit or loss and other comprehensive income (extract)**

Profit or loss for the year	**£m**
Investment income (10m × (6.5 – 5.0))	15
Dividend income (10m × 20p)	2
Transaction costs	(3)
Statement of financial position (extract)	
Investments in equity instruments (10m × 6.5) = 65	

(b) **Statement of profit or loss and other comprehensive income (extract)**

	£m
Profit or loss for the year	
Dividend income	2
Other comprehensive income	
Gain on investment in equity instruments (65 – (50 + 3))	12
Statement of financial position (extract)	
Investments in equity instruments (10m × 6.5)	65

(c) **Statement of profit or loss and other comprehensive income (extract)**

	£m
Profit or loss for the year	
Dividend income (5m × 20p)	1.0
Other comprehensive income	
Gain on revaluation of investment prior to disposal (5m × 6) – (25 + 1.5)	3.5
Gain on remaining investment in equity instruments (5m × 6.5) – (25 + 1.5)	6.0
Statement of financial position (extract)	32.5
Investments in equity instruments (5m × 6.5)	

Answer to Interactive question 9

The correct answer is:

B £80,613

The premium on redemption of the loan notes represents a finance cost. The effective rate of interest must be applied so that the debt is measured at amortised cost.

At the time of issue, the loan notes are recognised at their net proceeds of £599,800 (£600,000 – £200).

The finance cost for the year ended 31 December 20X4 is calculated as follows:

	B/f	Interest @ 12%	C/f
	£	£	£
20X3	599,800	71,976	671,776
20X4	671,776	80,613	752,389

Answer to Interactive question 10

The loans are now in Stage 3. Lifetime credit losses should continue to be recognised, and interest revenue should switch to a net interest basis, that is on the carrying amount net of allowance for credit losses.

Answer to Interactive question 11

The revenue associated with mortgage loan assets is the interest. Under IFRS 9, the variable element of the interest is accrued on a time basis, and the fixed element is reduced to reflect any initial transaction costs and to reflect a constant return on the balance outstanding.

Answer to Interactive question 12

The loan to Framlingham Inc was initially recognised on 1 May 20X8. At this point in time, 12-month expected credit losses would be recognised to reflect the probability of default within the next 12 months multiplied by total lifetime expected credit losses.

In the year ended 31 December 20X9, the terms of the loan to Framlingham Inc were modified. Southwold Co must assess whether it is a substantial change in terms. Given the delayed repayment is compensated by the continued accrual of interest and no other terms have changed, it looks likely that the initial loan will not be derecognised.

Therefore, Southwold Co must consider whether there has been a significant increase in credit risk since 1 May 20X8 and, if so, the loan moves to Stage 2 and Southwold Co recognises lifetime expected credit losses ie, the lifetime probability of default multiplied by total lifetime expected credit losses.

If interest did not accrue in the fourth and fifth years of the extended loan, Southwold Co may consider that the forbearance terms constitute a substantial change to the original terms. If so, the original loan is derecognised and a new loan recognised in Stage 1 with 12-month expected credit loss on initial recognition.

Answer to Interactive question 13

The fair value of Morden Co's promise is approximately £152,000. This is the present value of £200,000 in seven years' time at 4% (£200,000 × $1/1.04^7$).

The fair value of Merton Co's promise is approximately £116,700. This is the present value of £200,000 in seven years' time at 8% (£200,000 × $1/1.08^7$).

These two values are different, even though the amount and period are the same, due to the different risk profiles of the two companies.

Answer to Interactive question 14

In substance, the effect of the two transactions is an interest rate swap with no initial investment. This therefore meets the definition of a derivative since there is no initial net investment, an underlying variable is present and future settlement will take place. This would be the same even if no netting agreement existed because the definition of a derivative does not include a requirement for net settlement.

Answer to Interactive question 15

In Contract (1) there is an embedded derivative which should be separated from the host contract.

In Contract (2) there is an embedded derivative but this should not be separated from the host contract.

An embedded derivative should be separated from the host contract if:

- the economic characteristics and risks are not closely related to the host contract;
- a separate instrument with the same terms would meet the definition of a derivative; and
- the combined instrument has not been designated as at fair value through profit or loss.

Contract (1): The derivative embedded in this contract must be separated out, because the INR, the currency in which the contract is denominated, is not the functional currency of either party, nor is it the currency used internationally as the measure of contract prices so it is not closely related.

Contract (2): The cap is an embedded derivative that is closely related to the host debt contract because, at the time of the issuance of the debt, the cap is out-of-the-money. The embedded derivative does not require separation as it is closely related.

Answers to Self-test questions

1 Stripe Co

The asset is initially recognised at the fair value of the consideration, being £500,000.

At the period end it is remeasured to £520,000.

This results in the recognition of £20,000 in other comprehensive income.

2 Trowbridge Co

Trowbridge Co has purchased the bonds to hold until maturity. The interest payments are solely payments of principal and interest on the principal amount outstanding and they are held within a business model to collect contractual cash flows. The bonds are initially measured at fair value, the purchase price of £97,327, and are subsequently measured at amortised cost.

The company has purchased bonds at discount of £(100,000 – 97,327) = £2,673. This discount is amortised over the remaining expected life of the bond using the effective interest method.

The bond amortisation schedule is as follows:

Year	Opening balance £	Interest income at EIR = 6% £	Cash received £	Closing balance £
20X8	97,327	5,840	(5,000)	98,167
20X9	98,167	5,890	(5,000)	99,057
20Y0	99,057	5,943	(105,000)	–

The closing balance at the end of each year is the amortised cost of the bonds which is recognised in the statement of financial position. This is the present value of the estimated future cash flows discounted at the original effective interest rate.

Journal entries (amounts are rounded to nearest £): 1 January 20X8

		£	£
DEBIT	Financial asset	97,327	
CREDIT	Cash		97,327
(Purchase of bond asset at discount)			

31 December 20X8

		£	£
DEBIT	Financial asset	840	
DEBIT	Cash	5,000	
CREDIT	Interest income (6% of £97,327)		5,840
(Recognise interest income and bond amortisation based on EIR)			

31 December 20X9

		£	£
DEBIT	Financial asset	890	
DEBIT	Cash	5,000	
CREDIT	Interest income (6% of £98,167)		5,890
	(Recognise interest income and bond amortisation based on EIR)		

31 December 20Y0

		£	£
DEBIT	Financial asset	943	
DEBIT	Cash	5,000	
CREDIT	Interest income (6% of £99,057)		5,943
	(Recognise interest income and bond amortisation based on EIR)		
DEBIT	Cash	100,000	
CREDIT	Financial asset		100,000

(Amounts received on redemption of bond at par value)

3 Purple Company

The shares are initially measured at fair value (the purchase price here) plus transaction costs:

(80,000 × £4.54) = £363,200 + (£363,200 × 1%) = £366,832

The investment is derecognised on 31 December. The fact that the same quantity of shares are repurchased the next day does not prevent derecognition as the company has no obligation to repurchase them, therefore the risks and rewards of ownership are not retained.

Immediately before derecognition a loss is recognised in other comprehensive income as the company elected to hold the investment at fair value through other comprehensive income and the investment must be remeasured to fair value at the date of derecognition (IFRS 9.3.2.12a):

(80,000 × £4.22 bid price) = £337,600 – £366,832 = £29,232 loss

The transaction costs on sale of £3,376 (£337,600 × 1%) are recognised in profit or loss.

4 Marland

The trade receivable of £128.85 million should be recognised in the statement of financial position as at 30 April 20X5 as a financial asset.

Interest on the trade receivable is £128.85m × 8% = £10.308 million. This should be recognised in the statement of profit or loss as interest income.

A loss allowance for the trade receivable should be recognised at an amount equal to 12 months' expected credit losses. Although IFRS 9, *Financial Instruments* offers an option for the loss allowance for trade receivables with a financing component to always be measured at the lifetime expected losses, Marland has chosen instead to follow the three-stage approach of IFRS 9.

The 12-month expected credit losses are calculated by multiplying the probability of default in the next 12 months by the lifetime expected credit losses that would result from the default. Here this amounts to £18.822 million (£75.288m × 25%). Because this allowance is recognised at 1 May 20X4, the discount must be unwound by one year: £18.822m × 8% = £1.506m.

Overall adjustment:

		£m	£m
DEBIT	Finance costs (impairment of receivable) (18.822 + 1.506)	20.328	
CREDIT	Loss allowance		20.328

5 Longridge Co

Longridge Co classifies the bond at fair value through other comprehensive income (FVOCI). It recognises a loss allowance equal to 12-month expected credit losses on origination. This is recognised in profit or loss. Any subsequent increase in the loss allowance is also recognised in profit or loss. The fair value movements on the bond are recognised in other comprehensive income until sale when they are recycled to profit or loss.

1 January 20X8		**£**	**£**
DEBIT	Debt instrument – FVOCI	100,000	
CREDIT	Cash		100,000
	(Purchase of bond at FVOCI)		
DEBIT	Impairment allowance – profit or loss	1,000	
CREDIT	Other comprehensive income		1,000
	(12-month expected credit losses on orig.)		

Expected credit losses are not recognised in the statement of financial position for debt instruments measured at FVOCI, as the carrying amount of this asset should be the fair value. The impairment allowance is instead recognised in other comprehensive income as the accumulated impairment amount.

31 December 20X8		**£**	**£**
DEBIT	Cash	5,000	
CREDIT	Interest income		5,000
	(Interest income at 5% of £100,000)		
DEBIT	Impairment allowance – profit or loss	500	
CREDIT	Other comprehensive income		500
	(Increase in 1-month expected credit losses)		
DEBIT	Other comprehensive income	4,000	
CREDIT	Debt instrument – FVOCI		4,000
	(Fair value adj. to reduce asset value to £96,000)		
1 January 20X9		**£**	**£**

DEBIT	Cash	96,000	
CREDIT	Debt instrument – FVOCI		96,000
DEBIT	Loss on sale – profit or loss	2,500	
CREDIT	Other comprehensive income		2,500

(Proceeds from sale and recycling of fair value loss to profit or loss)

The gross carrying amount of the debt instrument may be reduced through use of an allowance account.

6 Gaia Bank

The credit card loans and advances are classified as amortised cost because the cash flows are solely payments of principal and interest on the principal amount outstanding, and the business model within which the balances are held is to collect the contractual cash flows rather than to sell the balances.

Under IFRS 9, the credit card loans are initially measured at fair value and are subsequently measured at amortised cost using the effective interest rate over the life of the loan.

Impairment allowances must be recognised on initial recognition of the credit card loans when they are classified in Stage 1. The impairment allowance is equal to the 12-month probability of default multiplied by lifetime expected credit losses. Expected credit losses on credit cards will be relatively high because the lending is unsecured and the loss given default is, on average, 90%.

Gaia Bank must assess at each reporting date whether the credit risk has significantly increased. It may do this on a collective basis because there is a very large number of customers (approximately 400,000 customers) but the grouping for collective assessment must be based on shared credit risk characteristics. Therefore, this could reflect the length of time a customer is past due, credit rating, past behaviour in terms of repayment, forbearance offered etc.

If credit risk has significantly deteriorated, the loan is moved to Stage 2 and lifetime expected credit losses must be recognised. The fact that the probability of default does not reduce on the credit cards as time goes on, is potentially an indicator of increased credit risk. However, spending on credit cards can vary, and the credit limit may be used to a greater or lesser extent over the product life. Behaviour on credit cards can be difficult to model because of this. It is also unclear how long customers intend to use the credit card and therefore what the expected product life is.

Gaia Bank does not appear to be defining Stage 3 accurately under IFRS 9. There is a rebuttable presumption that a loan is in default and moves to Stage 3 if payments are more than 90 days past due. The expected credit losses are still recognised on a lifetime basis, but interest is calculated differently. The effective interest rate is applied to the net balance, after impairment allowances, for loans in Stage 3. Therefore Gaia Bank could be overstating its interest income in the statement of profit or loss.

7 Pike

£8 million

IAS 32.31 requires that any derivative features embedded within a compound financial instrument (such as the call option) are 'included' in the liability component. The value of the option (£3 million), which is an asset for the company as it enables it to buy back the bonds when it wants to, is deducted from the liability element of the compound instrument (£11 million).

8 Mullet

£4 million

IAS 32.31 requires that any derivative features embedded within a compound financial instrument (such as the call option) are included in the liability component. The value of the option (£1 million) is deducted from the liability element of the compound instrument (£52 million) giving a final liability element of £51 million.

The equity element is the fair value of the compound instrument (£55 million) less the liability element after taking account of the derivative (£51 million), as IAS 32.31 requires that no gain or loss should arise on initial recognition of the component elements. The equity element is therefore £4 million.

Chapter 17

Financial instruments: hedge accounting

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Hedge accounting: the main points
2 Hedged items
3 Hedging instruments
4 Conditions for hedge accounting
5 Fair value hedge
6 Cash flow hedge
7 Hedge of a net investment
8 Disclosures
9 IAS 39 requirements on hedge accounting
10 Audit focus: fair value
11 Auditing financial instruments
12 Auditing derivatives

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Identify and explain current and emerging issues in corporate reporting
- Determine and calculate how different bases for recognising, measuring and classifying financial assets and financial liabilities can impact upon reported performance and position
- Evaluate the impact of accounting policies and choice in respect of financing decisions for example hedge accounting and fair values
- Explain and appraise accounting standards that relate to an entity's financing activities which include: financial instruments; leasing; cash flows; borrowing costs; and government grants
- Justify and conclude for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**Hedge accounting: the main points** Many entities which are exposed to various financial risks employ hedging techniques to reduce fluctuations in the value or cash flows of financial instruments.	**Approach** Hedge accounting can appear complicated, so it is essential to work through this overview to get the big picture of what hedge accounting is aiming to do. Work through the basic hedge examples. **Stop and think** Key point: hedge accounting reduces or eliminates the volatility in profit or	Hedge accounting is tested very regularly, often in a question that covers a number of other issues relating to financial instruments.	**IQ2: Simple fair value hedge** As the title suggests, this question is relatively straightforward for a difficult topic. Make sure you understand it fully before moving on.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		loss which would arise if the items were not linked for accounting purposes.		
2	**Hedged items** Hedge accounting attempts to reflect the economic aspects of hedging in the accounting treatment of such relationships.	**Approach** Study the definition of a hedged item, then look at paragraph 2.2 to see the forms a hedged item may take. **Stop and think** Can a group of assets be a hedged item?	Where there is hedge accounting there will be a hedging instrument and a hedged item.	N/A
3	**Hedging instruments** Nearly all derivatives and some non-derivatives can be designated as a hedging instrument.	**Approach** Focus on options (section 3.3), as this is most likely to come up in the exam. **Stop and think** Purchased options, whether call options or put options, have the potential to hedge price, currency and interest rate risks and can always qualify as hedging instruments.	The most common hedging instruments you will meet in your exam are options, forwards and futures.	N/A
4	**Conditions for hedge accounting** Under IFRS 9 the hedge effectiveness criteria are principles-based and aligned with risk management activities.	**Approach** This is a short section, but you should be very familiar with the hedge accounting conditions, so take your time over it. **Stop and think** What is rebalancing?	Part of a past exam question involved a Finance Director who was unsure if hedge accounting could be applied, so familiarity with this topic could be a good source of easy marks.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
5	**Fair value hedge** A fair value hedge is an investment position taken by a company or an investor aiming to protect the fair value of a specific asset, liability or unrecognised company commitment from risks that can affect their profit and loss accounts.	**Approach** You have been introduced to fair value hedging in the overview section. This section focuses on application and examples. **Stop and think** What would be a suitable hedging instrument to hedge changes in fair value of a fixed rate bond?	Fair value hedge accounting could be examined together with cash flow hedge accounting in a scenario where you are advising a finance director to choose between alternatives.	**IQ4: Fair value hedge** This question involves a futures contract and requires accounting with and without hedge accounting, illustrating the purpose of hedge accounting clearly.
6	**Cash flow hedge** A cash flow hedge is an investment position taken by a company or an investor aiming to offset the potential impact of a particular risk on the cash flows of an asset, liability or another sort of exposure. They are quite common among export companies or those with an international supply chain, who tend to use financial derivatives to protect their costs and revenues from exchange rate volatility.	**Approach** You met cash flow hedging in the overview section. This section focuses on application and practical examples. Key point: The part of the gain or loss arising from an effective hedge of a hedging instrument is recognised in OCI while the ineffective portion should be recognised in profit or loss. Learn the summary table in section 7.6 – it is essential for your exam. **Stop and think** What is a forecast transaction?	Questions could cover anything from the choice of cash flow hedging or fair value hedging to journal entries and adjustments of financial statements to use of cash flow hedging that is not appropriate. Entities are not always permitted a choice between fair value hedging and cash flow hedging.	**IQ7: Swap in cash flow hedge 2** This is a short question, but the answer is not obvious.
7	**Hedge of a net investment** An overseas equity investment, ie a subsidiary, associate	**Approach** Come back to this section after you have studied Chapter 21 on	Hedges of a net investment have not come up as frequently as fair value and cash flow	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	or joint venture is hedged by a derivative or non-derivative contract. The net investment in a foreign operation under IAS 21, is the amount of the reporting entity's interest in the net assets of the operation, including any recognised goodwill.	foreign currency transactions. The treatment is similar to a cash flow hedge as regards the treatment of effective and ineffective portions. **Stop and think** Why do you think hedging might be used for an overseas equity investment?	hedges. When they have come up they tend to be poorly answered, so work through this section carefully.	
8	**Disclosures** Certain disclosures must be made separately for each type of hedge described in IFRS 9 (ie, fair value hedges, cash flow hedges and hedges of net investments in foreign operations). Cash flow hedges have extra disclosures.	**Approach** Read through and highlight. **Stop and think** Why do cash flow hedges have additional disclosures?	You are unlikely to be asked to list these disclosures, although in an auditing context you may need to verify that they have been made.	N/A
9	**IAS 39 requirements on hedge accounting** IFRS 9 is now the examinable standard, so you should focus on the IFRS 9 hedging rules. However, entities may apply the hedge accounting rules of IAS 39 to all of its hedging relationships while following the classification and measurement rules of IFRS 9.	**Approach** You should work through section 9 for an overview of what these rules are, focusing on the differences from IFRS 9. The table in section 9.3 will be useful for this. The main difference relates to the hedge effectiveness criteria. Under IFRS 9 this is principles-based and aligned with risk management activities. IAS 39, in contrast, has a	You will not be examined solely on the IAS 39 hedging rules except in overview by way of contrast with the IFRS 9 rules. No aspects of IAS 39 other than hedging will be examined.	N/A There are no interactive questions just on the IAS 39 rules, but you can test yourself using the table in section 9.3 by covering up the IAS 39 column.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		quantitative hedge effectiveness test under which hedge effectiveness must fall in the range 80%–125%. **Stop and think** Why would companies apply the IAS 39 rules?		
10	**Audit focus: fair value** This section applies more broadly than just to financial instruments and fair value hedging. Fair value is a key issue to investment property, pension costs, share-based payments and many other areas of financial accounting.	**Approach** Read carefully and consider why auditing fair value might be more risky than historical cost. **Stop and think** Professional scepticism is needed.	Any of the areas in the left-hand column could give rise to a question in which fair value must be considered in the audit process.	N/A
11	**Auditing financial instruments** The key issue when auditing financial instruments is risk, in particular credit risk and market risk.	**Approach** Work through section 11.1.2 very carefully, especially the section on assessing and responding to the risk of material misstatement. **Stop and think** How might valuation of financial instruments be tested?	Risk is likely to feature in any audit/integrated questions on financial instruments.	**IQ14: Convertible debenture** This is a short 'issues' and 'procedures' question.
12	**Auditing derivatives** Derivatives are risky instruments and much can go wrong. An understanding of the business process in derivatives trading	**Approach** Pay particular attention to section 12.1 and the worked example. **Stop and think** Derivative traders	Particular risks apply to derivatives. Risks (credit, market, legal, settlement, solvency) and controls relating to those risks have been examined.	**IQ15: Derivatives** This is a comprehensive question such as you might get as Question 3, integrating financial reporting and

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	is necessary to audit derivatives. Adequate internal controls must be in place.	are the sort of people who thrive on risk. Check out Nick Leeson.		auditing. The audit part is mainly concerned with risk, but also with the audit of IFRS 7 disclosures.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Hedge accounting: the main points

Section overview

Pay particular attention to this first section, as it contains the main points you need to know.

1.1 Introduction

In earlier levels of your study for the ACA qualification, such as Financial Management, you have covered the way hedging is an important means by which a business can **manage the risks** it is exposed to.

As an example, a manufacturer of chocolate can fix now the price at which it buys a specific quantity of cocoa beans at a predetermined future date by arranging a forward contract with the cocoa beans producer.

The forward price specified in the forward contract may be higher or lower than the spot price at the time the contract is agreed, depending on seasonal and other factors. But by agreeing the forward contract both the manufacturer and the producer have removed the risk they otherwise would face of unfavourable price movements (price increases being unfavourable to the chocolate manufacturer and price decreases unfavourable to the cocoa beans producer) between now and the physical delivery date. Equally, they have removed the possibility of favourable price movements (price decreases being favourable to the chocolate manufacturer and price increases favourable to the cocoa beans producer) over the period.

Another way of achieving the same effect would be for the chocolate manufacturer to purchase cocoa bean futures on a recognised trading exchange. On the delivery date the manufacturer would close out the futures in the futures market and then buy the required quantity in the spot market. The profit/(loss) on the futures transaction should offset the increase/(decrease) in the spot price over the period.

Hedge accounting is the accounting process which reflects in financial statements the commercial substance of hedging activities. It results in the gains and losses on the linked items (eg, the purchase of coffee beans and the futures market transactions) being recognised in the same accounting period and in the same section of the statement of profit or loss and other comprehensive income ie, both in profit or loss, or both in other comprehensive income.

Hedge accounting reduces or eliminates the volatility in profit or loss which would arise if the items were not linked for accounting purposes.

> **Tutorial note**
>
> The forward contract is a derivative. Without hedge accounting, the profit/(loss) in the futures market would be recognised as the contract is remeasured to fair value at each reporting date, but the increased/(decreased) cost of the cocoa beans would be recognised at the later date when the chocolate is sold. Both would be recognised in profit or loss, but possibly in different accounting periods.

In the previous chapter the point was made that financial assets should be classified or designated at the time of their initial recognition, not at any later date. This is to prevent businesses making classifications or designations with the benefit of hindsight so as to present figures to their best advantage. Similarly, hedge accounting is only permitted by IFRS 9, *Financial Instruments* if the hedging relationship between the two items (the cocoa beans and the futures contract in the above example) is designated at the inception of the hedge. Designation is insufficient by itself; there must be formal documentation, both of the hedging relationship and of management's objective in undertaking the hedge.

1.1.1 IFRS 9

IFRS 9 provides guidance relating to hedging and allows hedge accounting where there is a designated hedging relationship between a hedging instrument and a hedged item. It is prohibited otherwise. **Hedge accounting is therefore not mandatory.**

1.1.2 IAS 39

The IASB currently allows an **accounting policy choice** to apply either the IFRS 9 hedging model or the IAS 39 model, with an additional option to use IAS 39 for macro hedging (currently a separate project) if using IFRS 9 for general hedge accounting (IFRS 9.7.2.21).

For this reason, the IAS 39 rules are covered in overview in section 9. However, **IFRS 9 is the examinable standard. For examination purposes you only need an awareness of the differences between IAS 39 and IFRS 9 with regard to hedging.**

1.2 Overview

In simple terms the main components of hedge accounting are as follows:

(a) The **hedged item** is an asset, a liability, a firm commitment (such as a contract to acquire a new oil tanker in the future) or a forecast transaction (such as the issue in four months' time of fixed rate debt) which exposes the entity to risks of fair value/cash flow changes. The hedged item generates the risk which is being hedged.

(b) The **hedging instrument** is a derivative or other financial instrument whose fair value/cash flow changes are expected to offset those of the hedged item. The hedging instrument reduces/eliminates the risk associated with the hedged item.

(c) There is a **designated relationship** between the item and the instrument which is documented.

(d) At inception the hedge must be expected to be **highly effective** and it must turn out to be highly effective over the life of the relationship.

(e) To qualify for hedging, the changes in fair value/cash flows must have the **potential to affect profit or loss**.

(f) There are two main types of hedge:

1. The **fair value hedge**: the gain and loss on such a hedge are recognised in profit or loss.
2. The **cash flow hedge**: the gain and loss on such a hedge are initially recognised in other comprehensive income and subsequently reclassified to profit or loss.

Notes

1 The key reason for having the two types of hedge is that profits/losses are initially recognised in different places.

2 In some circumstances the entity can choose whether to classify a hedge as a fair value or a cash flow hedge.

3 There is a third type of hedge: the **hedge of a net investment** in a foreign operation, such as the hedge of a loan in respect of a foreign currency subsidiary. This is accounted for similarly to cash flow hedges.

1.3 Effectiveness of the hedge

The effectiveness of the hedge is measured as the extent to which the change in the hedging instrument offsets the change in the hedged item. IFRS 9 has three **hedge effectiveness tests,** which will be covered in more detail later.

Note: Here are three definitions you may need for the illustration that follows.

Definitions

Forward contract: A commitment to undertake a future transaction at a set time and at a set price.

Future: This represents a **commitment** to an **additional transaction** in the future that limits the risk of existing commitments.

Option: This represents a commitment by a seller to undertake a future transaction, where the buyer has the option of not undertaking the transaction.

Worked example: Basic hedging 1

1 January

On 1 January the price (the spot price) of a consignment of cocoa beans is £1,000.

You know you will need to buy a consignment of cocoa beans on 28 February, as they will be needed to fulfil a customer order. You are afraid that the price of cocoa beans will rise significantly between 1 January and 28 February.

You therefore contract with a cocoa beans supplier to buy a consignment of cocoa beans at £1,050 on 28 February.

28 February

The price of a consignment of cocoa beans is now £1,100.

You nevertheless can hold the supplier to the forward contract and can buy the cocoa beans at £1,050.

However, if the market had not behaved as predicted and the price of cocoa beans was £980 on 28 February, you would still be obliged to buy the cocoa beans at the price of £1,050.

Similarly, if the customer had pulled out of the transaction, you would still have to buy the consignment of cocoa beans and dispose of them as best you could.

Hedging deals with the bad news you do not expect! 1 January

On 1 January the price (the spot price) of a consignment of cocoa beans is £1,000.

You have already agreed to buy a consignment of cocoa beans for £1,200 on 28 February, which means you appear to be at risk of paying too much for the cocoa beans.

You buy a three-month cocoa futures contract at £1,100 that expires on 31 March. This means you are committing to buying an additional consignment of cocoa beans, not at today's spot price, but at the futures price of £1,100. £1,100 represents what the market thinks the spot price will be on 31 March.

28 February

You buy the consignment of cocoa beans at £1,200.

You are still committed to buying the consignment at £1,100 on 31 March, but that will mean that you have two consignments of cocoa beans rather than just the one you need. You therefore sell the futures contract you bought on 1 January to eliminate this additional commitment. The futures contract is now priced at £1,233, as the market now believes that £1,233 will be the spot price on 31 March.

Because you have sold the contract for more than the purchase price, you have made a gain on the futures contract of £1,233 – £1,100 = £133. This can be set against the purchase you made.

Net cost = £1,200 – £133 = £1,067; the cost of paying more for the cocoa beans has been offset by the profit made on the futures contract.

1 January

On 1 January the price (the spot price) of a consignment of cocoa beans is £1,000.

You know you will need to buy a consignment of cocoa beans on 28 February, as they will be needed to fulfil a customer order. You think it is likely that the price of cocoa beans will rise significantly between 1 January and 28 February, but you believe that with current market uncertainty, the price of cocoa beans could even fall.

You therefore take out an option to buy cocoa beans at £1,050 on 28 February. Because you are being given the privilege of choosing whether or not to fulfil the option contract, you have to pay a premium of £30.28 February.

Requirements

1 What if the price of the cocoa beans has now risen to £1,100?

2 What if the price of cocoa beans has fallen to £980? You could let the option contract lapse and buy cocoa beans at £980.

3 What if your customer pulls out of the contract?

Solution

1 You can hold the supplier to the option contract and buy the cocoa beans at £1,050. Total cost = 1,050 + 30 = £1,080.

2 Total cost = 980 + 30 = £1,010.

3 You would not have to buy the cocoa beans and the only cost to you will be the premium of £30.



Worked example: Basic hedging 2

Red, whose functional currency is the pound (£), has invested €4.75 million in purchasing a majority shareholding in Blue. The investment in Blue is entirely financed by a loan in euros. The directors of Red decide to designate the loan as a hedging instrument and the investment as the related hedged item.

Requirement

Describe the accounting treatment of any gains or losses arising on the investment and the loan, assuming that the hedging relationship meets all the conditions required by IFRS 9, *Financial Instruments* to qualify for hedge accounting.

Solution

The transaction entered into by Red is a hedging transaction of a net investment in a foreign entity. The loan is the hedging instrument and the investment in Blue is the hedged item.

As the loan has been designated as the hedging instrument at the outset, and the transaction meets the hedging criteria of IFRS 9, the exchange movements in both items should be recognised in other comprehensive income. Any ineffective portion of the hedge should be recognised in profit or loss for the year.

Below are two simple illustrative examples of accounting for a fair value hedge and accounting for a cash flow hedge. The definitions and rules for a fair value hedge and a cash flow hedge are covered in greater detail in sections 5 and 6 of this chapter.

1.4 Accounting for a simple fair value hedge

On 1 August 20X5, an entity owned 50,000 litres of vegetable oil which had cost it £5 per litre and which had a selling price (spot price = fair value) of £6 per litre. The entity was concerned that the fair value might fall over the next three months, so it took out a three-month future to sell at £6 per litre. On 31 October the spot price of the oil had fallen to £5.60. On that date the entity closed out its future and sold its inventory, both transactions being at the spot price.

The sale of 50,000 litres at £5.60 generates revenue of £280,000; deducting the cost of £250,000, the profit recognised in profit or loss should be £30,000.

The entity also makes a profit of £0.40 (£6.00 – £5.60) per litre in the futures market, so on 50,000 litres a profit of £20,000 should be recognised in profit or loss.

Subject to any futures market transaction costs, the entity has protected itself against a fall in fair value below the £6 fair value at 1 August 20X5.

1.5 Accounting for a simple cash flow hedge

On 1 November 20X5 an entity, whose functional currency is the pound (£), entered into a contract to sell goods on 30 April 20X6 for $300,000. In fixing this dollar ($) price it worked on the basis of the spot exchange rate of $1.50 = £1, so that revenue would be £200,000. To ensure it received £200,000, on 30 June 20X6 the entity took out a six-month future to sell $300,000 for £200,000.

On 31 December 20X5 (which is the company's reporting date) an equivalent futures contract traded at a value of £18,182. This may be taken as an acceptable approximation to fair value. The future was therefore worth £18,182 and the entity recognised that amount as a financial asset and as a profit in other comprehensive income.

On 30 April 20X6, the spot exchange rate was $1.75 = £1 and the future was worth £28,571 (£200,000 – £(300,000/1.75)). The entity closed out its future position at the then spot price and sold the goods. The accounting entries should be:

DEBIT	Customer £(300,000/1.75)	£171,429	
DEBIT	Financial asset (28,571 – 18,182)	£10,389	
DEBIT	Other comprehensive income (Reclassification of gain to profit or loss)	£18,182	
CREDIT	Revenue		£200,000

The customer account and the financial asset are then cleared by cash receipts. Note that revenue is measured at the amount fixed as a result of the hedging transaction.



Interactive question 1: Simple derivative and hedging

BCL entered into a forward contract on 31 July 20X0 to purchase $2 million at a contracted rate of £1: $0.64 on 31 October 20X0. The contract cost was nil. BCL prepares its financial statements to 31 August 20X0. At 31 August 20X0, an equivalent contract for the purchase of $2 million could be acquired at a rate of £1 : $0.70.

Requirements

1.1 Explain how this financial instrument should be classified and prepare the journal entry required for its measurement as at 31 August 20X0.

1.2 Assume now that the instrument described above was designated as a hedging instrument in a cash flow hedge, and that the hedge was 100% effective.

(a) Explain how the gain or loss on the instrument for the year ended 31 August 20X0 should now be recorded and why different treatment is necessary.

(b) Prepare an extract of the statement of profit or loss and other comprehensive income for BCL for the year ended 31 August 20X0, assuming the profit for the year of BCL was £1 million, before accounting for the hedging instrument.

See **Answer** at the end of this chapter.

Interactive question 2: Simple fair value hedge

VB acquired 40,000 shares in another entity, JK, in March 20X3 for £2.68 per share. An irrevocable election was made under IFRS 9 to record changes in fair value in other comprehensive income. The shares were trading at £2.96 per share on 31 July 20X3. Commission of 5% of the value of the transaction is payable on all purchases and disposals of shares.

Requirement

(a) Prepare the journal entries to record the initial recognition of this financial asset and its subsequent measurement at 31 July 20X3 in accordance with IFRS 9, *Financial Instruments*.

(b) The directors of VB are concerned about the value of VB's investment in JK and, in an attempt to hedge against the risk of a fall in its value, are considering acquiring a derivative contract. The directors wish to use hedge accounting in accordance with IFRS 9.

Requirement

Discuss how both the investment in equity instruments and any associated derivative contract would be subsequently accounted for, assuming that the criteria for hedge accounting were met, in accordance with IFRS 9.

See **Answer** at the end of this chapter.

2 Hedged items

Section overview

This section deals with detailed issues related to hedged items in a hedging relationship.

Definitions

Hedged item: An asset, liability, firm commitment, highly probable forecast transaction or net investment in a foreign operation that:

- exposes the entity to risk of changes in fair value or future cash flows; and
- is designated as being hedged.

Firm commitment: A binding agreement for the exchange of a specified quantity of resources at a specified price on a specified future date or dates.

Forecast transaction: An uncommitted but anticipated future transaction.

Note: Neither firm commitments nor forecast transactions are normally recognised in financial statements. As is explained in more detail in a later part of this chapter, it is only when they are designated as hedged items that they are recognised.

2.1 Financial risks

Hedged items as defined above are **exposed to a variety of risks** that affect the value of their fair value or cash flows. For hedge accounting, these risks **need to be identified** and **hedging instruments which modify the identified risks** selected and designated. The risks for which the above items can be hedged are normally classified as follows:

- Market risk

Which can be made up of:

  - price risk
  - interest rate risk
  - currency risk

- Credit risk
- Liquidity risk

IFRS 9 allows for a portion of the risks or cash flows of an asset or liability to be hedged. For example, the hedged item may be as follows:

- Oil inventory (which is priced in $) for a UK company, where the fair value of foreign currency risk is being hedged but not the risk of a change in $ market price of the oil
- A fixed rate liability, exposed to foreign currency risk, where only the interest rate and currency risk are hedged but the credit risk is not hedged

2.2 Nature of hedged items

In this section we discuss some of the **key aspects** of the definition of a hedged item.

(a) The hedged item can be:

  (1) a single asset, liability, unrecognised firm commitment, highly probable forecast transaction or net investment in a foreign operation;

  (2) a group of assets, liabilities, firm commitments, highly probable forecast transactions or net investments in foreign operations with similar risk characteristics; or

  (3) a portion of a portfolio of financial assets or financial liabilities which share exposure to interest rate risk. In such a case, the portion of the portfolio that is designated as a hedged item is a hedged item with regard to interest rate risk only.

(b) Assets and liabilities designated as hedged items can be **either financial or non-financial items**.

  (1) **Financial items** can be designated as hedged items for the risks associated with only a **portion** of their cash flows or fair values. So a fixed rate liability that is exposed to foreign currency risk can be hedged in respect of currency risk, leaving the credit risk not hedged.

  (2) IFRS 9 allows **separately identifiable and reliably measurable risk components of non-financial items** to be designated as hedged items.

(c) **Unrecognised** assets and liabilities **cannot be designated as hedged items**. So unrecognised intangibles cannot be hedged items.

(d) **Only** assets, liabilities, firm commitments or highly probable transactions that involve a party external to the entity can be **designated as hedged items**. The effect is that hedge accounting can be applied to transactions between entities or segments in the same group only in the individual or separate financial statements of those entities or segments, and not in the consolidated financial statements.

(e) As an exception, an **intra-group monetary item** qualifies as a hedged item in the consolidated financial statements if it results in an exposure to foreign exchange rate gains and losses that are not eliminated on consolidation.



Context example: Intra-group hedge

The functional currency of Anson Co and its subsidiary Benson Co are the pound (£) and dollar ($) respectively. Benson Co sells $100 of goods to Anson Co just before the year end. The amount remains unsettled at the year end.

While the intercompany balances are eliminated on consolidation, the exchange differences that arise in Anson Co from the retranslation of the monetary liability are not eliminated on consolidation. Hence, the intercompany monetary item can be designated as a hedged item in a foreign currency hedge.

2.3 Designation of a group of assets as hedged items

IFRS 9 permits the designation of a group of assets as a hedged item provided that the following **conditions** are met.

(a) It consists of items which are individually eligible as hedged items.

(b) The items in the group are managed together on a group basis for risk management purposes.

(c) In the case of a cash flow hedge of a group of items whose variabilities in cash flows are not expected to be approximately proportional to the overall variability in cash flows of the group:

  (1) it is a hedge of foreign currency risk; and

  (2) the designation of that net position specifies the reporting period in which the forecast transactions are expected to affect profit or loss, as well as their nature and volume.

For a hedge of a net position whose hedged risk affects different line items in the statement of profit or loss and other comprehensive income, any hedging gains or losses in that statement are presented in a separate line from those affected by the hedged items.

(IFRS 9.6.6.4)



Worked example: Group of assets

An entity constructs a portfolio of shares to replicate a stock index and uses a put option on the index to protect itself from fair value losses.

Requirement

Can the portfolio of shares be designated as a hedged item?

Solution

The portfolio **cannot** be designated as a hedged item. Similar financial instruments should be aggregated and hedged as a group only if the change in fair value attributable to the hedged risk for each individual item in the group is expected to be approximately proportional to the overall change in fair value attributable to the hedged risk of the group. In the scenario above, the change in the fair value attributable to the hedged risk for each individual item in the group (individual share prices) is not expected to be approximately proportional to the overall change in fair value attributable to the hedged risk of the group; even if the index rises, the price of an individual share may fall.

2.4 Hedging an overall net position

IFRS 9 allows hedge accounting to be applied to **groups of items and net positions** if the group consists of individually eligible hedged items and those items are managed together on a group basis for risk management purposes.

For a **cash flow hedge of a group of items**, if the variability in cash flows is not expected to be approximately proportional to the group's overall variability in cash flows, the **net position is eligible** as a hedged item only **if it is a hedge of foreign currency risk**. In addition, the designation must specify the reporting period in which forecast transactions are expected to affect profit or loss, including the nature and volume of these transactions.

2.5 Risk components of non-financial items

IFRS 9 allows **separately identifiable and reliably measurable risk components of non-financial items**
to be designated as hedged items.

2.6 Components

IFRS 9 allows a component that is a proportion of an entire item or a **layer component** to be designated as a hedged item in a hedging relationship. A layer component may be specified from a defined, but open, population or a defined nominal amount. For example, an entity could designate 20% of a fixed rate bond as the hedged item, or the top layer of £20 principal from a total amount of £100 (defined nominal amount) of fixed-rate bond. It is necessary to track the fair value movements of the nominal amount from which the layer is defined.

2.7 Aggregate exposure

IFRS 9 allows **aggregated exposures** that include a derivative to be an **eligible hedged item**.



Context example: Aggregated exposure as hedged item

Zeta Bank has a fixed rate foreign currency loan which exposes it to both foreign exchange rate risk and fair value risk due to changes in interest rates. The bank enters into a cross-currency interest rate swap to eliminate the foreign exchange risk and fair value risk due to changes in interest rates but is now exposed to variable functional currency interest payments.

Zeta Bank may hedge the aggregated exposure (foreign currency loan + cross-currency interest rate swap) by using, for example, a pay fixed and receive floating interest rate swap in its own functional currency.

2.8 Equity investments at fair value through other comprehensive income

IFRS 9 allows an entity to classify equity investments not held for trading, at fair value through other comprehensive income through an irrevocable option on origination of the instrument. The gains and losses are recognised in other comprehensive income and never reclassified to profit or loss.

IFRS 9 allows these equity investments at fair value through other comprehensive income to be designated as hedged items. In this case, both the effective and ineffective portion of the fair value changes in the hedging instruments are recognised in other comprehensive income.

2.9 Fair value designation for credit exposures

Many banks use credit derivatives to manage credit risk exposures arising from their lending activities. The hedges of credit risk exposure allow banks to transfer the risk of credit loss to a third party. This may also reduce regulatory capital requirements.

IFRS 9 allows **credit exposure or part of the credit exposure** to be measured at **fair value through profit or loss** if an entity uses a **credit derivative** measured at **fair value through profit or loss** to manage the credit risk of all, or part of, the credit exposure. In addition, an entity may make the designation at initial recognition or subsequently, or while the financial instrument is unrecognised.

Interactive question 3: Credit derivative and credit exposures

Excel Bank extends a fixed rate loan commitment of £1 million to a customer. The bank's risk management strategy is to hedge the credit risk exposure of any individual loan commitment to the extent that it exceeds £500,000. As a result, Excel Bank enters into a credit default swap of £500,000 in relation to this loan commitment to the customer.

Requirement

Explain the accounting for the credit default swap and the loan commitment under IFRS 9.

See **Answer** at the end of this chapter.

2.10 Firm commitments as hedged items

Firm commitments (as defined above) are the result of legally binding contracts which normally specify penalties for non-performance. A firm commitment can be a hedged item.

2.11 Forecast transactions as hedged items

A forecast transaction (as defined above) **qualifies as a hedged item only if the transaction is highly probable**. Examples of forecast transactions that qualify as a hedged item include the following:

(a) The anticipated issue of fixed rate debt. This can be recognised as a hedged item under a cash flow hedge of a highly probable forecast transaction that will affect profit or loss.

(b) Expected, but not contractual, future foreign currency revenue streams, provided that the revenues are highly probable. A hedge of an anticipated sale can qualify as a cash flow hedge.

Context example: Forecast transaction

An airline may use models based on past experience and historical economic data to project its revenues in various currencies. If it can demonstrate that forecast revenues for a period of time into the future in a particular currency are highly probable, it may designate a currency borrowing as a cash flow hedge of the currency risk of the future revenue stream. The portion of the gain or loss on the borrowing that is determined to be an effective hedge is recognised in other comprehensive income until the revenues occur.

It is unlikely that an entity can reliably predict 100% of revenues for a future year. On the other hand, it is possible that a portion of predicted revenues, normally those expected in the short term, will meet the 'highly probable' criterion.

Because forecast transactions can only be hedged under cash flow hedges, the ways to assess the probability of a future transaction are covered below under cash flow hedges.

2.12 Intra-group and intra-entity hedging transactions

It has already been noted that hedged items have to involve a party external to the entity, with the result that intra-group transactions can be designated as hedged items only in the

individual or separate financial statements and not in consolidated financial statements. There are only two cases, both involving foreign exchange translation, where intra-group transactions will be recognised in the consolidated financial statements.

The first case is a result of IAS 21, *The Effects of Changes in Foreign Exchange Rates* under which foreign exchange gains and losses on an intra-group monetary asset or liability between entities with different functional currencies are not fully eliminated in the consolidated profit or loss. This is because a foreign currency monetary item represents a commitment to convert one currency into another one and exposes the reporting entity to a gain or loss through currency fluctuations. Because such exchange differences are not fully eliminated on consolidation, they will affect profit or loss in the entity's consolidated financial statements and hedge accounting may be applied.

The second case arises because IFRS 9 permits the foreign currency risks of a highly probable forecast intra-group transaction to be designated as a hedged item in the consolidated financial statements provided the following two conditions are met.

(a) The highly probable forecast intra-group transaction is denominated in a currency other than the functional currency of the group member entering into that transaction.

(b) The foreign currency risk will affect the group's consolidated profit or loss.

Worked example: Hedging intra-group monetary items

An Australian company, whose functional currency is the Australian dollar, has forecast purchases in Japanese yen that are highly probable. The Australian entity is wholly owned by a Swiss entity that prepares consolidated financial statements (which include the Australian subsidiary) in Swiss francs. The Swiss parent entity enters into a forward contract to hedge the change in yen relative to the Australian dollar.

Requirement

Explain whether the hedge can qualify for hedge accounting in the Swiss entity's consolidated financial statements.

Solution

The hedge can qualify for hedge accounting. Since the Australian entity did not hedge the foreign currency exchange risk associated with the forecast purchases in yen, the effects of exchange rate changes between the Australian dollar and the yen will affect the Australian entity's profit or loss and, therefore, would also affect consolidated profit or loss. IFRS 9 does not require the operating unit that is exposed to the risk being hedged to be a party to the hedging instrument.

3 Hedging instruments

Section overview

This section considers in detail the financial instruments that can be designated as hedging instruments for hedge accounting purposes.

3.1 Hedging instruments

Contracts that can be designated as hedging instruments include:

- derivatives; and
- non-derivative financial assets or liabilities measured at fair value through profit or loss (with the exception of financial liabilities designated at fair value through profit or loss, for which changes in fair value attributable to credit risk are presented in other comprehensive income).

For the hedge of a foreign currency risk, the foreign currency risk component of a non-derivative financial asset or liability may be designated as a hedging instrument, provided that it is not an investment in an equity instrument measured at fair value through comprehensive income by election.

An implication of this definition is that financial assets and liabilities whose fair value cannot be reliably measured cannot be designated as hedging instruments.

An entity may exclude the following from hedging relationships:

- Time value of purchased options
- Forward element of forward contracts and foreign currency basis spreads

3.2 Derivatives

Any derivative financial instrument, with the exception of written options to which special rules apply, **can be designated as a hedging instrument**. It is important to note that the fair value of derivative instruments correlates highly with that of the underlying.

3.3 Options

Options provide a more flexible way of hedging risks compared to other derivative instruments such as forwards, futures and swaps, because they give to the holder the choice as to whether or not to exercise the option.

When an entity purchases a put option, it buys the right to sell the underlying at the strike price. If the price of the underlying falls below the strike price, the entity exercises its option and receives the strike price; it has protected the value of its position. Similarly, if an entity needs to buy an asset in the future, it can purchase a call option on the asset that gives the entity the right to purchase the asset at the strike price, protecting it from a rise in the price of the asset in the future.

The difference between the purchased option and a forward contract is that under a forward contract the entity is obliged to buy or sell at the strike price, whereas under a purchased option it has the right, but not the obligation, to buy or sell at the strike price.

Purchased options, whether call options or put options, have the potential to hedge **price**, **currency** and **interest rate risks** and can always qualify as hedging instruments.

Examples of purchased options include options on equities, options on currencies and options on interest rates. An interest rate floor is achieved through a put option on an interest rate, and an interest rate cap is achieved through a call option on an interest rate.

In **IFRS 9**, an entity may **designate only the change in intrinsic value** of a purchased option as the hedging instrument in a fair value or cash flow hedge. The change in **fair value of the time value of the option** is recognised in **other comprehensive income** to the extent it relates to the hedged item. This change in IFRS 9 makes options more attractive as hedging instruments.

The method used to reclassify the amounts from equity to profit or loss is determined by whether the hedged item is transaction-related or time period-related.

The time value of a purchased option relates to **a transaction-related hedged item** if the nature of the hedged item is a transaction for which the time value has the character of costs of the transaction. For example, future purchase of a commodity or non-financial asset.

The **change in fair value of the time value of an option (transaction-related hedged item)** is accumulated in other comprehensive income over the term of the hedge, to the extent it relates to the hedged item. It is then treated as follows:

- If the hedged item results in the recognition of a **non-financial asset or liability or firm commitment** for a non-financial asset or liability, the amount accumulated in equity is removed and included in the **initial cost or carrying amount** of the asset or liability.
- For **other hedging relationships**, the amount accumulated in equity is **reclassified to profit or loss** as a reclassification adjustment in the period(s) in which the hedged expected cash flows affect profit or loss.

The time value of a purchased option relates to a time period-related hedged item if the following apply:

(a) The nature of the hedged item is such that the time value has the character of the cost for obtaining protection against a risk over a particular time period.

(b) The hedged item does not result in a transaction that involves the notion of a transaction cost.

The **change in fair value of the time value of an option (time period-related hedged item)** is accumulated in other comprehensive income over the term of the hedge, to the extent it relates to the hedged item. The time value of the option at the date of designation is amortised on a straight- line or other systematic and rational basis, and the amortisation amount is reclassified to profit or loss as a reclassification adjustment.

4 Conditions for hedge accounting



Section overview

Hedge accounting is permitted in certain circumstances, provided the hedging relationship is clearly defined, measurable and actually effective.

4.1 Hedge accounting conditions

Before a hedging relationship qualifies for hedge accounting, all of the following conditions must be met:

(a) The hedging relationship must consist only of eligible hedging instruments and eligible hedged items.

(b) At the inception of the hedge, there must be formal designation and documentation of the hedging relationship and the entity's risk management objective and strategy for undertaking the hedge. Documentation must include identification of the hedged item, the hedging instrument, the nature of the hedged risk and how the entity will assess whether the hedging relationship meets the hedge effectiveness requirements (including its analysis of the sources of hedge ineffectiveness and how it determines the hedge ratio).

(c) The hedging relationship meets all of the hedge effectiveness requirements:

(1) An economic relationship exists between the hedged item and the hedging instrument ie, the hedging instrument and the hedged item are expected to have offsetting changes in value;

(2) The effect of credit risk does not dominate the value changes ie, the value changes due to credit risk are not a significant driver of the value changes of either the hedging instrument or the hedged item; and

(3) The hedge ratio of the hedging relationship (quantity of hedging instrument vs quantity of hedged item) is the same as that resulting from the quantity of the hedged item that the entity actually hedges and the quantity of the hedging instrument that the entity actually uses to hedge that quantity of hedged item. (IFRS 9.6.4.1)

Definition

Hedge ratio: The relationship between the quantity of the hedging instrument and the quantity of the hedged item in terms of their relative weighting.

IFRS 9 requires that the hedge ratio used for accounting purposes is the same as that used for risk management purposes. This ensures that amounts are not manipulated in order to achieve a particular accounting outcome.

Professional skills focus: Applying judgement

The hedge effectiveness criteria are an example of an area where judgement must be applied. Judgement is generally needed where risk management is concerned, but having arbitrary percentage ranges, as IFRS 9's predecessor did, does not make the criteria any more precise, and could be misleading.

Context example: Hedge ratio

CoffeeBar plc buys coffee in a part-processed state, prior to the milling stage. As a result, it pays a discounted price compared to that for fully processed coffee. The price of fully processed coffee on the commodities market is on average 1.7 times the raw material price that CoffeeBar pays.

Therefore CoffeeBar uses a notional 1,000 pounds of forward contract for the commodity to hedge a highly probable forecast purchase of 1,700 pounds of raw material. The hedge ratio is therefore 1 : 1.7.

4.2 Rebalancing

Rebalancing refers to adjustments to the designated quantities of the hedged item, or the hedging instrument of an already existing hedging relationship for the purpose of maintaining a hedge ratio that complies with the hedge. This may be achieved by increasing or decreasing the volume of either hedged item or hedging instrument.

The standard requires rebalancing to be undertaken if the risk management objective remains the same, but the hedge effectiveness requirements are no longer met. Where the risk management objective for a hedging relationship has changed, rebalancing does not apply and the hedging relationship must be discontinued.

Context example: Rebalancing

Continuing with the previous example, assume that the relationship between the commodity price and raw material price changes over Year 1 such that by the end of the year the commodity price has decreased to an average 1.3 times the raw material price. In this case,

management of the reporting entity might reset the hedge ratio to 1 : 1.3. Therefore to rebalance the hedging relationship management can either:

- decrease the raw material that forms the hedged item so that the hedged item is 1.31 pounds (1/1.3 × 1.7); or
- increase the commodity that forms the hedging instrument, so that the hedging instrument is 1.3 pounds.

5 Fair value hedge

Section overview

The application of fair value hedge accounting is discussed through a number of practical examples.

5.1 Fair value hedges

A fair value hedge is a hedge of an entity's exposure to changes in fair value of a recognised asset or liability or an unrecognised firm commitment, or a part thereof, that is **attributable to a particular risk and could affect profit or loss**. Examples of fair value hedges include the hedge of exposures to changes in fair value of fixed rate debt using an interest rate swap and the use of an oil forward contract to hedge movements in the price of oil inventory.

5.1.1 Examples of fair value hedging

Hedged item	Risk exposure	Type of risk	Example of hedging instrument
Commodity inventory	Change in value due to changes in the price of commodity	Market risk (price risk)	Forward contract
Equities	Change in the value of the investments due to changes in the price of equity	Market risk (price risk)	Purchase put option
Issued fixed rate bond	Change in the value of the bond as interest rates change	Market risk (interest rate risk)	Interest rate swap
Purchase of materials denominated in foreign currency in three months	Depreciation of the local currency and increase in the cost of material	Market risk (foreign currency)	Forward contract

5.1.2 Hedge accounting and risk reduction

IFRS 9 does not require that in order for a hedging relationship to qualify for hedge accounting, it should lead to a reduction in the overall risk of the entity. A hedging relationship that satisfies the conditions for hedge accounting may be designed to protect the value of a particular asset. The following example illustrates the point.

Worked example: Qualification for hedge accounting

An entity has a fixed rate financial asset and a fixed rate financial liability, each having the same principal amount. Under the terms of the instruments, interest payments on the asset and liability occur in the same period and the net cash flow is always positive because the interest rate on the asset exceeds the interest rate on the liability. The entity wishes to hedge the financial asset and enters into an interest rate swap to receive a floating interest rate and pay a fixed interest rate on a notional amount equal to the principal of the asset. It designates the interest rate swap as a fair value hedge of the fixed rate asset.

Requirement

Does the hedging relationship qualify for hedge accounting even though the effect of the interest rate swap on an entity-wide basis is to create an exposure to interest rate changes that did not previously exist?

Solution

Yes. IFRS 9 does not require risk reduction on an entity-wide basis as a condition for hedge accounting. Exposure is assessed on a transaction basis and, in this instance, the asset being hedged has a fair value exposure* to interest rate increases that is offset by the interest rate swap.

* The fair value of a loan is the present value of the cash flows. A fixed rate loan has constant cash flows so the fair value is directly affected by a change in the discount rate (ie, the market interest rate).

5.2 Fair value hedge accounting

In a fair value hedge:

(a) the hedged item is a recognised asset or liability. This is re-measured to fair value at the end of the reporting period and the gain or loss on the hedged item attributable to the **hedged risk** is **recognised in profit or loss**;

(b) the hedging instrument is a derivative. The **gain or loss** resulting from **re-measuring** the hedging instrument at fair value is also **recognised in profit or loss**;

(c) if the hedged item is an **equity investment measured at fair value through OCI**, the gains and losses on the hedged investment and hedging instrument are both recognised in other comprehensive income rather than profit or loss; and

(d) if the hedged item is an **unrecognised firm commitment**, the cumulative change in the fair value of the hedged item after it has been designated as a hedged item is recognised as an asset or liability with a corresponding amount recognised in profit or loss.

Context example: Gain or loss on hedged item recognised in profit or loss

At 1 November 20X5, an entity held inventory with a cost of £400,000 and a fair value of £600,000. The entity acquired a derivative to hedge against a fall in the fair value of its inventory below £600,000. At its year end two months later, the fair value of its inventory had fallen by £20,000 and the derivative it holds had a value of £20,000.

The journals required at the year end are as follows.

DEBIT	Financial asset	£20,000	
CREDIT	Profit or loss		£20,000

To recognise the gain on the derivative hedging instrument

DEBIT	Profit or loss	£20,000	
CREDIT	Inventories		£20,000

To adjust the carrying amount of inventories by the loss in its fair value (because closing inventories reduce cost of sale, a decrease in their carrying amount increases cost of sales and reduces profit).

The effect is as follows:

(a) The loss on the hedged item has been recognised in profit or loss.

(b) There is a nil net effect in profit or loss, because the hedge has been 100% effective.

(c) Inventories are carried at £380,000. This is neither cost (£400,000) nor fair value (£580,000).

(d) The entity has been protected against loss of profit. If it had sold the inventory on 1 November, it would have made a profit of £200,000 (£600,000 – £400,000); if it sells the inventory on 1 January 20X6, it will make a profit of £200,000 (£580,000 – £380,000).



Interactive question 4: Fair value hedge

A company owns inventories of 40,000 gallons of oil which cost £800,000 on 1 December 20X3. In order to hedge the fluctuation in the market value of the oil, on 1 December 20X3 the company signs a futures contract to deliver 40,000 gallons of oil on 31 March 20X4 at the futures price of £22 per gallon.

The market price of oil on 31 December 20X3 is £22.25 per gallon, and the futures price at that date for delivery on 31 March 20X4 is £24 per gallon.

Assume that the IFRS 9 hedging criteria are met.

Requirements

Explain how these transactions should be accounted for at 31 December 20X3:

(a) without hedge accounting

(b) with hedge accounting

See **Answer** at the end of this chapter.

If only particular risks attributable to a hedged item are hedged, recognised changes in the hedged item's fair value unrelated to the hedged risk are recognised as normal. This means that changes in fair value of a hedged financial asset or liability that is not part of the hedging relationship would be accounted for as follows:

(a) For instruments measured at amortised cost, such changes would not be recognised.

(b) For instruments measured at fair value through profit or loss, such changes would be recognised in profit or loss in any event.

(c) For equity instruments in respect of which another comprehensive income election has been made, such changes would be recognised in other comprehensive income, as explained above. However, exceptions to this would include foreign currency gains and losses on monetary items and impairment losses, which would be recognised in profit or loss in any event.

If the fair value hedge is 100% effective (as in the above example), then the change in the fair value of the hedged item will be wholly offset by the change in the fair value of the hedging instrument and there will be no effect in profit or loss. Whenever the hedge is not perfect and the change in the fair value of the hedged item is not fully cancelled by change in the fair value of the hedging instrument, the resulting difference will be recognised in profit or loss. This difference is referred to as hedge ineffectiveness.



Worked example: Fair value hedge of inventory

On 1 July 20X6, a jewellery trader acquired 10,000 ounces of a material which it held in its inventory. This cost £200 per ounce, so a total of £2 million. The trader was concerned that the price of this inventory would fall, so on 1 July 20X6 he sold 10,000 ounces in the futures market for £210 per ounce for delivery on 30 June 20X7. On 1 July 20X6, the conditions for hedge accounting were all met, and these continued to be met throughout the hedging period.

At 31 December 20X6, the end of the trader's reporting period, the fair value of the inventory was £220 per ounce while the futures price for 30 June 20X7 delivery was £227 per ounce. On 30 June 20X7 the trader sold the inventory and closed out the futures position at the then spot price of £230 per ounce.

Requirement

Set out the accounting entries in respect of the above transactions.

Solution Entries

	Debit	Credit
1 July 20X6	**£**	**£**
Inventory Cash	2,000,000	
(To record the initial purchase of material)		2,000,000

At 31 December 20X6 the increase in the fair value of the inventory was £200,000 (10,000 × (£220 – £200)) and the increase in the forward contract liability was £170,000 (10,000 × (£227 – £210)). The IFRS 9 hedge accounting criteria have been met, so hedge accounting was permitted.

	Debit	Credit
31 December 20X6	**£**	**£**
Profit or loss	170,000	
Financial liability		170,000
(To record the loss on the forward contract)		
Inventories	200,000	
Profit or loss		200,000
(To record the increase in the fair value of the inventories)		

At 30 June 20X7 the increase in the fair value of the inventory was another £100,000 (10,000 × (£230 – £220)) and the increase in the forward contract liability was another £30,000 (10,000 × (£230 – £227)).

	Debit	Credit
30 June 20X7	**£**	**£**
Profit or loss	30,000	
Financial liability		30,000
(To record the loss on the forward contract)		
Inventories	100,000	
Profit or loss		100,000
(To record the increase in the fair value of the inventories)		
Profit or loss	2,300,000	
Inventories		2,300,000
(To record the inventories now sold)		
Cash	2,300,000	
Profit or loss – revenue		2,300,000
(To record the revenue from the sale of inventories)		
Financial liability	200,000	
Cash		200,000
(To record the settlement of the net balance due on closing the financial liability)		

Note that because the fair value of the material rose, the trader made a profit of only £100,000 on the sale of inventories. Without the forward contract, the profit would have been £300,000 (£2,300,000 – £2,000,000). In the light of the rising fair value, the trader might in practice have closed out the futures position earlier, rather than waiting until the settlement date.

5.2.1 Interest rate futures

An interest rate futures contract has interest-bearing instruments as its underlying asset. Futures contracts are available in relation to short-term interest rates in major currencies like sterling, euros, yen and Swiss francs. These derivatives can be used to gain exposure to, or hedge exposure against, interest rate movements. Three-month sterling future (short sterling) is a 90-day sterling SONIA interest rate future traded on ICE Futures Europe with the following characteristics:

Unit of trade: £500,000 (this is a notional amount on which interest effect is measured)

Quote: 100 minus interest rate

Tick size: 0.01 (smallest permitted quote movement ie, one basis point)

Tick value: £12.50 (£500,000 × 0.01% × 3/12)

The contract is cash settled ie, one party pays to the other the difference in value between the interest for three months at the rate agreed when the contract was originated and actual rate on maturity.

Note that SONIA, Sterling Overnight Index Average is a benchmark rate that replaced LIBOR in 2021.



Worked example: Fair value hedge using interest rate futures

Zeta Bank has a fixed rate financial asset of £10 million and is concerned that interest rates will increase from the current levels.

Requirement

Explain how Zeta Bank can hedge the fair value of the fixed rate financial asset of £10 million against an increase in interest rates using interest rate futures.

Solution

If interest rates increase, the fair value of the fixed rate financial asset will decrease. Zeta Bank requires a futures position that will yield profits when interest rates increase to offset this loss. It should therefore sell £(10,000,000/500,000) = 20 futures contracts. If the interest rate increases, the gain on the futures position will offset the loss on the fixed rate financial asset.

Zeta Bank should designate the futures contract as the hedging instrument and the fixed rate financial asset as the hedged item in a fair value hedge. If the IFRS 9 conditions for hedge accounting are met, the fair value movements on the futures contract and the financial asset will be recognised and offset in profit or loss.

5.3 Hedging of firm commitments

The hedging of a **firm commitment** should be treated as a **fair value hedge, except** that a firm commitment with a **price fixed** in **foreign currency** may be treated as **either a fair value hedge or a cash flow hedge** of the foreign currency risk.

When an unrecognised firm commitment to acquire an asset or to assume a liability is designated as a hedged item in a fair value hedge, the accounting treatment is as follows:

(a) The **subsequent cumulative change in the fair value of the firm commitment** attributable to the hedged risk since inception of the hedge is recognised as an asset or liability with a corresponding gain or loss **recognised in profit or loss**.

(b) The **changes in the fair value of the hedging instrument** are also **recognised in profit or loss**.

(c) When the firm commitment is **fulfilled**, the initial carrying amount of the asset or liability is **adjusted** to include the cumulative change in the firm commitment that has been recognised in the statement of financial position (SOFP) under the first point above.

5.4 Discontinuing fair value hedge accounting

Fair value hedge accounting should be discontinued if the hedging instrument expires or is sold, terminated or exercised, if the criteria for hedge accounting are no longer met, or if the entity revokes the designation.

The discontinuance should be accounted for prospectively ie, the previous accounting entries are not reversed. The hedged item is not adjusted for any further changes in its fair value and adjustments already made are recognised in profit or loss over the life of the item.

6 Cash flow hedge



Section overview

The application of cash flow hedge accounting is discussed in this section through a series of practical examples.

6.1 Cash flow hedge

A cash flow hedge is a **hedge of the variability in an entity's cash flows**. The variability should be attributable to a particular risk associated with a recognised asset or liability, or a highly probable forecast transaction and **could affect profit or loss**.

Examples of cash flow hedges include the following:

(a) The use of interest rate swaps to change floating rate debt into fixed rate debt. The entity is hedging the risk of variability in future interest payments which may arise for instance from changes in market interest rates. The fixed rate protects this cash flow variability (but with the consequence that the fair value of the instrument may now vary in response to market interest movements).

(b) The use of a commodity forward contract for a highly probable sale of the commodity in future. The entity is hedging the risk of variability in the cash flows to be received on the sale, due to changes in the market price of the goods.

The hedge of foreign currency assets and liabilities using forward exchange contracts can be treated as either a fair value or a cash flow hedge. This is because movements in exchange rates change both the fair value of such assets and liabilities and ultimate cash flows arising from them. Similarly, a hedge of the foreign currency risk of a firm commitment may be designated as either a fair value or a cash flow hedge.

6.2 Forecast transaction

A forecast transaction is an uncommitted but anticipated future transaction. To qualify for cash flow hedge accounting, the forecast transaction should be:

- **specifically identifiable** as a single transaction or a group of individual transactions which share the same risk exposure for which they are designated as being hedged;
- **highly probable**. The factors to be taken into account when assessing the probability of the transaction are discussed further below; and
- with a party that is **external** to the entity.

6.2.1 Specifically identifiable

Identification of hedged forecast transaction

A forecast transaction such as the purchase or sale of the **last 15,000 units** of a product in a specified period, or as a percentage of purchases or sales during a specified period, **does not qualify as a hedged item**.

This is because the hedged forecast transaction must be identified and documented with sufficient specificity so that when the transaction occurs, it is clear whether the transaction is or is not the hedged transaction. Therefore, a forecast transaction may be identified as the sale of the first 15,000 units of a specific product during a specified three-month period, but it could not be identified as the last 15,000 units of that product sold during a three-month period because the last 15,000 units cannot be identified when they are sold. For the same reason, a forecast transaction cannot be specified solely as a percentage of sales or purchases during a period.

Documentation of timing of forecast transaction

For a hedge of a forecast transaction, the documentation of the hedge relationship that is established at inception of the hedge should identify the date on which, or time period in which, the forecast transaction is expected to occur. This is because the hedge must relate to a specific identified risk and it must be possible to measure its effectiveness reliably. In addition, the hedged forecast transaction must be highly probable.

To meet these criteria, an entity is not required to predict and document the exact date a forecast transaction is expected to occur. However, it is required to identify and document the time period during which the forecast transaction is expected to occur within a reasonably specific and generally narrow range of time from a most probable date, as a basis for assessing hedge effectiveness. To determine that the hedge will be effective, it is necessary to ensure that changes in the fair value of the expected cash flows are offset by changes in the fair value of the hedging instrument, and this test may be met only if the timing of the cash flows occur within close proximity to each other.

6.2.2 What is 'highly probable'?

The term 'highly probable' indicates a much greater likelihood of happening than the term 'more likely than not'. An assessment of the likelihood that a forecast transaction will take place is not based solely on management's intentions because intentions are not verifiable. A transaction's probability should be supported by observable facts and the attendant circumstances.

In assessing the likelihood that a transaction will occur, an entity should consider the following circumstances:

(a) The frequency of similar past transactions

(b) The financial and operational ability of the entity to carry out the transaction

(c) Substantial commitments of resources to a particular activity (for example, a manufacturing facility that can be used in the short run only to process a particular type of commodity)

(d) The extent of loss or disruption of operations that could result if the transaction does not occur

(e) The likelihood that transactions with substantially different characteristics might be used to achieve the same business purpose (for example, an entity that intends to raise cash may have several ways of doing so, ranging from a short-term bank loan to an offering of ordinary shares)

(f) The entity's business plan

6.2.3 Further matters to consider

The length of time until a forecast transaction is projected to occur is also a factor in determining probability. Other factors being equal, the more distant a forecast transaction is, the less likely it is that the transaction would be regarded as highly probable and the stronger the evidence that would be needed to support an assertion that it is highly probable.

For example, a transaction forecast to occur in five years may be less likely to occur than a transaction forecast to occur in one year. However, forecast interest payments for the next 20 years on variable rate debt would typically be highly probable if supported by an existing contractual obligation.

In addition, other factors being equal, the greater the physical quantity or future value of a forecast transaction in proportion to the entity's transactions of the same nature, the less likely it is that the transaction would be regarded as highly probable and the stronger the evidence that would be required to support an assertion that it is highly probable. For example, less evidence generally would be needed to support forecast sales of at least 100,000 units in the next month than 950,000 units in that month when recent sales have averaged 950,000 units per month for the past three months.

A history of having designated hedges of forecast transactions and then determining that the forecast transactions are no longer expected to occur would call into question both an entity's ability to predict forecast transactions accurately and the propriety of using hedge accounting in the future for similar forecast transactions.

Professional skills focus: Applying judgement

With so many factors to consider, some of which may point to different conclusions, a considerable amount of judgement is needed in both financial reporting and auditing of probable forecast transactions.

6.3 Cash flow hedge accounting

A cash flow hedge involves hedging future cash flows. Initially therefore, only the hedging instrument (the derivative) is recognised.

The part of the gain or loss arising from an **effective** hedge of a hedging instrument is **recognised in other comprehensive income** while the **ineffective portion** of the gain or loss on the hedging instrument should **be recognised in profit or loss**.

Amounts recognised in other comprehensive income are accumulated in a cash flow hedge reserve. At a given reporting date this will be the lower of:

(a) the cumulative gain or loss on the hedging instrument from the inception of the hedge; and

(b) the cumulative change in fair value (present value) of the expected future cash flows on the hedged item from inception of the hedge.

When the separate component of equity has been adjusted to this amount, any remaining gain or loss on the hedging instrument is recognised in profit or loss.

In the period in which the hedged expected future cash flows affect profit or loss, the amount accumulated in the cash flow hedge reserve is reclassified to profit or loss with the following exception:

If the hedged transaction results in the recognition of a non-financial asset or liability, the amount accumulated in the cash flow hedge reserve is transferred to be included in the initial cost or carrying amount of the non-financial item. This is not a reclassification adjustment and does not affect the other comprehensive income of the period.

Interactive question 5: Cash flow hedge

A company enters into a hedge in order to protect its future cash inflows relating to a recognised financial asset held at amortised cost.

At inception the value of the hedging instrument was £0, but by the year end a gain of £8,800 was made when measured at market value. The corresponding loss in respect of the future cash flows amounted to £9,100 in fair value terms.

Requirement

How should the transaction be accounted for?

See **Answer** at the end of this chapter.

Interactive question 6: Swap in cash flow hedge 1

An entity issues a fixed rate debt instrument and enters into a receive-fixed, pay-variable, interest rate swap to offset the exposure to interest rate risk associated with the debt instrument.

Requirement

Can the entity designate the swap as a cash flow hedge of the future interest cash outflows associated with the debt instrument?

See **Answer** at the end of this chapter.

Interactive question 7: Swap in cash flow hedge 2

An entity manages interest rate risk on a net basis. On 1 January 20X6, it forecasts aggregate cash inflows of £1 million on a fixed rate financial asset and aggregate cash outflows of £900,000 on a fixed rate financial liability in the first quarter of 20X7. For risk management purposes it uses a receive-variable, pay-fixed, forward rate agreement (FRA) to hedge the forecast net cash inflow of £100,000. The entity designates as the hedged item the first £100,000 of cash inflows on fixed rate assets in the first quarter of 20X7.

Requirement

Can it designate the receive-variable, pay-fixed FRA as a cash flow hedge of the exposure to variability to cash flows in the first quarter of 20X7 associated with the fixed rate assets?

See **Answer** at the end of this chapter.

Worked example: Cash flow hedge – inventories

Bets Co is a manufacturer and retailer of gold jewellery.

On 31 October 20X1, the cost of Bets's inventories of finished jewellery was £8.280 million with a gold content of 24,000 troy ounces. At that date their sales value was £9.938 million.

The selling price of gold jewellery is heavily dependent on the current market price of gold (plus a standard percentage for design and production costs).

Bets's management wished to reduce their business risk of fluctuations in future cash inflow from sale of the jewellery by hedging the value of the gold content of the jewellery. In the past this has proved to be an effective strategy.

Therefore, it sold futures contracts for 24,000 troy ounces of gold at £388 per troy ounce at 31 October 20X1. The contracts mature on 30 October 20X2.

On 30 September 20X2 the fair value of the jewellery was £9.186 million and the forward price of gold per troy ounce for delivery on 30 October 20X2 was £352.

Requirement

Explain how the above transactions would be treated in Bets's financial statements for the year ended 30 September 20X2.

Solution

Bets is hedging the volatility of the future cash inflow from selling the gold jewellery. The futures contracts can be accounted for as a cash flow hedge in respect of those inflows, providing the criteria for hedge accounting are met.

The gain on the forward contract should be calculated as:

	£
Forward value of contract at 31.10.X1 (24,000 × £388)	9,312,000
Forward value of contract at 30.9.X2 (24,000 × £352)	8,448,000
Gain on contract	864,000

The change in the fair value of the expected future cash flows on the hedged item (which is not recognised in the financial statements) should be calculated as:

	£
At 31.10.X1	9,938,000
At 30.9.X2	9,186,000
	752,000

As this change in fair value is less than the gain on the forward contract, the hedge is not fully effective and only £752,000 of the gain on the forward should be recognised in other comprehensive income. The remainder should be recognised in profit or loss:

		£	£
DEBIT	Financial asset (Forward a/c)	864,000	
CREDIT	Other comprehensive income		752,000
CREDIT	Profit or loss		112,000

A hedging relationship continues to qualify for hedge accounting if it is effective. In this case:

- an economic relationship continues to exist between the hedged item and hedging instrument (since they are both gold); and
- the effect of credit risk does not dominate the value changes that result from the economic relationship.

The third criterion for hedge effectiveness is that the hedge ratio of the hedging relationship is the same as that resulting from the quantity of hedged item that the entity actually hedges and the quantity of hedging instrument that the entity actually uses to hedge that quantity of hedged items.

Since this hedge relationship results in a gain on futures contract of £864,000 but a loss on hedged item of only £752,000, it appears that the relationship should be rebalanced.

The current hedge ratio is 1:1 (with hedged item and hedging instrument both based on 24,000 troy ounces of gold); to maintain 100% effectiveness this should be reset by reducing the quantity of hedging instrument to 20,889 troy ounces (752/864 × 24,000) or increasing the quantity of hedged item to 27,574 troy ounces (864/752 × 24,000).



Interactive question 8: Foreign currency hedge 1

Allison Co has a foreign currency liability payable in six months' time and it wishes to hedge the amount payable on settlement against foreign currency fluctuations. To that end, it takes out a forward contract to buy the foreign currency in six months' time. The conditions for hedge accounting were met.

Requirements

8.1 Should the hedge be treated as a fair value hedge of the foreign currency liability or as a cash flow hedge of the amount to be settled in the future?

8.2 How should gains and losses on the liability and the forward contract be accounted for?

See **Answer** at the end of this chapter.

Interactive question 9: Foreign currency hedge 2

An entity exports a product at a price denominated in a foreign currency. At the date of the sale, the entity obtains a receivable for the sale price payable in 90 days and takes out a 90-day forward exchange contract in the same currency as the receivable to hedge its foreign currency exposure.
The conditions for hedge accounting were met.

Under IAS 21, the sale is recorded at the spot rate at the date of sale, and the receivable is restated during the 90-day period for changes in exchange rates with the difference being taken to profit or loss (IAS 21.23 and IAS 21.28).

Requirement

If the foreign exchange forward contract is designated as a hedging instrument, does the entity have a choice whether to designate it as a fair value hedge of the foreign currency exposure of the receivable, or as a cash flow hedge of the collection of the receivable?

How should gains and losses on the receivable and the forward contract be accounted for?

See **Answer** at the end of this chapter.

Interactive question 10: Cash flow hedge

RapidMart is a company that operates a chain of large out of town supermarkets. It has expanded rapidly over the last 10 years, opening new stores in its home country and overseas. It has also moved into a wide range of non-food sales and the provision of services, such as opticians. The company is currently preparing its consolidated financial statements for the year ending 30 September 20X5.

During the last year, RapidMart began to operate an online retail division, RapidMart Direct, as a pilot scheme. The service uses a fleet of delivery vans. This has proved to be very popular with customers and the company wants to expand this operation. The finance director identified a key risk of volatility of diesel prices and has taken out a forward contract to hedge against this.

On 1 August 20X5, RapidMart entered into a forward contract to hedge its expected fuel requirements for the second quarter of the next financial year for delivery of 1 million litres of diesel on 31 December 20X5 at a price of £2.04 per litre.

The company intended to settle the contract net in cash and purchase the actual required quantity of diesel in the open market on 31 December 20X5.

At the company's year end the forward price for delivery on 31 December 20X5 had risen to £2.16 per litre of fuel.

Requirement

How should the above transaction be accounted for in the financial statements of RapidMart for the year ending 30 September 20X5?

See **Answer** at the end of this chapter.

Professional skills focus: Assimilating and using information

The above question contains quite a lot of information, but you should quickly have spotted the fact that RapidMart is expanding and moving into new areas of business. Cash flow is likely to be an issue, as is volatility in prices.

Interactive question 11: Foreign currency receivables and forward contract

Armada is a public limited company reporting under IFRS Standards. It is preparing the financial statements as at 31 December 20X1. Included in trade receivables is an amount due from a customer located abroad. The amount (30.24 million coronna) was initially recognised when the exchange rate was £1 = 5.6 coronna.

At 31 December 20X1, the exchange rate was £1 = 5.4 coronna. No adjustment has been made to the trade receivable since it was initially recognised.

Given the size of the exposure, the company entered into a forward contract, at the same time as the receivable was initially recognised, in order to protect cash flows from fluctuations in the exchange rate. The forward contract is to sell 30.24 million coronna and it satisfies the necessary criteria to be accounted for as a hedge.

In the period between inception of the forward contract and the year end, the loss in fair value of the forward contract was £220,000. The company elected to designate the spot element of the hedge as the hedging relationship. The difference between the change in fair value of the receivable and the change in fair value of the forward contract since inception is the interest element of the forward contract.

Requirement

Show how this transaction should be accounted for in the financial statements of Armada for the year ended 31 December 20X1.

See **Answer** at the end of this chapter.

Interactive question 12: Cash flow hedge

On 1 November 20X2, Blenheim entered into a contract to purchase 3,000 tonnes of refined sunflower oil. The contract is for delivery in February 20X3 at a price of £1,440 per tonne. Blenheim uses sunflower oil to make its products.

At 31 December 20X2, an equivalent new contract for delivery of 3,000 tonnes of refined sunflower oil in February 20X3 could be entered into at £1,400 per tonne.

Blenheim does not intend to take delivery of the sunflower oil and instead intends to settle the contract net in cash, then purchase the actual required quantity based on demand at the time.

The contract is designated as a cash flow hedge of the highly probable forecast purchase of sunflower oil. All necessary documentation was prepared to treat the contract as a cash flow hedge. No accounting entries have been made.

Tax rules follow accounting rules in respect of financial instruments in the tax jurisdiction (with both profit and other comprehensive income items subject to tax at 30%) in which Blenheim operates. No current or deferred tax adjustments have been made for this transaction.

Requirement

Show how this transaction should be accounted for in the financial statements of Blenheim for the year ended 31 December 20X2.

See **Answer** at the end of this chapter.

6.4 Discontinuing cash flow hedge accounting

Cash flow hedge accounting should be discontinued if the hedging instrument expires or is sold, terminated or exercised, if the criteria for hedge accounting are no longer met, a forecast transaction is no longer expected to occur or if the entity revokes the designation.

The discontinuance should be accounted for prospectively ie, the previous accounting entries are not reversed. The cumulative gain or loss on the hedging instrument should be reclassified to profit or loss, as the hedged item is recognised in profit or loss.

7 Hedge of a net investment



Section overview

This section discusses issues specific to the accounting treatment of hedge of net investments.

7.1 Definition

Hedges of a net investment arise in the consolidated accounts where a parent company takes a foreign currency loan in order to buy shares in a foreign subsidiary. The loan and the investment need not be denominated in the same currency, however, assuming that the currencies perform similarly against the parent company's own currency, it should be the case that fluctuations in the exchange rate affect the asset (the net assets of the subsidiary) and the liability (the loan) in opposite ways, hence gains and losses are hedged.

In this type of accounting hedge, the hedging instrument is the foreign currency loan rather than a derivative.

7.2 Accounting treatment

You may understand this type of hedge better after studying consolidation of foreign operations, but in a simple sense, without applying hedging rules:

(a) The loan would be retranslated to the parent's own currency at the year end using the spot exchange rate; any resultant gain or loss would be recognised in profit or loss.

(b) Prior to consolidation, the subsidiary's accounts would be translated into the parent's own currency with any gain or loss recognised in other comprehensive income.

(c) On consolidation, the gain or loss on the loan would affect consolidated profit or loss and the loss or gain on the translation of the subsidiary's net assets would affect consolidated reserves.

The net investment hedge ensures that the gains and losses are both recognised in other comprehensive income and accumulated in reserves by:

- recognising the portion of the gain or loss on the hedging instrument that is determined to be effective in other comprehensive income; and
- recognising the ineffective portion in profit or loss.

Any gain or loss recognised in other comprehensive income is reclassified to profit or loss on the disposal or partial disposal of the foreign operation.

7.3 Hedging with a non-derivative financial instrument

A foreign currency borrowing can be designated as a hedge of a net investment in a foreign operation, with the result that any translation gain or loss on the borrowing should be recognised in other comprehensive income to offset the translation loss or gain on the investment. (For hedges other than hedges of foreign currency risk, when an entity designates a non-derivative financial asset or a non-derivative financial liability measured at fair value through profit or loss as a hedging instrument, it may only designate the non-derivative financial instrument in its entirety or a proportion of it.)

Context example: Use of non-derivative to hedge a net investment

Jenkins Co, whose functional currency is the pound (£), has a subsidiary in France. The subsidiary was purchased on 30 June 20X6 for €20 million and the acquisition was financed with a loan of €20 million. The carrying amount of the subsidiary in the consolidated financial statements (including goodwill acquired in the business combination) is €20 million. Jenkins Co has designated the foreign currency loan of €20 million as a hedge of its net investment in the foreign subsidiary.

Jenkins Co has a 30 June year end. The foreign currency rates at 30 June 20X6 and 30 June 20X7 were £1 = €1.52 and £1 = €1.48 respectively. In the year ended 30 June 20X7, the exchange difference on the **opening** net investment should be calculated as:

	€	£
At 30 June 20X6	20,000,000	13,157,895
At 30 June 20X7	20,000,000	13,513,514
Exchange gain		355,619

There is a corresponding loss on the foreign currency loan of £355,619. Because the hedge is perfectly effective, both the gain and the entire loss will be recognised in other comprehensive income. There is no ineffective portion of the loss on the hedging instrument to be recognised in profit or loss.

7.4 Hedging with derivatives

A net investment can be hedged with a derivative instrument, such as a currency forward contract. In this case, however, it would be necessary to designate at inception that effectiveness can be measured by reference to changes in spot exchange rates or changes in forward exchange rates.

7.5 The effect of re-balancing on hedge accounting

As we saw in section 4, in order to maintain a hedge ratio that complies with hedge effectiveness requirements, a hedge relationship may have to be rebalanced. There are two ways of achieving this.

(a) Increasing or decreasing the volume of the hedging instrument; or

(b) Increasing or decreasing the volume of the hedged item

The effect of a rebalancing on the accounting treatment described in the sections above is as follows:

Volume of hedging instrument increased	No effect on measurement of changes in fair value of original volume of hedging instrument. No effect on measurement of changes in fair value of hedged item. Changes in value of additional volume of hedging instrument are measured starting from the date of rebalancing rather than from the date on which the hedging relationship was designated.
Volume of hedging instrument decreased	No effect on measurement of changes in fair value of volume of hedging instrument that continues to be designated part of the hedge. No effect on measurement of changes in fair value of hedged item. The volume by which the hedging instrument was decreased is no longer part of the hedging relationship. The reporting entity therefore retains a derivative, but only part of it is a hedging instrument in the hedge relationship. The undesignated part of the derivative is accounted for in accordance with normal IFRS 9 requirements ie, at fair value through profit or loss.
Volume of hedged item increased	No effect on measurement of changes in fair value of hedging instrument. No effect on measurement of changes in fair value of original volume of hedged item. Changes in value of additional volume of hedged item are measured starting from the date of rebalancing rather than from the date on which the hedging relationship was designated.
Volume of hedged item decreased	No effect on measurement of changes in fair value of hedging instrument. No effect on measurement of changes in fair value of volume of hedged item that continues to be designated part of the hedge. The volume by which the hedged item was decreased is not part of the hedge from the rebalancing date, and is accounted for as a discontinuation of hedge accounting.

7.6 Summary of hedge accounting

The following table summarises the accounting treatment under IFRS 9 of the main types of hedges.

	Fair value hedge	**Cash flow hedge**
Gain or loss on hedging instrument	Profit or loss (Or OCI if the hedged item is an equity instrument for which an entity has elected to present changes in fair value in OCI)	Other comprehensive income
Adjustment to hedged item	Profit or loss (Or OCI if an equity instrument and OCI election made)	N/A
Hedge ineffectiveness is recorded in profit or loss	Yes	Yes

	Fair value hedge	Cash flow hedge
Gains or losses reclassified to profit or loss later	N/A	Yes

7.7 Types of hedge and their treatment

The following grid will be useful in distinguishing the types of hedge and their treatment.

	Firm commitment	Forecast transaction (highly probable)
Foreign currency	Either fair value hedge or cash flow hedge	Cash flow hedge
Other	Fair value hedge	Cash flow hedge

Professional skills focus: Structuring problems and solutions

It may not be immediately obvious what type of hedge is being used. The above tables sum up the treatment and can be used as an overview in order to structure your thinking about the information in the question.

Interactive question 13: Comprehensive fair value hedge

Toprate Exports, whose functional currency is the dollar (USD), has significant receipts in pounds sterling (GBP). In order to protect itself from currency fluctuations relating to its foreign currency receivables, it frequently enters into contracts to sell GBP forward. On 31 October 20X1 the company recognised a receivable of GBP 1 million, due on 31 January 20X2.

On 31 October 20X1 the company entered into a three-month forward contract for settlement on 31 January 20X2 to sell GBP 1 million at USD 1 = GBP 0.6202. The spot rate on 31 October 20X1 was USD 1 = GBP 0.6195.

At 31 December 20X1, the forward rate for settlement on 31 January 20X2 was USD 1 = GBP 0.6440 (spot rate on 31 December 20X1 was USD 1 = GBP 0.6435).

The applicable dollar yield curve gives the following (annualised) rate for discounting a cash flow occurring on 31 January 20X2:

At 31 December 20X1 0.325%

The company set up the appropriate documentation on 31 October 20X1 to treat the forward contract as a fair value hedge and designated the hedging relationship as being changes in the spot element of the forward exchange contract.

Requirement

Explain, showing relevant financial statement extracts, the accounting treatment of these transactions in Toprate Exports's financial statements (insofar as the information provided permits) for the year ended 31 December 20X1. (Notes to the financial statements are not required.)

You should perform any discounting necessary to the nearest month and work to the nearest $1.

See **Answer** at the end of this chapter.

8 Disclosures

Section overview

This section covers the disclosures required in respect of hedging.

Under IFRS 7, *Financial Instruments: Disclosures* an entity should disclose the following separately for each type of hedge described in IFRS 9 (ie, fair value hedges, cash flow hedges and hedges of net investments in foreign operations):

- A **description** of each type of hedge
- A **description** of the financial instruments designated as hedging instruments and their fair values at the reporting date
- The **nature of the risks** being hedged

For **cash flow hedges**, an entity should disclose the following:

- The periods when the cash flows are expected to occur and when they are expected to affect profit or loss
- A description of any forecast transaction for which hedge accounting had previously been used, but which is no longer expected to occur
- The amount that was recognised in other comprehensive income during the period
- The amount that was reclassified from equity to profit or loss for the year, showing the amount included in each line item in the statement of comprehensive income
- The amount that was reclassified from equity during the period and included in the initial cost or other carrying amount of a non-financial asset or non-financial liability whose acquisition or incurrence was a hedged highly probable forecast transaction

An entity should **disclose the following separately**:

- In fair value hedges, gains or losses:
  - on the hedging instrument; and
  - on the hedged item attributable to the hedged risk
- The ineffectiveness recognised in profit or loss that arises from cash flow hedges
- The ineffectiveness recognised in profit or loss that arises from hedges of net investments in foreign operations

9 IAS 39 requirements on hedge accounting

Section overview

- This section provides an overview of the hedging rules in IFRS 9's predecessor, IAS 39, *Financial Instruments: Recognition and Measurement*.
- Entities may apply the new IFRS 9 rules in their entirety, or entities may apply the hedge accounting rules of IAS 39 to all of their hedging relationships while following the classification and measurement rules of IFRS 9.
- Entities undertaking macro hedging activities may apply the new general hedge accounting model in IFRS 9 while continuing to apply the specific macro hedging requirements of IAS 39.
- IFRS 9 is the default standard for your exam.

Note: For the purpose of the exam, the candidate would be expected to use IFRS 9, but must understand the main differences in IAS 39, which can still be applied with regard to hedging.

9.1 What is macro hedging?

Macro hedging, also known as portfolio hedging, is a technique whereby financial instruments with similar risks are grouped together and the risks of the portfolio are hedged together. Often this is done on a net basis with assets and liabilities included in the same portfolio. For example, instead of using interest rate swaps to hedge interest rate exposure on a loan by loan basis, banks hedge the risk of their entire loan book or specific portions of the loan book.

Currently, IFRS 9 does not address macro hedging.

In general, IAS 39 does not permit an overall net position to be designated as a hedged item, for example a UK entity that has to make a purchase of £10 million in 30 days and a sale of £2 million in 30 days cannot designate the net purchase of £8 million as the hedged item. The exception is that IAS 39 permits macro hedging for the interest rate risk associated with a portfolio of financial assets or liabilities. There are, however, clearly prescribed procedures that must be followed in order to do so.

9.2 IAS 39 hedging differences

A number of aspects of hedge accounting are different under IAS 39; however, the **following are the same as IFRS 9**:

- The terminology used in IAS 39 and IFRS 9 is generally the same.
- The three types of hedges – fair value hedge, cash flow hedge and net investment hedge are the same.
- Hedge ineffectiveness is recognised in profit or loss except for the other comprehensive income option for equity investments.
- Hedge accounting with written options is prohibited.

9.3 IFRS 9 hedging vs IAS 39 hedging: summary of key differences

The IFRS 9 model for hedge accounting differs from that in IAS 39 in the following key areas:

	IFRS 9	IAS 39
Eligibility of hedging instruments	Any financial instrument may be a hedging instrument if it is measured at fair value through profit or loss.	Derivatives may be designated as hedging instruments. Non-derivatives may be designated as hedging instruments only for hedge of foreign currency risk.
Therefore, under IAS 39 non-derivative items are less widely used as hedging instruments than under IFRS 9.		
Eligibility of hedged items	In addition to IAS 39 eligible hedged items, IFRS 9 allows a risk component of a non-financial asset or liability to be designated as a hedged item in some circumstances.	Recognised assets, liabilities, firm commitments, highly probable forecast transactions and net investments in foreign operations may be designated as hedged items. In some circumstances, risk components of a financial asset or liability may be designated as a hedged item.
Therefore fewer items can be designated as hedged items under IAS 39.		

	IFRS 9	IAS 39
Qualifying criteria for applying hedge accounting	Hedge effectiveness criteria are principles-based and aligned with risk management activities.	A hedging relationship only qualifies for hedge accounting if certain criteria are met, including a quantitative hedge effectiveness test under which hedge effectiveness must fall in the range 80% – 125%.
Therefore genuine hedging relationships captured by IFRS 9 may be missed when applying IAS 39 rules.		
Rebalancing	Rebalancing is permitted by IFRS 9 in some circumstances (see above).	The concept of rebalancing does not exist within IAS 39.
Lack of guidance on rebalancing means that hedge accounting needs to be discontinued under IAS 39, while it could continue under IFRS 9.		
Discontinuation of hedging relationships	Hedge accounting may not be discontinued where the hedging relationship continues to meet qualifying criteria. Can only discontinue when qualifying criteria are no longer met.	Hedge accounting may be discontinued at any time.
The absence of strict discontinuation rules means that IAS 39 was less precise than IFRS 9.		
Accounting for the time value component of options and forward contracts	The time value component of an option is a cost of hedging presented in OCI.	The forward element of a forward contract may also be presented in OCI. The part of an option that reflects time value and the forward element of a forward contract are treated as derivatives held for trading purposes.
IAS 39 therefore led to greater volatility in profit or loss than IFRS 9 does.		

10 Audit focus: fair value

Section overview

- Fair value measurements of assets, liabilities and components of equity may arise from both the initial recording of transactions and later changes in value.
- Auditing fair value requires both the assessment of risk and evaluating the appropriateness of the fair value and how it is disclosed.
- Fair value is a key issue to investment property, pension costs, share-based payments and many other areas of financial accounting.

10.1 Audit issues around fair value

For the auditor, the use of fair values will raise a number of issues. The determination of fair value will generally be **more difficult** than determining historical cost. It will be more difficult to establish whether fair value is reasonable for complex assets and liabilities than for more straightforward assets or liabilities which have an actively traded market and therefore a market value.

Generally speaking, the trend towards fair value accounting will increase audit work required, not only because determining fair values is more difficult, but also because **fair values fluctuate** in a way that historical costs do not, and will need vouching each audit period. Fair value will, for the same reasons, **increase audit risk**.

ISA 540 (Revised), *Auditing Accounting Estimates and Related Disclosures* addresses the ongoing **complexity** and **subjectivity** of accounting estimates in company accounts (including those related to financial instruments) by considering factors such as **estimation uncertainty** for management and **professional scepticism** for auditors when assessing such estimates. ISA 540 (revised) is covered in more detail in earlier chapter.

10.2 Auditing fair values

ISA 540 (Revised) considers all accounting estimates, including those that relate to fair value. In overview, the standard requires the following when auditing fair values.

- When assessing the risk of material misstatement, ISA 540 (Revised) requires auditors to consider both the **entity and its environment** and any **internal controls** in place at the entity (ISA 540.13).
- **Risk assessment** is usually split into **inherent** and **control risks** (ISA 540.16):
  - **Inherent risks** would include the inherent complexity and subjectivity associated with fair values, such as changes in the marketplace (ISA 540.A59).
  - **Control risk** would consider the techniques used by management in creating accounting estimates, especially if models are required (ISA 540.A39) or the use of an expert for calculations requiring complicated level 3 fair values (ISA 540.A31).
- The audit of fair value estimates is a contest between auditor and management, where **professional scepticism** is used by the former to counter the risk of **bias** by the latter (ISA 540.32).
  - For example, when reviewing ranges subject to estimation uncertainty, auditors should consider the reasonableness of ranges used by management to reach such amounts (ISA 540.A47).
  - Auditors should also consider if suitable **segregation of duties** exists in departments who are responsible for deriving fair value estimates for which they would receive a bonus (ISA 540.A51).
- **Subsequent events** are used to assess the reasonableness of fair value estimates (ISA 540. A92).
- The auditor should adopt suitable procedures for determining adequate **disclosure** (ISA 540.31) especially in relation to the various **quantitative** and **qualitative** risks involved.

11 Auditing financial instruments



Section overview

- Financial instruments include items such as cash, accounts receivable and payable, loans receivable and payable, debt and equity investments, and derivatives.

- Financial instruments should be classified as either financial assets, financial liabilities or equity instruments.
- The key audit issue with these instruments is risk and IAS 32; IFRS 9 and IFRS 7 deal with the accounting/disclosure related to these instruments.
- Guidance for the auditor is provided by IAPN 1000.

11.1 *IAPN 1000*

IAPN 1000, Special Considerations in Auditing Financial Instruments was developed to help auditors with different levels of familiarity with financial instruments. It is therefore structured in two sections. Section I provides background and educational material to help those less familiar with financial instruments to understand some of the common features, and how they are used, managed and controlled by entities and particular financial reporting issues. Section II provides guidance on the relevant auditing considerations.

The introduction of the IAPN sets out the scope. It explains that the IAPN does not address the simplest financial instruments such as cash, simple loans, trade accounts receivable and trade accounts payable, nor the most complex ones. It also does not address specific accounting requirements, such as those relating to hedge accounting, offsetting or impairment.

11.1.1 Section I – Background information about financial instruments Purpose and risks of using financial instruments

Financial instruments are used for the following purposes:

- Hedging purposes (ie, to change an existing risk profile to which an entity is exposed)
- Trading purposes (ie, to enable an entity to take a risk position to benefit from long-term investment returns or from short-term market movements)
- Investment purposes (eg, to enable an entity to benefit from long-term investment returns)

Management and those charged with governance might not:

- fully understand the risks of using financial instruments
- have the expertise to value them appropriately
- have sufficient controls in place over financial instrument activities

Business risk and the risk of material misstatement also increase when management inappropriately hedge risk or speculate.

In particular the entity may be exposed to the following types of risk:

(a) Credit risk (the risk that one party will cause a financial loss to another party by failing to discharge an obligation)

(b) Market risk (the risk that the fair value or future cash flow of a financial instrument will fluctuate because of changes in market prices eg, currency risk, interest rate risk, commodity and equity price risk)

(c) Liquidity risk (includes the risk of not being able to buy or sell a financial instrument at an appropriate price in a timely manner due to a lack of marketability for that financial instrument)

(d) Operational risk (related to the specific processing required for financial instruments)

The risk of fraud may also be increased where an employee in a position to perpetrate a financial fraud understands both the financial instruments and the process for accounting for them, but management and those charged with governance have a lesser degree of understanding.

Professional skills focus: Applying judgement

Fair value is an area where judgement must be applied, particularly in the context of financial instruments where some instruments, such as derivatives and options, are high risk.

Controls relating to financial instruments

The level of sophistication of an entity's internal control will be affected by the size of the entity and the extent and complexity of the financial instruments used. An entity's internal control over financial instruments is more likely to be effective when management and those charged with governance have:

(a) established an appropriate control environment;

(b) established a risk management process;

(c) established information systems that provide an understanding of the nature of the financial instrument activities and the associated risks; and

(d) designed, implemented and documented a system of internal control to:

  (1) provide reasonable assurance that the use of financial instruments is within the entity's risk management policies;

  (2) properly present financial instruments in the financial statements;

  (3) ensure that the entity is in compliance with applicable laws and regulations; and

  (4) monitor risk.

The Appendix to IAPN 1000 provides examples of controls that may exist in an entity that deals with a high volume of financial instrument transactions. These include authorisation, segregation of duties (particularly of those executing the transaction (dealing) and those initiating cash payments and receipts (settlements)) and reconciliations of the entity's records to external banks' and custodians' records.

Completeness, accuracy and existence

The IAPN discusses a number of practical issues. For example, it explains that where transactions are cleared through a clearing house the entity should have processes to manage the information delivered to the clearing house. Adequate IT controls must also be maintained.

It also explains that in financial institutions where there is a high volume of trading, a senior employee typically reviews daily profits and losses on individual traders' books to evaluate whether they are reasonable based on the employee's knowledge of the market. Doing so may enable management to determine that particular trades were not completely or accurately recorded, or may identify fraud by a particular trader.

Valuation of financial instruments

Section I also provides material on financial reporting requirements. It explains that many financial reporting frameworks require financial instruments, including embedded derivatives, to be measured at fair value. In general, the objective of fair value is to arrive at the price at which an orderly transaction would take place between market participants at the measurement date under current market conditions; that is, it is not the transaction price for a forced liquidation or distressed sale. In meeting this objective all relevant market information is taken into account. It also explains that fair value measurement may arise at both the initial recording of transactions and later when there are changes in value. Changes in fair value measurement may be treated differently depending on the reporting framework. The IAPN then explores features of different financial reporting frameworks, including the following:

- The fair value hierarchy (as adopted by IFRS 13)
- The effects of inactive markets
- Management's valuation processes
- The use of models, third-party pricing sources and experts (entities often make use of a third party to obtain fair value information, particularly when expertise or data are required that management does not possess)

11.1.2 Section II – Audit considerations relating to financial instruments

IAPN 1000 identifies certain factors that make auditing complex financial instruments particularly challenging:

- Management and the auditors may find it difficult to understand the nature of the instruments and the risks to which the entity is exposed.
- Markets can change quickly, placing pressure on management to manage their exposures effectively.
- Evidence supporting valuation may be difficult to obtain.
- Individual payments may be significant, which may increase the risk of misappropriation of assets.
- The amount recorded may not be significant, but there may be significant risks and exposures associated with these complex financial instruments.
- A few employees may exert significant influence on the entity's financial instruments transactions, in particular where compensation arrangements are tied to revenue from these.

Professional scepticism

The need for professional scepticism increases with the complexity of the financial instruments, for example with regard to the following:

- Evaluating whether sufficient appropriate audit evidence has been obtained (which can be particularly challenging when models are used or in determining if markets are inactive)
- Evaluating management's judgements and potential for management bias in applying the applicable financial reporting framework (eg, choice of valuation techniques, use of assumptions in valuation techniques)
- Drawing conclusions based on the audit evidence obtained (for example assessing the reasonableness of valuations prepared by management's experts)

Planning considerations

The auditor's focus in planning is particularly on the following:

- Understanding the accounting and disclosure requirements

  ISA 540 requires the auditor to obtain an understanding of the requirements of the applicable financial reporting framework relevant to accounting estimates.
- Understanding the complex financial instruments This helps the auditor to identify whether:
  - important aspects of a transaction are missing or inaccurately recorded;
  - a valuation appears appropriate;
  - the risks inherent in them are fully understood and managed by the entity; or
  - the financial instruments are appropriately classified into current and non-current assets and liabilities.

Understanding management's process for identifying and accounting for embedded derivatives will help the auditor to understand the risks to which the entity is exposed.

- Determining whether specialised skills and knowledge are needed in the audit.

  The engagement partner must be satisfied that the engagement team and any auditor's experts collectively have the appropriate competence and capabilities.
- Understanding and evaluating the system of internal control in the light of the entity's financial instrument transactions and the information systems that fall within the scope of the audit.

  This understanding must be obtained in accordance with ISA 315. This understanding enables the auditor to identify and assess the risks of material misstatement at the financial statement and assertions levels, providing a basis for designing and implementing responses to the assessed risks of material misstatement.
- Understanding the nature, role and activities of the internal audit function

  Areas where the work of the internal audit function may be particularly relevant are as follows:
  - Developing a general overview of the extent of use of financial instruments
  - Evaluating the appropriateness of policies and procedures and management's compliance with them
  - Evaluating the operating effectiveness of financial instrument control activities
  - Evaluating systems relevant to financial instrument activities
  - Assessing whether new risks relating to financial instruments are identified, assessed and managed
- Understanding management's process for valuing financial instruments, including whether management has used an expert or a service organization.

  Again this understanding is required in accordance with ISA 540.
- Assessing and responding to the risk of material misstatement (see below)

Assessing and responding to the risk of material misstatement

Factors affecting the risk of material misstatement include the following:

- The volume of financial instruments to which the entity is exposed
- The terms of the financial instruments
- The nature of the financial instruments
- Fraud risk factors (eg, where there are employee compensation schemes, difficult financial market conditions, ability to override controls)

The assessment of risk will determine the appropriate audit approach in accordance with ISA 330, *The Auditor's Responses to Assessed Risks*, including substantive procedures and tests of controls. Where the entity is involved in a high level of trading and use of financial instruments, it is unlikely that sufficient evidence will be obtained through substantive testing alone. Where there are relatively few transactions of this nature a substantive approach may be more efficient. In reaching a decision on the nature, timing and extent of testing of controls the auditor may consider factors such as:

- the nature, frequency and volume of financial instrument transactions;
- the strength of controls including design;
- the importance of controls to the overall control objectives;
- the monitoring of controls and identified deficiencies in control procedures;
- the issues controls are intended to address;
- frequency of performance of control activities;
- level of precision the controls are intended to achieve;

- evidence of performance of control activities; and
- timing of key financial instrument transactions (eg, whether they are close to the period end).

Designing substantive procedures will include consideration of the following:

- Use of analytical procedures – they may be less effective as substantive procedures when performed alone, as the complex drivers of valuation often mask unusual trends
- Non-routine transactions – this applies to many financial instrument transactions and a substantive approach will normally be the most effective means of achieving the planned audit objectives
- Availability of evidence – eg, when the entity uses a third-party pricing source, evidence may not be available from the entity
- Procedures performed in other audit areas – these may provide evidence about completeness of financial instrument transactions eg, tests of subsequent cash receipts and payments, and the search for unrecorded liabilities
- Selection of items for testing – where the financial instrument portfolio comprises instruments with varying complexity and risk judgemental sampling may be useful

In some cases 'dual-purpose' tests may be used ie, it may be efficient to perform a test of controls and a test of details on the same transaction eg, testing whether a signed contract has been maintained (test of controls) and whether the details of the financial instrument have been appropriately captured in a summary sheet (test of details). Areas of significant judgement would normally be tested close to, or at, the period end.

Procedures relating to completeness, accuracy, existence, occurrence and rights and obligations may include the following:

- Remaining alert during the audit when inspecting records or documents (eg, minutes of meetings of those charged with governance, specific invoices and correspondence with the entity's professional advisers)
- External confirmation of bank accounts, trades and custodian statements
- Reconciliation of external data with the entity's own records
- Reading individual contracts and reviewing support documentation
- Reviewing journal entries or the internal control over the recording of such entries to determine if entries have been made by employees other than those authorised to do so
- Testing controls eg, by reperforming controls

Valuation of financial instruments

Management is responsible for the valuation of complex financial instruments and must develop a valuation methodology. In testing how management values the financial instrument, the auditor should undertake one or more of the following procedures:

(a) Test how management made the accounting estimate and the data on which it is based (including any models)

(b) Test the operating effectiveness of controls over how management made the accounting estimate, together with appropriate substantive procedures

(c) Develop a point estimate or a range to evaluate management's point estimate

(d) Determine whether events occurring up to the date of the auditor's report provide audit evidence regarding the accounting estimate

Audit procedures may include the following:

- Reviewing and assessing the judgements made by management
- Considering whether there are any other relevant price indicators or factors to take into account
- Obtaining third-party evidence of price indicators eg, by obtaining a broker quote
- Assessing the mathematical accuracy of the methodology employed
- Testing data to source materials

Significant risk

The auditor's risk assessment may lead to the identification of one or more significant risks relating to valuation. The following circumstances would be indicators that a significant risk may exist:

- High measurement uncertainty
- Lack of sufficient evidence to support management's valuation
- Lack of management understanding of its financial instruments or expertise to value these correctly
- Lack of management understanding of the complex requirements of the applicable financial reporting framework
- The significance of valuation adjustments made to model outputs when the applicable reporting framework requires or permits such adjustments

Where significant risks have been identified the auditor is required to evaluate how management has considered alternative assumptions or outcomes and why it has rejected them, or how management has addressed estimation uncertainty in making the accounting estimate. The auditor must also evaluate whether the significant assumptions used by management are reasonable. To do this the auditor must exercise professional judgement.

The IAPN also considers audit considerations for valuation in three specific circumstances: when management uses a third-party pricing source, when management estimates fair value using a model and when a management's expert is used.

Possible approaches to gathering evidence regarding information from third-party pricing sources may include the following:

- For Level 1 inputs, comparing the information from third-party pricing sources with observable market prices
- Reviewing disclosures provided by third-party pricing sources about their controls and processes, valuation techniques, inputs and assumptions
- Testing the controls management has in place to assess the reliability of information from third- party pricing sources
- Performing procedures at the third-party pricing source to understand and test the controls and processes, valuation techniques, inputs and assumptions used for asset classes or specific financial instruments of interest
- Evaluating whether the prices obtained from third-party pricing sources are reasonable in relation to prices from other third-party pricing sources, the entity's estimate or the auditor's own estimate
- Evaluating the reasonableness of valuation techniques, assumptions and inputs
- Developing a point estimate or a range for some financial instruments priced by the third-party pricing source and evaluating whether the results are within a reasonable range of each other
- Obtaining a service auditor's report that covers the controls over validation of the prices

When management estimates fair value using a model IAPN 1000 states that testing the model can be accomplished by two main approaches:

(a) The auditor can test management's model, by considering the appropriateness of the model used by management, the reasonableness of the assumptions and data used, and the mathematical accuracy.

(b) The auditor can develop their own estimate and then compare the auditor's valuation with that of the entity.

When a management expert is used the requirements which must be applied are the basic requirements of ISA 500 as discussed in earlier chapter. Procedures would include evaluating the competence, capabilities and objectivity of the management's expert, obtaining an understanding of their work and evaluating the appropriateness of that expert's work as audit evidence.

Presentation and disclosure

Audit procedures around presentation and disclosure are designed in consideration of the assertions of occurrence and rights and obligations, completeness, classification and understandability, and accuracy and valuation.

Professional skills focus: Concluding, recommending and communicating

As was mentioned in connection with derivatives in the previous chapter, the complexity of hedging makes clear and transparent communication essential.

Other relevant audit considerations

Written representations should be sought from management in accordance with ISA 540 (revised) and ISA 580, *Written Representations*.

11.1.3 Practice Note 23

Practice Note 23 *Special Considerations in Auditing Financial Instruments* was revised in July 2013. The revised Practice note is based on IAPN 1000 discussed above. Some additional points are however included as follows:

Section 1

- Operational risk includes model risk (the risk that imperfections and subjectivity of valuation models are not properly understood, accounted for or adjusted for) (PN23.18)
- Complete and accurate recording of financial instruments is an essential core objective (PN23.24- 1)
- When quoted prices are used as evidence of fair value, the source should be independent and where possible more than one quote (PN23.44-1)
- Where a price has been obtained from a pricing service and that price has been challenged, when considering whether the corrected price is a suitable basis for valuation, consideration should be given to how long the challenge process has taken and whether the underlying data remains valid (PN23.56-1)
- A key control over management's valuation process may be an independent price verification function which forms part of internal control (PN23.62-1)

Section 2

- Although it is not part of the auditor's role to determine the amount of risk an entity should take on, obtaining an understanding of the risk management process may identify risks of material misstatement (PN23.70-1)
- Assertions about valuation may be based on highly subjective assumptions, therefore evaluating audit evidence in respect of these requires considerable judgment (PN23.71-2)
- Determining materiality for financial instruments may be particularly difficult (PN23.73-1)
- When deciding which audits other than those of listed entities require an engagement quality control review, the existence of financial instruments may be a relevant factor (PN23.73-2)
- When obtaining an understanding of the entity's financial instruments, the auditor will consider the view of any correspondence with regulators in accordance with the FCA Code (PN23.76-2)
- The involvement of experts or specialists may be needed (PN23.79)
- The auditor may need to consider the control environment applicable to those responsible for functions dealing with financial instruments (PN23.89-2)
- Substantive procedures will include reviewing operational data such as reconciling differences (PN23.104)
- The auditor may use information included in a Prudent Valuation Return (prepared by UK banks and other regulated entities in the financial sector) to understand the uncertainties associated with the financial instruments used and disclosed by these entities (PN23.108-1)
- Tests of valuation include: verifying the external prices used to value financial instruments, confirming the validity of valuation models, and evaluating the overall results and reserving for residual uncertainties (PN23.113-1)
- The auditor must consider whether management has given proper consideration to the models used (PN23.134-1)
- When evaluating the amount of an adjustment that might be required, the auditor considers all factors taken into account in the valuation process and uses experience and judgment (PN23.137- 1)



Interactive question 14: Convertible debenture

On 1 January 20X8 Berriman plc issued a £10 million debenture at par. The debenture has a nominal rate of interest of 4% and is redeemable on 1 January 20Y3. On this date, the holder has the option to convert the debenture to 6 million £1 ordinary shares in Berriman plc. The financial statements currently show a long-term liability which represents the net proceeds of the debenture. The first payment of interest on 31 December 20X8 has also been recorded.

Requirements

14.1 Identify the issues surrounding this debenture.

14.2 List the audit procedures you would perform.

See **Answer** at the end of this chapter.



Context example: Royal Bank of Scotland

In the Royal Bank of Scotland (RBS)'s financial statements for the year ended 31 December 2014, Deloitte, RBS's statutory auditors, noted that the valuation of complex or illiquid financial

instruments was an area of audit risk which had merited specific audit focus. Their auditor's report described the risk, and the audit team's responses to the risk, as follows:

Risk

The valuation of the Group's financial instruments was a key area of focus of our audit given the degree of complexity involved in valuing some of the financial instruments and the significance of the judgements and estimates made by the directors. As set out in Note 11 of the consolidated financial statements, financial instruments held at fair value comprised assets of £534 billion and liabilities of £497 billion. In the Group's accounting policies, the directors have described the key sources of estimation involved in determining the valuation of financial instruments and in particular when the fair value is established using a valuation technique due to the instrument's complexity or due to the lack of availability of market-based data.

Our audit has focused on testing the valuation adjustments including those for credit risk, funding related and own credit. A particular area of focus of our audit has been in testing the valuation of the more illiquid financial instruments disclosed as Level 3 instruments which comprised assets of £5 billion and liabilities of £5 billion.

How the scope of our audit responded to the risk

We tested the design and operating effectiveness of the key controls in the Group's financial instrument valuation processes including the controls over data feeds and other inputs into valuation models and the controls over testing and approval of new models or changes to existing models.

Our audit work also included testing a sample of the underlying valuation models and the assumptions used in those models using a variety of techniques. This work included valuing a sample of financial instruments using independent models and source data and comparing the results to the Group's valuations and the investigation of any significant differences.

For instruments with significant, unobservable valuation inputs, we used our own internal valuation experts to assess and challenge the valuation assumptions used, including considering alternative valuation methodologies used by other market participants.

12 Auditing derivatives



Section overview

It is necessary for auditors to understand the process of derivative trading in order to audit derivatives successfully.

12.1 Auditing derivatives in the modern world

The key to using derivatives as part of an overall investment strategy is to have **adequate internal controls** in place and trained personnel handling the investments. Derivatives, which have been around for a very long time in one form or another, have been put to good use by transferring risk from one party, the hedger, to another, the speculator. There are many factors in today's world which can cause derivative investment strategies to go wrong. As we have seen, such factors can include the following:

- A lack of internal controls
- A *laissez-faire* management

- Greed
- Ineffective systems to identify and monitor risk
- Inexperience

An understanding of the business process involved in derivatives trading is necessary in order to audit derivatives successfully. The steps in a typical process are as follows:

(a) Entering the deal in the trader's deal sheet

(b) Trader types the deal into the system and sends an email

(c) The back office include the deal in reports

(d) Back office process the deal using market quote information from agencies

(e) Enter details into a 'pre-programmed' Excel sheet and/or other processing package

(f) Confirm deal with brokers/counterparties

(g) Carry out monthly settlement/processing

(h) Net off between Accounts Payable and Accounts Receivable and wire the payment as necessary

Each type of derivative will be different and non-standard derivatives will be **unique**. This poses challenges for the auditor.

Generally, however, the auditor should seek to:

(a) **understand the client's business** in order to establish the real role played by, and the risks that are inherent in, the derivatives activity;

(b) **document the system**. This would involve documenting various processes;

(c) **identify the controls** in each process in order to establish the risk passed to the client by inadequate or missing controls; and, therefore, to establish the audit risk and thus the audit work that needs to be performed;

(d) carry out the **appropriate control and substantive audit procedures**; and

(e) make **conclusions and report** on the outcome of the audit of derivatives.

Obviously the exact nature of what is to be done is dependent on the circumstances of the client. Ensuring that the information has been captured **completely and accurately** in each case is important.



Worked example: Systems and processes

Typically a large oil refining company in one country will obtain oil from different sources and refine it, producing various petroleum products for its customers.

Imagine the company is involved in trading in crude oil futures. The traders will take forward positions strategically with the information relating to future oil price movement available at the time. If the traders expect the price to rise, they will take a long position (buy the commodity forward) and if they expect it to fall they will take a short position (sell the commodity forward).

Requirement

Identify the key processes that the auditor would seek evidence for in connection with this.

Solution

Capture of information: The primary source document is the trader's deal sheet. This document should contain the date, time, oil index, quantity traded, position (long or short), nature of trade (hedge or speculation) and rationale for the trade.

Processing of information: The back office report should contain the same information as in the deal sheet.

Confirmation of information: There should be a statement from the clearing agents (since these are futures) confirming the details. (**Note:** Swaps transactions would be confirmed differently, via counterparty and broker confirmations and options are confirmed in the same way that futures are.)

Depositing of margin money: There should be evidence that margin money had been deposited with the exchange as required (in case the mark to market crosses the exchange's threshold limits).

Settlement: There will be clearing statements from clearing agents. These should be used in collaboration with internally generated information to confirm that the appropriate settlement amounts changed hands.

Accounting: The deals have been accounted for correctly.

In all these processes controls will have been implemented and the auditor should identify these and assess their utility.

Interactive question 15: Derivatives

You are the auditor in charge of the audit of Johannes plc, a UK company with the pound (£) as its functional currency.

On 1 January 20X7, Johannes plc (J) entered into a forward contract to purchase 40,000 barrels of crude oil at $70 per barrel on 1 January 20X9. J is not using this as a hedging instrument and is speculating that the price of oil will rise and plans to net settle the contract if the price rises. J does not pay anything to enter into this forward contract. At 31 December 20X7, the fair value of the forward contract has increased to £500,000. At 31 December 20X8, the fair value of the forward contract has declined to £400,000.

Requirements

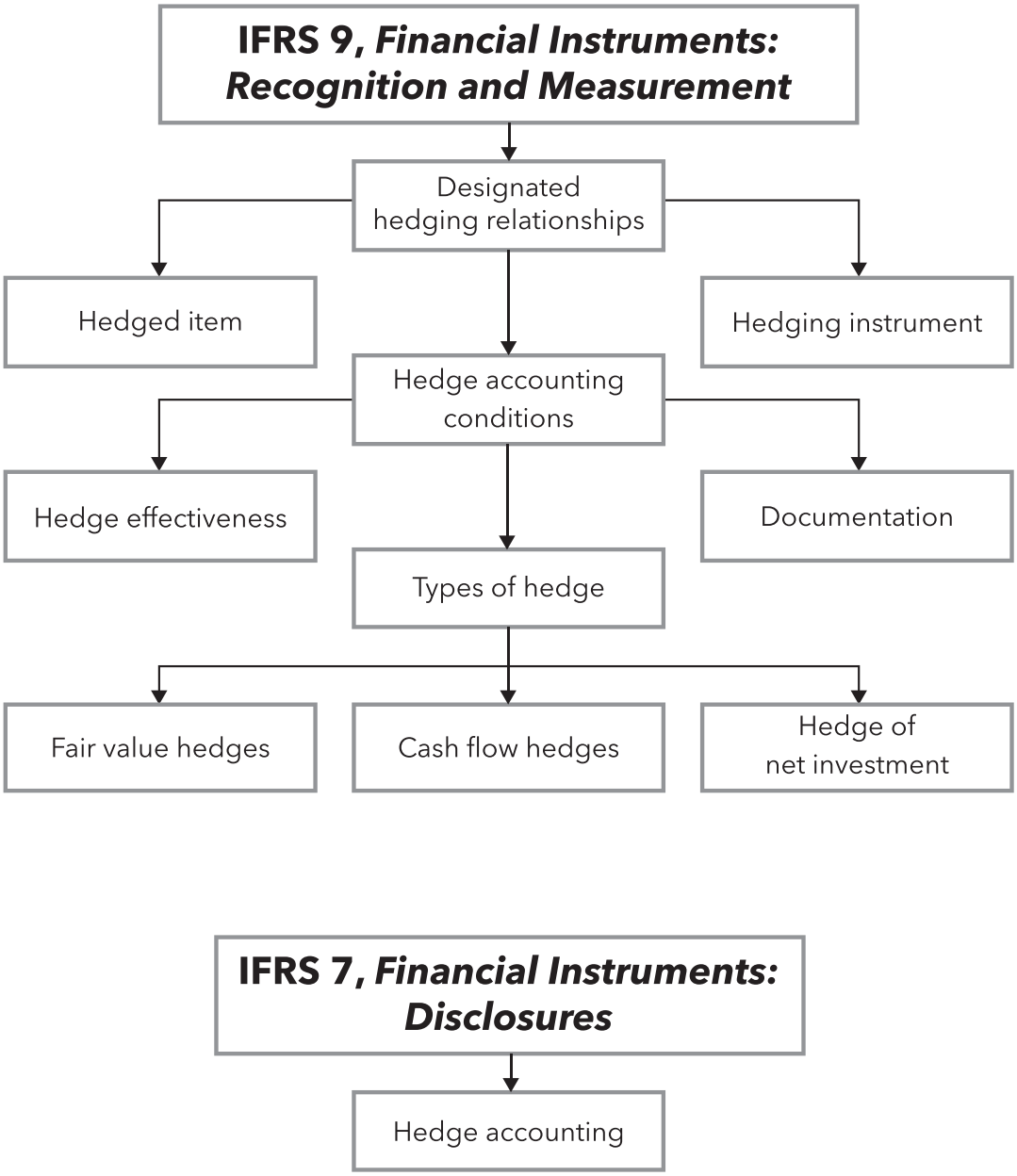
15.1 Identify the accounting entries you would expect to see at the inception of the contract, at 31 December 20X7, and at 31 December 20X8.

15.2 Identify the risks you would expect to find in this arrangement, and the audit procedures that you would carry out.

15.3 Outline the steps that you would take to ensure compliance with IFRS 7, *Financial Instruments: Disclosures*.

See **Answer** at the end of this chapter.

Summary


IFRS 9, Financial Instruments: Recognition and Measurement
Designated hedging relationships
Hedged item
Hedging instrument
Hedge accounting conditions
Hedge effectiveness
Documentation
Types of hedge
Fair value hedges
Cash flow hedges
Hedge of net investment
IFRS 7, Financial Instruments: Disclosures
Hedge accounting

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	What are the three types of hedge dealt with in IFRS 9, *Financial Instruments*? (Topic 1)
2.	What is a hedged item? (Topic 2)
3.	Which contracts can be designated as hedging instruments? (Topic 3)
4.	Do you understand the conditions that must be met for hedge accounting to be applied? (Topic 4)
5.	How should a hedge of a firm commitment be treated? (Topic 6)
6.	What are the main issues for the auditor when auditing derivatives? (Topic 7)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Hedging	This is a short, straightforward introduction to a complex topic.
JayGee	This question requires explanation rather than calculation, which could happen in an exam, particularly in the context of decision making for the future.
Anew	This is an integrated question, which revises derivatives, covered in an earlier chapter and tests the audit, trading and hedging aspects.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Kime (note 2 only)	This part of the question deals with hedging a trade receivable from a customer located abroad using a forward contract.
Johnson Telecom	This is a must-do full question on financial instruments, integrating FR and audit. Instruments covered are: investments in equity instruments, derivatives (with hedging), debt investments and a loan note.

Tydaway (forward contract with China only)	You are asked to consider four hedging scenarios relating to how the forward contract should be treated.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries.

Once you have attempted these questions, you can continue your studies by moving onto the next chapter.

Technical reference

1 Hedging relationships – IFRS 9

- Types of hedging relationships – **IFRS 9.6.1**
- Examples – **IFRS 9. IE**

2 Hedge accounting

- Definition – **IFRS 9 Appendix A**
- Conditions – **IFRS 9 6.4.1**

3 Hedging instruments

- Qualifying instruments – **IFRS 9 6.2.1**
- Written and purchased options – **IFRS 9 6.2.1**
- Non-qualifying instruments – **IFRS 9 6.4.1**
- Designations of hedging instruments – **IFRS 9 6.2.1**

4 Fair value hedges

- Definition – **IFRS 9 Appendix A**
- Recognition of gains or losses – **IFRS 9 6.5.1**
- Discontinuing fair value hedge accounting – **IFRS 9 6.5.1**

5 Cash flow hedges

- Definition – **IFRS 9 Appendix A**
- Recognition of gains or losses – **IFRS 9 6.5.1**
- Hedge of a forecast transaction – **IFRS 9 6.5.1**
- Discontinuing cash flow hedge accounting – **IFRS 9 6.5.1**

6 Hedges of a net investment – IFRS 9 6.5.1

7 Hedge effectiveness

- Criteria – **IFRS 9 6.4.1**
- Timing of assessment and methods of assessing effectiveness – **IFRS 9 6.4.1**

8 Auditing fair value measurements and disclosures

- Understanding the entity's process – **ISA 540.8**
- Evaluating reasonableness – **ISA 540.18, A116–A119**
- Audit procedures – **ISA 540.13**
- Written representations – **ISA 540.22**

9 Auditing financial instruments

- Professional scepticism – **IAPN 1000.71**
- Planning – **IAPN 1000.73–.84**
- Substantive procedures – **IAPN 1000.96**
- Valuation – **IAPN 1000.114, .118**

Self-test questions

Answer the following questions

1 Hedging

A company owns 100,000 barrels of crude oil which were purchased on 1 July 20X2 at a cost of $26.00 per barrel.

In order to hedge the fluctuation in the market value of the oil, the company signs a futures contract on the same date to deliver 100,000 barrels of oil on 31 March 20X3 at a futures price of $27.50 per barrel. The conditions for hedge accounting were met.

Due to unexpected increases in production, the market price of oil on 31 December 20X2 was $22.50 per barrel and the futures price for delivery on 31 March 20X3 was $23.25 per barrel at that date.

Requirement

Explain the impact of the transactions on the financial statements of the company for the year ended 31 December 20X2.

2 Columba

The Columba Company has hedged the cash flows relating to its interest rate risk by purchasing an interest rate cap. The conditions for hedge accounting were met.

Additional interest charges up to the end of the financial year amount to £17,000 while the fair value of the interest rate cap increased by £20,000.

Requirement

What amount relating to the interest rate cap should be recorded in profit or loss?

3 Pula

The Pula Company manufactures heavy engineering equipment which it sells in many countries throughout the world. The functional currency of Pula is the pound (£).

On 1 November 20X7 Pula entered into a contract with the Roadmans Company, whose functional currency is the N$, to sell a bulldozer for delivery on 1 April 20X8. The contract price is fixed in N$.

Also on 1 November 20X7, Pula entered into a foreign currency forward contract to hedge its future exposure to changes in the £:N$ exchange rate, arising from the contract with Roadmans.

The conditions for hedge accounting were met.

Requirement

What designations are available to Pula in respect of the hedging arrangement?

4 JayGee

On 1 August 20X1, JayGee entered into a non-cancellable purchase order to acquire equipment from Zenda Corporation, a Ruritanian entity, at 300 million rurits (Ru). Payment is

made upon delivery of the equipment to the UK on 31 December 20X1. On 30 September 20X1, JayGee entered into a forward contract to exchange Ru 300 million at a pre-determined exchange rate between the rurit and pound sterling on 31 December 20X1. The functional currency of JayGee is the pound sterling (£).

Requirement

Discuss the accounting implications for the purchase contract and forward contract if fair value hedge accounting is adopted.

5 Tried & Tested plc

You are a senior in the firm that acts as auditors and advisers to Tried & Tested plc (T&T). The finance director of T&T is proposing to restructure the company's loans to find cheaper sources of finance, because she thinks that the company's financial gearing is too high in relation to its uncertain cash inflows. The engagement partner has attended a meeting with the finance director and has obtained the following information.

T&T: Notes of meeting

- T&T has completed 2 years of a 10-year 8.5% fixed-rate mortgage on its premises amounting to £2.5 million, and it is on this loan that the finance director is looking to reduce interest costs.
- T&T's bank has indicated that it could swap the interest charges on its loan with another of its customers, LeytonPlus plc (LP), which wishes to fix its interest liabilities in view of uncertain interest rate prospects.
- LP has a floating rate loan of £3.8 million on which it currently pays SONIA + 2.0%.

The swap agreement

Because of its higher credit rating T&T has negotiated that it should pay LP at a rate of SONIA + 1.0% as its contribution to the swap. For its part LP would pay 8.8% fixed to T&T. The bank will charge T&T 0.04% per annum of the swap value for the service. The swap agreement would last for 5 years and would be for £2.5 million (assume the current SONIA to be 6%).

Other credit facilities

LP and T&T have additional loan facilities available to them as follows:

- LP can obtain fixed rate loans at 10% per annum for up to 20 years.
- T&T can obtain floating rate loans at SONIA + 1.0%.

You have been sent the following note by the engagement partner.

On the basis of my meeting notes please draft notes covering the following issues:

- The audit and assurance issues regarding the swap arrangement including an assessment of whether the proposed swap may be designated a hedge.
- The control structure you would expect to be in place to manage the risks associated with the swap.
- The finance director has had little experience of swap agreements, and requires confirmation of the net reduction in annual interest costs to T&T if the swap is agreed. Please produce this calculation.

Requirement

Respond to the partner's note.

6 Anew plc

Anew plc (Anew), a client of your firm, has recently established a treasury and investment division within its existing business. The division deals in derivatives, in addition to other treasury and investment instruments, for both trading and hedging purposes. Most of the company's derivative trading activities relate to sales, positioning and arbitrage. Sales activities involve offering products to customers and banks in order to enable them to transfer, modify or reduce current and future risks. Positioning involves managing market risk positions with the expectation of profiting from favourable movements in prices, rates or indices. Arbitrage involves profiting from price differentials between markets or products.

The company has adopted a comprehensive system for the measurement and management of risk. Part of the risk management process involves managing the company's exposure to fluctuations in foreign exchange and commission rates, to reduce exposure to currency and commission rate risks to acceptable levels as determined by the board of directors within the guidelines issued by the Central Bank. The company uses different types of derivatives, including swaps, forwards and futures, forward rate agreements, options and swaptions.

Requirements

6.1 In planning the audit of Anew, identify and explain five key risks that may arise from the derivatives trading activities that the newly formed division is involved in.

6.2 Identify and explain the general controls and application controls which you consider necessary for ensuring that these risks are controlled appropriately.

6.3 You have as one of the assistants on the audit Melanie, who has just completed the professional level examinations. She has asked you for a briefing note on "how to distinguish derivatives activity for trading purposes from derivatives activity for hedging purposes, and how these are accounted for and audited within the international accounting and auditing framework".

Requirement

In response to this request and considering that Melanie's main interest is in the audit of these instruments, draft a memorandum to Melanie providing her with the advice she requires, clarifying any ambiguous phrases in her request.

Your memorandum should be structured under the following headings.

- Derivative instruments
- Use of derivatives for trading purposes
- Use of derivatives for hedging purposes
- The audit of derivative instruments

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 **Explanation**

This forward contract is a **derivative**. It is a **financial liability** because it is **unfavourable** at the year end.

Under the forward contract, BCL has to pay £3.125 million ($2m ÷ 0.64).

At the year end, an equivalent contract would only have cost £2.857 million ($2m ÷ 0.7).

Therefore, the contract is standing at a loss of £0.268 million (£3.125m – £2.857m) at the year end. This is why it is a financial liability.

Normally derivatives are treated as being **held for trading**, so this contract will be treated as a **financial liability at fair value through profit or loss**. Journal entry:

DEBIT	Profit or loss	£0.268m	
CREDIT	Financial liability		£0.268m

Being the recognition of the liability and loss on forward contract

1.2

(a) **Recording the gain or loss**

If the forward contract is to be treated as a hedging instrument, it should still be measured at its fair value of £0.268 million but the loss should be recognised in **other comprehensive income** instead of profit or loss.

Why different treatment is necessary

The reason for hedging is to try to **offset the gain/loss on the hedged item** with the **corresponding loss/gain on the hedging instrument**.

With a **cash flow hedge**, the hedged item is often a future or forecast transaction that has not yet been recorded in the financial statements. If the normal accounting treatment was applied, the loss on this hedging instrument would be recognised in profit or loss in one period and the gain on the hedged item would be recognised in profit or loss in a later period, so the offsetting effect would not be reflected.

When the gain on the hedged item occurs and is recognised in profit or loss, the loss on the forward contract should be **reclassified** from other comprehensive income to profit or loss. This matches the gain and loss and better reflects the offsetting that was the purpose of the transaction.

(b) **Extract from statement of profit or loss and other comprehensive income for the year ended 31 August 20X0**

	£m
Profit for the year	1.000
Other comprehensive income	
Loss on forward contract	(0.268)
Total comprehensive income	0.732

Answer to Interactive question 2

(a) **Initial recognition**

DEBIT	Investment in equity instruments	£112,560	
CREDIT	Bank		£112,560

Being the initial recognition of investment in equity instruments at fair value, including transaction costs (W1).

Measurement at 31 July 20X3

DEBIT	Investment in equity instruments	£5,840	
CREDIT	Other comprehensive income		£5,840

Being the gain on remeasurement of the investment in equity instruments (W2). (The gain will be recognised in other components of equity.)

WORKINGS

(1) **Fair value March 20X3**

	£
Fair value (40,000 shares @ £2.68)	107,200
Commission (5% × 107,200)	5,360
	112,560

(2) **Gain to 31 July 20X3**

	£
Fair value (40,000 shares @ £2.96)	118,400
Previous value	(112,560)
	5,840

(b) Discussion as follows:

Fair value hedge

The entity has elected under IFRS 9 to recognise gains and losses on investments in equity instruments in **other comprehensive income** and the gain or loss on a derivative is recognised in **profit or loss**.

However, assuming that the derivative meets the criteria to be treated as a hedging instrument, it would be treated as a **fair value hedge.** This means that:

- the gain or loss on the investment in equity instruments (the 'hedged item') would be taken to **other comprehensive income**; and
- this would be offset by the corresponding loss or gain on the derivative, also in other comprehensive income (IFRS 9.5.8).

This treatment is a fair reflection of the economic substance of the hedging arrangement, where the intention is that the changes in value of the derivative will cancel out the changes in value of the hedged item.

Answer to Interactive question 3

The credit default swap (CDS) is recognised as a derivative at fair value through profit or loss. IFRS 9 allows fair value option for a proportion of the loan commitment. If this option is elected, then £500,000 of the loan commitment is accounted for at fair value through profit or loss and, as a result, provides an offset to the fair value through profit or loss on the CDS.

Answer to Interactive question 4

(a) The futures contract was intended to protect the company from a fall in oil prices (which would have reduced the profit when the oil was eventually sold). However, oil prices have actually risen, so that the company has made a loss on the contract.

Without hedge accounting

The futures contract is a derivative and therefore should be remeasured to fair value under IFRS 9. The loss on the futures contract should be recognised in profit or loss:

DEBIT	Profit or loss (40,000 × [£24 – £22])	£80,000	
CREDIT	Financial liability		£80,000

(b) **With hedge accounting**

The loss on the futures contract should be recognised in profit or loss, as before. There is an increase in the fair value of the inventories:

	£
Fair value at 31 December 20X3 (40,000 × £22.25)	890,000
Fair value at 1 December 20X3 = cost	(800,000)
Gain	90,000

The gain should also be recognised in profit or loss and adjusted against the carrying amount of the inventories:

DEBIT	Inventory	£90,000	
CREDIT	Profit or loss		£90,000

The net effect on profit or loss is a gain of £10,000 compared with a loss of £80,000 without hedging.

Answer to Interactive question 5

The hedge is fully effective, as the gain on the hedging instrument is less than the loss on the cash flows. The total gain of £8,800 is therefore recognised in other comprehensive income. The double entry is:

DEBIT	Hedging instrument (SOFP)	£8,800	
CREDIT	Other comprehensive income		£8,800

Answer to Interactive question 6

No. A cash flow hedge is defined as a hedge of the exposure to variability in cash flows attributable to a particular risk. In this case, the issued debt instrument does not give rise to any exposure to volatility in cash flows since the interest is calculated at a fixed rate.

The entity may designate the swap as a fair value hedge of the debt instrument, but it cannot designate the swap as a cash flow hedge of the future cash outflows of the debt instrument.

Answer to Interactive question 7

No. The FRA does not qualify as a cash flow hedge of the cash flow relating to the fixed rate assets because they do not have a cash flow exposure.

The entity could, however, designate the FRA as a hedge of the fair value exposure that exists before the cash flows are remitted.

Answer to Interactive question 8

8.1 IFRS 9 allows both of these two methods.

If the hedge is treated as a fair value hedge, the gain or loss on the fair value remeasurement of the hedging instrument and the gain or loss on the fair value remeasurement of the hedged item for the hedged risk should be recognised immediately in profit or loss.

8.2 If the hedge is treated as a cash flow hedge, the portion of the gain or loss on remeasuring the forward contract that is an effective hedge should be recognised in other comprehensive income. The amount should be reclassified in profit or loss in the same period or periods during which the hedged item (the liability) affects profit or loss ie, when the liability is remeasured for changes in foreign exchange rates. Therefore, if the hedge is effective, the gain or loss on the derivative is released to profit or loss in the same periods during which the liability is measured, not when the payment occurs.

Answer to Interactive question 9

The entity does have this choice.

If the entity designates the foreign exchange contract as a fair value hedge, the gain or loss from remeasuring the forward exchange contract at fair value is recognised immediately in profit or loss, and the gain or loss on remeasuring the receivable is also recognised in profit or loss.

If the entity designates the foreign exchange contract as a cash flow hedge of the foreign currency risk associated with the collection of the receivable, the portion of the gain or loss that is determined to be an effective hedge should be recognised in other comprehensive income, and the ineffective portion in profit or loss. The amount held in equity should be reclassified to profit or loss in the same period or periods during which changes in the measurement of the receivable affect profit or loss.

Answer to Interactive question 10

Given that RapidMart is hedging the volatility of the future cash outflow to purchase fuel, the forward contract is accounted for as a cash flow hedge, assuming all the criteria for hedge accounting are met (ie, the hedging relationship consists of eligible items, designation and documentation at inception as a cash flow hedge, and the hedge effectiveness criteria are met).

At inception, no entries are required as the fair value of a forward contract at inception is zero. However, the existence of the hedge is disclosed under IFRS 7, *Financial Instruments: Disclosures*.

At the year end, the forward contract must be valued at its fair value of £0.12 million as follows. The gain is recognised in other comprehensive income (items that may subsequently be reclassified to profit or loss) in the current year as the hedged cash flow has not yet occurred. This will be reclassified to profit or loss in the next accounting period when the cost of the diesel purchase is recognised.

WORKING

	£m
Market price of forward contract for delivery on 31 December (1m × £2.16)	2.16
RapidMart's forward price (1m × £2.04)	(2.04)
Cumulative gain	0.12

The gain is recognised in other comprehensive income as the cash flow has not yet occurred:

DEBIT	Forward contract (financial asset in SOFP)	£0.12m
CREDIT	Other comprehensive income	£0.12m

Answer to Interactive question 11

Foreign currency receivables

	£'000
Receivable originally recorded (30,240/5.6)	5,400
Receivable at year end (30,240/5.4)	5,600
Exchange gain	200

		£'000	£'000
DEBIT	Trade receivables	200	
CREDIT	Profit or loss (other income)		200

Forward contract

This is a cash flow hedge:		£'000	£'000
DEBIT	Other comprehensive income	200	
DEBIT	Finance cost (forward points)	20	
CREDIT	Financial liability		220

As the change in cash flow affects profit or loss in the current period, a reclassification adjustment is required (£'000):

		£'000	£'000
DEBIT	Profit or loss	200	
CREDIT	Other comprehensive income		200

Answer to Interactive question 12

Cash flow hedge

Value of contract:	£'000
Price at 31 December 20X2 (3,000 × 1,400)	4,200
Price at 1 November 20X2 (3,000 × 1,440)	(4,320)
Loss	(120)

		£'000	£'000
DEBIT	Other comprehensive income	120	
CREDIT	Financial liability		120

The tax treatment follows the treatment in the IFRS Standard. However, the current tax credit has not yet been recorded. This is credited to other comprehensive income rather than profit or loss, as the loss itself on the contract is recognised in other comprehensive income (IAS 12.61A):

		£'000	£'000
DEBIT	Current tax liability (SOFP) (120 × 30%)	36	
CREDIT	Income tax credit (OCI)		36

Answer to Interactive question 13

The purpose of the forward contract is to hedge the fair value of the recognised receivable due to fluctuations in exchange rates. It is therefore a fair value hedge.

Providing relevant documentation has been set up, which appears to be the case here, hedge accounting rules can be used provided that:

- the hedge is expected to be highly effective at achieving offsetting changes in fair value;
- the effectiveness of the hedge can be reliably measured; and
- the hedge is assessed to actually have been highly effective.

The foreign currency receivable will initially be recognised at the spot rate at the date of the transaction ie, at USD1,614,205 (GBP 1 million/0.6195).

At 31 December 20X1, the receivable is restated in accordance with IAS 21 to USD1,554,002 (GBP 1 million/0.6435). A loss of USD60,203 ($1,614,205 – $1,554,002) is therefore recognised in profit or loss.

The forward contract is recognised in the financial statements at 31 October 20X1. However, no double entries are recorded, as the value of a forward contract at inception is zero.

However, recognition of the hedge will trigger disclosure under IFRS 7 as follows:

- A description of the hedge
- A description of the forward contract designated as a hedging instrument
- The nature of the risk being hedged (ie, change in exchange rates affecting the fair value of the receivable)
- Gains and losses on the hedging instrument and the hedged item

At 31 December 20X1, the change in fair value of the forward contract is recognised in profit or loss as this is a fair value hedge:

	$
[(–1mGBP/0.6440 + (1mGBP/0.6202)) × (1/1.00325$^{1/12}$)]	59,572
At 31 October 20X1 (zero at inception)	(0)
Change in fair value of forward contract (gain)	59,572

The company has designated changes in the spot element of the forward contract as the hedge. The change in the spot element is:

	$
[(–1mGBP/0.6435 + (1mGBP/0.6195)) × (1/1.00325$^{1/12}$)]	60,187
At 31 October 20X1 (zero at inception)	(0)
Change in fair value of spot element of forward contract (gain)	60,187

Effectiveness of the hedge is calculated as:

(Cumulative change in fair value of spot element of hedging instrument/Cumulative change in fair value of hedged item)

= ($60,187/($60,203*) = 99.97% (or 100.03% if measured the other way around)

* If the effect of discounting short-term receivables to obtain a more precise fair value is taken into account, this could be measured at $60,187 giving effectiveness of exactly 100%.

The hedge is measurable and effective. Therefore hedge accounting can be used, assuming the hedge is expected to be highly effective until 31 January 20X2.

The interest element (which arises due to different interest rates between the currencies of the forward contract) is excluded from the hedging relationship and recognised as a finance cost:

		$	$
DEBIT	Forward contract	59,572	
DEBIT	Finance costs (P/L) (60,187 – 59,572)	615	
CREDIT	Profit or loss		60,187

Profit or loss:	$
Loss on foreign currency receivable	(60,203)
Gain on hedging instrument	60,187
Finance costs	(615)

Statement of financial position

Current assets	$
Trade receivables (1,614,205 – 60,203)	1,554,002
Forward contract hedging instrument	59,572

Answer to Interactive question 14

14.1 **Issues**

The treatment of the debenture does not appear to comply with accounting standards. It should be treated as a hybrid instrument, split into its equity and liability components. Normally the liability component should be calculated as the discounted present value of the cash flows of the debenture, discounted at the market rate of interest for a comparable borrowinWg with no conversion rights. The remainder of the proceeds represents the fair value of the right to convert and this element should be reclassified as equity.

14.2 **Procedures**

- Obtain a copy of the debenture deed and agree the nominal interest rate and conversion terms

- Assuming the revised treatment is adopted, review schedule calculating the fair value of the liability at the date of issue. Confirm that an appropriate discount rate has been used (ie, market rate of interest for a comparable borrowing with no conversion rights)
- Agree initial proceeds and interest payment to cash book and bank statement
- Review adequacy of disclosures in accordance with accounting standards

Answer to Interactive question 15

15.1 There are no accounting entries at the inception of the forward contract.

On 31 December 20X7, there is an increase in derivative asset (increase in fair value of forward contract) of £500,000 and this is reflected in profit or loss as a gain.

On 31 December 20X8, there is a decrease in derivative asset of £100,000 and this is reflected as a loss in profit or loss for the year.

15.2 As illustrated in part 1, one of the risks is that the fair value of the asset will go down. This is referred to as market risk – a risk relating to the adverse changes in the fair value of the derivative; in this case the forward contract. This is a very real risk for J.

There is foreign exchange risk. This is the risk that J's earnings will be affected as a result of fluctuations in currency exchange rates. J's functional currency is GBP but crude oil prices are quoted in the USD. The movement in the £/$ exchange rate will affect J's earnings arising from this contract.

There is credit risk – the risk that the counterparty will not settle the obligation at full value.

There is the related settlement risk – the risk that settlement will take place without J receiving value from the counterparty.

Solvency risk is the risk that J will not have the funds to settle when the payment for the barrels becomes due. This may be related to the market risk described above.

There is also interest rate risk. This is the risk that J will suffer loss as a result of fluctuations in the value of the forward contract due to changes in market interest rates. If the movement in interest rates is such that the price of crude goes down then J will be affected adversely.

As auditor, I would need to do the following:

- Assess the audit risk and design audit procedures to ensure that risk is reduced to an acceptable level.
- Understand J's accounting and internal control system to enable me to assess whether it is adequate to deal with forward contracts of this type specifically, but also with any type of derivatives J carries out, generally. I would need to assess the control environment to ensure that it is strong enough and that J has clear control objectives in place. Control objectives would include authorised execution of the deal, checking completeness and accuracy of the information, prevention and detection of errors, appropriate accounting for changes in the value of the derivative (the forward contract), and general ongoing monitoring.
- Confirm that appropriate reconciliations are carried out and that there are appropriate controls around the reconciliations. The reconciliations would include:
  - the one between the dealer's deal sheet and the back office records used for the ongoing monitoring process; the one between the clearing and bank accounts and the broker statements to ensure that all outstanding items are identified and promptly cleared; and the one between J and the appropriate brokers and agents.
- Verify that data security procedures are adequate to ensure recovery in the case of disaster by reference to either IT staff or via observation of suitable back-up procedures.

- I would carry out procedures to ensure that the amounts recorded at the year ends (31 December 20X7 and 31 December 20X8) are appropriate. These would include:
  - inspecting the agreement for the forward contract and the supporting documentation to ensure that the agreement occurred (at 31 December 20X7 only) and confirming that the situation has not changed subsequently;
  - inspecting documentation for evidence of the purchase price (at 31 December 20X7 only); and
  - obtaining evidence collaborating the fair value of the forward contract; for example quoted market prices.

15.3 I would confirm that the following IFRS 7 disclosures have been made.

- The accounting policy for financial instruments including forwards, especially how fair value is measured.
- Net gains to be recorded in profit or loss (£500,000 for year ending 31 December 20X7) and net losses (£100,000 for year ending 31 December 20X8).
- The fair value of the asset category which includes the forward contract. The disclosure should be such that it permits the information to be compared with the corresponding carrying amount.
- The nature and extent of risks arising from financial instruments, including forward contracts. The disclosures should be both qualitative and quantitative.

Answers to Self-test questions

1 Hedging

The futures contract was entered into to protect the company from a fall in oil prices and hedge the value of the inventories. It is therefore a fair value hedge.

The inventories should be recorded at their cost of $2,600,000 (100,000 barrels at $26) on 1 July 20X2.

The futures contract has a zero value at the date it is entered into, so no entry is made in the financial statements.

Note: However, the existence of the contract and associated risk would be disclosed from that date in accordance with IFRS 7.

At the year end the inventories should be measured at the lower of cost and net realisable value. Hence they should be measured at $2,250,000 (100,000 barrels at $22.50) and a loss of $350,000 recognised in profit or loss.

However, a gain has been made on the futures contract:

	$
The company has a contract to sell 100,000 barrels on 31 March 20X3 at $27.50	2,750,000
A contract entered into at the year end would sell these barrels at $23.25 on 31 March 20X3	2,325,000
Gain (= the value the contract could be sold on for to a third party)	425,000

The gain on the futures contract should also be recognised in profit or loss:

DEBIT	Future contract asset	$425,000	
CREDIT	Profit or loss		$425,000

The net effect on profit or loss is a gain of $75,000 ($425,000 less $350,000), whereas without the hedging contract there would have been a loss of $350,000.

Note: If the fair value of the inventories had increased, the carrying amount of the inventories should have been increased by the same amount and this gain also recognised in profit or loss (normally gains on inventories are not recognised until the goods are sold). A loss would have occurred on the futures contract, which should also have been recognised in profit or loss.

2 Columba

A gain of £3,000 should be recognised in profit or loss.

The ineffective portion of the gain or loss on the hedging instrument should be recognised in profit or loss. In a cash flow hedge the amount to be recognised in other comprehensive income is the lower of:

- the cumulative gain/loss on the hedging instrument ie, £20,000; and
- the cumulative change in fair value of the hedged item ie, £17,000.

So £17,000. This leaves £3,000 of the increase in the fair value of the cap to be recognised in profit or loss.

3 Pula

The hedging relationship may be designated **either** a fair value hedge **or** a cash flow hedge.

The contract to sell the bulldozer represents a firm commitment with Roadmans, not merely a proposed transaction, and it is expressed in a currency other than Pula's functional currency. A hedge of the foreign currency risk of a firm commitment may be accounted for as a fair value hedge or as a cash flow hedge.

4 JayGee

If JayGee designates the hedge as a fair value hedge, the non-cancellable purchase order in rurits is considered as a firm commitment to be hedged (hedged item) in connection with the spot foreign currency risk.

The rurit forward contract is considered to be the hedging instrument.

As a financial derivative, the rurit forward contract will have been reported at fair value on each reporting date, with gains or losses reported in profit or loss.

Under a fair value hedge, the change in fair value of the firm commitment related to the hedged risk will also be recognised in profit or loss and adjusts the carrying amount of the hedged item. This applies if the hedged item is otherwise measured at cost. For JayGee's hedged item, which is an unrecognised firm commitment, its cumulative change in the fair value attributable to the hedged risk is recognised as an asset or liability.

5 Tried & Tested plc

Audit and assurance issues Assessment of proposed swap

- The hedging relationship qualifies for hedge accounting only if all the following conditions are met:
  - (a) At the inception of the hedge there is formal designation and documentation of the hedging relationship, and the entity's risk management objective and strategy for undertaking the hedge. That documentation shall include identification of the hedging instrument, the hedged item or transaction, the nature of the risk being hedged and how the entity will assess the hedging instrument's effectiveness in offsetting the exposure to changes in the hedged item's fair value or cash flows attributable to the hedged risk.
  - (b) The hedge is expected to be highly effective in achieving offsetting changes in fair value or cash flows attributable to the hedged risk, consistently with the originally documented risk management strategy for that particular hedging relationship.
  - (c) For cash flow hedges, a forecast transaction that is the subject of the hedge must be highly probable and must present an exposure to variations in cash flows that could ultimately affect profit or loss.
  - (d) The effectiveness of the hedge can be reliably measured ie, the fair value or cash flows of the hedged item that are attributable to the hedged risk and the fair value of the hedging instrument can be reliably measured.
  - (e) The hedge is assessed on an ongoing basis and determined actually to have been highly effective throughout the financial reporting periods for which the hedge was designated.

Other audit and assurance issues

- This is a relatively complex transaction which will increase audit risk, particularly as the finance director has had little experience of this type of transaction before.

- T&T is exposed to interest rate risk if the swap goes ahead. T&T has changed its fixed interest commitments to variable commitments. It is possible that interest rates will rise, particularly as interest rates are said to be uncertain.
- Since T&T has a better credit rating, it is possible that it may face risk of default from LP on the loan payments under the swap agreement.
- The finance director is concerned about the level of gearing within the company. While steps are being taken to deal with the situation, this issue will need to be considered as part of the going concern review.
- The extent to which the bank will be expecting to place reliance on financial information as part of the process of agreeing the swap and any specific assurances which we might be required to provide.

6 Anew plc

6.1 Reduction in annual interest costs

Variable rate interest liabilities and income for T&T

	%
Interest on fixed loan currently paid	8.5
Received from LP under swap agreement	(8.8)
Difference	(0.3)
Payment by T&T to LP under swap agreement (SONIA + 1)	6 + 1.0
Total variable payments by T&T	6.7

This is a 0.3% reduction on its SONIA facilities.

	£	£
T&T currently pays – fixed (8.5% × £2.5m)		212,500
Will pay		
Interest (6.7% × £2.5m)	167,500	
Bank charge (0.04% × £2.5m)	1,000	
		(168,500)
Net annual saving		44,000

6.2 Necessary general controls and application controls

Note: This answer assumes that a computer system is used in processing trades involving derivatives.

General controls

A number of general controls may be relevant in this case, for example the following:

- For credit risk, general controls may include ensuring that off-market derivative contracts are only entered into with counterparties from a specific list and establishing credit limits for all customers.
- For legal risk, a general control may be to ensure that all transactions are reviewed by properly qualified lawyers and regulation specialists.
- For market risk, a general control may be to set strict investment acceptance criteria and ensure that these are adhered to.
- For settlement risk, a general control may be to set up a third party through whom settlement takes place, ensuring that the third party is instructed not to give value until value has been received.

- For solvency (liquidity) risk, general controls may include having diversified funding sources, managing assets with liquidity in mind, monitoring liquidity positions, and maintaining a healthy cash and cash equivalents balance.

Application controls

These include the following:

- A computer application may identify the credit risk. In this case, an appropriate control may be monitoring credit exposure, limiting transactions with an identified counterparty and stopping any further risk-increasing transactions with that counterparty.
- For legal risk, an application control may be the system insisting that it will not process a transaction/trade until an authorised person has signed into the system to give the authority. Such an authorised person may be different depending on the nature and type of transaction. In some cases it may be the company specialist solicitor; yet in other cases it may just be the dealer's supervisor.
- For market risk, an application control may carry out mark to market activity frequently and the production of timely exception management reports.
- For settlement risk, an application control may be a computer settlement system refusing to release funds/assets until the counterparty's value has been received, or an authorised person has confirmed to the system that there is evidence that value will be received.
- For solvency risk, an application control may be that the system will produce a report for management informing management that there needs to be a specific amount of funds available on a given date to settle the trades coming in for settlement on that date.

6.3 **Memorandum**

To Melanie

From Jane Chadge

Date 12 November 20X7

Subject Briefing note on all manner of things derivative

Thank you for your recent email asking me to explain how to distinguish derivatives activity for trading purposes from derivatives activity for hedging purposes. In the same email you asked me to explain how these instruments are accounted for and audited within the international accounting and auditing framework. In this briefing note, I am assuming that your reference to international accounting and auditing framework is to IFRS Standards/IAS Standards (International Financial Reporting Standards/International Accounting Standards) and ISAs (International Standards on Auditing UK) for Accounting and Auditing respectively. I am also assuming that '… **for trading purposes** 'refers to engagement in derivatives activity for speculative purposes.

Derivative instruments

- These are financial contracts between two parties where payments are dependent on movements in price in one or more underlying financial instruments, reference rates or indices.
- They are financial instruments or other contracts within the scope of relevant accounting standards. IAS 32, IFRS 9 and IFRS 7 deal with the accounting and disclosure requirements for derivatives.
- IFRS 9 requires that derivatives be measured at fair value in the statement of financial position unless they are linked to, and must be settled by, an investment in an unquoted equity instrument that cannot be reliably measured at fair value.
- In general, derivatives can be used either for speculation (trading) or for hedging (offsetting risk). How the derivative is accounted for and disclosed will depend on whether it is for speculative or hedging purposes.

- Generally, a derivative instrument can be any instrument that has the three characteristics stated in IFRS 9. Derivative instruments include swaps, options, swaptions, forwards and futures. A derivative instrument can be embedded in another (non-derivative) instrument; for example, a company issuing a bond whose interest is linked to the USD price of crude oil, such that the interest payments increase and decrease with this price.

Use of derivatives for trading purposes

- Speculators may trade with other speculators as well as with hedgers. In most financial derivatives markets, the value of speculative trading is far higher than the value of true hedge trading.
- As well as outright speculation, derivatives traders may look for arbitrage opportunities between different derivatives on identical or closely related underlying securities.
- Derivatives such as options, futures or swaps generally offer the greatest possible reward for betting on whether the price of an underlying asset will go up or down. For example, a person may believe that an oil company may find more oil reserves in the next year. If the person bought the stock (share) for £10, and it went to £20 after the discovery was announced, the person would have made a profit and have a return on his investment.
- Other uses of derivatives are to gain an economic exposure to an underlying security in situations where direct ownership of the underlying security is too costly or is prohibited by legal or regulatory restrictions, or to create a synthetic short position. In addition to directional plays (ie, simply betting on the direction of the underlying security), speculators can use derivatives to place bets on the volatility of the underlying security. This technique is commonly used when speculating with traded options.

The audit of derivative instruments

- For many entities, the use of derivatives has reduced exposure to changes in exchange rates, interest rates and commodity prices and other risks.
- The inherent characteristics of derivatives usually result in increased financial risk, in turn increasing audit risk and presenting the auditor with new challenges.
- To audit derivatives adequately, the auditor needs to:
  - have specialist skills and knowledge. In the case of Anew the auditor will need to understand the operating characteristics and risk profile of the industry in which Anew operates, the characteristics of the specific derivative used, Anew's derivatives information system, how Anew values derivatives and reporting requirements of IFRS 9 and IFRS 7;
  - know the business including industrial sector issues, political, economic, sociocultural and technological factors, the strengths and weaknesses of Anew (eg, experience of management and those charged with governance), and the reasons for using specific derivatives (in this case both speculative and risk management);
  - identify the key risks (as already identified);
  - assess the risk and the internal controls in accordance with ISA (UK) 315 (Revised July 2020), *Identifying and Assessing the Risks of Material Misstatement*;
  - identify internal controls (and test those controls identified to check if they are 'fit for purpose';
  - carry out substantive procedures to ensure that key assertions, as identified in ISA 315, are met. These key assertions relate to existence and occurrence, rights and obligations, completeness, accuracy, valuation and allocation, and presentation. These would have to be considered for all material derivatives; and
  - handle non-standard derivatives according to the circumstances ensuring that control and substantive tests have been carried out as appropriate.

I trust that the above points have provided you with the information you require. Please do not hesitate to seek further clarification on any of these points.

Chapter 18

Employee benefits

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Objectives and scope of IAS 19, Employee Benefits

2 Short-term employee benefits

3 Post-employment benefits overview

4 Defined contribution plans

5 Defined benefit plans – recognition and measurement

6 Defined benefit plans – other matters

7 Defined benefit plans – disclosure

8 Other long-term employee benefits

9 Termination benefits

10 IAS 26, Accounting and Reporting by Retirement Benefit Plans

11 Audit focus

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Explain how different methods of providing remuneration for employees may impact upon reported performance and position
- Explain and appraise accounting standards that relate to employee remuneration which include different forms of short-term and long-term employee compensation; retirement benefits; and share-based payment
- Justify and conclude for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Question
1	**Objectives and scope of IAS 19, *Employee Benefits*** This very short section introduces the standard and states that it is structured as follows: • Short-term employee benefits • Post- employment benefits • Other long-term employee benefits • Termination benefits	**Approach** Employee benefits is one of two chapters addressing employee remuneration. It is a new and complex topic to which you will need to devote a significant amount of time. Section 1 gives a summary of the types of benefits that the standard addresses. **Stop and think** Share-based payments are excluded because they are the subject of a separate standard.	Although IAS 19 is frequently examined, you will not be tested just on this introductory material.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Question
2	**Short-term employee benefits** Employee benefit costs can be a very significant proportion of total expenses. Accounting for short-term benefits presents few, if any, problems.	**Approach** The accounting for short-term employee benefits is relatively straightforward. The application of the accruals concept in relation to liabilities means that a short-term benefit should be recognised as an employee provides the services to the entity on which the benefits are payable. **Stop and think** Consider what short-term benefits are offered in your workplace.	The most common short-term benefits that have been tested are bonuses and holiday pay. Both could be tested in an integrated way. In the case of bonuses, ethical issues could arise.	N/A
3	**Post-employment benefits overview** This section introduces the concept of defined contribution (money purchase) and defined benefit (final salary) plans.	**Approach** Read carefully, using the diagrams in sections 3.1 and 3.2 to aid your understanding. **Stop and think** Why are defined benefit plans more risky than defined contribution plans?	This overview is unlikely to be tested directly in an exam – it is there for background only.	N/A
4	**Defined contribution plans** You will need to be able to apply the requirements of IAS 19 to the relatively straightforward defined contribution plans.	**Approach** The key points are: Recognise contributions payable as an expense in the period in which the employee provides services Recognise a liability where contributions remain unpaid at the period end. Study the worked example carefully. **Stop and think**	Defined contribution plans have been tested occasionally but you are more likely to be asked about defined benefit plans.	am

Topic	Practical significance	Study approach	Exam approach	Interactive Question
		Why are defined contribution plans becoming more common?		
5	**Defined benefit plans – recognition and measurement** The cost of long-term and post-employment benefits, which include pension plans, medical benefits and life insurance, is difficult to estimate. The entity must rely on the expertise of actuaries.	**Approach** Make sure you understand the main issue in 5.1 – the plan may be in surplus or deficit and the contributions will vary accordingly. Paragraphs 5.4 (movement in defined benefit plan) and 5.5 (four- step method) are the most important. **Stop and think** What discount rate should be used?	IAS 19 is new at Advanced Level and is examined very frequently. The movement on the asset or liability is often tested in a scenario where the company has failed to do this properly and treated the plan like a defined contribution plan.	N/A
6	**Defined benefit plans – other matters** This section deals with past service costs, curtailments, settlements and the asset ceiling test.	**Approach** Past service costs arise from plan amendments, for example increasing the benefits payable. They usually increase the obligation, but may reduce it. Remeasure the obligation first. Learn the suggested approach in Section 6.5 – this will cover most situations in the exam. **Stop and think** What is the asset ceiling?	Curtailments and past service costs have been examined regularly, sometimes interacting with deferred tax.	**IQ1: Defined benefit 1** This is a comprehensive question covering the movement on the asset or liability over three years with a settlement and an additional payment. You should do all four interactive questions in this section.
7	**Defined benefit plans – disclosure** The main disclosures include a description of the plan, a reconciliation of the fair value of plan assets from the	**Approach** Read through this section, but remember you can always look it up in the open book, should it be examined.	This topic has yet to be examined.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Question
	opening to closing position, the actual return on plan assets, a reconciliation of movements in the present value of the defined benefit obligation during the period, an analysis of the total expense recognized in profit or loss, and the principal actuarial assumptions made.	**Stop and think** Why do we have to disclose key assumptions and methods used in the accounting for defined benefit pension plans?		
8	**Other long-term employee benefits** Examples include paid sabbaticals and long-term sick leave.	**Approach** Read through quickly. **Stop and think** Where are actuarial gains and losses recognised?	This area has not yet been examined.	N/A
9	**Termination benefits** These are payments to encourage voluntary redundancy or payments before the normal retirement date.	**Approach** Read through quickly. **Stop and think** Recognise a termination benefit when there is a firm commitment to end the employment.	This has briefly been tested in the context of audit. Disclosures could also be tested.	N/A
10	**IAS 26, *Accounting and Reporting by Retirement Benefit Plans*** Retirement benefit plans must also prepare financial statements.	**Approach** Skim through quickly – this is examinable only at Level D. **Stop and think** What are net assets available for benefits?	This standard has not yet been examined.	**IQ5: Scope** This is a quick multiple choice question which should not take long.
11	**Audit focus** Retirement benefits are a challenge for auditors because of the degree of judgement and estimate involved, and the consequent risk.	**Approach** Do not neglect the audit focus section – pensions are a popular topic and as likely to come up in an integrated question as in a single silo FR	The audit of defined benefit schemes has been tested regularly, partly in the context of risk.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Question
		question. Learn the table in section 11.1. **Stop and think** The auditor must evaluate the appropriateness of fair value measurements, particularly in relation to plan assets.		

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Objectives and scope of IAS 19, *Employee Benefits*

Section overview

IAS 19 considers the following employee benefits:

- Short-term employee benefits
- Post-employment benefits
- Other long-term employee benefits
- Termination benefits

IAS 19, *Employee Benefits* should be applied by all entities in accounting for the provision of all employee benefits, except those benefits which are equity based and to which IFRS 2, *Share-based Payment* applies. The standard applies regardless of whether the benefits have been provided as part of a formal contract or an informal arrangement.

Employee benefits are all forms of consideration, for example cash bonuses, retirement benefits and private health care, given to an employee by an entity in exchange for the employee's services.

A number of accounting issues arise due to:

- the valuation problems linked to some forms of employee benefits; and
- the timing of benefits, which may not always be provided in the same period as the one in which the employee's services are provided.

IAS 19 is structured by considering the following employee benefits:

- Short-term employee benefits; such as wages, salaries, bonuses and paid holidays
- Post-employment benefits; such as pensions and post-retirement health cover
- Other long-term employee benefits; such as sabbatical and long-service leave
- Termination benefits; such as redundancy and severance pay

2 Short-term employee benefits

Section overview

Short-term employee benefits are those that fall due within 12 months from the end of the period in which the employees provide their services. The required accounting treatment is to recognise the benefits to be paid in exchange for the employee's services in the period on an accruals basis.

2.1 All short-term benefits

Definition

Short-term employee benefits: Employee benefits (other than termination benefits) that fall due within 12 months from the end of the period in which the employees provide their services.

Short-term employee benefits include the following:

(a) Wages, salaries and social security contributions

(b) Short-term absences where the employee continues to be paid, for example paid annual vacation, paid sick leave and paid maternity/paternity leave. To fall within the definition, the absences should be expected to occur within 12 months of the end of the period in which the employee services were provided

(c) Profit sharing and bonuses payable within 12 months of the end of the period

(d) Non-monetary benefits, for example private medical care, company cars and housing

The application of the accruals concept in relation to liabilities means that a short-term benefit should be recognised as an employee provides his services to the entity on which the benefits are payable. The benefit will normally be treated as an expense, and a liability should be recognised for any unpaid balance at the year end.

2.2 Short-term compensated absences



Definition

Short-term compensated absences: Compensated absences are periods of absence from work for which the employee receives some form of payment and which are expected to occur within 12 months of the end of the period in which the employee renders the services.

Examples of short-term compensated absences are paid annual vacation and paid sick leave.

Short-term compensated absences fall into two categories:

- **Accumulating absences**. These are benefits, such as paid annual vacation, that accrue over an employee's period of service and can potentially be carried forward and used in future periods; and
- **Non-accumulating absences**. These are benefits that an employee is entitled to, but are not normally capable of being carried forward to the following period if they are unused during the period, for example paid sick leave, maternity leave and compensated absences for jury service.

The cost of providing compensation for accumulating absences should be recognised as an expense as the employee provides the services on which the entitlement to such benefits accrues. Where an employee has an unused entitlement at the end of the reporting period and the entity expects to provide the benefit, a liability should be created.

The cost of providing compensation for non-accumulating absences should be expensed as the absences occur.



Worked example: Paid vacation

An entity has five employees and each is entitled to 20 days' paid vacation per year, at a rate of £50 per day. Unused vacation is carried forward to the following year.

At the year end, four of the employees have used their full holiday entitlement; the remaining one has four days' holiday to carry forward.

All five employees work for the entity throughout the year and are therefore entitled to their 20 days of vacation.

Requirement

How should the expense be recognised in the financial statements?

Solution

An expense should be recognised as part of staff costs for:

5 employees × 20 days × £50 = £5,000

Four of the employees use their complete entitlement for the year and the other, having used 16 days, is permitted to carry forward the remaining four days to the following period. A liability will be recognised at the period end for:

1 employee × 4 days × £50 = £200

2.3 Profit sharing and bonus plans

An entity should recognise an expense and a corresponding liability for the cost of providing profit- sharing arrangements and bonus payments when:

(a) the entity has a present legal or constructive obligation. The legal obligation arises when payment is part of an employee's employment contract. The constructive obligation arises where past performance has led to the expectation that benefits will be payable in the current period; and

(b) a reliable estimate of the obligation can be made.



Worked example: Bonus plan

An entity has a contractual agreement to pay a total of 4% of its net profit each year as a bonus. The bonus is divided between the employees who are with the entity at its year end. The following data is relevant:

Net profit	£120,000
Average employees	5
Employees at start of year	6
Employees at end of year	4

Requirement

How should the expense be recognised?

Solution

An expense should be recognised for the year in which the profits were made and therefore the employees' services were provided, for:

£120,000 × 4% = £4,800

Each of the four employees remaining with the entity at the year end is entitled to £1,200. A liability of £4,800 should be recognised if the bonuses remain unpaid at the year end.

Conditions may be attached to such bonus payments; commonly, the employee must still be in the entity's employment when the bonus becomes payable. An estimate should be made based on the expectation of the level of bonuses that will ultimately be paid. IAS 19 sets out that a reliable estimate for bonus or profit-sharing arrangements can be made only when:

- there are formal terms setting out determination of the amount of the benefit;
- the amount payable is determined by the entity before the financial statements are authorised for issue; or
- past practice provides clear evidence of the amount of a constructive obligation.

Worked example: Annual bonus

An entity with a 30 June year end has a past practice of paying an annual bonus to employees, although it has no contractual obligation to do so. Its practice is to appropriate 4% of its pre-tax profits, before charging the bonus, to a bonus pool and pay it to those employees who remain in employment on the following 30 September.

The total bonus is allocated to employees in proportion to their 30 June salaries, and amounts due to those leaving over the next three months are retrieved from the bonus pool for the benefit of the entity.

Past experience is that employees with salaries representing 8% of annual salaries leave employment by 30 September. The entity's pre-tax profits for the year ended 30 June 20X5 were £4 million.

Requirement

How should the bonus be recognised in the financial statements?

Solution

The bonus to be recognised as an expense in the year ended 30 June 20X5 is:

£4m × 4% × (100 – 8)% = £147,200

3 Post-employment benefits overview

Section overview

Post-employment benefits are employee benefits which are payable after the completion of employment.

These can be in the form of either of the following:

- Defined contribution schemes where the future pension depends on the value of the fund.
- Defined benefit schemes where the future pension depends on the final salary and years worked.

Definition

Post-employment benefits: Employee benefits (other than termination benefits) which are payable after the completion of employment. The benefit plans may have been set up under formal or informal arrangements.

Post-employment benefits include retirement benefits such as:

- pensions
- continued private medical care
- post-employment life assurance

There are two main types of post-employment benefit schemes:

- **Defined contribution schemes (money purchase schemes)**
- **Defined benefit schemes (final salary schemes)**

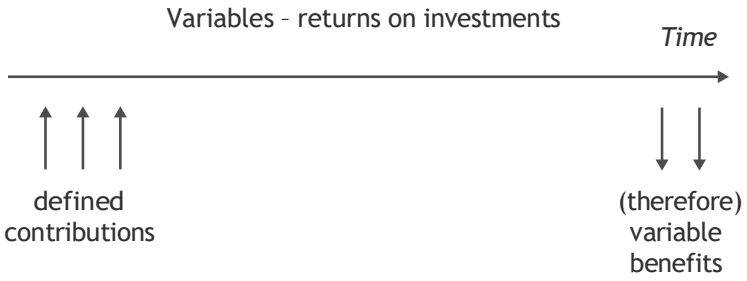
These two alternative schemes are discussed in more detail below.

A pension scheme will normally be held in the form of a trust separate from the sponsoring employer. Although the directors of the sponsoring company may also be trustees of the pension scheme, the sponsoring company and the pension scheme are separate legal entities that are accounted for separately. IAS 19 covers accounting for the pension scheme in the sponsoring company's accounts.

3.1 Defined contribution plans

Characteristics of a defined contribution plan are as follows:

- Contributions into the plan are fixed, normally at a percentage of an employee's salary.
- The amount of pension paid to retirees is not guaranteed and will depend on the size of the plan, which in turn depends on the performance of the pension fund investments.



Risk associated with defined contribution schemes

Contributions are usually paid into the plan by both the employer and the employee. The expectation is that the investments made will grow through capital appreciation and the reinvestment of returns and that, on a member's retirement, the plan should have grown to be sufficient to provide the anticipated benefits.

If the investments have not performed as anticipated, the size of the plan will be smaller than initially anticipated and therefore there will be insufficient assets to meet the expected benefits. This insufficiency of assets is described as the **investment risk** and is carried by the employee.

The other main risk with retirement plans is that a given amount of annual benefit will cost more than expected if, for example, life expectancy has increased markedly by the time benefits come to be drawn; this is described as the **actuarial risk** and, in the case of defined contribution plans, this is also carried by the employee.

Definitions

Investment risk: This is the risk that, due to poor investment performance, there will be insufficient funds in the plan to meet the expected benefits.

Actuarial risk: This is the risk that the actuarial assumptions such as those on employee turnover, life expectancy or future salaries vary significantly from what actually happens.

3.2 Defined benefit plans

Professional skills focus: Applying judgement

In the case of pension plans, particularly defined benefit plans, accountants and auditors need to rely on the judgement of actuaries, whose calculations may lie outside their field of

expertise. In turn, the actuaries themselves may rely on assumptions from external parties about the future direction of the economy and other matters.

These are defined by IAS 19 as all plans other than defined contribution plans. Characteristics of a defined benefit plan are as follows:

- The amount of pension paid to retirees is defined by reference to factors such as length of service and salary levels (ie, it is guaranteed).
- Contributions into the plan are therefore variable depending on how the plan is performing in relation to the expected future obligation (ie, if there is a shortfall, contributions will increase and vice versa).

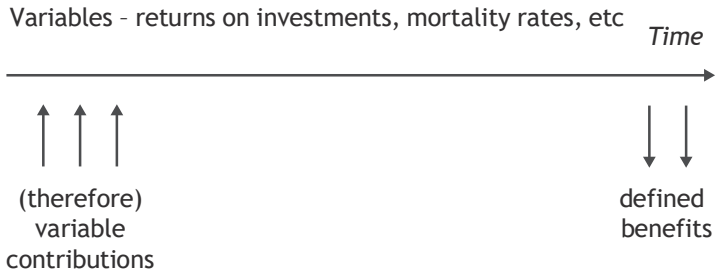


Figure 18.1: Defined benefit plans

Contribution levels

The actuary advises the company on contributions necessary to produce the defined benefits ('the funding plan'). It cannot be certain in advance that contributions plus returns on investments will equal benefits to be paid.

Formal actuarial valuations will be performed periodically (eg, every three years) to reveal any surplus or deficit on the scheme at a given date. Contributions may be varied as a result; for example, the actuary may recommend a contribution holiday (a period during which no contributions are made) to eliminate a surplus.

Risk associated with defined benefit schemes

As the employer is obliged to make up any shortfall in the plan, it is effectively underwriting the **investment** and **actuarial risk** associated with the plan. Thus in a defined benefit plan, the employer carries both the investment and the actuarial risk.

3.2.1 Types of defined benefit plan

There are two types of defined benefit plan:

(a) **Funded plans**

These plans are set up as separate legal entities and are managed independently, often by trustees. Contributions paid by the employer and employee are paid into the separate legal entity. The assets held within the separate legal entities are effectively ring-fenced for the payments of benefits.

Funded plans, illustrated diagrammatically below, represent the most common arrangement.

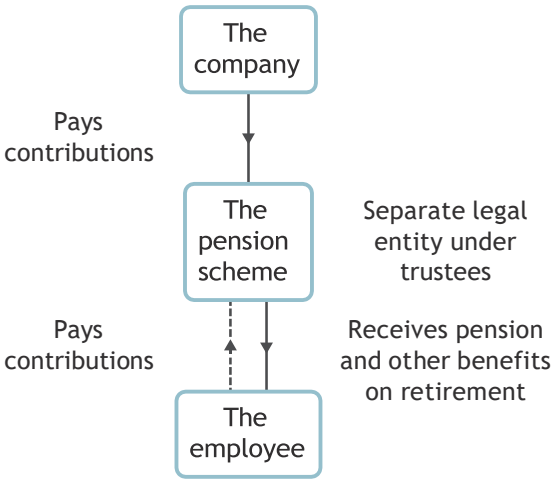


Figure 18.2: Funded plans

(b) **Unfunded plans**

These plans are held within employer legal entities and are managed by the employers' management teams. Assets may be allocated towards the satisfaction of retirement benefit obligations, although these assets are not ring-fenced for the payment of benefits and remain the assets of the employer entity. In the UK and the US, unfunded plans are common in the public sector but rare in the private sector. However, unfunded plans are the normal method of pension provision in many European countries (eg, Germany and France) and also in Japan.

3.2.2 Plans with promised returns on contributions

IAS 19 gives a number of examples of plans that would be deemed to be defined benefit plans even though on the face of it these may appear to be defined contribution. Examples include circumstances where an entity's obligation is not limited to an agreed level of contributions through either a legal or a constructive obligation ie, through an entity's past practices. Examples include:

- where there is a guaranteed level of return on contributions made or on the assets of the plan; in practical terms this means that the employee benefits from the upside potential on the investment but has a level of protection from downside risk;
- where a plan's level of benefits is not linked solely to the amount of contributions made into the plan; or
- where informal practices have led to the entity having a constructive obligation to provide additional benefits under a plan. A past practice of increasing benefits over and above the level due from the plan, to protect the retired person against inflation for example, would create a constructive obligation, even if the entity has no legal requirement to increase benefits.



Context example: Defined contribution or defined benefit?

Scenario 1

Entity ABC has a separately constituted retirement benefit plan for its employees which sets out that both ABC and its employees contribute 7% of annual salaries into the plan; contributions in respect of an individual employee create a right to a specified proportion of the plan assets, which on retirement is then used to buy the employee an annuity.

This is a defined contribution plan, because there appears to be no obligation on the part of ABC, other than to pay its annual 7% contribution.

Scenario 2

Entity DEF has a separately constituted retirement benefit plan for its employees; the plan is the same as the ABC plan, set out above, except that DEF has a contractual obligation to top up the plan assets if the return (calculated according to the rules) on these assets in any year is below 5%.

This is a defined benefit plan, because DEF has provided a guarantee over and above its obligation to make contributions.

Scenario 3

Entity GHI has a separately constituted retirement benefit plan for its employees; the plan is the same as the ABC plan, set out above. For some years GHI has made additional payments directly to retired ex-employees if the increase in the general price index exceeds 7% in any year. Such payments are at the discretion of GHI.

This is a defined benefit plan, because over and above its obligation to make contributions GHI has a past practice of increasing benefits in payment over and above the level due from the plan. This creates a constructive obligation that the entity will continue to do so.

4 Defined contribution plans



Section overview

Accounting for defined contribution plans is straightforward, as the obligation is determined by the amount paid into the plan in each period.

4.1 Recognition and measurement

Contributions into a defined contribution plan by an employer are made in return for services provided by an employee during the period. The employer has no further obligation for the value of the assets of the plan or the benefits payable.

- The entity should recognise contributions payable as an expense in the period in which the employee provides services (except to the extent that labour costs may be included within the cost of assets).
- A liability should be recognised where contributions arise in relation to an employee's service, but remain unpaid at the period end.
  In the unusual situation where contributions are not payable during the period (or within 12 months of the end of the period) in which the employee provides his or her services on which they accrue, the amount recognised should be discounted to reflect the time value of money.
- Any excess contributions paid should be recognised as an asset (prepaid expenses) but only to the extent that the prepayment will lead to a reduction in future payments or a cash refund.



Worked example: Defined contribution plan

Mouse Co agrees to contribute 5% of employees' total remuneration into a post-employment plan each period.

In the year ended 31 December 20X9, the company paid total salaries of £10.5 million. A bonus of £3 million based on the income for the period was paid to the employees in March 20Y0. The company had paid £510,000 into the plan by 31 December 20X9.

Requirement

Calculate the total expense for post-employment benefits for the year and the accrual which will appear in the statement of financial position at 31 December 20X9.

Solution

Total expense

		£
Salaries		10,500,000
Bonus		3,000,000
		13,500,000 × 5% = £675,000

		£	£
DEBIT	Staff costs expenses	675,000	
CREDIT	Cash		510,000
CREDIT	Accruals		165,000

4.2 Disclosure requirements

Where an entity operates a defined contribution plan during the period, it should disclose:

- the amount that has been recognised as an expense during the period in relation to the plan; and
- a description of the plan.

5 Defined benefit plans – recognition and measurement

Section overview

The accounting treatment for defined benefit plans is more complex than that applied to defined contribution plans:

- The value of the pension plan is recognised in the sponsoring employer's statement of financial position.
- Movements in the value of the pension plan are broken down into constituent parts and accounted for separately.

5.1 The problem

As we have seen, contributions to defined benefit schemes will vary depending on whether the actuary assesses the value of the plan to be adequate to meet future obligations.

In some instances there will be a shortfall, in which case the actuary will advise increased contributions. In other instances there may be a surplus, in which case the actuary may recommend a contributions holiday. Contributions will therefore vary substantially from year to year.

For this reason, it is inappropriate to apply the accounting treatment for defined contribution schemes and expense contributions through profit or loss.

5.2 Introduction to accounting for defined benefit plans

IAS 19 instead requires that the defined benefit plan is recognised in the sponsoring entity's statement of financial position as either a liability or asset depending on whether the plan is in deficit or surplus.

The value of the pension plan is calculated in its simplest form as:

	£
Present value of the defined benefit obligation at the reporting date	X
Fair value of plan assets at the reporting date	(X)
Plan deficit/surplus	X/(X)

5.2.1 Present value of the defined benefit obligation



Definition

Defined benefit obligation: The defined benefit obligation is the present value of all expected future payments required to settle the obligation resulting from employee service in the current and prior periods.

Expected future payments

Expected future payments are based on a number of assumptions and estimates, such as:

- the final benefits payable under the plan (often dependent on future salaries, as benefits are often quoted as a percentage of the employee's final salary); and
- the number of members who will draw benefits (this will in turn depend on employee turnover and mortality rates).

Discounting to present value

- Once determined, the expected future benefits should be discounted to present value (including those which may become payable within 12 months) using a discount rate determined by reference to:
  - market yields on high-quality fixed-rate corporate bonds at the reporting date, or where there is no market in such bonds; and
  - market yields on government bonds.

The corporate or government bonds should be denominated in the same currency as the defined benefit obligation, and be for a similar term.

Note: The examples of discount rates used in this chapter are merely to illustrate relevant calculations and may therefore be rather higher than would currently be found in practice.

Performance of valuations

IAS 19 encourages the use of a qualified actuary to measure the defined benefit obligation. However, this is not a requirement.

Frequency of valuations

Valuations are not required at each reporting date; however, they should be carried out sufficiently regularly to ensure that amounts recognised are not materially different from those which would be recognised if they were valued at the reporting date.

5.2.2 Fair value of plan assets



Definition

Plan assets: Those assets held by a long-term benefit fund and those insurance policies which are held by an entity, where the fund/entity is legally separate from the employer and assets/policies can only be used to fund employee benefits.

Investments owned by the employer which have been earmarked for employee benefits but which the employer could use for different purposes are not plan assets.

Fair value is the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date (IFRS 13).

Guidance on fair value is given in IFRS 13, *Fair Value Measurement*. Under IFRS 13, fair value is a market-based measurement, not an entity-specific measurement. It focuses on assets and liabilities and on exit (selling) prices. It also takes into account market conditions at the measurement date.

IFRS 13 states that valuation techniques must be those which are appropriate and for which sufficient data are available. Entities should maximise the use of relevant **observable inputs** and minimise the use of **unobservable inputs**.

5.3 Actuarial assumptions

Actuarial assumptions are needed **to estimate the size of the future (post-employment) benefits** that will be payable under a defined benefit plan. The main categories of actuarial assumptions are as follows.

(a) **Demographic assumptions** are about mortality rates before and after retirement, the rate of employee turnover, early retirement, claim rates under medical plans for former employees, and so on.

(b) **Financial assumptions** include future salary levels (allowing for seniority and promotion as well as inflation) and the future rate of increase in medical costs (not just inflationary cost rises, but also cost rises specific to medical treatments and to medical treatments required given the expectations of longer average life expectancy).

The standard requires actuarial assumptions to be neither too cautious nor too imprudent: they should be **'unbiased'**. They should also be based on **'market expectations'** at the year end, over the period during which the obligations will be settled.

5.4 Accounting for the movement in defined benefit plans

Both the present value of the defined benefit obligation and the fair value of plan assets, and therefore the overall plan surplus or deficit, will change from year to year. This movement is broken down into its constituent parts and each is accounted for separately.

The opening and closing obligation and plan assets can be reconciled as follows:

	PV of defined benefit obligation	FV of plan assets
	£	£
B/f at start of year (advised by actuary)	(X)	X
Retirement benefits paid out	X	(X)
Contributions paid into plan		X

	benefit obligation	FV of plan assets
	£	£
Interest on plan assets		X
Interest cost on obligation	(X)	
Current service cost	(X)	
	(X)	X
Gains/losses on remeasurement (balancing figure)	X/(X)	X/(X)
C/f at end of year (advised by actuary)	(X)	X

Note that while the interest on plan assets and interest on obligation are calculated separately, they are presented net and the same rate is used for both.

5.5 Outline of the method

There is a **four-step method** for recognising and measuring the expenses and liability of a defined benefit pension plan.

An outline of the method used by an employer to account for the expenses and obligation of a defined benefit plan is given below. The stages will be explained in more detail later.

Step 1

(a) An **actuarial technique** (the **projected unit credit method**) should be used to make a reliable estimate of the amount of future benefits employees have earned from service in relation to the current and prior years. The entity must determine how much benefit should be attributed to service performed by employees in the current period, and in prior periods. Assumptions include, for example, levels of employee turnover, mortality rates and future increases in salaries (if these will affect the eventual size of future benefits such as pension payments).

(b) The benefit should be **discounted** to arrive at the present value of the defined benefit obligation and the current service cost.

(c) The **fair value** of any **plan assets** should be deducted from the present value of the defined benefit obligation.

Step 2 The surplus or deficit measured in Step 1 may have to be adjusted if a net benefit asset has to be restricted by the **asset ceiling** (see section 6.2).

Step 3 Determine the amounts to be recognised in profit or loss:

(a) Current service cost

(b) Any past service cost and gain or loss on settlement

(c) Net interest on the net defined benefit liability (asset)

Step 4 Determine the remeasurements of the net defined benefit liability (asset), to be recognised in other comprehensive income (items that will not be reclassified to profit or loss):

(a) Actuarial gains and losses

(b) Return on plan assets (excluding amounts included in net interest on the net defined benefit liability (asset))

(c) Any change in the effect of the asset ceiling (excluding amounts included in net interest on the net defined benefit liability (asset))

5.5.1 Retirement benefits paid out

During an accounting year, some of the plan assets will be paid out to retirees, thus discharging part of the benefit obligation. This is accounted for by:

DEBIT	PV of defined benefit obligation	X	
CREDIT	FV of plan assets		X

Note that there is no cash entry, as the pension plan itself rather than the sponsoring employer pays the money out.

5.5.2 Contributions paid into plan

Contributions will be made into the plan as advised by the actuary. This is accounted for by:

DEBIT	FV of plan assets	X	
CREDIT	Cash		X

5.5.3 Return on plan assets



Definition

Return on plan assets: Defined as interest, dividends and other revenue derived from plan assets together with realised and unrealised gains or losses on the plan assets, less any costs of administering the plan and less any tax payable by the plan itself.

Accounting for the return on plan assets is explained in more detail below.

5.6 The statement of financial position

In the statement of financial position, the amount recognised as a **defined benefit liability** (which may be a negative amount ie, an asset) should be the following.

(a) The **present value of the defined obligation** at the year end; **minus**

(b) The **fair value of the assets of the plan** as at the year end (if there are any) out of which the future obligations to current and past employees will be directly settled.

The earlier parts of this section have looked at the recognition and measurement of the defined benefit obligation. Now we will look at issues relating to the assets held in the plan.

5.7 Plan assets

Plan assets are:

(a) assets such as stocks and shares, held by a fund that is legally separate from the reporting entity, which exists solely to pay employee benefits; and

(b) insurance policies, issued by an insurer that is not a related party, the proceeds of which can only be used to pay employee benefits.

Investments which may be used for purposes other than to pay employee benefits are not plan assets.

The standard requires that the plan assets are measured at fair value, as 'the price that would be received to sell an asset in an orderly transaction between market participants at the measurement date'. This is consistent with IFRS 13, *Fair Value Measurement*.

IAS 19 includes the following **specific requirements**:

(a) The plan assets should exclude any contributions due from the employer but not yet paid.

(b) Plan assets are reduced by any liabilities of the fund that do not relate to employee benefits, such as trade and other payables.

5.8 The statement of profit or loss and other comprehensive income

All the gains and losses that affect the plan obligation and plan assets must be recognised. The **components of defined benefit cost must be recognised as follows** in the statement of profit or loss and other comprehensive income:

Component	Recognised in
(1) **Service cost**	Profit or loss
(2) **Net interest on the net defined benefit liability**	Profit or loss
(3) **Remeasurements of the net defined benefit liability**	Other comprehensive income (not reclassified to profit or loss)

5.9 Service costs

These comprise the following:

(a) **Current service cost**; this is the increase in the present value of the defined benefit obligation resulting from employee services during the period. The measurement and recognition of this cost was introduced in section 5.5.

(b) **Past service cost**; this is the change in the obligation relating to service in **prior periods**. This results from amendments or curtailments to the pension plan, and

(c) Any **gain or loss on settlement**.

The detail relating to points (b) and (c) above will be covered in a later section. First, we will continue with the basic elements of accounting for defined benefit pension costs.

5.10 Net interest on the net defined benefit liability (asset)

In section 5.5 we looked at the recognition and measurement of the defined benefit obligation. This figure is the discounted **present value** of the future benefits payable. Every year the discount must be 'unwound', increasing the present value of the obligation as time passes through an interest charge.

5.10.1 Interest calculation

IAS 19 requires that the interest should be calculated on the **net defined benefit liability (asset)**. This means that the amount recognised in profit or loss is the net of the interest charge on the obligation and the interest income recognised on the assets.

The **net defined benefit liability/(asset)** should be measured as at the **start** of the accounting period, taking account of changes during the period as a result of contributions paid into the scheme and benefits paid out.

Many exam questions include the assumption that all payments into and out of the scheme take place at the end of the year, so that the interest calculations can be based on the opening balances.

5.10.2 Discount rate

The **discount rate** adopted should be determined by reference to **market yields on high-quality fixed-rate corporate bonds**. In the absence of a 'deep' market in such bonds, the yields

on comparable government bonds should be used as reference instead. The maturity of the corporate bonds that are used to determine a discount rate should have a term to maturity that is consistent with the expected maturity of the post-employment benefit obligations, although a single weighted average discount rate is sufficient.

Professional skills focus: Structuring problems and solutions

While the discount rate will generally be given to you in the exam, in practice it changes, and this can have an impact on the net present value. In the July 2021 exam candidates were asked to consider how a change in discount rate for a pension liability would affect the net present value. Excel file can be used to try out different discount rates and see their effect on the NPV.

Worked example: Interest cost

In 20X8, an employee leaves a company after working there for 24 years. The employee chooses to leave their accrued benefits in the pension scheme until they retire in seven years' time (they now work for another company).

At the time of their departure, the actuary calculates that it is necessary at that date to have a fund of £296,000 to pay the expected pensions to the ex-employee when they retire.

At the start of the year, the yield on high quality corporate debt was 8%, and remained the same throughout the year and the following year.

Requirement

Calculate the interest cost to be debited to profit or loss in Years 1 and 2.

Solution

Interest cost to be debited to profit or loss	**£**
Year 1:	
Discounted cost b/f	296,000
Interest cost (profit or loss) (8% × £296,000)	23,680
Obligation c/f (statement of financial position)	319,680
Year 2:	
Interest cost (profit or loss) (8% × £319,680)	25,574
Obligation c/f (statement of financial position)	345,254

5.11 Remeasurements of the net defined benefit liability

Remeasurements of the net defined benefit liability/(asset) comprise the following:

(a) Actuarial gains and losses

(b) The return on plan assets (excluding amounts included in net interest on the net defined benefit liability/(asset))

(c) Any change in the effect of the asset ceiling (excluding amounts included in net interest on the net defined benefit liability/(asset))

The gains and losses relating to points (a) and (b) above will arise in every defined benefit scheme so we will look at these in this section. The asset ceiling is a complication that is not relevant in every case, so it is dealt with separately, later in the chapter.

5.11.1 Actuarial gains and losses

Actuarial gains and losses arise for several reasons, and IAS 19 requires these to be recognised in full in other comprehensive income.

At the end of each accounting period, a new valuation, using updated assumptions, should be carried out on the obligation. Actuarial gains or losses arise because of the following.

- Actual events (eg, employee turnover, salary increases) differ from the actuarial assumptions that were made to estimate the defined benefit obligations
- The effect of changes to assumptions concerning benefit payment options
- Estimates are revised (eg, different assumptions are made about future employee turnover, salary rises, mortality rates, and so on)
- The effect of changes to the discount rate

Actuarial gains and losses are recognised in **other comprehensive income**. They are **not reclassified to profit or loss** under the 2011 revision to IAS 1 (see Chapter 9).

5.11.2 Return on plan assets

The return on plan assets must be calculated.

A new valuation of the plan assets is carried out at each period end, using current fair values. Any difference between the new value, and what has been recognised up to that date (normally the opening balance, interest, and any cash payments into or out of the plan) is treated as a 'remeasurement' and recognised in other comprehensive income.

Note: In the examples in this chapter, it is assumed that cash from contributions is received and pensions paid out at the end of the year, as no interest arises on it. In practice, it is more likely that contributions would be paid in part way through the year, and pensions paid out part way through the year or evenly over the year.



Worked example: Remeasurement of the net defined benefit liability

At 1 January 20X2 the fair value of the assets of a defined benefit plan was £1,100,000 and the present value of the defined benefit obligation was £1,250,000. On 31 December 20X2, the plan received contributions from the employer of £490,000 and paid out benefits of £190,000.

The current service cost for the year was £360,000 and a discount rate of 6% is to be applied to the net liability/(asset).

After these transactions, the fair value of the plan's assets at 31 December 20X2 was £1.5 million. The present value of the defined benefit obligation was £1,553,600.

Requirement

Calculate the gains or losses on remeasurement through other comprehensive income (OCI) and the return on plan assets and illustrate how this pension plan will be treated in the statement of profit or loss and other comprehensive income and statement of financial position for the year ended 31 December 20X2.

Solution

It is always useful to set up a working reconciling the assets and obligation:

	Assets	Obligation
	£	£
Fair value/present value at 1.1.X2	1,100,000	1,250,000
Interest (1,100,000 × 6%)/(1,250,000 × 6%)	66,000	75,000
Current service cost		360,000
Contributions received	490,000	
Benefits paid	(190,000)	(190,000)
Return on plan assets excluding amounts in net interest (balancing figure) (OCI)	34,000	–
Loss on remeasurement (balancing figure) (OCI)		58,600
	1,500,000	1,553,600

The following accounting treatment is required.

(1) In the **statement of profit or loss and other comprehensive income**, the following amounts will be recognised.

	£
In **profit or loss**:	
Current service cost	360,000
Net interest on net defined benefit liability (75,000 – 66,000)	9,000
In **other comprehensive income** (34,000 – 58,600)	24,600

(2) In the **statement of financial position**, the net defined benefit liability of £53,600 (1,553,600 – 1,500,000) will be recognised.

5.12 Section summary

The recognition and measurement of defined benefit plan costs are complex issues.

- Learn and understand the definitions of the various elements of a defined benefit pension plan
- Learn the outline of the method of accounting (see section 5.5)
- Learn the recognition method for the:
  - statement of financial position
  - statement of profit or loss and other comprehensive income

6 Defined benefit plans – other matters



Section overview

We have now covered the basics of accounting for defined benefit plans. This section looks at the special circumstances of:

- past service costs
- curtailments
- settlements
- asset ceiling test

6.1 Past service cost and gains and losses on settlement

In section 5.9 we identified that the total service cost may comprise not only the current service cost but also other items, past service cost and gains and losses on settlement. This section explains these issues and their accounting treatment.

6.1.1 Past service cost

Past service cost is the change in the present value of the defined benefit obligation resulting from a plan **amendment** or **curtailment**.

A plan **amendment** arises when an entity either introduces a defined benefit plan or **changes the benefits payable** under an existing plan. As a result, the entity has taken on additional obligations that it has not hitherto provided for. For example, an employer might decide to introduce a medical benefits scheme for former employees. This will create a new defined benefit obligation that has not yet been provided for.

A **curtailment occurs when an entity significantly reduces the number of employees covered by a plan**. This could result from an isolated event, such as closing a plant, discontinuing an operation or the termination or suspension of a plan.

Past service costs can be either **positive** (if the changes increase the obligation) or **negative** (if the change reduces the obligation).

6.1.2 Gains and losses on settlement

A **settlement** occurs when an employer enters into a transaction to eliminate part or all of its post- employment benefit obligations (other than a payment of benefits to or on behalf of employees under the terms of the plan and included in the actuarial assumptions).

A curtailment and settlement might **happen together**, for example when an employer brings a defined benefit plan to an end by settling the obligation with a one-off lump-sum payment and then scrapping the plan.

The gain or loss on a settlement is the difference between:

(a) the **present value of the defined benefit obligation** being settled, as valued on the date of the settlement; and

(b) the **settlement price**, including any plan assets transferred and any payments made by the entity directly in connection with the settlement.

6.1.3 Accounting for past service cost and gains and losses on settlement

An entity should **remeasure the obligation** (and the related plan assets, if any) using current actuarial assumptions, before determining past service cost or a gain or loss on settlement.

The rules for recognition for these items are as follows.

Past service costs are recognised at the earlier of the following dates:

(a) When the plan amendment or curtailment occurs

(b) When the entity recognises related restructuring costs (in accordance with IAS 37) or termination benefits

All gains and losses arising from past service costs or settlements must be recognised immediately in profit or loss.

6.2 Asset ceiling test

When we looked at the recognition of the net defined benefit liability/(asset) in the statement of financial position at the beginning of section 5 the term 'asset ceiling' was mentioned. This term relates to a threshold established by IAS 19 to ensure that any defined benefit asset (ie, a pension surplus) is carried at **no more than its recoverable amount**. In simple terms, this

means that any net asset is restricted to the amount of cash savings that will be available to the entity in future.

6.3 Net defined benefit assets

A net defined benefit asset may arise if the plan has been overfunded or if actuarial gains have arisen. This meets the definition of an asset (as stated in the *Conceptual Framework*) because **all** the following apply.

(a) The entity **controls a resource** (the ability to use the surplus to generate future benefits).

(b) That control is the **result of past events** (contributions paid by the entity and service rendered by the employee).

(c) **Future benefits** are available to the entity in the form of a reduction in future contributions or a cash refund, either directly or indirectly to another plan in deficit.

The **asset ceiling** is the **present value** of those future benefits. The **discount rate used is the same** as that used to calculate the net interest on the net defined benefit liability/(asset). The net defined benefit asset would be reduced to the asset ceiling threshold. Any related write-down would be treated as a **remeasurement** and recognised in **other comprehensive income**.

If the asset ceiling adjustment was needed in a subsequent year, the changes in its value would be treated as follows:

(a) **Interest** (as it is a discounted amount) recognised in profit or loss as part of the net interest amount

(b) **Other changes** recognised in profit or loss

6.4 Other issues

6.4.1 Multiple plans and offsetting

Where a sponsoring employer runs more than one defined benefit scheme, each must be accounted for separately and a plan deficit in one cannot be set off against a plan surplus in another unless there is a legal right of offset and the entity intends to settle on a net basis.

6.4.2 Projected unit credit method

The projected unit credit method is the method required by IAS 19 to be used in determining the present value of the defined benefit obligation and current service cost.

This method sees each period of service giving rise to an additional unit of benefit entitlement (ie, for each extra year worked by an employee, their pension increases).

Each of these units is measured separately and the total of all units to date (both current year and previous years) is the final obligation under the plan.

The total of the current year units is the current service cost.

Attribution of benefit to period of service

In order to apply the projected unit credit method, a unit, or amount of future benefit, must be attributed to each period of service.



Worked example: Projected unit credit method 1

A defined benefit plan provides a lump-sum benefit of £100 per year of service payable on retirement.

Requirement

How is this benefit attributed to periods of service and how is the resulting current service cost and defined benefit obligation calculated?

Solution

A benefit of £100 is attributed to each year.

The current service cost = the present value of £100.

The present value of the defined benefit obligation = the present value of £100 × number of years of service to reporting date.



Worked example: Projected unit credit method 2

A company operates a defined benefit scheme that pays a lump-sum benefit equal to £500 for each year of service.

An employee joins the company at the beginning of Year 1 and is due to retire after five years of service.

For the sake of simplicity ignore the possibility of the employee leaving the company before the expected date.

The discount rate is 10%.

Requirement

Calculate the current service cost to be debited to profit or loss in Years 1 to 5, and the present value of the defined benefit obligation in each of these years.

Solution

Current service cost to be debited to profit or loss

Year	**1**	**2**	**3**	**4**	**5**
	£	**£**	**£**	**£**	**£**
Current year benefit	500	500	500	500	500
Current service cost	500 ÷ (1.1)4	500 ÷ (1.1)3	500 ÷ (1.1)2	500 ÷ (1.1)	
	= 342	= 376	= 413	= 455	500
PV of defined benefit obligation	342	376 × 2	413 × 3	455 × 4	500 × 5
		= 752	= 1,239	= 1,820	= 2,500

Note: Previously we have said that the present value of the obligation moves from year to year due to: payments out to retirees, the unwinding of one year's discount, and the current service cost.

This can be applied to Year 2 as follows:

	£
PV of defined benefit obligation b/f	342
Unwinding of discount (342 × 10%)	34
Current service cost	376
PV of defined benefit obligation c/f	752

6.5 Suggested approach

The suggested approach to defined benefit schemes is to deal with the change in the obligation and asset in the following order.

Step	Item	Recognition
1	**Record opening figures:** Asset Obligation	
2	**Interest cost on obligation** Based on discount rate and PV obligation at start of period. Should also reflect any changes in obligation during period.	DEBIT – Interest cost (P/L) (x% × b/d obligation) CREDIT – PV defined benefit obligation (SOFP)
3	**Interest on plan assets** Based on discount rate and asset value at start of period. Technically, this interest is also time apportioned on contributions less benefits paid in the period.	DEBIT – Plan assets (SOFP) CREDIT – Interest cost (P/L) (x% × b/d assets)
4	**Current service cost** Increase in the present value of the obligation resulting from employee service in the current period.	DEBIT – Current service cost (P/L) CREDIT– PV defined benefit obligation (SOFP)
5	**Contributions** As advised by actuary.	DEBIT – Plan assets (SOFP) CREDIT – Company cash
6	**Benefits** Actual pension payments made.	DEBIT – PV defined benefit obligation (SOFP) CREDIT – Plan assets (SOFP)
7	**Past service cost** Increase/decrease in PV obligation as a result of introduction or improvement of benefits.	**Positive (increase in obligation):** DEBIT – Past service cost (P/L) CREDIT – PV defined benefit obligation (SOFP) **Negative (decrease in obligation):** DEBIT – PV defined benefit obligation (SOFP) CREDIT – Past service cost (P/L)

Step	Item	Recognition
8	**Gains and losses on settlement** Difference between the value of the obligation being settled and the settlement price.	**Gain** DEBIT – PV defined benefit obligation (SOFP) CREDIT – Service cost (P/L) **Loss** DEBIT – Service cost (P/L) CREDIT – PV defined benefit obligation (SOFP)
9	**Remeasurements: actuarial gains and losses** Arising from annual valuations of obligation. On obligation, differences between actuarial assumptions and actual experience during the period, or changes in actuarial assumptions.	**Gain** DEBIT – PV defined benefit obligation (SOFP) CREDIT – Other comprehensive income **Loss** DEBIT – Other comprehensive income CREDIT – PV defined benefit obligation (SOFP)
10	**Remeasurements: return on assets (excluding amounts in net interest)** Arising from annual valuations of plan assets	**Gain** DEBIT – FV plan assets (SOFP) CREDIT – Other comprehensive income **Loss** DEBIT – Other comprehensive income CREDIT – FV plan assets (SOFP)
11	**Disclose in accordance with the standard**	See section 7.

Professional skills focus: Structuring problems and solutions

The good news is that exam pension questions usually give rise to the structured approach suggested in the above table. However, the question is unlikely to have all the elements in the table, for example, a settlement.

Interactive question 1: Defined benefit plan 1

For the sake of simplicity and clarity, all transactions are assumed to occur at the year end.

The following data applies to the post-employment defined benefit compensation scheme of BCD Co.

Discount rate: 10% (each year)

Present value of obligation at start of 20X2: £1m Market value of plan assets at start of 20X2: £1m The following figures are relevant.

	20X2	20X3	20X4
	£'000	£'000	£'000
Current service cost	140	150	150
Benefits paid out	120	140	150
Contributions paid by entity	110	120	120
Present value of obligation at year end	1,200	1,650	1,700
Fair value of plan assets at year end	1,250	1,450	1,610

Additional information:

(1) At the end of 20X3, a division of the company was sold. As a result of this, a number of the employees of that division opted to transfer their accumulated pension entitlement to their new employer's plan. Assets with a fair value of £48,000 were transferred to the other company's plan and the actuary has calculated that the reduction in BCD's defined benefit liability is £50,000. The year-end valuations in the table above were carried out **before** this transfer was recorded.

(2) At the end of 20X4, a decision was taken to make a one-off additional payment to former employees currently receiving pensions from the plan. This was announced to the former employees before the year end. This payment was not allowed for in the original terms of the scheme. The actuarial valuation of the obligation in the table above **includes** the additional liability of £40,000 relating to this additional payment.

Requirement

Show how the reporting entity should account for this defined benefit plan in each of years 20X2, 20X3 and 20X4.

See **Answer** at the end of this chapter.



Interactive question 2: Defined benefit plan 2

Peters operates a defined benefit pension plan for its employees. At 1 January 20X5 the fair value of the pension plan assets was £5,200,000 and the present value of the plan liabilities was £5,800,000.

The actuary estimates that the current and past service costs for the year ended 31 December 20X5 are £900,000 and £180,000 respectively. The past service cost is caused by an increase in pension benefits. The plan liabilities at 1 January and 31 December 20X5 correctly reflect the impact of this increase.

The yield on high-quality corporate bonds is estimated at 8% and the expected return on plan assets at 5%.

The pension plan paid £480,000 to retired members in the year to 31 December 20X5. Peters paid £1,460,000 in contributions to the pension plan and this included £180,000 in respect of past service costs.

At 31 December 20X5 the fair value of the pension plan assets is £6,800,000 and the present value of the plan liabilities is £7,000,000.

In accordance with IAS 19, *Employee Benefits,* Peters recognises gains and losses on remeasurement of the defined benefit asset/liability in other comprehensive income in the period in which they occur.

Requirement

Calculate the actuarial gains or losses on pension plan assets and liabilities that will be included in other comprehensive income for the year ended 31 December 20X5. (Round all figures to the nearest £'000.)

See **Answer** at the end of this chapter.

Interactive question 3: Defined benefit plan 3

The defined benefit pension plan of Leadworth plc was formed on 1 January 20X3. The following details relate to the scheme at 31 December 20X3.

	£m
Present value of obligation	208
Fair value of plan assets	200
Current service cost for the year	176
Contributions paid	160
Interest cost on obligation for the year	32
Interest on plan assets for the year	16

The directors are aware that IAS 19 has been revised but are unsure how to treat any gain or loss on remeasurement of the plan asset or liability.

Requirement

Show how the defined benefit pension plan should be dealt with in the financial statements for the year ended 31 December 20X3.

See **Answer** at the end of this chapter.

Interactive question 4: Defined benefit plan 4

Baseline plc has a defined benefit pension scheme and wishes to recognise the full deficit in its statement of financial position.

Requirement

Using the information below, prepare extracts from the statement of financial position and the statement of comprehensive income, together with a reconciliation of scheme movements for the year ended 31 January 20X8. Ignore taxation.

(1) The opening scheme assets were £3.6 million on 1 February 20X7 and scheme liabilities at this date were £4.3 million.

(2) Company contributions to the scheme during the year amounted to £550,000.

(3) Pensions paid to former employees amounted to £330,000 in the year.

(4) The yield on high-quality corporate bonds was 8% and the actual return on plan assets was £295,000.

(5) During the year, five staff were made redundant, and an extra £58,000 in total was added to the value of their pensions.

(6) Current service costs as provided by the actuary are £275,000.

(7) The actuary valued the plan liabilities at 31 January 20X8 as £4.54 million.

See **Answer** at the end of this chapter.

Professional skills focus: Assimilating and using information

As you will have seen in the above interactive questions, all the information given in the pensions element of a question is used in the answer. Assimilating it and using it is therefore the straightforward bit, at least in the exam. In practice, this may be much harder.

7 Defined benefit plans – disclosure

Section overview

- The disclosure requirements for defined benefit plans are extensive and detailed in order to enable users to understand the plan and the nature and extent of the entity's commitment.

Detailed disclosure requirements are set out in IAS 19 in relation to defined benefit plans, to provide users of the financial statements with information that enables an evaluation of the nature of the plan and the financial effect of any changes in the plan during the period.

Requirements for disclosures include a description of the plan, a reconciliation of the fair value of plan assets from the opening to closing position, the actual return on plan assets, a reconciliation of movements in the present value of the defined benefit obligation during the period, an analysis of the total expense recognised in profit or loss, and the principal actuarial assumptions made.

Additional disclosures set out in the amendment to IAS 19 include:

- an analysis of the defined benefit obligation between amounts relating to unfunded and funded plans;
- a reconciliation of the present value of the defined benefit obligation between the opening and closing statement of financial position, separately identifying each component in the reconciliation;
- a reconciliation of the present value of the defined benefit obligation and the fair value of the plan assets to the pension asset or liability recognised in the statement of financial position;
- a breakdown of plan assets for the entity's own financial instruments, for example an equity interest in the employing entity held by the pension plan and any property occupied by the entity or other assets used by the entity;
- for each major category of plan assets the percentage or amount that it represents of the total fair value of plan assets;
- the effect of a one percentage point increase or decrease in the assumed medical cost trend rate on amounts recognised during the period, such as service cost and the pension obligation relating to medical costs;

- amounts for the current annual period and the previous four annual periods of the present value of the defined benefit obligation, fair value of plan assets and the resulting pension surplus or deficit, and experience adjustments on the plan liabilities and assets in percentage or value terms; and
- an estimate of the level of future contributions to be made in the following reporting period.

8 Other long-term employee benefits

Section overview

The accounting treatment for other long-term employee benefits is a simplified version of that adopted for defined benefit plans.

Definition

Other long-term employee benefits: Employee benefits (other than post-retirement benefit plans and termination benefits) which do not fall due wholly within 12 months after the end of the period in which the employees render the service.

Examples of other long-term employee benefits include **long-term disability benefits** and **paid sabbatical leave**.

Although such long-term benefits have many of the attributes of a defined benefit pension plan, they are not subject to the same level of uncertainty. Furthermore, the introduction of such benefits or changes to these benefits rarely causes a material amount of past service cost. As a consequence, the accounting treatment adopted is a simplified version of that for a defined benefit plan. The only difference is that all actuarial gains and losses are recognised immediately in profit or loss.

9 Termination benefits

Section overview

Termination benefits are recognised as an expense when the entity is committed to either:

- terminating the employment before normal retirement date; or
- providing termination benefits in order to encourage voluntary redundancy.

Definition

Termination benefits: Employee benefits payable on the termination of employment, through voluntary redundancy or as a result of a decision made by the employer to terminate employment before the normal retirement date.

Where voluntary redundancy has been offered, the entity should measure the benefits based on an expected level of take-up. If, however, there is uncertainty about the number of employees who will accept the offer, then there may be a contingent liability, requiring disclosure under IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*.

An entity should recognise a termination benefit when it has made a firm commitment to end the employment. Such a commitment will exist where, for example, the entity has a detailed formal plan for the termination and it cannot realistically withdraw from that commitment.

Where termination benefits fall due more than 12 months after the reporting date they should be discounted.



Worked example: Termination benefits

The directors of an entity met on 23 July 20X3 to discuss the need to decrease costs by reducing the number of employees. On 17 August 20X3 they met again to agree a plan. On 6 September 20X3 other members of the management team were informed of the plan. On 7 October 20X3 the plan was announced to the employees affected and implementation of the formalised plan began.

Requirement

When should the entity recognise the liability?

Solution

The entity should only recognise the liability for the termination benefits when it is demonstrably committed to terminating the employment of those affected. This occurred on 7 October 20X3 when the formal plan was announced and it is at this date that there is no realistic chance of withdrawal.

10 IAS 26, Accounting and Reporting by Retirement Benefit Plans



Section overview

IAS 26 applies to the preparation of financial reports by retirement benefit plans which are either set up as separate entities and run by trustees or held within the employing entity.

10.1 Objectives, scope and definitions of IAS 26

The objective of IAS 26 is to provide useful and consistently produced information on retirement benefit plans for members of the plans and other interested parties.

IAS 26 should be applied in the preparation of financial reports by retirement benefit plans.

Although it is commonplace for a retirement benefit plan to be set up as a separate legal entity run by independent trustees, plan assets may be held within the entity employing the plan's members. IAS 26 applies in both sets of circumstances. In the latter case IAS 26 still regards the retirement benefit plan as a reporting entity separate from the employing entity.

The preparation of a retirement benefit plan's financial report should be in accordance with not only IAS 26 but also all other international standards to the extent that they are not overridden by IAS 26.

IAS 26 does not cover:

- the preparation of reports to individual participants about their retirement benefit rights; or
- the determination of the cost of retirement benefits in the financial statements of the employer having pension plans for its employees and providing other employee benefits.



Definition

Retirement benefit plans: Arrangements whereby an entity provides benefits for its employees on or after termination of service (either in the form of an annual income or as a lump sum), when such benefits, or the employer's contributions towards them, can be determined or estimated in advance of retirement from the provisions of a document or from the entity's practices.

There are two main types of retirement benefit plan, both discussed in section 3 of this chapter.

(a) **Defined contribution plans** (sometimes called 'money purchase schemes'). These are retirement plans under which payments into the plan are fixed. Subsequent payments out of the plan to retired members will therefore be determined by the value of the investments made from the contributions that have been made into the plan and the investment returns reinvested.

(b) **Defined benefit plans** (sometimes called 'final salary schemes'). These are retirement plans under which the amount that a retired member will receive from the plan during retirement is fixed. Contributions are paid into the scheme based on an estimate of what will have to be paid out under the plan.



Interactive question 5: Scope

To which of the following does IAS 26, *Accounting and Reporting by Retirement Benefit Plans* apply?

A The general purpose financial reports of pension schemes

B The cost to companies of employee retirement benefits

C The financial statements relating to an actuarial business

D Reports to individuals of their future retirement benefits

See **Answer** at the end of this chapter.

10.2 Key concepts

'**Funding**' represents the employer's contributions paid to the fund in order to meet the future obligations under the plan for the payment of retirement benefits.

'**Participants**' are those employees who will benefit under the plan (ie, employees and retired employees).

The '**net assets available for benefits**' are the assets less liabilities of the plan that are available to generate future investment income that will increase the plan's assets. These net assets are calculated before the deduction of the actuarial assessment of promised retirement benefits.

10.3 Key requirements

The following summarises the key requirements of IAS 26.

Key requirements	Defined contribution plans	Defined benefit plans	All retirement plans
Investments to be carried at fair value wherever possible			Yes
Recognition of the actuarial present value of promised retirement benefits		Yes	
A statement of changes in net assets available for benefits			Yes
No requirement for an actuarial report	Yes		

10.4 Disclosure

The report of all retirement benefit plans should include the following information.

- A statement of changes in the net assets that are available in the fund to provide future benefits
- A summary of the plan's significant accounting policies

The statement of changes in the net assets available to provide future benefits should disclose a full reconciliation showing movements during the period, for example contributions made to the plan split between employee and employer, investment income, expenses and benefits paid out.

Information should be provided on the plan's funding policy, the basis of valuation for the assets in the fund and details of significant investments that exceed a 5% threshold of net assets in the fund available for benefits. Any liabilities that the plan has other than those of the actuarially calculated figure for future benefits payable and details of any investment in the employing entity should also be disclosed.

General information should be included about the plan, such as the names of the employing entities, the groups of employees that are members of the plan, the number of participants receiving benefits under the plan and the nature of the plan i.e., defined contribution or defined benefit. If employees contribute to the plan, this should be disclosed along with an explanation of how the promised benefits are calculated and details of any termination terms of the plan. If there have been changes in any of the information disclosed then this fact should be explained.

Professional skills focus: Concluding, recommending and communicating

Users of IAS 26 reports include plan members, who do not usually have specialist knowledge. The general information referred to above should therefore be clearly written, avoiding special jargon.

11 Audit focus

Section overview

- The estimation of pension costs, particularly those for defined benefit pension schemes, involves a high level of uncertainty.
- The auditor must evaluate the appropriateness of the fair value measurements.

Fair value accounting applies to pension costs, so auditors must be aware of the issues around auditing fair value when auditing this area. Please refer back to earlier chapters for further details on the IAASB's guidance on auditing fair value.

Context example: British Home Stores (BHS)

Established UK high street retailer BHS went into administration in 2016, ceasing to trade later in the same year. It had been sold by its previous owner Arcadia to Retail Acquisitions Ltd in 2015 for £1 in the hope that the company's underlying trading position could be improved. However, in addition to this problem, BHS had also been attempting to improve its pension scheme which was displaying significant deficits: by 2016, the funding level of the scheme was as low as 65% showing a combined deficit that was valued as high as £571 million. In 2017, the 19,000 members received news of a settlement worth £363 million from Sir Philip Green, the previous owner, who had agreed to address the deficit. In 2018, the remaining liabilities of the fund were sold to an insurance company.

While most of the blame associated with the demise of BHS has been attributed to both Philip Green and the person to whom he sold BHS, Dominic Chappell, and their combined failure to address the problems faced by BHS, the importance of the pension scheme cannot be overestimated, given the sums required to address its liabilities and the impact they would have had on its already strained financial position.

(Source: The Pensions Regulator (2017) *TPR publishes report on BHS case*. [Online]. Available from: www.thepensionsregulator.gov.uk/en/media-hub/press-releases/tpr-publishes-report-on-bhs-case [Accessed 3 October 2019]

Auditors need to ensure they have reviewed the various assumptions used by management when accounting for pension schemes and have assessed their sensitivity to change in the current economic climate using suitable levels of professional scepticism. The issues discussed in auditing chapters on ISA 540 (Revised) are extremely relevant here: **complexity**, **subjectivity** and **estimation uncertainty**.

11.1 Auditing pension costs

The table below summarises the areas of audit focus, and the audit evidence required, when auditing pension costs.

Issue	Evidence
Scheme assets (including quoted and unquoted securities, debt instruments, properties)	• Ask directors to reconcile the scheme assets valuation at the scheme year-end date with the assets valuation at the reporting entity's date being used for IAS 19 purposes. • Obtain direct confirmation of the scheme assets from the investment custodian.

Issue	Evidence
	• Consider requiring scheme auditors to perform procedures.
Scheme liabilities	• Auditors must follow the principles relating to work done by a management's expert as defined in ISA (UK) 500, *Audit Evidence* (and covered in Chapter 6) to assess whether it is appropriate to rely on the actuary's work. • Specific matters would include: - the source data used; - the assumptions and methods used; and - the results of actuaries' work in the light of auditors' knowledge of the business and results of other audit procedures. • Actuarial source data is likely to include: - scheme member data (for example, classes of member and contribution details); and - scheme asset information (for example, values and income and expenditure items).
Actuarial assumptions (for example, mortality rates, termination rates, retirement age and changes in salary and benefit levels)	Auditors will not have the same expertise as actuaries and are unlikely to be able to challenge the appropriateness and reasonableness of the assumptions. They should nevertheless ascertain the qualifications and experience of the actuaries. Auditors can, also, through discussion with directors and actuaries: • obtain a general understanding of the assumptions and review the process used to develop them; • consider whether assumptions comply with IAS 19 requirements ie, are unbiased and based on market expectations at the year end, over the period during which obligations will be settled; • compare the assumptions with those which directors have used in prior years; • consider whether, based on their knowledge of the reporting entity and the scheme, and on the results of other audit procedures, the assumptions appear to be reasonable and compatible with those used elsewhere in the preparation of the entity's financial statements; and • obtain **written representations** from directors confirming that the assumptions are consistent with their knowledge of the business.
Items charged to profit or loss (current service cost, past service cost, gains and losses on settlements and	• Discuss with directors and actuaries the factors affecting current service cost (for example, a scheme closed to new entrants may see an increase year on year as a percentage of pay with the average age of the workforce increasing).

Issue	Evidence
curtailments)	• Confirm that net interest cost has been based on the discount rate determined by reference to market yields on high-quality fixed-rate corporate bonds.
Items recognised in other comprehensive income	• Check basis of updated assumptions used to calculate actuarial gains/losses. • Check basis of calculation of return on plan assets ie, using current fair values. Fair values must be measured in accordance with IFRS 13.
Contributions paid into plan (Retirement benefits paid out are paid by the pension plan itself so there is no cash entry in the entity's books)	Agree cash payments to cash book/bank statements.

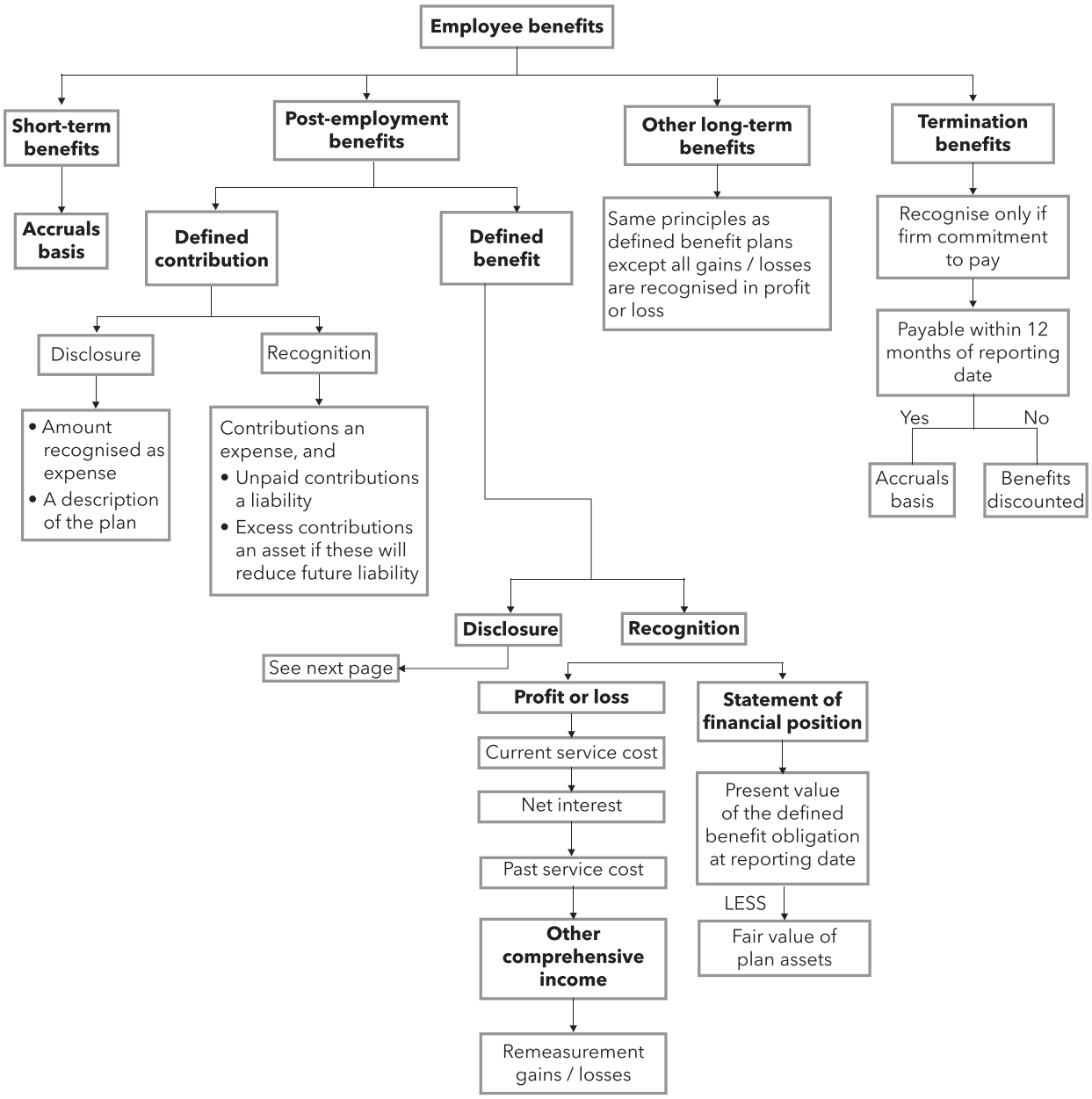
Where the results of auditors' work are inconsistent with those reached by directors and actuaries, additional procedures, such as requesting directors to obtain evidence from another actuary, may help in resolving the inconsistency.

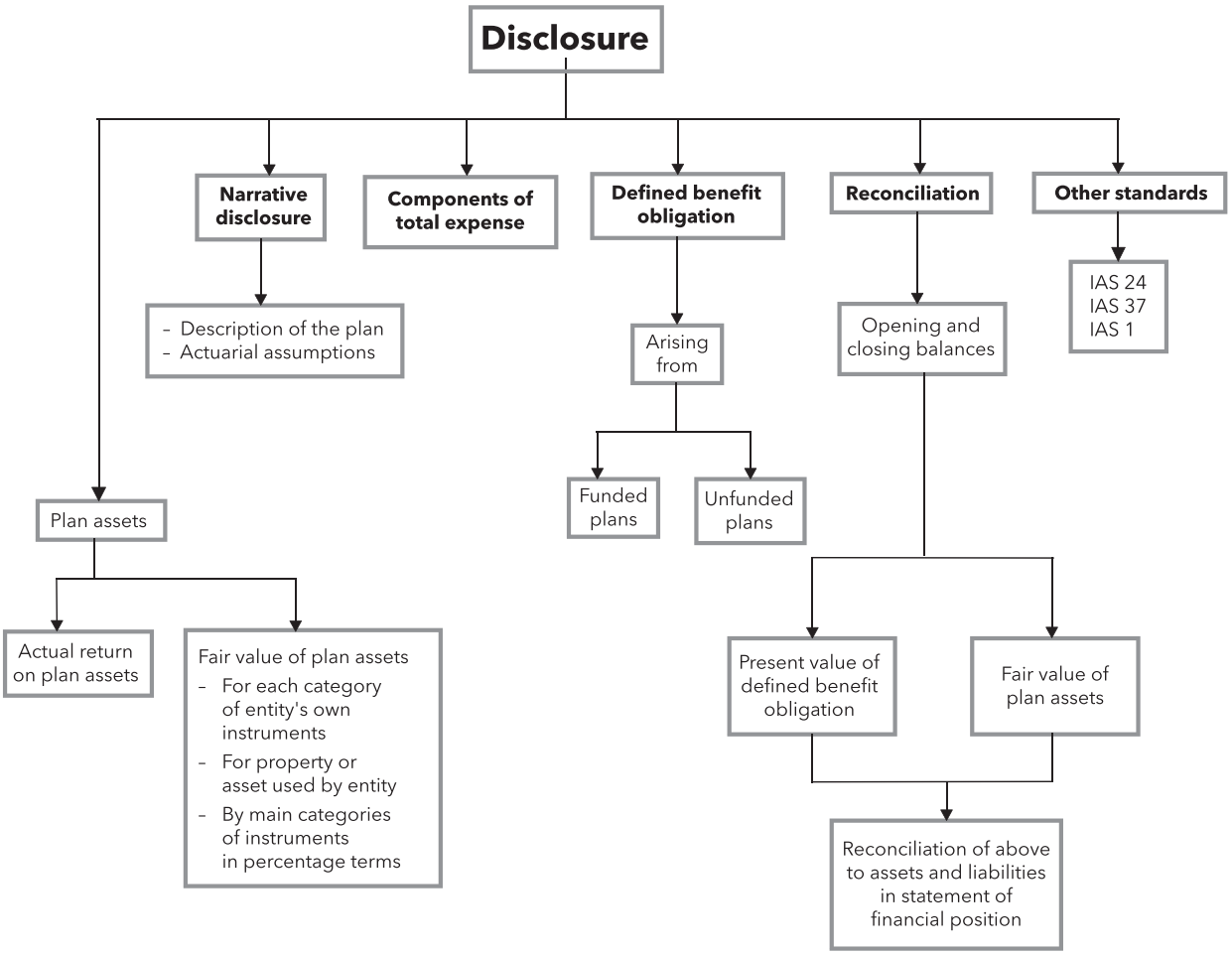


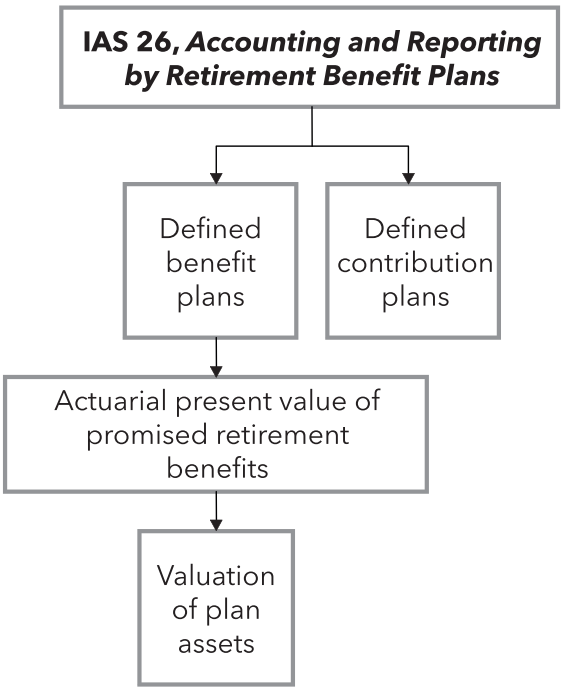
Professional skills focus: Concluding, recommending and communicating

Employee benefits, particularly retirement benefits, are a sensitive area of any audit given the widely publicised, if rare, scandals regarding appropriation. Communication skills are of paramount importance for the auditor.

Summary


Employee benefits
Short-term benefits
Accruals basis
Post-employment benefits
Defined contribution
Disclosure
• Amount recognised as expense
• A description of the plan
Recognition
Contributions an expense, and
• Unpaid contributions a liability
• Excess contributions an asset if these will reduce future liability
Defined benefit
Other long-term benefits
Same principles as defined benefit plans except all gains / losses are recognised in profit or loss
Termination benefits
Recognise only if firm commitment to pay
Payable within 12 months of reporting date
Yes
No
Accruals basis
Benefits discounted
Disclosure
See next page
Recognition
Profit or loss
Current service cost
Net interest
Past service cost
Other comprehensive income
Remeasurement gains / losses
Statement of financial position
Present value of the defined benefit obligation at reporting date
LESS
Fair value of plan assets


Disclosure
Narrative disclosure
Components of total expense
Defined benefit obligation
Reconciliation
Other standards
– Description of the plan
– Actuarial assumptions
Arising from
Opening and closing balances
IAS 24
IAS 37
IAS 1
Funded plans
Unfunded plans
Plan assets
Actual return on plan assets
Fair value of plan assets
– For each category of entity's own instruments
– For property or asset used by entity
– By main categories of instruments in percentage terms
Present value of defined benefit obligation
Fair value of plan assets
Reconciliation of above to assets and liabilities in statement of financial position


IAS 26, Accounting and Reporting by Retirement Benefit Plans
Defined benefit plans
Defined contribution plans
Actuarial present value of promised retirement benefits
Valuation of plan assets

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	What is the IAS 19 accounting treatment for holiday pay? (Topic 2)
2.	Can you distinguish a defined contribution pension plan from a defined benefit plan? (Topic 3)
3.	How is the value of a defined benefit pension plan calculated? (Topic 3)
4.	Can you account for the movement in a defined benefit plan? (Topic 4)
5.	What is the IAS 19 treatment of gains and losses on remeasurement of the net defined benefit asset or liability? (Topic 5)
6.	Can you account for past service costs, curtailments and settlements? (Topic 6)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Lampard	A short question, this will test whether you have understood the concept of past service costs.
Interest	This question focuses on the discount factor to be used in accounting for employee benefits in accordance with IAS 19.
Straw Holdings	The focus of this question is on giving advice to directors unsure how to apply IAS 19. You are also asked for the benefits of moving to a defined contribution plan, which is happening in practice.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Aytace (notes 2 and 3 only)	Note 2 requires the calculation (with explanation) of the defined benefit expense, including an improvement to the scheme. Note 3 requires calculation of a holiday pay accrual.

Maykem	This part of the question is based around correcting an incorrect treatment of the pension contribution and requires calculation of the loss on remeasurement and the required IAS 19 presentation in the financial statements.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries.

Once you have attempted these questions, you can continue your studies by moving onto the next chapter.

Technical reference

IAS 19, *Employee Benefits*

1 Four categories of employee benefits – IAS 19.4

- Short-term employee benefits
- Post-employment benefits
- Other long-term employee benefits
- Termination benefits

2 Short-term employee benefits – IAS 19.7, 19.8

- Wages, salaries and social security contributions falling due within 12 months of employee service
- Short-term compensated absences such as vacation entitlement and paid sick leave
- Profit-sharing and bonuses
- Non-monetary benefits

3 Accounting for short-term employee benefits – IAS 19.10

- Short-term employee benefits are recognised as an expense and a corresponding liability and are accounted for on an undiscounted basis

4 Post-employment benefits

- Classified as either:
- Defined contribution plans
- Where entity's legal or constructive obligation is limited to the amount it agrees to contribute to the fund and consequently bears no actuarial or investment risk
- Defined benefit plans
- Where entity provides agreed benefits and bears both actuarial and investment risk – **IAS 19.25, 19.26, 19.27**

5 Multi-employer plans – IAS 19.7

- Defined contribution or defined benefit plans that:
- Pool assets contributed by entities not under common control
- Provide benefits to employees of more than one entity with benefits determined without regard to the identity of the entity
- Defined benefit plans that share risks between entities under common control are not multi- employer plans – **IAS 19.34**

6 Recognition and measurement of defined contribution plans

Recognise as a liability and expense unless another standard allows inclusion in asset (eg, IAS 2 or IAS 16) – **IAS 19.44**

Disclose expense and required disclosure under – **IAS 24 IAS 19.46, 19.47**

7 Recognition and measurement of defined benefit plans

- Entity underwriting both investment and actuarial risk – **IAS 19.50**
- Accounting involves following steps:
  - Using actuarial techniques make reliable estimate of the amount of benefit employees earned in current and prior periods
  - Discount benefit using projected unit credit method
  - Determine fair value of plan assets
  - Determine total amount of remeasurement gains and losses to be recognised
  - Where plan introduced or changed determine resulting past service cost
  - Where plan has been curtailed or settled calculate resulting gain or loss – **IAS 19.50**

8 Defined benefit scheme

- Recognised in statement of financial position as the net of:
  - Present value of defined benefit obligation at reporting date, minus
  - Fair value at the reporting date of plan assets – **IAS 19.54**
- Recognised in profit or loss
  - Current service cost
  - Net interest on the net defined benefit liability (asset)
  - Past service cost
  - Effect of curtailments or settlements – **IAS 19.61**
- Recognised in other comprehensive income
  - Actuarial gains and losses
  - Returns on plan assets (excluding amounts in net interest) – **IAS 19.61**

9 Actuarial assumptions

- Shall be unbiased and mutually compatible
  - Demographic assumptions
  - Financial assumptions – **IAS 19.72**

10 Discount rate – IAS 19.78

- Rate used to discount post-employment obligations shall be determined by reference to market yields at reporting date on high quality corporate bonds

11 Actuarial gains and losses

- Gains and losses on remeasurement of plan assets and liabilities must be recognised in other comprehensive income (not reclassified to profit or loss) in the period in which they occur – **IAS 19.93 B**

12 Past service cost

- Arises when entity introduces defined benefit plan or changes benefit under an existing defined benefit plan – **IAS 19.97**
- Entity shall recognise past service cost as an expense in profit or loss – **IAS 19.96**

13 Reimbursements – IAS 19.104A

- An entity shall recognise its right to reimbursement as a separate asset only when it is virtually certain that another party will reimburse some or all of the expenditure required to settle a defined benefit obligation

14 Business combinations – IAS 19.108

- In a business combination (see IFRS 3, *Business Combinations*) an entity shall recognise assets and liabilities arising from post-employment benefits at the:
  - Present value of the obligation, less
  - Fair value of any plan assets
- The present value of the obligation includes all of the following even if not recognised by acquiree:
  - All actuarial gains and losses
  - Past service cost before acquisition date
  - Amounts not recognised under transitional provisions

15 Other long-term employee benefits

- Examples include sabbatical leave and long-term disability benefits – **IAS 19.126**
- Amount recognised as a liability is the net of the following:
  - Present value of the defined benefit obligation at the reporting date, minus
  - The fair value of plan assets at the reporting date – **IAS 19.128**
- Amount recognised in profit or loss is as for defined benefit schemes except that all actuarial gains and losses are recognised immediately in profit or loss – **IAS 19.127**

16 Termination benefits

- Termination benefits are recognised as an expense when the entity is committed to
  - Terminate the employment before normal retirement date, or
  - Provide termination benefits as a result of an offer for voluntary redundancy – **IAS 19.133**
- Where termination benefits fall due more than 12 months after the reporting date they shall be discounted – **IAS 19.139**

17 IAS 26, *Accounting and Reporting by Retirement Benefit Plans*

- Scope – **IAS 26.1**
- Definitions – **IAS 26.8**
- Defined contribution plans – **IAS 26.13**
- Defined benefit plans – **IAS 26.17–19**
- Frequency of actuarial valuations – **IAS 26.27**
- Financial statement content – **IAS 26.28–31**
- All plans:
  - Valuation of plan assets – **IAS 26.32**
  - Disclosure – IAS 26.34

Self-test questions

Answer the following questions

1 Employee benefits

Under which category of employee benefits should the following items be accounted for according to IAS 19, *Employee Benefits*?

Item	Category
Paid annual leave	
Lump-sum benefit of 1% of the final salary for each year of service	
Actuarial gains	

2 Lampard

The Lampard company operates two major benefit plans on behalf of its employees under which the amounts of benefit payable depend on a number of factors, the most important of which is length of service. The plans are:

(1) Plan Deben, a post-retirement defined benefit plan

(2) Plan Limen, a long-term disability benefits plan

Changes to the terms of these plans coming into effect from 31 December 20X7 will result in past service cost attributable to unvested benefits, to the extent of £500,000 on Plan Deben and £220,000 on Plan Limen. Within each plan the average period until benefits become vested is five years. There are no past service costs brought forward on either plan.

Requirement

Under IAS 19, *Employee Benefits*, what is the total amount of past service costs which must be recognised by Lampard in the year ended 31 December 20X7?

3 Tiger

The Tiger company operates a post-retirement defined benefit plan under which post-retirement benefits are payable to ex-employees and their partners.

For all the years this plan has been in operation, Tiger has used the market yield on its own corporate bonds as the rate at which it has discounted its defined obligation, because the yield on its own bonds has been the same as that on high-quality corporate bonds. In the current year Tiger has experienced a sharp downgrading in its credit rating, such that the yield on its own bonds at the year end is 8% while that on high-quality corporate bonds is 6%. Tiger is proposing to use the yield on its own bonds as the discount rate, to reflect the greater risk

Requirement

Indicate whether Tiger's approach is correct according to IAS 19, *Employee Benefits*.

4 Interest

An entity's defined benefit net liability at 31 December 20X4 and 20X5 is measured as follows

	20X4	20X5
	£	£
Defined benefit obligation	950,000	1,150,000

The discount rates used for calculating the defined benefit obligation were 6.5% at 31 December 20X4 and 6% at 31 December 20X5.

Requirements

4.1 Calculate the interest cost to be charged to profit or loss for 20X5.

4.2 How should the discount factor that is used to discount post-employment benefit obligations be determined?

4.3 What elements should the discount rate specifically not reflect according to IAS 19?

5 Straw Holdings plc

John Cork, financial director of Straw Holdings plc, your audit client, has recently written to you for advice on pension scheme accounting.

The company's defined benefit pension scheme has net assets valued at £20.2 million. Scheme assets of £19.4 million at the beginning of the year were expected to be enhanced by a cash contribution to the scheme of £0.4 million greater than payments to pensioners. An appropriate discount rate of 10% has been identified.

Based on these figures Mr Cork has prepared the following reconciliation.

Assets at 31 December 20X1

	£m
Opening scheme net assets	19.40
Add interest on assets @ 10%	1.94
Net contributions received	0.40
Less actuarial deficit (balancing figure)	(1.54)
Closing scheme net assets	20.20

The deficit of £1.54 million has come as a surprise to Mr Cork. He is unsure how to treat this deficit in the financial statements and is concerned about the impact it will have on the company's profits.

Requirements

5.1 Explain the impact of the actuarial valuation of the scheme's assets and the resultant deficit on the financial statements of Straw Holdings plc for the year ended 31 December 20X1.

5.2 Identify **two** benefits to Straw Holdings plc of moving from a defined benefit to a defined contribution scheme.

6 IAS 26

Answer the following questions in accordance with IAS 26, *Accounting and Reporting by Retirement Benefit Plans*.

6.1 How should a defined contribution retirement benefit plan carry property, plant and equipment used in the operation of the fund?

6.2 Is a defined contribution retirement benefit plan permitted to use a constant rate redemption yield to measure any securities with a fixed redemption value which are acquired to match the obligations of the plan?

6.3 Does IAS 26 specify a minimum frequency of actuarial valuations?

7 Commercial Properties plc

Commercial Properties plc is a construction company based in Leeds which specialises in the construction of manufacturing units and warehouses. You are conducting the audit for the year ended 31 December 20X8 and have obtained the following information.

The company has two warehouses which it lets to commercial tenants, one located in York and the other in Huddersfield. The property in York has been held for a number of years while construction of the property in Huddersfield was completed on 1 January 20X8 and then subsequently let.

The policy of the company in respect of investment properties is to carry them in the statement of financial position at open market value. They are revalued annually on the advice of professional surveyors at eight times the aggregate rental income.

In March 20X9 a rent review on the warehouse in York was implemented. The directors intend to base the valuation of this warehouse for the year ended 31 December 20X8 on this revised figure on the basis that this represents a more up to date assessment of the market value of the property.

The property in Huddersfield has been let on special terms to another company in which one of the directors holds an interest.

Commercial Properties plc also operates a defined benefit pension scheme on behalf of its employees. The actuary has performed an annual review of funding and based on these figures the statement of financial position is showing the pension fund as a net liability. There is an actuarial surplus on liabilities of £375,000 and a deficit on assets of £525,000. The discount rate determined by reference to market yields on high-quality fixed-rate corporate bonds is 12%.

Requirements

7.1 Identify and explain the audit issues regarding the two investment properties.

7.2 List the audit procedures you would perform to confirm the valuation of the properties.

7.3 List the audit procedures relating to the above pension scheme to be carried out as part of the 20X8 audit.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

The actuarial gain or loss is established as a balancing figure in the calculations, as follows.

Present value of obligation

	20X2 £'000	20X3 £'000	20X4 £'000
PV of obligation at start of year	1,000	1,200	1,600
Interest cost (10%)	100	120	160
Current service cost	140	150	150
Past service cost	–	–	40
Benefits paid	(120)	(140)	(150)
Settlements	–	(50)	–
Actuarial (gain)/loss on obligation: balancing figure	80	320	(100)
PV of obligation at end of year	1,200	*1,600	1,700

* (1,650 – 50)

Market value of plan assets

	20X2 £'000	20X3 £'000	20X4 £'000
Market value of plan assets at start of year	1,000	1,250	1,402
Interest on plan assets (10%)	100	125	140
Contributions	110	120	120
Benefits paid	(120)	(140)	(150)
Settlements	–	(48)	–
Gain on remeasurement through OCI: balancing figure	160	95	98
Market value of plan assets at year end	1,250	*1,402	1,610

* (1,450 – 48)

In the statement of financial position, the liability that is recognised is calculated as follows.

	20X2 £'000	20X3 £'000	20X4 £'000
Present value of obligation	1,200	1,600	1,700
Market value of plan assets	1,250	1,402	1,610
Liability/(asset) in statement of financial position	(50)	198	90

The following will be recognised in profit or loss for the year:

	20X2	20X3	20X4
	£'000	£'000	£'000
Current service cost	140	150	150
Past service cost	–	–	40
Net interest on defined benefit liability (asset)	–	(5)	20
Gain on settlement of defined benefit liability		(2)	
Expense recognised in profit or loss	140	143	210

The following remeasurements will be recognised in other comprehensive income for the year:

	20X2	20X3	20X4
	£'000	£'000	£'000
Actuarial (gain)/loss on obligation	80	320	(100)
Return on plan assets (excluding amounts in net interest)	(160)	(95)	(98)

Answer to Interactive question 2

Gains or losses on plan assets

	£'000
Fair value of plan assets at 1.1.20X5	5,200
Interest on plan assets (8% × £5,200)	416
Contributions	1,460
Benefits paid	(480)
Remeasurement gain to OCI (balancing figure)	204
Fair value of plan assets at 31.12.20X5	6,800

Gains or losses on obligation

	£'000
Present value of obligation at 1.1.20X5	5,800
Current service cost	900
Past service cost	180
Interest cost (8% × £5,800)	464
Benefits paid	(480)
Remeasurement loss to OCI (balancing figure)	136
Present value of obligation at 31.12.20X5	7,000

Answer to Interactive question 3

The defined benefit pension plan is treated in accordance with IAS 19, *Employee Benefits,* as revised in 2011.

The pension plan has a deficit of liabilities over assets.

	£m
Fair value of plan assets	200
Less present value of obligation	(208)
	(8)

The deficit is reported as a liability in the statement of financial position. Profit or loss for the year includes:

	£m
Current service cost	176
Net interest on net defined benefit liability (32 – 16)	16
	192

The company is required by the revised IAS 19 to recognise the £24,000,000 remeasurement gain (see working) immediately in other comprehensive income.

WORKING

	PV of obligation £m	FV of plan assets £m
b/f	Nil	Nil
Contributions paid		160
Interest on plan assets		16
Current service cost	176	
Interest cost on obligation	32	
Actuarial difference (bal fig)		24
c/f	208	200

Answer to Interactive question 4

Extracts as follows:

Statement of financial position extract	**£'000**
Non-current liabilities (4,115 – 4,540)	425

Statement of comprehensive income extract	**£'000**
Charged to profit or loss	
Current service cost	275
Net interest on net defined benefit liability (344 – 288)	56
Curtailment cost	58
	389

Other comprehensive income	**£'000**
Actuarial gain on obligation	107
Return on plan assets (excluding amounts in net interest)	7

Changes in the present value of the defined benefit obligation	**£'000**
Defined benefit obligation at 1 Feb 20X7	4,300
Interest cost @ 8%	344
Pensions paid	(330)
Curtailment	58
Current service cost	275
Actuarial gain (residual)	(107)
Defined benefit obligation at 31 Jan 20X8	4,540

Changes in the fair value of plan assets	**£'000**
Fair value of plan assets at 1 Feb 20X7	3,600
Contributions	550
Pensions paid	(330)
Interest on plan assets 8% × 3,600	288
Remeasurement gain (295 – 288)	7
Fair value of plan assets at 31 Jan 20X8 (residual)	4,115

Answer to Interactive question 5

The correct answer is:

A The general purpose financial reports of pension schemes

Answers to Self-test questions

1 Employee benefits

Item	Category
Paid annual leave	Short term employee benefits
Lump-sum benefit of 1% of the final salary for each year of service	Defined benefit plans
Actuarial gains	Defined benefit plans

IAS 19.9 highlights paid annual leave as a short-term benefit.

The actuarial gains and the lump-sum benefit relate to defined benefit plans (per IAS 19.24–27 and 54).

2 Lampard

£720,000 (£500,000 + £220,000)

Under IAS 19, *Employee Benefits* (revised 2011), past service cost attributable to all benefits, whether vested or not within a post-retirement defined benefit plan, must be recognised immediately in profit or loss. Similarly, past service cost attributable to all benefits, whether vested or not, within a long- term disability benefits plan must be recognised immediately in profit or loss, so the full £720,000 must be recognised.

3 Tiger

Tiger should not be using 8% as its discount rate.

Under IAS 19.78 the yield on high-quality corporate bonds must be used as the discount rate for the defined benefit obligation. (Using a higher rate would result in a lower obligation, which would not reflect greater risk.)

4 Interest

4.1 The interest cost for 20X5 is calculated by multiplying the defined benefit obligation at the start of the period by the discount rate at the start of the period, so: £950,000 × 6.5% = £61,750

4.2 The discount factor should be determined by reference to high-quality corporate bonds with similar currency and maturity as the benefit obligations.

Where no market in corporate bonds exists the discount rate should be determined by reference to government debt.

Where there is no deep market in corporate bonds with sufficiently long maturities the standard requires the use of current market rates of appropriate term to discount short-term payments and the estimation of the rate for longer maturities by extrapolating current market rates on the yield curves.

4.3 The discount rate should not reflect:

- investment risk
- actuarial risk
- specific risk relating to the entity's business

5 Straw Holdings plc

5.1 **Impact of actuarial valuation**

The plan assets are less than expected by £1.54 million, this being a remeasurement loss. Under IAS 19, *Employee Benefits* (revised 2011), such losses must be recognised in other comprehensive income in the period in which they occur. This would result in the net actuarial difference (the £1.54 million loss on the assets combined with any actuarial gain or loss on the obligation) being presented as other comprehensive income and debited directly to reserves. There would therefore be no direct impact on profit or loss.

5.2 Two benefits of moving to defined contribution scheme:

(1) Defined contribution schemes are easier for the company to administer and manage. A fixed level of contributions is paid in monthly instalments for each employee. The risk resulting from the variable returns achieved by funds invested is then borne by the employee. This risk is borne by the employer with a defined benefit scheme.

(2) Defined contribution schemes are easier to account for. Contributions are charged to profit or loss on a systematic basis.

Provided the company is not in arrears on contributions, the monthly double entry required is therefore:

		£	£
DEBIT	Staff costs	X	
CREDIT	Cash		X

6 IAS 26

6.1 Under IAS 26.33 all types of retirement benefit plan should account for assets used in the operation of the plan under the applicable standards. IAS 16 is applicable in this case and either the cost model or the revaluation model may be used.

6.2 All types of retirement plan are permitted by IAS 26.33 to use this method of measuring such securities.

6.3 No minimum frequency of actuarial valuation is specified in IAS 26.27 or elsewhere.

7 Commercial Properties plc

7.1 As follows:

- **Audit issues**
  - Whether the two warehouses fall within the IAS 40, *Investment Property* definition of an investment property
  - Based on the following evidence, it would appear that the definition is satisfied:
    - Both properties are held for the purposes of generating rental income
    - They are not owner occupied but let to third parties

- Whether the initial recognition of the newly completed warehouse is in accordance with IAS 40

  The property would normally be measured at cost during the period of construction (although IAS 40 does allow construction cost to be measured at fair value if it can be measured reliably).

- Whether the accounting treatment adopted for the subsequent measurement of the investment properties is in accordance with IAS 40

  Under IAS 40, a company may choose to adopt the cost model or the fair value model. Commercial Properties plc has adopted the fair value model. As a result of this, no depreciation should be charged and any changes in fair value should be recognised in profit or loss for the period.

- Whether the basis used as fair value is appropriate

Rental income has been used as a basis of calculating an estimated fair value. This is a suitable basis, provided rentals are an indication of the market-based exit value at the measurement date.

IFRS 13 states that fair value should reflect market conditions at the measurement date. The use of the revised rentals following the rent review therefore does not seem to be appropriate. This is because the rent reviews took place after the year end date and, at the end of the reporting period, would not have been guaranteed. As a result, they could not have been reflected in a fair value at 31 December 20X8.

In accordance with IFRS 13, the market-based current exit price is based on the concept that an exchange between unrelated knowledgeable and willing parties has taken place. The warehouse in Huddersfield is being let to another company with which a director is associated. As a result of the relationship between the two parties the rent may not reflect market conditions and therefore may not be a suitable estimation of fair value.

- Whether there is a related party relationship

  The relationship between the director and the company leasing the warehouse in Huddersfield may constitute a related party transaction, which may require disclosure in the financial statements.

- Materiality

  The materiality of any adjustments required as a result of any changes to the fair values of the investment properties will need to be considered. As any changes in fair value are reflected in the statement of profit or loss and other comprehensive income, materiality will need to be assessed both in terms of the impact on the statement of financial position and on the statement of profit or loss and other comprehensive income.

Tutorial Note

This question contains a revision of the audit of investment properties, covered in earlier chapters.

7.2 As follows:

- **Audit procedures: investment properties**
  - Obtain the report produced by the professional surveyors to confirm their valuation at eight times aggregate rental income, and to support the assumption that rentals provide an indication of an appropriate exit value in accordance with IFRS 13.
  - Consider the extent to which their expertise can be relied on eg, reputable firm, experience etc.

- Obtain a copy of the rental agreement for the warehouse in York both before and after the rent review. Confirm that the year-end value has been based on the revised rent and calculate the adjustment which would be required if based on the initial agreement.
- Discuss the nature of the special terms on which the warehouse in Huddersfield has been let and the nature of the relationship between the director and the entity.
- Compare fair values as calculated by the directors with current prices for similar properties in similar locations.
- Where cash flows have been discounted, review whether any assumptions built in to the calculation are reasonable eg, discount rates used.
- Compare any proposed adjustments with materiality levels set for the audit.

7.3 As follows:

- **Audit procedures: pension scheme**

- Obtain the client's permission to liaise with the actuary, and review the actuary's professional qualification.
- Agree the validity and accuracy of the actuarial valuation.
  - Agree that the actuarial valuation method satisfies the accounting objectives of IAS 19.
  - Confirm that net interest cost has been based on the discount rate determined by reference to market yields on high-quality fixed-rate bonds.
- Agree the completeness of the actuarial valuation.
  - Identify major events that should have been taken into account.
  - Scrutinise relevant correspondence eg, between the client, the actuary, the solicitor.
  - Review minutes of board meetings.
- Agree opening balances to last year's working papers.
- Reconcile closing balance provision to opening statement of financial position.
- Agree contributions paid to the cash book and to the funding rate recommended by the actuary in the most recent actuarial valuation.
- Check that disclosures comply with the requirements of IAS 19.

- If fair values have been based on discounted cash flows, ie, future rentals, determine whether this is the most appropriate estimate of a market-based exit value. Compare predicted cash flows with rental agreements. Review the basis of the interest rate applied.
- Review documentation to support method used.
- Recalculate the gain or loss on change in fair value and agree to amount recognised in profit or loss.
- If fair value cannot be measured reliably confirm use of cost model.
- Agree disclosure is in accordance with IAS 40 and IFRS 13.

Chapter 19

Share-based payment

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Background

2 Objective and scope of IFRS 2, Share-based Payment

3 Share-based transaction terminology

4 Equity-settled share-based payment transactions

5 Cash-settled share-based payment transactions

6 Share-based payment with a choice of settlement

7 Group and treasury share transactions

8 Disclosure

9 Distributable profits and purchase of own shares

10 Audit focus

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Appraise corporate reporting regulations, and related legal requirements, with respect to presentation, disclosure, recognition and measurement
- Explain how different methods of providing remuneration for employees may impact upon reported performance and position
- Explain and appraise accounting standards that relate to employee remuneration which include different forms of short-term and long-term employee compensation; retirement benefits; and share-based payment
- Justify and conclude for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**Background** In a working environment you may have to assess the effect of granting stock options as a remuneration policy on the entity's financial statements and advise on how these should be reported and disclosed. In major capital markets the potential gains that senior management can achieve through share-based payment transactions are so significant that they may dwarf other	**Approach** Read through this section quickly – it gives the background behind the publication of IFRS 2. **Stop and think** If the granting of equity instruments does not require the entity to part with assets, why should a charge be recognised in profit or loss?	Share-based payment is new to Advanced Level and is examinable at Level A. The reasons why IFRS 2 was needed will not be tested directly, but it helps to understand them in order to understand IFRS 2 itself.	N/A

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	elements of remuneration packages.			
2	**Objective and scope of IFRS 2, *Share-based Payment*** The general principle of IFRS 2 is that an entity recognises an expense for goods or services with the credit entry recognised either in equity or as a liability depending on the classification of the share-based payment award.	**Approach** This is a short section with the key point being that there are three types of share-based payment: Equity-settled share-based payment transactions Cash-settled share-based payment transactions Transactions with a choice of settlement **Stop and think** Why has this been a controversial standard polarising views at the highest political level?	In your exam you will need to recognise and distinguish between these three types of share-based payment in order to get the financial reporting treatment right.	N/A
3	**Share-based transaction terminology** IFRS 2 has distinctive terminology that must be learned.	**Approach** Use the diagram in section 3 to understand what is going on before you approach the definitions. **Stop and think** What are market-based vesting conditions?	An exam will not ask for these definitions directly, but you will need to know them when tackling scenario-based questions.	N/A
4	**Equity-settled share-based payment transactions** The entity receives goods or services in exchange for equity instruments of the entity (including shares or share options).	**Approach** The key point is that the fair value of the transaction is recognised in profit or loss, spread over the vesting period. **Stop and think** Fair value has a different definition from the IFRS 13	This topic is regularly examined. Typically you would need to identify that share options represent an equity-settled share-based payment and account for the share-based payment correctly.	**IQ2: Non-market-based vesting conditions** A short question in which the condition attached to the grant relates to employees still being in service at the vesting date.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		one, because share-based payments are outside the scope of IFRS 13.		
5	**Cash-settled share-based payment transactions** The entity receives goods or services in exchange for amounts of cash that are based on the price (or value) of the entity's shares or other equity instruments of the entity.	**Approach** The key points are that: The credit entry in respect of a cash-settled share-based payment transaction is reported as a liability. The fair value of the liability should be remeasured at each reporting date until settled. Changes in the fair value are recognised in profit or loss. **Stop and think** Can you think of any examples of cash-settled share based payment transactions?	Cash-settled share-based payment transactions are examined as frequently as equity-settled. An exam scenario arose with a company deciding whether to use cash-based or equity-settled SBP.	**IQ6: Share-based payment** This question asks you to show the double entry for equity-settled SBP and to explain how this would be different if it were cash-settled.
6	**Share-based payment with a choice of settlement** The entity receives goods or services and either the entity or the supplier has a **choice** as to whether the entity settles the transaction in cash (or other assets) or by issuing equity instruments.	**Approach** The key point is that the treatment depends on who has the choice. If the issuing entity has the choice recognise a liability to the extent that it has a present obligation to deliver cash. If the counterparty has the choice, use split accounting. **Stop and think** What should the treatment be if the issuing entity does not have a present obligation to deliver	This is less frequently examined than when there is no choice.	**IQ7: Share-based payment** This is a more complex question with several 'what if' scenarios.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		cash?		
7	**Group and treasury share transactions** This is a fairly marginal topic.	**Approach** Read through quickly. **Stop and think** What are treasury shares?	This topic has not yet been examined.	N/A
8	**Disclosure** In a working environment you may need to advise on how the impact of share-based payment on an entity's financial performance and position should be reported and disclosed.	**Approach** Pay particular attention to section 8.4 (effect on financial performance and position). **Stop and think** How do share options affect earnings per share?	Disclosures don't tend to get tested in detail.	N/A There are no interactive questions relating to this section.
9	**Distributable profits and purchase of own shares** Dividends may only be paid out of distributable profits. There are strict rules relating to reduction of a company's capital.	**Approach** Distributable profits is revision from your earlier studies, so work though this quickly. Purchase of own shares is new to Advanced Level. **Stop and think** Why might a company wish to reduce its capital?	Distributable profits came up recently, when candidates were required to calculate the amount of a company's legally distributable reserves. In the past it has also come up as a combined FR/audit issue.	**IQ8: Main rules** A straightforward, bookwork question. If you're in a hurry, just highlight the main points in the answer.
10	**Audit focus** As is often the case, the auditor must focus on fair value: is the fair value of the payment appropriate?	**Approach** Use the table to determine the evidence required for each issue. **Stop and think** How would you audit the fair value of cash-settled SBP transactions?	As indicated earlier, share-based payment transactions can be tested from both the FR and the auditing standpoint.	**IQ10: Share-based payments** A short, audit-based question focusing on compliance with IFRS 2 and fair value and disclosure.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Background

Section overview

Companies sometimes pay for goods and services provided to them in the form of shares or share options. This raises the issue of how such payments should be accounted for, and in particular whether they should be expensed in profit or loss.

1.1 Introduction

Share-based payment occurs when an entity purchases goods or services from another party such as a supplier or employee and rather than paying directly in cash, settles the amount owing in shares, share options or future cash amounts linked to the value of shares. This is common:

- in e-businesses which do not tend to be profitable in early years and are cash poor;
- within all sectors where a large part of the remuneration of directors is provided in the form of shares or options. Employees may also be granted share options.

1.2 The accounting problem

Pre-IFRS 2

Before the publication of IFRS 2, *Share-based Payment* there appeared to be an anomaly to the extent that if a company paid its employees in cash, an expense was recognised in profit or loss, but if the payment was in share options, no expense was recognised.

IFRS 2 requirements

As will be seen throughout the chapter, IFRS 2 requires an expense to be recognised in profit or loss in relation to share-based payments.

The publication of IFRS 2 in 2004 and introduction of this requirement to recognise an expense caused huge controversy, with opposition especially strong among high tech companies. The arguments over expensing share-based payments polarised opinion, especially in the US.

- The main argument against recording an expense was that no cash changes hands as part of such transactions, and it is claimed therefore that there is no true expense.
- The main argument for was that share-based payments are simply another form of compensation that should go into the calculation of earnings for the sake of transparency for investors and the business community. It was also argued that recording an expense better reflects the accruals basis of financial statement preparation.

Practical application of IFRS 2

In practice, the implementation of IFRS 2 has resulted in earnings being reduced, sometimes significantly. It is generally agreed that as a result of this standard companies focus more on the earnings effect of different rewards policies.

Following the adoption of IFRS 2, some companies have admitted that they are re-evaluating the use of share options as part of employee remuneration.

2 Objective and scope of IFRS 2, *Share-based Payment*

Section overview

A share-based payment transaction is one in which the entity transfers equity instruments, such as shares and share options, in exchange for goods and services supplied by employees or third parties.

2.1 Transactions within the scope of IFRS 2

IFRS 2 applies to all share-based payment transactions. The standard recognises and addresses **three** types of transactions according to the method of settlement.

- **Equity-settled share-based payment transactions**

  The entity receives goods or services in exchange for equity instruments of the entity (including shares or share options).

- **Cash-settled share-based payment transactions**

  The entity receives goods or services in exchange for amounts of cash that are based on the price (or value) of the entity's shares or other equity instruments of the entity.

- **Transactions with a choice of settlement**

  The entity receives goods or services and either the entity or the supplier has a **choice** as to whether the entity settles the transaction in cash (or other assets) or by issuing equity instruments.

IFRS 2 requires an entity to recognise share-based payment transactions in its financial statements. Transactions in which an entity receives goods or services as consideration for equity instruments of the entity (including shares or share options) are share-based payment transactions. Such transactions give rise to expenses (or, if applicable, assets) that should be measured at fair value.

IFRS 2 was amended in 2009 to address situations in those parts of the world where, for public policy or other reasons, companies give their shares or rights to shares to individuals, organisations or groups that have not provided goods or services to the company. An example is the issue of shares to a charitable organisation for less than fair value, where the benefits are more intangible than usual goods or services.

Note that the requirements of IFRS 13, *Fair Value Measurement* do not apply to share-based payment transactions within the scope of IFRS 2.

2.1.1 Share-based payments among group entities

Payment for goods or services received by an entity within a group may be made in the form of granting equity instruments of the parent company, or equity instruments of another group company.

IFRS 2.3 states that this type of transaction qualifies as a share-based payment transaction within the scope of IFRS 2.

In 2009, the standard was amended to clarify that it applies to the following arrangements:

- Where the entity's suppliers (including employees) will receive cash payments that are linked to the price of the equity instruments of the entity

- Where the entity's suppliers (including employees) will receive cash payments that are linked to the price of the equity instruments of the entity's parent

  Under either arrangement, the entity's parent had an obligation to make the required cash payments to the entity's suppliers. The entity itself did not have any obligation to make such payments. IFRS 2 applies to arrangements such as those described above even if the entity that receives goods or services from its suppliers has no obligation to make the required share-based cash payments.

2.2 Transactions outside the scope of IFRS 2

The following are outside the scope of IFRS 2:

- Transactions with employees and others in their capacity as a **holder of equity instruments** of the entity (for example, where an employee receives additional shares in a rights issue to all shareholders)
- The issue of equity instruments in exchange for control of another entity in a **business combination**
- Contracts to buy or sell non-financial items that may be settled **net** in shares or rights to shares are outside the scope of IFRS 2 and are addressed by IAS 32, *Financial Instruments: Presentation* and IFRS 9, *Financial Instruments*.

Context example: Transactions within and outside the scope of IFRS 2

Scenario 1: Entity A grants share warrants and its own equity to its external consultants. The warrants become exercisable once an initial public offering (IPO) is made and on condition that the consultants continue to provide agreed services to Entity A until the date the IPO is made.

This transaction is a share-based payment within the scope of IFRS 2.

Scenario 2: Entity B buys back some of its own shares from employees in their capacity as shareholders for the market value of those shares.

This transaction is a simple purchase of treasury shares and is outside the scope of IFRS 2, being governed instead by IAS 32.

Scenario 3: Entity C buys back some of its own shares but pays an amount in excess of their market value only to shareholders who are employees.

The excess over market value to employees only would be considered as a compensation expense within the scope of IFRS 2.

Scenario 4: Entity D enters into a contract to buy a commodity for use in its business for cash, at a price equal to the value of 1,000 shares of Entity D at the date the commodity is delivered. Although Entity D can settle the contract net, it does not intend to do so, nor does it have a past practice of doing so.

This transaction is within the scope of IFRS 2, as it meets the definition of a cash-settled share-based payment transaction. Entity D will be acquiring goods in exchange for a payment, the amount of which will be based on the value of its shares.

If, however, Entity D has a practice of settling these contracts net, or did not intend to take physical delivery, then the forward contract would be within the scope of IAS 32 and IFRS 9 and outside the scope of IFRS 2.

3 Share-based transaction terminology

Section overview

Share-based transactions are agreed between an entity and counterparty at the grant date; the counterparty becomes entitled to the payment at the vesting date.

Before considering the accounting treatment of share-based payment transactions, it is important to understand the terminology used within the topic.

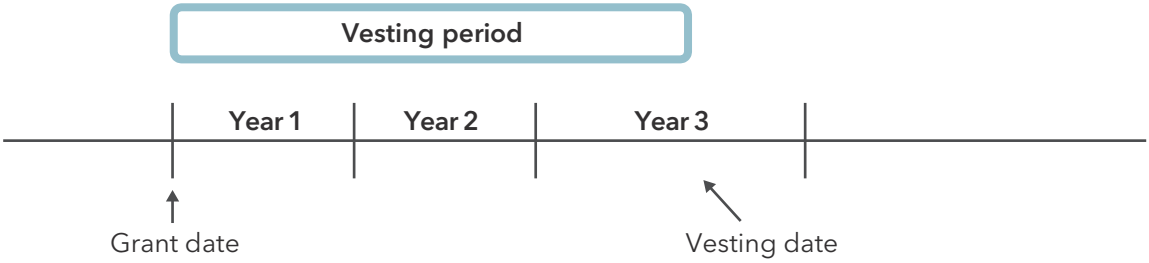


Figure 19.1: Vesting period

Definitions

Grant date: The date at which the entity and other party agree to the share-based payment arrangement. At this date the entity agrees to pay cash, other assets or equity instruments to the other party, provided that specified **vesting conditions**, if any, are met. If the agreement is subject to shareholder approval, then the approval date becomes the grant date.

Vesting conditions: The conditions that must be satisfied for the other party to become entitled to receive the share-based payment.

Vesting period: The period during which the vesting conditions are to be satisfied.

Vesting date: The date on which all vesting conditions have been met and the employee/third party becomes entitled to the share-based payment.

In some cases the grant date and vesting date are the same. This is the case where vesting conditions are met immediately and therefore there is no vesting period.

3.1 Vesting conditions

IFRS 2 recognises two types of vesting conditions:

Non-market-based vesting conditions

These are conditions other than those relating to the market value of the entity's shares. Examples include vesting dependent on:

- the employee completing a minimum period of service (also referred to as a service condition)
- achievement of minimum sales or earnings target
- achievement of a specific increase in profit or earnings per share
- successful completion of a flotation
- completion of a particular project

Market-based vesting conditions

Market-based performance or vesting conditions are conditions linked to the market price of the shares in some way. Examples include vesting dependent on achieving:

- a minimum increase in the share price of the entity
- a minimum increase in shareholder return
- a specified target share price relative to an index of market prices

Restricted definition of vesting conditions

The definition of vesting conditions:

- is restricted to service conditions and performance conditions; and
- excludes other features such as a requirement for employees to make regular contributions into a savings scheme.

Under IFRS 2, features of a share-based payment that are not vesting conditions should be included in the grant date fair value of the share-based payment. The fair value also includes market-related vesting conditions.

Vesting conditions and cancellations

Under IFRS 2, a failure to meet a condition, other than a vesting condition, is treated as a cancellation. Following a 2008 amendment, IFRS 2 specifies the accounting treatment of cancellations by the entity and gives guidance on the treatment of cancellations by parties other than the entity. The amendment requires cancellations by parties other than the entity to be accounted for in the same way as cancellations by the entity.

January 2018 amendment

IFRS 2 was amended in 2018. The 2018 amendments to IFRS 2 have a significant influence on the way companies measure and account for share-based payment transactions. Their purpose is to clarify issues which were not unambiguously defined in the existing standard and, in so doing, to reduce complexity in relation to measurement and classification. There are three areas of amendment:

(a) Classification of share-based payments that have a net settlement feature within the framework of an equity-settled plan

(b) Accounting for modifications that change the classification of payments from cash-settled to equity-settled

(c) The effects of vesting/non-vesting conditions on cash-settled share-based payments

4 Equity-settled share-based payment transactions



Section overview

Where payment for goods or services is in the form of shares or share options, the fair value of the transaction is recognised in profit or loss, spread over the vesting period.

4.1 Introduction

If goods or services are received in exchange for shares or share options, the transaction is accounted for by:

		£	£
DEBIT	Expense/Asset	X	
CREDIT	Equity		X

IFRS 2 does not stipulate which equity account the credit entry is made to. It is normal practice to credit a separate component of equity, although an increasing number of companies are crediting retained earnings.

We must next consider:

(a) Measurement of the total expense taken to profit or loss

(b) When this expense should be recorded

4.2 Measurement

When considering the total expense to profit or loss, the basic principle is that equity-settled share- based transactions are measured at fair value, as the amount for which an asset could be exchanged, a liability settled, or an equity instrument granted could be exchanged, between knowledgeable, willing parties in an arm's length transaction.

Note that this definition is still applicable, rather than the definition in IFRS 13, because IFRS 13 does not apply to transactions within the scope of IFRS 2.

Fair value will depend on who the transaction is with:

(a) There is a rebuttable presumption that the fair value of goods/services received from a **third party** can be measured reliably.

(b) It is not normally possible to measure services received when the shares or share options form part of the remuneration package of employees.

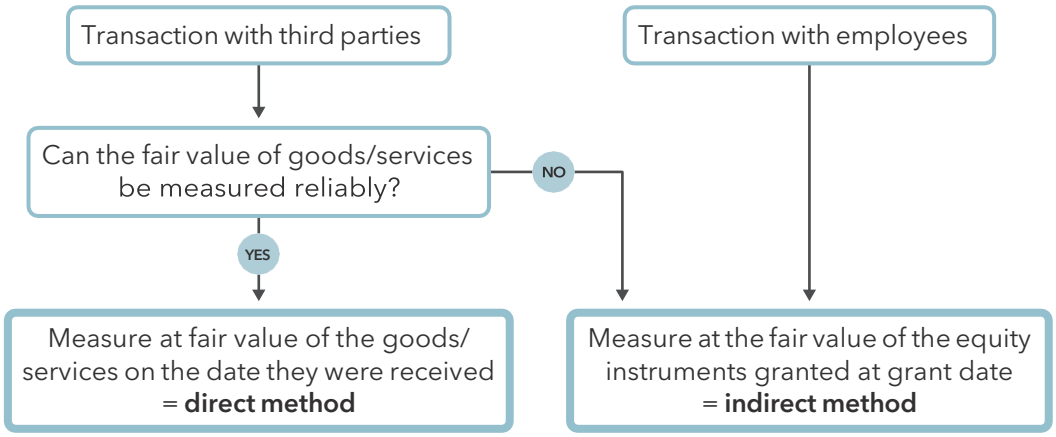


Figure 19.2: Measurement of equity-settled share-based transactions

Professional skills focus: Structuring problems and solutions

The above decision tree is one way to structure a problem if you are diagrammatically minded. This is particularly useful because IFRS 2 does not follow the usual fair value rules in IFRS 13.

4.3 Allocation of expense to financial years

Immediate vesting

Where the instruments granted vest immediately, i.e., the recipient party becomes entitled to them immediately, then the transaction is accounted for in full on the grant date.

Vesting period exists

Where entitlement to the instruments granted is conditional on vesting conditions, and these are to be met over a specified vesting period, the expense is spread over the vesting period.

4.4 Transactions with third parties (non-employees)

Applying the rules seen in the sections above, transactions with third parties are normally:

- measured at the fair value of goods/services received
- recorded when the goods/services are received

Worked example: Third-party transactions – direct method

Entity A has been paying Entity B, a corporate finance consultancy, in cash at the rate of £600 per hour for advice. Entity B is proposing to increase its fees by 5% per annum. Entity A is experiencing cash flow pressures, so it has persuaded Entity B to accept payment in the form of shares with effect from 1 July 20X5. The initial arrangement is for two years with Entity A agreeing to issue 6,000 of its shares to Entity B every six months in exchange for Entity B providing 300 hours of advice evenly over the six-month period.

Requirement

What is the expense in profit or loss and the corresponding increase in equity?

Solution

The services received and the shares issued by Entity A are measured at the fair value of the services received. For the first year, the hourly rate will be measured at that originally proposed by Entity B, 105% of £600. Entity B plans to increase that rate by another 5% for the second year.

The expense in profit or loss and the increase in equity associated with these arrangements will be:

		£
July – December 20X5	300 × £630	189,000
January – December 20X6	(300 × £630) + (300 × £630 × 1.05)	387,450
January – June 20X7	300 × £630 × 1.05	198,450

4.5 Transactions with employees

Applying the rules seen in the sections above, transactions with employees are normally:

- measured at the fair value of equity instruments granted **at grant date**; and
- spread over the vesting period (often a specified period of employment).

Worked example: Employee transactions – indirect method

A company provides each of 10 key employees with 1,000 share options on 1 January 20X7. Each option has a fair value of £9 at the grant date, £11 on 1 January 20X8, £14 on 1 January 20X9 and £12 on 31 December 20X9.

The options do not vest until 31 December 20X9 and are dependent on continued employment. All 10 employees are expected to remain with the company.

Requirement

What are the accounting entries to be recorded in each of the years 20X7, 20X8 and 20X9?

Solution

The changes in the value of equity instruments after grant date do not affect the charge to profit or loss for equity-settled transactions.

Based on the fair value at grant date, the remuneration expense is calculated as follows.

Number of employees × Number of equity instruments × Fair value of equity instruments at grant date

= 10 × 1,000 × £9 = £90,000

The remuneration expense should be recognised over the vesting period of three years. An amount of £30,000 should be recognised for each of the three years 20X7, 20X8 and 20X9 in profit or loss with a corresponding credit to equity.



Interactive question 1: Employee transactions

An entity provides each of its employees with 10 share options at 1 July 20X5, but the options do not vest until 30 June 20X7. The share options may be exercised after vesting date provided that the employees remain in the entity's employment.

The fair value of the share options is £20 on grant date and there are 1,500 employees in the entity's employment at 1 July 20X5.

Requirement

How should the entity account for the transaction if all employees remain in the entity's employment?

See **Answer** at the end of this chapter.



Professional skills focus: Assimilating and using information

Most problems involving share-based payments will be like the above example, with relatively small amounts of information. However, some will be longer, including complications like deferred tax.

4.5.1 Immediate vesting

Some share-based transactions with employees vest immediately. In this case, it is assumed that the relevant services have already been received and so the transaction is recognised on the grant date.



Worked example: Share options vest immediately

An entity issues 10 share options to each of its employees on 1 July 20X5. The share options vest immediately and there is a two-year period over which the employees may exercise the share options. Employees are entitled to exercise the options regardless of whether or not they remain in the entity's employment during the period of exercise. The fair value of the share options is £10 on grant date and there are 1,500 employees in the entity's employment at 1 July 20X5.

Requirement

How should the transaction be accounted for?

Solution

The total fair value for the share options issued at grant date is:

£10 × 1,500 employees × 10 options = £150,000

The entity should therefore charge £150,000 to profit or loss as employee remuneration on 1 July 20X5 and the same amount will be recognised as part of equity on that date.

4.6 The impact of different types of vesting conditions

As we have seen earlier, vesting conditions may be:

- non-market-based ie, not relating to the market value of the entity's shares; or
- market-based ie, linked to the market price of the entity's shares in some way.

4.6.1 Non-market-based vesting conditions

- These conditions are taken into account when determining the expense which must be taken to profit or loss in each year of the vesting period.
- Only the number of shares or share options expected to vest will be accounted for.
- At each period end (including interim periods), the number expected to vest should be revised as necessary.
- On the vesting date, the entity should revise the estimate to equal the number of shares or share options that do actually vest.



Worked example: Non-market-based vesting conditions

On 1 January 20X1 an entity grants 100 share options to each of its 400 employees. Each grant is conditional upon the employee working for the entity until 31 December 20X3. The fair value of each share option at the grant date is £20.

During 20X1 20 employees leave and the entity estimates that a total of 20% of the employees will leave during the three-year period.

During 20X2 a further 25 employees leave and the entity now estimates that 25% of its employees will leave during the three-year period.

During 20X3 a further 10 employees leave.

Requirement

Calculate the remuneration expense that will be recognised in respect of the share-based payment transaction for each of the three years ended 31 December 20X3.

Solution

IFRS 2 requires the entity to recognise the remuneration expense, based on the fair value of the share options granted, as the services are received during the three-year vesting period.

In 20X1 and 20X2 the entity estimates the number of options expected to vest (by estimating the number of employees likely to leave) and bases the amount that it recognises for the year on this estimate.

In 20X3 it recognises an amount based on the number of options that actually vest. A total of 55 employees left during the three-year period and therefore 34,500 options (400 – 55) × 100 vested.

The amount recognised as an expense for each of the three years is calculated as follows.

Cumulative expense

	at year end	Expense for year
	£	£
20X1 100 options × 400 employees × 80% × £20 × 1/3	213,333	213,333
20X2 100 options × 400 employees × 75% × £20 × 2/3	400,000	186,667
20X3 34,500 × £20 × 3/3	690,000	290,000



Interactive question 2: Non-market-based vesting conditions

On 1 January 20X3 an entity grants 500 share options to each of its 400 employees. The only condition attached to the grant is that the employees should continue to work for the entity until 31 December 20X6. 10 employees leave during the year, and it is expected that a further 10 will leave each year.

The market price of each option was £10 at 1 January 20X3 and £12 at 31 December 20X3.

Requirement

Show how this transaction will be reflected in the financial statements for the year ended 31 December 20X3.

See **Answer** at the end of this chapter.



Worked example: Non-market-based vesting conditions

On 1 January 20X4 an entity granted options over 10,000 of its shares to Sally, one of its senior employees. One of the conditions of the share option scheme was that Sally must work for the entity for three years. Sally continued to be employed by the entity during 20X4, 20X5 and 20X6.

A second condition for vesting is that the costs for which Sally is responsible should reduce by 10% per annum compound over the three-year period. At the date of grant, the fair value of each share option was estimated at £21.

At 31 December 20X4 Sally's costs had reduced by 15% and therefore it was estimated that the performance condition would be achieved.

Due to a particularly tough year of trading for the year ended 31 December 20X5 Sally had only reduced costs by 3% and it was thought at that time that she would not meet the cost reduction target by 31 December 20X6.

At 31 December 20X6, the end of the performance period, Sally did meet the overall cost reduction target of 10% per annum compound.

Requirement

How should the transaction be recognised?

Solution

The cost reduction target is a non market performance condition which is taken into account in estimating whether the options will vest. The expense recognised in profit or loss in each of the three years is:

	Cumulative	Charge in the year
	£	£
20X4 (10,000 × £21)/3 years	70,000	70,000
20X5 Assumed performance would not be achieved	0	(70,000)
20X6 10,000 × £21	210,000	210,000

Professional skills focus: Structuring problems and solutions

Share-based payment is a good example of where an answer can be structured and the structure applied routinely, so is a good source of relatively easy marks. The topic is likely to appear fairly regularly, as it is new to Advanced Level.

4.6.2 Market-based vesting conditions

- These conditions are taken into account when calculating the fair value of the equity instruments at the grant date.
- They are not taken into account when estimating the number of shares or share options likely to vest at each period end.
- If the shares or share options do not vest, any amount recognised in the financial statements will remain.

4.6.3 Market and non-market-based vesting conditions

Where equity instruments are granted with both **market** and **non-market** vesting conditions, paragraph 21 of IFRS 2 requires an entity to recognise an expense irrespective of whether market conditions are satisfied, provided **all other vesting conditions** are satisfied.

In summary, where **market** and **non-market** conditions coexist, it makes no difference whether the **market conditions** are achieved. The possibility that the target share price may not be achieved has already been taken into account when estimating the fair value of the options at grant date.

Therefore, the amounts recognised as an expense in each year will be the same regardless of what share price has been achieved. This is the case only with equity-settled share-based payment, not cash-settled share-based payment.

Worked example: Market and non-market vesting conditions

On 1 January 20X4 an entity granted options over 10,000 of its shares to Jeremy, one of its senior employees. One of the conditions of the share option scheme was that Jeremy must work for the entity for three years. Jeremy continued to be employed by the entity during 20X4, 20X5 and 20X6. A second condition for vesting is that the share price increases at 25% per annum compound over the three-year period. At the date of grant the fair value of each share option was estimated at £18 taking into account the estimated probability that the necessary share price growth would be achieved at 25%.

During the year ended 31 December 20X4 the share price rose by 30% and by 26% per annum compound over the two years to 31 December 20X5. For the three years to 31 December 20X6 the increase was 24% per annum compound.

Requirement

How should the transaction be recognised?

Solution

Jeremy satisfied the service requirement but the share price growth condition was not met. The share price growth is a market condition and is taken into account in estimating the fair value of the options at grant date. No adjustment should be made if there are changes from that estimated in relation to the market condition. There is no write-back of expenses previously charged, even though the shares do not vest.

The expense recognised in profit or loss in each of the three years is one-third of 10,000 × £18 = £60,000.



Interactive question 3: Market and non-market performance conditions

Company B issued 100 share options to certain employees, that will vest once revenues reach £1 billion and its share price equals £50. The employee will have to be employed with Company B at the time the share options vest in order to receive the options. The share options had a fair value of £20 at the grant date and will expire in 10 years.

Requirement

How should the expense be recorded under each of the following different scenarios?

(1) All options vest.

(2) Revenues have reached £1 billion, all employees are still employed and the share price is £49.

(3) The share price has reached £50, all employees are still employed but revenues have not yet reached £1 billion.

(4) Revenues have reached £1 billion, the share price has reached £50 and half the employees who received options left the company before the vesting date.

See **Answer** at the end of this chapter.

4.7 Other issues

4.7.1 Transactions during the year

Where the grant date arises mid year, the calculation of the amount charged to profit or loss must be pro-rated to reflect that fact.



Worked example: Options issued during the year

Yarex plc is proposing to award share options to directors and senior employees during the accounting year ending 31 December 20X6.

The proposal is to issue 100,000 options to each of the 50 directors and senior managers on 1 November 20X6.

The exercise price of the options would be £5 per share. The scheme participants will need to have been with the company for at least three years before being able to exercise their options.

It is estimated that 75% of the current directors and senior managers will remain with the company for three years or more.

	1 November 20X6 £	Average 20X6 £	31 December 20X6 £
Price per share	5.00	5.50	5.80 (estimated)
Fair value of each option	2.00	2.40	3.80 (estimated)

Requirement

Show how the proposed scheme would be reflected in the financial statements on 31 December 20X6 and 31 December 20X7. Ignore taxation.

Solution

The three-year service condition specified by the options contract is a non-market vesting condition which should be taken into account when estimating the number of options which will vest at the end of each period. Therefore the proportion of directors expected to remain with the company is relevant in determining the remuneration charge arising from the options.

The fair value for the options used is the fair value at the grant date ie, the fair value of £2 on 1 November 20X6.

The remuneration expense in respect of the options for the year ended 31 December 20X6 is calculated as follows:

Fair value of options expected to vest at grant date:

(75% × 50 employees) × 100,000 options × £2 = £7,500,000 Annual charge to profit or loss therefore £7.5m/3 years = £2.5m

Charge to profit or loss for y/e 31 December 20X6 = £2.5m × 2/12 months = £416,667 The accounting entry for the year ending 31 December 20X6 is:

		£	£
DEBIT	Remuneration expense	416,667	
CREDIT	Equity		416,667

In 20X7 the remuneration charge is for the whole year. Assuming there is no change in the estimated retention rate for employees, the accounting entry is:

		£	£
DEBIT	Remuneration expense	2,500,000	
CREDIT	Equity		2,500,000

4.7.2 Vested options not exercised

If, after the vesting date, options are not exercised or the equity instrument is forfeited, there will be no impact on the financial statements. This is because the holder of the equity instrument has effectively made that decision as an investor.

The services for which the equity instrument remunerated were received by the entity and the financial statements reflect the substance of this transaction. IFRS 2 does, however, permit a transfer to be made between reserves in such circumstances to avoid an amount remaining in a separate equity reserve where no equity instrument will be issued.

4.7.3 Variable vesting date

Where the vesting date is variable depending on non-market-based vesting conditions, the calculation of the amount expensed in profit or loss must be based on the best estimate of when vesting will occur.

Interactive question 4: Variable vesting date

At the beginning of Year 1, Kingsley grants 100 shares each to 500 employees, conditional upon the employees remaining in the entity's employ during the vesting period. The shares will vest at the end of Year 1 if the entity's earnings increase by more than 18%; at the end of Year 2 if the entity's earnings increase by more than an average of 13% per year over the two-year period; and at the end of Year 3 if the entity's earnings increase by more than an average of 10% per year over the three- year period. The shares have a fair value of £30 per share at the start of Year 1, which equals the share price at grant date. No dividends are expected to be paid over the year period.

By the end of Year 1, the entity's earnings have increased by 14%, and 30 employees have left. The entity expects that earnings will continue to increase at a similar rate in Year 2, and therefore expects that the shares will vest at the end of Year 2. The entity expects, on the basis of a weighted average probability, that a further 30 employees will leave during Year 2, and therefore expects that 440 employees will vest in 100 shares at the end of Year 2.

By the end of Year 2, the entity's earnings have increased by only 10% and therefore the shares do not vest at the end of Year 2. 28 employees have left during the year. The entity expects that a further 25 employees will leave during Year 3, and that the entity's earnings will increase by more than 6%, thereby achieving the average of 10% per year.

By the end of Year 3, 23 employees have left and the entity's earnings had increased by 8%, resulting in an average increase of 10.64% per year. Therefore 419 employees received 100 shares at the end of Year 3.

Requirement

Show the expense and equity figures which will appear in the financial statements in each of the three years.

See **Answer** at the end of this chapter.

4.7.4 Modifications and repricing

Equity instruments may be modified before they vest.

Eg, a downturn in the equity market may mean that the original option exercise price set is no longer attractive. Therefore the exercise price is reduced (the option is 'repriced') to make it valuable again.

Such modifications will often affect the fair value of the instrument and therefore the amount recognised in profit or loss.

The accounting treatment of modifications and repricing is as follows:

(a) Continue to recognise the **original** fair value of the instrument in the normal way (even where the modification has reduced the fair value).

(b) Recognise any **increase** in fair value at the modification date (or any increase in the number of instruments granted as a result of modification) spread over the period between the modification date and vesting date.

(c) If modification occurs after the vesting date, then the additional fair value must be recognised immediately unless there is, for example, an additional service period, in which case the difference is spread over this period.

Worked example: Repricing of share options

An entity granted 1,000 share options at an exercise price of £50 to each of its 30 key management personnel on 1 January 20X4. The options only vest if the managers were still

employed on 31 December 20X7. The fair value of the share options was estimated at £20 and the entity estimated that the options would vest with 20 managers. This estimate was confirmed on 31 December 20X4.

The entity's share price collapsed early in 20X5. On 1 July 20X5 the entity modified the share options scheme by reducing the exercise price to £15. It estimated that the fair value of an option was £2 immediately before the price reduction and £11 immediately after. It retained its estimate that options would vest with 20 managers.

Requirement

How should the modification be recognised?

Solution

The total cost to the entity of the original option scheme was:

1,000 shares × 20 managers × £20 = £400,000

This was being recognised at the rate of £100,000 each year.

The cost of the modification is:

1,000 × 20 managers × (£11 – £2) = £180,000

This additional cost should be recognised over 30 months, being the remaining period up to vesting, so £6,000 a month.

The total cost to the entity in the year ended 31 December 20X5 is:

£100,000 + (£6,000 × 6) = £136,000.



Interactive question 5: Repricing of share options

At the beginning of Year 1, an entity grants 100 share options to each of its 500 employees. Each grant is conditional upon the employee remaining in service over the next three years. The entity estimates that the fair value of each option is £15. On the basis of a weighted average probability, the entity estimates that 100 employees will leave during the three-year period and therefore forfeit their rights to the share options.

During the first year, 40 employees leave. By the end of the first year, the entity's share price has dropped, and the entity reprices its share options. The repriced share options vest at the end of Year 3. The entity estimates that a further 70 employees will leave during Years 2 and 3, and hence the total expected employee departures over the three-year vesting period is 110 employees. During Year 2 a further 35 employees leave, and the entity estimates that a further 30 employees will leave during Year 3, to bring the total expected employee departures over the three-year vesting period to 105 employees. During Year 3, a total of 28 employees leave, and hence a total of 103 employees ceased employment during the vesting period. For the remaining 397 employees, the share options vested at the end of Year 3.

The entity estimates that, at the date of repricing, the fair value of each of the original share options granted (ie, before taking into account the repricing) is £5 and that the fair value of each repriced share option is £8.

Requirement

What are the amounts that should be recognised in the financial statements for Years 1 to 3?

See **Answer** at the end of this chapter.

4.7.5 Cancellations and settlements

An entity may settle or cancel an equity instrument during the vesting period. Where this is the case, the correct accounting treatment is as follows:

(a) To immediately charge any remaining fair value of the instrument that has not been recognised in profit or loss (the cancellation or settlement accelerates the charge and does not avoid it).

(b) Any amount paid to the employees by the entity on settlement should be treated as a buyback of shares and should be recognised as a deduction from equity. If the amount of any such payment is in excess of the fair value of the equity instrument granted, the excess should be recognised immediately in profit or loss.



Worked example: Cancellation

An entity granted 2,000 share options at an exercise price of £18 to each of its 25 key management personnel on 1 January 20X4. The options only vest if the managers are still employed by the entity on 31 December 20X6. The fair value of the options was estimated at £33 and the entity estimated that the options would vest with 23 managers. This estimate was confirmed on 31 December 20X4.

In 20X5 the entity decided to base all incentive schemes around the achievement of performance targets and to abolish the existing scheme for which the only vesting condition was being employed over a particular period. The scheme was cancelled on 30 June 20X5 when the fair value of the options was £60 and the market price of the entity's shares was £70. Compensation was paid to the 24 managers in employment at that date, at the rate of £63 per option.

Requirement

How should the entity recognise the cancellation?

Solution

The original cost to the entity for the share option scheme was:

2,000 shares × 23 managers × £33 = £1,518,000

This was being recognised at the rate of £506,000 in each of the three years.

At 30 June 20X5 the entity should recognise a cost based on the amount of options it had vested on that date. The total cost is:

2,000 × 24 managers × £33 = £1,584,000

After deducting the amount recognised in 20X4, the 20X5 charge to profit or loss is £1,078,000. The compensation paid is:

2,000 × 24 × £63 = £3,024,000

Of this, the amount attributable to the fair value of the options cancelled is:

2,000 × 24 × £60 (the fair value of the option, not of the underlying share) = £2,880,000 This is deducted from equity as a share buyback. The remaining £144,000 (£3,024,000 less £2,880,000) is charged to profit or loss.

4.7.6 Cancellation and reissuance

Where an entity has been through a capital restructuring or there has been a significant downturn in the equity market through external factors, an alternative to **repricing** the **share options** is to **cancel** them and issue new options based on revised terms. The end result is essentially the same as an entity modifying the original options and therefore should be recognised in the same way.

4.7.7 Cancellation by parties other than the entity

As well as the entity, other parties may cancel an equity instrument, for example the counterpart (eg, the employee) may cancel.

Cancellations by the employee must be treated in the same way as cancellations by the employer, resulting in an **accelerated charge to profit or loss of the unamortised balance of the options granted**.

4.7.8 Repurchase after vesting

Where equity instruments are repurchased by an employing entity following vesting, this is similar to the entity providing the employee with cash remuneration in the first instance. The reporting therefore reflects this, with the payment being recognised as a deduction from equity. The charge recognised in profit or loss will remain, as this reflects the services for which the employee has been remunerated. If the payment made is in excess of the fair value of the instruments granted, then this is recognised immediately in profit or loss reflecting that this is payment for additional services beyond what was originally agreed.

4.8 Determining the fair value of equity instruments granted

Where a transaction is measured by reference to the fair value of the equity instruments granted, **fair value** is based on **market prices** where available.

If **market prices** are not available, the entity should estimate the **fair value** of the equity instruments granted using a suitable **valuation technique**. These techniques are covered at Professional Level and in your Strategic Business Management Workbook.

4.8.1 Calculating fair value

Share options granted to employees do not generally have a readily obtainable market price because the conditions attached to the options usually make them different from other options that an entity may trade on the open market.

Where there is no readily obtainable market price an entity should use an **option pricing model** to calculate fair value. The type of option pricing model should reflect the nature of the options.
Employee options often have long lives and are generally exercisable between the vesting date and the end of the life of the option; the model used should allow for such circumstances.

All **option pricing models** take into account a number of factors as set out below.

(a) The **exercise price** of the option – a known amount.

(b) The **life of the option** – although the maximum life of the option is a known quantity, this input requires an estimate of the expected life of the individual option. Employee options, for example, are typically exercised soon after vesting date, as this is generally the only way that an employee can crystallise any gain.

(c) The **current price** of **the underlying shares** – this is generally a known amount (the listed market value of the shares).

(d) The **expected volatility of the share price** – volatility is typically expressed in annualised terms for ease of comparability. The expected annualised volatility of a share is expressed in terms of the compounded annual rate of return that is expected to arise approximately two-thirds of the time. Volatility (also discussed in detail in your Strategic Business Management Workbook) is a measure of the amount by which a share price is expected to fluctuate during a period. For example, a share worth £100 with a volatility of 40% would suggest that it will be worth between £60 and £140 approximately two-thirds of the time between the grant date and the exercise of the options.

(e) The **dividend expected** on the shares – this should only be included where the employee is not entitled to dividends on the underlying options granted.

(f) The **risk-free interest rate** for the life of the option – this is typically "the implied yield currently available on zero-coupon government issues of the country in whose currency the exercise price is expressed".

Factors which are typically used to assess volatility include the **historical volatility** of the entity's share price and the length of time that the shares have been publicly traded.

4.8.2 Which valuation model?

The determination of an appropriate model for the valuation of share-based payment transactions is an accounting policy choice and should be applied consistently to similar types of transactions.

In choosing an appropriate model the key question is whether the model used to estimate fair value represents the economics of the instruments and whether the inputs represent the attributes being measured.

The Black-Scholes model

The Black-Scholes-Merton formula (normally referred to as the Black-Scholes model) is a popular model and one that is easy to use. However, the model is based on the assumption that options are not exercised before the end of their lives and therefore may not be suitable for valuing employee share options. Where such options have a relatively short life and the period after vesting is short, the Black-Scholes model may provide a reasonable approximation of fair value.

Strengths

- The formula required to calculate fair value is relatively straightforward and can be easily used in excel.
- Its wide use enhances comparability.

Weaknesses

The inputs and assumptions of the Black-Scholes model are designed to cover the entire period the option is outstanding. Because of this feature the model is described as a 'closed form solution'.

- The model cannot therefore be adjusted to take into account anticipated changes in market conditions or incorporate different values for variables (such as volatility) over the term of the option as perceived at the grant date. The Black-Scholes model assumes constant volatility and cannot therefore be adjusted to take into account the empirical observation that the implied volatility of a share option changes as the intrinsic value of the option changes.
- The model cannot take into account the possibility of early exercise. This is less of an issue when options have to be exercised on or shortly after vesting.

The Binomial model

The Binomial model applies the same principles as decision tree analysis to the pricing of an option. Based on the relative probabilities of each path, an expected outcome is estimated.

In contrast to the Black-Scholes model, the Binomial model can incorporate different values for the variables over the term of the option. Therefore it is described as an 'open form solution' and it can be adjusted to take into account changes in market conditions and the input variables.

Strengths

- The Binomial model is generally accepted as a more flexible alternative to the Black-Scholes model.
- The inputs into the model are more suitable for an option with a longer term.

Weaknesses

- In practice, the application of the model is more complex than the Black-Scholes model. Whereas the Black-Scholes model allows the value of an option to be calculated using a relatively simple Formulae, the Binomial model requires a more complex formulae in excel or program to calculate the option value.
- The calculation of the probabilities of particular price movements is highly subjective.

Monte-Carlo simulation

Monte-Carlo simulation extends the Binomial model by undertaking thousands of simulations of potential future outcomes for key assumptions and calculating the option value under each scenario. Monte-Carlo models can incorporate very complex performance conditions and exercise patterns.

They are generally considered the best type of model for valuing employee share-based payments, although they are also affected by the subjectivity of probabilities.

5 Cash-settled share-based payment transactions



Section overview

- The credit entry in respect of a cash-settled share-based payment transaction is reported as a liability.
- The fair value of the liability should be remeasured at each reporting date until settled. Changes in the fair value are recognised in profit or loss.

5.1 Introduction

Cash-settled share-based payment transactions are transactions where the amount of cash paid for goods and services is based on the value of an entity's equity instruments.

Examples of this type of transaction include:

(a) share appreciation rights (SARs): the employees become entitled to a future cash payment (rather than an equity instrument), based on the increase in the entity's share price from a specified level over a specified period of time; or

(b) an entity might grant to its employees a right to receive a future cash payment by granting to them a right to shares that are redeemable.

5.2 Accounting treatment

If goods or services are received in exchange for cash amounts linked to the value of shares, the transaction is accounted for by:

		£	£
DEBIT	Expense/Asset	X	
CREDIT	Liability		X

Allocation of expense to financial years

The expense should be recognised as services are provided. For example, if share appreciation rights do not vest until the employees have completed a specified period of

service, the entity should recognise the services received and the related liability, over that period.

Measurement

The goods or services acquired and the liability incurred are measured at the **fair value** of the **liability**.

The entity should **remeasure** the **fair value** of the liability at each reporting date and at the date of settlement. Any changes in fair value are recognised in profit or loss for the period.

Vesting conditions

Vesting conditions should be taken into account in a similar way as for equity-settled transactions when determining the number of rights to payment that will vest.



Worked example: Cash-settled share-based payment transaction

On 1 January 20X1 an entity grants 100 cash SARs to each of its 500 employees, on condition that the employees continue to work for the entity until 31 December 20X3.

During 20X1 35 employees leave. The entity estimates that a further 60 will leave during 20X2 and 20X3.

During 20X2 40 employees leave and the entity estimates that a further 25 will leave during

20X3. During 20X3 22 employees leave.

At 31 December 20X3 150 employees exercise their SARs. Another 140 employees exercise their SARs at 31 December 20X4 and the remaining 113 employees exercise their SARs at the end of 20X5.

The fair values of the SARs for each year in which a liability exists are shown below, together with the intrinsic values at the dates of exercise.

	Fair value	Intrinsic value
	£	£
20X1	14.40	
20X2	15.50	
20X3	18.20	15.00
20X4	21.40	20.00
20X5		25.00

Requirement

Calculate the amount to be recognised in profit or loss for each of the five years ended 31 December 20X5 and the liability to be recognised in the statement of financial position at 31 December for each of the five years.

Solution

For the three years to the vesting date of 31 December 20X3 the expense is based on the entity's estimate of the number of SARs that will actually vest (as for an equity-settled transaction). However, the fair value of the liability is remeasured at each year end.

The intrinsic value of the SARs at the date of exercise is the amount of cash actually paid.

	Liability at year end		Expense for year
	£	£	£
20X1 Expected to vest (500 – 95):			
405 × 100 × £14.40 × 1/3	194,400		194,400
20X2 Expected to vest (500 – 100):			
400 × 100 × £15.50 × 2/3	413,333		218,933
20X3 Exercised:			
150 × 100 × £15.00		225,000	
Not yet exercised (500 – 97 – 150):			
253 × 100 × £18.20	460,460	47,127	
			272,127
20X4 Exercised:			
140 × 100 × £20.00		280,000	
Not yet exercised (253 – 140):			
113 × 100 × £21.40	241,820	(218,640)	
			61,360
20X5 Exercised:			
113 × 100 × £25.00		282,500	
	Nil	(241,820)	
			40,680
			787,500



Interactive question 6: Share-based payment

J&B granted 200 options on its £1 ordinary shares to each of its 800 employees on 1 January 20X1. Each grant is conditional upon the employee being employed by J&B until 31 December 20X3.

J&B estimated at 1 January 20X1 that:

- the fair value of each option was £4 (before adjustment for the possibility of forfeiture); and
- approximately 50 employees would leave during 20X1, 40 during 20X2 and 30 during 20X3 thereby forfeiting their rights to receive the options. The departures were expected to be evenly spread within each year.

The exercise price of the options was £1.50 and the market value of a J&B share on 1 January 20X1 was £3.

In the event, only 40 employees left during 20X1 (and the estimate of total departures was revised down to 95 at 31 December 20X1), 20 during 20X2 (and the estimate of total departures was revised to 70 at 31 December 20X2) and none left during 20X3. The departures were spread evenly during each year.

The directors of J&B have asked you to illustrate how the scheme is accounted for under IFRS 2, *Share-based Payment*.

Requirements

6.1 Show the double entries for the charge to profit or loss for employee services over the three years and for the share issue, assuming all employees entitled to benefit from the scheme exercised their rights and the shares were issued on 31 December 20X3.

6.2 Explain how your solution would differ had J&B offered its employees cash, based on the share value rather than share options.

See **Answer** at the end of this chapter.

6 Share-based payment with a choice of settlement

Section overview

Accounting for share-based transactions with a choice of settlement depends on which party has the choice.

- Where the counterparty has a choice of settlement, a liability component and an equity component are identified.
- Where the entity has a choice of settlement, the whole transaction is treated either as cash-settled or as equity-settled, depending on whether the entity has an obligation to settle in cash.

6.1 Counterparty has the choice

Where the counterparty or recipient, rather than the issuing entity, has the right to choose the form settlement will take, IFRS 2 regards the transaction as a compound financial instrument to which split accounting must be applied.

This means that the entity has issued an instrument with a debt component in so far as the recipient may demand cash and an equity component to the extent that the recipient may demand settlement in equity instruments.

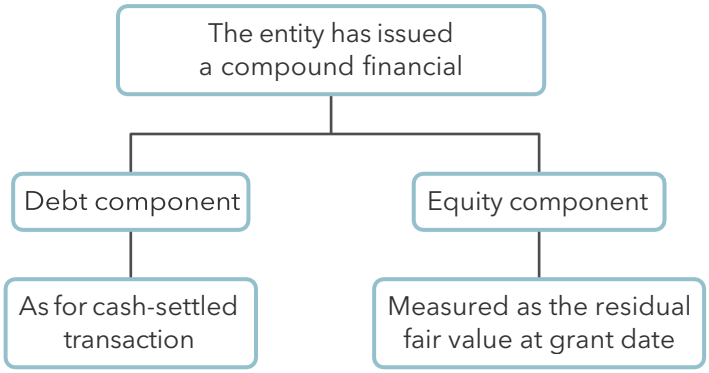


Figure 19.3: Choice of settlement

IFRS 2 requires that the value of the debt component is established first. The equity component is then measured as the residual between that amount and the value of the instrument as a whole. In this respect IFRS 2 applies similar principles to IAS 32, *Financial Instruments: Presentation* where the value of the debt components is established first. However, the method used to value the constituent parts of the compound instrument in IFRS 2 differs from that of IAS 32.



Figure 19.4: Compound instrument and IFRS 2

For transactions in which the fair value of goods or services is measured directly (that is, normally where the recipient is not an employee of the company), the fair value of the equity component is measured as the difference between the fair value of the goods or services required and the fair value of the debt component.

For other transactions including those with employees where the fair value of the goods or services is measured indirectly by reference to the fair value of the equity instruments granted, the fair value of the compound instrument is estimated as a whole.

The debt and equity components must then be valued separately. Normally transactions are structured in such a way that the fair value of each alternative settlement is the same.

Worked example: Choice of settlement

On 1 January 20X4 an entity grants an employee a right under which she can, if she is still employed on 31 December 20X6, elect to receive either 8,000 shares or cash to the value, on that date, of 7,000 shares.

The market price of the entity's shares is £21 at the date of grant, £27 at the end of 20X4, £33 at the end of 20X5 and £42 at the end of 20X6, at which time the employee elects to receive the shares. The entity estimates the fair value of the share route to be £19.

Requirement

Show the accounting treatment.

Solution

This arrangement results in a compound financial instrument.

The fair value of the cash route is:

7,000 × £21 = £147,000

The fair value of the share route is:

8,000 × £19 = £152,000

The fair value of the equity component is therefore:

£5,000 (£152,000 less £147,000)

The share-based payment is recognised as follows:

		Liabilities £	Equity £	Expense £
20X4	1/3 × 7,000 × £27	63,000		63,000
	£5,000 × 1/3		1,667	1,667
20X5	2/3 × 7,000 × £33	154,000		91,000
	£5,000 × 1/3		1,667	1,667
20X6	7,000 × £42	294,000		140,000
	£5,000 × 1/3		1,667	1,667

As the employee elects to receive shares rather than cash, £294,000 is transferred from liabilities to equity at the end of 20X6. The balance on equity is £299,000.

Tutorial Note

In this example, which is based on the one in IFRS 2, there is only one employee. The standard is therefore unclear on whether the equity element should be adjusted for the number of employees. If there were 700 employees meeting the vesting conditions at grant date, expected to fall to 600 a year later for example. Both answers i.e., adjusting the equity element and not adjusting equity element would be accepted.

6.2 Entity has the choice

Where the entity may choose what form the settlement will take, it should recognise a liability to the extent that it has a present obligation to deliver cash. Such circumstances arise where, for example, the entity is prohibited from issuing shares or where it has a stated policy, or past practice, of issuing cash rather than shares. Where a present obligation exists, the entity should record the transaction as if it is a cash-settled share-based payment transaction. If no present obligation exists, the entity should treat the transaction as if it was purely an equity-settled transaction. On settlement, if the transaction was treated as an equity-settled transaction and cash was paid, the cash should be treated as if it was a repurchase of the equity instrument by a deduction against equity.

Interactive question 7: Share-based payment

Woodley plc is one of your assurance clients, and has asked you to advise on how to apply IFRS 2 to its new share option scheme. The company's reporting period is to 31 December.

The scheme is open to all 450 employees and all options are granted on 1 January 20X7. The fair value of each option is £15 on 1 January 20X7. The company estimates that this fair value will rise by approximately £5 per year. Each employee is given 100 options.

The vesting of the share option depends on achieving two independent targets. The first target is that the share price must have increased by a total of at least 10% in order for the options to vest.

The second target is that shares can vest when profits increase by 15% in any year, or by an average of 12% in any two years. The scheme will be cancelled after four years.

In 20X7, profits increase by 10% and the share price has risen by 5%. The shares do not vest, but at 31 December 20X7 the forecast increase in profits for 20X8 is 14%, and the forecast increase in share price for 20X8 is 5%. It is therefore anticipated the shares will vest in 20X8.

30 employees leave in 20X7 and it is estimated that a further 25 will leave before the options vest.

In 20X8, profits increase by 13% so the shares do not vest, although the share price target has now been achieved. 15 employees left during the year and it is anticipated that a further 26 will leave before the scheme is expected to vest in 20X9 (forecast profit increase for 20X9 is 12%).

In 20X9, profits increased by the forecast 12%, so the options vest. 390 employees ultimately received their options.

Requirements

7.1 Explain the principles of how this scheme should be measured and recognised. Calculate the IFRS 2 expense and set out the double entries required for 20X7, 20X8 and 20X9.

7.2 Describe how your answer would be different if in 20Y0, 100 employees allowed their vested share options to lapse.

7.3 How would your answer to part 1 be different if the actual increase in profits for the year to 31 December 20X9 was 10% and, at that date, it was forecast that profits for 20Y0 were to increase by 15%, but the actual increase achieved in 20Y0 was 9%? If the targets related to share price and not profits, describe how to account for failing to meet the targets set.

See **Answer** at the end of this chapter.

7 Group and treasury share transactions



Section overview

IFRS 2 gives guidance on group and treasury share transactions.

7.1 Background

IFRS 2 gives guidance on group and treasury shares in three circumstances:

- Where an entity grants rights to its own equity instruments to employees, and then either chooses or is required to buy those equity instruments from another party, in order to satisfy its obligations to its employees under the share-based payment arrangement
- Where a parent company grants rights to its equity instruments to employees of its subsidiary
- Where a subsidiary grants rights to equity instruments of its parent to its employees

7.2 Accounting treatment

7.2.1 Entity chooses or is required to purchase its own shares

These transactions should always be accounted for as equity-settled share-based payment transactions under IFRS 2.

7.2.2 Parent grants rights to its equity instruments to employees of its subsidiary

Assuming the transaction is accounted for as equity-settled in the consolidated financial statements, the subsidiary must measure the services received using the requirements for equity-settled transactions in IFRS 2, and must recognise a corresponding increase in equity as a contribution from the parent.

7.2.3 Subsidiary grants rights to equity instruments of its parent to its employees

The subsidiary accounts for the transaction as a cash-settled share-based payment transaction.

Therefore, in the subsidiary's individual financial statements, the accounting treatment of transactions in which a subsidiary's employees are granted rights to equity instruments of its parent would differ, depending on whether the parent or the subsidiary granted those rights to the subsidiary's employees. This is because in the former situation, the subsidiary has not incurred a liability to transfer cash or other assets of the entity to its employees, whereas it has incurred such a liability in the latter situation (being a liability to transfer equity instruments of its parent).

8 Disclosure

Section overview

The disclosures of IFRS 2 are extensive and require the analysis of share-based payments made during the year, their impact on earnings and the financial position of the company and the basis on which fair values were calculated.

8.1 Nature and extent of share-based payment arrangements in the period

(a) A **description** of each type of share-based payment arrangement that existed at any time during the period, including the general terms and conditions of each arrangement.

The disclosure requirements of IFRS 2 are designed to enable the user of financial statements to understand:

- The nature and extent of share-based payment arrangements that existed during the period
- The basis upon which fair value was measured
- The impact of share-based transactions on earnings and financial position

(b) The **number and weighted average exercise prices** of share options for each of the following groups of options.

- Outstanding at the beginning of the period
- Granted during the period
- Forfeited during the period
- Exercised during the period (plus the weighted average share price at the time of exercise)
- Expired during the period
- Outstanding at the end of the period
- Exercisable at the end of the period

(c) For share options **exercised** during the period, the **weighted average share price** at the date of exercise.

(d) For share options **outstanding** at the end of the period, the range of exercise prices and weighted average remaining contractual life.

8.2 Disclosable transactions under IAS 24

Share-based payments in respect of key management personnel and related parties have to be disclosed in accordance with IAS 24, *Related Party Disclosures*.

8.3 Basis of fair value measurement

IFRS 2 requires disclosure of information that enables users of the financial statements to **understand how the fair value** of the goods or services received, or the fair value of the equity instruments granted, during the period **was determined**.

For equity instruments the disclosure of fair value methodology applies to both:

- new instruments issued during the reporting period; and
- existing instruments modified during the reporting period.

The entity must disclose the option pricing model used and the inputs to that model. These will include at least:

- the weighted average share price
- the exercise price
- the expected volatility of the share price
- the life of the option
- the expected dividends on the underlying share
- the risk-free interest rate over the life of the option
- the method used and the assumptions made to incorporate the effect of early exercise

8.4 Impact on earnings and financial position

Entities should also disclose information that enables users of the financial statements to understand the effect of share-based payment transactions on the entity's **profit or loss for the period** and on its **financial position**.

(a) The **total expense recognised for the period** arising from share-based payment transactions, including separate disclosure of that portion of the total expense that arises from transactions accounted for as equity-settled share-based payment transactions

(b) For **liabilities** arising from share-based payment transactions:

  (1) The **total carrying amount** at the end of the period

  (2) The **total intrinsic value** at the end of the period of liabilities for which the counterparty's right to cash or other assets had vested by the end of the period

The disclosure requirements of IFRS 2 are illustrated by an example below.

8.5 Model disclosures

(Note that comparative figures have been omitted from the disclosure.)

Share options are granted to both directors and employees, with the exercise price always set at the market price at the date of grant. Conditions of the options typically include sales growth and cost- reduction targets over a three-year period from the date of grant. Options which vest are exercisable over the subsequent four years.

For the year the number and weighted average exercise prices were as follows.

	Options	Average exercise price
		£
At start of year	163,000	2.00
Granted	50,000	3.00
Forfeited	(8,000)	–
Exercised	(30,000)	1.80
Expired	(6,000)	–
At end of year	169,000	2.40

Details of the 169,000 outstanding options are as follows:

- The range of exercise prices is £1.50 to £3.00 and the weighted average remaining life is 5.1 years.
- 72,000 are currently exercisable.

The weighted average share price at the date the 30,000 options were exercised during the year was £2.95.

The fair value of the options granted, all of which were granted on 18 June, was £5.60, based on the Black-Scholes model. The key inputs to that model were a weighted average share price of £3.50, an exercise price of £3.00, expected volatility (based on historic volatility) of 28% and a risk-free interest rate of 4% per annum.

The total expense for share options recognised in the year was £280,000.

8.6 Impact of share-based payments on earnings per share (EPS)

IAS 33, *Earnings per Share* requires that for calculating diluted EPS all dilutive options need to be taken into account. Employee share options with fixed terms and non-vested ordinary shares are treated as options outstanding on grant date even though they may not have vested on the date the diluted EPS is calculated. All awards which do not specify performance criteria are treated as options.

Employee share options contingent on performance-related conditions are treated as contingently issuable shares and are dealt with in detail in earlier chapters.

9 Distributable profits and purchase of own shares

Section overview

Various rules have been created to ensure that dividends are only paid out of **distributable profits**.

Definition

Dividend: An amount payable to shareholders from profits or other distributable reserves.

9.1 Power to declare dividends

A company may only pay dividends out of **profits available for the purpose.**

The power to declare a dividend is given by the articles which often include the following rules.

Rules related to the power to declare a dividend
The **company** in **general meeting** may declare dividends.
No dividend may exceed the **amount recommended** by the directors who have an implied power in their discretion to set aside profits as reserves.
The directors may declare such **interim dividends** as they consider justified.
Dividends are normally declared payable on the **paid-up amount** of **share capital**. For example, a £1 share which is fully paid will carry entitlement to twice as much dividend as a £1 share 50p paid.
A dividend may be paid **otherwise than in cash**.
Dividends may be paid by **cheque** or **warrant** sent through the post to the shareholder at his registered address or in any **electronic mode**. If shares are held jointly, payment of dividend is made to the first-named joint holder on the register.

Listed companies generally pay two dividends; an **interim dividend** based on interim profit figures, and a **final dividend** based on the annual accounts and approved at the AGM.

A **dividend becomes a debt** when it is **declared** and **due for payment**. A shareholder is not entitled to a dividend unless it is declared in accordance with the procedure prescribed by the articles and the declared date for payment has arrived.

This is so even if the member holds **preference shares** carrying a priority entitlement to receive a specified amount of dividend on a specified date in the year. The directors may decide to withhold profits and cannot be compelled to recommend a dividend.

If the articles refer to 'payment' of dividends this means **payment in cash**. A power to pay dividends **in specie** (otherwise than in cash) is not implied but may be expressly created. **Scrip dividends** are dividends paid by the issue of additional shares.

Any provision of the articles of association for the declaration and payment of dividends is subject to the overriding rule that **no dividend may be paid except out of profits distributable by law**.

9.2 Distributable profits

Tutorial note

Distributable profits may be defined as 'accumulated realised profits ... less accumulated realised losses'. **'Accumulated'** means that any losses of previous years must be included in reckoning the current distributable surplus. **'Realised'** profits are determined in accordance with generally accepted accounting principles.

Definition

Profits available for distribution: Accumulated realised profits (which have not been distributed or capitalised) less accumulated realised losses (which have not been previously written off in a reduction or reorganisation of capital).

The word **'accumulated'** requires that any **losses** of **previous years** must be included in reckoning the current distributable surplus.

A profit or loss is deemed to be **realised** if it is treated as realised in accordance with generally accepted accounting principles (GAAP). Hence, financial reporting and accounting standards in issue, plus GAAP, should be taken into account when determining realised profits and losses.

Depreciation must be treated as a **realised loss**, and debited against profit, in determining the amount of distributable profit remaining.

However, a **revalued asset** will have depreciation charged on its historical cost and the increase in the value in the asset.

Effectively there is a cancelling out, and at the end **only depreciation that relates to historical cost will affect dividends**.

Context example: Depreciation charge

Suppose that an asset purchased for £20,000 has a 10-year life. Provision is made for depreciation on a straight-line basis. This means the annual depreciation charge of £2,000 must be deducted in reckoning the company's realised profit less realised loss.

Suppose now that after five years the asset is revalued to £50,000 and in consequence the annual depreciation charge is raised to £10,000 (over each of the five remaining years of the asset's life).

The effect of the Act is that £8,000 of this amount may be reclassified as a realised profit. The net effect is that realised profits are reduced by only £2,000 in respect of depreciation, as before.

If, on a general revaluation of all fixed assets, it appears that there is a diminution in value of any one or more assets, then any related provision(s) need **not** be treated as a realised loss.

9.3 Dividends of public companies

A public company may only make a distribution if its **net assets are**, at the time, **not less than the aggregate of its called-up share capital and undistributable reserves**. The dividend which it may pay is limited to such amount as will leave its net assets at not less than that aggregate amount.

Undistributable reserves are defined as follows:

(a) Share premium account

(b) Capital redemption reserve

(c) Any surplus of accumulated unrealised profits over accumulated unrealised losses (known as a revaluation reserve). However, a deficit of accumulated unrealised profits compared with accumulated unrealised losses must be treated as a realised loss

(d) Any reserve which the company is prohibited from distributing by statute or by its constitution or any law



Context example: Permissible dividend

Suppose that a public company has an issued share capital (fully paid) of £800,000 and £200,000 on share premium account (which is an undistributable reserve). If its assets less liabilities are less than £1 million it may not pay a dividend. However, if its net assets are, say, £1,250,000 it may pay a dividend but only of such amount as will leave net assets of £1 million or more, so its maximum permissible dividend is £250,000.

The dividend rules apply to every form of distribution of assets except the following:

- The **issue of bonus shares** whether fully or partly paid
- The **redemption** or **purchase** of the company's **shares** out of **capital** or **profits**
- A **reduction** of **share capital**
- A **distribution** of **assets** to members in a **winding up**

You must appreciate how the rules relating to public companies in this area are more stringent than the rules for private companies.



Interactive question 8: Main rules

What are the main rules affecting a company's ability to distribute its profits as dividends?

See **Answer** at the end of this chapter.

9.4 Relevant accounts

Tutorial note

The profits available for distribution are generally determined from the **last annual accounts** to be prepared.

Whether a company has profits from which to pay a dividend is determined by reference to its

'relevant accounts', which are generally the last annual accounts to be prepared.

If the auditor has qualified their report on the accounts they must also state in writing whether, in their opinion, the subject matter of their qualification is **material** in determining whether the dividend may be paid. This statement must have been circulated to the members (for a private company) or considered at a general meeting (for a public company).

A company may produce **interim accounts** if the latest annual accounts do not disclose a sufficient distributable profit to cover the proposed dividend. It may also produce **initial accounts** if it proposes to pay a dividend during its first accounting reference period or before its first accounts are laid before the company in general meeting. These accounts may be unaudited, but they must suffice to permit a proper judgement to be made of amounts of any of the relevant items.

If a **public** company has to produce initial or interim accounts, which is unusual, they must be full accounts such as the company is required to produce as final accounts at the end of the year. They need not be audited. However, the auditors must, in the case of initial accounts, satisfy themselves that the accounts have been 'properly prepared' to comply with the Act. A copy of any such accounts of a public company (with any auditors' statement) must be delivered to RJSC. Refer to Professional level Corporate Laws & Practices workbook for regulatory requirements relating to dividend.

If an unlawful dividend is paid by **reason of error** in the **accounts** the company may be unable to claim against either the directors or the members. The company might then have a claim against its **auditors** if the undiscovered mistake was due to negligence on their part.

Re London & General Bank (No 2) 1895

The facts: The auditor had drawn the attention of the directors to the fact that certain loans to associated companies were likely to prove irrecoverable. The directors refused to make any provision for these potential losses. They persuaded the auditor to confine his comments in his audit report to the uninformative statement that the value of assets shown in the statement of financial position 'is dependent on realisation'. A dividend was paid in reliance on the apparent profits shown in the accounts. The company went into liquidation and the liquidator claimed compensation from the auditor for loss of capital due to his failure to report clearly to members what he well knew was affecting the reliability of the accounts.

Decision: The auditor has a duty to report what he knows of the true financial position: otherwise his audit is 'an idle farce'. He had failed in this duty and was liable.



Professional skills focus: Concluding, recommending and communicating

The above case, in which the auditor was found wanting, shows that communication can fail by being too vague or incomplete, as well as by stating something misleading.

9.5 Purchase of own shares

Tutorial note

You must be able to carry out **simple calculations** showing the amounts to be transferred to the **capital redemption reserve** on purchase or redemption of own shares and how the amount of any **premium** on redemption would be treated.

Any limited company is permitted without restriction to cancel unissued shares and in that way to reduce its authorised share capital. That change does not alter its financial position. Refer to Professional level Corporate Laws & Practices workbook for regulatory requirements relating to reduction of share capital. There are **three basic methods of reducing share capital** specified.

(a) **Extinguish or reduce liability on partly paid shares**. A company may have issued £1 (nominal) shares 75p paid up. The outstanding liability of 25p per share may be eliminated altogether by reducing each share to 75p (nominal) fully paid or some intermediate figure eg, 80p (nominal) 75p paid. Nothing is returned to the shareholders but the company gives up a claim against them for money which it could call up whenever needed.

(b) **Cancel paid-up share capital which has been lost or which is no longer represented by available assets.** Suppose that the issued shares are £1 (nominal) fully paid but the net assets now represent a value of only 50p per share. The difference is probably matched by a debit balance on retained earnings (or provision for fall in value of assets). The company could reduce the nominal value of its £1 shares to 50p (or some intermediate figure) and apply the amount to write off the debit balance or provision wholly or in part. It would then be able to resume payment of dividends out of future profits without being obliged to make good past losses. The resources of the company are not reduced by this procedure of part cancellation of nominal value of shares but it avoids having to rebuild lost capital by retaining profits.

(c) **Pay off part of the paid-up share capital out of surplus assets.** The company might repay to shareholders, say, 30p in cash per £1 share by reducing the nominal value of the share to 70p. This reduces the assets of the company by 30p per share.

9.6 Share premium account

Whenever a company obtains for its shares a consideration in excess of their nominal value, it must transfer the excess to a share premium account. The general rule is that the **share premium account is subject to the same restriction as share capital**. **However, a bonus issue can be made using the share premium account** (reducing share premium in order to increase issued share capital).

The **other permitted uses of share premium** are to pay the following:

(a) Capital expenses such as preliminary expenses of forming the company

(b) Discount on the issue of shares or debentures

(c) Premium (if any) paid on redemption of debentures

Private companies (but not public companies) may also use a share premium account in purchasing or redeeming their own shares out of capital. Refer to Professional level Corporate Laws & Practices workbook for regulatory requirements relating to share premium account.

9.7 Practical reasons for purchase or redemption

Companies may wish to repurchase or redeem their issued shares for a variety of reasons.

(a) The company may have **surplus funds** for which it **cannot identify** sufficient **attractive business opportunities**.

(b) A reduction in the number of issued shares helps to **improve earnings per share (EPS)** and **return on capital employed** (ROCE).

(c) Dividend payments may be reduced, allowing the **cash** to be **used** for **other purposes**:

- funding operating activities
- capital expenditure
- repayment of debt

(d) The remaining shareholders' holdings will proportionately increase. Hence, even if the overall total dividends might not increase, some **shareholders** could receive **more cash individually**.

(e) Problem or **dissident shareholders** in private companies can be **paid off** and leave the company without spreading the membership of the company beyond the existing shareholders.

(f) It provides a potential **exit route** for **venture capitalists** who intend to be involved in the business for a limited period.

(g) It provides an **escape route** for **entrepreneurs** who have taken their companies to market to take them **back into private ownership**.

Refer to Professional level Corporate Laws & Practices workbook for regulatory requirements relating to Purchase or redemption by a company of its own shares

Note: Following IAS 32, **redeemable preference shares** are no longer classified as equity; they are classified as **financial liabilities**.

Context example: Capitalisation of profits

Suppose, for example, that Muffin Ltd decided to repurchase and cancel £100,000 of its ordinary share capital. A statement of financial position of the company is currently as follows.

	£
Assets	
Cash	100,000
Other assets	300,000
	400,000
Equity and liabilities	
Ordinary shares	130,000
Retained earnings	150,000
Trade payables	120,000
	400,000

Now if Muffin Ltd were able to repurchase the shares without making any transfer from the retained earnings to a capital redemption reserve, the effect of the share redemption on the statement of financial position would be as follows.

	£
Net assets	
Non-cash assets	300,000
Less trade payables	120,000
	180,000
Equity	
Ordinary shares	30,000
Retained earnings	150,000
	180,000

In this example, the company would still be able to pay dividends out of profits of up to £150,000. If it did, the creditors of the company would be highly vulnerable, financing £120,000 out of a total of £150,000 assets of the company.

The regulations in the Act are intended to prevent such extreme situations arising. On repurchase of the shares, Muffin Ltd would have been required to transfer £100,000 from its retained earnings to a non-distributable reserve, called a capital redemption reserve. The effect of the redemption of shares on the statement of financial position would have been:

	£	£
Net assets		
Non-cash assets		300,000
Less trade payables		120,000
		180,000
Equity		
Ordinary shares		30,000
Reserves		
Distributable (retained earnings)	50,000	
Non-distributable (capital redemption reserve)	100,000	
		150,000
		180,000

The maximum distributable profits are now £50,000. If Muffin Ltd paid all these as a dividend, there would still be £250,000 of assets left in the company, just over half of which would be financed by non-distributable equity capital.

When a company redeems some shares, or purchases some of its own shares, they **should be redeemed**:

(a) **out of distributable profits; or**

(b) **out of the proceeds of a new issue of shares.**

If there is any premium on redemption, **the premium must be paid out of distributable profits**, except that if the shares were issued at a premium, then any premium payable on their redemption may be paid out of the proceeds of a new share issue made for the purpose, up to an amount equal to the lesser of the following:

(a) The aggregate premiums received on issue of the shares

(b) The balance on the share premium account (including premium on issue of the new shares)



Context example: Repurchase of shares

A numerical example might help to clarify this point. Suppose that Just Desserts Ltd intends to repurchase 10,000 shares of £1 each at a premium of 5p per share. The redemption may be financed out of:

Distributable profits (10,000 × £1.05 = £10,500).

The proceeds of a new share issue (say, by issuing 10,000 new £1 shares at par). The premium of £500 must be paid out of distributable profits. Combination of a new share issue and distributable profits.

Out of the proceeds of a new share issue where the shares to be repurchased were issued at a premium. For example, if the shares had been issued at a premium of 3p per share, then (assuming that the balance on the share premium account after the new share issue was at least £300) £300 of the premium on redemption could be debited to the share premium account and only £200 need be debited to distributable profits.

(1) Where a company purchases its own shares wholly out of distributable profits, it must transfer to the capital redemption reserve an amount equal to the nominal value of the shares repurchased.

In this example, the accounting entries would be:

		£	£
DEBIT	Share capital account	10,000	
	Retained earnings (premium on redemption)	500	
CREDIT	Cash		10,500
DEBIT	Retained earnings	10,000	
CREDIT	Capital redemption reserve		10,000

(2) Where a company redeems shares or purchases its shares wholly or partly out of the proceeds of a new share issue, it must transfer to the capital redemption reserve an amount by which the nominal value of the shares redeemed exceeds the **aggregate** proceeds from the new issue (ie, nominal value of new shares issued plus share premium).

(3) In example (3) the accounting entries would be:

		£	£
DEBIT	Share capital account (redeemed shares)	10,000	
	Retained earnings (premium)	500	
CREDIT	Cash (redemption of shares)		10,500
DEBIT	Cash (from new issue)	10,000	
CREDIT	Share capital account		10,000

No credit to the capital redemption reserve is necessary because there is no decrease in the creditors' buffer.

(4) If the redemption in the same example were made by issuing 5,000 new £1 shares at par, and paying £5,500 out of distributable profits:

		£	£
DEBIT	Share capital account (redeemed shares)	10,000	
	Retained earnings (premium)	500	
CREDIT	Cash (redemption of shares)		10,500
DEBIT	Cash (from new issue)	5,000	
CREDIT	Share capital account		5,000
DEBIT	Retained earnings	5,000	
CREDIT	Capital redemption reserve		5,000

(5) In the example (4) above (assuming a new issue of 10,000 £1 shares at a premium of 8p per share) the accounting entries would be:

		£	£
DEBIT	Cash (from new issue)	10,800	
CREDIT	Share capital account		10,000
	Share premium account		800
DEBIT	Share capital account (redeemed shares)	10,000	
	Share premium account	300	
	Retained earnings	200	
CREDIT	Cash (redemption of shares)		10,500

No capital redemption reserve is required, as in (1) above. The redemption is financed entirely by a new issue of shares.

9.8 Commercial reasons for altering capital structure

These include the following:

- Greater security of finance
- Better image for third parties
- A 'neater' statement of financial position
- Borrowing repaid sooner
- Cost of borrowing reduced

Interactive question 9: Krumpet plc

Set out below is the summarised statement of financial position of Krumpet plc at 30 June 20X5.

	£'000
Net assets	520
Equity	
Called up share capital £1 ordinary shares	300
Share premium account	60
Retained earnings	160
	520

On 1 July 20X5 Krumpet plc purchased and cancelled 50,000 of its ordinary shares at £1.50 each.

The shares were originally issued at a premium of 20p. The redemption was partly financed by the issue at par of 5,000 new shares of £1 each.

Requirement

Prepare the summarised statement of financial position of Krumpet plc at 1 July 20X5 immediately after the above transactions have been effected.

See **Answer** at the end of this chapter.

10 Audit focus



Section overview

The auditor will need to evaluate whether the fair value of the share-based payment is appropriate.

The auditor will require evidence in respect of all the components of the estimated amounts, as well as reperforming the calculation of the expense for the current year.

Issue	Evidence
Number of employees in scheme/number of instruments per employee/length of vesting period	• Scheme details set out in contractual documentation
Number of employees estimated to benefit	• Inquire of directors • Compare to staffing numbers per forecasts and prediction
Fair value of instruments	• For equity-settled schemes confirm that fair value is estimated at the grant date • For cash-settled schemes confirm that the fair value is recalculated at the end of the reporting period and at the date of settlement • Confirm that model used to estimate fair value is in line with IFRS 2 and is appropriate to the conditions. Obtain expert advice on the valuation if appropriate
General	• Obtain representations from management confirming their view that: - the assumptions used in measuring the expense are reasonable; and - there are no share-based payment schemes in existence that have not been disclosed to the auditors.

Professional skills focus: Applying judgement

Fair value is an area where management exercises judgement and makes assumptions. The auditor needs to confirm that the assumptions are reasonable and the evidence sufficient.

Interactive question 10: Share-based payments

You are the auditor of Russell plc. The draft financial statements for the year ending 31 December 20X5 show a profit before tax of £400,000. Russell plc provided four of its directors with 3,000 share options each on 1 January 20X5 which vest on 31 December 20X7. The fair value of the options, determined by use of the Black-Scholes model, is as follows:

£10	At the grant date
£12	On 1 January 20X6
£15	On 1 January 20X7
£13	On 31 December 20X7

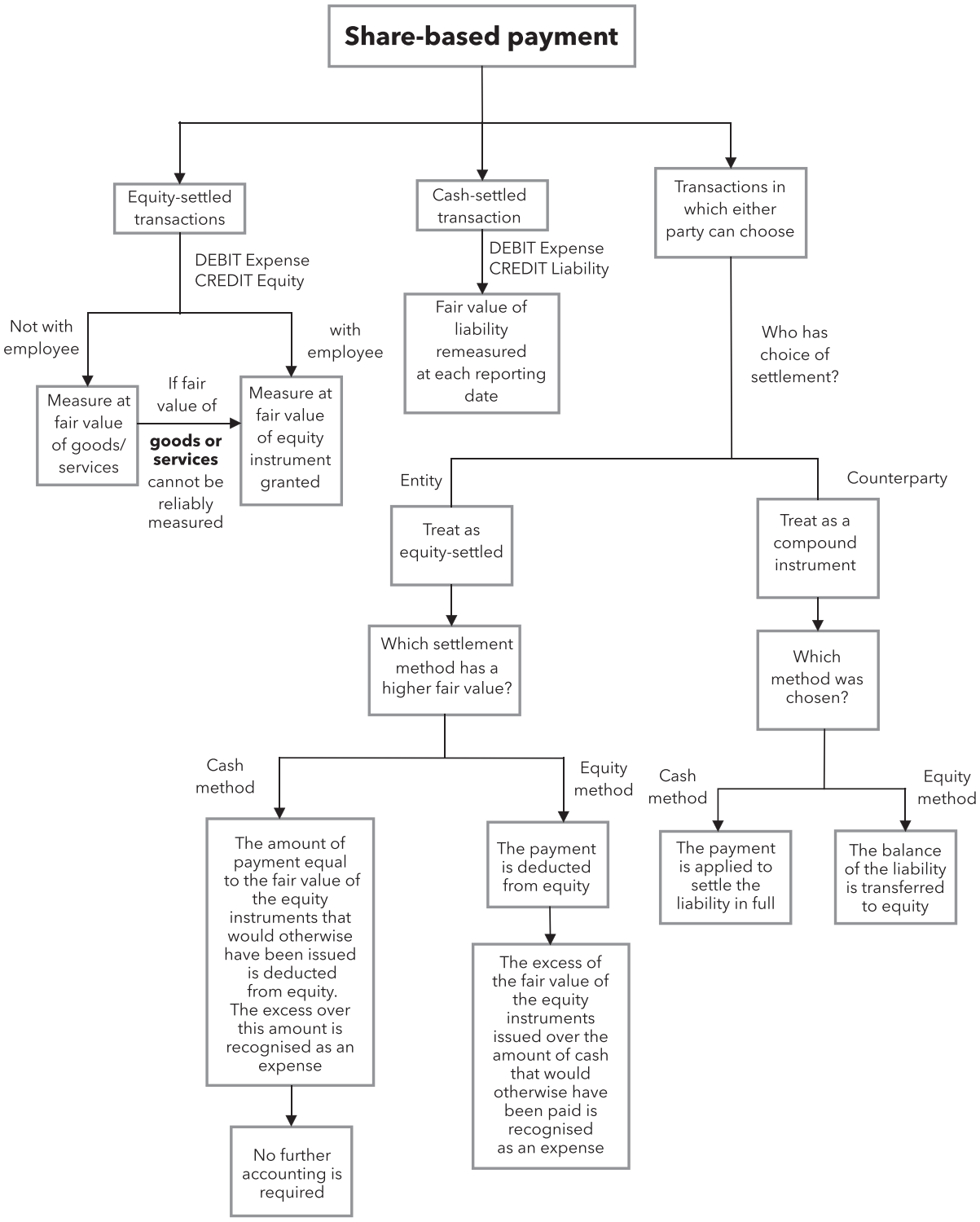
The options are dependent on continued employment. All four directors are expected to remain. No entry has been made in the financial statements of Russell plc in respect of the options on the basis that they do not vest until 31 December 20X7.

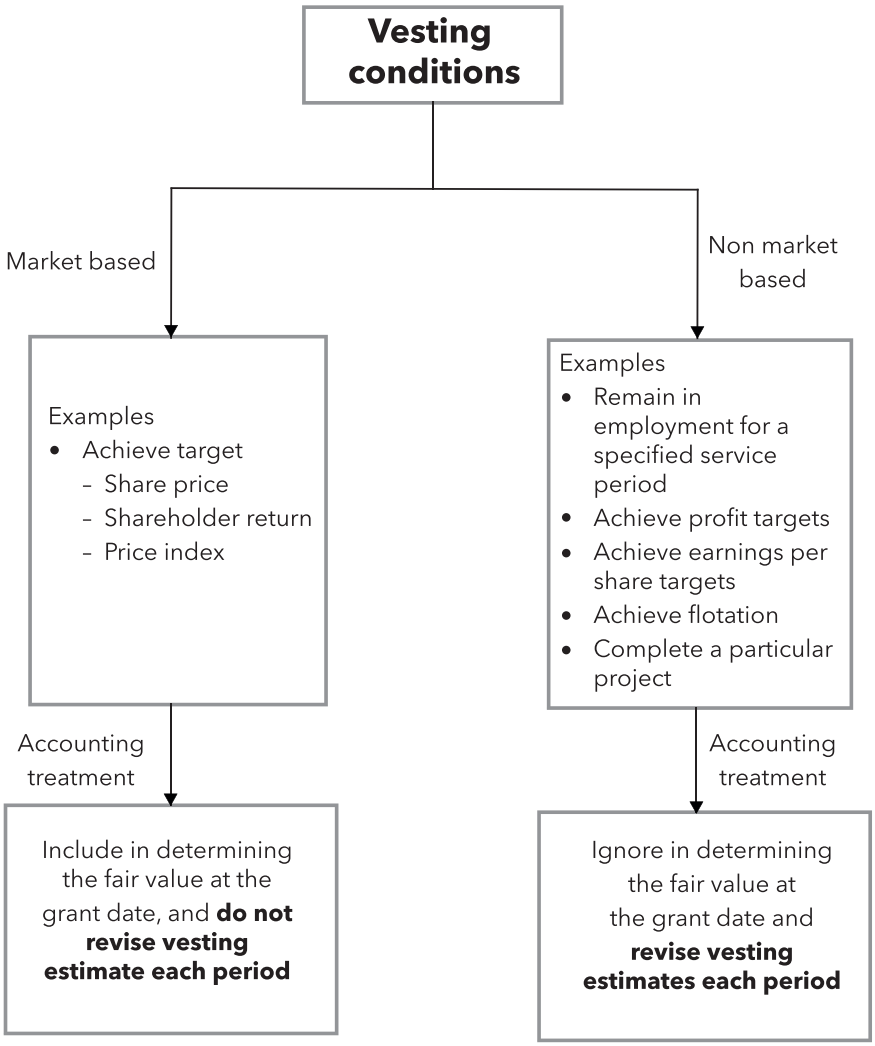
Requirement

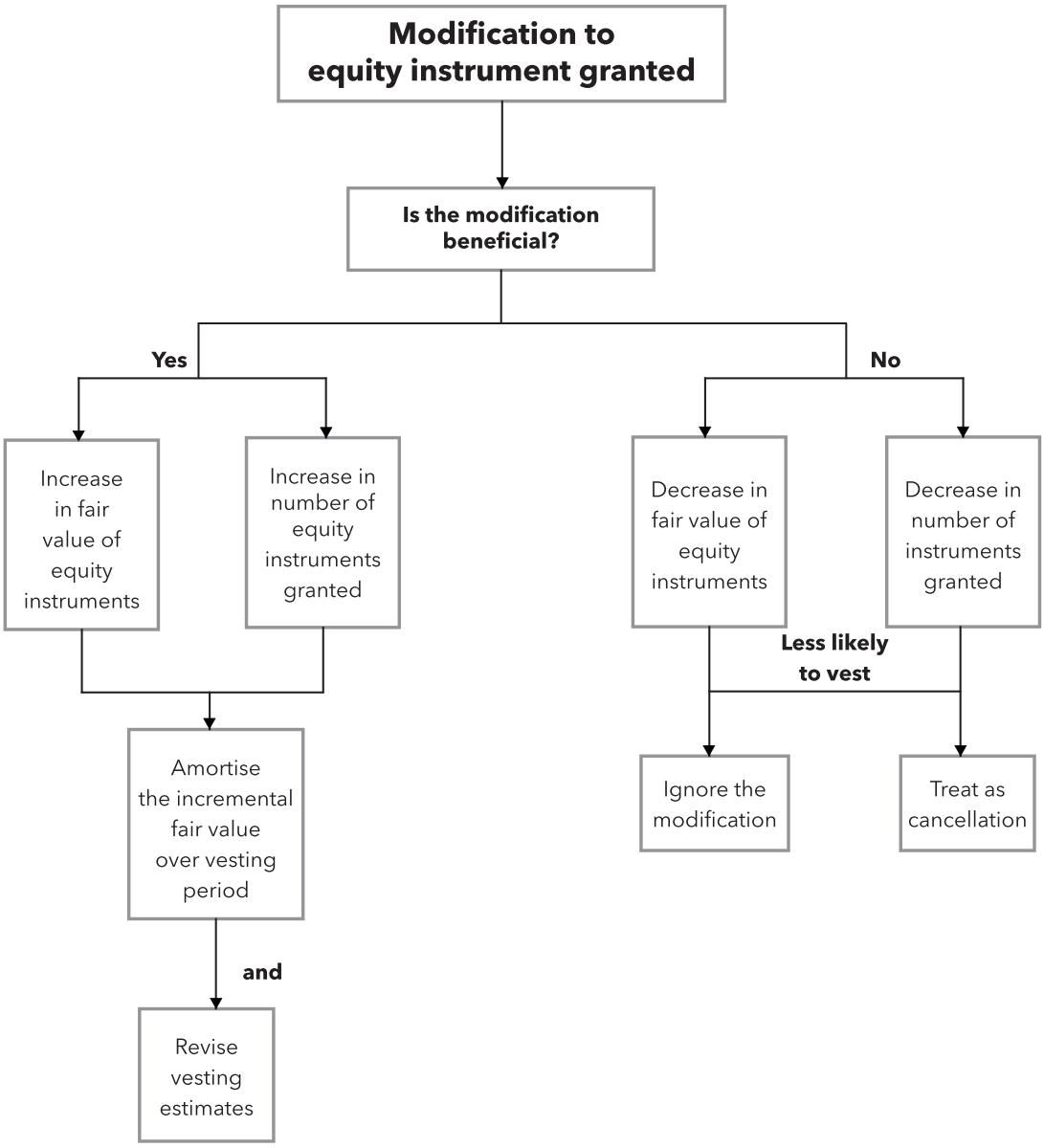
Identify the audit issues you would need to consider in respect of the share options.

See **Answer** at the end of this chapter.

Summary


Share-based payment
Equity-settled transactions
Cash-settled transaction
Transactions in which either party can choose
DEBIT Expense
CREDIT Equity
DEBIT Expense
CREDIT Liability
Not with employee
with employee
Measure at fair value of goods/ services
If fair value of goods or services cannot be reliably measured
Measure at fair value of equity instrument granted
Fair value of liability remeasured at each reporting date
Who has choice of settlement?
Entity
Counterparty
Treat as equity-settled
Treat as a compound instrument
Which settlement method has a higher fair value?
Which method was chosen?
Cash method
Equity method
Cash method
Equity method
The amount of payment equal to the fair value of the equity instruments that would otherwise have been issued is deducted from equity. The excess over this amount is recognised as an expense
The payment is deducted from equity
The payment is applied to settle the liability in full
The balance of the liability is transferred to equity
No further accounting is required
The excess of the fair value of the equity instruments issued over the amount of cash that would otherwise have been paid is recognised as an expense


Vesting conditions
Market based
Non market based
Examples
• Achieve target
– Share price
– Shareholder return
– Price index
Examples
• Remain in employment for a specified service period
• Achieve profit targets
• Achieve earnings per share targets
• Achieve flotation
• Complete a particular project
Accounting treatment
Accounting treatment
Include in determining the fair value at the grant date, and **do not revise vesting estimate each period**
Ignore in determining the fair value at the grant date and **revise vesting estimates each period**


Modification to equity instrument granted
Is the modification beneficial?
Yes
No
Increase in fair value of equity instruments
Increase in number of equity instruments granted
Decrease in fair value of equity instruments
Decrease in number of instruments granted
Less likely to vest
Amortise the incremental fair value over vesting period
and
Revise vesting estimates
Ignore the modification
Treat as cancellation

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	Under IFRS 2, *Share-based Payment* what is an equity-settled share-based payment transaction? (Topic 2)
2.	What is the vesting period? (Topic 2)
3.	Can you distinguish between market and non-market vesting conditions? (Topic 3)
4.	Can you account for equity-settled and cash-settled share-based payment transactions? (Topic 4)
5.	Can you account for share-based payment transactions with a choice of settlement? (Topic 6)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Definition	This question is on equity-settled share-based payment and focuses on the explanation.
BCN	This covers share appreciation rights (cash-settled share-based payment) and focuses on the calculations.
Mayflower	This is an integrated question, set in the context of audit, covering other issues as well as share-based payment, as is likely to happen in an exam. If you're short of time, just skim-read the introductory paragraph for context and go straight to the section on share options.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Upstart (share options only)	You have already attempted other parts of this question. This part requires calculation and explanation of the correct treatment of equity-settled share-based payment.
Your Nature plc (share options only)	This part of a much longer question deals with share appreciation rights, and is a cash-settled share-based payment. Remember to re-measure the fair value of the liability at each year end.
Eastoak (share options only)	This is a more complex test of share-based payment with employees having the right to choose the form of settlement. In such cases IFRS 2 regards the transaction as a compound financial instrument, which must be split into debt and equity.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving onto the next chapter.

Technical reference

1 Three specific types of transactions

- Equity-settled share-based payment – **IFRS 2.2(a)**
- Cash-settled share-based payment – **IFRS 2.2(b)**
- Share-based payment transactions with a cash alternative – **IFRS 2.2(c)**

2 Transactions excluded from scope

- Transactions with employees in their capacity as shareholders – **IFRS 2.4**
- Issues of shares in a business combination – **IFRS 2.5**
- Contracts that may be settled net in shares or rights to shares – **IFRS 2.6**

3 Recognition

- Goods or services received in share-based transaction to be recognised as expenses or assets – **IFRS 2.8**
- Entity shall recognise corresponding increase in equity for equity-settled transaction or a liability for cash-settled transactions – **IFRS 2.7**

4 Equity-settled share-based transactions

- Goods or services received and corresponding increase in equity shall be measured as fair value of goods or services received (direct method) – **IFRS 2.10**
- If fair value of goods or services cannot be measured reliably, then measure at fair value of instruments granted – **IFRS 2.10**
- For transactions with third parties there is a presumption that the fair value of the goods or services can be estimated reliably – **IFRS 2.13**
- Fair value shall be measured at the date entity obtains the goods or counterparty renders service – **IFRS 2.13**
- If equity instruments vest immediately entity will recognise services received and corresponding increase in equity immediately – **IFRS 2.14**
- If equity instruments will be received in future, services and increase in equity will be recognised during vesting period – **IFRS 2.15**
- Fair value of equity instruments granted to be based on market prices if available – **IFRS 2.16**
- If market prices are not available a generally acceptable valuation technique should be used – **IFRS 2.17**
- Non-market vesting conditions shall be taken into account by adjusting the number of equity instruments included in the measurement of the transactions at each reporting date – **IFRS 2.19**
- For non-market vesting conditions the amount ultimately recognised will be the number of equity instruments that actually vest – **IFRS 2.19, 2.20**
- For grants of equity instruments with market conditions the entity shall recognise the goods or services from counterparty who satisfies all the other vesting conditions irrespective of whether market conditions are satisfied – **IFRS 2.21**

- Market conditions will be part of fair value at grant date. This should not be revised at each reporting date and where options do not vest the charge should not be reversed – **IFRS 2.21**
- If vested equity instruments are forfeited, entity shall make no adjustment to total equity except a transfer from one equity component to another – **IFRS 2.23**

5 Cash-settled share-based payment transactions

- Goods or services acquired should be measured at fair value of liability – **IFRS 2.30**
- Liability should be remeasured at each reporting date – **IFRS 2.30**

6 Share-based payment transaction with cash alternative

- If counterparty has right to choose then entity has granted a compound financial instrument with debt component and equity component – **IFRS 2.34**
- For counterparties other than employees, the equity component is measured as the difference between the fair value of the goods or services received and the fair value of the debt component (the counterparty's right to demand payment in cash) – **IFRS 2.35**
- For transactions with employees, the entity shall measure the fair value of the debt component and the fair value of the equity component separately and account for the debt component as a cash-settled transaction and the equity component as an equity-settled transaction – **IFRS 2.36, 2.37, 2.38**
- At the date of settlement the entity shall remeasure the liability to its fair value – **IFRS 2.39**
- Where equity instruments are paid instead of cash, the liability shall be transferred to equity – **IFRS 2.39**
- Where entity pays cash on settlement rather than equity, the payment is applied to the liability. The equity components previously recognised will remain in equity and entity can make a transfer from one component of equity to another – **IFRS 2.40**
- Where entity has the choice of settlement, if present obligation exists to deliver cash, it should recognise and treat as cash-settled share-based payment transaction – **IFRS 2.41**
- If no present obligation exists to pay cash, entity should treat transaction as an equity-settled transaction. On settlement if cash was paid, cash should be treated as repurchase of equity by a deduction against equity – **IFRS 2.43**

Self-test questions

Answer the following questions.

1 Share-based transactions

Which are the three types of share-based transactions covered by IFRS 2?

2 Definition

Which of the following transactions are NOT within the definition of a share-based payment under IFRS 2?

(1) Employee share ownership plans (ESOPs)

(2) Transfers of equity instruments of the parent of the reporting entity to third parties that have supplied goods or services to the reporting entity

(3) The acquisition of property, plant and equipment as part of a business combination

(4) Share appreciation rights (SARs)

(5) The raising of funds through a rights issue to all shareholders including those who are employees

(6) Cash bonus to employees dependent on share price performance

(7) Employee share purchase plans

(8) Remuneration in non-equity shares

3 BCN

BCN Co grants 1,000 share options to each of its 300 staff to be exercised in two years' time at a price of £6.10. The current fair value of the option is £1.40 and the expected fair value in two years' time is £2.40 (adjusted for the possibility of forfeiture in both cases). Under IFRS 2, *Share-based Payment*, how much expense would be recognised in profit or loss at the date of issue of the options?

4 Condition 1

On 1 January 20X3 an entity grants 250 share options to each of its 200 employees. The only condition attached to the grant is that the employees should continue to work for the entity until 31 December 20X6. Five employees leave during the year.

The market price of each option was £12 at 1 January 20X3 and £15 at 31 December 20X3.

Requirement

Show how this transaction will be reflected in the financial statements for the year ended 31 December 20X3.

5 Annerly

On 1 July 20X4 Annerly Co granted 20 executives options to buy up to 10,000 shares each. The options only vest if the executives are still in the service of the company on 1 July 20X6. It

is estimated that 90% of the executives will remain with the company for the duration of the vesting period and exercise their options in full.

The following information is relevant.

- The exercise price of the option is £20 per share.
- The market value of each share was £15 on 1 July 20X4 and £18 on 30 June 20X5. It is £19 on 20 July 20X5, when the draft financial statements for the year to 30 June 20X5 are being reviewed.
- The market value of the option is £3 on 1 July 20X4, £3.20 on 30 June 20X5, and £2.50 on 20 July 20X5.

Requirement

How should the transaction be accounted for in the financial statements for the year to 30 June 20X5?

6 Condition 2

An entity grants 100 share options on its £1 shares to each of its 500 employees on 1 January 20X5. Each grant is conditional upon the employee working for the entity over the next three years. The fair value of each share option as at 1 January 20X5 is £15.

On the basis of a weighted average probability, the entity estimates on 1 January 20X5 that 20% of employees will leave during the three-year period and therefore forfeit their rights to share options.

Requirements

Show the accounting entries which will be required over the three-year period in the event of the following:

(a) 20 employees leave during 20X5 and the estimate of total employee departures over the three- year period is revised to 15% (75 employees).

(b) 22 employees leave during 20X6 and the estimate of total employee departures over the three- year period is revised to 12% (60 employees).

(c) 15 employees leave during 20X7, so a total of 57 employees left and forfeited their rights to share options. A total of 44,300 share options (443 employees × 100 options) vested at the end of 20X7.

7 Share appreciation rights

On 1 January 20X4 an entity grants 100 cash share appreciation rights (SARs) to each of its 500 employees on condition that the employees remain in its employ for the next two years. The SARs vest on 31 December 20X5 and may be exercised at any time up to 31 December 20X6. The fair value of each SAR at the grant date is £7.40.

Year ended	Leavers	No. of employees exercising rights	Outstanding SARs	Estimated further leavers	Fair value of SARs	Intrinsic value* (ie, cash paid)
					£	£
31 Dec 20X4	50	–	450	60	8.00	
31 Dec 20X5	50	100	300	–	8.50	8.10
31 Dec 20X6	–	300	–	–	–	9.00

* Intrinsic value is the fair value of the shares less the exercise price

Requirement

Show the expense and liability which will appear in the financial statements in each of the three years.

8 ZZX plc

ZZX plc has provided a share incentive scheme to a number of its employees on 1 January 20X4. This allows for a cash payment to be made to the individuals concerned equal to the share price at the end of a three-year period subject to the following conditions.

(1) Vesting will be after three years.

(2) The share price must exceed £2.

(3) The employee must be with the company on 31 December 20X6.

Each scheme issued will result in payment, subject to the conditions outlined, equal to the value of 10 shares at the end of the three-year period if the conditions are satisfied. The payments, once earned, are irrevocable.

The finance director has been issued 20 such schemes. The share prices over the next three years were as follows.

- 31 December 20X4 = £2.20
- 31 December 20X5 = £1.80
- 31 December 20X6 = £2.40

Requirements

8.1 Prepare the journal entries for the transactions of the share incentives issued to the finance director.

8.2 Assuming that on 1 January 20X4 nine other individuals were also granted equivalent rights to the finance director and that on 1 January 20X5 two of those individuals left the company, prepare the journal entries for the transactions relating to the incentive schemes.

9 Kapping

The directors of Kapping are adopting IFRS for the first time and are reviewing the impact of IFRS 2, *Share-based Payment* on the financial statements for the year ended 31 May 20X7. They require you to do the following:

Requirements

9.1 Explain why share options, although having no cost to the company, should be reflected as an expense in profit or loss.

9.2 Discuss whether the expense arising from share options under IFRS 2 actually meets the definition of an expense under the IASB's *Conceptual Framework*.

9.3 Explain the impact of IFRS 2 on earnings per share, given that an expense is shown in profit or loss and the impact of share options is recognised in the diluted earnings per share calculation.

9.4 Briefly discuss whether the requirements of IFRS 2 should lead them to reconsider their remuneration policies.

10 Mayflower plc

Your firm has been working on a new audit assignment, Mayflower plc (Mayflower), a listed, diversified group of companies. Its main business interests include construction, publishing, food processing and a restaurant chain.

Mayflower's draft profit before tax for the year ended 31 March 20X8 is £17.5 million (20X7 £16.3 million) and its revenue is £234.5 million (20X7 £197.5 million).

The senior in charge of the audit for the year ended 31 March 20X8 has fallen ill towards the end of the audit and you are now helping to complete it. There are several matters outstanding and you will need to consider their impact on the financial statements and the audit. Your line manager has asked to meet you to discuss the significant outstanding matters. He has asked you to prepare briefing notes for the meeting, including your views on the impact on the financial statements and any additional audit work that might be required, so that a way forward can be agreed.

On reviewing the previous senior's notes you find the following outstanding areas:

Forward contract

On 1 April 20X6 Mayflower entered into a forward contract to purchase a large quantity of sugar on 1 April 20Y0. As far as we can tell, this was purely speculative based on the expectation that the price of sugar would rise. Mayflower did not pay to enter this contract. The company has not accounted for this contract in the years ended 31 March 20X7 or 20X8.

The finance manager of Mayflower has told us that at 31 March 20X7 the value of the contract had risen to £800,000 and by 31 March 20X8 its fair value had risen further to £850,000.

Share options

On 1 April 20X5 Mayflower provided three of its directors with 4,000 share options each, which vest on 1 April 20X8, assuming the directors remain in employment.

The fair values of each option were:

1 April	£
20X5	12
20X6	15
20X7	17
20X8	16

We have established that no accounting entries have been made for the above.

Debenture issue

Mayflower issued a £5 million convertible debenture at par on 1 April 20X7. The debenture has an annual nominal rate of interest of 4.5% and is redeemable on 1 April 20Y7 at par. Alternatively, the holder has the option to convert the debenture to four million £1 shares in Mayflower.

The debenture is presented as a non-current liability at the net proceeds and this amount has not changed since issue.

The directors have agreed to identify a suitable comparable debenture with an observable market rate of interest, but they have not yet done so.

Properties

The audit has verified the following facts about properties held by Mayflower:

PROPERTY	DESCRIPTION
Exeter House	Used as the company's head office.
33–39 Reeves Road	A warehouse held as a right-of-use asset that, up to six months ago, was used by Mayflower but has since been rented out to a third party on a short-term lease.
41–51 Reeves Road	A warehouse facility adjacent to the existing warehouse. This was purchased in November 20X7 at a cost of £3.8 million. Professional fees of £60,000 were also incurred. It is currently empty and has been since purchase, but a tenant is actively being sought.
Falcon House	An office block owned by Mayflower for seven years. It is currently all rented to a non-group company apart from a couple of small rooms in the basement that are used by Mayflower as an external store for some of its records.

Mayflower has classified all these properties as investment properties and has adopted the fair value model in accordance with IAS 40.

Since 31 March 20X8 property values have dropped by 2% on average.

We need to finalise their treatment as investment properties and verify their valuation.

Requirement

Prepare the required briefing notes on the financial reporting and auditing implications of each of the outstanding areas for discussion with your line manager.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Fair value of options granted at grant date: 1,500 employees × 10 options × £20 = £300,000

This should be charged to profit or loss as employee remuneration evenly over the two-year period from 1 July 20X5 to 30 June 20X7.

£150,000 is recognised each year. A corresponding amount will be recognised as part of equity as the services are recognised.

Answer to Interactive question 2

The remuneration expense for the year is based on the fair value of the options granted at the grant date (1 January 20X3):

[400 employees – (10 leavers × 4 years)] × 500 options × £10 = £1,800,000.

Therefore, the entity recognises a remuneration expense of £450,000 (£1.8m/4 years) in profit or loss and a corresponding increase in equity of the same amount.

Answer to Interactive question 3

The total expense recorded over the expected vesting period would be as follows:

(1) All options vest: 100 options × £20 = £2,000 total expense

(2) All vesting conditions are met, except the market-based performance condition: 100 options × £20 = £2,000 total expense

(3) All vesting conditions are met, except the non-market-based performance condition: nil expense

(4) All vesting conditions are met, except half of the employees who received options left the company before the vesting date: 50 options × £20 = £1,000 total expense

Paragraph 21 of IFRS 2 states that the grant date fair value of the share-based payment with market- based performance conditions that has met all its other vesting conditions should be recognised, irrespective of whether that market condition is achieved. The company determines the grant date fair value of the share-based payment excluding the non-market-based performance factor, but including the market-based performance factor.

Answer to Interactive question 4

Expense/Equity figures

	Expense	Equity (per statement of financial position)
	£	£
Year 1	660,000	660,000
Year 2	174,000	834,000
Year 3	423,000	1,257,000

WORKINGS

(1) **Year 1**

Equity: (440 employees × 100 options × £30)/2 years = £660,000 (using original estimate of two-year period)

(2) **Year 2**

	£
Equity c/d [(500 – 30 – 28 – 25) employees × 100 × £30 × 2/3] (using revised estimate of three-year period)	834,000
Previously recognized	(660,000)
expense	174,000

(3) **Year 3**

	£
Equity c/d [(500 – 30 – 28 – 23) × 100 × £30]	1,257,000
Previously recognised	(834,000)
expense	423,000

Answer to Interactive question 5

The incremental value is £3 per share option (£8 – £5). This amount is recognised over the remaining two years of the vesting period, along with remuneration expense based on the original option value of £15.

The amounts recognised in Years 1–3 are as follows.

Year		£
1	Equity c/d [(500 – 110) × 100 × £15 × 1/3]	195,000
	DEBIT Expenses	£195,000
	CREDIT Equity	£195,000
2	Equity c/d [(500 – 105) × 100 × ((£15 × 2/3) + (£3 × 1/2))]	454,250
	Less previously recognised	(195,000)
		259,250
	DEBIT Expenses	£259,250
	CREDIT Equity	£259,250
3	Equity c/d [(500 – 103) × 100 × (£15 + £3)]	714,600
	Less previously recognised	(454,250)
		260,350
	DEBIT Expenses	£260,350
	CREDIT Equity	£260,350

Answer to Interactive question 6

6.1

Accounting entries

		£	£
	31.12.X1		
DEBIT	Staff costs expense Equity reserve ((800–95) × 200 × £4 ×	188,000	
CREDIT	1/3)		188,000
	31.12.X2		
DEBIT	Staff costs expense (W1)	201,333	
CREDIT	Equity reserve		201,333
	31.12.X3		
DEBIT	Staff costs expense (W2)	202,667	
CREDIT	Equity reserve		202,667
	Issue of shares:		
DEBIT	Cash ((800 – 40 – 20) × 200 × £1.50)	222,000	
DEBIT	Equity reserve	592,000	
CREDIT	Share capital (740 × 200 × £1)		148,000
CREDIT	Share premium (balancing figure)		666,000

WORKINGS

(1) **Equity reserve at 31.12.X2**

	£
Equity c/d ((800 – 70) × 200 × £4 × 2/3)	389,333
Less previously recognized	(188,000)
charge	201,333

(2) **Equity reserve at 31.12.X3**

	£
Equity c/d ((800 – 40 – 20) × 200 × £4 × 3/3)	592,000
Less previously recognized	(389,333)
charge	202,667

6.2 **Cash-settled share-based payment**

If J&B had offered cash payments based on the value of the shares at vesting date rather than options, in each of the three years an accrual would be shown in the statement of financial position representing the expected amount payable based on the following:

No of employees estimated at the year end to be entitled to rights at the vesting date	×	Number of rights each	×	Fair value of each right at year end	×	Cumulative proportion of vesting period elapsed

The movement in the accrual would be charged to profit or loss representing further entitlements received during the year and adjustments to expectations accrued in previous years.

The accrual would continue to be adjusted (resulting in an expense charge) for changes in the fair value of the rights over the period between when the rights become fully vested and are subsequently exercised. It would then be reduced for cash payments as the rights are exercised.

Answer to Interactive question 7

7.1 **Explanation**

- Employee services – no reliable fair value Use fair value of the equity instrument
- Fair value measured at grant date – and not subsequently changed Expense in P/L over vesting period
- If vesting period can vary as a result of non-market conditions, use best estimate of length of period
- Best estimate of number that will vest
- Credit entry to equity – separate component or retained earnings

20X7

Expense is at fair value £15 based on an expected two-year vesting period 450 employees – 30 leavers – 25 future leavers = 395 employees

Expense = 395 × 100 options × £15 × 1/2 years

= £296,250

20X8

450 employees – 30 left Year 1 – 15 left Year 2 – 26 future leavers = 379 employees

Expense is now spread over a three-year vesting period

Expense = £15 × 379 × 100 × 2/3 years	£379,000
Less recognised in Year 1	£296,250
Year 2 expense	£82,750

20X9

390 × 100 × £15 × 3/3 years =	£585,000
Less recognised previously	£379,000
Expense in Year 3	£206,000

Double entries

		£	£
20X7			
DEBIT	Employment costs	296,250	
CREDIT	Equity		296,250
20X8			
DEBIT	Employment costs	82,750	
CREDIT	Equity		82,750

20X9

DEBIT	Employment costs	206,000	
CREDIT	Equity		206,000

7.2 **20Y0**

If employees do not exercise their options, but allow them to lapse, the net expense recognised does not change. As long as the options vest, an expense will appear in the accounts.

7.3 In this case, the options would never vest. In 20X9, the expense would be extended to 20Y0 (effectively a four-year option scheme) before the scheme was cancelled in 20Y0 according to the initial details of the scheme. If the non-market condition was not achieved in 20Y0, the net expense recognised is reversed and a credit would appear in profit or loss for 20Y0 to the value of the previous cumulative expense recognised (in 20X9 this was £585,000). A market condition not being achieved would never affect the expense being recognised, as the share price movement is called 'volatility' which is included in the £15 fair value.

Answer to Interactive question 8

Dividends may only be paid by a company out of profits available for the purpose. There is a detailed code of statutory rules which determines what are distributable profits. The profits which may be distributed as dividend are accumulated realised profits, so far as not previously used by distribution or capitalisation, less accumulated realised losses, so far as not previously written off in a reduction or reorganisation of capital duly made.

The word 'accumulated' requires that any losses of previous years must be included in reckoning the current distributable surplus.

The word 'realised' presents more difficulties. It may prevent the distribution of an increase in the value of a retained asset at fair value through profit or loss. However, it does not prevent a company from transferring to retained earnings profit earned on an uncompleted contract, if it is in accordance with generally accepted accounting principles. In view of the authority of accounting standards, it is unlikely that profits determined in accordance with accounting standards would be considered unrealised. A realised capital loss will reduce realised profits.

The above rules on distributable profits apply to all companies, private or public. A public company is subject to an additional rule which may diminish but cannot increase its distributable profit as determined under the above rules.

A public company may only make a distribution if its net assets are, at the time, not less than the aggregate of its called-up share capital and undistributable reserves. The dividend which it may pay is limited to such amount as will leave its net assets at not less than that aggregate amount.

Answer to Interactive question 9

WORKINGS FOR KRUMPET PLC	**£**	**£**
		75,000
Cost of redemption (50,000 × £1.50)		
Premium on redemption (50,000 × 50p)		25,000
No premium arises on the new issue		
Distributable profits		
Retained earnings before redemption		160,000
Premium on redemption (25,000 – 5,000 charged to share premium account)		(20,000)
		140,000

Remainder of redemption costs	50,000	
Proceeds of new issue 5,000 × £1	(5,000)	
Remainder out of distributable profits		(45,000)
Balance on retained earnings		95,000

Statement of financial position of Krumpet plc as at 1 July 20X5

	£'000
Net assets (520 – 75 + 5)	450
Capital and reserves	
Ordinary shares (300 – 50 + 5)	255
Share premium: 60,000 less 5,000 (10,000 allowable, being premium on original issue of 50,000 × 20p, restricted to proceeds of new issue of 5,000)	55
Capital redemption reserve	45
	355
Retained earnings (W)	95
	450

Answer to Interactive question 10

Audit issues:

- Consider compliance with IFRS 2. Based on the fair value at grant date (as provided) the total remuneration expense would be as follows:

4 × 3,000 × £10 = £120,000

The expense would then be recognised over the vesting period of three years. An amount of £40,000 should be recognised in 20X5 as an expense in profit or loss for the year with a corresponding credit in equity.

- The expense of £40,000 represents 10% of the profit before tax and is therefore likely to be material to the financial statements.
- Whether the basis of valuing fair value is appropriate. Ideally fair value should be based on market price if available. As the options have a relatively short life the valuation method used may provide a reasonable approximation to fair value.
- Adequacy of disclosure in accordance with IFRS 2.

Answers to Self-test questions

1 Share-based transactions

Share-based transactions covered by IFRS 2:

(1) Equity-settled share-based payment transactions

(2) Cash-settled share-based payment transactions

(3) Transactions with a choice of settlement

2 Definition

The following transactions are not within the definition of a share-based payment under IFRS 2:

(3) The acquisition of property, plant and equipment as part of a business combination. This is within the scope of IFRS 3, *Business Combinations*.
(5) The raising of funds through a rights issue to all shareholders including those who are employees. IFRS 2 does not apply to transactions with employees in their capacity as shareholders.
(8) Remuneration in non-equity shares.

Payment in non-equity shares does not fall within the scope of IFRS 2 since it is a transaction in which the entity receives goods and services in return for a financial liability.

3 BCN

No expense is recognised at the issue date of the options, but the expected benefit is accrued as a cost over the life of the option:

300 employees × 1,000 options × £1.40 = £420,000

This is spread equally over the two-year period to vesting, resulting in an annual charge to profit or loss of £210,000.

This would not be adjusted for any changes in expected benefit due to changes in expected share price as the value is measured at grant date, but is adjusted for the numbers of employees entitled to options at each reporting date.

4 Condition 1

The remuneration expense for the year is based on the fair value of the options granted at the grant date (1 January 20X3). As five of the 200 employees left during the year it is reasonable to assume that 20 employees will leave during the four-year vesting period and that therefore 45,000 options (250 × 180) will actually vest.

Statement of profit or loss and other comprehensive income

Staff costs (45,000 × £12)/4 years	£135,000

Statement of financial position

Equity	£135,000

5 Annerly

IFRS 2 requires that share-based transactions made in return for goods or services are recognised in the financial statements. The granting of options to the senior executives is a

share-based payment under IFRS 2 and will need to be recognised as a remuneration expense. The amount to be charged as an expense is measured at the fair value of the goods or services provided as consideration for the share-based payment or at the fair value of the share-based payment, whichever can be more reliably measured.

In the case of employee share options, the market value of the options on the day these are granted is used, as this can be measured more reliably. The market value of the share options at the day these were granted, 1 July 20X4, was £3 each.

The exercise price for the option at £20 per share is above the market price on the date of issue on 1 July 20X4. This, however, does not mean that the option has zero market value. It has no intrinsic value, but it has what is referred to as time value relating to expectations of a share price increase over time.

The options vest at the end of the two-year period. The company expects that 90% of the options will vest, as it is estimated that 90% of the executives will remain in employment for the two-year period and thus be able to exercise their options in full.

The remuneration expense will be 10,000 × £3 × 20 × 90% = £540,000 and, as this vests over a two- year period, the entry to income for the current year to 30 June 20X5 will relate to half that amount: 1/2 × £540,000 = £270,000.

		£	£
DEBIT	Share-based payment remuneration expense	270,000	
CREDIT	Equity share-based payment reserve		270,000

When the shares are issued a transfer will be made from that reserve together with any further proceeds (if any) of the shares and will be credited to the share capital and share premium accounts.

6 Condition 2

(a) **Accounting entries**

			£
20X5 Equity c/d (500 × 85% × 100 × £15 × 1/3) =			212,500
DEBIT	Expenses	£212,500	
CREDIT	Equity		£212,500

(b) **Accounting entries**

			£
20X6 Equity c/d (500 × 88% × 100 × £15 × 2/3) =			440,000
Less previously recognised			(212,500)
			227,500
DEBIT	Expenses	£227,500	
CREDIT	Equity		£227,500

(c) **Accounting entries**

			£
20X7 Equity c/d (443 × 100 × £15) =			664,500
Less previously recognised			(440,000)
			224,500
DEBIT	Expenses	£224,500	
CREDIT	Equity		£224,500

7 Share appreciation rights

Expense and liability

	£
Year ended 31 December 20X4	
Expense and liability ((450 – 60) × 100 × £8.00 × 1/2)	156,000
Year ended 31 December 20X5	
Liability c/d (300 × 100 × £8.50)	255,000
Less b/d liability	(156,000)
	99,000
Plus cash paid on exercise of SARs by employees	
(100 × 100 × £8.10)	81,000
Expense	180,000
Year ended 31 December 20X6	
Liability c/d	–
Less b/d liability	(255,000)
	(255,000)
Plus cash paid on exercise of SARs by employees	
(300 × 100 × £9.00)	270,000
Expense	15,000

8 ZZX plc

8.1 **Journal entries for transactions: finance director**

The transactions are settled in cash and hence liabilities are created.

31 December 20X4		**£**	**£**
DEBIT	Expense	147	
CREDIT	Liability		147

It is assumed that the current share price is the best estimate of the final share price.

(Calculation note: 20 × 10 × £2.20 × 1/3)

31 December 20X5		**£**	**£**
DEBIT	Liability	147	
CREDIT	Expense		147

Reverses entries for 20X4 as share price is less than minimum

31 December 20X6		**£**	**£**
DEBIT	Expense	480	
CREDIT	Liability		480
DEBIT	Liability	480	
CREDIT	Cash		480

(20 × 10 × £2.40 × 3/3)

8.2 **Journal entries for transactions: other individuals**

		£	£
31 December 20X4			
DEBIT	Expense	1,467	
CREDIT	Liability		1,467
(Calculation note: 20 × 10 × 10 × £2.20 × 1/3)			
		£	**£**
31 December 20X5			
DEBIT	Liability	1,467	
CREDIT	Expense		1,467
31 December 20X6			
DEBIT	Expense	3,840	
CREDIT	Liability		3,840
DEBIT	Liability	3,840	
CREDIT	Cash		3,840

(Calculation note: 20 × 10 × 8 × £2.40 × 3/3)

9 Kapping

9.1 When shares are issued for cash or in a business combination, an accounting entry is needed to recognise the receipt of cash (or other resources) as consideration for the issue. Share options (the right to receive shares in future) are also issued in consideration for resources: services rendered by directors or employees. These resources are consumed by the company and it would be inconsistent not to recognise an expense.

9.2 The *Framework* defines an expense as a decrease in economic benefits in the form of outflows of assets or incurrences of liabilities. It is not immediately obvious that employee services meet the definition of an asset and therefore it can be argued that consumption of those services does not meet the definition of an expense. However, share options are issued for consideration in the form of employee services so that arguably there is an asset, although it is consumed at the same time that it is received. Therefore the recognition of an expense relating to share-based payment is consistent with the *Framework*.

9.3 It can be argued that to recognise an expense in profit or loss would have the effect of distorting diluted earnings per share as diluted earnings per share would then take the expense into account twice. This is not a valid argument. There are two events involved: issuing the options and consuming the resources (the directors' service) received as consideration.

The diluted earnings per share calculation only reflects the issue of the options; there is no adjustment to basic earnings. Recognising an expense reflects the consumption of services. There is no 'double counting'.

9.4 It is true that accounting for share-based payment reduces earnings. However, it improves the information provided in the financial statements, as these now make users aware of the true economic consequences of issuing share options as remuneration. The economic consequences are the reason why share option schemes may be discontinued. IFRS 2 simply enables management and shareholders to reach an informed decision on the best method of remuneration, weighing the advantages of granting these and their potential beneficial effect on motivation and corporate performance against the disadvantages of the impact on earnings.

10 Mayflower plc

Note: This question includes a revision of the audit of financial instruments and investment properties.

Forward contract Accounting issues

- At 31 March 20X7 there was an increase in the value of the derivative asset of £800,000 and this gain should have been included in profit or loss for the year ended 31 March 20X7.
- A prior year adjustment is needed in respect of the error of £800,000 not recognised in the opening balance. In accordance with IAS 8 this will be adjusted directly through reserves. The amount is likely to be material. Also the comparatives would need to be restated.
- The derivative will therefore be shown as an asset in the statement of financial position at £800,000 and £850,000 in the two years ended 31 March 20X7 and 20X8 respectively.

Audit procedures

- Review terms of the contract to determine the appropriate treatment ie, as a derivative financial asset
- The reason for the non-recognition of the fair value last year should be established and the systems for identifying and measuring forward contracts with zero initial values should be established
- General checks:
  - Obtain an understanding of the factors which affect the entity's derivative activities
  - Assess level of audit risk re the forward contract (and ensure audit procedures reduce the risk to an acceptable level)
  - Understand systems for dealing with forward contracts – ensure they cover authorisations, checking for completeness and accuracy, appropriate accounting for changes in values, ongoing monitoring, how errors are prevented and detected
  - Ensure data is secure
  - Check if there have been any margin settlements made
- Check the amounts at the year ends:
  - Inspect contract and confirm that Mayflower did not pay to enter the contract
  - Confirm no changes in contract
  - Assess the expertise and experience of the finance manager in determining the fair values and confirm that these have been derived in accordance with IFRS 13 (ie, Level 1 or Level 2 inputs)
  - Review documentation supporting information used by management including any assumptions made
  - Inquire whether management experts have been used to measure fair value and determine disclosures
  - Look for evidence of year-end fair values (quoted market price). The fair values of the forward contract would need to be attested. As sugar is traded on an active international market there should be sufficient publicly available information to observe the fair value of equivalent futures trades in an open market ie, in the principal market
  - Recalculate the value of the derivative asset
  - Obtain written representations from management

- Review events after the year end which may provide evidence about valuation at the period end

- Check IFRS 7 and IFRS 13 disclosure:
  - Accounting policy for financial instruments and how fair value is measured (valuation technique and inputs used)
  - Appropriate gains recognised in profit or loss for the year
  - Fair value ensuring the information can be compared with prior year
  - Nature and extent of risks arising

Share options

Accounting issues

- IFRS 2 deals with this area.
- The total remuneration expense, based on the fair value at the grant date, would be: 3 × 4,000 × £12 = £144,000. This should have been spread evenly over the vesting period of three years assuming that there were no changes in assessment of when the directors would leave the company (and assuming that none of them have left). If there have been any variations over time as to the likelihood of the three directors being in post at vesting then there may be an adjustment to the amounts below (in particular if there was any change in numbers of shares vesting between last year and this year).
- This will necessitate a prior period adjustment of £144,000 × 2/3 = £96,000 which would be debited to retained earnings and credited to a share option reserve (or other appropriate reserve). As the standard only requires the credit to be posted to equity it is possible for no double entry to be posted at all (as retained earnings may substitute for a share option reserve). Regardless of how the accounting is reflected, the comparatives will need to be adjusted.
- The current year expense of £48,000 will be treated in the same way as above except that it will be included in profit or loss for the current year – most appropriately as a staff cost.

Audit procedures

- View documentation to verify option terms such as exercise price and term to vesting date.
- Ensure relevant directors are still employed.
- Investigate whether there was any indication that they might leave in the past, as this could affect the estimation basis of the prior period adjustment.
- Determine the basis on which the fair value of the options has been calculated – should be based on appropriate valuation based on market prices wherever possible. If an option pricing model has been used, determine which method has been adopted and whether it is appropriate and reflects the nature of the options.
- Confirm that the fair value is calculated as at the grant date and that this is when offer and acceptance has taken place.
- Recalculate the fair value using the same inputs and consider obtaining expert advice if appropriate.
- Check volatility to published statistics.
- Agree share price at issue to market price on that date.
- Agree risk-free rate appropriate at time of granting options.
- Assess whether there are any terms of the share option which give the directors a choice of exercise through cash receipt as opposed to exercising the share options.

- Recalculate the expense spread over the vesting period.
- Obtain written representations from management confirming their view that any assumptions they have used in measuring fair value are reasonable and that there are no further share-based payment schemes in existence that have not been disclosed.
- Check disclosure is in accordance with IFRS 2.

Debenture

Accounting issues

- The current treatment of leaving the debenture at the value of the net proceeds does not comply with IFRS 9 and IAS 32, as there has been no accounting for the post-issue finance costs and the fact that the instrument is a hybrid instrument which needs to be split into liability and equity components.
- Calculate the liability element as the present value of the cash flows of the debenture, discounted at the market rate of interest for comparable borrowings with no conversion rights. This will require some attention on our part, to ensure that an appropriate interest rate has been selected.
- Classify as equity the remainder of the proceeds representing the fair value of the right to convert.
- The market rate of interest should then be used to unwind the discounting for the current year which will bring the liability closer to the final redemption value and allocate an appropriate finance charge to profit or loss. The annual interest payment will also be recorded as a cash flow and will reduce the liability.

Audit procedures

- Review debenture deed and agree nominal interest rates and conversion terms.
- Review calculation of the fair value of the liability at the date of issue.
- Confirm appropriate discount rate used by reference to external market while considering if market rates are representative of the instrument issued.
- We should discuss and assess the reasons for the selection of the rate by management and the assumptions made.
- We should review interest rates associated with companies in the industries which make up Mayflower's core interests and also some conglomerates (even though the mix is highly unlikely to be anything like Mayflower's) and compare these with the rate selected by management. We could also perform a sensitivity analysis, if a material risk is identified, in order to quantify the effect of changing the interest rate.
- Agree calculation of amortised cost after it has been performed by the directors.
- Agree initial proceeds and interest payments to records and bank statement.
- Review disclosures and ensure in accordance with IAS 32, IFRS 9, IFRS 7 and IFRS 13.

Properties

Accounting issues

Classification as investment property:

Exeter House	This property does not fall within the definition of investment property in accordance with IAS 40, as it is owner occupied. It should be accounted for as property, plant and equipment under IAS 16.

33–39 Reeves Road	Although this property is not legally owned, it is held as a right-of-use asset in accordance with IFRS 16, *Leases*, and can be treated as an investment property from the date of renting out to a third party.
41–51 Reeves Road	Although vacant it can be classified as an investment property as it is held for investment purposes.
Falcon House	This property is owned by Mayflower and the vast majority of it is let out to third parties. It appears unlikely that the two parts of the property (that rented to a third party and that used by Mayflower) could be sold or leased separately, particularly the part used by Mayflower, as it is a couple of small rooms in the basement. It is therefore inappropriate to treat the two parts separately as investment property and property, plant and equipment respectively. As the part used by Mayflower appears to be an insignificant portion of the whole, the whole property can be treated as an investment property.

Valuation

Exeter House	This should be valued according to IAS 16 (at cost or revalued amount less depreciation).
33–39 Reeves Road	This should have been recognised at the inception of the lease at the present value of the future lease payments. On being rented out it would be valued at fair value in accordance with company policy in respect of investment properties. The revaluation to fair value on transfer to investment properties is accounted for under IAS 16. Thereafter changes in fair value are recognised in profit or loss.
41–51 Reeves Road	This should be initially recognised at cost, including transaction costs. As the fair value model is being adopted the asset should be subsequently recognised at fair value. Changes in fair value are recognised in profit or loss.
Falcon House	Recognised at fair value, changes in fair value are recognised in profit or loss, as above.

IAS 40 requires that the fair value of the investment properties should be measured in accordance with IFRS 13.

Audit procedures

- Confirm that all investment properties are classified in accordance with IAS 40 definitions (see above).
- Ensure that Exeter House is reclassified as property, plant and equipment.
- Assess useful life of Exeter House and residual value in order to recalculate and agree depreciation charged.
- Determine the valuation policy to be used for Exeter House ie, cost or valuation and ensure that the policy is correctly applied.
- Evaluate the process by which Mayflower establishes fair values of investment properties and the control environment around such procedures.
- Determine basis for calculation of fair values (per IAS 40 (40) fair value must reflect rental income from current leases and other assumptions that market participants would use

when pricing investment property under current market conditions). Look for best evidence of fair value (for example, year-end prices in an active market for similar properties in the same location and condition).

- If external valuers have been used agree valuation to valuer's certificate and assess the extent that they can be relied on in accordance with ISA 620.
- If fair values have been based on discounted cash flows, ie, future rentals, determine whether this is the most appropriate estimate of a market-based exit value. Compare predicted cash flows with rental agreements. Review the basis of the interest rate applied.
- Review documentation to support method used.
- Recalculate the gain or loss on change in fair value and agree to amount recognised in profit or loss.
- If fair value cannot be measured reliably confirm use of cost model.
- Agree disclosure is in accordance with IAS 40 and IFRS 13.

Chapter 20

Groups: types of investment and business combination

Introduction

Learning outcomes

Chapter study guidance

Learning topics

1 Summary and categorisation of investments

2 IFRS 10, Consolidated Financial Statements

3 IFRS 3, Business Combinations

4 IFRS 13, Fair Value Measurement (business combination aspects)

5 IAS 28, Investments in Associates and Joint Ventures

6 IFRS 11, Joint Arrangements

7 Question technique and practice

8 IFRS 12, Disclosure of Interests in Other Entities

9 Step acquisitions

10 Disposals

11 Consolidated statements of cash flows

12 Audit focus: group audits

13 Auditing global enterprises

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Appraise and evaluate cash flow measures and disclosures in single entities and groups
- Analyse and evaluate the criteria used to determine whether and how different types of investment are recognised and measured as business combinations
- Calculate and produce, from financial and other data, the amounts to be included in an entity's consolidated financial statements in respect of its new, continuing and discontinued interests (which include situations when acquisitions occur in stages and in partial disposals) in subsidiaries, associates and joint ventures
- Justify and conclude for a particular scenario what comprises sufficient, appropriate audit evidence
- Design and determine audit procedures in a range of circumstances and scenarios, for example identifying an appropriate mix of tests of controls, analytical procedures and tests of details
- Demonstrate and explain, in the application of audit procedures, how relevant ISAs affect audit risk and the evaluation of audit evidence

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
1	**Summary and categorisation of investments** In a working context you will have to account for acquisitions, disposals and business combinations such as investments in associates, joint arrangements, or consolidations of subsidiaries.	**Approach** This chapter is revision from your studies at Professional Level. This introductory section starts with a useful summary table showing which standard to apply. **Stop and think** Can a 45% investment be a subsidiary?	Exam questions will be more complex than at Professional Level and you will need to exercise judgement more.	N/A
2	**IFRS 10, *Consolidated Financial Statements*** IFRS 10 will be familiar to you, but you should still	**Approach** The key to this section is **control.** Study the definition carefully, and make sure you understand	You may well have to determine the status of an investment by applying the definition of control.	N/A While there are no interactive questions in this section, you can test yourself by covering up the

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	approach this section as if you are studying it from scratch, because it is so fundamental to have a grasp of the brought forward knowledge.	power, returns and the link between power and returns. Then read through the sections on exemption, exclusion and uniform accounting policies. **Stop and think** Why would uniform accounting policies be preferable?		solution to the worked example which tests some rather complex aspects of control that you won't have met at earlier levels.
3	**IFRS 3, *Business Combinations*** Again this should be revision, but bear in mind that the level required is A rather than B, as before.	**Approach** Focus on the four-stage process in using the acquisition method: • Identify acquirer • Acquisition date • Recognise and measure assets • Recognise and measure goodwill (see worked examples) **Stop and think** What is a control premium?	At Advanced Level, the acquirer may not always be easy to identify.	**IQ1: Calculation of goodwill** This question focuses on the consideration aspect of the goodwill calculation. You should do both interactive questions in this section.
4	**IFRS 13, *Fair Value Measurement* (business combination aspects)** You have met IFRS 13 in many contexts, including investment property and financial instruments. This section focuses on the business combination issues, relating to valuing the assets and liabilities for the	**Approach** Recap the IFRS 13 rules on fair value, including the hierarchy. Then study the worked example carefully. **Stop and think** How is a decommissioning liability valued in a business combination?	At Advanced Level the valuation of the assets and liabilities in a business combination may not be straightforward.	**IQ3: Goodwill on consolidation** This focuses on the assets and liabilities, the other main part of the goodwill calculation.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	purposes of calculating goodwill.			
5	**IAS 28, *Investments in Associates and Joint Ventures*** Associates and joint ventures are both accounted for using the equity method.	**Approach** Make sure you know how the equity method is applied in the statement of financial position, the statement of profit or loss and in other comprehensive income. **Stop and think** What is the double entry for when an associate sells to the parent?	As well as being comfortable with equity accounting, you may come across associates in a step acquisition (associate to subsidiary) or disposal (subsidiary to associate), so you need to be clear about what is meant by significant influence. Joint ventures have been tested regularly, sometimes in an audit context or where the investment has been wrongly treated as a joint operation.	N/A While there are no interactive questions in this section, you can test yourself by covering up the solution to the worked example.
6	**IFRS 11, *Joint Arrangements*** Business combinations are becoming increasingly complex. The nature of purchase consideration and the legal form of such transactions are often driven by financial and tax features. Structures are carefully modelled to ensure that transactions optimise post-acquisition earnings as well as meeting corporate strategic objectives.	**Approach** Make sure you understand the distinction between joint ventures (equity accounted) and joint operations (line by line recognition of assets, liabilities, revenue and expenses). **Stop and think** How is a joint arrangement different from an associate?	Joint ventures are examined much more frequently than joint operations.	**IQ4: Joint arrangement** This question asks why joint operations are formed.
7	**Question technique and practice**	**Approach** Learn the step-by-	In the exam you will need to know all the techniques used in	**IQ6: Consolidation technique 2**

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
	This is an essential section – the brought forward knowledge should be second nature.	step process in sections 7.1 and 7.2. Then do both interactive questions. **Stop and think** Did you get them at least 80% right? If not, practise using the self-test questions at the beginning of the Chapter.	this section, but the consolidation will only be one of the complexities you will be facing.	You should really do both interactive questions, but if you're going to pick one, pick this one as it is more substantial than IQ5: Consolidation technique 1.
8	**IFRS 12, *Disclosure of Interests in Other Entities*** Users should review the disclosures included in the financial statements regarding business combinations. Extensive disclosures allow users of the financial statements to understand the effect of the combination on the group's operations during the period. This improves the predictive value of the financial information.	**Approach** Read through and highlight. **Stop and think** Extensive disclosures allow users of the financial statements to understand the effect of the combination on the group's operations during the period. This improves the predictive value of the financial information.	IFRS 12 has not been examined yet. It is more likely to come up in an auditing context.	N/A There are no interactive questions on this short section.
9	**Step acquisitions** This topic is new to Advanced Level, and happens quite a lot in practice.	**Approach** The key question is: has an accounting boundary been crossed? Whenever you cross the 50% boundary, you revalue, and a gain or loss is reported in profit or loss for the year. If you do not cross the 50% boundary, no	Step acquisitions are regularly tested, so it is important to get a good grasp of the technique in this section.	**IQ7: Control in stages – no previous significant influence** You don't have to prepare the consolidated financial statements in this question, but you do have to do the significant calculations relating to step acquisitions that will enable you to prepare them.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		gain or loss is reported; instead there is an adjustment to the parent's equity. **Stop and think** If you go from 60% to 80%, an accounting boundary is not crossed. If you go from 40% to 80% it is.		
10	**Disposals** While you have met disposals at Professional Level, you will not have come across partial disposals.	**Approach** As with step acquisitions, the key question is, was an accounting boundary crossed? **Stop and think** If you go from 80% to 60%, an accounting boundary is not crossed. If you go from 80% to 40% it is.	Partial disposals (subsidiary to associate or investments) are more likely to be tested at Advanced Level because complete disposals were tested at Professional Level.	**IQ8: All types of disposal** This covers 100% disposal, partial disposal without losing control, partial disposal to associate, control lost and partial disposal to asset at FVTOCI with control lost.
11	**Consolidated statements of cash flows** You may also be required to prepare consolidated statements of cash flows, which are likely to include the effect of acquisitions and disposals.	**Approach** You have already revised single company statements of cash flows, so focus on the aspects specific to groups: acquisition or disposal of an associate or subsidiary during the period. **Stop and think** When taking account of the disposal in calculating working capital movements, make sure you get the pluses and	Because single company statements of cash flow were examined at Professional Level, you are more likely to get a consolidated statement at Advanced Level.	**IQ10: Disposal** This question focuses on every calculation related to the group aspects, even though you don't need to prepare the statement itself.

Topic	Practical significance	Study approach	Exam approach	Interactive Questions
		minuses the right way round.		
12	**Audit focus: group audits** This is a long chapter, but do not neglect section 12, which deals with group audits, including foreign subsidiaries. You may wish to come back to that part once you have studied foreign currency transactions.	**Approach** Focus on sections 12.6 and 12.7 on group audit procedures. **Stop and think** What are component auditors?	You may well get a scenario where an overseas subsidiary company has been acquired during the year, audited by another firm overseas which raises technical audit issues regarding the audit approach and the application of ISA UK 600 (Revised November 2019).	**IQ11: Component auditors** This is a comprehensive question on a key issue in the audit of group financial statements.
13	**Auditing global enterprises** This section focuses on risk, which is generally greater when auditing a foreign company.	**Approach** Focus on the table in section 13.3. **Stop and think** Why is transfer pricing so difficult?	Risk is always important in auditing questions.	N/A

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Summary and categorisation of investments

Section overview

- This chapter revises IFRS 3, *Business Combinations*, which was covered at Professional Level. It also covers the following standards:
  - IFRS 13, *Fair Value Measurement* (business combination aspects)
  - IFRS 10, *Consolidated Financial Statements*
  - IAS 28, *Investments in Associates and Joint Ventures*
  - IFRS 11, *Joint Arrangements*
  - IFRS 12, *Disclosure of Interests in Other Entities*
- Some of the above IFRS® Standards, or the topics to which they relate, were covered at Professional Level, but a deeper knowledge is needed at Advanced Level.

A summary of the different types of investment and the required accounting for them is as follows.

Investment	Criteria	Required treatment in group accounts
Subsidiary	Control	Full consolidation (IFRS 10)
Associate	Significant influence	Equity accounting (IAS 28)
Joint venture	Contractual arrangement	Equity accounting (IAS 28), distinguish from joint operation (IFRS 11)
Investment which is none of the above	Asset held for accretion of wealth	As for single company accounts (per IFRS 9)

1.1 Investments in subsidiaries

The important point here is **control**. In most cases, this will involve the parent company owning a majority of the ordinary shares in the subsidiary (to which normal voting rights are attached). There are circumstances, however, when the parent may own only a minority of the voting power in the subsidiary, **but** the parent still has control.

IFRS 10, *Consolidated Financial Statements* has **control** as the key concept underlying the parent/subsidiary relationship but it has broadened the definition and clarified its application. This will be covered in more detail in section 2 below.

IFRS 10 states that an investor **controls** an investee if and only if it has all of the following.

(a) **Power** over the investee

(b) Exposure, or rights, to **variable returns** from its involvement with the investee (see section 2)

(c) The **ability to use its power** over the investee to affect the amount of the investor's returns (see section 2)

1.2 Investments in associates

The key criterion here is **significant influence**. This is defined as the 'power to participate', but **not** to 'control' (which would make the investment a subsidiary).

Significant influence is presumed to exist if an investor holds **20% or more of the voting power** of the investee, **unless** it can be clearly shown that this is not the case.

However, the **existence of significant influence** can also be evidenced in other ways.

- Representation on the **board of directors** of the investee
- Participation in the **policy making process**
- **Material transactions** between investor and investee
- Interchange of **management personnel**
- Provision of **essential technical information**

IAS 28, *Investments in Associates and Joint Ventures* requires the use of the **equity method** of accounting for investments in associates. This method will be explained in section 5.

1.3 Accounting for investments in joint arrangements

IFRS 11 classes joint arrangements as either **joint operations** or **joint ventures**. The classification of a joint arrangement as a joint operation or a joint venture depends on the rights and obligations of the parties to the arrangement.

The detail of how to distinguish between joint operations and joint ventures will be considered in section 6.

1.3.1 Accounting treatment in group accounts

IFRS 11 requires that a joint operator recognises line by line the following in relation to its interest in a **joint operation**:

- Its (the joint operation's) assets, including its (the investor's) share of any jointly held assets
- Its liabilities, including its share of any jointly incurred liabilities
- Its revenue from the sale of its share of the output arising from the joint operation
- Its share of the revenue from the sale of the output by the joint operation
- Its expenses, including its share of any expenses incurred jointly

This treatment is applicable in both the separate and consolidated financial statements of the joint operator.

In its consolidated financial statements, IFRS 11 requires that a joint venturer recognises its interest in a **joint venture** as an investment and accounts for that investment using the equity method in accordance with IAS 28, *Investments in Associates and Joint Ventures* unless the entity is exempted from applying the equity method (see section 5).

In its separate financial statements, a joint venturer should account for its interest in a joint venture in accordance with IAS 27, *Separate Financial Statements,* namely:

- at cost;
- in accordance with IFRS 9, *Financial Instruments*; or
- using the equity method specified in IAS 28.

1.4 Other investments

Investments which do not meet the definitions of any of the above should be accounted for according to IFRS 9, *Financial Instruments.* An example of such an investment would be a 15% shareholding in a company with no significant influence.

2 IFRS 10, *Consolidated Financial Statements*

Section overview

IFRS 10 covers the basic definitions and consolidation requirements and the rules on **exemptions** from preparing group accounts. The standard requires a parent to **present consolidated financial statements**, consolidating **all subsidiaries**, both foreign and domestic. The most important aspect is **control**.

2.1 Introduction

When a parent issues consolidated financial statements, it should consolidate **all subsidiaries**, both foreign and domestic. The first step in any consolidation is to identify the subsidiaries present in the group.

Definition

Consolidated financial statements: The financial statements of a group presented as those of a single economic entity. (IFRS 10)

You should make sure that you understand the various ways in which **control** can arise, as this is something that you may be asked to discuss in the context of a scenario in the exam.

2.1.1 Power

Power is defined as **existing rights that give the current ability to direct the relevant activities of the investee**. There is no requirement for that power to have been exercised.

Relevant activities may include:

- selling and purchasing goods or services
- managing financial assets
- selecting, acquiring and disposing of assets
- researching and developing new products and processes
- determining a funding structure or obtaining funding

In some cases assessing power is straightforward; for example, where power is obtained directly and solely from having the majority of voting rights or potential voting rights, and as a result the ability to direct relevant activities.

In other cases, assessment is more complex and more than one factor must be considered. IFRS 10 gives the following examples of **rights**, other than voting or potential voting rights, which individually, or alone, can give an investor power.

- Rights to appoint, reassign or remove key management personnel who can direct the relevant activities
- Rights to appoint or remove another entity that directs the relevant activities
- Rights to direct the investee to enter into, or veto changes to, transactions for the benefit of the investor
- Other rights, such as those specified in a management contract

Voting rights **in combination with other rights** may give an investor the current ability to direct the relevant activities. For example, this is likely to be the case when an investor holds 40% of the voting rights of an investee and holds substantive rights arising from options to acquire a further 20% of the voting rights.

IFRS 10 suggests that the **ability** rather than contractual right to achieve the above may also indicate that an investor has power over an investee.

An investor can have power over an investee even where other entities have significant influence or other ability to participate in the direction of relevant activities.

2.1.2 Returns

An investor must have exposure, or rights, to **variable returns** from its involvement with the investee in order to establish control.

This is the case where the investor's returns from its involvement have the potential to vary as a result of the investee's performance.

Returns may include the following:

- Dividends
- Remuneration for servicing an investee's assets or liabilities
- Fees and exposure to loss from providing credit support
- Returns as a result of achieving synergies or economies of scale through an investor combining use of their assets with use of the investee's assets

2.1.3 Link between power and returns

In order to establish control, an investor must be able to use its power to affect its returns from its involvement with the investee. This is the case even where the investor delegates its decision-making powers to an agent.



Professional skills focus: Structuring problems and solutions

When determining whether an entity has control, it often helps to draw a group structure diagram, which helps clarify what is going on. Such a diagram is also useful, together with a timeline, in questions involving mid-year acquisitions or disposals.



Worked example: Control

1 Twist holds 40% of the voting rights of Oliver and 12 other investors each hold 5% of the voting rights of Oliver. A shareholder agreement grants Twist the right to appoint, remove and set the remuneration of management responsible for directing the relevant activities. To change the agreement, a two-thirds majority vote of the shareholders is required. To date, Twist has not exercised its rights with regard to the management or activities of Oliver.

Requirement

Explain whether Twist should consolidate Oliver in accordance with IFRS 10.

2 Copperfield holds 45% of the voting rights of Spenlow. Murdstone and Steerforth each hold 26% of the voting rights of Spenlow. The remaining voting rights are held by three other shareholders, each holding 1%. There are no other arrangements that affect decision making.

Requirement

Explain whether Copperfield should consolidate Spenlow in accordance with IFRS 10.

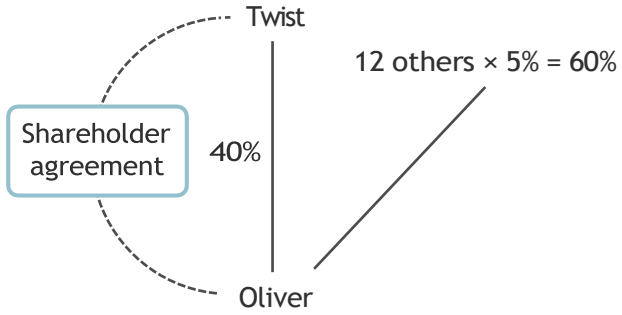
3 Scrooge holds 70% of the voting rights of Cratchett. Marley has 30% of the voting rights of Cratchett. Marley also has an option to acquire half of Scrooge's voting rights, which is exercisable for the next two years, but at a fixed price that is deeply out of the money (and is expected to remain so for that two-year period).

Requirement

Explain whether either of Scrooge or Marley should consolidate Cratchett in accordance with IFRS 10.

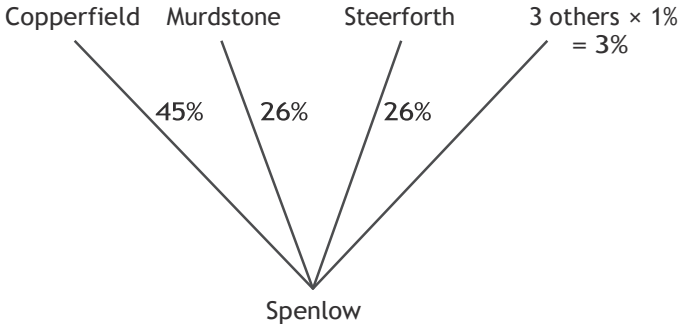
Solution

1 **Group structure 1**



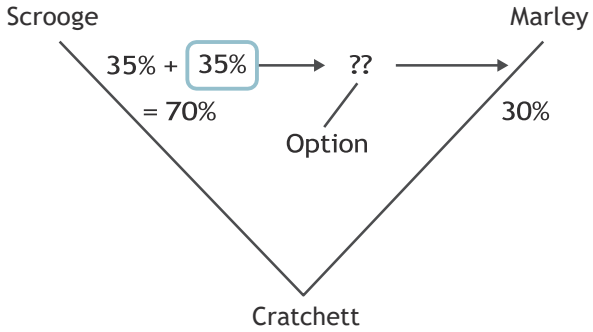
The absolute size of Twist's holding and the relative size of the other shareholdings alone are not conclusive in determining whether the investor has rights sufficient to give it power. However, the fact that Twist has **a contractual right to appoint, remove and set the remuneration of management** is sufficient to conclude that it **has power over Oliver**. The fact that Twist has not exercised this right is not a determining factor when assessing whether Twist has power. In conclusion, Twist does control Oliver, and should consolidate it.

2 **Group structure 2**



In this case, the size of Copperfield's voting interest and its size relative to the other shareholdings are sufficient to conclude that Copperfield **does not have power**. Only two other investors, Murdstone and Steerforth, would need to cooperate to be able to prevent Copperfield from directing the relevant activities of Spenlow.

3 **Group structure 3**



Scrooge holds a majority of the current voting rights of Cratchett, so is likely to meet the power criterion because it appears to have the current ability to direct the relevant activities. Although Marley has currently exercisable options to purchase additional voting rights (that, if exercised, would give it a majority of the voting rights in Cratchett), the terms and conditions associated with those options are such that the options are not considered substantive.

Thus voting rights, even combined with potential voting rights, may not be the deciding factor. Scrooge should consolidate Cratchett.

2.2 Exemption from preparing group accounts

A parent **need not present** consolidated financial statements if and only if all of the following hold:

(a) The parent is itself a **wholly owned subsidiary** or it is a **partially owned subsidiary** of another entity and its other owners, including those not otherwise entitled to vote, have been informed about, and do not object to, the parent not presenting consolidated financial statements.

(b) Its debt or equity instruments are **not publicly traded**.

(c) It is **not in the process of issuing securities** in public securities markets.

(d) The **ultimate or intermediate parent** publishes consolidated financial statements that comply with International Financial Reporting Standards.

A parent that does not present consolidated financial statements must comply with the IAS 27 rules on separate financial statements.

Although a parent may not have to prepare consolidated financial statements, it may wish to provide qualitative information about the nature and size of ownership of un-consolidated subsidiaries as this could be beneficial to the users of the financial statements.

2.3 Potential voting rights

An entity may own share warrants, share call options or other similar instruments that are **convertible into ordinary shares** in another entity. If these are exercised or converted they may give the entity voting power or reduce another party's voting power over the financial and operating policies of the other entity (potential voting rights). The **existence and effect** of potential voting rights, including potential voting rights held by another entity, should be considered when assessing whether an entity has control over another entity (and therefore has a subsidiary). Potential voting rights are considered only if the rights are **substantive** (meaning that the holder must have the practical ability to exercise the right).

In assessing whether potential voting rights give rise to control, the investor should consider the **purpose and design of the instrument**. This includes an assessment of the various terms and conditions of the instrument as well as the investor's apparent expectations, motives and reasons for agreeing to those terms and conditions.

2.4 Exclusion of a subsidiary from consolidation

Where a parent controls one or more subsidiaries, IFRS 10 requires that consolidated financial statements are prepared to include **all subsidiaries, both foreign and domestic**, other than:

- those held for sale in accordance with IFRS 5, *Non-current Assets Held for Sale and Discontinued Operations*; and
- those held under such long-term restrictions that control cannot be operated.

The rules on exclusion of subsidiaries from consolidation are necessarily strict, because this is a common method used by entities to manipulate their results. If a subsidiary which carries a

large amount of debt can be excluded, then the gearing of the group as a whole will be improved. In other words, this is a way of taking debt **out of the consolidated statement of financial position**.

IFRS 10 is clear that a subsidiary should not be excluded from consolidation simply because it is loss making or its business activities are dissimilar from those of the group as a whole. IFRS 10 rejects the latter argument: exclusion on these grounds is not justified because better information can be provided about such subsidiaries by consolidating their results and then giving additional information about the different business activities of the subsidiary, eg, under IFRS 8, *Operating Segments*.

2.5 Other matters

Different reporting dates

Where one or more subsidiaries prepare accounts to a different reporting date from the parent and the bulk of other subsidiaries in the group:

(a) the subsidiary may prepare additional statements to the reporting date of the rest of the group, for consolidation purposes; or

(b) if this is not possible, the subsidiary's accounts may still be used for consolidation **provided that** the gap between the reporting dates is **three months or less** and that **adjustments are made** for the effects of significant transactions or other events that occur between that date and the parent's reporting date.

Uniform accounting policies

Uniform **accounting policies** should be used and **adjustments** must be made where the subsidiary's policies differ from those of the parent.

Date of inclusion/exclusion

The results of subsidiary undertakings are included in the consolidated financial statements from:

(a) the date of 'acquisition' ie, the **date on which the investor obtains control**; to

(b) the date of 'disposal' ie, the **date when the investor loses control**.

Once an investment is no longer a subsidiary, it should be treated as an associate under IAS 28 (if applicable) or as an investment under IFRS 9.

Accounting for subsidiaries and associates in the parent's separate financial statements

A parent company will usually produce its own single company financial statements. In these statements, governed by IAS 27, *Separate Financial Statements,* investments in subsidiaries and associates included in the consolidated financial statements should be **either**:

- accounted for at **cost**;
- in accordance with **IFRS 9**; or
- using the equity method specified in IAS 28.

Where subsidiaries are **classified as held for sale** in accordance with IFRS 5 they should be accounted for in accordance with IFRS 5.

Non-controlling interest (NCI)

Within the statement of profit or loss and other comprehensive income, profit for the year and total comprehensive income must be split between the shareholders of the parent and the non- controlling interest.

IFRS 10 requires an entity to attribute their share of total comprehensive income to the non-controlling interest, even if this results in a negative (debit) NCI balance.

Acquisitions and disposals which do not result in a change of control

Acquisitions of further shares in an existing subsidiary or disposals of shares by a parent which do not result in a loss of control are accounted for within shareholders' equity.

No gain or loss is recognised and goodwill is not remeasured. This is explained further within sections 9 and 10 of this chapter.

Loss of control

Where a parent loses control of a subsidiary:

- assets, liabilities and the non-controlling interest must be derecognised;
- any interest retained is recognised at fair value at the date of loss of control; and
- a gain or loss on loss of control is recognised in profit or loss.

This is explained further within section 10 of this chapter.

3 IFRS 3, *Business Combinations*

Section overview

- IFRS 3 refers to business combinations as 'transactions or events in which an **acquirer** obtains control of one or more **businesses**'. In a straightforward business combination one entity acquires another, resulting in a parent/subsidiary relationship.
- Business combinations are accounted for using the **acquisition method**.

3.1 The acquisition method

All business combinations should be recognised using the **acquisition** method which involves the following steps.

(a) **Identifying an acquirer** – which obtains **control** of the other entity. If this cannot be established from shareholdings and other factors listed in IFRS 10 (see section 2.1 above), IFRS 3 provides additional indicators, such as the fact that the entity **with the larger fair value** is likely to be the acquirer.

(b) **Determining the acquisition date**. This is generally the date on which the parent company (acquirer) transfers consideration and acquires the net assets of the acquiree.

(c) **Recognising and measuring** the **identifiable assets** acquired, the **liabilities** assumed and any **non-controlling interest** in the acquiree.

(d) **Recognising and measuring goodwill** or a **gain from a bargain purchase**.

Measurement of the non-controlling interest in Step 3 along with the measurement of goodwill in Step 4 are covered in more detail in the next section.

Measurement of the identifiable assets and liabilities in Step 3 is covered in section 3.6 of this chapter.

3.2 Calculation of goodwill

IFRS 3 (Revised) requires goodwill acquired in a business combination (or a gain on a bargain purchase) to be measured as:

£

Consideration transferred: Fair value of assets given, liabilities assumed and equity instruments issued, including contingent amounts	X
Non-controlling interest at the acquisition date	X
	X
Less total fair value of net assets of acquiree	(X)
Goodwill/(gain from a bargain purchase)	X/(X)

This calculation includes the non-controlling interest and is therefore calculated based on the whole net assets of the acquiree.

Sections 3.3 and 3.4 consider the first two elements of the revised calculation – consideration transferred and the non-controlling interest – in more detail.



Worked example: Calculation of goodwill 1

On 1 January 20X7, Avon acquired 80% of the equity share capital of Tweed, for a total consideration of £670,000. The fair value of the net assets of Tweed at this date was £700,000. The non-controlling interest is measured at £140,000.

Requirement

What goodwill arises on the acquisition?

Solution

Goodwill is calculated as:

	£
Consideration transferred	670,000
Non-controlling interest at the acquisition date	140,000
	810,000
Less total fair value of net assets of acquiree	(700,000)
Goodwill	110,000

In this example the non-controlling interest has been measured as the relevant percentage of Tweed's acquisition date net assets ie, 20% × £700,000.

Note that the non-controlling interest is not necessarily calculated as a proportion of acquisition date net assets.

3.3 Consideration transferred

Contingent consideration – initial measurement

IFRS 3 requires the consideration to be measured at the fair value at the acquisition date. This includes any contingent consideration payable even if, at the date of acquisition, **it is not deemed probable** that it will be paid.

Contingent consideration – subsequent measurement

IFRS 3 requires contingent consideration to be classified as follows:

- A liability where contingent consideration is cash or shares to a specific value
- Equity where contingent consideration is a specified number of ordinary shares regardless of their value

Subsequent changes are then dealt with as follows:

(a) If the change is due to additional information obtained that affects the position at the acquisition date, goodwill should be remeasured.

(b) If the change is due to events which took place after the acquisition date, for example, meeting earnings targets:

  (1) where consideration is recorded as a liability (including a provision), any remeasurement is recorded in profit or loss (so an increase in the liability due to strong performance of the subsidiary will result in an expense and a decrease in the liability due to underperformance will result in a gain); and

  (2) where consideration is recorded in equity, remeasurement is not required.

The treatment means that group profits are now reduced where good performance of the subsidiary results in additional payments to the seller.

Acquisition-related costs

These costs do not form part of consideration within the above calculation. Instead, all finders' fees, legal, accounting, valuation and other professional fees must be expensed through profit or loss.

Costs incurred to issue securities will be dealt with in accordance with IFRS 9. The treatment will therefore reduce goodwill values and profits.

3.4 Non-controlling interest

The standard allows a **choice** in valuing the non-controlling interest within the goodwill calculation. It may be measured either:

- as a proportion of the identifiable net assets of the subsidiary at the acquisition date (as in the example above); or
- at fair value of the equity shares held by the non-controlling interest on the acquisition date.

The choice is available on a transaction by transaction basis.

The **proportion of net assets** method is used to value the non-controlling interest (NCI) at the reporting date. It requires identifiable assets to be recognised at the fair values of the individual assets. Where this method is used to value the non-controlling interest under IFRS 3 (Revised), the resulting goodwill corresponds only to the share of the entity held by the parent company.

The **fair value** method brings measurement of the non-controlling interest into line with measurement of consideration and the acquiree's net assets (ie, all at fair value). The fair value of the non-controlling interest (eg, the active market prices of the equity shares not held by the acquirer) is likely to exceed the proportion of net assets attributable to the non-controlling interest, this being by an amount which represents goodwill attributable to the non-controlling interest. Therefore goodwill on acquisition calculated using this method will represent 100% of goodwill in the acquiree.
Accordingly, this method is sometimes known as the '**full goodwill**' method.



Worked example: Calculation of goodwill 2

The consideration transferred by National plc when it acquired 80,000 of the 100,000 equity shares of Locale Ltd was £25 million. At the acquisition date the fair value of Locale Ltd's net assets was £21 million and the fair value of the 20,000 equity shares in Locale Ltd not acquired was £5 million.

Requirement

Calculate the goodwill acquired in the business combination on the basis that the non-controlling interest in Locale Ltd is measured at:

(1) Its share of identifiable net assets

(2) Fair value of the non-controlling interest's equity shares

Solution

Goodwill

	(1) NCI at share of net assets £'000	(2) NCI at fair value £'000
Consideration transferred	25,000	25,000
Non-controlling interest – 20% × £21m/fair value	4,200	5,000
	29,200	30,000
Total net assets of acquiree	(21,000)	(21,000)
Goodwill acquired in business combination	8,200	9,000

As the non-controlling interest is £0.8 million higher when measured at fair value, it follows that goodwill is also £0.8 million higher.

This amount is the goodwill relating to the non-controlling interest. The calculation of goodwill when the NCI is valued at fair value could be laid out as:

	Group £'000	NCI £'000
Consideration/fair value	25,000	5,000
Share of net assets 80%/20% × £21m	(16,800)	(4,200)
Goodwill	8,200	800

Total (or full) goodwill is £9 million; of this, the parent's share is £8.2 million and the non-controlling interest's share is £0.8 million.

Note that the goodwill is not split in the same proportion as ownership of the shares:

- National owns 80% of the shares but 91% of goodwill.
- The non-controlling interest owns 20% of shares but just 9% of goodwill.

This discrepancy is due to the 'control premium' paid by National.



Interactive question 1: Calculation of goodwill

On 1 January 20X5, ABC acquired 90% of DEF when the fair value of DEF's net assets was £18 million. The consideration was structured as follows.

- Three million ABC ordinary shares to be issued on the acquisition date.
- An additional one million ABC ordinary shares to be issued on 31 December 20X6 if DEF's revenue increases by 10% in the interim two years. This condition is likely to be achieved.

The market price of ABC ordinary shares is £7 at the acquisition date and has increased to £9 by 31 December 20X6.

ABC incurs professional acquisition fees amounting to £50,000.

It is ABC group policy to value the non-controlling interest using the proportion of net assets method.

Requirement

Calculate the consideration transferred and the goodwill arising on acquisition.

See **Answer** at the end of this chapter.

3.4.1 Non-controlling interest – subsequent valuation

Where the non-controlling interest is measured using the proportion of net assets method, the NCI at the reporting date is calculated as the non-controlling interest's share of the subsidiary's net assets.

Where the non-controlling interest is measured using the fair value method, a consolidation adjustment is required to recognise the additional goodwill in the consolidated statement of financial position. This is best achieved using the following calculation:

Non-controlling interest

	£	£
Share of net assets (NCI% × net assets at reporting date (W2))		X
Share of goodwill		
NCI at acquisition date at fair value (W3)	X	
NCI at acquisition date at share of net assets (NCI% × net assets at acquisition (W2))	(X)	
Difference, being goodwill attributable to NCI		X
		X

Note: The references are to standard consolidation workings:

W2 Net assets of the subsidiary

W3 Goodwill working



Interactive question 2: Non-controlling interest

Robson acquired 75% of the ordinary shares of Ives on 30 June 20X7. At this date Ives had net assets of £250,000, and the fair value of the 25% of Ives's equity shares not acquired by Robson was £90,000. Ives uses the fair value (full goodwill) method to measure non-controlling interests.

The abbreviated statement of financial position of Ives at 31 December 20X9 is as follows:

	£
Assets	440,000
Share capital	100,000
Retained earnings	245,000
Liabilities	95,000
	440,000

- At acquisition, the fair value of land owned by Ives was £50,000 greater than its carrying amount; Ives has subsequently sold the land to a third party.
- During the year ended 31 December 20X9, Ives sold goods to Robson, making a profit of £12,000. Half of these goods are included in Robson's inventory count at the year end.

Requirement

What is the value of the non-controlling interest in the consolidated statement of financial position at 31 December 20X9?

See **Answer** at the end of this chapter.

3.5 Step acquisitions

Under IFRS 3 and IFRS 10, acquisition accounting is only applied to business combinations when control is achieved.

Where a parent acquires control of a subsidiary in stages (a step acquisition), this means that goodwill is only calculated once, upon initially achieving control. It is not then recalculated in response to further acquisitions of shares in the same subsidiary.

Accounting for step acquisitions is covered in further detail within section 9 of this chapter.

3.6 Assets and liabilities acquired

In the previous sections, we discussed the first two elements of the goodwill calculation: consideration transferred and the non-controlling interest.

This section deals with the third element of goodwill – the net assets acquired.

3.6.1 Recognition

Assets and liabilities existing at the acquisition date, and meeting the *Conceptual Framework* definition of an asset or liability, should be recognised within the goodwill calculation.

(a) Only those liabilities which **exist at the date of acquisition** are recognised (so not future operating losses or reorganisation plans which will be put into effect after control is gained).

(b) Some assets not recognised by the acquiree in its individual company financial statements may be recognised by the acquirer in the consolidated financial statements. These include identifiable intangible assets, such as brand names. Identifiable means that these assets are separable or arise from contractual or other legal rights.

3.6.2 Measurement

The basic requirement of IFRS 3 is that the identifiable assets and liabilities acquired are measured at their acquisition date **fair value.**

To understand the importance of fair values in the acquisition of a subsidiary, consider again the definition of goodwill.



Definition

Goodwill: Any excess of the cost of the acquisition over the acquirer's interest in the fair value of the identifiable assets and liabilities acquired as at the date of the exchange transaction.

The **statement of financial position of a subsidiary company** at the date it is acquired may not be a guide to the fair value of its net assets. For example, the market value of a freehold

building may have risen greatly since it was acquired, but it may appear in the statement of financial position at historical cost less accumulated depreciation.

3.6.3 What is fair value?

Fair value is defined as follows by IFRS 13, *Fair Value Measurement* – it is an important definition.

"The price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date." (IFRS 13, Appendix A)

We will look at the requirements of IFRS 3 and IFRS 13 regarding fair value in more detail in section 4. First, let us look at some practical matters. The following example will remind you how to make a fair value adjustment, using the standard consolidation workings from your Professional Level studies.



Worked example: Fair value adjustment

P Co acquired 75% of the ordinary shares of S Co on 1 September 20X5. At that date the fair value of S Co's non-current assets was £23,000 greater than their carrying amount, and the balance of retained earnings was £21,000. The fair value of the non-controlling interest was £17,000. The statements of financial position of both companies at 31 August 20X6 are given below. S Co has not incorporated any revaluation in its financial statements.

P Co values the non-controlling interest using the fair value (full goodwill) method.

P Co statement of financial position as at 31 August 20X6

	£	£
Assets		
Non-current assets		
Tangible assets	63,000	
Investment in S Co at cost	51,000	
		114,000
Current assets		82,000
Total assets		196,000
Equity and liabilities		
Equity		
Ordinary shares of £1 each	80,000	
Retained earnings	96,000	
		176,000
Current liabilities		20,000
Total equity and liabilities		196,000

S Co statement of financial position as at 31 August 20X6

	£	£
Assets		
Tangible non-current assets		28,000
Current assets		43,000
Total assets		71,000
Equity and liabilities		
Equity		

Ordinary shares of £1 each	20,000	
Retained earnings	41,000	
		61,000
Current liabilities		10,000
Total equity and liabilities		71,000

If S Co had revalued its non-current assets at 1 September 20X5, an addition of £3,000 would have been made to the depreciation expense charged for 20X5/X6.

Requirement

Prepare P Co's consolidated statement of financial position as at 31 August 20X6.

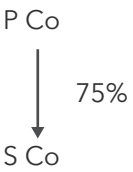
Solution

P Co consolidated statement of financial position as at 31 August 20X6

	£	£
Assets		
Non-current assets		
Tangible non-current assets £(63,000 + 28,000 + 23,000 – 3,000)	111,000	
Intangibles – goodwill (W3)	4,000	
		115,000
Current assets £(82,000 + 43,000)		125,000
Total assets		240,000
Equity and liabilities		
Capital and reserves		
Ordinary share capital	80,000	
Retained earnings (W5)	108,750	
Equity		188,750
Non-controlling interest (W4)		21,250
		210,000
Current liabilities £(20,000 + 10,000)		30,000
Total equity and liabilities		240,000

WORKINGS

(1) **Group structure**



(2) **Net assets**

	Reporting date	Acquisition	Post -acquisition
	£	£	£
Share capital	20,000	20,000	–
Retained earnings			
– per question	41,000	21,000	20,000
– additional depreciation	(3,000)		(3,000)
Fair value adjustment to PPE	23,000	23,000	
	81,000	64,000	17,000

(3) **Goodwill**

	£
Consideration transferred	51,000
Non-controlling interest	17,000
	68,000
Less net assets of acquiree (W2)	(64,000)
	4,000

(4) **Non-controlling interest**

	£	£
S Co (25% × £81,000 (W2))		20,250
NCI share of goodwill at acquisition		
FV of NCI at acquisition	17,000	
NCI share of net assets at acquisition (25% × £64,000)	(16,000)	
		1,000
Non-controlling interest		21,250

(5) **Retained earnings**

	£
P Co	96,000
S Co (£17,000 (W2) × 75%)	12,750
	108,750

Remember also that when preparing consolidated financial statements all **intra-group balances, transactions, profits and losses** need to be eliminated. Where there are provisions for unrealised profit and the parent is the seller the adjustment is made against the parent's retained earnings (in the retained earnings working). Where the subsidiary is the seller its retained earnings are adjusted (in the net assets working) thus ensuring that the non-controlling interest (ie, the minority interest) bear their share of the provision.

3.7 Impairments and the non-controlling interest

A subsidiary is subject to impairment review as a cash generating unit. The recoverable amount of the subsidiary is compared with its carrying amount to assess whether an impairment has occurred.

Notional grossing up of goodwill

In order to be comparable with the calculated recoverable amount, the carrying amount of the subsidiary must include 100% of both its net assets and goodwill. Therefore:

(a) where the proportion of net assets method is used to measure the NCI, goodwill must be notionally adjusted so that it represents both the parent and NCI share of goodwill (ie, 100% of goodwill). This involves grossing up the parent's goodwill according to percentage shareholdings; and

(b) where fair value is used to measure the NCI, goodwill already represents 100% of goodwill and no such adjustment is required.

Split of goodwill between the parent and NCI

Where goodwill requires grossing up according to ownership percentages, total goodwill is split in proportion to these ownership percentages.

Where goodwill does not require grossing up, total goodwill is not necessarily split in proportion to ownership percentages due to the control premium.



Context example: Notional goodwill

A acquires 80% of B for £120,000. The net assets at the date of acquisition were £130,000 and the fair value of the 20% non-controlling interest equity shares was £28,000.

Goodwill is calculated as:

	Proportion of net assets method	Fair value method
	£	£
Consideration transferred	120,000	120,000
Non-controlling interest (20% × £130,000)/fair value	26,000	28,000
	146,000	148,000
Net assets of acquiree	(130,000)	(130,000)
Goodwill	16,000	18,000
Attributable to:		
Parent	16,000	16,000
NCI [£28,000 – (20% × £130,000)]	n/a	2,000

For the purposes of an impairment review, the goodwill calculated using the proportion of net assets method is notionally adjusted as follows:

	£
Parent goodwill	16,000
Notional NCI goodwill (20%/80% × £16,000)	4,000
	20,000

In other words, the notional goodwill attributable to the non-controlling interest calculated here includes an element of control premium which is not evident when calculating goodwill attributable to the non-controlling interest using the fair value method.

Allocation of the impairment loss to parent / NCI

IFRS 3 has amended IAS 36, *Impairment of Assets* to require that any impairment loss is allocated 'on the same basis as that on which profit or loss is allocated'. This is likely to correspond to ownership percentages.

Therefore in the illustration above, any impairment loss to goodwill is allocated 80% to the parent and 20% to the NCI.

(a) Where the proportion of net assets method is used and goodwill is notionally calculated for the NCI, this split corresponds exactly to the split of goodwill. Assuming an impairment of £10,000:

	Parent	NCI
	£	£
Goodwill	16,000	4,000
Impairment (80%/20% × £10,000)	(8,000)	(2,000)
	8,000	2,000
Impairment of goodwill	50%	50%

Thus half the total goodwill has been impaired, being half of the parent's goodwill and half of the NCI's notional goodwill.

(b) Where the fair value method is used, and there is a control premium, such that the parent and NCI goodwill are not in proportion, then any impairment is not in proportion to the starting goodwill. This time assuming an impairment of £9,000:

	Parent	NCI
	£	£
Goodwill	16,000	2,000
Impairment (80%/20% × £9,000)	(7,200)	(1,800)
	8,800	200
Impairment of goodwill	45%	90%

3.8 Restructuring and future losses

An acquirer **should not recognise liabilities for future losses** or other costs expected to be incurred as a result of the business combination.

IFRS 3 explains that a plan to restructure a subsidiary following an acquisition is not a present obligation of the acquiree at the acquisition date. Neither does it meet the definition of a contingent liability. Therefore, an acquirer **should not recognise a liability for** such **a restructuring plan** as part of allocating the cost of the combination unless the subsidiary was already committed to the plan before the acquisition.

This **prevents creative accounting**. An acquirer cannot set up a provision for restructuring or future losses of a subsidiary and then release this to profit or loss in subsequent periods in order to reduce losses or smooth profits.

3.9 Intangible assets

The acquiree may have **intangible assets**, such as development expenditure. These can be recognised separately from goodwill only if they are **identifiable**. An intangible asset is identifiable only if it:

(a) is **separable** ie, capable of being separated or divided from the entity and sold, transferred or exchanged, either individually or together with a related contract, asset or liability; or

(b) arises from **contractual or other legal rights**.

3.10 Contingent liabilities

Contingent liabilities of the acquirer are **recognised** if their **fair value can be measured reliably**. A **contingent liability** must be recognised even if the outflow is not probable, provided there is a present obligation.

This is a departure from the normal rules in IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*; contingent liabilities are not normally recognised, but only disclosed.

After their initial recognition, the acquirer should measure contingent liabilities that are recognised separately at the higher of:

(a) the amount that would be recognised in accordance with IAS 37; and

(b) the amount initially recognised.

3.11 Other exceptions to the recognition or measurement principles

(a) **Deferred tax:** Use IAS 12 values

(b) **Employee benefits:** Use IAS 19 values

(c) **Indemnification assets:** Measurement should be consistent with the measurement of the indemnified item, for example an employee benefit or a contingent liability

(d) **Reacquired rights**: Value on the basis of the remaining contractual term of the related contract regardless of whether market participants would consider potential contractual renewals in determining its fair value

(e) **Share-based payment**: Use IFRS 2 values

(f) **Assets held for sale**: Use IFRS 5 values

3.12 Goodwill arising on the acquisition

Goodwill should be carried in the statement of financial position at **cost less any accumulated impairment losses**. The treatment of goodwill was covered in detail in your earlier studies.

3.13 Adjustments after the initial accounting is complete

Sometimes the fair values of the acquiree's identifiable assets, liabilities or contingent liabilities or the cost of the combination can only be measured **provisionally** by the **end of the period in which the combination takes place**. In this situation, the acquirer **should account for the combination using those provisional values**. The acquirer should **recognise any adjustments** to those provisional values as a result of completing the initial accounting:

(a) **within 12 months** of the acquisition date

(b) **from** the acquisition date (ie, retrospectively)

This means that:

(a) the **carrying amount** of an item that is recognised or adjusted as a result of completing the initial accounting shall be calculated **as if its fair value** at the acquisition date **had been recognised from that date**; and

(b) **goodwill should be adjusted** from the acquisition date by an amount equal to the adjustment to the fair value of the item being recognised or adjusted.

Any further adjustments after the initial accounting is complete should be **recognised only to correct an error** in accordance with IAS 8, *Accounting Policies, Changes in Accounting Estimates and Errors*. Any subsequent changes in estimates are dealt with in accordance with IAS 8 (ie, the effect is recognised in the current and future periods). IAS 8 requires an entity to

account for an error correction retrospectively, and to present financial statements as if the error had never occurred by restating the comparative information for the prior period(s) in which the error occurred.

3.14 Reverse acquisitions

IFRS 3 also addresses a certain type of acquisition, known as a **reverse acquisition** or **reverse takeover**. This is where Company A acquires ownership of Company B through a share exchange. (For example, a private entity may arrange to have itself 'acquired' by a smaller public entity as a means of obtaining a stock exchange listing.) The number of shares issued by Company A as consideration to the shareholders of Company B is so great that control of the combined entity after the transaction is with the shareholders of Company B.

In legal terms Company A may be regarded as the parent or continuing entity, but IFRS 3 states that, as it is the Company B shareholders who control the combined entity, **Company B should be treated as the acquirer**. Company B should apply the acquisition method to the assets and liabilities of Company A.

Context example: Reverse acquisition

On 1 January 20X5, ABC's share capital was 10,000 ordinary shares, quoted on a public exchange, and the unquoted share capital of DEF was 24,000 ordinary shares. On that date ABC's shares were quoted at £20. ABC issued 30,000 new shares, and then exchanged them for the entire share capital of DEF. The fair value of DEF's shares on 1 January 20X5 was agreed by the professional advisers to both ABC and DEF as £62.

After the acquisition, the relative interests of the two shareholder groups in ABC were as follows.

	Shares held	% of total
ABC shareholders	10,000	25
DEF shareholders	30,000	75
	40,000	100

As the DEF shareholder group controls the combined entities, DEF is treated as the acquirer and ABC as the acquiree.

If DEF had issued enough of its own shares to give ABC shareholders a 25% interest in DEF, it would have had to issue 8,000 shares (ie, 25/75 of 24,000 shares). DEF's share capital would then have been 32,000 (24,000 + 8,000) and the ABC shareholders' interest would have been 8,000 so 25%.

The consideration for the acquisition is £496,000, being 8,000 DEF shares at their agreed fair value of £62.

4 IFRS 13, *Fair Value Measurement* (business combination aspects)

Section overview

- The accounting requirements and disclosures of the **fair value exercise** are covered by **IFRS 3**. **IFRS 13**, *Fair Value Measurement* gives extensive guidance on how the fair value of assets and liabilities should be established.
- Business combinations are accounted for using the **acquisition method.**

CHAPTER 20

4.1 Fair value

The general rule under IFRS 3 is that the subsidiary's assets and liabilities **must be measured at fair value** except in **limited, stated cases**. The assets and liabilities must:

(a) meet the definitions of assets and liabilities in the IASB *Conceptual Framework*; and

(b) be part of what the acquiree (or its former owners) exchanged in the business combination rather than the result of separate transactions.

IFRS 13, *Fair Value Measurement* provides extensive guidance on how the fair value of assets and liabilities should be established.

This standard requires that the following are considered in measuring fair value:

(a) The asset or liability being measured

(b) The principal market (i.e., that where the most activity takes place) or where there is no principal market, the most advantageous market (i.e., that in which the best price could be achieved) in which an orderly transaction would take place for the asset or liability

(c) The highest and best use of the asset or liability and whether it is used on a standalone basis or in conjunction with other assets or liabilities

(d) Assumptions that market participants would use when pricing the asset or liability

Having considered these factors, IFRS 13 provides a hierarchy of inputs for arriving at fair value. It requires that Level 1 inputs are used where possible:

Level 1	Quoted prices in active markets for identical assets that the entity can access at the measurement date
Level 2	Inputs other than quoted prices that are directly or indirectly observable for the asset
Level 3	Unobservable inputs for the asset



Worked example: Examples of fair value and business combinations

For non-financial assets, fair value is decided based on the highest and best use of the asset as determined by a market participant. The following examples, adapted from the illustrative examples to IFRS 13, demonstrate what is meant by this.

1 **Example: Land (Based on IFRS 13: IE7)**

Anscome Co has acquired land in a business combination. The land is currently developed for industrial use as a site for a factory. The current use of land is presumed to be its highest and best use unless market or other factors suggest a different use. Nearby sites have recently been developed for residential use as sites for high-rise apartment buildings. On the basis of that development and recent zoning and other changes to facilitate that development, Anscome determines that the land currently used as a site for a factory could be developed as a site for residential use (ie, for high-rise apartment buildings) because market participants would take into account the potential to develop the site for residential use when pricing the land.

Requirement

How would the highest and best use of the land be determined?

2 **Example: Research and development project (Based on IFRS 13: IE9)**

Searcher acquires a research and development (R&D) project following a business combination. Searcher does not intend to complete the project. If completed, the project

would compete with one of its own projects (to provide the next generation of the entity's commercialised technology). Instead, the entity intends to hold (ie, lock up) the project to prevent its competitors from obtaining access to the technology. In doing this the project is expected to provide defensive value, principally by improving the prospects for the entity's own competing technology.

If it could purchase the R&D project, Developer Co would continue to develop the project and that use would maximise the value of the group of assets or of assets and liabilities in which the project would be used (ie, the asset would be used in combination with other assets or with other assets and liabilities). Developer Co does not have similar technology.

Requirement

How would the fair value of the project be measured?

3 **Example: Decommissioning liability (Based on IFRS 13: IE35)**

Deacon assumes a decommissioning liability in a business combination. It is legally required to dismantle a power station at the end of its useful life, which is estimated to be 20 years.

Requirement

How would the decommissioning liability be measured?

Solution

1 The highest and best use of the land would be determined by comparing both of the following:

(1) The value of the land as currently developed for industrial use (i.e., the land would be used in combination with other assets, such as the factory, or with other assets and liabilities)

(2) The value of the land as a vacant site for residential use, taking into account the costs of demolishing the factory and other costs (including the uncertainty about whether the entity would be able to convert the asset to the alternative use) necessary to convert the land to a vacant site (i.e., the land is to be used by market participants on a standalone basis)

The highest and best use of the land would be determined on the basis of the higher of those values.

2 The fair value of the project would be measured on the basis of the price that would be received in a current transaction to sell the project, assuming that the R&D would be used with its complementary assets and the associated liabilities and that those assets and liabilities would be available to Developer Co.

3 Because this is a business combination, Deacon must measure the liability at fair value in accordance with IFRS 13, rather than using the best estimate measurement required by IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*.

Deacon will use the expected present value technique to measure the fair value of the decommissioning liability. If Deacon were contractually committed to transfer its decommissioning liability to a market participant, it would conclude that a market participant would use all of the following inputs, probability weighted as appropriate, when estimating the price it would expect to receive.

(1) Labour costs

(2) Allocated overhead costs

(3) The compensation that a market participant would generally receive for undertaking the activity, including profit on labour and overhead costs and the risk that the actual cash outflows might differ from those expected

(4) The effect of inflation

(5) The time value of money (risk-free rate)

(6) Non-performance risk, including Deacon's own credit risk

As an example of how the probability adjustment might work, Deacon values labour costs on the basis of current marketplace wages adjusted for expected future wage increases. It determines that there is a 20% probability that the wage bill will be £15 million, a 30% probability that it will be £25 million and a 50% probability that it will be £20 million. Expected cash flows will then be (20% × £15m) + (30% × £25m) + (50% × £20m) = £20.5m. The probability assessments will be developed on the basis of Deacon's knowledge of the market and experience of fulfilling obligations of this type.



Interactive question 3: Goodwill on consolidation

Tyzo plc prepares its financial statements to 31 December. On 1 September 20X7 Tyzo plc acquired 6 million £1 shares in Kono Ltd at £2 per share. The purchase was financed by an additional issue of loan stock at an interest rate of 10%. At that date Kono Ltd produced the following interim financial information.

	£m
Non-current assets	
Property, plant and equipment (Note 1)	16.0
Current assets	
Inventories (Note 2)	4.0
Receivables	2.9
Cash and cash equivalents	1.2
	8.1
Total assets	24.1
Equity and liabilities	
Equity	
Share capital (£1 shares)	8.0
Reserves	4.4
	12.4
	£m
Non-current liabilities	
Long-term loan (Note 3)	4.0
Current liabilities	
Trade payables	3.2
Provision for taxation	0.6
Bank overdraft	3.9
	7.7
Total equity and liabilities	24.1

Notes

1 Please see the table below for information related to the property, plant and equipment of Kono Ltd at 1 September 20X7.

2 The inventories of Kono Ltd in hand at 1 September 20X7 consisted of raw materials at cost. They would have cost £4.2 million to replace at 1 September 20X7.

3 The long-term loan of Kono Ltd carries a rate of interest of 10% per annum, payable on 31 August annually in arrears. The loan is redeemable at par on 31 August 20Y1. The interest cost is representative of current market rates. The accrued interest payable by Kono Ltd at 31 December 20X7 is included in the trade payables of Kono Ltd at that date.

4 On 1 September 20X7 Tyzo plc took a decision to rationalise the group so as to integrate Kono Ltd. The costs of the rationalisation were estimated to total £3 million and the process was due to start on 1 March 20X8. No provision for these costs has been made in any of the financial statements given above.

5 Kono Ltd has disclosed a contingent liability of £200,000 in its interim financial statements relating to litigation.

6 Tyzo Group values the non-controlling interest using the proportion of net assets method.

Property, plant and equipment

	£m
Gross replacement cost	28.4
Net replacement cost	16.8
Economic value	18.0
Net realisable value	8.0

Requirement

Compute the goodwill on consolidation of Kono Ltd that will be included in the consolidated financial statements of Tyzo plc for the year ended 31 December 20X7, explaining your treatment of the items mentioned above. You should refer to the provisions of relevant accounting standards.

See **Answer** at the end of this chapter.



Professional skills focus: Assimilating and using information

There is a considerable amount of information to assimilate before you can proceed to the goodwill calculation. Consider doing the explanations first, then when you come to the calculation you will know what to do.

5 IAS 28, *Investments in Associates and Joint Ventures*



Section overview

- **IAS 28** deals with accounting for associates and joint ventures using the **equity method**.
- An associate exists where there is '**significant influence**'.

- The criteria for identifying a joint venture are contained in **IFRS 11**.
- The **accounting** for associates and joint ventures is **identical**.

IAS 28 does not apply to investments in associates or joint ventures held by venture capital organisations, mutual funds, unit trusts and similar entities that are measured at fair value in accordance with IFRS 9.

IAS 28 requires investments in associates to be accounted for using the equity method, **unless** the investment is classified as 'held for sale' in accordance with IFRS 5, in which case it should be accounted for under IFRS 5.

An investor is **exempt** from applying the equity method if (IAS 28.17):

(a) it is a parent exempt from preparing consolidated financial statements under IAS 27 (Revised); or

(b) all of the following apply:

  (1) The investor is a **wholly owned subsidiary** or it is a **partially owned subsidiary** of another entity and its other owners, including those not otherwise entitled to vote, have been informed about, and do not object to, the investor not applying the equity method.

  (2) Its securities are **not publicly traded**.

  (3) It is **not in the process of issuing securities** in public securities markets.

  (4) The **ultimate or intermediate parent** publishes consolidated financial statements that comply with International Financial Reporting Standards.

IAS 28 **does not allow** an investment in an associate to be excluded from equity accounting when an investee operates under severe long-term restrictions that significantly impair its ability to transfer funds to the investor. Significant influence must be lost before the equity method ceases to be applicable.

The use of the equity method should be **discontinued** from the date that the investor **ceases to have significant influence**.

From that date, the investor shall account for the investment in accordance with IFRS 9. The fair value of the retained interest must be regarded as its fair value on initial recognition as a financial asset under IFRS 9.

5.1 Separate financial statements of the investor

In accordance with IAS 27, *Separate Financial Statements* where an entity prepares separate financial statements it must account for associates (and joint ventures) in its separate financial statements in one of the following ways:

- Accounted for at **cost**
- In accordance with **IFRS 9**
- Using the equity method as described in IAS 28, *Investments in Associates and Joint Ventures*

5.2 Application of the equity method: consolidated accounts

The **equity method** should be applied in the consolidated accounts as follows:

(a) **Statement of financial position** includes investment in associate at cost plus (or minus) the group's share of the associate's post-acquisition profits (or losses) minus any impairments in the investment to date.

(b) **Profit or loss (statement of profit or loss and other comprehensive income):** group share of associate's profit after tax.

(c) **Other comprehensive income (statement of profit or loss and other comprehensive income):** group share of associate's other comprehensive income after tax.

The investment in the associate will also include any other long-term interests in the associate, for example preference shares or long-term receivables or loans.

IFRS 9 sets out a list of indications that a financial asset (including an associate) may have become impaired.

Many of the procedures required to apply the equity method are the same as are required for full consolidation. In particular, **fair value adjustments** are required and the group share of **intra-group unrealised profits** must be excluded.



Worked example: Associate

P Co, a company with subsidiaries, acquires 25,000 of the 100,000 £1 ordinary shares in A Co for £60,000 on 1 January 20X8. In the year to 31 December 20X8, A Co earns profits after tax of £24,000, from which it declares and pays a dividend of £6,000.

Requirement

How will A Co's results be accounted for in the individual and consolidated accounts of P Co for the year ended 31 December 20X8?

Solution

In the **individual accounts** of P Co, the investment will be recorded on 1 January 20X8 at cost. Unless there is an impairment in the value of the investment (see below), most companies will choose the policy that this amount (the cost) will remain in the individual statement of financial position of P Co permanently. The only entry in P Co's individual statement of profit or loss and other comprehensive income will be to record dividends received. For the year ended 31 December 20X8, P Co will:

DEBIT	Cash (£6,000 × 25%)	£1,500	
CREDIT	Income from shares in associated companies		£1,500

In the **consolidated accounts** of P Co equity accounting will be used. Consolidated profit after tax will include the group's share of A Co's profit after tax (25% × £24,000 = £6,000).

In the consolidated statement of financial position the non-current asset 'Investment in associates' will be stated at £64,500, being cost of £60,000 plus the group's share of post-acquisition retained profits of £4,500 ((24,000 – 6,000) × 25%).

5.3 Transactions between a group and its associate

Unlike for subsidiaries, trading transactions are not cancelled out. However, any **unrealised profits** on these transactions **should be eliminated**, but only to the extent of **the group's share**.

Where the **associate sells to the parent/subsidiary** the double entry is as follows, where A% is the parent's holding in the associate, and PURP is the provision for unrealised profit:

DEBIT	Retained earnings of parent/subsidiary	PURP × A%	
CREDIT	Group inventories/PPE		PURP × A%

Where the **parent/subsidiary sells to the associate**:

DEBIT	Retained earnings of parent/subsidiary	PURP × A%	
CREDIT	Investment in associate		PURP × A%

5.4 Associate's losses

When the equity method is being used and the investor's share of losses of the associate equals or exceeds its interest in the associate, the investor should **discontinue** including its share of further losses. The investment is reported at nil value. After the investor's interest is reduced to nil, **additional losses** should only be recognised where the investor has incurred obligations or made payments on behalf of the associate (for example, if it has guaranteed amounts owed to third parties by the associate).

5.5 Impairment losses

Any impairment loss is recognised in accordance with IAS 36, *Impairment of Assets* for each associate individually.

6 IFRS 11, *Joint Arrangements*



Section overview

- IFRS 11 classes joint arrangements as either **joint operations** or **joint ventures**.
- The classification of a joint arrangement as a joint operation or a joint venture depends on the **rights and obligations** of the parties to the arrangement.
- Joint arrangements are often found when each party can **contribute in different ways** to the activity. For example, one party may provide finance, another purchases or manufactures goods, while a third offers its marketing skills.

6.1 Definitions

The IFRS begins by listing some important definitions.



Definitions

Joint arrangement: An arrangement of which two or more parties have joint control.

Joint control: The contractually agreed sharing of control of an arrangement, which exists only when decisions about the relevant activities require the unanimous consent of the parties sharing control.

Joint operation: A joint arrangement whereby the parties that have joint control of the arrangement have rights to the assets and obligations for the liabilities relating to the arrangement.

Joint venture: A joint arrangement whereby the parties that have joint control of the arrangement have rights to the net assets of the arrangement. (IFRS 11, Appendix A)

6.2 Forms of joint arrangement

IFRS 11 classes joint arrangements as either joint operations or joint ventures. The classification of a joint arrangement as a joint operation or a joint venture depends on the rights and obligations of the parties to the arrangement.

A **joint operation** is a joint arrangement whereby the parties that have joint control (the joint operators) have rights to the assets, and obligations for the liabilities, of that joint

arrangement. A joint arrangement that is **not structured through a separate entity** is always a joint operation.

A **joint venture** is a joint arrangement whereby the parties that have **joint control** (the joint venturers) of the arrangement have **rights to the net assets** of the arrangement.

A **joint arrangement** that is structured through a **separate entity** may be either a joint operation or a joint venture. In order to ascertain the classification, the parties to the arrangement should assess the terms of the contractual arrangement together with any other facts or circumstances to assess whether they have:

- rights to the assets, and obligations for the liabilities, in relation to the arrangement (indicating a joint operation); and
- rights to the net assets of the arrangement (indicating a joint venture).

Detailed guidance is provided in the appendices to IFRS 11 in order to help this assessment, giving consideration to, for example, the wording contained within contractual arrangements.

IFRS 11 summarises the basic issues that underlie the classifications in the following diagram.

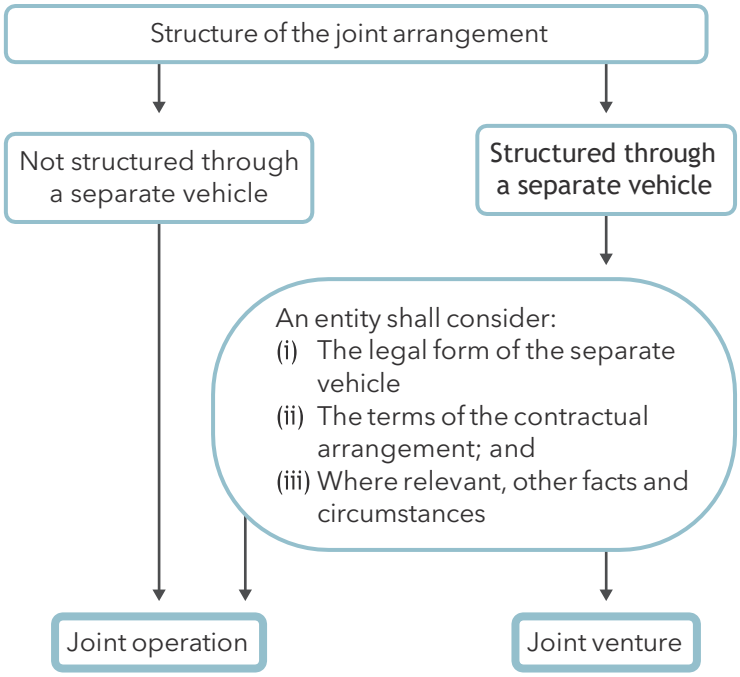


Figure 20.1: Joint Arrangements

(Source: IFRS 11: AG. B21)

6.2.1 Contractual arrangement

The existence of a contractual agreement distinguishes a joint arrangement from an investment in an associate. **If there is no contractual arrangement, then a joint arrangement does not exist**.

Evidence of a contractual arrangement could be in one of several forms.

- **Contract** between the parties
- **Minutes** of discussion between the parties
- Incorporation in the **articles or by-laws** of the joint venture

The contractual arrangement is usually **in writing**, whatever its form, and it will deal with the following issues surrounding the joint venture.

- **Its activity, duration and reporting obligations**
- The appointment of its **board of directors** (or equivalent) and the **voting rights** of the parties
- **Capital contributions** to it by the parties
- How its output, income, expenses or results are **shared** between the parties

It is the contractual arrangement which establishes **joint control** over the joint venture, so that no single party can control the activity of the joint venture on its own.

The terms of the contractual arrangement are key to deciding whether the arrangement is a joint venture or joint operation. IFRS 11 includes a table of issues to consider and explains the influence of a range of points that could be included in the contract. The table is summarised below.

	Joint operation	Joint venture
The terms of the contractual arrangement	The parties to the joint arrangement have rights to the assets, and obligations for the liabilities, relating to the arrangement.	The parties to the joint arrangement have rights to the net assets of the arrangement (ie, it is the separate vehicle, not the parties, that has rights to the assets, and obligations for the liabilities).
Rights to assets	The parties to the joint arrangement share all interests (eg, rights, title or ownership) in the assets relating to the arrangement in a specified proportion (eg, in proportion to the parties' ownership interest in the arrangement or in proportion to the activity carried out through the arrangement that is directly attributed to them).	The assets brought into the arrangement or subsequently acquired by the joint arrangement are the arrangement's assets. The parties have no interests (ie, no rights, title or ownership) in the assets of the arrangement.
Obligations for liabilities	The parties share all liabilities, obligations, costs and expenses in a specified proportion (eg, in proportion to their ownership interest in the arrangement or in proportion to the activity carried out through the arrangement that is directly attributed to them).	The joint arrangement is liable for the debts and obligations of the arrangement. The parties are liable to the arrangement only to the extent of their respective: • investments in the arrangement; or • obligations to contribute any unpaid or additional capital to the arrangement; or • both.
	The parties to the joint arrangement are liable for claims by third parties.	Creditors of the joint arrangement do not have rights of recourse against any party.

	Joint operation	Joint venture
Revenues, expenses, profit or loss	The contractual arrangement establishes the allocation of revenues and expenses on the basis of the relative performance of each party to the joint arrangement. For example, the contractual arrangement might establish that revenues and expenses are allocated on the basis of the capacity that each party uses in a plant operated jointly.	The contractual arrangement establishes each party's share in the profit or loss relating to the activities of the arrangement.
Guarantees	The provision of guarantees to third parties, or the commitment by the parties to provide them, does not, by itself, determine that the joint arrangement is a joint operation.	

6.2.2 Section summary

- There are two **types of joint arrangement**: joint ventures and jointly controlled operations.
- A **contractual arrangement** must exist which establishes joint control.
- **Joint control** is important: one **operator** must not be able to govern the financial and operating policies of the joint venture.

Professional skills focus: Concluding, recommending and communicating

The nature of a joint arrangement may not be immediately apparent. If asked to determine whether a joint arrangement is a joint operation or a joint venture, you should explain your reasons clearly.

6.3 Accounting treatment

The accounting treatment of joint arrangements depends on whether the arrangement is a joint venture or joint operation.

6.3.1 Accounting for joint operations

IFRS 11 requires that a joint operator recognises line by line the following in relation to its interest in a joint operation:

(a) Its assets, including its share of any jointly held assets

(b) Its liabilities, including its share of any jointly incurred liabilities

(c) Its revenue from the sale of its share of the output arising from the joint operation

(d) Its share of the revenue from the sale of the output by the joint operation

(e) Its expenses, including its share of any expenses incurred jointly

This treatment is applicable in both the separate and consolidated financial statements of the joint operator.

Interactive question 4: Joint arrangement

Can you think of examples of situations where this type of joint arrangement might take place?

See **Answer** at the end of this chapter.

6.3.2 Joint ventures

IFRS 11 and **IAS 28** require **joint ventures** to be accounted for using **the equity method**. The rules for equity accounting are included in IAS 28. These were covered in your Professional studies and are revised above.

Application of IAS 28 to joint ventures

The consolidated statement of financial position is prepared by:

- including the interest in the joint venture at cost plus share of post-acquisition total comprehensive income; and
- including the group share of the post-acquisition total comprehensive income in group reserves.

The consolidated statement of profit or loss and other comprehensive income will include:

- the group share of the joint venture's profit or loss; and
- the group share of the joint venture's other comprehensive income.

The use of the equity method should be **discontinued** from the date on which the joint venturer ceases to have joint control over, or have significant influence on, a joint venture.

Transactions between a joint venturer and a joint venture

Downstream transactions

A joint venturer may **sell or contribute assets** to a joint venture so making a profit or loss. Any such gain or loss should, however, only be recognised to the extent that it reflects the substance of the transaction.

Therefore:

- only the **gain** attributable to the interest of the other joint venturers should be recognised in the financial statements; and
- the full amount of any **loss** should be recognised when the transaction shows evidence that the net realisable value of current assets is less than cost, or that there is an impairment loss.

Upstream transactions

When a joint venturer purchases assets from a joint venture, the joint venturer should not recognise its share of the profit made by the joint venture on the transaction in question until it resells the assets to an independent third party ie, until the profit is realised.

Losses should be treated in the same way, **except** losses should be recognised immediately if they represent a reduction in the net realisable value of current assets, or a permanent decline in the carrying amount of non-current assets.

6.3.3 Section summary

- **Joint operations** are accounted for by including the investor's share of assets, liabilities, income and expenses as per the contractual arrangement.
- **Joint ventures** are accounted for using the **equity method** as under IAS 28.

7 Question technique and practice



Section overview

Although you have studied consolidation at Professional Level, it is vitally important that you have retained this knowledge and can put it into practice. This section summarises the basic question techniques and provides question practice before you move on to the more advanced topics of changes in group structure and foreign currency transactions. A number of standard workings should be used when answering consolidation questions.

7.1 Question technique for consolidated statement of financial position

As questions increase in complexity a formal pattern of workings is needed. Review the standard workings below.

(1) Establish group structure

P Ltd

↓ 80%

S Ltd

(2) **Set out net assets of S Ltd**

	At year end	At acquisition	Post-acquisition
	£	£	£
Share capital	X	X	X
Retained earnings	X	X	X
	X	X	X

(3) **Calculate goodwill**

	£
Consideration transferred	X
Plus Non-controlling interest at acquisition	X
	X
Less net assets at acquisition (see W2)	(X)
	X
Impairment to date	(X)
Balance c/f	X

(4) Calculate non-controlling interest (NCI) at year end

	£
At acquisition (NCI% × net assets (W2) or fair value)	X
Share of post-acquisition profits and other reserves (NCI% × post-acquisition (W2))	X
	X

(5) Calculate retained earnings

	£
P Ltd (100%)	X
S Ltd (share of post-acquisition retained earnings (see W2))	X
Goodwill impairment to date (see W3)	(X)
Group retained earnings	X

Note: You should use the proportionate basis for measuring the NCI at the acquisition date unless a question specifies the fair value basis.

7.2 Consolidated statement of profit or loss

As with the statement of financial position, as questions increase in complexity a formal pattern of workings is needed.

(1) **Establish group structure**

P Ltd



80%

S Ltd

(2) **Prepare consolidation schedule**

	P	S	Adj	Consol
	£	£	£	£
Revenue	X	X	(X)	X
Cost of sales – Per Q	(X)	(X)	X	(X)
– PURP (seller's books)	(X)	or (X)		
Expenses – Per Q	(X)	(X)		(X)
– Goodwill impairment (if any)*	(X)	(X)	(X)	
Tax – Per Q	(X)	(X)		(X)
Profit		X		

May need workings for (eg)

- PURPs
- Goodwill impairment

(3) **Calculate non-controlling interest**

				£
S PAT × NCI%	NCI% ×	X	=	X

* If the non-controlling interest is measured at fair value, then the NCI% of the impairment loss will be debited to the NCI. This is based on the NCI shareholding. For instance, if the parent has acquired 75% of the subsidiary and the NCI is measured at fair value, then 25% of any goodwill impairment will be debited to NCI.



Professional skills focus: Structuring problems and solutions

However complex consolidation questions may be at Advanced Level, following the above step-by- step structure will get you many of the marks. Practice is essential so that each step becomes routine.



Interactive question 5: Consolidation technique 1

At 1 July 20X8 Anima plc had investments in two companies: Orient Ltd and Oxendale Ltd. On 1 April 20X9 Anima plc purchased 85% of the ordinary share capital of Carnforth Ltd for £3 million.

Extracts from the draft individual financial statements of the four companies for the year ended 30 June 20X9 are shown below:

Statements of profit or loss

	Anima plc	Orient Ltd	Carnforth Ltd	Oxendale Ltd
	£	£	£	£
Revenue	1,410,500	870,300	640,000	760,090
Cost of sales	(850,000)	(470,300)	(219,500)	(345,000)
Gross profit	560,500	400,000	420,500	415,090
Operating expenses	(103,200)	(136,000)	(95,120)	(124,080)
Profit before taxation	457,300	264,000	325,380	291,010
Income tax expense	(137,100)	(79,200)	(97,540)	(86,400)
Profit for the year	320,200	184,800	227,840	204,610

Statements of financial position (extracts) at year end

	Anima plc	Orient Ltd	Carnforth Ltd	Oxendale Ltd
	£	£	£	£
Equity				
Ordinary share capital (£1 shares)	4,000,000	3,500,000	2,000,000	3,000,000
Retained earnings	1,560,000	580,000	605,000	340,000
	5,560,000	4,080,000	2,605,000	3,340,000

Additional information:

(1) A number of years ago Anima plc acquired 2.1 million of Orient Ltd's ordinary shares and 900,000 of Oxendale Ltd's ordinary shares. Balances on retained earnings at the date of acquisition were £195,000 for Orient Ltd and £130,000 for Oxendale Ltd. The non-controlling interest and goodwill arising on the acquisition of Orient Ltd were both calculated using the fair value method; the fair value of the non-controlling interest at acquisition was £1,520,000.

(2) At the date of acquisition the fair values of Carnforth Ltd's assets and liabilities were the same as their carrying amounts except for its head office (land and buildings) which had a fair value of £320,000 in excess of its carrying amount. The split of the value of land to buildings is 50:50 and the buildings had a remaining life of 40 years at 1 April 20X9. Carnforth Ltd's profits accrued evenly over the current year. The non-controlling interest and goodwill arising on the acquisition of Carnforth Ltd were both calculated using the proportionate method.

(3) During the year Anima plc sold goods to Orient Ltd and Oxendale Ltd at a mark-up of 15%. Anima plc recorded sales of £149,500 and £207,000 to Orient Ltd and Oxendale Ltd respectively during the year. At the year-end inventory count Orient Ltd was found still to be holding half these goods and Oxendale Ltd still held one-third.

(4) Anima plc has undertaken annual impairment reviews in respect of all its investments and at 30 June 20X9 an impairment loss of £10,000 had been identified in respect of Oxendale Ltd.

Requirement

Prepare the consolidated statement of profit or loss of Anima plc for the year ended 30 June 20X9 and an extract from the consolidated statement of financial position as at the same date showing all figures that would appear as part of equity.

See **Answer** at the end of this chapter.



Interactive question 6: Consolidation technique 2

Preston plc has investments in two companies, Longridge Ltd and Chipping Ltd. The draft summarised statements of financial position of the three companies at 31 March 20X4 are shown below:

	Preston plc £	Longridge Ltd £	Chipping Ltd £
Assets			
Non-current assets			
Property, plant and equipment	660,700	635,300	261,600
Intangibles	101,300	72,000	–
Investments	350,000		
	1,112,000	707,300	261,600
Current assets			
Inventories	235,400	195,900	65,700
Trade and other receivables	174,900	78,800	56,600
Cash and cash equivalents	23,700	11,900	3,400
	434,000	286,600	125,700
Total assets	1,546,000	993,900	387,300
Equity and liabilities			
Equity			
Ordinary share capital (£1 shares)	100,000	500,000	200,000
Revaluation surplus	125,000	–	–
Retained earnings	1,084,800	312,100	12,000
	1,309,800	812,100	212,000
Current liabilities			
Trade and other payables	151,200	101,800	137,400
Taxation	85,000	80,000	37,900
	236,200	181,800	175,300
Total equity and liabilities	1,546,000	993,900	387,300

Additional information:

(1) Preston plc acquired 75% of Longridge Ltd's ordinary shares on 1 April 20X2 for total cash consideration of £691,000. £250,000 was payable on the acquisition date and the remaining £441,000 two years later, on 1 April 20X4. The directors of Preston plc were unsure how to treat the deferred consideration and have ignored it when preparing the draft financial statements above.

On the date of acquisition Longridge Ltd's retained earnings were £206,700. The non-controlling interest and goodwill arising on the acquisition of Longridge Ltd were both calculated using the proportionate method.

(2) The intangible asset in Longridge Ltd's statement of financial position relates to goodwill which arose on the acquisition of an unincorporated business, immediately before Preston plc purchasing its shares in Longridge Ltd. Cumulative impairments of £18,000 in relation to this goodwill had been recognised by Longridge Ltd as at 31 March 20X4.

The fair values of the remaining assets, liabilities and contingent liabilities of Longridge Ltd at the date of its acquisition by Preston plc were equal to their carrying amounts, with the exception of a building purchased on 1 April 20X0, which had a fair value on the date of acquisition of £120,000. This building is being depreciated by Longridge Ltd on a straight-line basis over 50 years and is included in the above statement of financial position at a carrying amount of £92,000.

(3) Immediately after its acquisition by Preston plc, Longridge Ltd sold a machine to Preston plc. The machine had been purchased by Longridge Ltd on 1 April 20X0 for £10,000 and was sold to Preston plc for £15,000. The machine was originally assessed as having a total useful life of five years and that estimate has never changed.

(4) Chipping Ltd is a joint venture, set up by Preston plc and a fellow venturer on 30 June 20X2. Preston plc paid cash of £100,000 for its 40% share of Chipping Ltd.

(5) During the current year Preston plc sold goods to Longridge Ltd for £12,000 and to Chipping Ltd for £15,000, earning a 20% gross margin on both sales. All these goods were still in the purchasing companies' inventories at the year end.

(6) At 31 March 20X4 Preston plc's trade receivables included £50,000 due from Longridge Ltd. However, Longridge Ltd's trade payables included only £40,000 due to Preston plc. The difference was due to cash in transit.

(7) At 31 March 20X4 impairment losses of £25,000 and £10,000 respectively in respect of goodwill arising on the acquisition of Longridge Ltd and the carrying amount of Chipping Ltd need to be recognised in the consolidated financial statements.

In the next financial year, Preston plc decided to invest in a third company, Sawley Ltd. On 1 December 20X4 Preston plc acquired 80% of Sawley Ltd's ordinary shares for £385,000. On the date of acquisition Sawley Ltd's equity comprised share capital of £320,000 and retained earnings of £112,300. Preston plc chose to measure the non-controlling interest at the acquisition date at the non-controlling interest's share of Sawley Ltd's net assets. Goodwill arising on the acquisition of Sawley Ltd has been correctly calculated at £39,160 and will be recognised in the consolidated statement of financial position as at 31 March 20X5.

Requirements

6.1 Prepare the consolidated statement of financial position of Preston plc as at 31 March 20X4.

6.2 Set out the journal entries that will be required on consolidation to recognise the goodwill arising on the acquisition of Sawley Ltd in the consolidated statement of financial position of Preston plc as at 31 March 20X5.

Note: An appropriate discount rate is 5% p.a.

See **Answer** at the end of this chapter.

8 IFRS 12, *Disclosure of Interests in Other Entities*

Section overview

IFRS 12, *Disclosure of Interests in Other Entities* requires disclosure of a reporting entity's interests in other entities in order to help identify the profit or loss and cash flows available to the reporting entity and determine the value of a current or future investment in the reporting entity.

8.1 Objective

The objective of IFRS 12 is to require entities to disclose information that enables the user of the financial statements to evaluate the nature of, and risks associated with, interests in other entities, and the effects of those interests on its financial position, financial performance and cash flows.

This is particularly relevant in light of the financial crisis and recent accounting scandals. The IASB believes that better information about interests in other entities is necessary to help users to identify the profit or loss and cash flows available to the reporting entity and to determine the value of a current or future investment in the reporting entity.

8.2 Scope

IFRS 12 covers disclosures for entities which have interests in the following:

- Subsidiaries
- Joint arrangements (ie, joint operations and joint ventures, see above)
- Associates
- Unconsolidated structured entities

8.3 Structured entity

IFRS 12 defines a structured entity.

Definition

Structured entity: An entity that has been designed so that **voting or similar rights** are **not the dominant factor** in **deciding who controls** the entity, such as when any voting rights relate to administrative tasks only and the relevant activities are directed by means of contractual arrangements.

(IFRS 12, Appendix A)

8.4 Disclosure

IFRS 12, *Disclosure of Interests in Other Entities* provides guidance applicable to consolidated financial statements.

The standard requires disclosure of:

(a) the significant judgements and assumptions made in determining the nature of an interest in another entity or arrangement, and in determining the type of joint arrangement in which an interest is held; and

(b) information about interests in subsidiaries, associates, joint arrangements and structured entities that are not controlled by an investor.

8.4.1 Disclosure of subsidiaries

The following disclosures are required in respect of subsidiaries:

(a) The interest that non-controlling interests have in the group's activities and cash flows, including the name of relevant subsidiaries, their principal place of business, and the interest and voting rights of the non-controlling interests

(b) Nature and extent of significant restrictions on an investor's ability to use group assets and liabilities

(c) Nature of the risks associated with an entity's interests in consolidated structured entities, such as the provision of financial support

(d) Consequences of changes in ownership interest in subsidiary (whether control is lost or not)

8.4.2 Disclosure of associates and joint arrangements

The following disclosures are required in respect of associates and joint arrangements:

(a) Nature, extent and financial effects of an entity's interests in associates or joint arrangements, including name of the investee, principal place of business, the investor's interest in the investee, method of accounting for the investee and restrictions on the investee's ability to transfer funds to the investor

(b) Risks associated with an interest in an associate or joint venture

(c) Summarised financial information, with more detail required for joint ventures than for associates

9 Step acquisitions



Section overview

- Subsidiaries and associates are consolidated/equity accounted for from the date control/significant influence is gained.
- In some cases acquisitions may be achieved in stages. These are known as step acquisitions.

A step acquisition occurs when the parent entity acquires control over the subsidiary in stages, achieved by buying blocks of shares at different times.

Acquisition accounting is only applied when control is achieved.

The date on which control is achieved is the date on which the acquirer should recognise the acquiree's identifiable net assets and any goodwill acquired (or bargain purchase) in the business combination.

Until control is achieved, any pre-existing interest is accounted for in accordance with:

- IFRS 9 in the case of investments in equity instruments
- IAS 28 in the case of associates and joint ventures

9.1 Types of business combination achieved in stages

There are three possible types of business combinations achieved in stages:

(a) A previously held **investment**, say 10% of share capital, with **no significant influence** (accounted for under IFRS 9), is **increased to a controlling holding** of 50% or more.

(b) A **previously held equity investment**, say 35% of share capital, accounted for as an **associate** under IAS 28, is increased to a controlling holding of 50% or more.

(c) A **controlling holding** in a subsidiary is **increased**, say from 60% to 80%.

The first two transactions are treated in the same way, but the third is not. There is a reason for this.

9.2 General principle: 'crossing an accounting boundary'

Under IFRS 3 a business combination occurs only when one entity **obtains control over another**, which is generally when 50% or more has been acquired. The 2008 Deloitte guide: *Business Combinations and Changes in Ownership Interests* calls this '**crossing an accounting boundary**'.

When this happens, the original investment – whether an investment in equity instruments with no significant influence, or an associate – is treated as if it were **disposed of at fair value and reacquired at fair value**. This **previously held interest** at fair value, together with any consideration transferred, is the **'cost' of the combination** used in calculating the goodwill.

If the 50% **boundary is not crossed**, as when a shareholding in an existing subsidiary is increased, the event is treated as a **transaction between owners**.

Whenever you cross the 50% boundary, you revalue, and a gain or loss is reported in profit or loss for the year. If you do not cross the 50% boundary, no gain or loss is reported; instead there is an adjustment to the parent's equity.

The following diagram, adapted from the **Deloitte** guide to IFRS 3, may help you visualise the boundary:

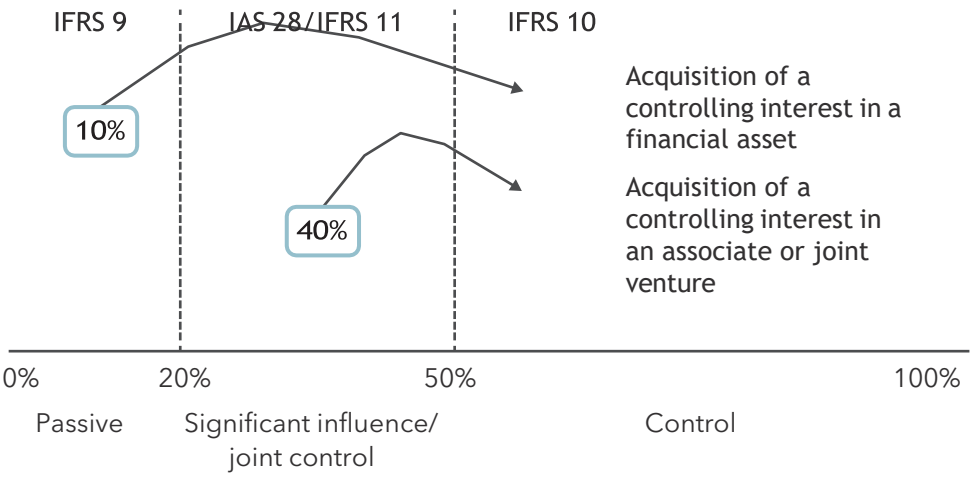


Figure 20.2: Transactions that trigger remeasurement of an existing interest

As you will see from the diagram, the third situation in section 9.1, where an interest in a subsidiary is increased from, say, 60% to 80%, does not involve crossing that all-important 50% threshold. Likewise, purchases of stakes up to 50% do not involve crossing the boundary, and therefore do not trigger a calculation of goodwill.

Professional skills focus: Assimilating and using information

When tackling exam questions that involve a step-acquisition or a disposal, this skill (assimilating and using information) is vital. You should be able to work out quickly whether an accounting boundary has been crossed, and this will determine the financial reporting treatment.

9.3 Achieving control

When control is achieved:

- any previously held equity shareholding should be treated as if it had been disposed of and then reacquired at fair value at the acquisition date; and
- any gain or loss on remeasurement to fair value should be recognised in profit or loss in the period.

Goodwill is calculated as:

	£
Fair value of consideration paid to acquire control	X
Non-controlling interest (valued using either fair value or the proportion of net assets method)	X
Fair value of previously held equity interest at acquisition date	X
	X
Fair value of net assets of acquiree	(X)
Goodwill	X

9.3.1 Gain or loss on derecognition

The gain or loss is recognised in profit or loss unless the equity interest previously held was an investment in equity instruments and an irrevocable election was made to hold the investment at fair value through other comprehensive income. In the latter case, in accordance with IFRS 9, *Financial Instruments*, the gain on derecognition of the investment is taken to other comprehensive income, or, in the SOFP, other components of equity.

As we saw in Chapter 16 on financial instruments, for an equity instrument at fair value through other comprehensive income (FVTOCI) the asset's carrying amount is remeasured to fair value at the date of derecognition (IFRS 9: para. 3.2.12) immediately prior to the disposal. A step acquisition from a financial asset to subsidiary is like a disposal of the financial asset. Any change resulting from such a remeasurement is recognised in OCI. Assuming the disposal is at fair value then, as a consequence of the revaluation there is nil profit/loss on disposal recorded in profit or loss. IFRS 9 interacts with IFRS 3, which requires that, in the case of a step acquisition going from a financial asset at FVTOCI to a subsidiary, the financial asset at FVTOCI is revalued in accordance with IFRS 3 para. 42:

"In a business combination achieved in stages, the acquirer shall remeasure its previously held equity interest in the acquiree at its acquisition-date fair value and **recognise the resulting gain or loss, if any, in profit or loss or other comprehensive income, as appropriate**."

It is therefore clear that the gain on revaluation of a financial asset in a step acquisition from financial asset to subsidiary would go through OCI in the case of financial assets at FVTOCI and to profit or loss in the case of financial assets at fair value through profit or loss (FVTPL).

Worked example: Control in stages – previous holding a simple investment

Bath Ltd has 1 million shares in issue. Bristol plc acquired 50,000 shares in Bath Ltd on 1 January 20X6 for £100,000. These shares were classified as a financial asset at fair value through other comprehensive income and on 31 December 20X8 their carrying amount was £230,000 and increases in fair value of £130,000 had been recognised in other comprehensive income and were held in equity. On 1 June 20X9 when the fair value of Bath Ltd's net assets was £4 million, Bristol plc acquired another 650,000 shares in Bath Ltd for £3.9 million. On 1 June 20X9 the fair value of the 50,000 shares already held was £250,000.

Requirement

Show the journal entry required in respect of the 50,000 shareholding on 1 June 20X9 and calculate the goodwill acquired in the business combination on that date assuming that goodwill is valued using the proportion of net assets method.

Solution

Journal entry

		£'000	£'000
DEBIT	Investment in Bath Ltd (250,000 – 230,000)	20	
CREDIT	Other comprehensive income and equity reserve		20

To recognise the gain on the deemed disposal of the shareholding in Bath Ltd existing immediately before control being obtained.

Calculation of goodwill in respect of 70% (5% + 65%) holding in Bath Ltd:

	£'000
Consideration transferred	3,900
Non-controlling interest (30% × £4m)	1,200
Acquisition-date fair value of previously held equity	250
	5,350
Net assets acquired	(4,000)
Goodwill	1,350

Interactive question 7: Control in stages – no previous significant influence

Good, whose year end is 30 June 20X9 has a subsidiary, Will, which it acquired in stages. The details of the acquisition are as follows.

Date of acquisition	Holding acquired %	Retained earnings at acq. $m	Purchase consideration $m
1 July 20X7	20	270	120
1 July 20X8	60	450	480

The share capital of Will has remained unchanged since its incorporation at $300 million. The fair values of the net assets of Will were the same as their carrying amounts at the date of the acquisition. Good did not have significant influence over Will at any time before gaining control of Will. The group policy is to measure non-controlling interest at its proportionate share of the fair value of the subsidiary's identifiable net assets.

Requirements

7.1 Calculate the goodwill on the acquisition of Will that will appear in the consolidated statement of financial position at 30 June 20X9.

7.2 Calculate the profit on the derecognition of any previously held investment in Will to be reported in group profit or loss for the year ended 30 June 20X9.

See **Answer** at the end of this chapter.

9.4 Acquisitions that do not result in a change of control

Where an entity increases its investment in an existing subsidiary:

- no gain or loss is recognised;
- goodwill is not remeasured; and
- the difference between the fair value of consideration paid and the change in the non-controlling interest is recognised directly in equity attributable to owners of the parent.



Worked example: Acquisitions that do not result in a change of control

On 1 June 20X6, Santander acquired 70% of the equity of Madrid in exchange for £760,000 cash and 100,000 Santander shares. At this date the fair value of the identifiable net assets of Madrid was £850,000 and the market value of Santander shares was £2.50.

On 31 December 20X8, Santander acquired a further 10% of the equity of Madrid at a cost of £105,000. On this date the identifiable net assets of Madrid were £970,000.

Santander measures the non-controlling interest using the proportion of net assets method.

Requirements

1 What goodwill is recorded in the consolidated statement of financial position at 31 December 20X8, assuming that there is no impairment?

2 What journal adjustment is required on the acquisition of the further 10% of shares?

Solution

1 The goodwill included in the statement of financial position at 31 December 20X8 is that goodwill calculated on the initial acquisition in June 20X6:

	£'000
Consideration (£760,000 + (100,000 × £2.50))	1,010
Non-controlling interest (30% × £850,000)	255
	1,265
Net assets of acquiree	(850)
Goodwill	415

2 The adjustment required is based on the change in the non-controlling interest at the acquisition date:

	£
NCI on 31 December 20X8 based on old interest (30% × £970,000)	291,000
NCI on 31 December 20X8 based on new interest (20% × £970,000)	194,000
Adjustment required	97,000

Therefore:

DEBIT	Non-controlling interest	97,000	
DEBIT	Shareholders' equity (bal fig)	8,000	
CREDIT	Cash		105,000

10 Disposals



Section overview

- Subsidiaries and associates are consolidated/equity accounted for until the date control/significant influence is lost therefore profits need to be time apportioned.
- A gain on disposal must also be calculated, by reference to the fair value of any interest retained in the subsidiary or associate.

An entity may sell all or some of its shareholding in another entity. Full disposals of subsidiaries and associates were covered in FR and are revised here. Other situations which may arise are as follows:

- The sale of shares in a subsidiary such that control is retained
- The sale of shares in a subsidiary such that the subsidiary becomes an associate
- The sale of shares in a subsidiary such that the subsidiary becomes an investment
- The sale of shares in an associate such that the associate becomes an investment

10.1 Crossing an accounting boundary revisited

Under IFRS 3 a gain on disposal occurs only when one entity loses control over another, which is generally when its holding is decreased to less than 50%. As noted above, the Deloitte guide: *Business Combinations and Changes in Ownership Interests* calls this 'crossing an accounting boundary' but in this case the investment is being **reduced**, rather than increased as in sections 9.1 and 9.2 above.

On disposal of a controlling interest, any retained shareholding (an associate or trade investment) is measured at fair value on the date that control is lost. This fair value is used in the calculation of the gain or loss on disposal, and also becomes the carrying amount for subsequent accounting for the retained shareholding.

If the **50%** boundary is **not crossed**, as when the shareholding in a subsidiary is reduced, but control is still retained, the event is treated as a **transaction between owners and no gain or loss is recognised**.

Whenever you cross the 50% boundary, you revalue, and a gain or loss is reported in profit or loss for the year. If you do not cross the 50% boundary, no gain or loss is reported; instead there is an adjustment to the parent's equity.

The following diagram, adapted from the **Deloitte** guide, may help you visualise the boundary:

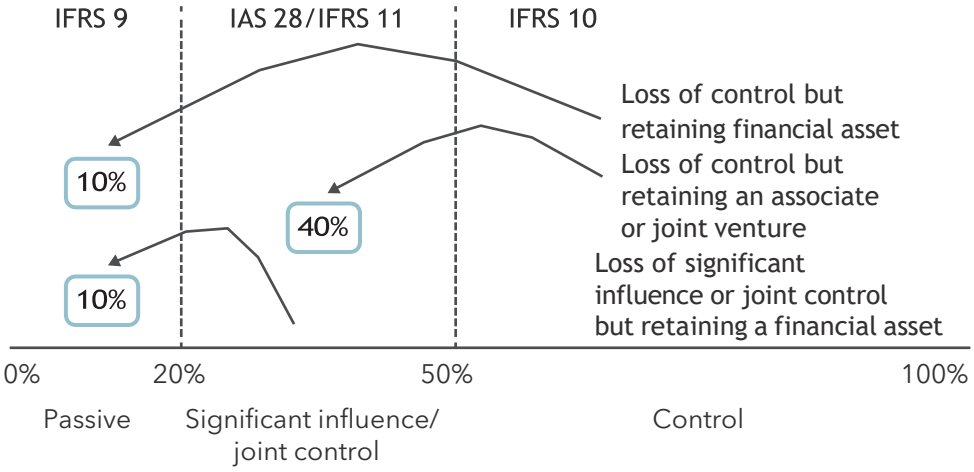


Figure 20.3: Transactions that require remeasurement of a retained interest

The situation where a shareholding in a subsidiary is reduced from, say, 80% to 60% – that is, where control is retained – does not involve crossing that all-important 50% threshold.

10.2 Disposal of a whole subsidiary or associate – revision

10.2.1 Parent company's accounts

In the **parent's individual financial statements** the **profit or loss on** disposal of a subsidiary or associate holding will be calculated as:

	£
Sales proceeds	X
Less carrying amount (cost in P's own statement of financial position)	(X)
Profit (loss) on disposal	X/(X)

10.2.2 Group accounts – disposal of subsidiary Gain or loss on disposal

In the **group financial statements** the **profit or loss on disposal** will be calculated as:

	£	£
Proceeds		X
Less: amounts recognised before disposal:		
net assets of subsidiary	X	
goodwill	X	
non-controlling interest	(X)	
		(X)
Profit/loss		X/(X)

Remember:

(a) If the disposal is mid year:

  (1) a working will be required to calculate both net assets and the non-controlling interest at the disposal date; and

  (2) any dividends declared or paid in the year of disposal and before the disposal date must be deducted from the net assets of the subsidiary if they have not already been accounted for.

(b) Goodwill recognised before disposal is original goodwill arising less any impairments to date.



Worked example: Group gain or loss on disposal

Kingdom acquired 80% of Westville on 1 January 20X5 for £280,000 when Westville had share capital of £100,000 and retained earnings of £188,000. On this date the fair value of the non-controlling interest was £67,000. Kingdom's policy is to value the non-controlling interest using the fair value (ie, full goodwill) method. Goodwill has suffered no impairment since acquisition.

On 30 June 20X8, Kingdom disposed of its investment in Westville, raising proceeds of £350,000. The following are extracts from the accounts of Westville for the year ended 31 December 20X8:

	£'000
Retained earnings b/f	215
Profit for the year	24

A final dividend for 20X7 of £10,000 was paid on 14 March 20X8. This has not yet been accounted for.

Requirement

What profit or loss on disposal of Westville is reported in the Kingdom group accounts for the year ended 31 December 20X8?

Solution

Profit on disposal

	£	£
Proceeds		350,000
Less amounts recognised before disposal		
Net assets of Westville (£100,000 + £215,000 + (1/2 × £24,000) – £10,000)	317,000	
Goodwill (£280,000 + £67,000) – (£100,000 + £188,000)	59,000	
NCI at disposal		
Share of net assets (20% × £317,000)	(63,400)	
Goodwill on acquisition (£67,000 – (20% × £288,000))	(9,400)	
		(303,200)
Profit on disposal		46,800

Consolidated statement of financial position

The statement of financial position does not include the subsidiary disposed of, as it is no longer controlled at the reporting date.

Consolidated statement of profit or loss and other comprehensive income

- The time-apportioned results of the subsidiary should be consolidated up to the date of disposal.
- The non-controlling interest must be time apportioned.
- The gain or loss on disposal forms part of the profit or loss for the year.

IFRS 5 discontinued operations

If the sale represents a **discontinued operation** per IFRS 5 the consolidated statement of profit or loss and other comprehensive income will reflect, **as one figure, 'Profit for the period from discontinued operations'**, being the group profit on disposal plus the subsidiary's profit for the year to disposal.

10.2.3 Group accounts – disposal of associate

In the **consolidated financial statements** the **profit or loss on disposal** should be calculated as:

	£	£
Proceeds		X
Less: cost of investment	X	
share of post-acquisition profits retained by associate at disposal	X	
impairment of investment to date	(X)	
		(X)
Profit/(loss)		X/(X)

The other effects of disposal are also similar to those of the disposal of a subsidiary:

- There is no holding in the associate at the end of the reporting period, so there is no investment to recognise in the consolidated statement of financial position.
- The associate's after-tax earnings should be included in consolidated profit or loss up to the date of disposal.

10.3 Part disposal from a subsidiary holding

10.3.1 Subsidiary to subsidiary

Shares may be disposed of such that a subsidiary holding is still retained eg, a 90% holding is reduced to a 70% holding.

In this case there is **no loss of control** and therefore:

- no gain or loss on disposal is calculated;
- no adjustment is made to the carrying value of goodwill; and
- the difference between the proceeds received and the change in the non-controlling interest is accounted for in shareholders' equity.

Consolidated statement of financial position

- Consolidate as normal with the NCI calculated by reference to the year-end shareholding.
- Calculate goodwill as at the original acquisition date.
- Record the difference between NCI and proceeds in shareholders' equity as above.

Consolidated statement of profit or loss and other comprehensive income

- Consolidate the subsidiary's results for the whole year.
- Calculate the NCI on a pro rata basis.



Worked example: Part disposal (subsidiary to subsidiary)

Express acquired 90% of Billings in 20X2 when Billings had retained earnings of £250,000. Goodwill was calculated as £45,000 using the proportion of net assets method to value the non-controlling interest. Goodwill has been impaired by £5,000 since acquisition.

At 31 December 20X8 the abbreviated statements of financial position of the two entities were as follows:

	Express	Billings
	£'000	£'000
Non-current assets	2,300	430
Investments	360	–
Current assets	1,750	220
	4,410	650
Share capital	1,000	100
Retained earnings b/f	1,190	304
Profit for the year	120	36
Liabilities	2,100	210
	4,410	650

Express disposed of a 10% holding in Billings on 31 August 20X8 for £70,000; this has not yet been recorded in Express's individual accounts.

Requirement

Prepare the consolidated statement of financial position as at 31 December 20X8.

Solution

Express Group statement of financial position at 31 December 20X8

	£'000
Non-current assets (£2,300,000 + £430,000)	2,730
Goodwill (£45,000 – £5,000)	40
Current assets (£1,750,000 + £220,000 + proceeds £70,000)	2,040
	4,810
Share capital	1,000
Retained earnings (W1)	1,412
Non-controlling interest 20% × (£650,000 – £210,000)	88
Liabilities (£2,100,000 + £210,000)	2,310
	4,810

WORKINGS

(1) **Retained earnings**

	£'000
Retained earnings of Express (£1,190,000 + £120,000) 1,310.0 Retained earnings of Billings	
Acquisition – 31 Aug 20X8 90% × (£304,000 + (8/12 × £36,000) –£250,000)	70.2
31 Aug 20X8 – 31 Dec 20X8 (80% × 4/12 × £36,000)	9.6
Impairment of goodwill	(5.0)
NCI adjustment on disposal (W2)	27.2
	1,412.0

(2) **NCI adjustment on disposal**

NCI in Billings at disposal:

	£'000
NCI at acquisition (10% × (100 + 250))	35.0
Share of post acqn. reserves (Working 3)	7.8
	42.8
Increase 10%/10%	42.8
NCI after disposal: 20%	85.6

At disposal date:

NCI based on old shareholding (10% × £428,000)		42.8
NCI based on new shareholding (20% × £428,000)		85.6
Adjustment required		42.8
DEBIT Proceeds	£70,000	
CREDIT NCI		£42,800
CREDIT Shareholders' equity (to Working 1)		£27,200

(3) **NCI share of post acqn. retained earnings**

	£'000
Retained earnings b/f (£304,000 – £250,000)	54.0
Profit to disposal (£36,000 × 8/12)	24.0
	78.0
NCI share 10%	7.8

10.3.2 Subsidiary to associate

Shares may be disposed of such that an associate holding is still retained, eg, a 90% holding is reduced to a 30% holding.

Gain or loss on disposal

In this case there is a loss of control, and so a gain or loss on disposal is calculated as:

	£	£
Proceeds		X
Fair value of interest retained		X
		X
Less net assets of subsidiary recognised before disposal:		
Net assets	X	
Goodwill	X	
Non-controlling interest	(X)	
		(X)
Profit/loss		X/(X)



Worked example: Part disposal (subsidiary to associate)

Allister Group acquired a 75% holding in Brown in 20X0, which it held until 31 December 20X8 when two-thirds of the investment were sold for £490,000.

At this date the net assets of Brown were £800,000, goodwill arising on acquisition had been fully impaired and the fair value of a 25% interest in Brown was £220,000.

The non-controlling interest is valued using the proportion of net assets method.

Requirement

What gain or loss arises on the disposal in the Allister Group accounts in the year ended 31 December 20X8?

Solution

Disposal

	£	£
Proceeds		490,000
Fair value of 25% interest retained		220,000
		710,000
Less amounts recognised before disposal:		
Net assets of Brown	800,000	
Goodwill (fully impaired)	–	
NCI at disposal (25% × £800,000)	(200,000)	
		(600,000)
Gain on disposal		110,000

Note: The disposal triggers remeasurement of the residual interest to fair value. The gain on disposal could be analysed as:

Realised gain	**£**	**£**
Proceeds on disposal	490,000	
Interest disposed of (50% × £800,000)	(400,000)	
Realised gain		90,000
Holding gain		
Retained interest at fair value	220,000	
Retained interest at carrying value (25% × £800,000)	(200,000)	
		20,000
Total gain		110,000

The retained 25% interest in Brown is included in the consolidated statement of financial position at 31 December 20X8 at the fair value of £220,000.

Consolidated statement of financial position

- Equity account by reference to the year-end holding. The carrying value of the associate is based on the fair value of the interest as included within the gain calculation.

Consolidated statement of profit or loss and other comprehensive income

- Consolidate results up to the date of disposal based on the pre-disposal holding
- Equity account for results after the date of disposal based on the post-disposal holding
- Include gain or loss on disposal as calculated above

Reclassification of other comprehensive income

At the date of disposal, amounts recognised in other comprehensive income in relation to the subsidiary should be accounted for in the same way as if the parent company had directly disposed of the assets that they relate to, for example, if the subsidiary holds revalued assets, the revaluation surplus previously recognised in consolidated other comprehensive income should be transferred to group retained earnings.

10.3.3 Subsidiary to investment

Shares may be disposed of such that an investment is still retained eg, a 90% holding is reduced to a 10% holding.

Gain or loss on disposal

The gain or loss on disposal is calculated as described in section 10.3.2. Therefore at the point of disposal, the retained interest is measured at fair value.

Statement of financial position

- The interest retained is initially recorded at fair value (as included within the gain calculation).

Statement of profit or loss and other comprehensive income

- Consolidate results up to the date of disposal based on the pre-disposal holding
- Include dividend income after the date of disposal
- Include gain or loss on disposal as calculated above

Reclassification of other comprehensive income

Upon loss of control, other comprehensive income recorded in relation to the subsidiary should again be accounted for in the same way as if the parent company had directly disposed of the assets that it relates to.

10.4 Part disposal from an associate holding

10.4.1 Associate to investment

Shares may be disposed of such that an investment is still retained eg, a 40% holding is reduced to a 10% holding.

Gain or loss on disposal

In this case there is a loss of significant influence, and a gain or loss on disposal is calculated as:

	£	£
Proceeds		X
Fair value of interest retained		X
		X
Less: Cost of investment	X	
Share of post-acquisition profits retained by associate at disposal	X	
Impairment of investment to date	(X)	
		(X)
Profit/(loss)		X/(X)

Statement of financial position

The interest retained is initially recorded at fair value (as included within the gain calculation).

Statement of profit or loss and other comprehensive income

- Equity account for results up to the date of disposal based on the pre-disposal holding
- Include dividend income after the date of disposal
- Include gain or loss on disposal as calculated above



Interactive question 8: All types of disposal

Streatham Co bought 80% of the share capital of Balham Co for £324,000 on 1 October 20X5. At that date Balham Co's retained earnings balance stood at £180,000. The statements of financial position at 30 September 20X8 and the summarised statements of profit or loss for the year to that date are given below:

	Streatham Co £'000	Balham Co £'000
Non-current assets	360	270
Investment in Balham Co	324	–
Current assets	370	370
	1,054	640
Equity		
Ordinary shares	540	180
Reserves	414	360

Current liabilities	100	100
	1,054	640
Profit before tax	153	126
Tax	(45)	(36)
Profit for the year	108	90

No entries have been made in the accounts for any of the following transactions. Assume that profits accrue evenly throughout the year.

It is the group's policy to value the non-controlling interests at its proportionate share of the fair value of the subsidiary's identifiable net assets.

Ignore tax on the disposal.

Requirements

8.1 Prepare the consolidated statement of financial position and statement of profit or loss at 30 September 20X8 in each of the following circumstances. (Assume no impairment of goodwill.)

(a) Streatham Co sells its entire holding in Balham Co for £650,000 on 30 September 20X8.

(b) Streatham Co sells one-quarter of its holding in Balham Co for £160,000 on 30 September 20X8.

8.2 In the following circumstances you are required to calculate the gain on disposal, group retained earnings and carrying value of the retained investment at 30 September 20X8.

(a) Streatham Co sells one-half of its holding in Balham Co for £340,000 on 30 June 20X8, and the remaining holding (fair value £250,000) is to be dealt with as an associate.

(b) Streatham Co sells one-half of its holding in Balham Co for £340,000 on 30 June 20X8, and the remaining holding (fair value £250,000) is to be dealt with as a financial asset at fair value through other comprehensive income.

See **Answer** at the end of this chapter.

11 Consolidated statements of cash flows



Section overview

- The consolidated statement of cash flows shows the impact of the acquisition and disposal of subsidiaries and associates.
- Exchange differences arising on the translation of the foreign currency accounts of group companies will also impact the consolidated statement of cash flows. This is covered in more detail in later Chapters.
- Both single company and consolidated statements of cash flow were covered at Professional Level. Single company statements were revisited in earlier Chapters of this Workbook. In this chapter we summarise the main points and provide two interactive questions and two comprehensive self- test questions. Please look back to your earlier study material if you have any major problems with these.

11.1 Consolidated statements of cash flows – revision

You should remember from your Professional Level studies that the consolidated statement of cash flows is put together from the consolidated financial statements themselves. Additional figures over and above a single company statement of cash flows will be as follows:

- Dividends paid to the non-controlling interest
- Dividend received from associates and joint ventures
- Acquisitions/disposals of subsidiaries
- Acquisitions/disposals of associates and joint ventures

11.1.1 Cash flows to the non-controlling interest

The non-controlling interest represents a third party so **dividends paid to the non-controlling interest are reflected as a cash outflow.** This payment should be presented separately and classified as '**Cash flows from financing activities**'.

Dividends paid to the non-controlling interest may be calculated using a T-account as follows:

NON-CONTROLLING INTEREST

	£		£
NCI dividend paid (balancing figure)	**X**	b/f NCI (CSFP)	X
c/f NCI (CSFP)	X	NCI (CIS)	X
	X		X

11.1.2 Associates and joint ventures

There are two issues to consider with regard to the associate/joint venture. (Note that joint ventures are treated like associates in that they are equity accounted, so the points below apply to both.)

(a) The aim of the statement of cash flows is to show the cash flows of the parent and any subsidiaries with third parties, therefore **any cash flows between the associate and third parties are irrelevant.** As a result, **the group share of profit of the associate must be deducted as an adjustment in the reconciliation of profit before tax to cash generated from operations**. This is because group profit before tax includes the results of the associate.

(b) **Dividends received** from the associate must be disclosed as a **separate cash flow** classified as **'Cash flows from investing activities'**. The cash receipt can be calculated as follows:

INVESTMENTS IN ASSOCIATES

	£		£
		b/f NCI (CSFP)	X
		NCI (CIS)	X
NCI dividend paid (balancing figure)	**X**		
c/f NCI (CSFP)	X		
	X		X

11.1.3 Acquisitions and disposals of subsidiaries

If a subsidiary is acquired or disposed of during the accounting period **the net cash effect of the purchase or sale transaction should be shown separately under 'Cash flows from investing activities'**. The net cash effect will be the cash purchase price/cash disposal proceeds net of any cash or cash equivalents acquired or disposed of.

As the cash effect of the acquisition/disposal of the subsidiary is dealt with in a single line item as we saw above, **care must be taken not to double count the effects of the acquisition/disposal when looking at the movements in individual asset balances**.

Each of the individual assets and liabilities of a subsidiary acquired/disposed of during the period must be excluded when comparing group statements of financial position for cash flow calculations as follows:

Subsidiary acquired in the period	**Subtract** PPE, inventories, payables, receivables etc, at the date of acquisition from the movement on these items.
Subsidiary disposed of in the period	**Add** PPE, inventories, payables, receivables etc, at the date of disposal to the movements on these items.

This would also affect the calculation of the **dividend paid to the non-controlling interest**. The T- account working is modified as follows:

NON-CONTROLLING INTEREST

	£		£
NCI in Subsidiary at disposal	**X**	b/f NCI (CSFP)	X
NCI dividend paid (balancing figure)	X	**NCI in Subsidiary at acquisition**	**X**
c/f NCI (CSFP)	X	NCI (CIS)	X
	X		X

11.1.4 Acquisitions and disposals of associates

If an **associate is acquired or disposed of** during the accounting period the payment or receipt of cash is classified as investing activities.



Interactive question 9: Acquisition of a subsidiary

On 1 October 20X8 Pippa plc acquired 90% of S Ltd by issuing 100,000 shares at an agreed value of £2 per share and paying £100,000 in cash.

At that time the net assets of S Ltd were as follows:

	£'000
Property, plant and equipment	190
Inventories	70
Trade receivables	30
Cash and cash equivalents	10
Trade payables	(40)
	260

The consolidated statements of financial position of Pippa plc as at 31 December were as follows:

	20X8 £'000	20X7 £'000
Non-current assets		
Property, plant and equipment	2,500	2,300
Goodwill	66	
	2,566	2,300
Current assets		
Inventories	1,450	1,200
Trade receivables	1,370	1,100
Cash and cash equivalents	76	50
	2,896	2,350
	5,462	4,650
Equity attributable to owners of the parent		
Ordinary share capital (£1 shares)	1,150	1,000
Share premium account	650	500
Retained earnings 1,791		1,530
	3,591	3,030
Non-controlling interest	31	
Total equity	3,622	3,030
Current liabilities		
- Trade payables	1,690	1,520
- Income tax payable	150	100
	1,840	1,620
	5,462	4,650

The consolidated statement of profit or loss for the year ended 31 December 20X8 was as follows:

	£'000
Revenue	10,000
Cost of sales	(7,500)
Gross profit	2,500
Administrative expenses	(2,080)
Profit before tax	420
Income tax expense	(150)
Profit for the year	270

Profit attributable to:	
Owners of Pippa plc	261
Non-controlling interest	9
	270

The statement of changes in equity for the year ended 31 December 20X8 (extract) was as follows:

	Retained earnings
	£'000
Balance at 31 December 20X7	1,530
Total comprehensive income for the year	261
Balance at 31 December 20X8	1,791

You are also given the following information:

(1) All other subsidiaries are wholly owned.

(2) Depreciation charged to the consolidated statement of profit or loss amounted to £210,000.

(3) There were no disposals of property, plant and equipment during the year.

(4) Goodwill is not impaired.

(5) Non-controlling interest is valued on the proportionate basis.

Requirement

Prepare a consolidated statement of cash flows for Pippa plc for the year ended 31 December 20X8 under the indirect method in accordance with IAS 7, *Statement of Cash Flows*. The only notes required are those reconciling profit before tax to cash generated from operations and a note showing the effect of the subsidiary acquired in the period.

See **Answer** at the end of this chapter.



Interactive question 10: Disposal

Below is the consolidated statement of financial position of the Caitlin Group as at 30 June 20X8 and the consolidated statement of profit or loss for the year ended on that date:

Consolidated statement of financial position as at 30 June

	20X8	20X7
	£'000	£'000
Non-current assets		
Property, plant and equipment	4,067	3,950
Current assets		
Inventories	736	535

Receivables	605	417
Cash and cash equivalents	294	238
	1,635	1,190
	5,702	5,140
Equity attributable to owners of the parent		
Share capital	1,000	1,000
Retained earnings	3,637	3,118
	4,637	4,118
Non-controlling interest	482	512
Total equity	5,119	4,630
Current liabilities		
Trade payables	380	408
Income tax payable	203	102
	583	510
	5,702	5,140

Consolidated statement of profit or loss for the year ended 30 June 20X8 (summarised)

	£'000
Continuing operations	
Profit before tax	862
Income tax expense	(290)
Profit for the year from continuing operations	572

	£'000
Discontinued operations	
Profit for the year from discontinued operations	50
Profit for the year	622
Profit attributable to:	
Owners of Caitlin plc	519
Non-controlling interest	103
	622

You are given the following information:

(1) Caitlin plc sold its entire interest in Desdemona Ltd on 31 March 20X8 for cash of £400,000. Caitlin plc had acquired an 80% interest in Desdemona Ltd on incorporation several years ago. The net assets at the date of disposal were:

(2)

	£'000
Property, plant and equipment	390
Inventories	50
Receivables	39
Cash and cash equivalents	20
Trade payables	(42)
	457

(3) The profit for the period from discontinued operations figure is made up as follows:

	£'000
Profit before tax	20
Income tax expense	(4)
Profit on disposal	34
	50

(4) The depreciation charge for the year was £800,000.

There were no disposals of non-current assets other than on the disposal of the subsidiary.

Requirements

With regard to the consolidated statement of cash flows for the year ended 30 June 20X8:

(a) Show how the disposal will be reflected in the statement of cash flows.

(b) Calculate additions to property, plant and equipment as they will be reflected in the statement of cash flows.

(c) Calculate dividends paid to the non-controlling interest.

(d) Prepare the note to the statement of cash flows required for the disposal of the subsidiary.

(e) Prepare the reconciliation of profit before tax to cash generated from operations.

Note: Work to the nearest £'000.

See **Answer** at the end of this chapter.

12 Audit focus: group audits

Section overview

- The group auditor has sole responsibility for the audit opinion on the group financial statements.
- The component auditors should co-operate with the group auditors. In some cases this will be a legal duty.
- The group auditor will need to assess the extent to which the work of the component auditors can be relied on.
- Specific audit procedures will be performed on the consolidation process.
- Where a group includes a foreign subsidiary, compliance with relevant accounting standards will need to be considered.

12.1 Introduction

Many of the basic principles applied in the audit of a group are much the same as the audit of a single company. However, there are a number of significant **additional considerations**.

The first area to consider is the use of another auditor. Often, one or more subsidiaries in the group will be audited by a different audit firm. Evaluating whether the component auditor's work can be relied on, and communicating effectively with the component auditor, therefore become important.

Another of the key issues will be the impact of the group structure on the **risk assessment**, including the process by which the existing structure has been achieved eg, acquisition and MBO, and/or changes to that structure. In many cases, the risk issues will be related to the accounting treatments adopted.

12.2 Acceptance as group auditors

ISA 600, *Special Considerations – Audits of Group Financial Statements (Including the Work of Component Auditors)* is the relevant standard.

ISA 600.12 states that the auditor must consider whether "sufficient appropriate audit evidence can reasonably be expected to be obtained in relation to the consolidation process and the financial information of the components on which to base the group audit opinion". For this purpose, the group engagement partner must obtain an understanding of the group. This may be obtained from the following sources.

In the case of **a new engagement**:

- Information provided by group management
- Communication with group management
- Where applicable, communication with the previous group engagement team, component manager or component auditors

Other matters to consider will include the following:

- The group structure
- Components' business activities including the industry and regulatory, economic and political environments in which those activities take place
- The use of service organisations

- A description of group-wide controls
- The complexity of the consolidation process
- Whether component auditors that are not from the group engagement partner's firm will perform work on the financial information of any of the components and group management's rationale for appointing more than one auditor
- Whether the group engagement team will have unrestricted access to those charged with governance of the group, group management, those charged with governance of the component, component management, component information and the component auditors (including relevant audit documentation sought by the group engagement team)
- Whether the group engagement team will be able to perform necessary work on the financial information of the components

(ISA 600.A10–.A11)

In the case of **continuing engagements**, the group engagement team's ability to obtain sufficient appropriate audit evidence may be affected by **significant changes**, for example changes in group structure, changes to group-wide controls or the applicable financial reporting framework, changes in business activities and concerns regarding the integrity and competence of group or component management.

(ISA 600.A12)

Where components within the group are likely to be **significant components** the group engagement partner evaluates the extent to which the group engagement team will be able to be involved in the work of those component auditors.

The group engagement partner must either refuse to accept, or resign from, the engagement if he concludes that it will not be possible to obtain sufficient appropriate audit evidence.

(ISA 600.12-.13)

Note: In addition to these points, the prospective group auditor should consider the general points relating to acceptance of appointment which you have covered in Professional level.

12.3 Using the work of another auditor

We have identified the fact that entities within the group may be audited by different auditors as a risk factor. Guidance on this aspect of the group audit is provided in ISA 600. Subsidiaries may be audited by firms other than the parent company auditor, meaning that the parent company auditor will have to express an opinion on financial information, some of which has been audited **by another party**. ISA 600 addresses this issue in particular and is summarised below.

12.3.1 Responsibility of group auditors



Definitions

Group audit: The audit of the group financial statements.

Group engagement partner: The partner or other person in the firm who is responsible for the group audit engagement and its performance and for the auditor's report on the group financial statements that is issued on behalf of the firm.

Group engagement team: Partners, including the group engagement partner, and staff who establish the overall group audit strategy, communicate with component auditors, perform work on the consolidation process, and evaluate the conclusions drawn from the audit evidence as the basis for forming an opinion on the group financial statements.

Component auditor: An auditor who, at the request of the group engagement team, performs work on financial information related to a component for the group audit.

Component: An entity or business activity for which the group or component management prepares financial information that should be included in the group financial statements.

Component materiality: The materiality for a component determined by the group engagement team.

Significant component: A component identified by the group engagement team: (a) that is of individual significance to the group, or (b) that, due to its specific nature or circumstances, is likely to include significant risks of material misstatement of the group financial statements.

(ISA 600.9)

The duty of the group auditors is to report on the **group accounts**, which includes balances and transactions of all the components of the group.

The group auditors have **sole responsibility** for this opinion even where the group financial statements include amounts derived from accounts which have not been audited by them. ISA 600 explains that even where an auditor is required by law or regulation to refer to the component auditors in the auditor's report on the group financial statements, the report must indicate that the reference does not diminish the group engagement partner's or the firm's responsibility for the group audit opinion. As a result, they cannot discharge their responsibility to report on the group financial statements by an unquestioning acceptance of component companies' financial statements, whether audited or not.

(ISA 600.11)

Note: For audits of group financial statements of PIEs the group engagement partner is also responsible for the additional report to the audit committee as required by ISA 260 (Revised)

(ISA 600.49).

12.3.2 Rights of group auditors

The group auditors of a parent company have the following rights:

- The right to require the information and explanations they require from auditors of subsidiaries
- The right to require the parent company to take all reasonable steps to obtain reasonable information and explanations from foreign subsidiaries

These rights are addressed in ISA 600 by para.19 the group engagement team shall request the agreement of the component auditor to the transfer of relevant documentation during the conduct of the group audit, a condition of the use by the group engagement team of the work of the component auditor.

Note: The group auditors also have all rights and powers in respect of their audit of the parent company that you should be familiar with from Professional Level for example, right of access at all times to the parent company's books, accounts and vouchers.

12.4 Planning and risk assessment as group auditor

The planning and risk assessment process will need to take into account the fact that all elements of the group financial statements are not audited by the group auditor directly. The group auditor will not be able to simply rely on the conclusions of the component auditor. ISA 600 requires the group auditor to evaluate the reliability of the component auditor and the work performed. This will then determine the extent of further procedures.

Understanding the component auditor

This involves an assessment of the following:

(a) Whether the component auditor understands and will comply with the ethical requirements that are relevant to the group audit and, in particular, is independent

(b) The component auditor's **professional competence**

(c) Whether the group engagement team will be able to be **involved in the work of the component auditor** to the extent necessary to obtain sufficient appropriate audit evidence

(d) Whether the component auditor operates in a **regulatory environment** that actively oversees auditors

(ISA 600.19)

The group engagement team's understanding of the component auditor's professional competence may include considering whether the component auditor:

- possesses an understanding of auditing and other standards applicable to the group audit;
- has sufficient resources to perform the work;
- possesses the special skills necessary to perform the work; and
- where relevant, possesses an understanding of the applicable financial reporting framework.

(ISA 600.A38)

The group engagement team may obtain this understanding in a number of ways. In the first year, for example, the component auditor may be visited to discuss these issues. Alternatively, the component auditor may be asked to confirm these matters in writing or to complete a questionnaire. Confirmations from professional bodies may also be sought and the reputation of the firm will be taken into account.

Materiality

The group auditor is responsible for setting the materiality level for the group financial statements as a whole. Materiality levels should also be set for components which are individually significant. These should be set at a lower level than the materiality level of the group as a whole.

(ISA 600.21)

Extent of work required on components

The ISA distinguishes between **significant components** and other components which are not individually significant to the group financial statements.

The group auditor should be involved in the assessment of risk in relation to significant components.

If a component is financially significant to the group financial statements then the group engagement team or a component auditor will perform a full audit based on the component materiality level.

If the component is likely to include significant risks of material misstatement of the group financial statements due to its nature or circumstances, the group auditors will require one of the following:

- A full audit using component materiality
- An audit of specified account balances related to identified significant risks
- Specified audit procedures relating to identified significant risks

(ISA 600.27)

Components that are not significant components will be subject to analytical review at a group level.

(ISA 600.28)

If the group engagement team does not consider that sufficient appropriate audit evidence on which to base the group audit opinion will be obtained from the work performed on significant components, on group-wide controls and the consolidation process and the analytical procedures performed at group level, then some components that are not significant components will be selected and one or more of the following will be performed (either by the group auditor or the component auditor):

- An audit using component materiality
- An audit of one or more account balances, classes of transactions or disclosures
- A review using component materiality
- Specified procedures

(ISA 600.29)

Involvement in the work of a component auditor

The extent of involvement by the group auditor at the planning stage will depend on the:

- significance of the component;
- identified significant risks of material misstatement of the group financial statements; and
- group auditor's understanding of the component auditor.

Based on these factors the group auditors may perform the following procedures:

(a) Meeting with the component management or the component auditors to obtain an understanding of the component and its environment

(b) Reviewing the component auditor's overall audit strategy and audit plan

(c) Performing risk assessment procedures to identify and assess risks of material misstatement at the component level. These may be performed with the component auditor or by the group auditor

Where the component is a **significant component** the nature, timing and extent of the group auditor's involvement is affected by their understanding of the component auditor but at a minimum should include the following procedures:

(a) Discussion with the component auditor or component management regarding the component's business activities that are significant to the group

(b) Discussing with the component auditor the susceptibility of the component to material misstatement of the financial information due to fraud or error

(c) Reviewing the component auditor's documentation of identified significant risks of material misstatements. This may be in the form of a memorandum including the conclusions drawn by the component auditors

12.4.1 Evaluating the work of the component auditor

(ISA 600.30)

For all companies in the group, the group auditor is required to perform a review of the work done by the component auditor (ISA 600.42). This is normally achieved by reviewing a report or questionnaire completed by the component auditor which highlights the key issues which have been identified during the course of the audit. The effect of any uncorrected misstatements and any instances where there has been an inability to obtain sufficient appropriate audit evidence should also be evaluated. On the basis of this review the group auditor then needs to determine whether any additional procedures are necessary. These may include the following:

- Designing and performing further audit procedures. These may be designed and performed with the component auditors, or by the group auditor

- Participating in the closing and other key meetings between the component auditors and component management
- Reviewing other relevant parts of the component auditors' documentation

12.4.2 Communication

The group engagement team shall communicate its requirements to the component auditor on a timely basis (ISA 600.40).

ISA 600 prescribes the types of information that must be sent by the group auditor to the component auditor and vice versa.

The group auditor must set out for the component auditor the work to be performed, the use to be made of that work and the form and content of the component auditor's communication with the group engagement team. According to para A35 of ISA 600 this includes the following:

- A request that the component auditor **confirms their cooperation** with the group engagement team.
- The **ethical requirements** that are relevant to the group audit and in particular independence requirements.
- In the case of an audit or review of the financial information of the component, **component materiality** and the threshold above which misstatements cannot be regarded as clearly trivial to the group financial statements.
- Identified **significant risks** of material misstatement of the group financial statements, due to fraud or error that are relevant to the work of the component auditor. The group engagement team requests the component auditor to communicate any other identified significant risks of material misstatement and the component auditor's responses to such risks.
- A list of **related parties** prepared by group management and any other related parties of which the group engagement team is aware. Component auditors are requested to communicate any other related parties not previously identified.

In addition to the above, the group auditor must ask the component auditor to communicate matters relevant to the group engagement team's conclusion with regard to the group audit. These include the following:

(a) Whether the component auditor has complied with **ethical requirements** that are relevant to the group audit, including independence and professional competence

(b) Whether the component auditor has complied with the **group engagement team's requirements**

(c) Identification of the **financial information** of the component on which the component auditor is reporting

(d) Information on instances of **non-compliance with laws and regulations** that could give rise to material misstatement of the group financial statements

(e) A list of **uncorrected misstatements** of the financial information of the component (the list need not include items that are below the threshold for clearly trivial misstatements)

(f) Indicators of **possible management bias**

(g) Description of any identified **significant deficiencies in internal control** at the component level

(h) **Other significant matters** that the component auditor communicated or expects to communicate to those charged with governance of the component, including fraud or suspected fraud involving component management, employees who have significant roles in internal control at the component level or others where the fraud resulted in a material misstatement of the financial information of the component

(i) **Any other matters** that may be relevant to the group audit or that the component auditor wishes to draw to the attention of the group engagement team, including exceptions noted in the written representations that the component auditor requested from component management

(j) The component auditor's **overall finding, conclusions or opinion**

This communication often takes the form of a **memorandum or report of work performed**.

12.4.3 Communicating with group management

ISA 600 states that the group engagement team will determine which of the identified deficiencies in internal control should be communicated to those charged with governance and group management. In making this assessment the following matters should be considered.

- Significant deficiencies in the design or operating effectiveness of group-wide controls
- Deficiencies that the group engagement team has identified in internal controls at components that are judged to be significant to the group
- Deficiencies that component auditors have identified in internal controls at components that are judged to be significant to the group
- Fraud identified by the group engagement team or component auditors or information indicating that a fraud may exist

(ISA 600.46-.47)

Where a component auditor is required to express an audit opinion on the financial statements of a component, the group engagement team will request group management to inform component management of any matters that they, the group engagement team, have become aware of that may be significant to the financial statements of the component. If group management refuses to pass on the communication, the group engagement team will discuss the matter with those charged with governance of the group. If the matter is still unresolved the group engagement team shall consider whether to advise the component auditor not to issue the audit report on the component financial statements until the matter is resolved.

12.4.4 Communication with those charged with governance of the group

(ISA 600.48)

The following matters should be communicated to those charged with governance of the group:

(a) An overview of the type of work to be performed on the financial statements of the component

(b) An overview of the nature of the group engagement team's planned involvement in the work to be performed by the component auditors on significant components

(c) Instances where the group engagement team's evaluation of the work of a component auditor gave rise to a concern about the quality of that auditor's work

(d) Any limitations on the group audit, for example, where the group engagement team's access to information may have been restricted

(e) Fraud or suspected fraud involving group management, component management, employees who have significant roles in group-wide controls or others where fraud resulted in a material misstatement of the group financial statements

12.4.5 Documentation

(ISA 600.49)

The group engagement team must include in the audit documentation the following matters:

- An analysis of components, indicating those that are significant

- The nature, timing and extent of the group engagement team's involvement in the work performed by the component auditors on significant components including, where applicable, the group engagement team's review of the component auditor's audit documentation
- Written communications between the group engagement team and the component auditors about the group engagement team's requirements

In the UK, the Companies Act 2006 requires group auditors to review the audit work conducted by other persons and to record that review. This requirement is now also specifically addressed in the revised ISA.

(ISA 600.50)



Interactive question 11: Component auditors

You are the main auditor of Mouldings Holdings, a listed company, which has subsidiaries in the country and overseas, many of which are audited by other firms. All subsidiaries are involved in the manufacture or distribution of plastic goods and have accounting periods coterminous with that of the parent company.

Requirements

11.1 State why you would wish to review the work of the auditors of the subsidiaries not audited by you.

11.2 Describe the key audit procedures you would carry out in performing such a review. See

Answer at the end of this chapter.

12.5 Risk assessment procedures

12.5.1 General accounting issues

Consolidated financial statements give rise to additional audit risks:

- Are **consolidated group accounts** correctly prepared?
- Where there have been any **disposals** of material business segments, have the requirements of IFRS 5, *Non-current Assets Held for Sale and Discontinued Operations* been satisfied?
- Have adequate **disclosures** of significant investments and financing been made?
- Where there have been acquisitions during the year, have **new assets and liabilities** been correctly brought into the financial statements?
- Where there have been acquisitions during the year, has **purchase consideration** been correctly accounted for and disclosed?
- Have **foreign taxes** (including corporate tax, income taxes for employees and capital gains tax) been accounted for correctly?
- Have transfer pricing issues around **intercompany transactions,** and their VAT impact, been considered?

12.5.2 Acquisition

Acquisitions can take many forms. The type of acquisition (eg, hostile, friendly) and future management of the subsidiary (fully integrated, autonomous) will also impact on risk.

Risk areas	Key issues
Valuation of assets and liabilities	These should be valued at **fair value** at the date of acquisition in accordance with IFRS 13.
Valuation of consideration	This should be at **fair value** and will include any **contingent consideration**. Any **deferred consideration** should be discounted.
Goodwill	This must be calculated and accounted for in accordance with IFRS 3.
Date of control	The results of any subsidiary should only be accounted for from the **date of acquisition**.
Level of control or influence	This will determine the nature of the investment and its subsequent treatment in the group financial statements eg, subsidiary, associate and should be determined in accordance with IFRS 10/IAS 28 (IFRS 10 retains control as the key concept underlying the parent/subsidiary relationship but has broadened the definition and clarified the application).
Accounting policies/reporting periods	Accounting policies and reporting periods must be **consistent** across the group.
Consolidation adjustments	The group must have **systems** which enable the **identification** of intra-group balances and accounts.
Adequacy of provisions in the target company	While the acquirer is likely to know its plans, **other provisions** may be necessary within the acquired entity. If such provisions are currently unrecognised and have never been recorded (eg, in board minutes), there is a clear risk that the acquiring entity will overpay.
Use of provisions to manipulate post- acquisition profits	Provisions may be recognised at the point of acquisition and then **released at some point in the future** in order to make post-change results appear impressive. This may imply that change was a correct business decision. The use of such provisions has been reduced by IAS 37.

12.5.3 Divestment and withdrawal

The withdrawal from a market will give rise to several audit issues:

Obsolete inventory

Inventory which is no longer required will need to be written down. When carrying out the inventory count, the auditor will need to ensure that all inventory relating to the discontinued activity is identified. The following procedures should take place.

- **Discussions with management** concerning net realisable value of inventory
- **Investigation of future costs** relating to any modifications needed for the inventory
- **Review of after-date sales** to assess likelihood of sale and sale proceeds

Obsolete non-current assets

If an operation has been withdrawn from, rather than sold as, a business segment, it is likely that some non-current assets such as plant and machinery and property will remain with the company. The following will need to be considered.

(a) **Net realisable value** of non-current assets (review of post year end sales invoices/review of trade magazines)

(b) **Write-down** of non-current assets to recoverable amount, surplus property to the market value

(c) **Impairment reviews** will need to be performed. This is likely to consider the remaining assets as one cash generating unit; lack of future revenue from this unit is likely to result in the assets being valued at their net realisable value (ie, fair value less costs of disposal) rather than their value in use

Disposal of investments

Disposal of a subsidiary, or other investment, will need to be accounted for in the group and parent company financial statements.

The auditor will need to ascertain the **date of disposal**, through inspection of sale agreements, to identify the correct period of allocation to the accounts.

(a) **Parent company** – The auditor will need to compare the sale proceeds (inspection of sale agreements/bank receipts) with the carrying amount of the investment in the statement of financial position to ensure that the correct profit or loss on disposal is accounted for.

(b) **Group accounts** – The auditor would need to ensure that the investment is not included in the year-end statement of financial position (unless some shareholding remains). Amounts in the statement of profit or loss and other comprehensive income should be pro-rated for the number of months held.

Chargeable gains and corporation tax

The disposal of a subsidiary or other investment is likely to have chargeable gains tax and corporation tax implications. These include:

(a) **Chargeable gains on disposal** – Chargeable gains or losses will arise on the disposal of subsidiaries or other investments. The auditor will need to discuss with management and inspect the sale agreement, in order to understand which party will bear the tax liability arising from chargeable gains. Where the tax liability falls on the seller, the chargeable gains tax calculations must be reviewed, and the utilisation of losses or tax relief verified. This should be done by a tax specialist if the matter is material.

(b) **Degrouping charges** – Degrouping charges will arise if any assets have previously been transferred to the subsidiary being sold at no gain/no loss. If the amount is material, the workings should be reviewed and recalculated.

(c) **Intra-group balances** – Existing loans and other outstanding balances with group companies may be written off when subsidiaries are disposed of or liquidated. The availability of corporation tax deductions on intercompany loan write-offs can be a complex issue and often constitutes a material matter. The corporation tax calculations should therefore be reviewed by a tax accounting specialist.

12.5.4 Joint arrangements

The example below illustrates some of the risk factors the auditor needs to consider in relation to joint arrangements.



Worked example: Joint arrangements

The Royal Bank of Edinburgh (RBE) has entered into several joint arrangements:

(1) "Ecost Personal Finance has made excellent progress since it was launched 18 months ago. To date, it has acquired over 700,000 customers.

In a relatively short time, in partnership with Ecost, we have established a significant and

innovative new force in UK banking. We remain very optimistic about the prospects for Ecost Personal Finance and our expectations remain that this business will move into profit in the near future.

(2) We have also received an encouraging response to the Branson One Account through our collaboration with Branson Direct Personal Financial Services."

Requirement

Examine the risk factors associated with the Royal Bank of Edinburgh joint arrangements.

Solution

Costs

The set-up costs of the two ventures will need financing. Will this be done from the existing funds within the companies, or will external finance be needed?

As RBE is a financial institution, is it providing the bulk of the finance with loan amount outstanding to the other parties?

How will the infrastructure be established? Who will pay for the website to be constructed and maintained? What is the split of these costs?

What is the profit forecast for the first periods? Initial expenses are likely to exceed revenues, therefore losses may be expected in the initial periods.

Accounting

RBE is already established in this market and is therefore likely to be providing the asset base to support its activities. How are the assets valued in the joint venture accounts?

Is there any payment to be made to RBE for the knowledge and experience that it brings to the joint arrangements?

What type of joint arrangement is it?

What is the agreement on profit sharing? The underlying elements will need to be audited and the profit share recalculated. The tax liability arising from RBE's share of the profits also needs to be audited.

How long is the joint arrangement agreement for? This will help ascertain the correct write-off period of assets.

If either of the joint arrangements is loss making, has consortium relief been assumed in RBE's accounts? Has this been correctly calculated?

Markets

The products are likely to be launched through the internet; this may expand the customer base of the companies. E-business has its own specific set of risks; these are covered in the Business Strategy section of Business Environment.

People

It is likely that there will be a combination of staff involved from each of the parties, plus some additional staff new to both organisations. The cultural and operational impacts (as explained in the main text) need to be considered.

Systems

If the arrangements described are joint ventures, a completely new set of systems will need to be established. Risk will be increased due to the unfamiliarity of the staff with these systems.

Responsibility for control

If the entity is a limited company then the directors will be responsible for ensuring proper controls.

12.5.5 Management buy-outs

Whether it is a management buy-out (MBO), a management buy-in (MBI) or a combination of the two (BIMBO), the most pressing risk is that one party is likely to have a **comparative advantage** on detailed financial information. In this respect, the current management team of a business have a major advantage over existing shareholders, interested investors/buyers and providers of finance.

For example, in an MBI where the outgoing management team own stakes in the entity, there is an inherent danger of overstating the value of the organisation. In an MBO, it may work both ways, with the MBO team looking for as low a price as possible, but also needing to convince outside providers of finance to invest.

In a BIMBO, the situation becomes even more complicated, with the existing management team looking to convince the additional management team that prospects are good in order to conclude the deal, while knowing that once the deal is finalised the two groups will have to work in tandem.

In addition to the above issues, an existing management team may find its role changing from manager to owner/manager, introducing new risks (conflict of interest, bias in preparing financial information etc).

In any buy-out scenario, both **historical and future financial information** will play a key role. **Consistency** must exist between accounting treatments in both past and future data to ensure **comparability**.

Historical data is likely to suffer from the problem that the audit was not designed with a business valuation in mind. Future data presents a more obvious problem for auditors and accountants, in that there is full knowledge of the reasons for preparation and, as such, liability for error and omission is clearer. The level of assurance, and therefore the level of detail of the assurance work, will necessarily be greater than for a statutory audit.

As a result of these issues, the external parties in a buy-out situation are likely to take **independent assurance advice**, rather than reliance being placed on the work of the company's usual auditors or accountants. While any deal remains incomplete, privacy is likely to ensure that external parties have to rely on complete and accurate information being presented. The **due diligence** exercise (see Chapter 25) provides a detailed verification, normally after the deal is complete, that the information relied upon was correct.

The greater the level of assurance that can be achieved, the easier it is likely to be to raise finance, whatever the source. The cost of finance may also be reduced if the assurance achieved is considered reliable. For this reason, many organisations will pay relatively high professional fees to the largest and most renowned firms of accountants and advisers.

While the initial investment in fees is high, the returns (greater probability of finance and at a lower cost) can easily make the decision valid.

12.6 Audit procedures

The diagram below summarises the key points in the context of the group audit.

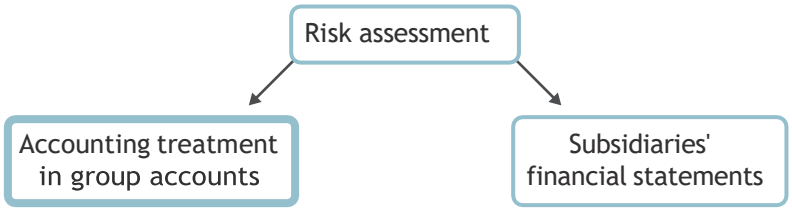


Figure 20.4: Audit procedures

Potential misstatement in group accounts due to	Factors affecting risk of material misstatement in subsidiary financial statements
• Misclassification of investments (subsidiary vs associate vs financial asset) • Inappropriate inclusion, or exclusion from, consolidation or incorrect treatment of excluded subsidiaries • Inappropriate consolidation method • Inappropriate translation method for overseas subsidiaries • Incorrect consolidation adjustments eg, failure to eliminate intra-group items properly eg, leading to potential overstatements of assets and profits • Inconsistent accounting policies for amounts included in consolidation • Incorrect calculation (fair values) or treatment of goodwill • Incorrect calculation of profit/loss on disposal or classification of results of subsidiaries disposed of (continuing vs discontinued) • Incorrect determination of date of acquisition • Deferred or contingent consideration; step acquisition	• Scope of component auditors' work (may not provide sufficient appropriate evidence that financial statements are free from material misstatement) • Past audit problems • Anticipated changes • Materiality • Sufficiency of evidence to confirm amounts • Overseas subsidiaries • Non-coterminous year ends • Existence of letter of comfort

12.6.1 Acquisition

If the group audit includes a **newly acquired subsidiary** or a subsidiary which is disposed of, compliance with IFRS 3 and IFRS 10 will be relevant. The auditor will need to consider the following issues in particular:

Issue	Audit consequence
Level of control	The auditor will need to consider whether the **appropriate accounting treatment** has been adopted depending on the level of control (per IFRS 10 an investor controls an investee if it has power over the investee, exposure or rights to variable returns and the ability to use power to affect returns). Procedures will be as follows: • Identify total number of shares held to calculate % holding. • Review contract or agreements between companies to identify key terms which may indicate control and any restriction on control eg, right of veto of third parties.
Date of control/change in stake	The auditor should: • Review **purchase agreement** to identify date of control. • Ensure consolidation has occurred from **date control achieved**. • Review consolidation schedules to ensure amounts have been **time apportioned** if appropriate.

Issue	Audit consequence
Valuation of assets and liabilities at fair value	A review will need to be carried out of the **fair value** of assets and liabilities at the date of acquisition, adjusted to the year end (in accordance with IFRS 13). Review of trade journals or specialist valuations may be required. Where specialist valuers have been used (eg, to value brands) an assessment will need to be made on the reliability of these valuations. Where **intangibles have been recognised on consolidation** which were not previously recognised in the individual financial statements of the company acquired the auditor will need to give careful consideration as to the justification of this and whether the treatment is in accordance with IFRS 3/IFRS 13. Estimates for provisions existing at the date of acquisition will need to be **assessed for reliability**.
Valuation of consideration	**Contingent consideration** should be included as part of the consideration transferred. It must be measured at fair value at the acquisition date. The **discount rate** used to discount deferred consideration should be validated.
Goodwill	The auditor will need to consider whether the initial calculation is correct in accordance with IFRS 3. Performance of the subsidiary company will need to be reviewed to identify whether any **impairment** is necessary.
Tax liabilities and assets	The amount of corporation tax liabilities provided for will need to be reviewed. Deferred tax assets and liabilities must also be reviewed. The impairment of assets or goodwill should be taken into account.
Prior year audit of subsidiary	As first year of inclusion of subsidiary, review last year's audit report for any **modification** and consider implications for this year's audit if necessary.
Planning issues	**Adjust audit plan** to ensure visit to subsidiary is included. If audited by another auditor contact secondary auditor to discuss the following: • Audit deadline • Type and quality of audit papers • Review of audit • Identification of consolidation adjustments

12.6.2 Disposal

Where the group includes a subsidiary which has been disposed of during the year, the following issues will be relevant:

- Identification of the **date of the change** in stake
- Assessment of the **remaining stake** to determine the appropriate accounting treatment post- disposal
- Assessment of the **fair value of the remaining stake**
- Whether the **profit or loss on disposal** has been calculated in accordance with IFRS Standards
- Whether amounts have been appropriately **time apportioned** eg, income and expense items

12.6.3 Auditing an ongoing group of companies

Certain issues will be relevant to the auditor each year **irrespective of whether there is any change in the structure of the group**. In particular, the auditor will need to ensure that IFRS 10 has been complied with. The following will be relevant.

Issue	Audit consequence
Accounting policies/reporting period	Identify subsidiary's **accounting policies** from review of financial statements, compare to parent company's and adjust for consistency where necessary. Further adjustments may be required if some group companies prepare financial statements in accordance with **IFRS Standards**. Ensure that subsidiary's reporting period is **consistent** with the parent company's or that interim accounts have been prepared where necessary. (If not possible subsidiary's accounts may still be used for consolidation provided that the gap between the reporting dates is three months or less.)
Consolidation adjustments	Review consolidation schedules, purchase, sales ledger and intra-group accounts to identify any intra-group transactions or outstanding balances, ensure these have been cancelled out in the group accounts. **Transactions involving group companies** Transactions involving group companies should be audited in the same way as other transactions with third parties. However, **systems should exist** to ensure all intra-group transactions are separately identified to ensure they are all appropriately eliminated on consolidation. **Intra-group balances** These should be audited in the same way as balances with third parties. In particular: • share certificates should be examined; • dividends should be verified; • intra-group balances should be verified including any security attaching thereto; • carrying amounts should be assessed in the same way as third-party investments; and • the need for transfer pricing adjustments assessed. **Intercompany guarantees** Any intercompany guarantees (eg, as surety for external loans) should be ascertained and consideration given to whether disclosure as a **contingent liability** is required.

12.6.4 Materiality

Where a subsidiary is immaterial, limited work will be performed. However, care should be taken with respect to the following:

- Apparently immaterial subsidiaries may be materially understated.
- Several small subsidiaries may cumulatively be material.
- Subsidiaries with a small asset base may engage in transactions of significant value and which may be relevant to understanding the group.

The following factors may influence component materiality levels:

- The fact that component materiality must always be lower than group materiality
- The size of the component
- Whether the component has a statutory audit
- The characteristics or circumstances that make the component significant
- The strength of the component's control environment
- The likely incidence of misstatements, taking account past experience

12.6.5 Understanding the group structure

There is a need to analyse the group structure. An understanding of the group structure allows the group auditor to:

- plan work to deal with different accounting frameworks or policies applied throughout the group;
- deal with differences in auditing standards; and
- integrate the group audit process effectively with local statutory audit requirements.

12.7 The consolidation: audit procedures

After receiving and reviewing all the subsidiaries' (and associates') financial statements, the **group** auditors will be in a position to audit the consolidated financial statements. The group auditors should not treat the consolidation as simply an arithmetical exercise. There are risks inherent in the consolidation process itself, for example:

- Consolidation adjustments are a major source of journal entries therefore procedures relating to the detection of fraud may be relevant.
- Risks may arise from incomplete information to support adjustments between accounting frameworks.

An important part of the work on the consolidation will be checking the consolidation adjustments. Consolidation adjustments generally fall into two categories:

- Permanent consolidation adjustments
- Consolidation adjustments for the current year

The audit steps involved in the consolidation process may be summarised as follows.

Step 1 Compare the audited accounts of each subsidiary/associate to the consolidation schedules to ensure figures have been transposed correctly.

Step 2 Review the adjustments made on consolidation to ensure they are appropriate and comparable with the previous year. This will involve the following:

- **Recording** the **dates** and **costs** of **acquisitions** of subsidiaries and the assets acquired
- **Calculating goodwill** and **pre-acquisition reserves** arising on consolidation
- **Preparing** an overall **reconciliation** of movements on reserves and NCIs
- **Adjusting** the individual subsidiary financial statements for **differences in accounting policies** compared to the parent. This may include compliance with the accounting regulations of a different jurisdiction (eg, where the individual subsidiary is UK GAAP compliant and the group reports under IFRS Standards)

Step 3 For business combinations, determine the following:

- Whether combination has been **appropriately** treated as an acquisition
- The **appropriateness** of the **date** used as the date of combination

- The **treatment** of the **results** of **investments** acquired during the year
- If acquisition accounting has been used, that the **fair value** of acquired **assets** and **liabilities** is in accordance with IFRS 13
- **Goodwill** has been **calculated correctly** and impairment adjustment made if necessary

Step 4 For disposals:

- agree the **date** used as the date for disposal to sales documentation; and
- review management accounts to ascertain whether the **results** of the **investment** have been **included** up to the date of disposal, and whether figures used are reasonable.

Step 5 **Consider** whether **previous treatment** of **existing subsidiaries** or **associates** is still **correct** (consider level of influence, degree of support).

Step 6 Verify the **arithmetical accuracy** of the consolidation workings by recalculating them.

Step 7 **Review** the **consolidated accounts** for **compliance** with the legislation, accounting standards and other relevant regulations. Care will need to be taken in the following circumstances:

- Where group companies do not have coterminous accounting periods
- Where subsidiaries are not consolidated
- Where accounting policies of group members differ because foreign subsidiaries operate under different rules, especially those located in developing countries
- Where elimination of intra-group balances, transactions and profits is required

Other important areas include the following:

- Treatment of associates
- Treatment of goodwill and intangible assets
- Foreign currency translation
- Taxation and deferred tax
- Treatment of loss-making subsidiaries
- Treatment of restrictions on distribution of profits of a subsidiary
- Share options

Step 8 **Review** the **consolidated accounts** to confirm that they give a true and fair view in the circumstances (including subsequent event reviews from all subsidiaries updated to date of audit report on consolidated accounts).

It is important for group auditors to considering the process used to perform the consolidation process. Where excel spreadsheets are used it is not enough to check the data that has been entered. Auditors also need to check that the consolidation spreadsheets are actually working properly.



Interactive question 12: Intra-group balances/profits

Your firm is the auditor of Beeston Industries, a limited company, which has a number of subsidiaries in your country (and no overseas subsidiaries), some of which are audited by other firms of professional accountants. You have been asked to consider the work which should be carried out to ensure that intra-group transactions and balances are correctly treated in the group accounts.

Requirements

12.1 Describe the audit procedures you would perform to check that intra-group balances agree, and state why intra-group balances should agree and the consequences of them not agreeing.

12.2 Describe the audit procedures you would perform to verify that intra-group profit in inventory has been correctly accounted for in the group accounts.

See **Answer** at the end of this chapter.

12.8 Overseas subsidiaries

The inclusion of one or more foreign subsidiaries within a group introduces additional risks, including the following:

- Non-compliance with the **accounting requirements** of IAS 21, *The Effects of Changes in Foreign Exchange Rates*
- Potential misstatement due to the effects of **high inflation**
- Possible difficulty in the parent being able to exercise **control**, for example due to political instability
- Currency restrictions limiting payment of profits to the parent
- There may be threats to **going concern** due to economic and/or political instability
- Non-compliance with local taxes or misstatement of local tax liabilities

Audit procedures should include the following:

- Confirm that the balances of the subsidiary have been appropriately translated to the group reporting currency:
  - Assets and liabilities at the **closing rate** at the end of the reporting period
  - Income and expenditure at the **rate ruling at the transaction date**. An average would be a suitable alternative provided there have been no significant fluctuations
- Confirm **consistency** of treatment of the translation of equity (closing rate or historical rate)
- Confirm that the **consolidation process** has been performed correctly eg, elimination of intra- group balances
- Recalculate the **non-controlling interest**
- Confirm that **goodwill** has been translated at the closing rate
- Confirm the **disclosure of exchange differences** as a separate component of equity
- Assess whether **disclosure requirements** of IAS 21 have been satisfied
- If the foreign operation is operating in a **hyperinflationary economy** confirm that the financial statements have been adjusted under IAS 29, *Financial Reporting in Hyperinflationary Economies* before they are translated and consolidated
- Involve a specialist tax audit team to review the calculation of tax balances against submitted and draft tax returns

12.9 Other considerations in the group context

Other considerations include the following.

12.9.1 Related parties

Remember that when auditing a consolidation, the relevant related parties are those related to the **consolidated group**. Transactions with consolidated subsidiaries need **not** be disclosed, as

they are incorporated in the financial statements. However, related party relationships where there is control (eg, a parent) need to be disclosed even where there are no transactions with this party.

The group auditors are often requested to carry out the consolidation work even where the accounts of the subsidiaries have been prepared by the client. In these circumstances the auditors are, of course, acting as accountants and auditors and care must be taken to ensure that the **audit** function is carried out and evidenced.

Transfer pricing issues around transactions with related parties must also be considered. Transfer pricing adjustments, when they are required, are often material to the financial statements.

12.9.2 Support (comfort) letters

It is sometimes the case that a subsidiary, when considered in isolation, does not appear to be a going concern. In the context of group accounts, the parent and the subsidiary are seen to be a complete entity, so if the group as a whole is a going concern, that is sufficient.

When auditing the incorporation of the single company into the group accounts, however, the auditor will need assurance that the subsidiary is a going concern.

In such a case, the auditor may request a 'support letter' from the directors of the parent company. This letter states that the intention of the parent is to continue to support the subsidiary, which makes it a going concern. A support letter may be sufficient appropriate audit evidence on this issue, but further evidence would be bank guarantees.

12.10 Promoting best practice in group audits

The following practical guidance can be used for each stage of the group audit:

Understand group management's process and timetable to produce consolidated accounts	Understand the group structure and the nature of the components of the group Consider whether to accept an engagement where the group auditor is only directly responsible for a minority of the total group Understand the accounting framework applicable to each component and any local statutory reporting requirements Understand the component auditors – consider their qualifications, independence and competence For unrelated auditors or related auditors where the group auditor is unable to rely on common policies and procedures, consider the following: • Visiting the component auditor • Requesting that the component auditor completes a questionnaire or representation • Obtaining confirmation from a relevant regulatory body • Discussing the component auditor with colleagues from their own firm For component auditors based overseas consider whether they have enough knowledge and experience of ISAs

Design group audit process to match management's process and timetable	Get involved early. Talk to group management while they are planning the consolidation Draft instructions to component auditors allocating work and be clear as to deadlines required Focus the group audit on high risk areas Consider risks arising from the consolidation process itself: • Consolidation adjustments • Incomplete information to support adjustments between accounting frameworks eg, where a subsidiary prepares its local accounts under US GAAP and the parent is preparing IFRS financial statements Discuss fraud with component auditors and consider the following: • Business risks • How and where the group financial statements may be susceptible to material misstatement due to fraud or error • How group management and component management could perpetrate and conceal fraudulent financial reporting and how assets of the components could be misappropriated • Known factors affecting the group that may provide the incentive or pressure for group or component management or others to commit fraud or indicate a culture or environment that enables those people to rationalise committing fraud • The risk that group or component management may override controls Understand internal control across the group: • Request details of material weaknesses in internal controls identified by component auditors • Communicate material weaknesses in group-wide controls and significant weaknesses in internal controls of components to group management
Clearly communicate expectations and information required including timetable	Explain the extent of the group auditors' involvement in the work of the component auditors: • Make it clear what the component auditors are being asked to perform eg, a full audit, a review or work on specific balances or transactions • Clarify the timetable and format of reporting back Review completed questionnaires and other deliverables from component auditors carefully Decide whether and when to visit component auditors and when to request access to their working papers Get group management to obtain the consent of subsidiary management to communicate with the group auditor to deal with concerns about client confidentiality and sensitivity Consider whether holding discussions with or visiting component auditors could deal with secrecy and data-protection issues

Obtain information early where practicable	There is often only a short time for group auditors to resolve any issues arising from the report they receive from component auditors Request some information early, such as copies of management letter points from component auditors carrying out planning and control testing before the year end
Keep track of whether reports have been received and respond to any issues in a timely fashion	Where component auditors indicate up front that they will not be able to provide the information requested, consider alternatives rather than waiting until the sign-off deadline Put in place a system to monitor responses to instructions and follow up on non-submission
Conclude on the audit and consider possible improvements for the next year's process including management letter issues	The group auditors should be in a position to form their opinion on the group financial statements The group auditors will consider the need for a group management letter and reporting to those charged with governance of the group Debrief the team and consider whether the process worked as well as it could have done, along with any changes to future accounting and auditing requirements, and whether there are any issues that should be communicated to management and those charged with governance, or any changes to next year's audit strategy

13 Auditing global enterprises



Section overview

- Global enterprises are particularly affected by the following risks:
  - financial risks
  - political risks
  - regulatory risks
- Internal control will have to have regard to a variety of local requirements.
- Compliance will be a key feature of international business strategy.
- In response to the trend towards globalisation the Forum of Firms has been founded.

13.1 Introduction

Large businesses are increasingly becoming global organisations. This has implications for the business itself and the way in which the audit is conducted. In the remainder of this section we will look at a number of key issues affecting global organisations.

13.2 Inherent risk

13.2.1 Financial risk

By its very nature a global organisation will have businesses in countries all over the world. At any point in time these countries may be experiencing **diverse economic circumstances**. This may impact on key aspects of **financial management**, including the effects of the following:

- inflation
- interest rates
- exchange rates
- currency restrictions

In some instances the impact of these can be significant. For example, Venezuela experienced inflation in excess of 1,000,000% in 2019 (hyperinflation will be considered further in next Chapter).

Overseas financial risk

The **financial strategy** of the company may be one of **overseas financing**. This could be in two situations.

(a) The raising of finance overseas but remittance of funds back to the country – probably to take advantage of more reasonable terms

(b) The raising of finance overseas for the purpose of providing capital for an overseas subsidiary, branch or significant equity investment

Identifying overseas financial risks

The **business** choice between the above transactions can be summarised as follows.

- The legal requirements for raising loan capital in the country of origin may be more complex than the BDT equivalent.
- The tax implications may be diverse and difficult to manage.
- The country of origin may have strict rules on remittance of proceeds back to Bangaldesh.
- The company may have a cultural image or mission statement with which overseas financing conflicts.
- The decision to finance overseas may have been made on factors that can easily be distorted in the short term. The factors likely to have been considered would have been:
  - the prevailing rate of exchange
  - the cost of financing interest or dividend payments

Foreign exchange and interest rates are outside the control of the individual entity. Therefore, if there is an unfavourable movement in the foreign exchange or interest rates, it could result in a significant change in **cash flow** in both the short and long term.

Mitigating overseas financial risks

The business may choose to mitigate risk by adopting the following or similar procedures.

- The company should **obtain legal, taxation and accounting advice** before the overseas financing commencing.
- The company should **hedge** any overseas transactions to reduce the risk of currency and interest rate movements significantly affecting its assets and liabilities.
- The company at board level should consider the **cultural and political implications** of an investment overseas.

Assessing overseas financial risks

The assurance adviser can approach the assignment initially by focusing on business risk and, where necessary, with a more **substantive approach**.

The assurance adviser will therefore be addressing the issue of changing exchange and interest rates, where material, in the relevant accounting period. He would discuss with

management any difficulties that had arisen over the legal requirements and whether remittance from overseas had proved straightforward.

The advisor would also have gained assurance that the company had identified the reporting requirements of the country.

Having addressed business risk, the following steps may need to be taken:

- Examination of the loan capital terms and contractual liabilities of the company
- Checking the remittance of proceeds between the country of origin and the company by reference to bank and cash records
- Reviewing the movement of exchange and interest rates, and discussing their possible impact with the directors
- Obtaining details of any hedging transaction and ensuring that exchange rate movements on the finance had been offset
- Examining the financial statements to determine accurate disclosure of accounting policy and accounting treatment conforming to locall requirements
- Evaluating whether the directors had satisfied themselves as to the company's conforming status as a going concern

13.2.2 Political risk

Political issues, particularly political unrest, may have implications for both the business and the way in which its results are reported. For example, restrictions imposed by foreign governments may call into question the ability of a parent to control its subsidiary. This may raise questions about **possible impairment** of the value of the parent's investment in the subsidiary and may affect the way in which the investment is recorded in the group financial statements.

13.2.3 Regulatory risk

A global organisation will need to be equipped to deal with a range of local legislation. Key areas include the following:

- health and safety
- environmental legislation
- trade descriptions
- consumer protection
- data protection
- employment issues

Failure to comply with these may result in **financial or other penalties**, having to spend money and resources in fighting litigation and loss of reputation.

In this area, **governance codes** will be particularly important examples of best practice which should be adopted worldwide, and organisations must consider the risks of breaching provisions relating to integrity and objectivity, and also control over the organisation as a whole.



Professional skills focus: Applying judgement

As we have often seen, the higher the risk of an audit assignment, the more important it is for the auditor to exercise judgement. For example, just because a firm has a code of governance does not mean that it will be complied with.

Audit impact

The variation in local regulation may have an impact on the audit itself. For example, some subsidiaries may be in countries which do not have accounting and auditing standards developed to the same extent as those in the country of Parent company. As a result, the financial statements and the audit work carried out on them by local practitioners of subsidiary companies **may not conform to the standards of the local requirements of parent company**.

In this situation, the group auditor may need to:

- request adjustments to be made to the financial statements of the subsidiary; and
- request additional audit procedures to be performed.

The increasing acceptance of international accounting and auditing standards and ongoing convergence between these and local regulation means that this problem should become less significant over time.

13.3 Internal control

Many of the internal control issues stem from the **inherent risks** identified above. For example, geographical, cultural and regulatory differences may result in a variety of internal control mechanisms being adopted by the organisation. The entity will need to ensure that these mechanisms satisfy not only local requirements but also the internal control objectives of the entity as a whole. The following specific points should be noted.

Risk management

The management of a global enterprise will need to have regard to the **corporate governance requirements** (see Chapter 4) as follows:

Control environment	This sets the tone from the top of the organisation and will need to be applicable at both a local and global level. Factors to consider include the following: • Organisational structure of the group • Level of involvement of the parent company in components • Degree of autonomy of management of components • Supervision of components' management by parent company • Information systems, and information received centrally on a regular basis
Risk assessment	The nature of a global organisation increases risk. Management need to ensure that a process is in place to identify the risks at the global level and assess their impact
Information systems	Information systems will need to be designed so that accurate and timely information is available both at the local level and on an entity basis. Compatibility of systems and processes will be important
Control procedures	While there may be local variations, minimum entity-wide standards must be established to ensure that there are adequate controls throughout the organisation
Monitoring	In organisations of this size audit committees and the internal audit function will have a crucial role to play

13.3.1 Transfer pricing

Transfer pricing addresses the need to value the use of goods and services of one division by another. A well thought-out system may allow accurate **divisional performance measurement**

while ensuring that divisions are not motivated to make decisions adverse for the company as a whole.

Transfer pricing systems have four primary aims:

(a) To enable the **realistic measurement of divisional profit**

(b) To provide producer and receiver with **realistic income and cost**

(c) To avoid taking **too much autonomy** from managers

(d) To ensure **goal congruence and profit maximisation** for the company as a whole

Transfer pricing is notoriously difficult to get right. One particular problem associated with transfer prices is in valuing a company or division as an independent unit. For example, a division of a major company is likely to find major costs (eg, rent) to be much higher without the economic support of a parent.

Where the product or service has a readily available outside market, internal transfers are unlikely to occur unless the transfer price is similar to the market price. The transfer price may be slightly below the market price to recognise the reduction in risk (eg, no cash changes hands so risk of bad debts may be reduced) and likely savings in packing and delivery.

Another method used to set transfer prices is **'cost plus'**. The transferring division should be able to recover its variable costs and any contribution lost because it diverted resources in order to fulfil the internal transfer. The costs involved should be standard rather than actual, otherwise inefficiencies in the selling division will be passed on to the buying division.

The choice of the most appropriate transfer pricing method depends on the nature of the business and the level of risks borne by the selling division.

Transfer pricing can be used to **manipulate profits for tax purposes, rather than to measure performance.** Consequently, transfer pricing issues have come to the top of the agenda for tax authorities worldwide, and become the focus of an increasing number of NBR and Customs duty inquiries (particularly where they involve transactions between divisions which are resident in different countries).



Context example: Divisional performance

Company C is organised into two divisions, Division A and Division B. Division A can sell its product outside the company at £20 per unit or transfer internally to Division B at £20 per unit.

Division B buys the product from Division A and develops it into a higher-margin finished product. The division's usual selling price for its finished product is £70.

Division A's variable costs are £10 per unit and its fixed costs are £5 per unit. Division B's variable costs are £15 per unit and its fixed costs are £10 per unit.

If Division B received an offer from a customer of £30 per unit for its final product, it would not accept the offer. This is because, taking the transfer price of £20 into account, its accounts would show a negative contribution per unit of (30 – 20 – 15) £5.

However, assuming that Division A has surplus capacity, Company C would accept the offer. From the point of view of the company as a whole, the contribution would be positive: (30 – 10 – 15) £5 per unit.

If Division A is already operating at full capacity then there would be a lost external sales contribution in A of (20 – 10) £10. Therefore, in this situation, the company as a whole should also reject the offer because the deal would represent a loss in potential contribution of £5 per unit.

If challenged by the tax authorities, transfer pricing adjustments can have a material impact on the selling and buying divisions' corporation tax expense and tax liabilities. It may also change the recognition of intercompany revenue in the individual companies' financial statements.

Auditing the transfer pricing status of large multinational groups often requires the involvement of tax specialists. Issues that the auditor should consider when reviewing the company's transfer pricing policies include the following:

(a) Are there any unresolved tax enquiries/tax audits relating to transfer pricing?

(b) Has an Advance Transfer Pricing Agreement been signed between the group and the tax authorities? If so, does the transfer pricing policy applied in the period conform to the Agreement?

(c) Is the transfer pricing method adopted appropriate for the type of transaction, and the nature of the selling division's business?

(d) Do the transfer prices appear reasonable, compared to existing benchmarks (for example, is the percentage of mark-up in a cost-plus policy in line with the mark-ups applied by comparable companies in the industry)?

(e) Have there been any changes in the divisions' business, which may require the transfer pricing policy to be revised?

13.4 Compliance

A key feature of any international business strategy is that it is likely to involve compliance with overseas accounting and auditing regulations of the host countries in which an entity does business. The most important piece of recent legislation in this respect has been the **Sarbanes–Oxley Act which** is covered in detail in earlier Chapter of this Workbook.

13.5 Transnational audits

13.5.1 The Forum of Firms

In response to the trend towards globalisation an international grouping, the Forum of Firms (FoF), was founded by the following networks: BDO, Deloitte Touche Tohmatsu, Ernst & Young, Grant Thornton, KPMG and PricewaterhouseCoopers.

Membership is open to firms and networks that have transnational audit appointments or are interested in accepting such appointments.

These firms have a voluntary agreement to meet certain requirements that are set out in their constitution. These relate mainly to the following:

- Promoting the use of high-quality audit practices worldwide, including the use of ISAs
- Maintaining quality management standards in accordance with International Standards on Quality Management issued by the IAASB, and conducting globally co-ordinated internal quality assurance reviews

The **Transnational Auditors Committee** (TAC) provides guidance to the members of the FoF and provides the official linkage between the FoF and the International Federation of Accountants (IFAC).

The TAC has issued the following definition of transnational audit.



Definition

Transnational audit: An audit of financial statements which are or may be relied upon outside the audited entity's home jurisdiction for purposes of significant lending, investment or regulatory decisions; this will include audits of all financial statements of companies with listed equity or debt and other public interest entities which attract particular public attention because of their size, products or services provided.

Audits of entities with listed equity or debt are always transnational audits, as their financial statements are or may be relied upon outside their home jurisdiction. Other audits that are transnational audits include audits of those entities in either the public or the private sectors where there is a reasonable expectation that the financial statements of the entity may be relied upon by a user outside the entity's home jurisdiction for purposes of significant lending, investment or regulatory decisions, whether or not the entity has listed equity or debt or where entities attract particular attention because of their size, products or services provided. (These would include, for example, large charitable organisations or trusts, major monopolies or duopolies, providers of financial or other borrowing facilities to commercial or private customers, deposit-taking organisations and those holding funds belonging to third parties in connection with investment or savings activities.)

In principle, the definition of transnational audit should be applied to the whole group audit, including the individual components comprising the consolidated entity.

Examples to illustrate the definition:

Example	Explanation
Audit of a private company in the US raising debt finance in Canada	This would qualify as a transnational audit, as it is reasonable to expect that the financial statements of the company would be used across national borders in obtaining the debt financing.
Audit of a private savings and loans business operating entirely in the US (ie, only US depositors and US investments)	Although it could be considered a public interest entity, this would not qualify as a transnational audit assuming it can be demonstrated that there are no transnational users. In applying the definition of transnational audit, there should be a rebuttable presumption that all banks and financial institutions are included, unless it can be clearly demonstrated that there is no transnational element from the perspective of a financial statement user and that there are no operations across national borders. Potential transnational users would include investors, lenders, governments, customers and regulators.
Audit of an international charity taking donations through various national branches and making grants around the world	This entity can clearly be considered a public interest entity and operating across borders. Further, the international structure would create a reasonable expectation that the financial statements could be used across national borders by donors in other countries if not by others for purposes of significant lending, investment or regulatory decisions. The audit is likely to qualify as transnational.

13.5.2 Features of transnational audits

In the globalised business and financial environment, many audits are clearly transnational, and this produces a number of specific problems which can limit the reliability of the audited financial statements:

- Regulation and oversight of auditors differs from country to country
- Differences in auditing standards from country to country
- Variability in audit quality in different countries

13.5.3 Role of the international audit firm networks

The 'Big 4' and other international networks of firms can be seen as being ahead of governments and institutions in terms of their global influence. They are in a position to establish consistent practices worldwide in areas such as:

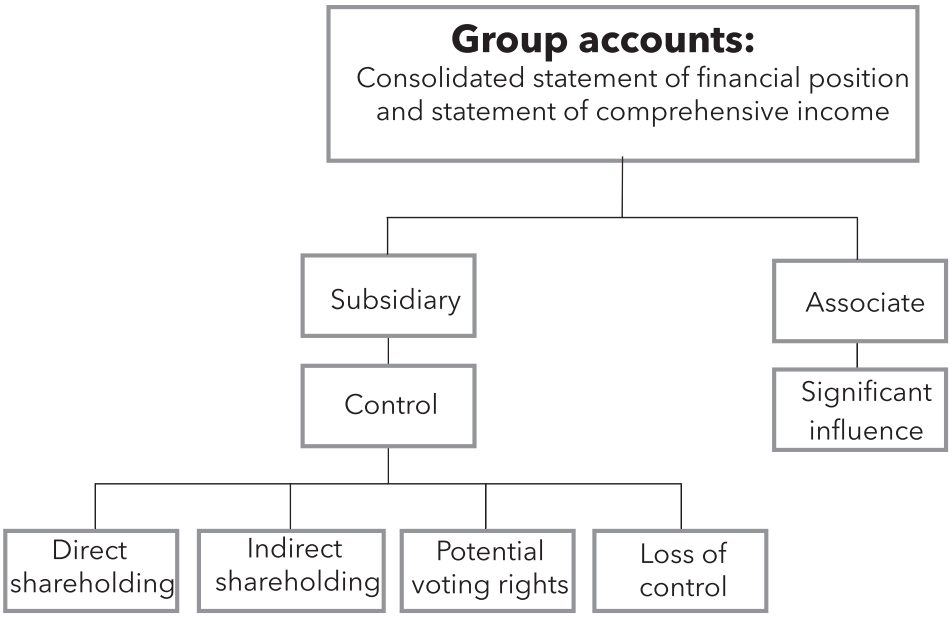
- training and education
- audit procedures
- quality control procedures

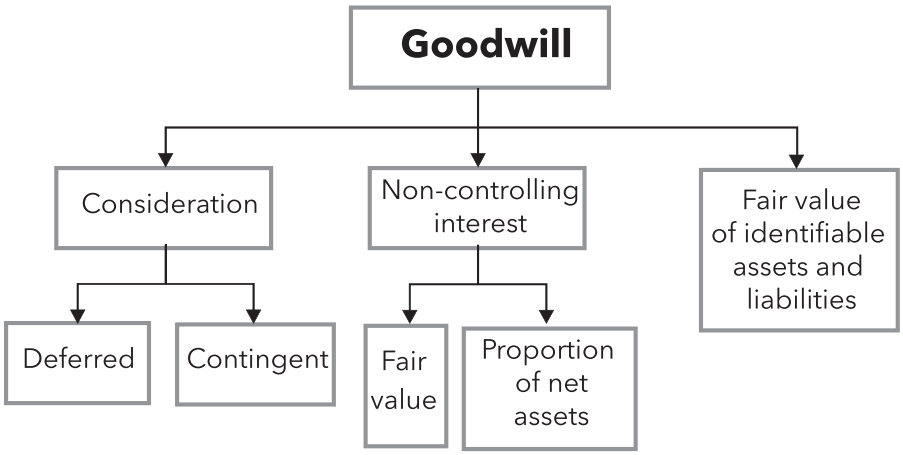
These firms may as a result be in a better position than national regulators to ensure consistent implementation of high-quality auditing standards.

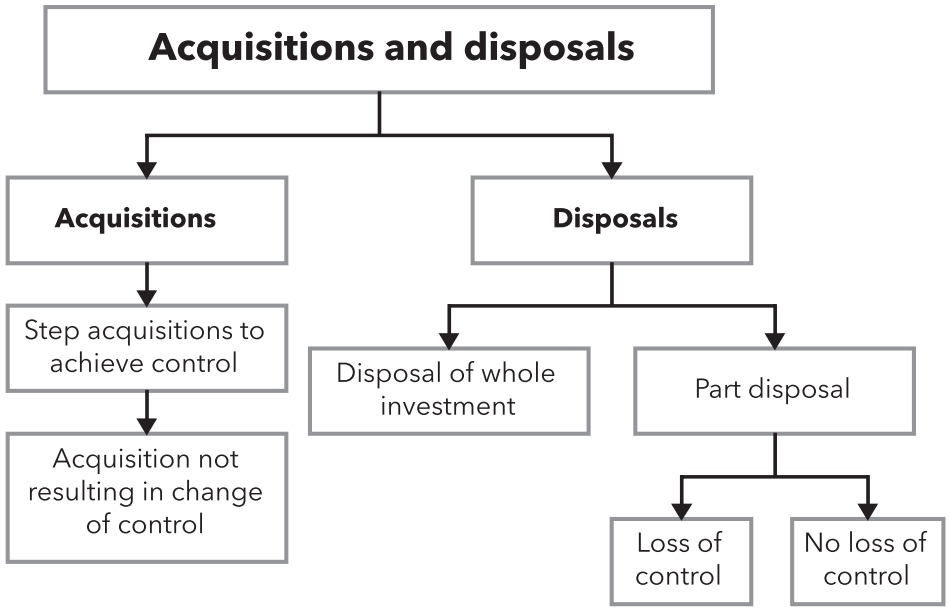
Membership of the Forum of Firms imposes commitments and responsibilities, namely to:

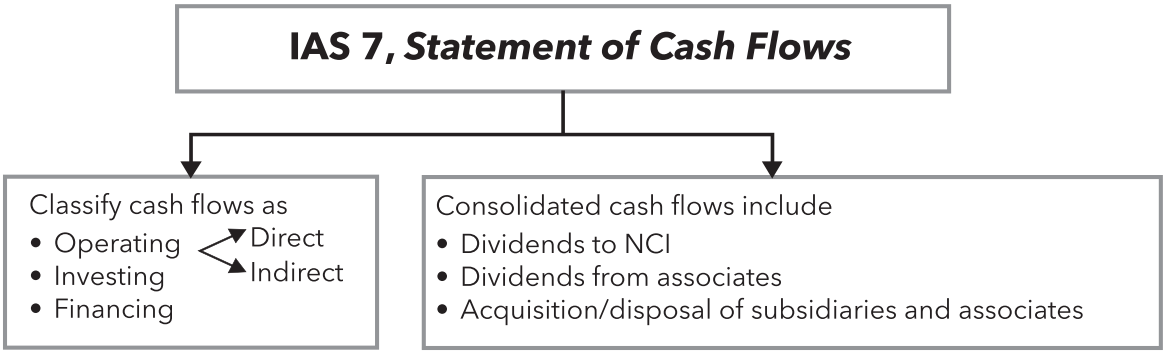
- perform transnational audits in accordance with ISAs;
- comply with the IESBA Code of Ethics; and
- be subject to a programme of quality assurance.

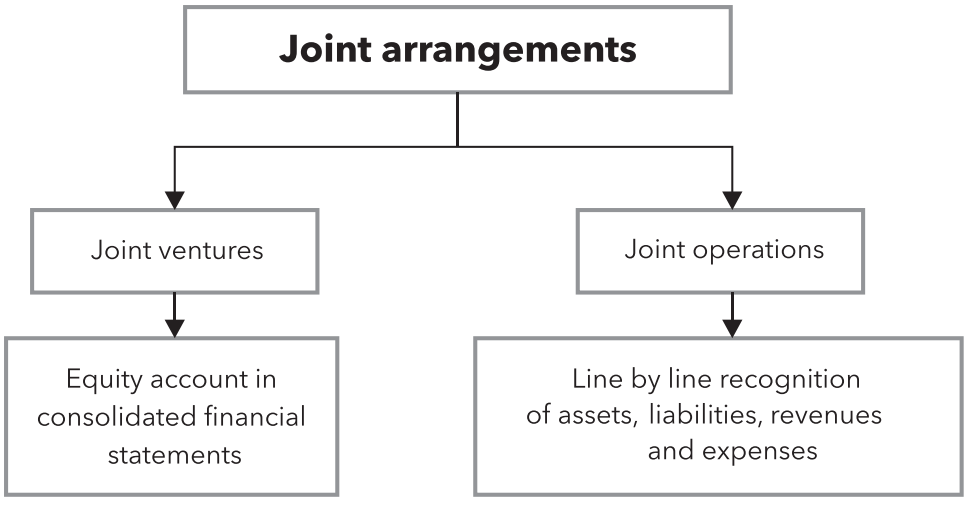
Summary


Group accounts:
Consolidated statement of financial position and statement of comprehensive income
Subsidiary
Associate
Control
Significant influence
Direct shareholding
Indirect shareholding
Potential voting rights
Loss of control


Goodwill
Consideration
Non-controlling interest
Fair value of identifiable assets and liabilities
Deferred
Contingent
Fair value
Proportion of net assets


Acquisitions and disposals
Acquisitions
Disposals
Step acquisitions to achieve control
Acquisition not resulting in change of control
Disposal of whole investment
Part disposal
Loss of control
No loss of control


IAS 7, Statement of Cash Flows
Classify cash flows as
• Operating
• Investing
• Financing
Direct
Indirect
Consolidated cash flows include
• Dividends to NCI
• Dividends from associates
• Acquisition/disposal of subsidiaries and associates


Joint arrangements
Joint ventures
Joint operations
Equity account in consolidated financial statements
Line by line recognition of assets, liabilities, revenues and expenses

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	Can you apply the IFRS 10 definition of control? (Topic 2)
2.	How should goodwill be calculated? How should it be treated subsequent to the acquisition? (Topic 3)
3.	How should contingent consideration be measured subsequent to the acquisition? (Topic 3)
4.	Can you apply IFRS 3, *Fair Value Measurement* to business combinations? (Topic 4)
5.	Where an entity prepares separate financial statements how should it account for associates and joint ventures in its separate financial statements? (Topic 5)
6.	Can you account for disposals (Topic 10) and business combinations achieved in stages (Topic 9)?
7.	Do you understand the potential risks faced by the auditor when auditing group financial statements? (Topic 13)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Burdett	This is a straightforward calculation of goodwill, but also requires you to know the treatment of a gain on a bargain purchase.
Gibston	A further calculation of goodwill, this tests deferred and contingent consideration and the net asset calculation is more complicated than for Burdett.
Green and Yellow	An explanation of joint arrangements under IFRS 11 is required.
Fleurie	This question contains a disposal where control is retained.
Chianti	In contrast, in this disposal, control is lost.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Upstart Records requirement (1)	This requires a detailed explanation with calculations of the treatment of goodwill and non-controlling interest on consolidation.
Poe, Whitman & Co (disposal of Scherzo only)	This focuses on audit and ethical issues relating to the partial disposal of a subsidiary.
EyeOp requirement (1)	This is a calculation of goodwill with a business combination achieved in stages (financial asset to subsidiary).

Refer back to the learning in this chapter for any questions which you did not answer correctly, or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving onto the next chapter.

Technical reference

1 IFRS 3, Business Combinations

Basics

- Definitions: control, parent, subsidiary, acquisition date, goodwill – **IFRS 3 (App A)**
- Acquisition method: acquirer, acquisition date, recognising and measuring assets, liabilities, non- controlling interest and goodwill – **IFRS 3.5**

Measurement of identifiable assets acquired

- At fair value – **IFRS 3.18**
- Non-controlling interest measured at fair value or as a proportionate share of the acquiree's net assets – **IFRS 3.19**
- Exceptions to measurement principles
  - Contingent liabilities recognised if fair value measured reliably regardless of probable outcome – **IFRS 3.23**
  - Income taxes in accordance with IAS 12 – **IFRS 3.24**
  - Employee benefits in accordance with IAS 19 – **IFRS 3.26**
  - Assets held for sale in accordance with IFRS 5 – **IFRS 3.31**

Consideration transferred

- Fair value of assets transferred, liabilities incurred and equity instruments issued – **IFRS 3.37**
- Contingent consideration accounted for at fair value – **IFRS 3.39**
- Subsequent accounting for contingent consideration – **IFRS 3.58**
- Acquisition related costs – **IFRS 3.53**

Goodwill

- Calculation – **IFRS 3.32**

Bargain purchases

- Reassess identification and measurement of the net assets acquired, the non-controlling interest, if any, and consideration transferred – **IFRS 3.36**
- Any remaining amount recognised in profit or loss in period the acquisition is made – **IFRS 3.34**

Business combination achieved in stages (step acquisition)

- Remeasure previously held interest to fair value at acquisition date – **IFRS 3.42**

Measurement period

- Adjustment to amounts only within 12 months of acquisition date – **IFRS 3.45**
- Subsequently: errors accounted for retrospectively, everything else prospectively – **IFRS 3.50**

Disclosures

- Business combinations occurring in the accounting period or after its finish but before financial statements authorised for issue (in the latter case, by way of note) – **IFRS 3.59**

2 IFRS 10, Consolidated Financial Statements

Basic rule

- Parent must prepare CFS to include all subsidiaries – **IFRS 10.2**

Exception

- No need for CFS if wholly owned or all non-controlling shareholders have been informed of and none have objected to the plan that CFS need not be prepared – **IFRS 10.4**

Control – IFRS 10.7

- Power over the investee
- Exposure or rights to variable returns
- Ability to use its power
- Power is existing rights that give the current ability to direct the relevant activities of the investee

Procedures

- Non-controlling interest shown as a separate figure:
  - In the statement of financial position, within total equity but separately from the parent shareholders' equity – **IFRS 10.22**
  - In the statement of profit or loss and other comprehensive income, the share of the profit after tax and share of the total comprehensive income
- Accounting dates of group companies to be no more than three months apart – **IFRS 10.B93**
- Uniform accounting policies across group or adjustments to underlying values – **IFRS 10.19**
- Bring in share of new subsidiary's income and expenses: – **IFRS 10.20**
  - From date of acquisition, on acquisition
  - To date of disposal, on disposal
- Changes that do not result in a loss of control accounted for as equity transactions – **IFRS 10.B96**

Loss of control – IFRS 10.B97–99

- Calculation of gain
- Account for amounts in other comprehensive income as if underlying assets disposed of
- Retained interest accounted for in accordance with relevant standard based on fair value

Parent's separate financial statements

- Account for subsidiary on basis of cost and distributions declared – **IAS 27.12**

3 IAS 28, Investments in Associates and Joint Ventures

Definitions

- The investor has significant influence, but not control – **IAS 28.2**
- Significant influence is the power to participate in financial and operating policy decisions of the investee, but is not control over those policies (if the investor had control, then under IFRS 10 the investee would be its subsidiary) – **IAS 28.3**
- Presumptions re less than 20% and 20% or more – **IAS 28.5**
- Can be an associate, even if the subsidiary of another investor

Equity method

- In statement of financial position: non-current asset = cost plus share of post-acquisition change in A's net assets – **IAS 28.10**
- In statement of profit or loss: share of A's post-tax profits less any impairment loss – **IAS 28.32**
- In statement of changes in equity: share of A's changes – **IAS 28.10**
- Use IAS 27 to account in investor's separate financial statements – **IAS 28.44**
- Also applies to joint ventures – **IAS 28.10**

Disclosures

These are specified in IFRS 12, *Disclosure of Interests in Other Entities*.

4 Key areas of consolidations

- Control still possible if less than 50% of the voting rights owned – **IAS 27.13**
- Potential voting rights – **IAS 27.14–15**
- Loss of control – **IAS 27.32–34**
- Contingent consideration – **IFRS 3.39–40,58**

5 IFRS 11, Joint Arrangements

- Definitions – **Appendix A**
- Two forms of joint arrangement – **IFRS 11.6**
- Contractual arrangement – **IFRS 11.5**
- Joint ventures – **IFRS 11.24**
  - Equity method
- Joint operations – **IFRS 11.20**
  - What a joint operation is
  - Line by line recognition

6 Consolidated statements of cash flows

- Only cash flows between the group and an associate are reported in the group statement of cash flows with respect to associates – **IAS 7.37**
- Aggregate cash flows from acquisitions and disposals of subsidiaries presented as investing activities – **IAS 7.39**

7 Audit of groups

- Definitions:
  - Group audit partner and group engagement team – **ISA 600.9**
  - Component auditor – **ISA 600.9**
- Responsibility – **ISA 600.11**
- Acceptance considerations – **ISA 600.12–.13**
- Procedures to assess the extent to which the component auditor can be relied upon – **ISA 600.19–.20**
- Significant components – **ISA 600.26–.27**
- Communication with component auditors – **ISA 600.40–.41**
- Communication with group management and those charged with governance – **ISA 600.46–.49**

Self-test questions

Answer the following questions

1 Burdett

The Burdett Company acquired an 80% interest in The Swain Company for £1,340,000 when the fair value of Swain's identifiable assets and liabilities was £1,200,000.

Burdett acquired a 60% interest in The Thamin Company for £340,000 when the fair value of Thamin's identifiable assets and liabilities was £680,000.

Neither Swain nor Thamin had any contingent liabilities at the acquisition date and the above fair values were the same as the carrying amounts in their financial statements. Annual impairment reviews have not resulted in any impairment losses being recognised. The Burdett Company values the non-controlling interest as the proportionate interest in the identifiable net assets at acquisition.

Requirement

Under IFRS 3, *Business Combinations*, what figures in respect of goodwill and of the excess of assets and liabilities acquired over the cost of combination should be included in Burdett's consolidated statement of financial position?

2 Sheliak

The Sheliak Company acquired equipment on 1 January 20X3 at a cost of £1,000,000, depreciating it over eight years with a nil residual value.

On 1 January 20X6 The Parotia Company acquired 100% of Sheliak and estimated the fair value of the equipment at £575,000, with a remaining life of five years. This fair value was not incorporated into Sheliak's books and subsequent depreciation charges continued to be made by reference to original cost.

Requirement

Under IFRS 3, *Business Combinations* and IFRS 10, *Consolidated Financial Statements*, what adjustments should be made to the depreciation charge for the year and the SFP carrying amount in preparing the consolidated financial statements for the year ended 31 December 20X7?

3 Finch

On 1 January 20X6 The Finch Company acquired 85% of the ordinary share capital and 40% of the irredeemable preference share capital of The Sequoia Company for consideration totalling £5.1 million. At the acquisition date Sequoia had the following statement of financial position.

	£'000
Non-current assets	5,500
Current assets	2,600
	8,100

Ordinary shares of £1	1,000
Preference shares of £1	2,500
Retained earnings	3,300
Current liabilities	1,300
	8,100

Included in non-current assets of Sequoia at the acquisition date was a property with a carrying amount of £600,000 that had a fair value of £900,000.

Sequoia had been making losses, so Finch created a provision for restructuring of £800,000 under plans announced on 2 January 20X6.

Sequoia has disclosed in its accounts a contingent liability with a reliably estimated value of £20,000. Sequoia has not recorded this as a provision, as payment is not considered probable.

Finch values the non-controlling interest using the proportion of net assets method.

Requirement

Under IFRS 3, *Business Combinations*, what goodwill arises at the time of the acquisition of Sequoia?

4 Maackia

On 31 December 20X7 The Maackia Company acquires 65% of the ordinary share capital of The Sorbus Company for £4.8 million. Sorbus is incorporated in Flatland, which has not adopted IFRS for its financial statements. At the acquisition date the fair value of a 35% interest in Sorbus is £1.8 million and net assets of Sorbus have a carrying amount of £4.6 million before taking into account the following.

Included in the net assets of Sorbus is a business that Maackia has put on the market for immediate sale. This business has a carrying amount of £500,000, a fair value of £760,000 and a value in use of £810,000. Costs to sell are estimated at £40,000.

Sorbus also has a defined benefit pension plan which has a plan asset of £1.6 million, and an obligation with present value of £1,280,000. Sorbus has recognised the net plan asset of £320,000 in its statement of financial position. Maackia does not anticipate any reduction in contributions as a result of acquiring Sorbus.

Requirement

Under IFRS 3, *Business Combinations*, what (to the nearest £1,000) is the goodwill arising on the acquisition of Sorbus, assuming that Maackia measures the non-controlling interest using the fair value method?

5 Gibbston

On 1 May 20X7 The Gibbston Company acquired 70% of the ordinary share capital of The Crum Company for consideration of £15 million cash payable immediately and £5.5 million payable on 1 May 20X8. Further consideration of £13.3 million cash is payable on 1 May 20Y0 dependent on a range of contingent future events. The management of Gibbston believe there is a 45% probability of paying the amount in full.

The equity of Crum had a carrying amount of £20 million at 1 January 20X7. Crum incurred losses of £3 million evenly over the year ended 31 December 20X7. The carrying amount and fair values of the assets of Crum are the same except that at the acquisition date the fair value relating to plant is £9 million higher than carrying amount. The weighted average remaining useful life of the plant is three years from the acquisition date.

Requirements

Calculate the following amounts (to the nearest £0.1m) in accordance with IFRS 3, *Business Combinations* and IFRS 10, *Consolidated Financial Statements* that would be included in the consolidated financial statements of Gibbston for the year ended 31 December 20X7.

(a) Goodwill

(b) Non-controlling interest in profit/loss

(c) Non-controlling interest included in equity

Note: A rate of 10% is to be used in any discount calculations. The non-controlling interest is measured using the proportion of net assets method.

6 Supermall

This question is based on Illustrative example 2 from IFRS 11.

Two real estate companies (the parties) set up a separate vehicle (Supermall) for the purpose of acquiring and operating a shopping centre. The contractual arrangement between the parties establishes joint control of the activities that are conducted in Supermall. The main feature of Supermall's legal form is that the entity, not the parties, has rights to the assets, and obligations for the liabilities, relating to the arrangement. These activities include the rental of the retail units, managing the car park, maintaining the centre and its equipment, such as lifts, and building the reputation and customer base for the centre as a whole.

The terms of the contractual arrangement are such that:

(1) Supermall owns the shopping centre. The contractual arrangement does not specify that the parties have rights to the shopping centre.

(2) The parties are not liable in respect of the debts, liabilities or obligations of Supermall. If Supermall is unable to pay any of its debts or other liabilities or to discharge its obligations to third parties, the liability of each party to any third party will be limited to the unpaid amount of that party's capital contribution.

(3) The parties have the right to sell or pledge their interests in Supermall.

(4) Each party receives a share of the income from operating the shopping centre (which is the rental income net of the operating costs) in accordance with its interest in Supermall.

Requirement

Explain how Supermall should be classified in accordance with IFRS 11, *Joint Arrangements*.

7 Green and Yellow

Green entered into an agreement with Yellow, a public limited company, on 1 December 20X8. Each of the companies holds one-half of the equity in an entity, Orange, a public limited company, which operates offshore oil rigs. The contractual arrangement between Green and Yellow establishes joint control of the activities that are conducted in Orange. The main feature of Orange's legal form is that Orange, not Green or Yellow, has rights to the assets, and obligations for the liabilities, relating to the arrangement.

The terms of the contractual arrangement are such that:

(1) Orange owns the oil rigs. The contractual arrangement does not specify that Green and Yellow have rights to the oil rigs.

(2) Green and Yellow are not liable in respect of the debts, liabilities or obligations of Orange. If Orange is unable to pay any of its debts or other liabilities or to discharge its obligations to third parties, the liability of each party to any third party will be limited to the unpaid amount of that party's capital contribution.

(3) Green and Yellow have the right to sell or pledge their interests in Orange.

(4) Each party receives a share of the income from operating the oil rig in accordance with its interest in Orange.

Green wants to account for the interest in Orange by using the equity method, and wishes for advice on the matter.

Green also owns a 10% interest in a pipeline, which is used to transport the oil from the offshore oil rig to a refinery on the land. Green has joint control over the pipeline and has to pay its share of the maintenance costs. Green has the right to use 10% of the capacity of the pipeline. Green wishes to show the pipeline as an investment in its financial statements to 30 November 20X9.

Requirement

Discuss how the above arrangements would be accounted for in Green's financial statements.

8 Stuhr

On 1 January 20X6 The Stuhr Company acquired 30% of the ordinary share capital of the Bismuth Company by the issue of one million shares with a fair value of £4.00 each. From that date Stuhr did not exercise significant influence over Bismuth, and accounted for the investment at fair value with changes in value included in profit or loss. At 1 January 20X6 the net assets of Bismuth had a carrying amount of £6.4 million and a fair value of £7.2 million.

On 1 March 20X7, Stuhr bought a further 50% of the ordinary share capital of Bismuth by issuing a further 2 million shares with a fair value of £5.00 each. At that date the net assets of Bismuth had a carrying amount of £7.8 million and a fair value of £8.4 million. The non-controlling interest had a fair value of £2 million, and the 30% interest already held by Stuhr had a fair value of £3 million.

Stuhr values the non-controlling interest using the full goodwill method.

Requirement

What is the goodwill figure in the consolidated statement of financial position at 31 December 20X7, in accordance with IFRS 3, *Business Combinations*?

9 Fleurie

Fleurie bought 85% of Merlot on 30 June 20X3, providing consideration in the form of £2 million cash immediately and a further £1.5 million cash conditional upon earnings targets being met.

At acquisition, the net assets of Merlot were £2.2 million and the fair value of the 15% not purchased was £350,000. Fleurie measures the non-controlling interest using the full goodwill method.

On 31 December 20X7, as part of a long-term strategy, Fleurie sold a 15% stake from its 85% holding in Merlot, realising proceeds of £560,000. At this date the net assets of Merlot were £3.5 million.

Requirement

What impact does the disposal have on the financial statements of the Fleurie Group?

10 Chianti

Chianti purchased 95% of the 100,000 £1 ordinary share capital of Barolo on 1 January 20X2, giving rise to goodwill of £70,000. At this date Barolo's retained earnings were £130,000. There were no other reserves.

On 30 September 20X2, when the net assets of Barolo were £320,000, Chianti disposed of the majority of its holding in Barolo, retaining just 5% of share capital, with a fair value of £20,000. Chianti received £340,000 consideration for the sale.

The following information is relevant:

- Goodwill in Barolo has been impaired by £20,000.
- In April 20X2, Barolo revalued a plot of land from £50,000 to £75,000. This land was held with the intention of building a new head office on it.
- The non-controlling interest is valued using the proportion of net assets method.

Requirement

What is the impact of the disposal on the Chianti Group statement of profit or loss and other comprehensive income?

11 Porter

The following consolidated financial statements relate to Porter, a public limited company:

Porter group: statement of financial position as at 31 May 20X6

	20X6 £m	20X5 £m
Non-current assets		
Property, plant and equipment	958	812
Goodwill	15	10
Investment in associate	48	39
	1,021	861
Current assets		
Inventories	154	168
Trade receivables	132	112
Financial assets at fair value through profit or loss	16	0
Cash and cash equivalents	158	48
	460	328
	1,481	1,189
Equity attributable to owners of the parent		
Share capital (£1 ordinary shares)	332	300
Share premium account	212	172
Retained earnings	188	165
Revaluation surplus	101	54
	833	691
Non-controlling interests	84	28
	917	719

Non-current liabilities		
Long-term borrowings	380	320
Deferred tax liability	38	26
	418	346
Current liabilities		
Trade and other payables	110	98
Interest payable	8	4
Current tax payable	28	22
	146	124
	1,481	1,189

Porter group: statement of profit or loss and other comprehensive income for the year ended 31 May 20X6

	£m
Revenue	956
Cost of sales	(634)
Gross profit	322
Other income	6
Distribution costs	(97)
Administrative expenses	(115)
Finance costs	(16)
Share of profit of associate	12
Profit before tax	112
Income tax expense	(34)

	£m
Profit for the year	78
Other comprehensive income (items that will not be reclassified to profit or loss):	
Gains on property revaluation	58
Share of other comprehensive income of associate	8
Income tax relating to items of other comprehensive income	(17)
Other comprehensive income for the year, net of tax	49
Total comprehensive income for the year	127

	£m
Profit attributable to:	
Owners of the parent	68
Non-controlling interests	10
	78

Total comprehensive income attributable to:	
Owners of the parent	115
Non-controlling interests	12
	127

The following information relates to the consolidated financial statements of Porter:

(1) During the period, Porter acquired 60% of a subsidiary. The purchase was effected by issuing shares of Porter on a 1 for 2 basis, at their market value on that date of £2.25 per share, plus £26 million in cash.

A statement of financial position of the subsidiary, prepared at the acquisition date for consolidation purposes, showed the following position:

	£m
Property, plant and equipment	92
Inventories	20
Trade receivables	16
Cash and cash equivalents	8
	136
Share capital (£1 shares)	80
Reserves	40
	120
Trade payables	12
Income taxes payable	4
	136

An impairment test conducted at the year end resulted in a write down of goodwill relating to another wholly owned subsidiary. This was charged to cost of sales.

Group policy is to value non-controlling interests at the date of acquisition at the proportionate share of the fair value of the acquiree's identifiable assets acquired and liabilities assumed.

(2) Depreciation charged to the consolidated profit or loss amounted to £44 million. There were no disposals of property, plant and equipment during the year.

(3) Other income represents gains on financial assets at fair value through profit or loss. The financial assets are investments in quoted shares. They were purchased shortly before the year end with surplus cash, and were designated at fair value through profit or loss as they are expected to be sold shortly after the year end. No dividends have yet been received.

(4) Included in 'trade and other payables' is the £ equivalent of an invoice for 102 million shillings for some equipment purchased from a foreign supplier. The asset was invoiced on 5 March 20X6, but had not been paid for at the year end, 31 May 20X6.

Exchange gains or losses on the transaction have been included in administrative expenses. Relevant exchange rates were as follows:

Shillings to £1

5 March 20X6 = 6.8

31 May 20X6 = 6.0

(5) Movement on retained earnings was as follows:

	£m
At 31 May 20X5	165
Total comprehensive income	68
Dividends paid	(45)
At 31 May 20X6	188

Requirement

Prepare a consolidated statement of cash flows for Porter for the year ended 31 May 20X6 in accordance with IAS 7, *Statement of Cash Flows*, using the indirect method.

Notes to the statement of cash flows are not required.

12 Tastydesserts

The following are extracts from the financial statements of Tastydesserts and one of its wholly owned subsidiaries, Custardpowders, the shares in which were acquired on 31 October 20X2.

Statements of financial position

	Tastydesserts and subsidiaries		Custardpowders
	31 December 20X2 £'000	31 December 20X1 £'000	31 October 20X2 £'000
Non-current assets			
Property, plant and equipment	4,764	3,685	694
Goodwill	42	–	–
Investment in associates	2,195	2,175	
	7,001	5,860	694
Current assets			
Inventories	1,735	1,388	306
Receivables	2,658	2,436	185
Bank balances and cash	43	77	7
	4,436	3,901	498

	Tastydesserts and subsidiaries		Custardpowders
	31 December 20X2 £'000	31 December 20X1 £'000	31 October 20X2 £'000
	11,437	9,761	1,192
Equity			
Share capital	4,896	4,776	400
Share premium	216	–	–
Retained earnings	2,540	2,063	644
	7,652	6,839	1,044

Non-current liabilities			
Loans	1,348	653	–
Deferred tax	111	180	
	1,459	833	
Current liabilities			
Payables	1,915	1,546	148
Bank overdrafts	176	343	–
Current tax payable	235	200	
	2,326	2,089	148
	11,437	9,761	1,192

Consolidated statement of profit or loss and other comprehensive income for the year ended 31 December 20X2

	£'000
Profit before interest and tax	546
Finance costs	–
Share of profit of associates	120
Profit before tax	666
Income tax expense	126
Profit/total comprehensive income for the year	540
Attributable to:	
Owners of the parent	540
Non-controlling interests	-
	540

The following information is also given:

(1) The consolidated figures at 31 December 20X2 include Custardpowders.

(2) The amount of depreciation on property, plant and equipment during the year was £78,000. There were no disposals.

(3) The cost on 31 October 20X2 of the shares in Custardpowders was £1,086,000 comprising the issue of £695,000 unsecured loan stock at par, 120,000 ordinary shares of £1 at a value of 280p each and £55,000 in cash.

(4) No write-down of goodwill was required during the period.

(5) Total dividends paid by Tastydesserts (parent) during the period amounted to £63,000.

Requirement

Prepare a consolidated statement of cash flows for Tastydesserts and subsidiaries for the year ended 31 December 20X2 using the indirect method.

Notes to the statement of cash flows are not required.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Goodwill on acquisition of DEF

		£'000
Consideration	3m shares issued at £7	21,000
	1m additional shares at £7	7,000
Non-controlling interest	10% × £18m	1,800
		29,800
Net assets acquired		(18,000)
Goodwill		11,800

Answer to Interactive question 2

£99,750

Net assets of Ives

	At acquisition	At reporting date
	£	£
Share capital	100,000	100,000
Retained earnings	150,000	245,000
FV adjustment	50,000	–
PURP in inventory		(6,000)
	300,000	339,000

Non-controlling interest

	£	£
Share of net assets (25% × 339,000)		84,750
Share of goodwill:		
NCI at acquisition date at fair value	90,000	
NCI share of net assets at acquisition date (25% × £300,000)	(75,000)	
		15,000
		99,750

Answer to Interactive question 3

Goodwill on consolidation of Kono Ltd

	£m	£m
Consideration (£2.00 × 6m)		12.0
Non-controlling interest Share capital	8.0	
Pre-acquisition retained earnings	4.4	
Fair value adjustments		
Property, plant and equipment (16.8 – 16.0)	0.8	
Inventories (4.2 – 4.0)	0.2	
Contingent liability	(0.2)	
	13.2	
Non-controlling interest (25%)		3.3
		15.3
Net assets acquired		(13.2)
Goodwill		2.1

Notes

1 It is assumed that the market value (ie, fair value) of the loan stock issued to fund the purchase of the shares in Kono Ltd is equal to the price of £12 million. IFRS 3 requires goodwill to be calculated by comparing the consideration transferred plus the non-controlling interest, valued either at fair value or, in this case, as a percentage of net assets, with the fair value of the identifiable net assets of the acquired business or company.

2 Share capital and pre-acquisition profits represent the book value of the net assets of Kono Ltd at the date of acquisition. Adjustments are then required to this book value in order to give the fair value of the net assets at the date of acquisition. For short-term monetary items, fair value is their carrying value on acquisition.

3 The fair value of property, plant and equipment should be determined by market value or, if information on a market price is not available (as is the case here), then by reference to depreciated replacement cost, reflecting normal business practice. The net replacement cost (ie, £16.8m) represents the gross replacement cost less depreciation based on that amount, and so further adjustment for extra depreciation is unnecessary.

4 Raw materials are valued at their replacement cost of £4.2 million.

5 The rationalisation costs cannot be reported in pre-acquisition results under IFRS 3, as they are not a liability of Kono Ltd at the acquisition date.

6 The fair value of the loan is the present value of the total amount payable (principal and interest). The present value of the loan is the same as its par value.

7 The contingent liability should be included as part of the acquisition net assets of Kono even though it is not deemed probable and therefore has not been recognised in Kono's individual accounts. However, the disclosed amount is not necessarily the fair value at which a third party would assume the liability. If the probability is low, then the fair value is likely to be lower than £200,000.

Answer to Interactive question 4

IFRS 11 gives examples in the oil, gas and mineral extraction industries. In such industries companies may, say, jointly control and operate an oil or gas pipeline. Each company transports its own products down the pipeline and pays an agreed proportion of the expenses of operating the pipeline (perhaps based on volume). In this case the parties have rights to assets (such as exploration permits and the oil or gas produced by the activities).

A further example is a property which is jointly controlled, each venturer taking a share of the rental income and bearing a portion of the expense.

Answer to Interactive question 5

Consolidated statement of profit or loss for the year ended 30 June 20X9

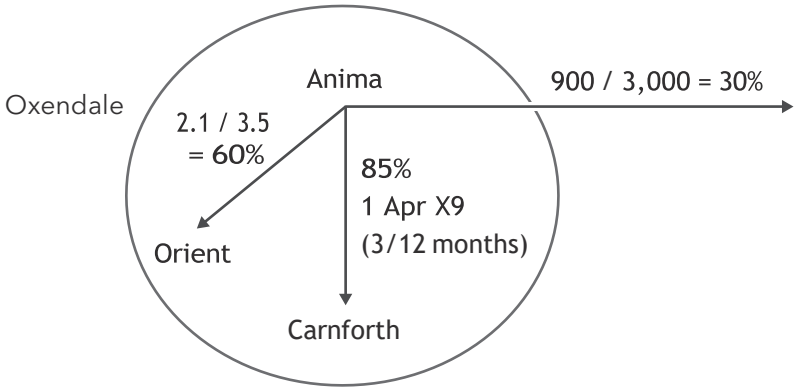
	£
Revenue (W2)	2,291,300
Cost of sales (W2)	(1,238,125)
Gross profit	1,053,175
Operating expenses (W2)	(263,980)
Profit from operations	789,195
Share of profits of associate (W6)	51,383
Profit before tax	840,578
Income tax expense (W2)	(240,685)
Profit for the year	599,893
Profit attributable to:	
Owners of Anima plc (Bal)	517,579
Non-controlling interest (W5)	82,314
	599,893

Consolidated statement of financial position (extract)

	£
Equity attributable to owners of Anima plc	
Ordinary share capital	4,000,000
Retained earnings (W7)	1,879,116
	5,879,116
Non-controlling interest (W8)	2,112,600
Total equity	7,991,716

WORKINGS

(1) **Group structure**



(2) **Consolidation schedule**

	Anima	Orient	Carnforth	Adjustments	Total
			3/12		
Revenue	1,410,500	870,300	160,000	(149,500)	2,291,300
Cost of sales					
Per question	(850,000)	(470,300)	(54,875)	149,500	(1,238,125)
PURP (W4)	(9,750)				
PURP (W4)	(2,700)				
Operating expenses					
Per question	(103,200)	(136,000)	(23,780)		(263,980)
Fair value adj (dep) (W3)			(1,000)		
Tax	(137,100)	(79,200)	(24,385)		(240,685)
PAT		184,800	55,960		

(3) **Fair value adjustment**

Additional fair value £320,000

Buildings £320,000 × 50% = £160,000

Additional depreciation charge in year £160,000 / 40 years × 3/12 months = £1,000

(4) **Unrealised profit**

Oxendale	Orient	%
207,000	149,500	115
(180,000)	(130,000)	(100)
27,000	19,500	15

Orient – £19,500 × ½ = £9,750

Oxendale – £27,000 × 1/3 = £9,000

Anima share of Oxendale PURP – £9,000 × 30% = £2,700

(5) Non-controlling interest

Orient Ltd (40% × £184,800 (W2)) = £73,920

Carnforth Ltd (15% × £55,960 (W2)) = £8,394

Non-controlling interest = £73,920 + £8,394 = £82,314

(6) Share of profits of associate

	£
Profit for the year	204,610
Anima share × 30%	61,383
Less impairment for year	(10,000)
	51,383

(7) **Consolidated retained earnings**

	£
Anima plc – c/fwd	1,560,000
Less PURP with Orient (W4)	(9,750)
Less PURP with Oxendale (W4)	(2,700)
Orient Ltd (60% × (580 – 195))	231,000
Carnforth Ltd (85% × 55,960) (W2)	47,566
Oxendale Ltd ((30% × (340 – 130)) – 10 (impairment))	53,000
	1,879,116

(8) **Non-controlling interest – SFP**

	£	£
Orient Ltd		
FV of NCI at acquisition date	1,520,000	
Share of post-acquisition reserves ((580 – 195) × 40%)	154,000	
		1,674,000
Carnforth Ltd		
Net assets per question	2,605,000	
Fair value adjustment (increase)	320,000	
Less extra depreciation on FV adj	(1,000)	
	2,924,000	
NCI – 2,924,000 × 15%		438,600
		2,112,600

Answer to Interactive question 6

6.1 **Consolidated statement of financial position as at 31 March 20X4**

Assets	**£**	**£**
Non-current assets		
Property, plant and equipment		
(660,700 + 635,300 + 24,000 – 1,000 (W1) – 3,000 (W7))		1,316,000
Intangibles (101,300 + 144,475 (W2))		245,775
Investment in joint venture (W6)		93,600
		1,655,375
Current assets		
Inventories (235,400 + 195,900 – 2,400 (W5))	428,900	
Trade and other receivables (174,900 + 78,800 – 50,000)	203,700	
Cash and cash equivalents (23,700 + 11,900 + 10,000)	45,600	
		678,200
Total assets		2,333,575
Equity and liabilities		
Equity attributable to owners of Preston plc		
Ordinary share capital		100,000
Revaluation surplus		125,000
Retained earnings (W4)		1,099,550
		1,324,550
Non-controlling interest (W3)		190,025
Total equity		1,514,575
Current liabilities		
Trade and other payables (151,200 + 101,800 – 40,000)	213,000	
Taxation (85,000 + 80,000)	165,000	
Deferred consideration	441,000	
		819,000
Total equity and liabilities		2,333,575

WORKINGS

(1) **Net assets – Longridge Ltd**

	Year end £	Acquisition £	Post acq £
Share capital	500,000	500,000	
Retained earnings			
Per Q	312,100	206,700	
Less intangible (72,000 + 18,000)	(72,000)	(90,000)	
Fair value adj re PPE (120,000 – (92,000 × 48/46))	24,000	24,000	
Dep thereon (24,000 × 2/48)	(1,000)	–	
PPE PURP (W7)	(3,000)		
	760,100	640,700	119,400

(2) **Goodwill – Longridge Ltd**

	£
Consideration transferred (250,000 + (441,000 – 41,000 (W4)))	650,000
Non-controlling interest at acquisition (640,700 (W1) × 25%)	160,175
	810,175
Net assets at acquisition (W1)	(640,700)
	169,475
Impairment to date	(25,000)
	144,475

(3) **Non-controlling interest – Longridge Ltd**

	£
Non-controlling interest at acquisition (W2)	160,175
Share of post-acquisition reserves (119,400 (W1) × 25%)	29,850
	190,025

(4) **Retained earnings**

	£
Preston plc	1,084,800
Unwinding of discount on deferred consideration: Two years (441,000 – (441,000 / 1.05^2))	(41,000)
Less PURP (Longridge Ltd) (W5)	(2,400)
Longridge Ltd (119,400 (W1) × 75%)	89,550
Chipping Ltd (W6)	3,600
Less impairments to date (25,000 + 10,000)	(35,000)
	1,099,550

(5) **Inventory PURPs**

		Chipping Ltd	Longridge Ltd
	%	£	£
SP	100	15,000	12,000
Cost	(80)	(12,000)	(9,600)
GP	20	3,000	2,400

(6) **Investment in joint venture – Chipping Ltd**

	£	£
Cost		100,000
Add post-acquisition profits	12,000	
Less PURP (W5)	(3,000)	
	9,000	
× 40%		3,600
		103,600
Less impairment to date		(10,000)
		93,600

(7) **PPE PURP – Longridge Ltd.**

	£
Asset now in Preston plc's books at 15,000 × 1/3	5,000
Asset would have been in Longridge Ltd's books at 10,000 × 1/5	(2,000)
	3,000

6.2 **Goodwill journal entries**

		£	£
DEBIT	Intangibles – goodwill	39,160	
DEBIT	Share capital	320,000	
DEBIT	Retained earnings	112,300	
CREDIT	Investments		385,000
CREDIT	Non-controlling interest (320,000 + 112,300) × 20%		86,460

Answer to Interactive question 7

7.1 Goodwill (at date control obtained)

	$m	$m
Consideration transferred		480
NCI (20% × 750)		150
Fair value of previously held equity interest ($480m × 20/60)		160
Fair value of identifiable assets acquired and liabilities assumed		
Share capital	300	
Retained earnings	450	
		(750)
		40

7.2 Profit on derecognition of investment

	$m
Fair value at date control obtained (see part 1)	160
Cost	(120)
	40

Answer to Interactive question 8

8.1 Answer as follows:

(a) **Complete disposal at year end (80% to 0%)**

Consolidated statement of financial position as at 30 September 20X8

	Streatham	Balham	Adj 1	Adj 2	Disposal	Consolidated
	£'000	£'000	£'000	£'000	£'000	£'000
Non-current assets	360	270			(270)	360
Investment	324		(324)			
Goodwill			36		(36)	
Current assets	370	370			370 + 650	1,020
						1,380
Share capital	540	180	(180)			540
Reserves	414	360	(180)	(36)	182	740
NCI			72	36	(108)	–
Current liabilities	100	100			(100)	100
						1,380

Consolidated statement of profit or loss for the year ended 30 September 20X8

	Streatham	Balham	Adj 1	Adj 2	Disposal	Consolidated
	£'000	£'000	£'000	£'000	£'000	£'000
Profit before tax	153	126				279
Profit on disp'l					182	182
Tax	(45)	(36)				(81)
	108	90				380
Owners of parent	108	90		(18)	182	362
NCI				18		18
						380

Notes

1 The above consolidation schedules are prime examples of where the excel file should be used. For full marks to be awarded, the schedule must be added across correctly, and it is easier to do this using an excel file. It is important that you know which way round the adjustments go. You need to copy the schedule into the word-processing area and show your workings. The examiners have made it clear that the markers will not check formulae in the excel file cells so it is vital to make it clear where your figures have come from.

2 The above recommendation of using an excel file for consolidation adjustments also applies to questions which ask you to adjust financial statements to correct financial reporting treatments. Adjustments and totals are shown in rows and columns. Such questions are often a feature of Question 2 in the exam.

WORKINGS

(1) **Goodwill**

	£'000
Consideration transferred	324
NCI: 20% × (180 + 180)	72
	396
Net assets (180 + 180)	(360)
	36

Consolidation adjustment journal		£'000	£'000
DEBIT	Goodwill	36	
DEBIT	Share capital	180	
DEBIT	Reserves	180	
CREDIT	Investment		324
CREDIT	Non-controlling interest		72

To recognise the acquisition and related goodwill and non-controlling interest.

(2) **Allocate profits between acquisition date and disposal date to NCI**

	£
Post-acquisition profits (360 – 180)	180,000
NCI share (20%)	36,000

Of these, £18,000 (90,000 × 20%) relate to the current year.

Consolidation adjustment journal (SOFP)		**£'000**	**£'000**
DEBIT	Reserves	36	
CREDIT	Non-controlling interest		36

To allocate the NCI share of post-acquisition profits.

In addition, 20% of the profits of Balham Co arising in the year are allocated to the NCI:

Consolidation adjustment journal (SPL)		**£'000**	**£'000**
DEBIT	Profits attributable to owners of parent	18	
CREDIT	Non-controlling interest in profit		18

To allocate the NCI share of post-acquisition profits in the current year.

(3) **Profit on disposal of Balham Co**

	£'000	**£'000**
Fair value of consideration received		650
Less:		
net assets at disposal	540	
goodwill	36	
NCI (540 × 20%)	(108)	
		(468)
		182

Consolidation adjustment journal (SPL)		**£'000**	**£'000**
DEBIT	Cash	650	
DEBIT	NCI	108	
DEBIT	Disposal date liabilities of Balham	100	
CREDIT	Goodwill		36
CREDIT	Disposal date current assets of Balham		370
CREDIT	Disposal date non-current assets of Balham		270
CREDIT	Reserves		182

To recognise the group gain on disposal of Balham Co.

(b) **Partial disposal: subsidiary to subsidiary (80% to 60%)**

Consolidated statement of financial position as at 30 September 20X8

	Streatham	Balham	Adj 1 (part (a))	Adj 2 (part (a))	Disposal	Consolid'd
	£'000	£'000	£'000	£'000	£'000	£'000
Non-current assets	360	270				630
Investment	324		(324)			
Goodwill			36			36
Current assets	370	370			160	900
						1,566
Share capital	540	180	(180)			540
Reserves	414	360	(180)	(36)	52	610
NCI			72	36	108	216
Current liabilit's	100	100				200
						1,566

Consolidated statement of profit or loss for the year ended 30 September 20X8

	Streatham	Balham	Adj 1	Adj 2	Disposal	Consolid'd
	£'000	£'000	£'000	£'000	£'000	£'000
Profit before tax	153	126				279
Profit on disp'l						–
Tax	(45)	(36)				(81)
	108	90				198
Owners of parent	108	90		(18)		180
NCI				18		18
						198

WORKING

Disposal

Adjustment is made to equity as control is not lost.

	£'000
NCI before disposal 80% (360 + 180)	432
NCI after disposal 60% (360 + 180)	(324)
Required adjustment	108

Consolidation adjustment journal (SOFP)		£'000	£'000
DEBIT	Current assets (cash)	160	
CREDIT	NCI		108
CREDIT	Reserves		52

8.2 Answer as follows:

(a) **Partial disposal: subsidiary to associate (80% to 40%) Profit on disposal**

	£'000	£'000
Fair value of consideration received		340
Fair value of 40% investment retained		250
Less: Net assets when control lost		
(540 – (90 × 3/12))	517.5	
Goodwill (part (a))	36	
NCI (517.5 × 20%)	(103.5)	
		(450)
		140

Group reserves

	Streatham reserves £'000	Balham £'000	Balham 40% reserve £'000
At date of disposal	414		
Group profit on disposal (W1)	140		
Balham: share of post-acquisition earnings (157.5 × 80%)	126		
Balham: share of post-acquisition earnings (22.5 × 40%)	9		
	689		
At date of disposal (360 – (90 × 3/12))/per question	414	337.5	360.0
Retained earnings at acquisition/on disposal	414	(180.0)	(337.5)
		157.5	22.5

Investment in associate

	£'000
Fair value at date control lost (new 'cost')	250
Share of post-acquisition reserves (90 × 3/12 × 40%)	9
	259

(b) **Partial disposal: subsidiary to financial asset (80% to 40%) Profit on disposal** – as in part (a)

Group reserves

	£'000
Streatham Co's reserves	414
Group profit on disposal (W1)	140
Balham: share of post-acquisition reserves (157.5 (see below) × 80%)	126
	680

Balham

At date of disposal (360 – (90 × 3/12))	337.5
Reserves at acquisition	(180.0)
	157.5

Retained investment – at £250,000 fair value

Answer to Interactive question 9

Consolidated statement of cash flows for the year ended 31 December 20X8

	£'000	£'000
Cash flows from operating activities		
Cash generated from operations (Note 1)	340	
Income taxes paid (W4)	(100)	
Net cash from operating activities		240
Cash flows from investing activities		
Acquisition of subsidiary S Ltd, net of cash acquired (Note 2)	(90)	
Purchase of property, plant and equipment (W1)	(220)	
Net cash used in investing activities		(310)
Cash flows from financing activities		
Proceeds from issue of share capital (1,150 + 650 – 1,000 – 500 – (100 × £2))	100	
Dividend paid to non-controlling interest (W3)	(4)	
Net cash from financing activities		96
Net increase in cash and cash equivalents		26
Cash and cash equivalents at the beginning of period		50
Cash and cash equivalents at the end of period		76

Notes to the statement of cash flows

(1) **Reconciliation of profit before tax to cash generated from operations**

	£'000
Profit before taxation	420
Adjustments for:	
Depreciation	210
	630
Increase in trade receivables (1,370 – 1,100 – 30)	(240)
Increase in inventories (1,450 – 1,200 – 70)	(180)
Increase in trade payables (1,690 – 1,520 – 40)	130
Cash generated from operations	340

(2) **Acquisition of subsidiary**

During the period the group acquired subsidiary S Ltd. The fair values of assets acquired and liabilities assumed were as follows:

	£'000
Cash and cash equivalents	10
Inventories	70
Receivables	30
Property, plant and equipment	190
Trade payables	(40)
Non-controlling interest	(26)
	234
Goodwill	66
Total purchase price	300
Less cash of S Ltd	(10)
Less non-cash consideration	(200)
Cash flow on acquisition net of cash acquired	90

WORKINGS

(1) **PROPERTY, PLANT AND EQUIPMENT**

	£'000		£'000
b/f	2,300	Depreciation	210
On acquisition	190	c/f	2,500
Additions (balancing figure)	220		
	2,710		2,710

(2) **GOODWILL**

	£'000		£'000
b/f	–		

	£'000		£'000
Additions (300 – (90% × 260))	66	Impairment losses (balancing figure)	0
		c/f	66
	66		66

(3) **NON-CONTROLLING INTEREST**

	£'000		£'000
Dividend (balancing figure)	4	b/f	–
c/f	31	On acquisition	26
		CSPL	9
	35		35

(4) **INCOME TAX PAYABLE**

	£'000		£'000
		b/f	100
Cash paid (balancing figure)	100	CSPL	150
c/f	150		
	250		250

Answer to Interactive question 10

(a) **Cash flows from investing activities**

	£'000
Disposal of subsidiary Desdemona Ltd, net of cash disposed of (400 – 20)	380

(b) **Cash flows from investing activities**

	£'000
Purchase of property, plant and equipment (W1)	(1,307)

(c) **Cash flows from financing activities**

	£'000
Dividends paid to non-controlling interest (W2)	(42)

(d) **Note to the statement of cash flows**

During the period the group disposed of subsidiary Desdemona Ltd. The book values of assets and liabilities disposed of were as follows:

	£'000
Cash and cash equivalents	20
Inventories	50
Receivables	39
Property, plant and equipment	390
Payables	(42)

	£'000
Non-controlling interest (W2)	(91)
	366
Profit on disposal	34
Total sale proceeds	400
Less cash of Desdemona Ltd disposed of	(20)
Cash flow on disposal net of cash disposed of	380

(e) **Reconciliation of profit before tax to cash generated from operations**

	£'000
Profit before tax (862 + 20)	882
Adjustments for:	
Depreciation	800
	1,682
Increase in receivables (605 – 417 + 39)	(227)
Increase in inventories (736 – 535 + 50)	(251)
Increase in payables (380 – 408 + 42)	14
Cash generated from operations	1,218

WORKINGS

(1) **PROPERTY, PLANT AND EQUIPMENT – CARRYING AMOUNT**

	£'000		£'000
b/f	3,950	c/f	4,067
Additions (balancing figure)	1,307	Disposal of sub	390
		Depreciation charge	800
	5,257		5,257

(2) **NON-CONTROLLING INTEREST**

	£'000		£'000
c/f	482	b/f	512
Disposal of sub (457 × 20%)	91	CSPL	103
Dividends to NCI (balancing figure)	42		
	615		615

Answer to Interactive question 11

11.1 Reasons for reviewing the work of component auditors

The main consideration which concerns the audit of all group accounts is that the holding company's auditors (the 'group' auditors) are responsible to the members of that company for the audit opinion on the whole of the group accounts.

It may be stated (in the notes to the financial statements) that the financial statements of certain subsidiaries have been audited by other firms, but this does not absolve the group auditors from any of their responsibilities.

The auditors of a holding company have to report to its members on the truth and fairness of the view given by the financial statements of the company and its subsidiaries dealt with in the group accounts. The group auditors should have powers to obtain such information and explanations as they reasonably require from the subsidiary companies and their auditors, or from the parent company in the case of overseas subsidiaries, in order that they can discharge their responsibilities as holding company auditors.

The auditing standard ISA (UK) 600 (Revised November 2019), *Special Considerations – Audits of Group Financial Statements (Including the Work of Component Auditors)* clarifies how the group auditors can carry out a review of the audits of subsidiaries in order to satisfy themselves that, with the inclusion of figures not audited by themselves, the group accounts give a true and fair view.

The scope, standard and independence of the work carried out by the auditors of subsidiary companies (the 'component' auditors) are the most important matters which need to be examined by the group auditors before relying on financial statements not audited by them. The group auditors need to be satisfied that all material areas of the financial statements of subsidiaries have been audited satisfactorily and in a manner compatible with that of the group auditors themselves.

11.2 Procedures to be carried out by group auditors in reviewing the component auditors' work

(1) Send a questionnaire to all other auditors requesting detailed information on their work, including:

(a) An explanation of their general approach (in order to make an assessment of the standards of their work)

(b) Details of the accounting policies of major subsidiaries (to ensure that these are compatible within the group)

(c) The component auditors' opinion of the subsidiaries' overall level of internal control, and the reliability of their accounting records

(d) Any limitations placed on the scope of the auditors' work

(e) Any qualifications, and the reasons for them, made or likely to be made to their audit reports

(2) Carry out a detailed review of the component auditors' working papers on each subsidiary whose results materially affect the view given by the group financial statements. This review will enable the **group** auditors to ascertain whether (inter alia):

a. An up to date permanent file exists with details of the nature of the subsidiary's business, its staff organisation, its accounting records, previous year's financial statements and copies of important legal documents.

b. The systems examination has been properly completed, documented and reported on to management after discussion.

c. Tests of controls and substantive procedures have been properly and appropriately carried out, and audit programmes properly completed and signed.

d. All other working papers are comprehensive and explicit.

e. The overall review of the financial statements has been adequately carried out, and adequate use of analytical procedures has been undertaken throughout the audit.

f. The financial statements agree in all respects with the accounting records and comply with all relevant legal requirements and accounting standards.

g. Minutes of board and general meetings have been scrutinised and important matters noted.

h. The audit work has been carried out in accordance with approved auditing standards.

i. The financial statements agree in all respects with the accounting records and comply with all relevant legal and professional requirements.

j. The audit work has been properly reviewed within the firm of auditors and any laid-down quality control procedures adhered to.

k. Any points requiring discussion with the holding company's management have been noted and brought to the **group** auditors' attention (including any matters which might warrant a modification in the audit report on the subsidiary company's financial statements).

l. Adequate audit evidence has been obtained to form a basis for the audit opinion on both the subsidiaries' financial statements and those of the group.

If the **group** auditors are not satisfied as a result of the above review, they should arrange for further audit work to be carried out either by the component auditors on their behalf, or jointly with them. The component auditors are fully responsible for their own work; any additional tests are those required for the purpose of the audit of the group financial statements.

Answer to Interactive question 12

12.1 Intra-group balances should agree because, in the preparation of consolidated accounts, it is necessary to cancel them out. If they do not cancel out then the group accounts will be displaying an item which has no value outside of the group and profits may be correspondingly under- or overstated. The audit procedures required to check that intra-group balances agree would be as follows.

(1) Obtain and review a copy of the parent company's instructions to all group members relating to the procedures for reconciliation and agreement of year-end intra-group balances. Particular attention should be paid to the treatment of 'in transit' items to ensure that there is a proper cut-off.

(2) Obtain a schedule of intra-group balances from all group companies and check the details therein to the summary prepared by the parent company. The details on these schedules should also be independently confirmed in writing by the other auditors involved.

(3) Nil balances should also be confirmed by both the group companies concerned and their respective auditors.

(4) The details on the schedules in (2) above should also be agreed to the details in the financial statements of the individual group companies which are submitted to the parent company for consolidation purposes.

12.2 Where one company in a group supplies goods to another company at cost plus a percentage, and such goods remain in inventory at the year end, then the group inventory will contain an element of unrealised profit. In the preparation of the group accounts, best accounting practice requires that an allowance should be made for this unrealised profit.

In order to verify that intra-group profit in inventory has been correctly accounted for in the group accounts, the audit procedures required would be as follows.

(1) Confirm the group's procedures for identification of such inventory and their notification to the parent company who will be responsible for making the required provision.

(2) Obtain and review schedules of intra-group inventory from group companies and confirm that the same categories of inventory have been included as in previous years.

(3) Select a sample of invoices for goods purchased from group companies and check to see that these have been included in year-end intra-group inventory as necessary and obtain confirmation from component auditors that they have satisfactorily completed a similar exercise.

(4) Check the calculation of the allowance for unrealised profit and confirm that this has been arrived at on a consistent basis with that used in earlier years, after making due allowance for any known changes in the profit margins operated by various group companies.

(5) Check the schedules of intra-group inventory against the various inventory sheets and consider whether the level of intra-group inventory appears to be reasonable in comparison with previous years, ensuring that satisfactory explanations are obtained for any material differences.

Answers to Self-test questions

1 Burdett

Goodwill: £380,000

Excess of assets/liabilities over cost of combination: Nil

	Swain	Thamin
	£'000	£'000
Consideration transferred	1,340	340
NCI (20% × £1.2m) / (40% × £680,000)	240	272
	1,580	612
Net assets of acquiree	(1,200)	(680)
Goodwill / Gain on bargain purchase	380	(68)

See IFRS 3.32 and 34. The second acquisition resulted in an excess over the cost of combination which should be recognised immediately in profit.

2 Sheliak

Depreciation charge: decrease by £10,000

Carrying amount: decrease by £30,000

Depreciation charge: Sheliak (£1m / 8 years)	£125,000
Parotia (£575,000 / 5 years)	£115,000
Decrease	£10,000
Carrying amount: Sheliak (£1m × 3/8 years)	£375,000
Parotia (£575,000 × 3/5 years)	£345,000
Decrease	£30,000

Fair value adjustments not reflected in the books must be adjusted for on consolidation. In this example the depreciation is decreased by the difference between Sheliak's depreciation charge and Parotia's fair value depreciation calculation. The carrying amount is decreased by the difference between Sheliak's carrying value at 31 December 20X7 and Parotia's carrying value at 31 December 20X7.

3 Finch

£207,000

	£'000
Net assets per SFP (£8.1m – £1.3m)	6,800
Fair value adjustment to property	300
Contingent liability	(20)
Adjusted net assets	7,080

Net assets is increased by the fair value adjustment of £300,000 and decreased by the contingent liability, which is recognised in accordance with IFRS 3.23.

Goodwill is therefore calculated as:

	£'000
Consideration transferred	5,100
Non-controlling interest	
Ordinary shares 15% × (£7.08m – £2.5m)	687
Preference shares 60% × £2.5m	1,500
	7,287
Net assets of acquiree	(7,080)
Goodwill	207

4 Maackia

£1,780,000

	£'000	£'000
Consideration transferred		4,800
Non-controlling interest (fair value)		1,800
		6,600
Net assets of acquiree		
Draft	4,600	
Business held for sale ((£760,000 – £40,000) – £500,000)	220	
		(4,820)
Goodwill		1,780

In allocating the cost of the business combination to the assets and liabilities acquired, the business that is on the market for immediate sale should be valued in accordance with IFRS 5 at fair value less costs to sell (IFRS 3.31).

The defined benefit plan net asset should be recognised in accordance with IAS 19 (IFRS 3.26).

5 Gibbston

(a) £4.9m

	£'000	£'000
Consideration transferred: Cash		15,000
Deferred cash (£5.5m/1.1)		5,000
Contingent cash (£13.3m/1.1_3) × 45%		4,500
		24,500
Non-controlling interest: Net assets at 1 Jan 20X7	20,000	
Loss to acquisition (£3m × 4/12)	(1,000)	
Fair value adjustment	9,000	
30% ×	28,000	8,400
		32,900

	£'000	£'000
Net assets of acquiree		(28,000)
Goodwill		4,900

IFRS 3.39 requires contingent consideration capable of reliable measurement to be included at fair value regardless of whether payment is probable.

The net assets at the acquisition date are those at the start of the year adjusted for the fair value increase of plant (£9m) less the losses between the start of the year and the acquisition date.

(b) £1.2m share of loss

	£m
Non-controlling interest share of loss:	
Loss: Acquisition to 31 December (8/12 × £3m)	2
Extra depreciation (£9m/3 years × 8/12)	2
	4 × 30% = £1.2m

(c) £7.2m

	£m
Non-controlling interest in SFP	
Share of the net assets at the date of acquisition (£28m × 30%)	8.4
Share of loss since acquisition (part (b))	(1.2)
	7.2

6 Supermall

Supermall has been set up as a separate vehicle. As such, it could be either a joint operation or joint venture, so other facts must be considered.

There are no facts that suggest that the two real estate companies have rights to substantially all the benefits of the assets of Supermall or an obligation for its liabilities.

Each party's liability is limited to any unpaid capital contribution.

As a result, each party has an interest in the net assets of Supermall and should account for it as a joint venture using the equity method in accordance with IFRS 11, *Joint Arrangements*.

7 Green and Yellow

Orange

Green wishes to account for its arrangement with Yellow using the equity method. It can only do so if the arrangement meets the criteria in IFRS 11, *Joint Arrangements* for a **joint venture**.

A **joint arrangement** is an arrangement, as here, of which two or more parties have joint control. A **joint venture** is a joint arrangement whereby the parties that have control of the arrangement have **rights to the net assets** of the arrangement.

Orange is a **separate vehicle**. As such, it could be either a joint operation or joint venture, so other facts must be considered.

There are no facts that suggest that Green and Yellow have rights to substantially all the benefits of the assets of Orange or an obligation for its liabilities.

Each party's liability is limited to any unpaid capital contribution.

As a result, each party has an interest in the **net assets** of Orange and should account for it as a **joint venture** using the **equity method.**

Pipeline

Since Green has joint control over the pipeline, even though its interest is only 10%, it would not be appropriate to show the pipeline as an investment. This is a **joint arrangement** under IFRS 11.

The pipeline is a **jointly controlled asset**, and it is **not structured through a separate vehicle**. Accordingly, the arrangement is a **joint operation**.

IFRS 11, *Joint Arrangements* requires that a joint operator **recognises line by line the following** in relation to its interest in a joint operation:

(1) Its **assets**, including its share of any jointly held assets

(2) Its **liabilities**, including its share of any jointly incurred liabilities

(3) Its **revenue** from the sale of its share of the output arising from the joint operation

(4) Its **share of the revenue from the sale of the output** by the joint operation, and

(5) Its **expenses**, including its share of any expenses incurred jointly

This treatment is applicable in both the separate and consolidated financial statements of the joint operator.

8 Stuhr

£6,600,000

	£'000
Consideration transferred	10,000
Non-controlling interest (fair value)	2,000
Fair value of retained interest	3,000
	15,000
Fair value of net assets of Bismuth	(8,400)
Goodwill	6,600

9 Fleurie

There is no loss of control and therefore:

- no gain or loss arises on disposal of the 15% holding in Merlot; and
- there is no change to the carrying value of goodwill.

Instead, the transaction is accounted for in shareholders' equity, with any difference between proceeds and the change in the non-controlling interest being recognised in retained earnings.

The non-controlling interest before disposal is the fair value at acquisition plus the fair value of post- acquisition reserves to the date of disposal, as represented by change in net assets:

	£
Fair value at acquisition	350,000
Share of post-acquisition reserves [(£3.5m – £2.2m) × 15%)]	195,000
	545,000
Increase in NCI: 15%/15%	545,000
NCI after change: 30%	1,090,000

The non-controlling interest has therefore increased by £545,000, recorded by:

DEBIT	Cash proceeds	£560,000	
CREDIT	NCI		£545,000
CREDIT	Retained earnings		£15,000

10 Chianti

On disposal of Barolo, the consolidated statement of profit or loss and other comprehensive income will include a gain on disposal of £6,000 (as calculated below).

The group share of the revaluation surplus recognised by Barolo is not reclassified into profit or loss but is transferred from the group revaluation reserve to group retained earnings.

	£	£
Proceeds		340,000
Fair value of interest retained		20,000
		360,000
Net assets	320,000	
Goodwill (£70,000 – £20,000)	50,000	
NCI (5% × 320,000)	(16,000)	
		(354,000)
Gain on disposal		6,000

11 Porter

Porter Group statement of cash flows for the year ended 31 May 20X6

	£m	£m
Cash flows from operating activities		
Profit before taxation	112	
Adjustments for:		
Depreciation	44	
Impairment losses on goodwill (W2)	3	
Foreign exchange loss (W8)	2	

	£m	£m
Investment income – share of profit of associate	(12)	
Investment income – gains on financial assets at fair value through profit or loss	(6)	
Interest expense	16	
	159	
Increase in trade receivables (132 – 112 – 16)	(4)	
Decrease in inventories (154 – 168 – 20)	34	
Decrease in trade payables (110 – 98 – 12 – (W8) 17 PPE payable)	(17)	
Cash generated from operations	172	
Interest paid (W7)	(12)	
Income taxes paid (W6)	(37)	

	£m	£m
Net cash from operating activities		123
Cash flows from investing activities		
Acquisition of subsidiary, net of cash acquired (26 – 8)	(18)	
Purchase of property, plant and equipment (W1)	(25)	
Purchase of financial assets (W4)	(10)	
Dividend received from associate (W3)	11	
Net cash used in investing activities		(42)
Cash flows from financing activities		
Proceeds from issuance of share capital (332 + 212 – 300 – 172 – (80 × 60%/2 × £2.25))	18	
Proceeds from long-term borrowings (380 – 320)	60	
Dividend paid	(45)	
Dividends paid to non-controlling interests (W5)	(4)	
Net cash from financing activities		29
Net increase in cash and cash equivalents		110
Cash and cash equivalents at the beginning of the year		48
Cash and cash equivalents at the end of the year		158

WORKINGS

(1) **Additions to property, plant and equipment**

PROPERTY, PLANT AND EQUIPMENT

	£m		£m
b/d			
Revaluation	58	Depreciation	44
Acquisition of subsidiary	92		
Additions on credit (W8)	15		
∴ Additions for cash	25	c/d	958
	1,002		1,002

(2) **Impairment losses on goodwill**

GOODWILL

	£m		£m
b/d	10		
Acq'n of subsidiary [(80 × 60%/2 × 2.25) + 26 + (120 × 40%) – 120 net assets]	8	∴ Impairment losses	3
		c/d	15
	18		18

(3) **Dividends received from associate**

INVESTMENT IN ASSOCIATE

	£m		£m
b/d	39		
P/L	12	∴ Dividends received	11
OCI	8		
		c/d	48
	59		59

(4) **Purchase of financial assets**

FINANCIAL ASSETS AT FAIR VALUE THROUGH PROFIT OR LOSS

	£m		£m
b/d	0		
P/L	6		
∴ Additions	10		
		c/d	16
	16		16

(5) **Dividends paid to non-controlling interests**

NON-CONTROLLING INTERESTS

	£m		£m
		b/d	28
∴ Dividends paid	4	TCI	12
c/d	84	Acquisition of subsidiary (120 × 40%)	48
	88		88

(6) **Income taxes paid**

INCOME TAX PAYABLE

	£m		£m
		b/d (deferred tax)	26
		b/d (current tax)	22
		P/L	34
∴ Income taxes paid	37	OCI	17
c/d (deferred tax)	38	Acquisition of subsidiary	4
c/d (current tax)	28		
	103		103

(7) **Interest paid**

INTEREST PAYABLE

	£m		£m
		b/d	4
		P/L	16
∴ Interest paid	12		
c/d	8		
	20		20

(8) **Foreign currency transaction**

Transactions recorded on:

		£m	£m
(1) **5 March**			
DEBIT	Property, plant and equipment (102m/6.8)	15	
CREDIT	Payables		15
(2) **31 May**	Payable = 102m/6.0 = £17m		
DEBIT	P/L (Admin expenses)	2	
CREDIT	Payables (17 – 15)		2

12 Tastydesserts

Consolidated statement of cash flows for the year ended 31 December 20X2

	£'000	£'000
Cash flows from operating activities		
Profit before taxation	666	
Adjustments for:		
Depreciation	78	
Share of profit of associates	(120)	
Interest expense		
	624	
Increase in receivables (2,658 – 2,436 – 185)	(37)	
Increase in inventories (1,735 – 1,388 – 306)	(41)	
Increase in payables (1,915 – 1,546 – 148)	221	
Cash generated from operations	767	
Interest paid	–	
Income taxes paid (W5)	(160)	
Net cash from operating activities		607
Cash flows from investing activities		
Acquisition of subsidiary Custardpowders net of cash acquired (55 – 7)	(48)	
Purchase of property, plant and equipment (W1)	463)	
Dividends received from associates (W3)	100	
Net cash used in investing activities		(411)
Cash flows from financing activities		
Dividends paid	(63)	
Net cash used in financing activities		(63)
Net increase in cash and cash equivalents		133
Cash and cash equivalents at beginning of year		(266)
Cash and cash equivalents at end of year		(133)

WORKINGS

(1) **Purchase of property, plant and equipment**

PROPERTY, PLANT AND EQUIPMENT

	£'000		£'000
b/d	3,685		
Acquisition of Custardpowders	694	Depreciation	78
∴ Cash additions	463	c/d	4,764
	4,842		4,842

(2) **Goodwill**

GOODWILL

	£'000		£'000
b/d	-		
Acquisition of Custardpowders (1,086 - (1,044 × 100%))	42	∴ Impairment losses	0
		c/d	42
	42		42

(3) **Dividends received from associates**

INVESTMENT IN ASSOCIATE

	£'000		£'000
b/d	2,175		
Share of profit	120	∴ Dividends received	100
		c/d	2,195
	2,295		2,295

(4) **Reconciliation of share capital**

	£'000
Share capital plus premium b/d	4,776
Issued to acquire sub (120,000 × £2.80)	336
	£'000
Share capital plus premium c/d (4,896 + 216)	5,112

∴ no shares have been issued for cash during the year.

(5) **Income taxes paid**

INCOME TAX PAYABLE

	£'000		£'000
		b/d - current tax	200
		- deferred tax	180
∴ Cash paid	160	P/L	126
c/d- current tax	235		
- deferred tax	111		
	506		506

CA BANGLADESH

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